Shared priority, distributed memory and task-dependent readout: synthesis of nineteen computational experiments

Visual Attention and Working Memory Project

18 October 2026

1 The experimental program

This volume treats the previously trained visual model as one computational subject and applies a common experimental logic to every task. It combines fresh psychophysics, threshold-defined commitment times, spatial-priority perturbations, memory resets, external decoders and population-state analyses. The following nineteen papers contain task-specific hypotheses, methods, results, figures, native condition tables and every-frame representative attention atlases.

The main checkpoint is 204,289, selected under the prior monitoring criterion. The final checkpoint at 220,637 supplies same-movie robustness measurements. Native behavior covers all 437 conditions with 128 trials per condition (55,936 presentations). Additional experiments examine weaker signals, causal manipulations, independent fitting/test populations and explicit VDA counterfactuals. The source data distinguish presentations from unique procedural scenes. Timing manipulations that reuse a scene are clustered together in uncertainty calculations.

2 Behavior across the entire suite

Task	Conditions	Trials	Accuracy	AUC
motion direction	1	128	100.0%	1.000
orientation	1	128	100.0%	1.000
contrast	1	128	100.0%	1.000
spatial frequency	1	128	100.0%	1.000
chromatic increment	1	128	100.0%	1.000
contour	1	128	99.2%	1.000
natural spectrum	1	128	100.0%	1.000
orientation ring	1	128	100.0%	1.000
orientation cued	4	512	99.8%	1.000
motion duration cued	4	512	97.5%	0.999
krauzlis cued motion	3	384	95.8%	0.995
spatial binding	4	512	100.0%	1.000
image recognition	12	1536	94.3%	0.963
vda4	3	384	68.0%	0.755
vda9	3	384	68.5%	0.760
vda16	3	384	64.8%	0.723
memory guided search	288	36864	53.1%	0.724
popout search	24	3072	86.4%	0.938

Task	Conditions	Trials	Accuracy	AUC
motion memory search	81	10368	97.7%	0.999

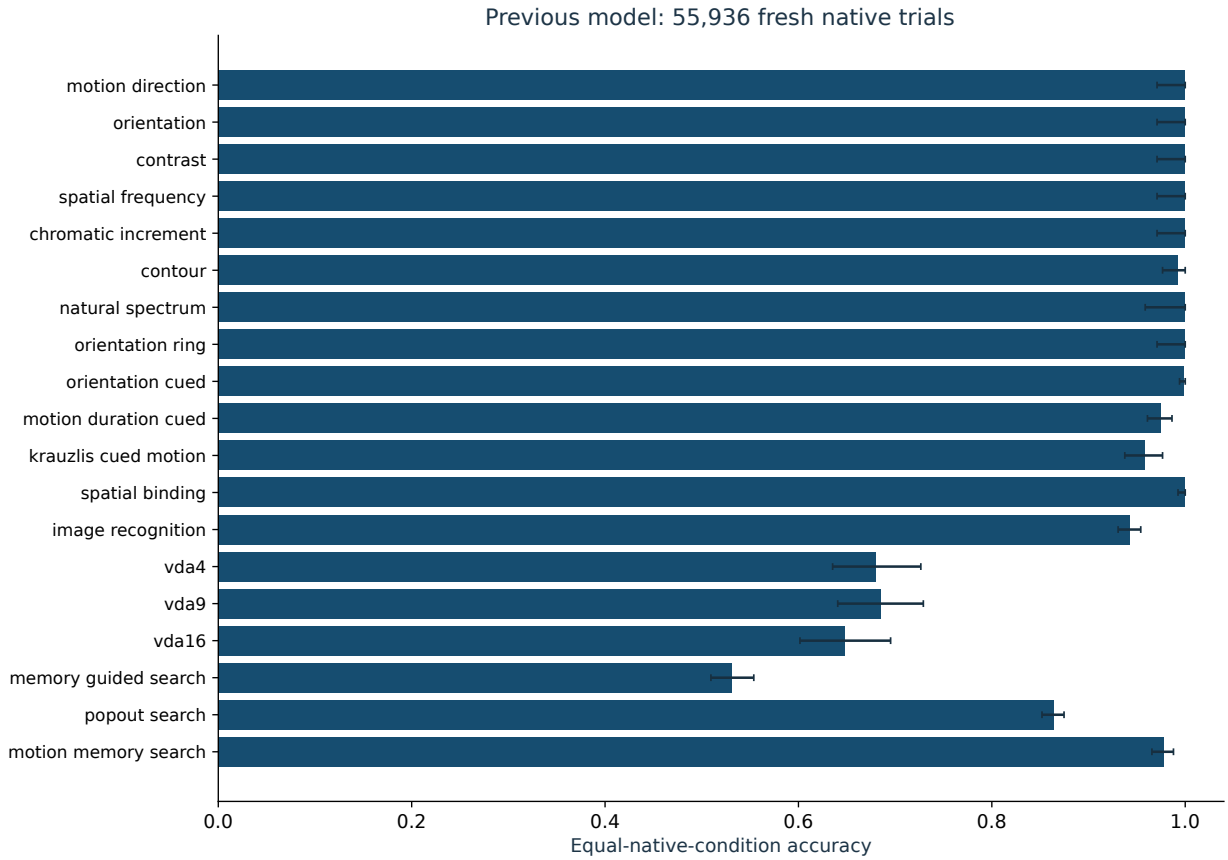


Figure 1: Accuracy of the selected previous model on fresh native trials. Each task averages its native conditions equally; the number of conditions differs across tasks. Intervals preserve procedural scene clusters across timing conditions.

3 Causal organization

Across 108 predesignated dose-four evidence comparisons, 41 survived the cross-task Holm correction at 0.05: 5 benefits and 36 costs. This family includes target and alternative-site stimulation and inhibition in each selected condition. The full exploratory dose and epoch sweeps are reported separately in the task papers.

3.1 Spatial memory and global memory support distinct tests

orientation cued: cls: +0.0 percentage points, spatial: -46.1 percentage points, comparison: +0.0 percentage points.

spatial binding: cls: -0.8 percentage points, spatial: -43.0 percentage points, comparison: +0.0 percentage points.

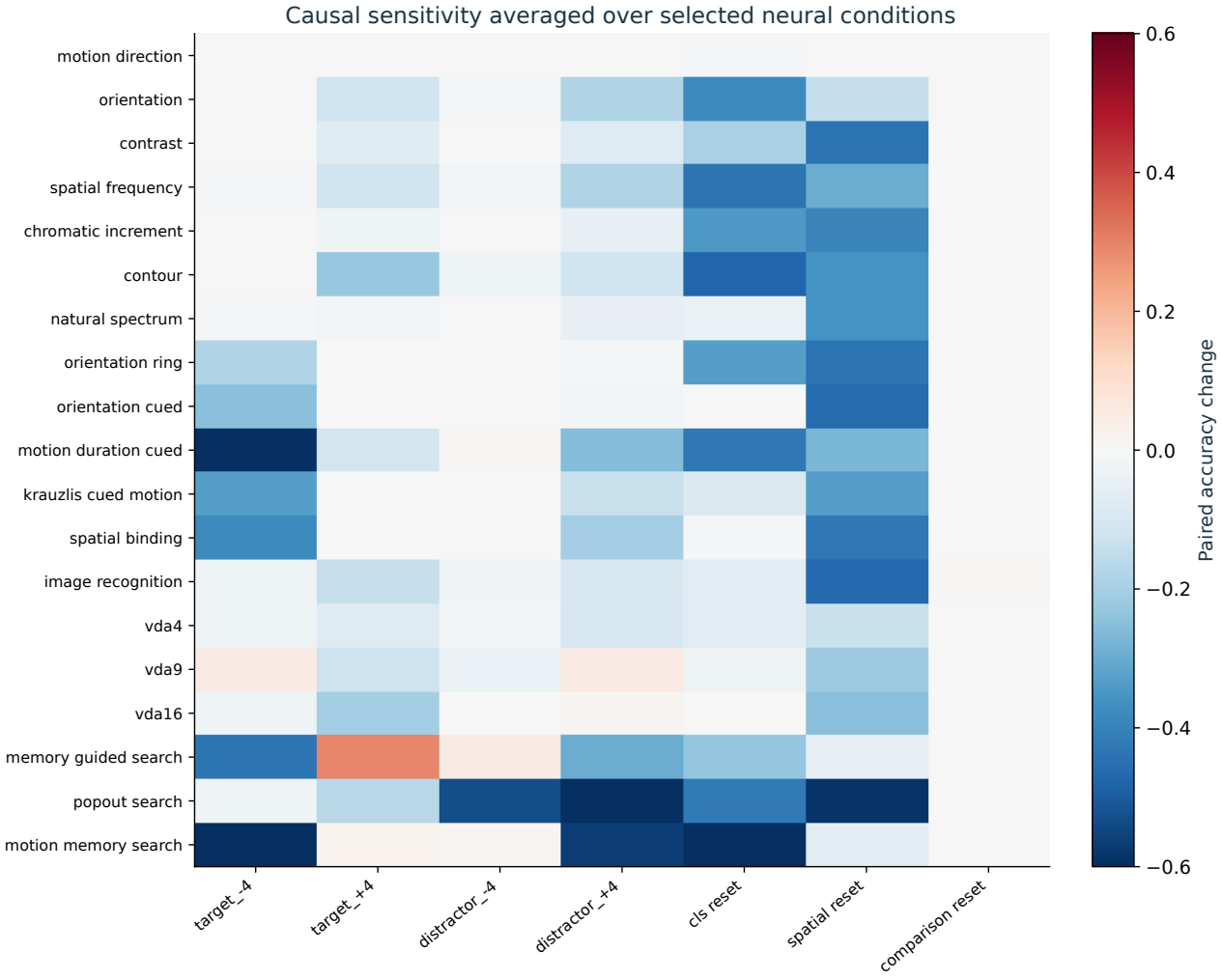


Figure 2: Causal effects of four spatial interventions and three memory resets. Values average the task-specific selected neural conditions, whose full identities are retained in the protocol. Positive values indicate improved paired accuracy; negative values indicate impairment.

image recognition: cls: -6.2 percentage points, spatial: -46.5 percentage points, comparison: +0.8 percentage points.

memory guided search: cls: -23.3 percentage points, spatial: -5.1 percentage points, comparison: +0.0 percentage points.

motion memory search: cls: -72.7 percentage points, spatial: -6.6 percentage points, comparison: +0.0 percentage points.

These resets measure dependence at a specified instant, followed by normal processing of the remaining movie. A state may contain readable content and still be recoverable from other states after a reset. Conversely, a reset can disrupt subsequent computation even when a restricted linear probe reads the state poorly. Pairing causal and representational results addresses this distinction directly.

3.2 The VDA counterfactual

VDA4: catch accuracy change +9.0 points, paired interval [3.1%, 14.8%]; cued accuracy change -53.9 points, paired interval [-60.2%, -48.0%]; uncued accuracy change +0.8 points, paired interval [-1.2%, 3.1%].

VDA9: catch accuracy change +6.6 points, paired interval [1.2%, 12.5%]; cued accuracy change -59.0 points, paired interval [-64.8%, -52.7%]; uncued accuracy change +3.1 points, paired interval [0.4%, 5.9%].

VDA16: catch accuracy change +13.7 points, paired interval [9.4%, 18.4%]; cued accuracy change -27.3 points, paired interval [-32.8%, -21.9%]; uncued accuracy change -2.7 points, paired interval [-5.9%, 0.4%].

These trials pair a certain bottom-right cue with a controlled event at the cue, at top left, or nowhere. The event sequence deliberately tests an expectation violation. All three VDA variants are short enough that the delayed comparison memory is zero; the shared visual-attention and recurrent pathways therefore carry the measured effect. The paired catch condition distinguishes improved detection from increased positive-response tendency.

3.3 Testing the dual effect away from ceiling

The exploratory weak-signal follow-up tested 2, 4 and 8 degrees in every VDA size. A joint dual-effect criterion requires both a cued detection cost and an uncued detection benefit under cue inhibition, with both paired tests surviving Holm correction across all 147 follow-up comparisons from the three VDA tasks. This criterion was met in 0 of nine task-strength settings. The complete corrected family is supplied as `vda_followup_family.json`.

3.4 Holding the array fixed while changing the remembered target

memory guided search: features_distinct_S8_D12_H8: 99.6% balanced query accuracy, template probability gain 0.991; features_similar_S8_D12_H8: 86.7% balanced query accuracy, template probability gain 0.827; fractals_distinct_S8_D12_H8: 45.3% balanced query accuracy, template probability gain 0.250; fractals_similar_S8_D12_H8: 29.3% balanced query accuracy, template probability gain 0.049; objects_distinct_S8_D12_H8: 32.4% balanced query accuracy, template probability gain 0.061; objects_similar_S8_D12_H8: 26.2% balanced query accuracy, template probability gain 0.013.

motion memory search: motion_A20_S12_D12_H12: 95.3% balanced query accuracy, template probability gain 0.925; motion_A90_S12_D12_H12: 99.2% balanced query accuracy, template probability gain 0.987.

The four target queries share identical search pixels within each scene. A query-dependent change therefore reflects the influence of the preceding sample and resulting recurrent state on the same current evidence. The accompanying attention contrast measures whether that influence also changes spatial allocation toward the newly queried location.

3.5 Usable object information can exceed the learned choice

Separate linear probes reconstruct the remembered sample from middle-delay spatial memory and the current candidates from local appearance features at the first array frame. An external squared-distance comparison of those reconstructions then chooses a location. Candidate fitting preserves whole scenes in inner validation, and evaluation uses 128 independent scenes per condition. Probe pairs are fitted separately within each family/similarity condition; the original shared agent remains frozen.

Family	Agent	Joint readout	Paired Holm p
features_distinct	100.0%	100.0%	1
features_similar	88.3%	100.0%	0.0001831
objects_distinct	32.0%	100.0%	7.755e-26
objects_similar	20.3%	93.0%	2.553e-24
fractals_distinct	46.1%	86.7%	1.076e-09
fractals_similar	29.7%	52.3%	0.0002272

The two object conditions show especially large improvements: the available recurrent and visual representations support nearly perfect external matching, while the learned agent makes many errors. This localizes a major component of the failure to how available information is used for comparison and selection. Similar fractals remain more difficult, even though their joint-readout performance also improves. The scientific distinction is between storing information, representing current candidates, and implementing a successful comparison between them.

4 Neuroscientific interpretation

Causal attention experiments motivate manipulating a spatial selection signal while measuring perception. Reversible superior-colliculus and frontal-eye-field inactivation can impair cued motion-change performance (Bollimunta, Bogadhi and Krauzlis, 2018). The model permits a corresponding computational question: how does changing priority at a retinotopic location alter the contribution of evidence there and elsewhere? Because the weights form a normalized distribution, this intervention intrinsically redistributes relative influence. The separate location, event and catch measurements make that redistribution behaviorally interpretable.

The memory analyses distinguish content from use. Visual working-memory experiments motivate measuring a sample representation during stimulus absence (Miller, Erickson and Desimone, 1996). Memory-guided search additionally asks how that representation selects a new target among alternatives (Chelazzi and colleagues, 1993). In this architecture, storage can occur across the spatial and global recurrent states, while the query that selects current evidence is computed from the preceding global state. Testing preserved sample information, subsequent target allocation and memory-reset effects therefore provides three complementary observations of the storage-to-selection process.

Population geometry supplies the temporal link. Dynamic prefrontal coding shows why a changing activity pattern can support ongoing computation (Stokes and colleagues, 2013). Temporal generalization asks whether the same fitted linear rule reads information at different times (King and Dehaene, 2014). Here, a shared training-derived projection prevents the axes from changing merely because each time point was analyzed separately. Cross-time readout, state displacement and continuous remembered-feature reconstruction then jointly characterize what persists and how its code evolves.

5 Using the papers and source data

Each task paper is self-contained. The HTML index links its PDF, editable LaTeX source, behavioral data, causal data and neural summaries. CSV source tables expose individual native conditions, psychometric points, threshold statistics and intervention effects. Raw trial arrays and metadata remain in the local Results directory. The portable paper archive includes reports, figures, code and summary source data; it excludes model checkpoints and raw stimulus/video datasets.

The analysis is conditional on one trained model. The next replication axis is independently trained models. The fixed threshold procedure supplies computational commitment times, and the intervention definitions specify priority-logit and memory-tensor operations precisely. Those definitions support reproducible comparisons to subsequent architectures without requiring a change in the task suite.

References

[1] Bollimunta A, Bogadhi AR, Krauzlis RJ (2018). Comparing frontal eye field and superior colliculus contributions to covert spatial attention. Nature Communications 9:3553. https://www.nature.com/articles/s41467-018-06042-2.

[2] Miller EK, Erickson CA, Desimone R (1996). Neural mechanisms of visual working memory in prefrontal cortex of the macaque. Journal of Neuroscience 16:5154-5167. https://pubmed.ncbi.nlm.nih.gov/8756444/.

[3] Chelazzi L, Miller EK, Duncan J, Desimone R (1993). A neural basis for visual search in inferior temporal cortex. Nature 363:345-347. https://pubmed.ncbi.nlm.nih.gov/8497317/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[6] Wolfe JM (2021). Guided Search 6.0: An updated model of visual search. Psychonomic Bulletin and Review 28:1060-1092. https://pubmed.ncbi.nlm.nih.gov/33547630/.

From local correspondence to cardinal motion decisions

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized motion direction in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 1 native conditions and 128 fresh trials, equal-condition accuracy was 100.0% (95% interval [97.1%, 100.0%]). For the representative condition mixed, the strongest final answer readout among the tested regions was fusion, with held-out macro AUC 1.000. The largest observed impairment in that condition was target_+8_evidence, changing accuracy by -0.8 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

A direction judgement requires relating structure across observations. An isolated random-dot image carries no unique displacement vector. This task therefore provides a direct test of whether a shared recurrent visual system transforms temporal correspondence into a categorical motion decision. Its four alternatives distinguish direction coding from a simple detector of visual change.

The primary psychometric variable is dot displacement in input pixels. Small translations challenge spatial correspondence; large translations within the correspondence bank should produce a concentrated population over candidate offsets. We ask where the direction becomes linearly readable: the appearance encoder, the fused visual tokens, spatial recurrent memory, or the global recurrent state. A strong separation between appearance and fused representations would localize the useful computation to the temporal pathway.

The biological motivation is developed through [1]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Salzman and colleagues showed that stimulating direction-selective MT populations can bias monkeys toward the preferred motion direction of those populations. This motivates linking a representation to behavior by intervention as well as by decoding. In the present architecture, temporal correspondence is first represented locally and then selected spatially. The two operations predict different perturbation signatures: a change in selected location changes which evidence is pooled, whereas a direction-selective change would alter the content of that evidence. Our spatial experiment tests the first causal step. The appearance-to-fusion decoding comparison identifies where the temporal relationship first becomes accessible to a simple readout.

1.2 Biological experiments motivating this task

Experimental question. How does a population representation of image displacement become a direction judgment?

Britten, Shadlen, Newsome and Movshon (1992) In stochastic-motion discrimination, sensitivity of many individual MT neurons approached behavioral sensitivity when stimuli were tailored to the recorded neuron. The study compared neuronal and behavioral discrimination on the same task. [4]

Salzman, Britten and Newsome (1990) Electrical stimulation of direction-selective MT clusters biased reported motion toward the preferred direction of the stimulated population, linking a characterized sensory code to a perceptual decision. [5]

Ling, Liu and Carrasco (2009) External-noise motion experiments separated spatial from feature-based attention. Threshold-versus-noise functions supported different combinations of population gain and tuning, giving behavioral tests of attentional mechanisms. [6]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Which cardinal direction links the two random-dot frames?

Response labels are 0: right; 1: up; 2: left; 3: down. The native suite contains 1 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use mixed.

1.4 Stimulus sequence and spatial layout

For the representative native condition mixed, the sequence has 2 frames, indexed from zero: frame 0: reference; frame 1: comparison. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Dot displacement (pixels)	1.0, 2.0, 3.0
Stimulus family	motion_direction

The movie has two observations. The terminal decision therefore supplies only one post-evidence sample. Confidence thresholds characterize certainty at that sample; they cannot yield a richly resolved latency distribution. The mechanistic comparison is the increase in direction information between the first and second observations. Opposite-quadrant priority changes test how much distributed dot evidence is required for a correct judgement.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 100.0%, with macro AUC 1.000. The matched final checkpoint changed accuracy by +0.00 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was mixed at 100.0%; the highest was mixed at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (0.1) produced 45.3% correct responses, compared with 100.0% at the largest strength (3). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

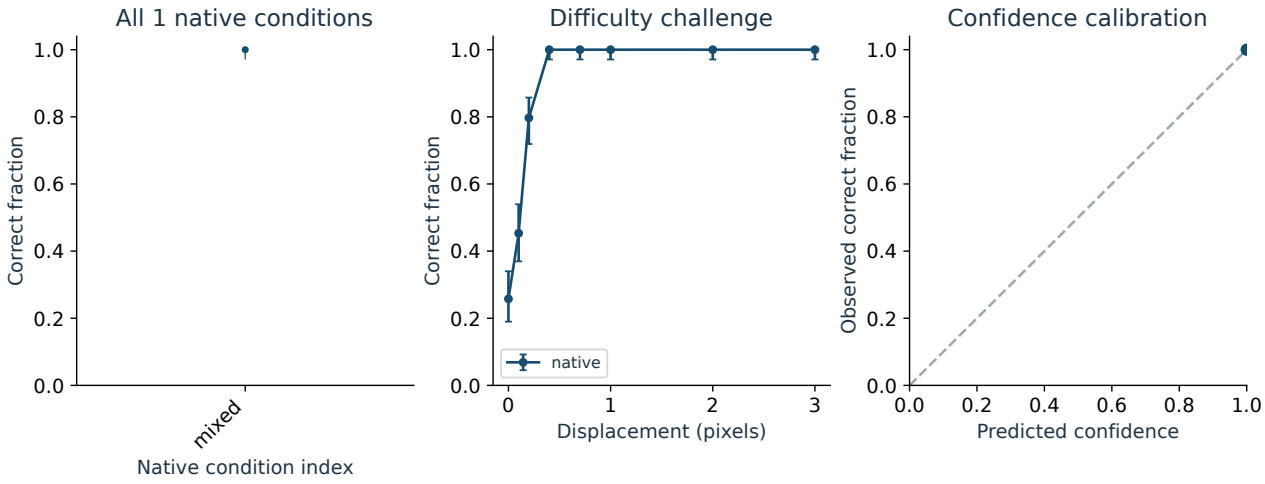


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)[1 - \exp(-(x/a)^b)]$. Here x is signal strength, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 0.1476 task units (bootstrap interval [0.1255, 0.1711]), with slope 3.901 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.013.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	100.0%	32.8%
0.7	100.0%	1.0	100.0%	1.6%
0.8	100.0%	1.0	100.0%	0.0%
0.9	100.0%	1.0	100.0%	0.0%

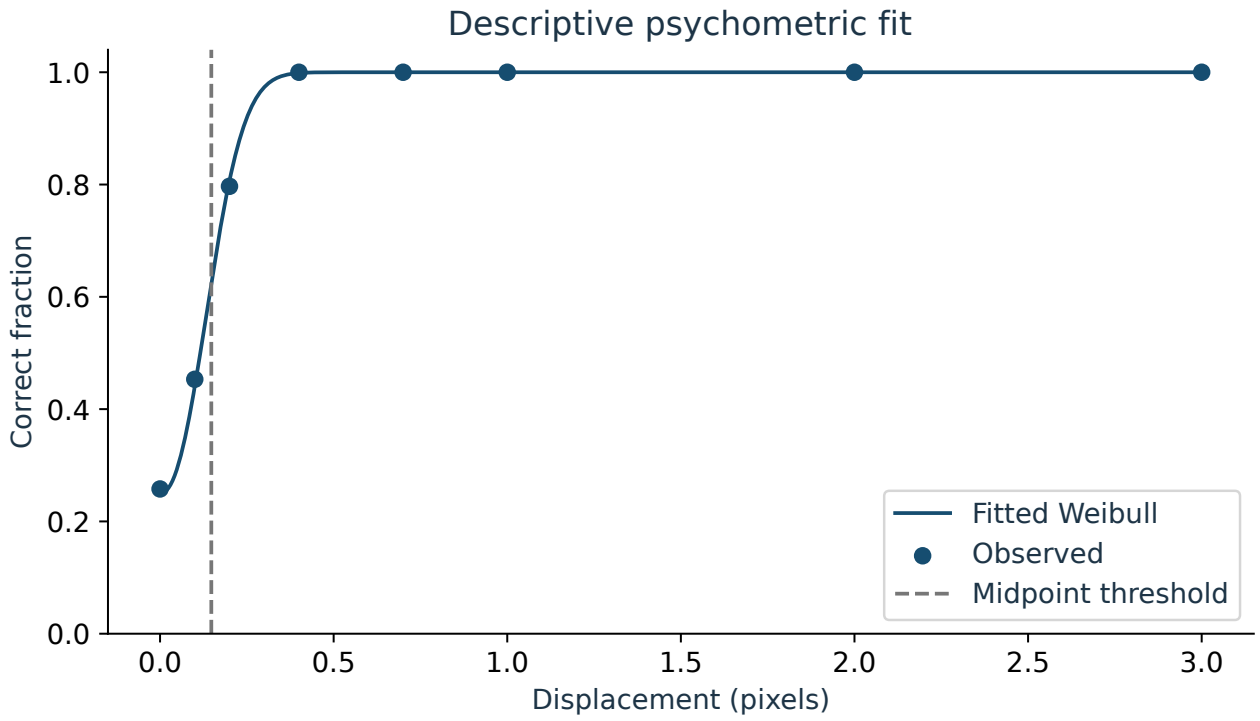


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

The representative condition mixed provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 128 of 128 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

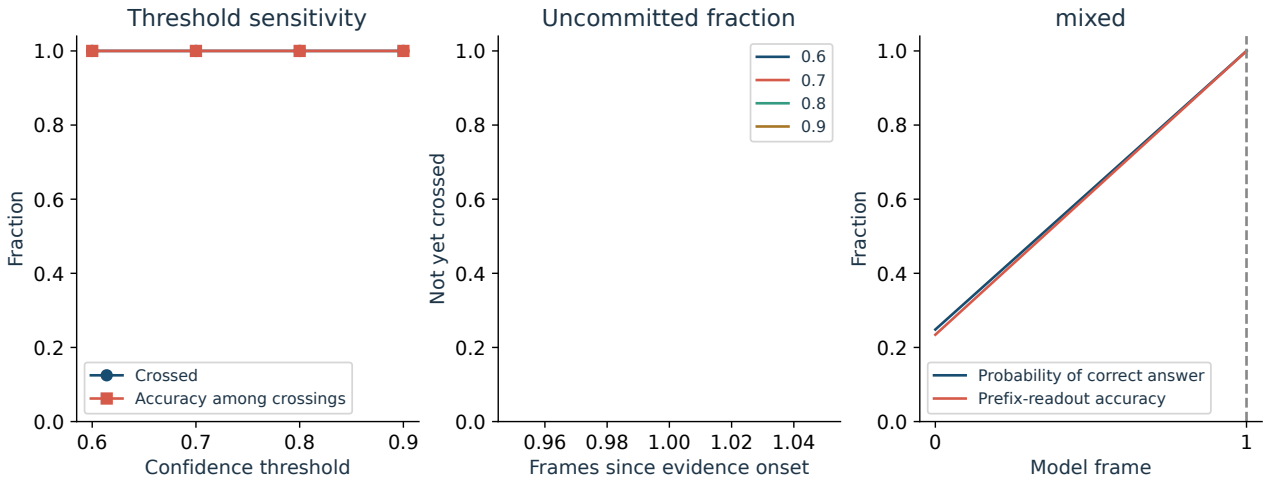


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

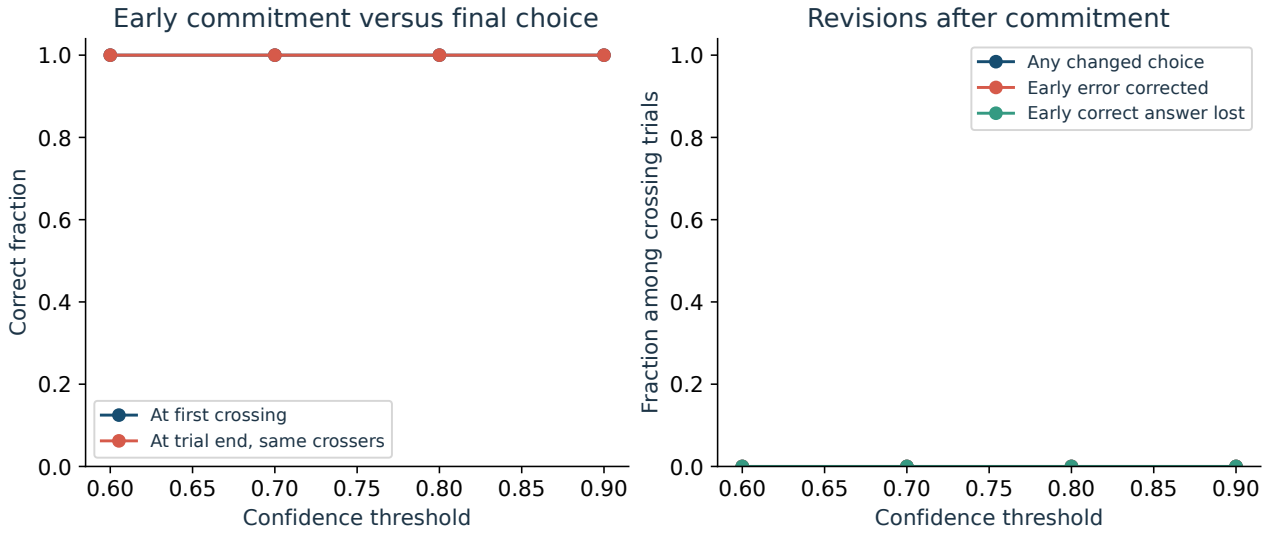


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

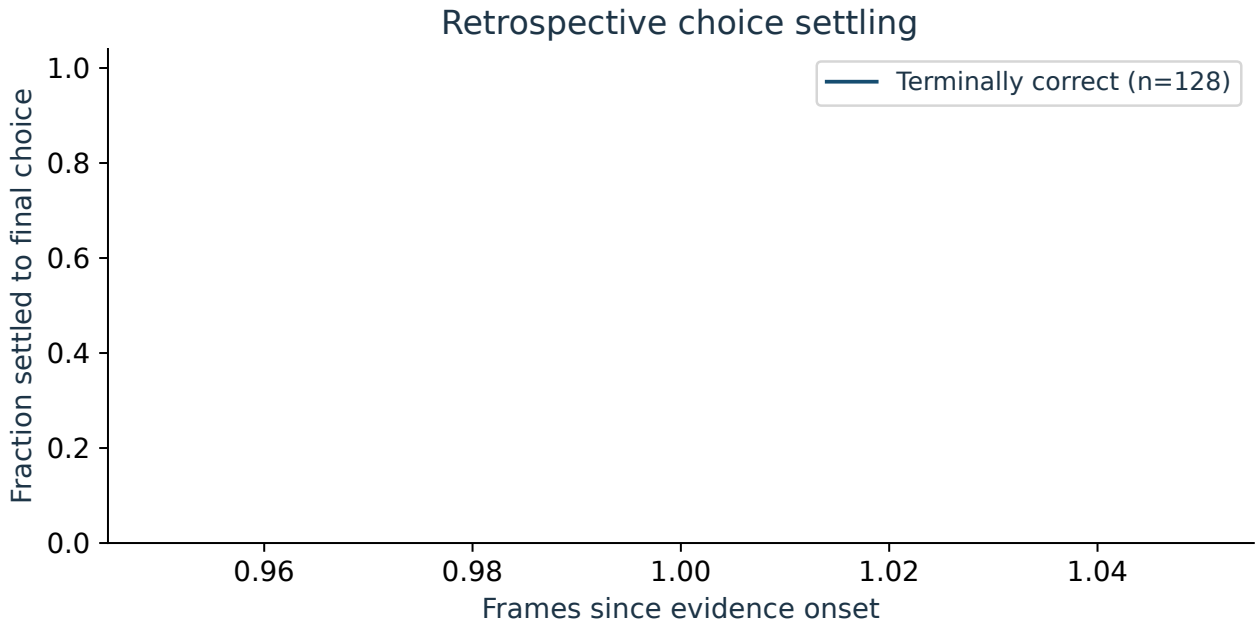


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

Rows separate the four motion directions and pool the tested dot speeds. The four line traces refer to fixed image quadrants.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Right	32	32
Up	32	32
Left	32	32
Down	32	32

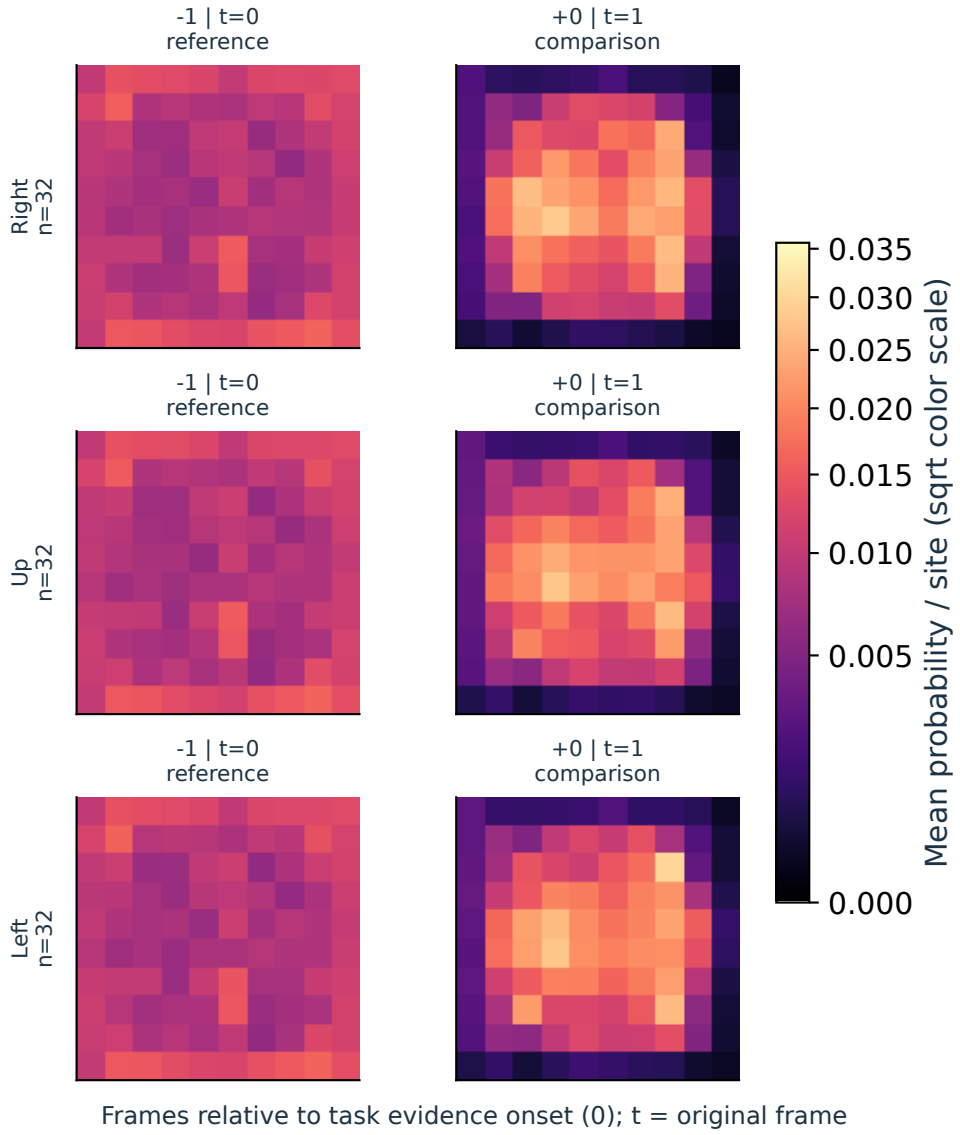


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

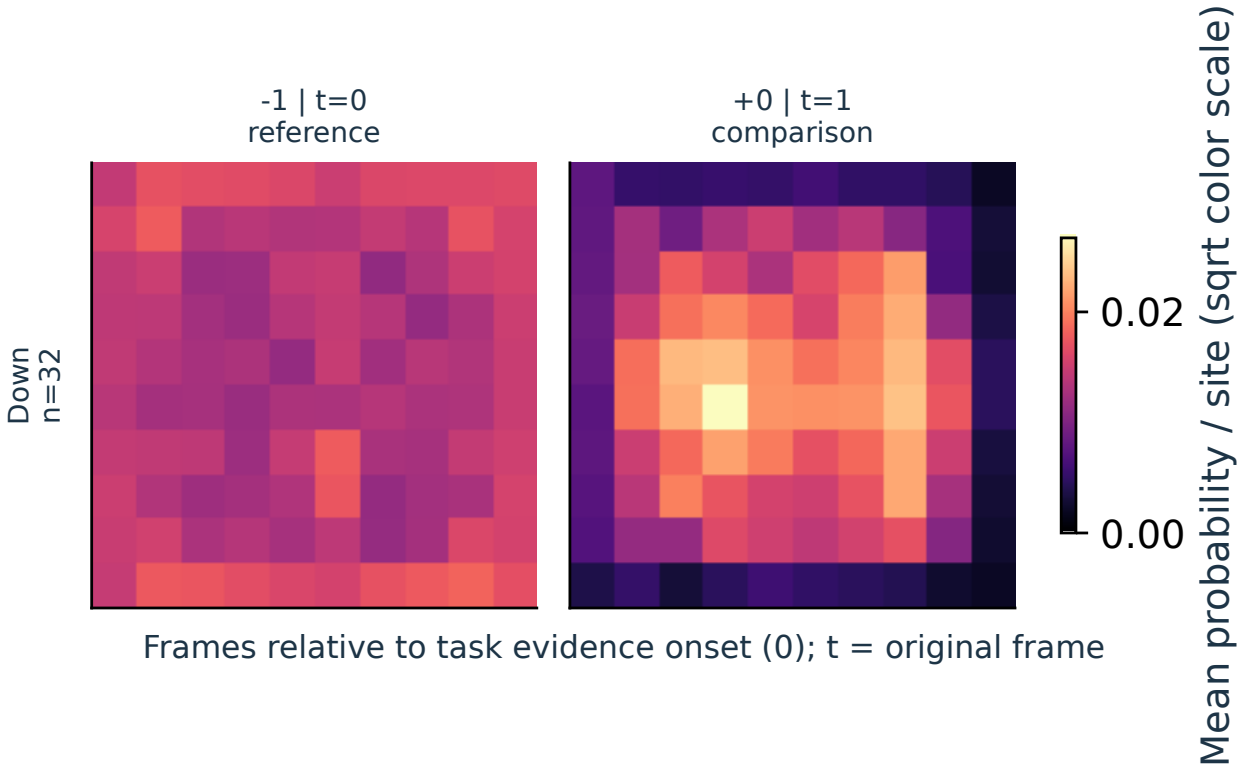


Figure 7: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

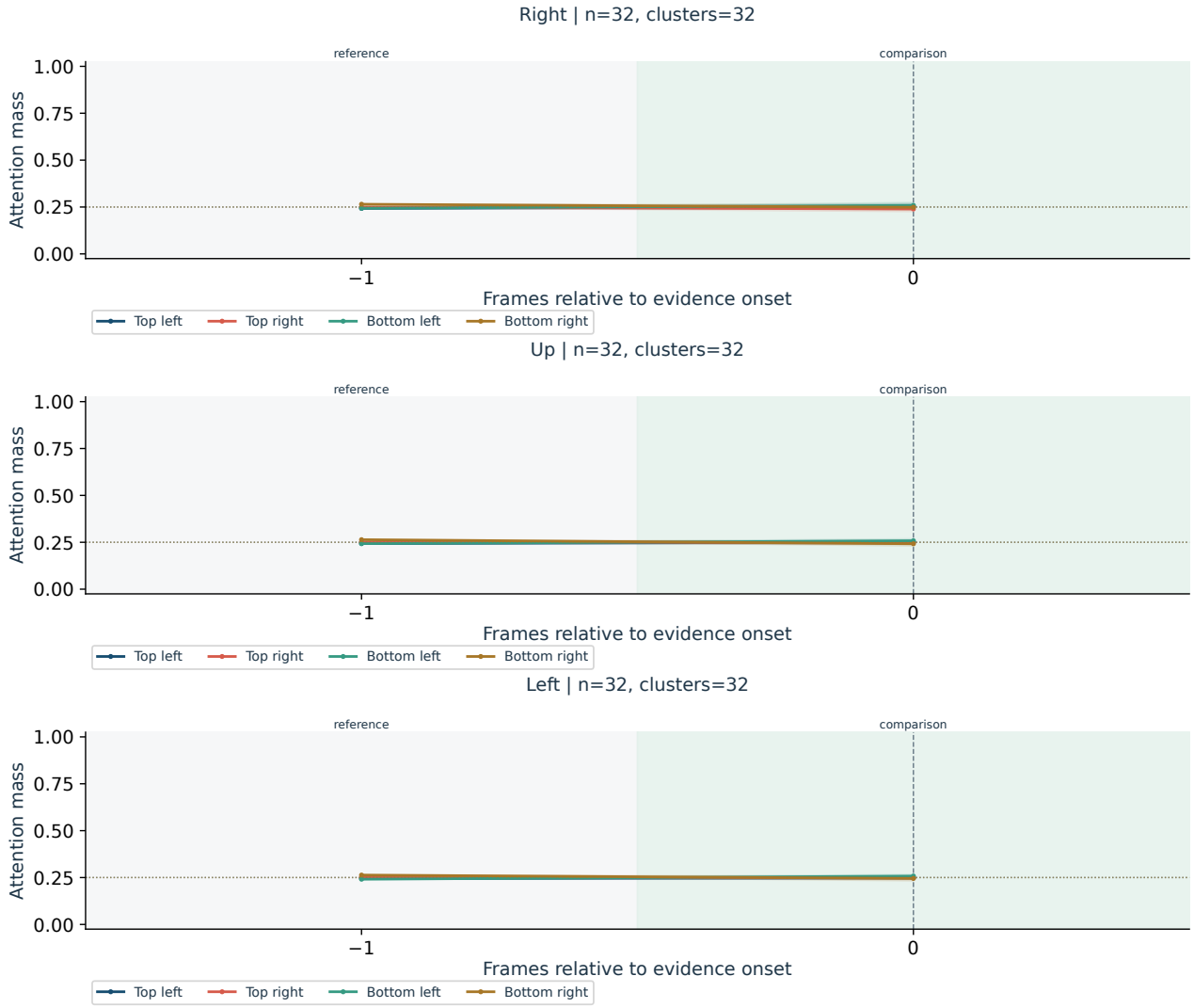


Figure 8: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

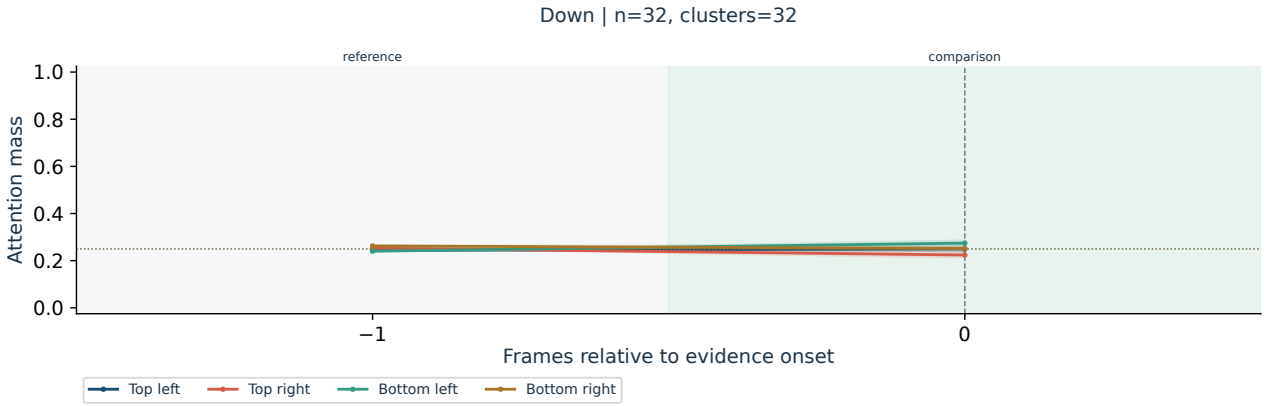


Figure 9: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

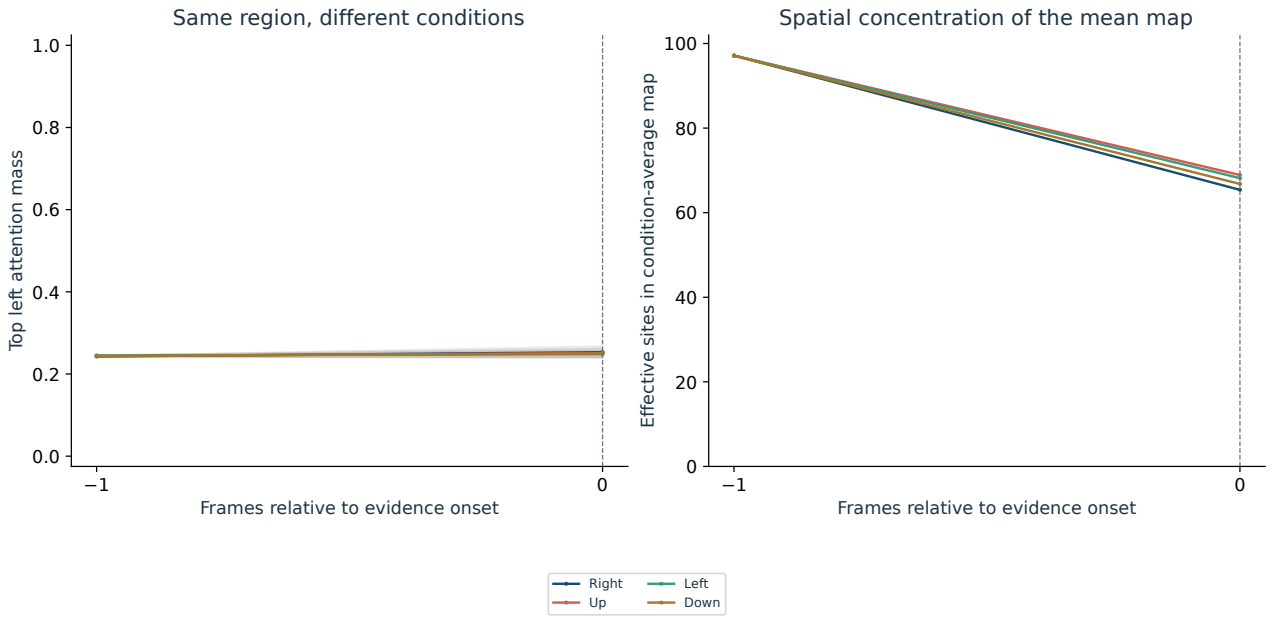


Figure 10: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Right: top left mass peaks at 0.253 at original frame 1 (comparison), and is 0.253 on the final frame. Up: top left mass peaks at 0.251 at original frame 1 (comparison), and is 0.251 on the final frame. Left: top left mass peaks at 0.248 at original frame 1 (comparison), and is 0.248 on the final frame. Down: top left mass peaks at 0.250 at original frame 1 (comparison), and is 0.250 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For target_+8_evidence, paired accuracy changed by -0.8 percentage points (paired 95% interval [-2.3%, 0.0%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 1 newly incorrect trials. For target_-8_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

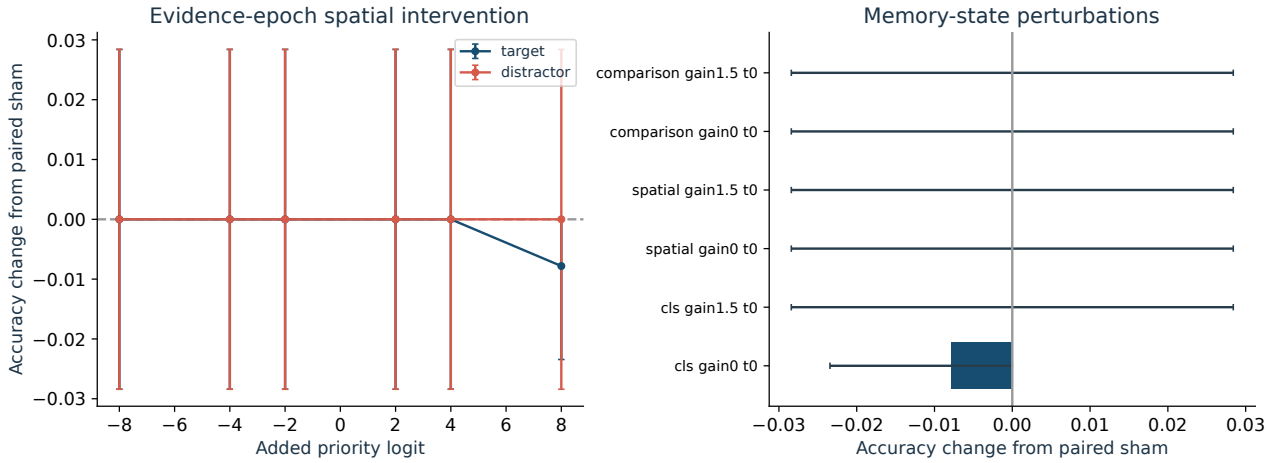


Figure 11: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
mixed	target_-4	+0.0	1
mixed	target_+4	+0.0	1
mixed	distractor_-4	+0.0	1
mixed	distractor_+4	+0.0	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.538	28.1%	0.255
fusion	1.000	100.0%	0.005
spatial	1.000	100.0%	0.005
cls	1.000	100.0%	0.005
comparison	Constant zero on this condition		
attention	0.548	27.3%	0.280

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

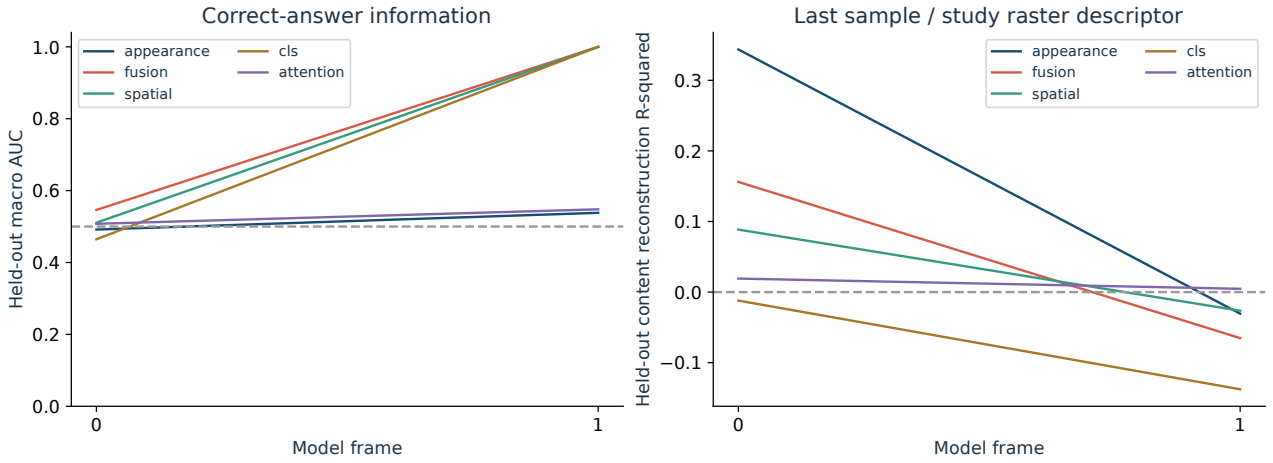


Figure 12: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 4.34 to 22.71 within the retained subspace. The final step displacement was 1.432 standardized RMS units. In spatial memory, the participation ratio ranged from 17.05 to 20.96 within the retained subspace. The final step displacement was 1.025 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 0, coarse sample-content reconstruction had R-squared -0.012 from CLS memory and 0.089 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -0.8 percentage points, spatial, +0.0 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.456	0.492	3200
fusion	1.000	0.503	6400
spatial	1.000	0.499	6400
cls	1.000	0.473	64
comparison	unavailable	unavailable	–
attention	0.633	0.504	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 64 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

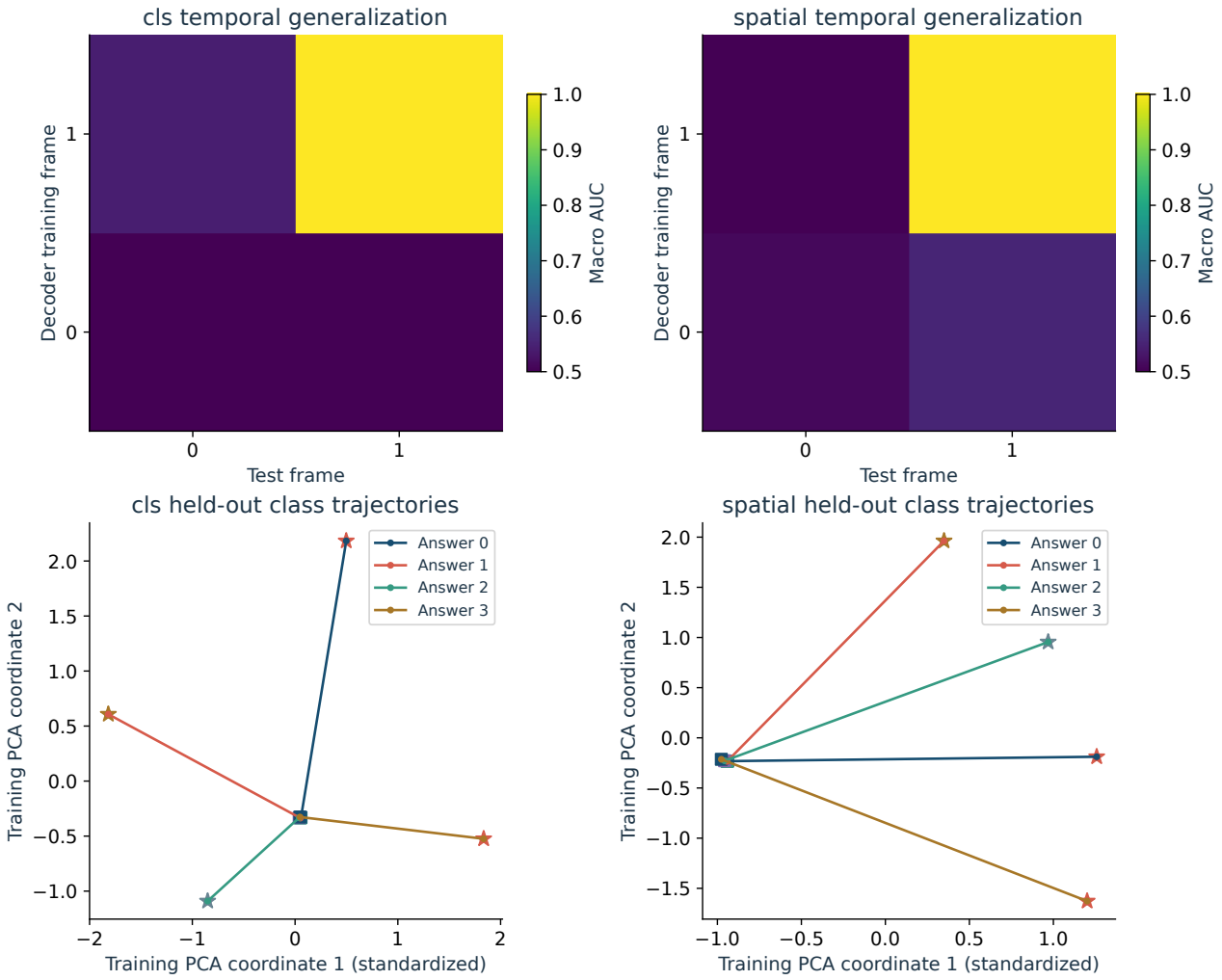


Figure 13: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

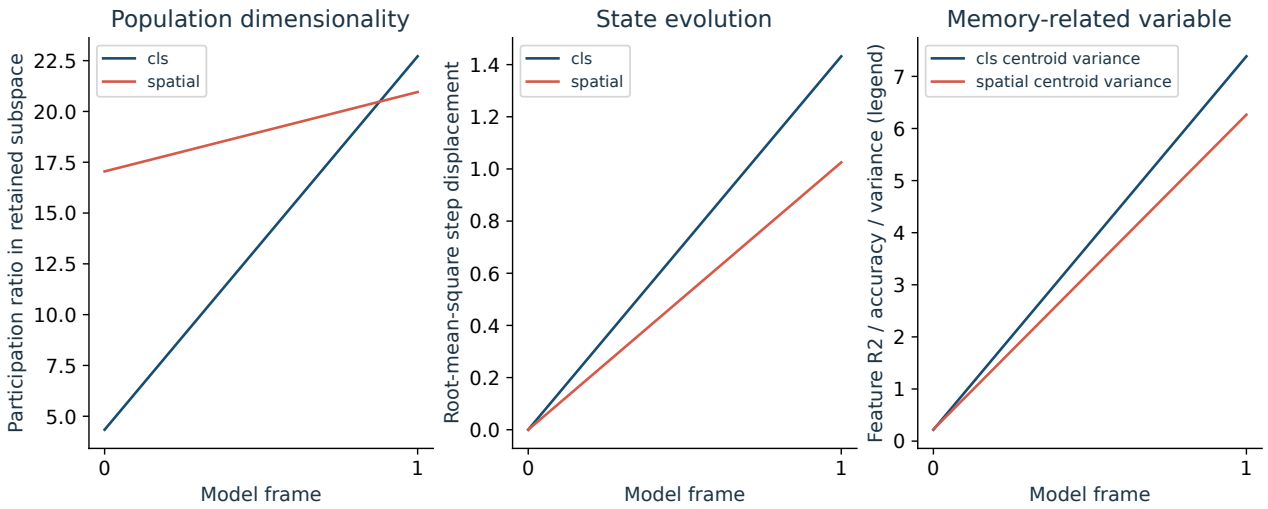


Figure 14: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

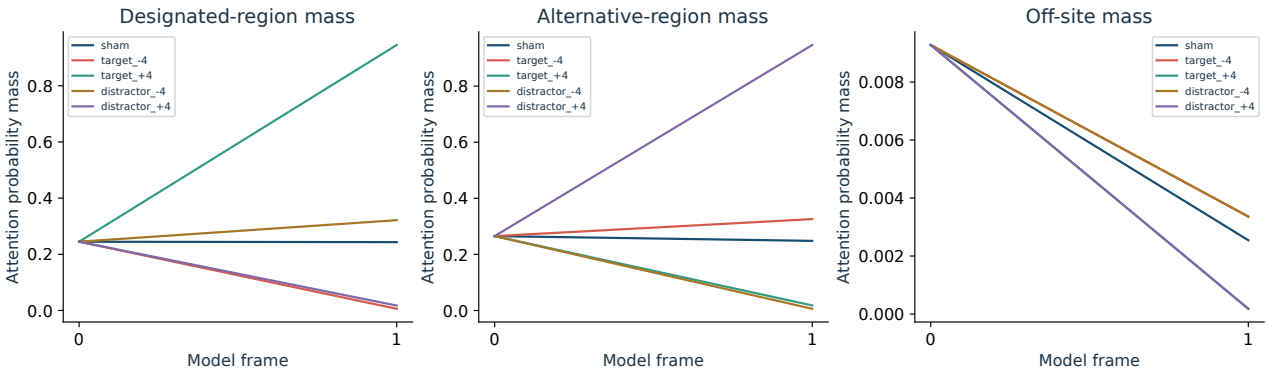


Figure 15: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

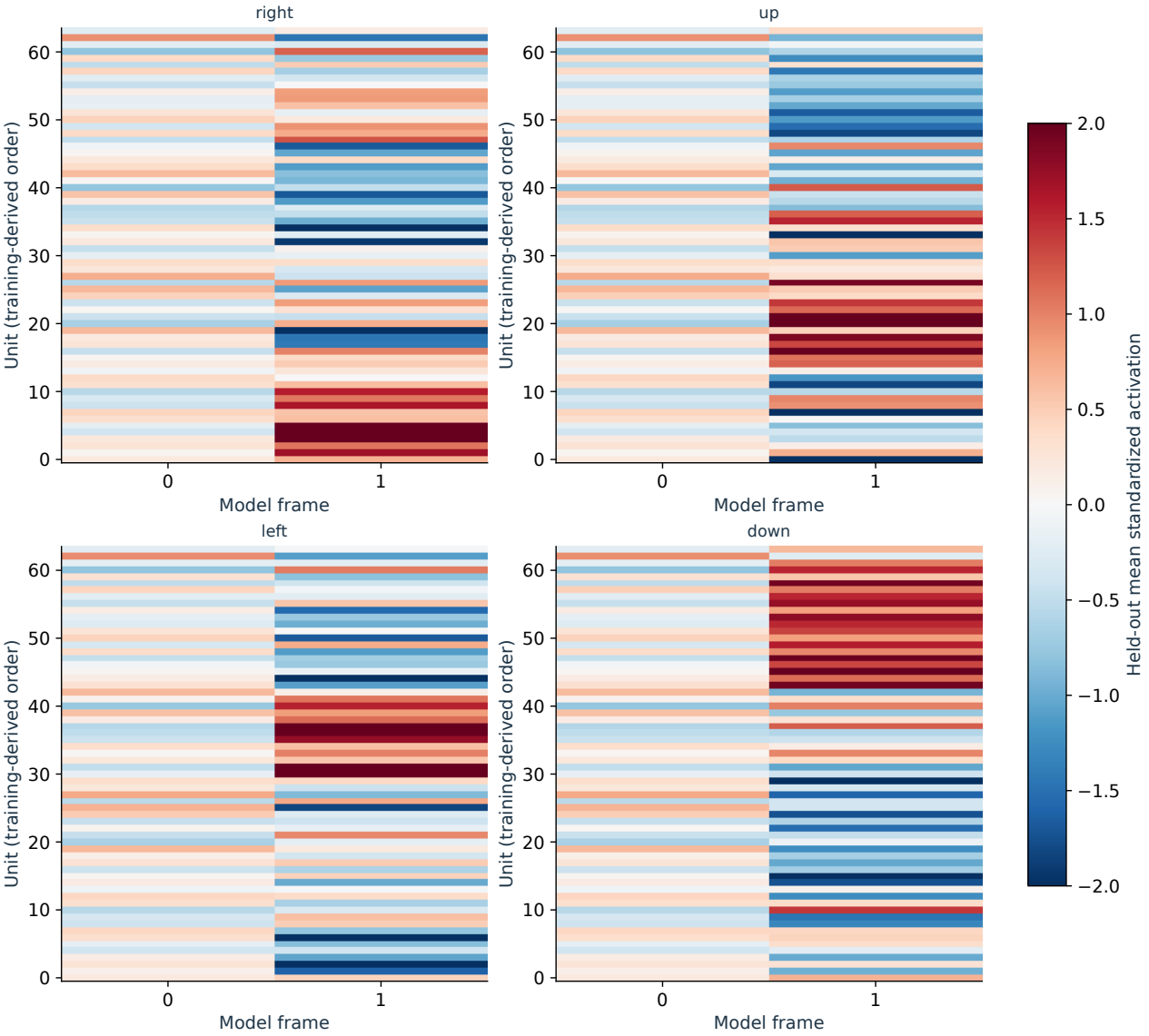


Figure 16: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

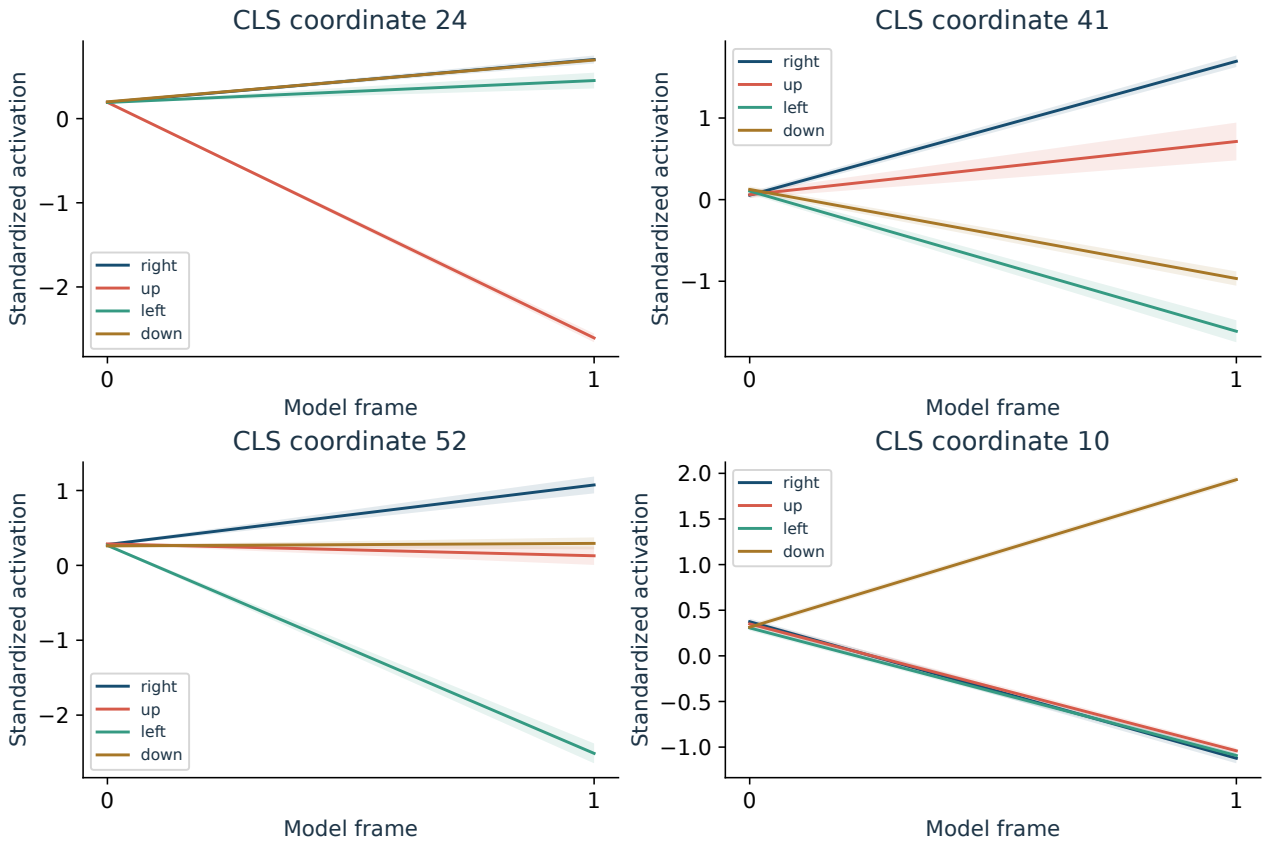


Figure 17: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Region	Correct answer	Actual choice	Cue/target location
appearance	28.1%	28.1%	unavailable
fusion	100.0%	100.0%	unavailable
spatial	100.0%	100.0%	unavailable
cls	100.0%	100.0%	unavailable
attention	27.3%	27.3%	unavailable

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

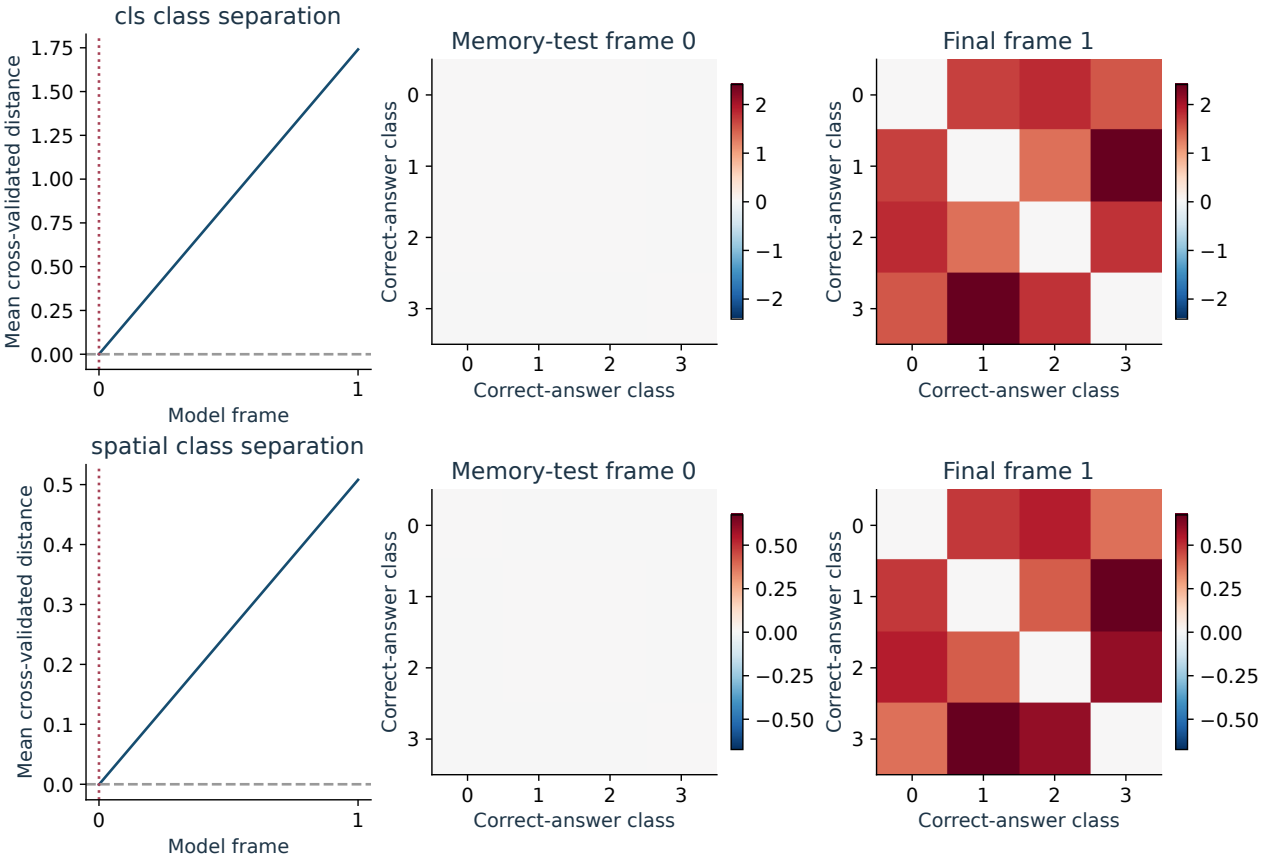


Figure 18: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

The large increase in direction decodability between appearance and fusion places an accessible direction signal upstream of the recurrent decision states. The neural-state resets produced little impairment at the native displacement strengths, and the long-window comparison state was zero. Together, these observations favor a computation in which the short-lag correspondence representation already supplies sufficient directional evidence for the downstream classifier. The correspondence bank itself uses preceding pixels; resetting a recurrent neural state leaves that causal visual history available. The weak-displacement curve supplies the more sensitive test of the front end than native ceiling performance. A spatial redistribution can also leave many informative dots available, explaining why strong direction information and small quadrant-perturbation effects can coexist.

The behavioral result establishes 100.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies fusion as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies target_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

A useful follow-up would vary coherence and dot lifetime independently of displacement. That would distinguish tolerance of weak correspondence from tolerance of directional noise. Direction-selective perturbations of the motion population would additionally test feature-specific causal influence, complementing the spatial interventions reported here.

4.1 Interpreting these results alongside biological experiments

4.1.1 Britten, Shadlen, Newsome and Movshon (1992)

Our visual-to-fusion decoding increase supplies a computational counterpart to the distinction between sensory evidence and a report. A useful comparison is the entire neurometric-versus-psychometric curve as displacement weakens. Native accuracy at ceiling contributes little to that comparison. The animal experiment varied stochastic-motion strength; our existing challenge varies pixel displacement, which changes correspondence ambiguity and rasterization. These manipulate different sources of uncertainty. [4]

4.1.2 Salzman, Britten and Newsome (1990)

The present quadrant intervention changes where evidence is sampled. MT microstimulation changes the balance of direction-selective evidence. A small quadrant effect in a full-field movie therefore fits distributed evidence: useful dots remain elsewhere. A closer causal experiment would identify direction-preferring fused coordinates on independent fitting trials, perturb those coordinates during weak motion, and ask whether choices shift toward the fitted preferred direction. [5]

4.1.3 Ling, Liu and Carrasco (2009)

The human noise manipulation provides a way to distinguish stronger evidence from rejection of irrelevant directions. Our correspondence front end already compares local image shifts; successful direction decoding can follow from that operation. Varying dot-direction noise while holding speed, contrast and dot count fixed would reveal whether the learned readout pools direction evidence robustly or depends on nearly uniform displacement. [6]

4.2 A computational account of the observed behavior

The relevant theory is population decoding. A candidate direction is supported by a pattern across channels, and a downstream computation must compare competing patterns. In this model, the fixed correspondence calculation supplies local displacement alternatives and learned fusion converts them into task-useful features. Spatial attention then determines which sites contribute. These are three distinct operations: estimate local correspondence, represent a direction, and pool it for a decision. Their separation explains why strong direction information can coexist with weak effects of redistributing attention across a uniformly informative display. The previous-frame pixel history also survives a reset of neural recurrent memory, so the reset experiment tests the added recurrent contribution after short-lag visual evidence is already available.

4.3 Where this task agrees with the biological evidence

The agreement is a readable sensory code that supports accurate direction decisions. The strongest mechanistic distinction is the engineered correspondence bank preceding the learned network. Current observations support efficient use of that bank; the relation to MT becomes more specific when channel tuning and direction-selective perturbations are measured together.

4.4 A follow-up experiment motivated by these results

Cross displacement with motion coherence and contrast, fit direction tuning on independent movies, and perturb matched versus mismatched direction populations. Predict a horizontal choice-curve shift for a directional evidence bias and a threshold change for reduced evidence reliability.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Salzman CD, Britten KH, Newsome WT (1990). Cortical microstimulation influences perceptual judgements of motion direction. Nature 346:174-177. https://pubmed.ncbi.nlm.nih.gov/

2366872/.

[2] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[3] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[4] Britten, Shadlen, Newsome and Movshon (1992). The analysis of visual motion: a comparison of neuronal and psychophysical performance. Primary source.

[5] Salzman, Britten and Newsome (1990). Cortical microstimulation influences perceptual judgements of motion direction. Primary source.

[6] Ling, Liu and Carrasco (2009). How spatial and feature-based attention affect the gain and tuning of population responses. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
mixed	128	100.0%	1.000	0.000

B Every-frame attention atlas

Condition mixed. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 32 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

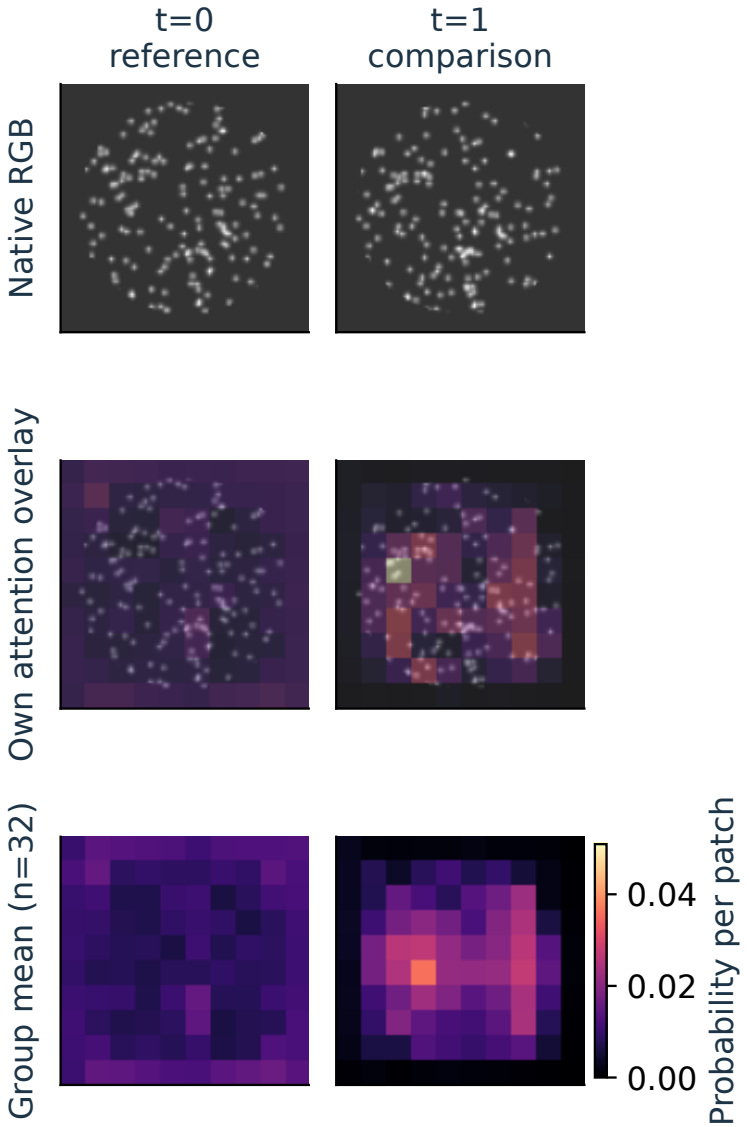


Figure 19: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

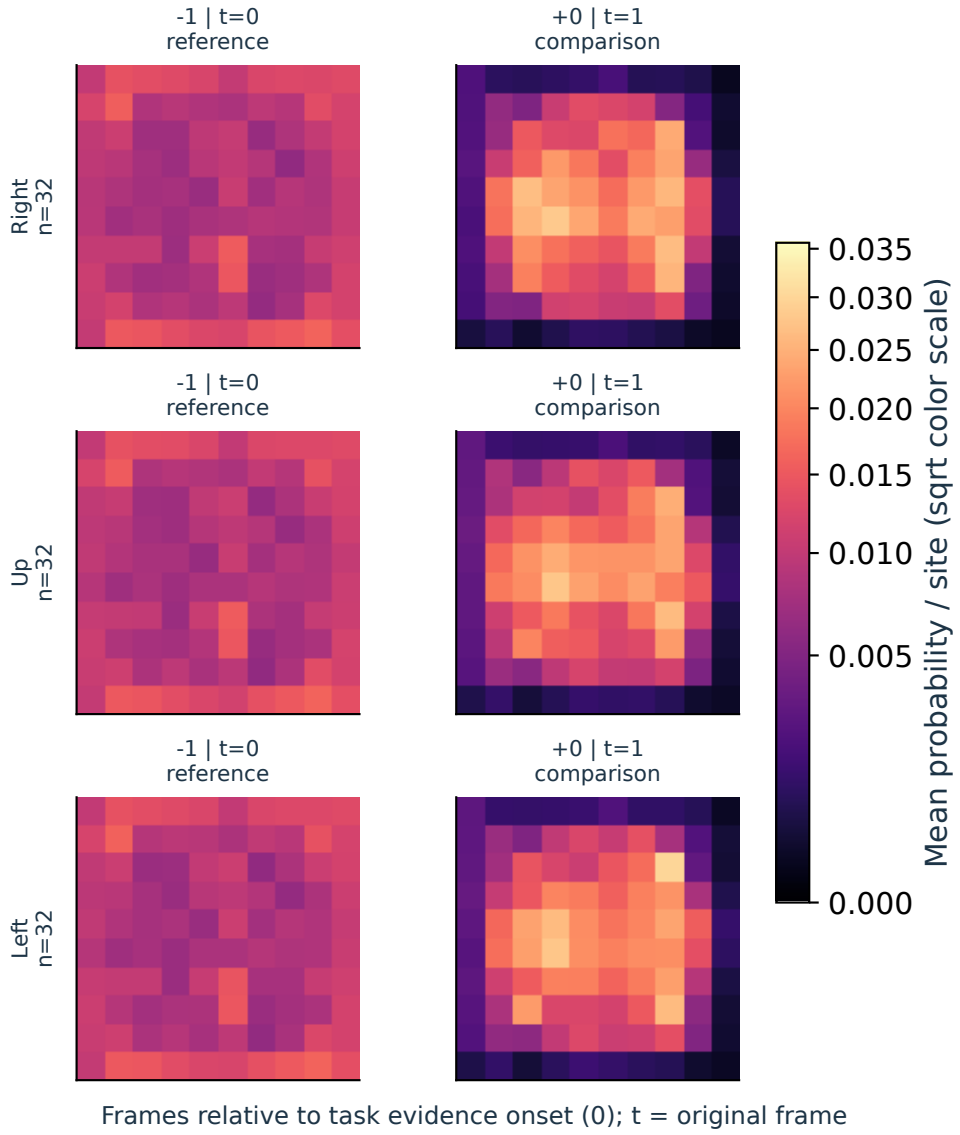


Figure 20: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

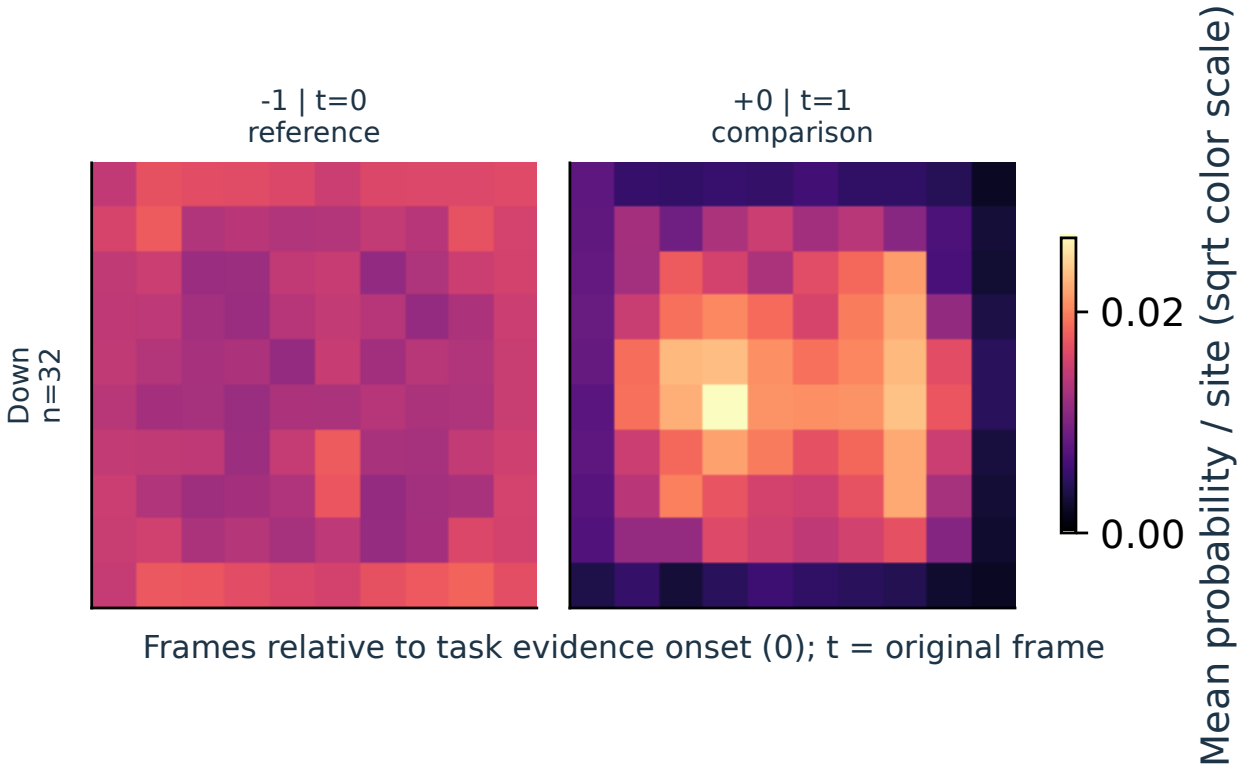


Figure 21: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Signed orientation comparison through recurrent visual memory

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized orientation in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 1 native conditions and 128 fresh trials, equal-condition accuracy was 100.0% (95% interval [97.1%, 100.0%]). For the representative condition mixed, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was cls_gain0_t0, changing accuracy by -38.3 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The observer must determine the sign of an orientation change across two gratings. The answer is relational: the absolute orientation of either grating alone is insufficient. The same positive and negative rotations must be interpreted across different starting angles and phases, creating a compact test of visual comparison.

We vary rotation magnitude and measure correct sign reports. Orientation is axial: angles separated by 180 degrees describe the same grating. Consequently, a continuous neural target should use cosine and sine of twice the angle. The model need not explicitly represent an angle to solve the task; successful readout from the fused or recurrent representations can instead arise from distributed image features.

The biological motivation is developed through [1]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Neurons selective for oriented structure provide a useful conceptual starting point for an orientation-comparison task. The computational requirement extends beyond detecting an edge: the observer must preserve a description of one pattern and compare it with another. A neuroscientific analysis therefore asks whether response selectivity emerges during the comparison or is already present during the reference. Memory reset separates these alternatives experimentally. If a correct answer survives the removal of one recurrent component, another route may carry the reference or sufficient comparison information. Simultaneous recordings across the architecture make those routes directly inspectable.

1.2 Biological experiments motivating this task

Experimental question. How is the orientation of one observation retained and compared with a later observation?

De Valois, Yund and Hepler (1982) Striate neurons exhibited a broad distribution of orientation bandwidths and both direction-selective and direction-insensitive responses. Orientation and motion selectivity were measured as tuning properties of individual cells. [4]

Harrison and Tong (2009) Human fMRI patterns in V1-V4 identified the remembered grating orientation during a blank delay, including cases with little sustained average activation. Information in a population pattern could persist despite low mean activity. [5]

McAdams and Maunsell (1999) During an orientation attention task, V4 responses increased with attention largely through scaling of tuning curves, without a systematic change in tuning width. V1 modulation was smaller. [6]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

What is the sign of the grating orientation change from frame 0 to frame 1?

Response labels are 0: negative axial angle change; 1: positive axial angle change. The native suite contains 1 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use mixed.

1.4 Stimulus sequence and spatial layout

For the representative native condition mixed, the sequence has 2 frames, indexed from zero: frame 0: reference; frame 1: comparison. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Signed grating rotation (degrees)	-22.0, -10.0, -4.0, 4.0, 10.0, 22.0
Stimulus family	orientation

Spatial attention in a full-field grating determines which image regions contribute to the comparison. The fixed quadrant manipulations test that spatial selection. Memory perturbations between the two observations test whether a preserved first-frame description is necessary. A fall in accuracy after a recurrent-state reset, together with strong final decoding, would support a sequential comparison computation.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 100.0%, with macro AUC 1.000. The matched final checkpoint changed accuracy by +0.00 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was mixed at 100.0%; the highest was mixed at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (0.5) produced 78.9% correct responses, compared with 100.0% at the largest strength (22). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

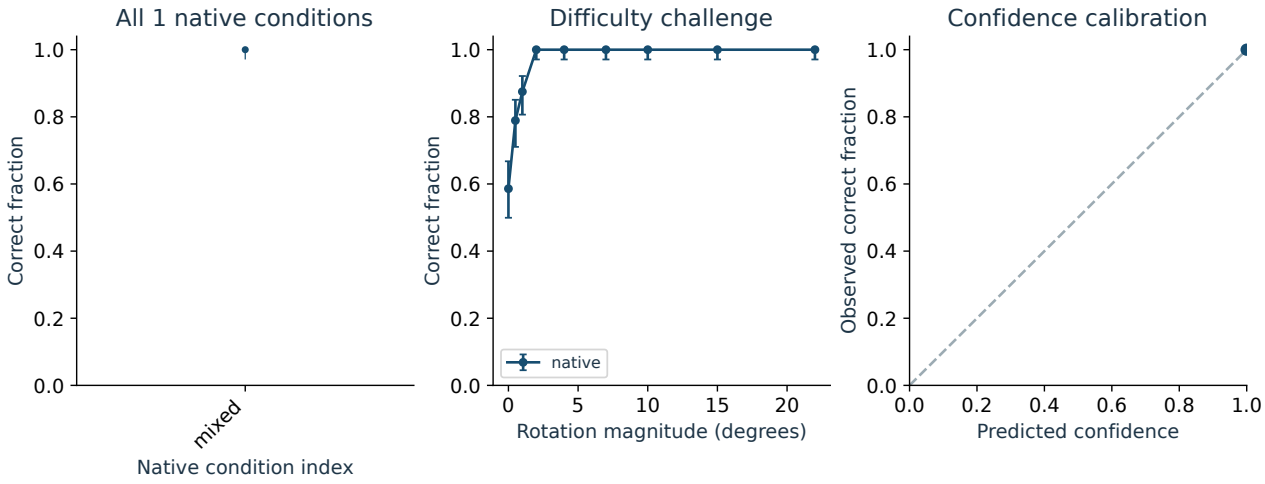


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)[1 - \exp(-(x/a)^b)]$. Here x is signal strength, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 0.4917 task units (bootstrap interval [0.3209, 0.6791]), with slope 0.45 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.086.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	100.0%	95.3%
0.7	100.0%	1.0	100.0%	93.0%
0.8	100.0%	1.0	100.0%	88.3%
0.9	100.0%	1.0	100.0%	83.6%

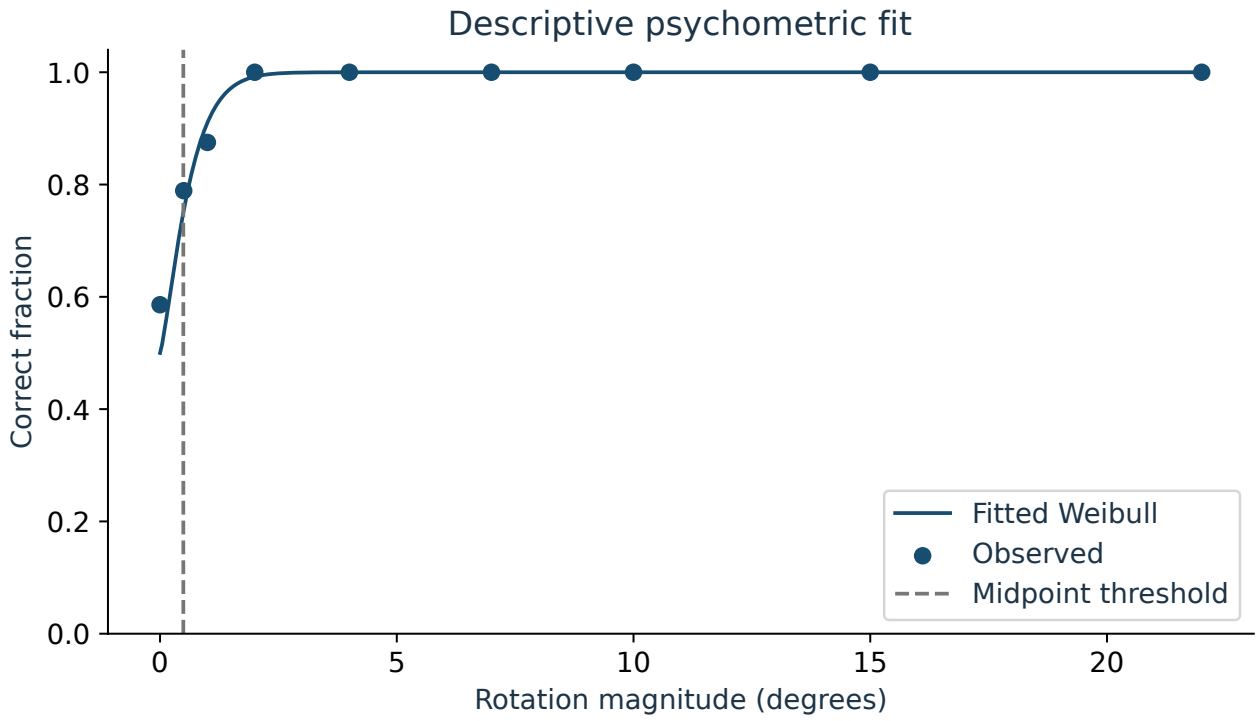


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

The representative condition mixed provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 128 of 128 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

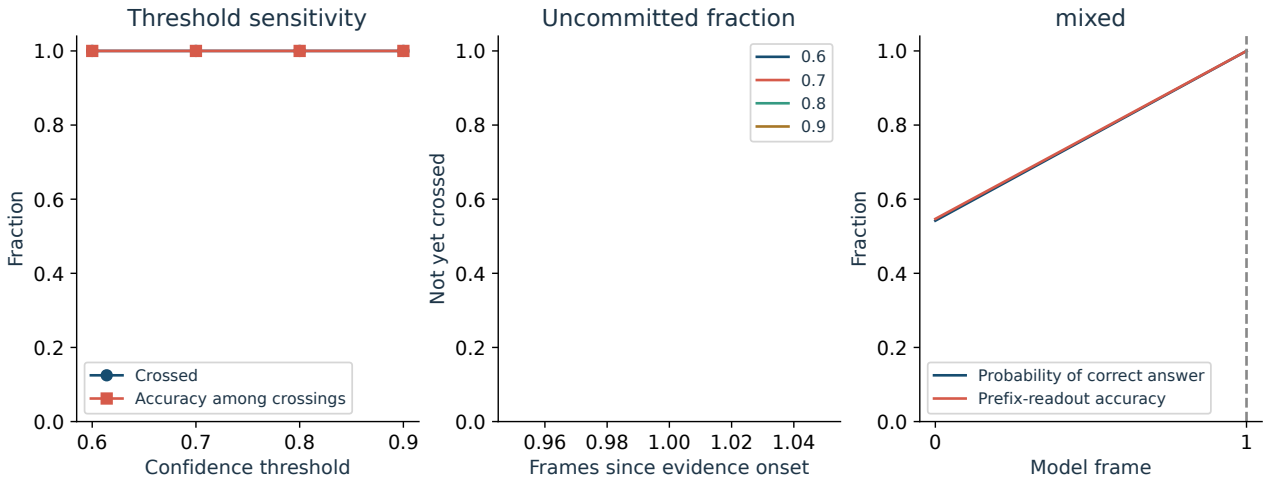


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

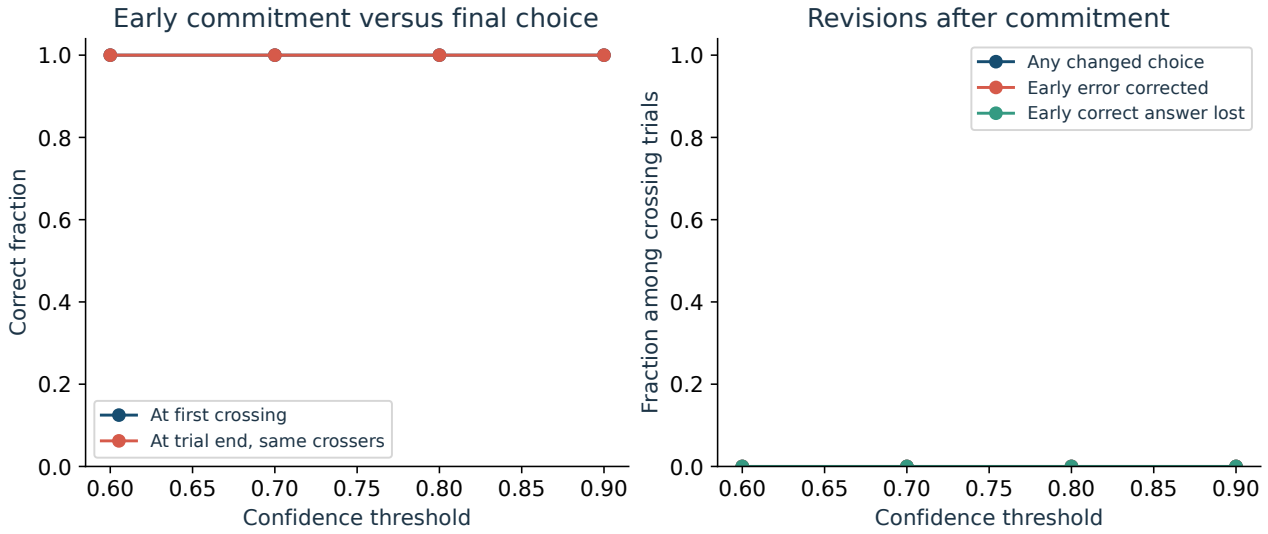


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

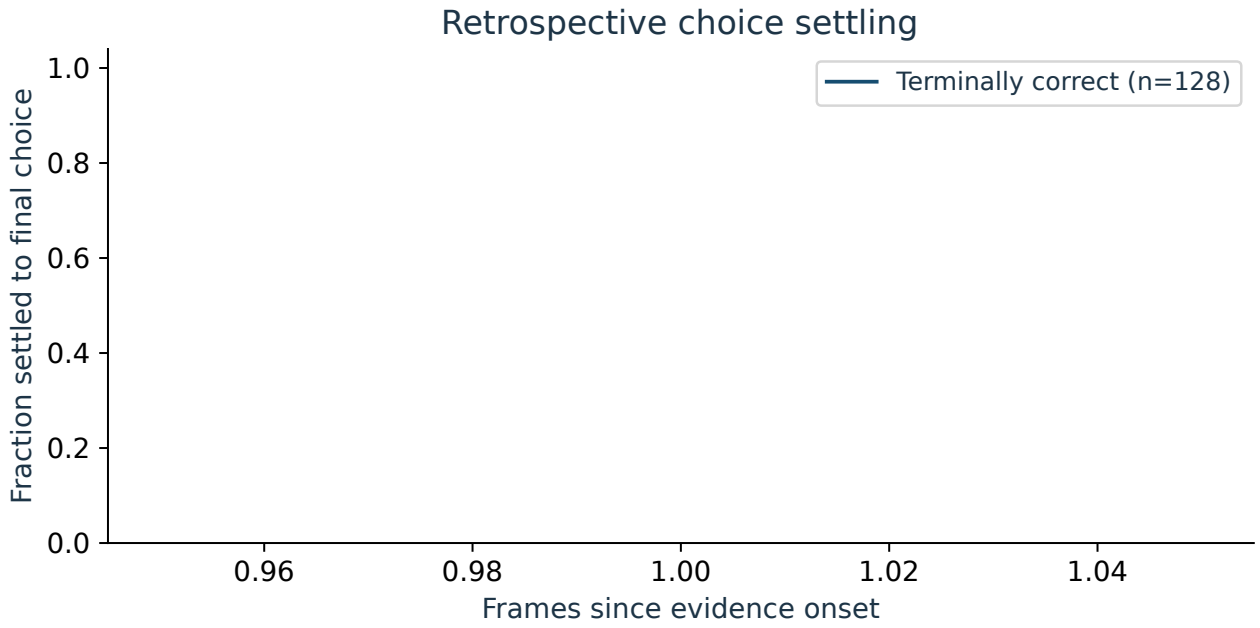


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

Rows separate absolute orientation differences and pool rotation signs and nuisance orientations. Traces refer to fixed image quadrants.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
signed orientation degrees: 4	44	44
signed orientation degrees: 10	47	47
signed orientation degrees: 22	37	37

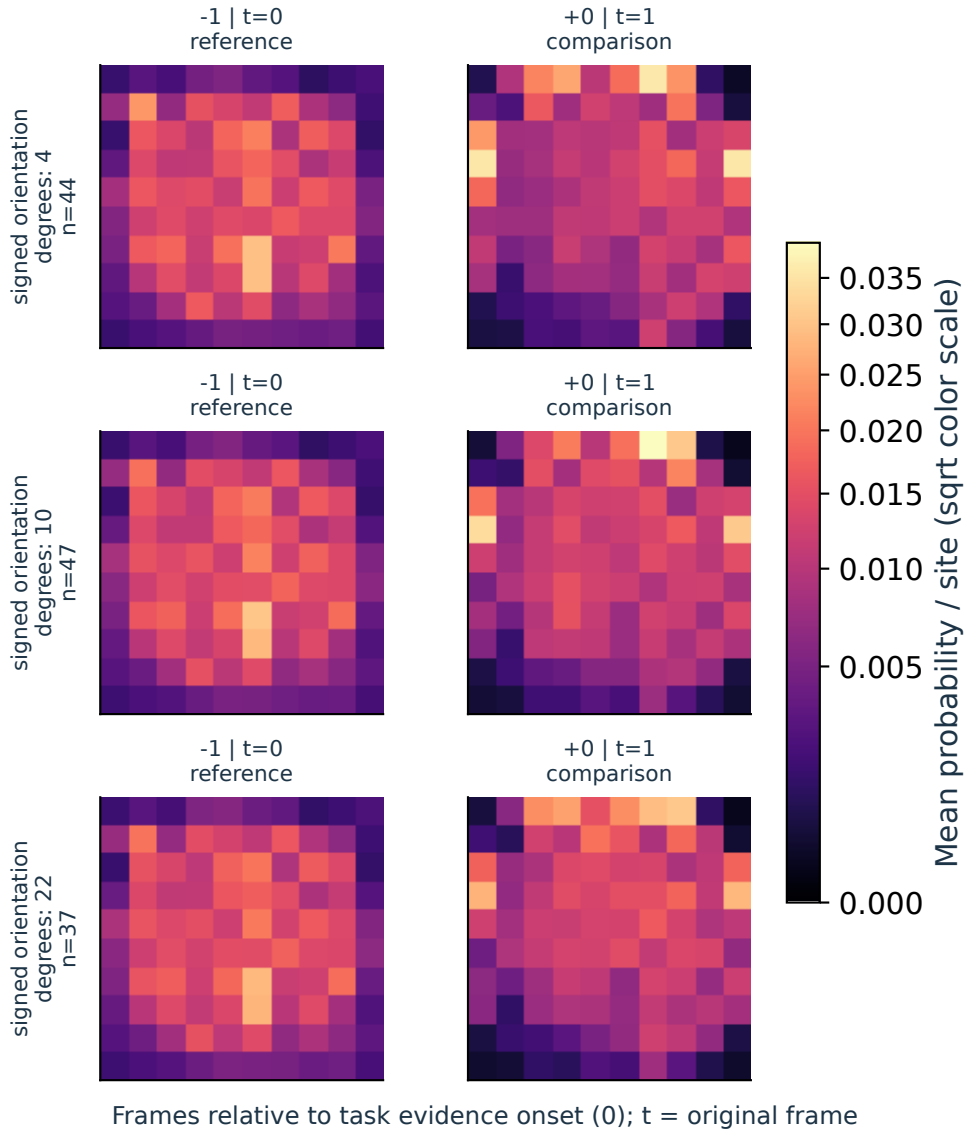


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

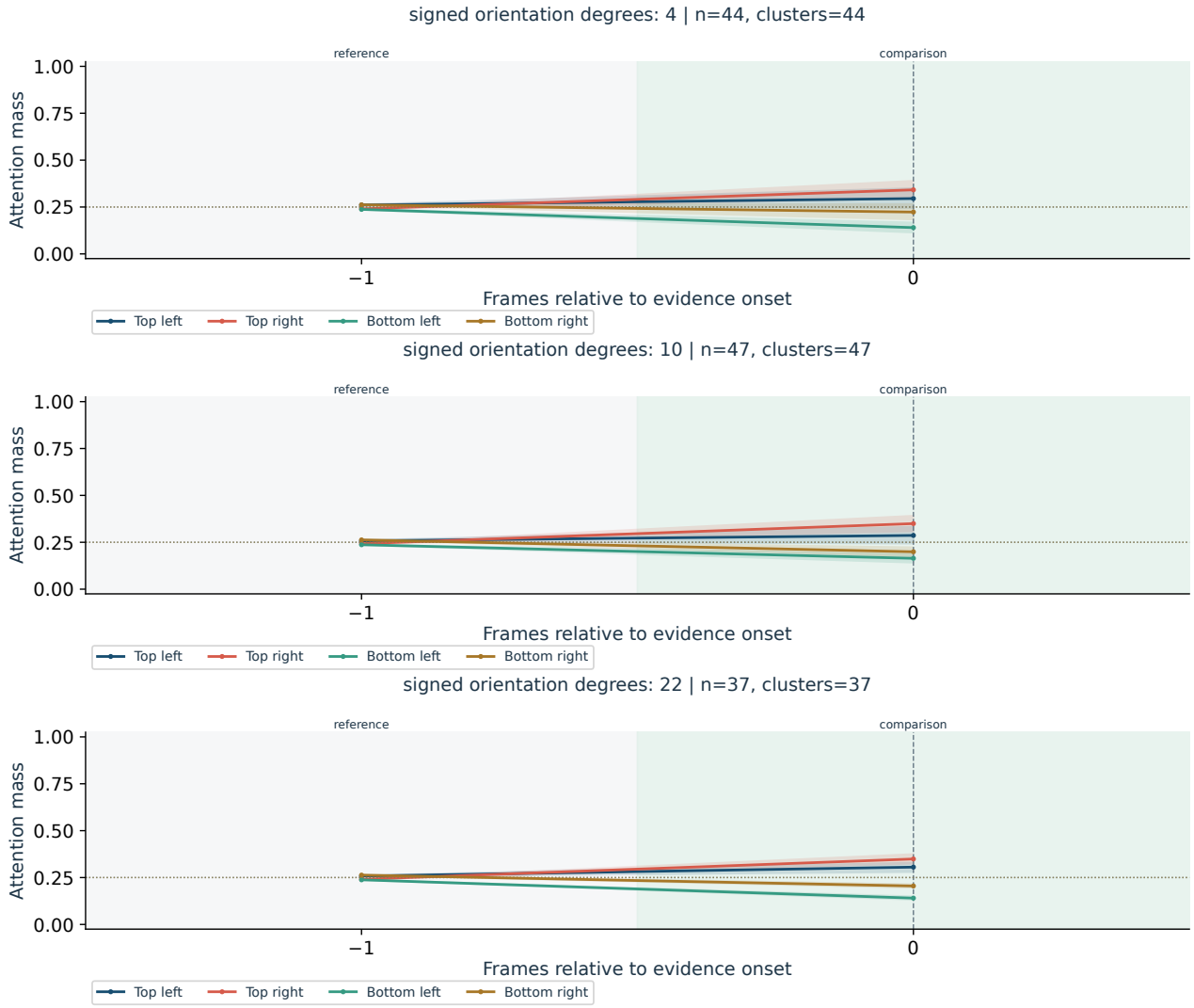


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

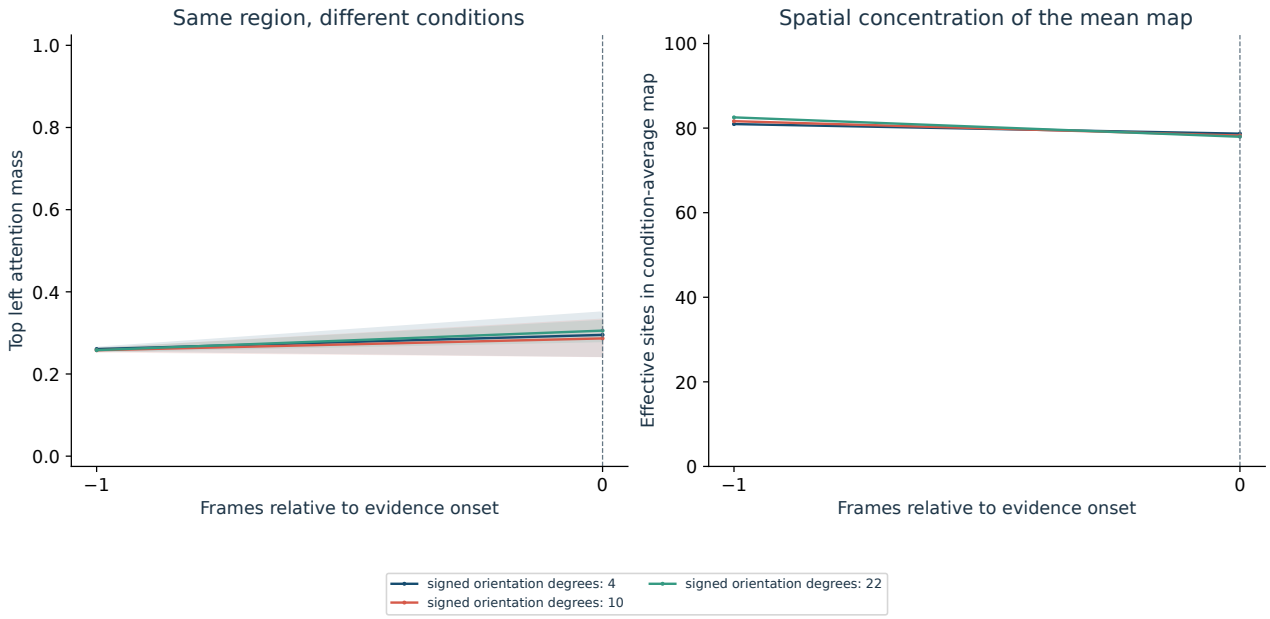


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. signed orientation degrees: 4: top left mass peaks at 0.295 at original frame 1 (comparison), and is 0.295 on the final frame. signed orientation degrees: 10: top left mass peaks at 0.287 at original frame 1 (comparison), and is 0.287 on the final frame. signed orientation degrees: 22: top left mass peaks at 0.306 at original frame 1 (comparison), and is 0.306 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For cls_gain0_t0, paired accuracy changed by -38.3 percentage points (paired 95% interval [-46.9%, -29.7%]; paired p=3.553e-15; within-task Holm p=1.101e-13). There were 0 corrected trials and 49 newly incorrect trials. For target_-8_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

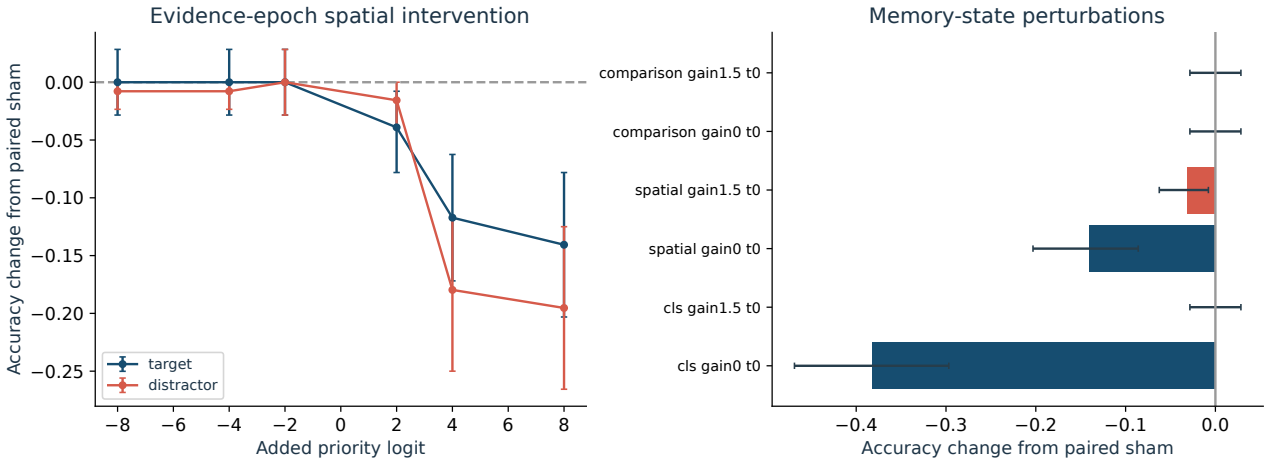


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
mixed	target_-4	+0.0	1
mixed	target_+4	-11.7	0.004517
mixed	distractor_-4	-0.8	1
mixed	distractor_+4	-18.0	1.955e-05

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.487	48.4%	0.725
fusion	0.521	49.2%	0.610
spatial	0.887	80.5%	0.005
cls	1.000	100.0%	0.005
comparison	Constant zero on this condition		
attention	0.548	53.1%	0.240

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

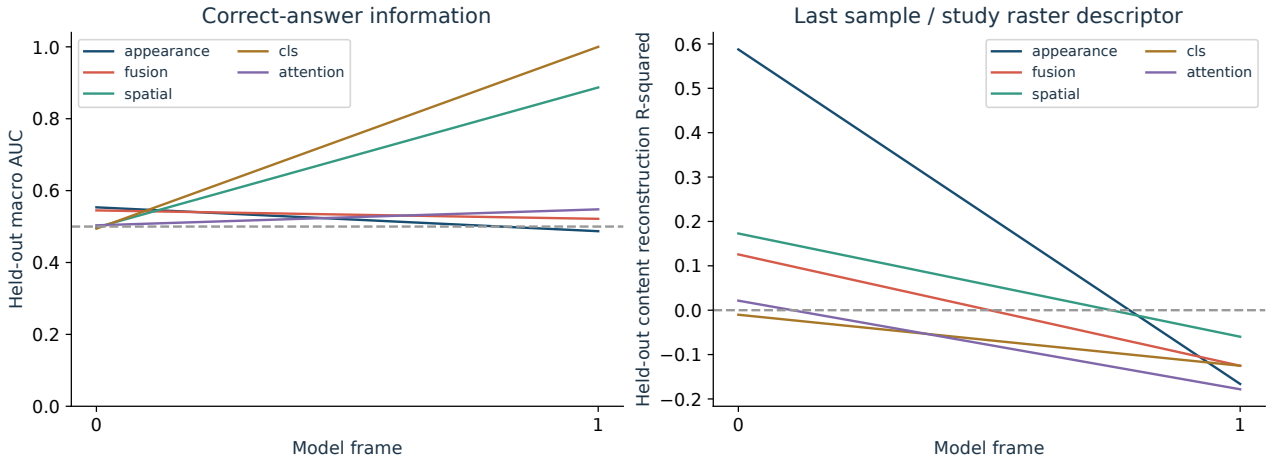


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 8.45 to 20.11 within the retained subspace. The final step displacement was 1.391 standardized RMS units. In spatial memory, the participation ratio ranged from 10.32 to 21.69 within the retained subspace. The final step displacement was 1.086 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 0, coarse sample-content reconstruction had R-squared -0.010 from CLS memory and 0.173 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -38.3 percentage points, spatial, -14.1 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.446	0.481	3200
fusion	0.521	0.549	6400
spatial	0.971	0.573	6400
cls	1.000	0.493	64
comparison	unavailable	unavailable	–
attention	0.598	0.528	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 28 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

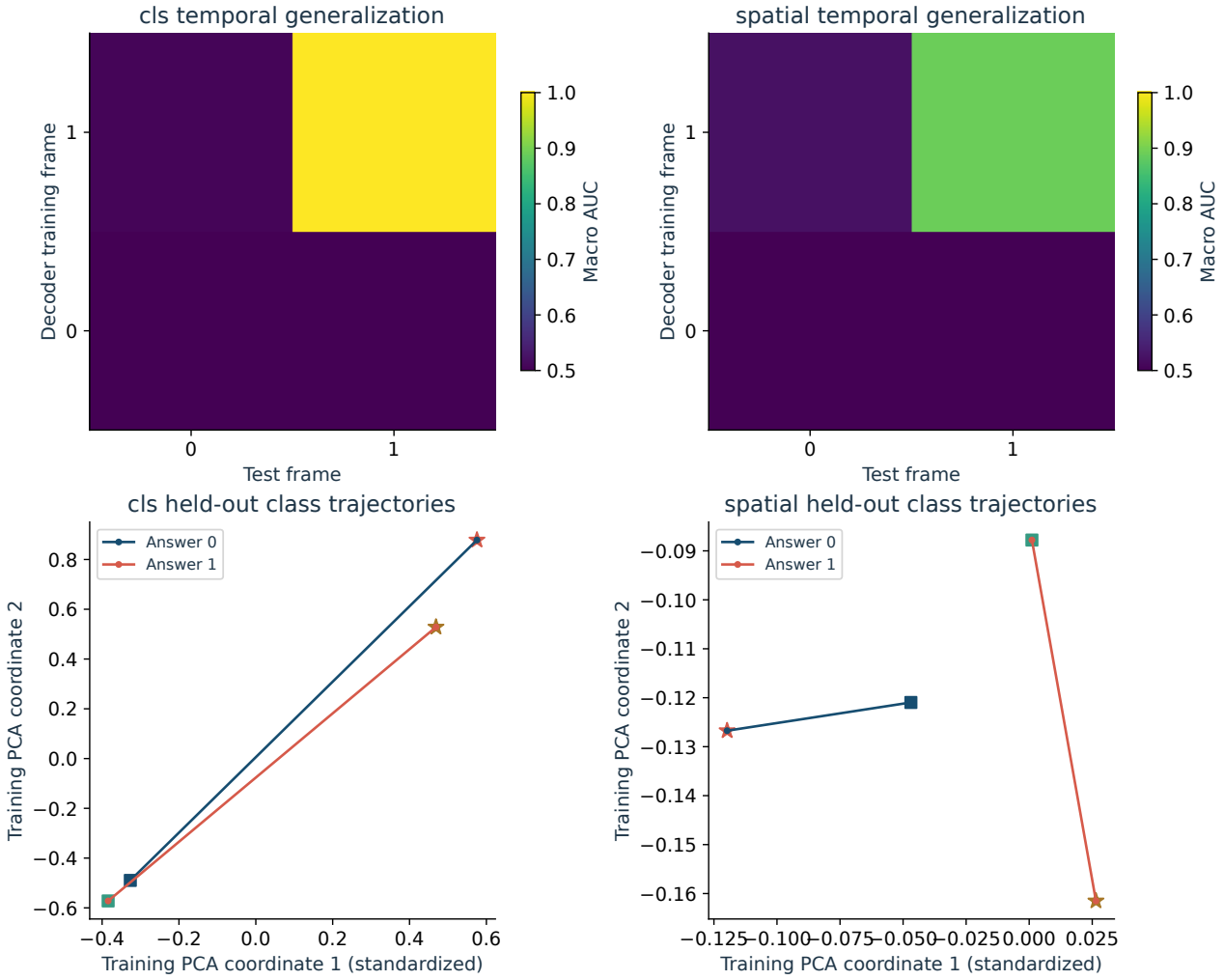


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

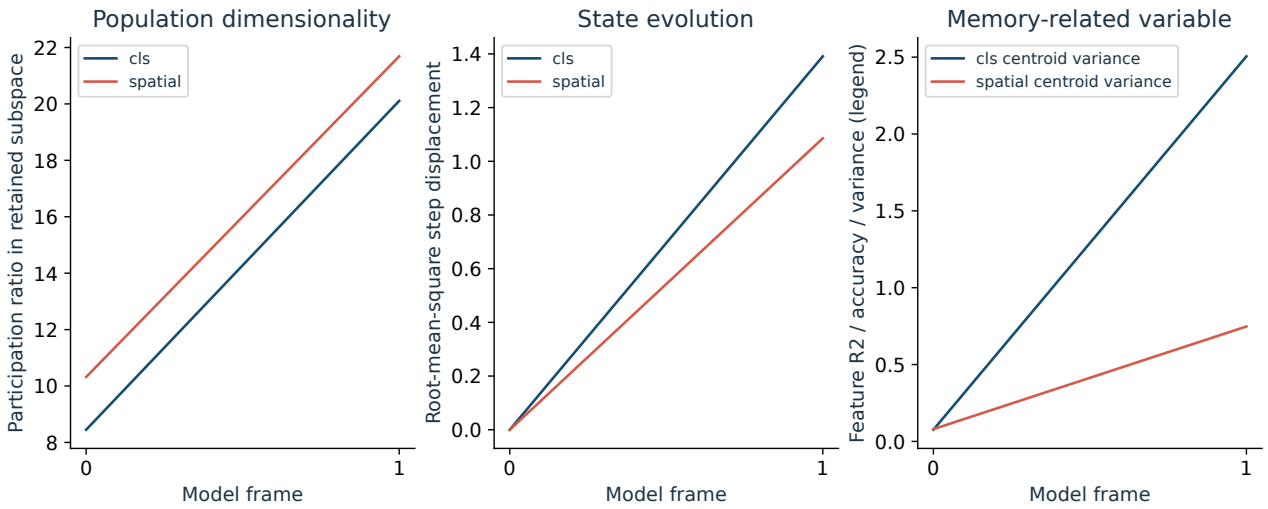


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

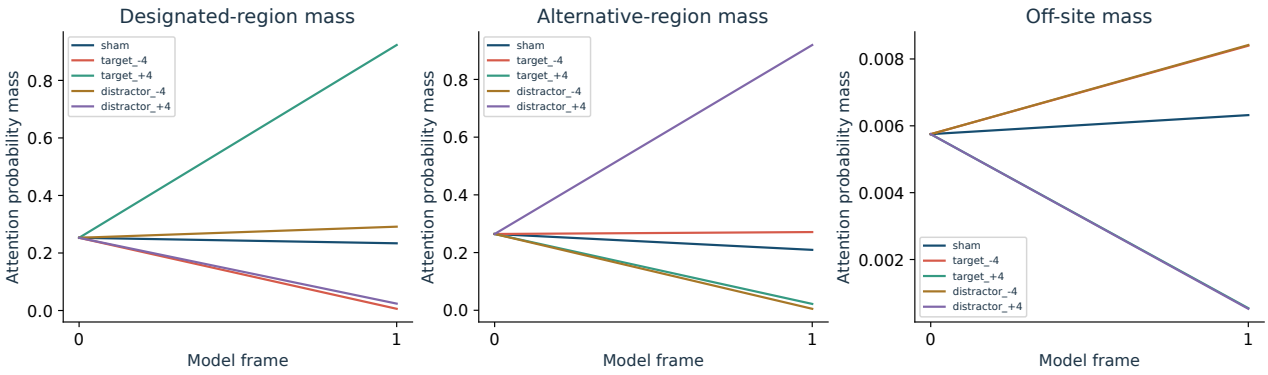


Figure 13: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

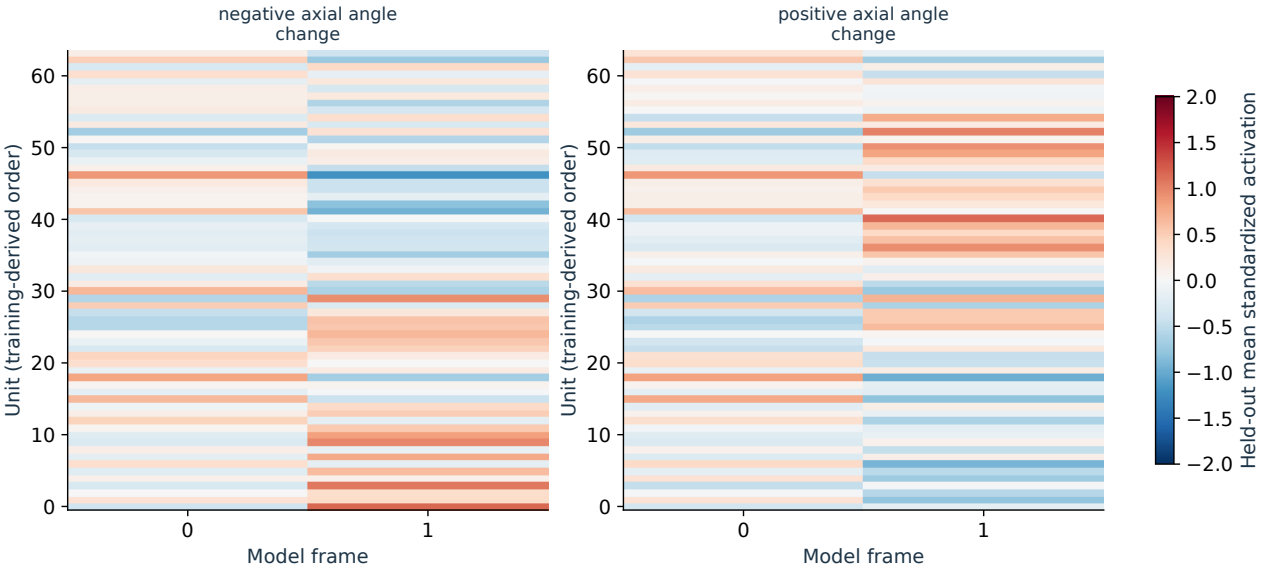


Figure 14: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	48.4%	48.4%	unavailable
fusion	49.2%	49.2%	unavailable
spatial	80.5%	80.5%	unavailable
cls	100.0%	100.0%	unavailable
attention	53.1%	53.1%	unavailable

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

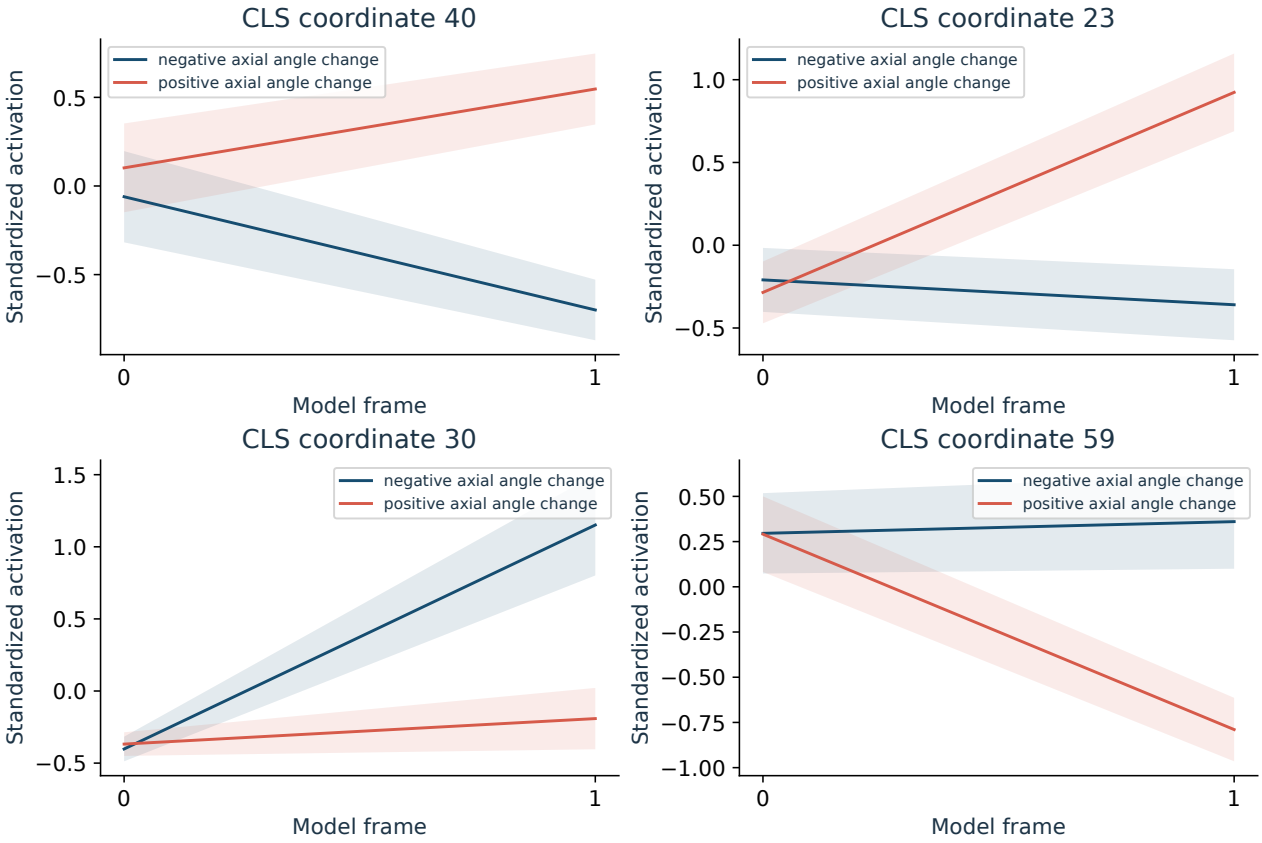


Figure 15: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

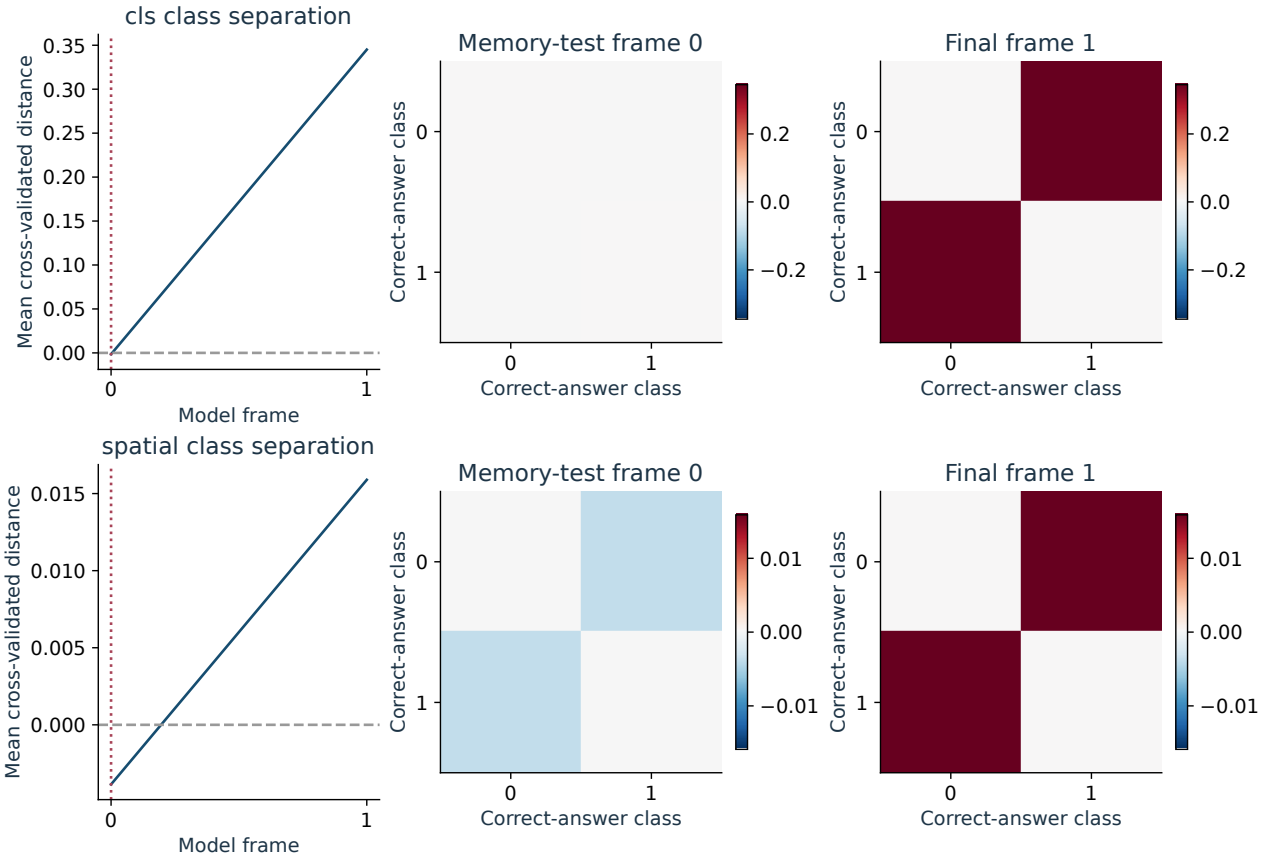


Figure 16: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

Orientation ordering becomes much more readable in recurrent states than in the capacity-matched appearance and fused representations. Resetting the global reference-period state produced a substantial subsequent error increase, with an additional cost from resetting spatial memory. This pattern links the final response to information carried across the two observations. Concentrating priority in a fixed quadrant also impaired performance, even though the grating occupies the full image. The relevant spatial question is therefore how the selected sample supports a stable comparison, rather than whether an isolated target object receives more attention. The combination of perturbation and decoding localizes a comparison computation more strongly than final CLS decodability alone.

The behavioral result establishes 100.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies cls_gain0_t0 as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Future tests should rotate the entire coordinate frame, alter spatial frequency, and independently change contrast. Those transfers would assess whether the learned sign judgement follows orientation itself across nuisance transformations.

4.1 Interpreting these results alongside biological experiments

4.1.1 De Valois, Yund and Hepler (1982)

The biological measurement concerns a neuron's response as absolute orientation changes. Our answer decoder concerns the sign of a difference between two orientations. Those quantities should be separated experimentally: first map absolute-angle tuning, then measure whether comparison information appears only after the second image. Strong late answer decoding can arise even when early features represent both angles in a distributed form. [4]

4.1.2 Harrison and Tong (2009)

Our recurrent-state reset costs connect the first observation to the later decision. The human result motivates decoding the retained angle itself during the intervening state. An answer label alone is insufficient before the second angle defines clockwise or counterclockwise. Testing angle reconstruction and then resetting its carrier would link represented content to comparison behavior. [5]

4.1.3 McAdams and Maunsell (1999)

Concentrating priority in a quadrant harms this model's full-field comparison. This is compatible with losing useful distributed evidence, whereas the V4 experiment measures gain for an attended stimulus.

To compare them, hold the grating and response rule fixed and measure coordinate tuning with and without a spatial-priority bias. The shape of that tuning curve matters in addition to its amplitude. [6]

4.2 A computational account of the observed behavior

A natural computational account has an encoding stage, a retained reference, and a signed comparison. Because orientation repeats after 180 degrees, a stable representation uses two coordinates, cosine and sine of twice the angle. A readout comparing those coordinates can distinguish nearby rotations while respecting the wrap at 180 degrees. The existing state-reset results support a role for retained reference information: removing global or spatial memory after the reference impairs the final report. The mechanism could preserve an angle, a richer image representation, or both. The two-frame native sequence makes these hypotheses especially testable through controlled changes in phase, frequency and contrast between the observations.

4.3 Where this task agrees with the biological evidence

Behavior agrees with accurate sequential orientation discrimination, and recurrent perturbation identifies a causal dependence on reference-period state. Agreement with a specific cortical tuning mechanism remains an empirical question about absolute-angle response profiles and generalization across nuisance features.

4.4 A follow-up experiment motivated by these results

Independently randomize grating phase and contrast across the two observations. Decode the first angle before the probe and compare preferred-angle tuning across states. If a transferable orientation reference drives the task, signed judgment should survive nuisance changes that preserve the two orientations.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] De Valois RL, Albrecht DG, Thorell LG (1982). Spatial frequency selectivity of cells in macaque visual cortex. Vision Research 22:545-559. https://pubmed.ncbi.nlm.nih.gov/7112954/.

[2] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[3] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[4] De Valois, Yund and Hepler (1982). The orientation and direction selectivity of cells in macaque visual cortex. Primary source.

[5] Harrison and Tong (2009). Decoding reveals the contents of visual working memory in early visual areas. Primary source.

[6] McAdams and Maunsell (1999). Effects of attention on orientation-tuning functions of single neurons in macaque cortical area V4. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
mixed	128	100.0%	1.000	0.000

B Every-frame attention atlas

Condition mixed. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 64 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

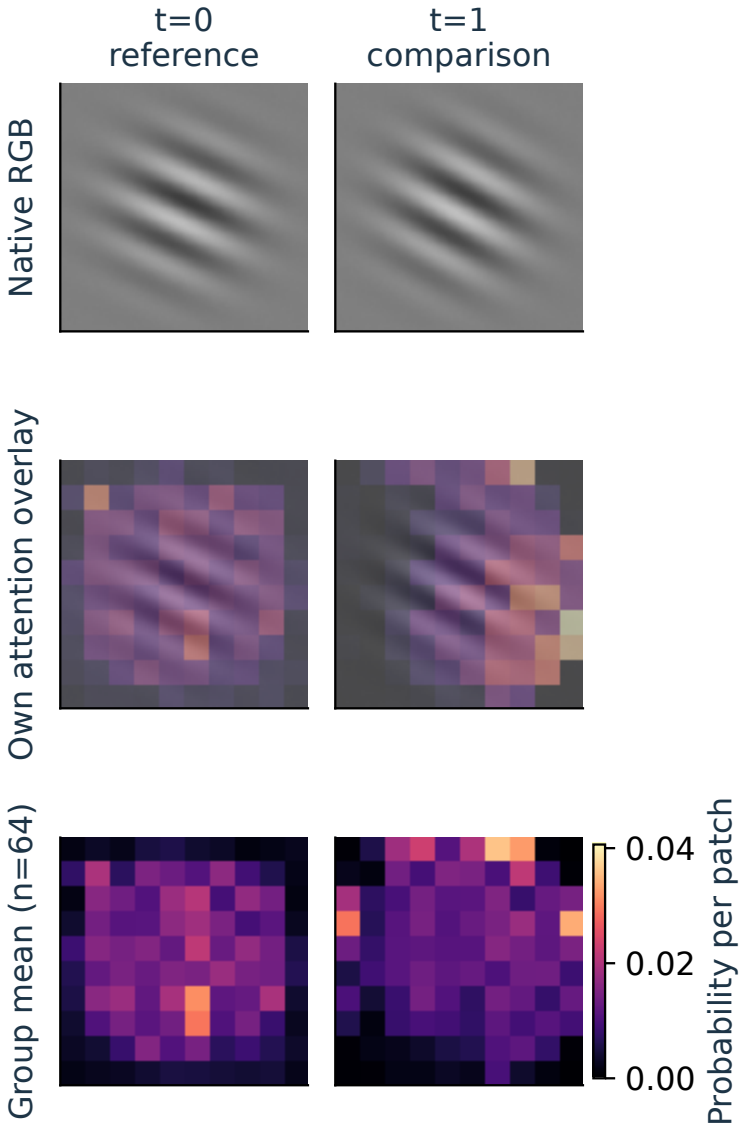


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

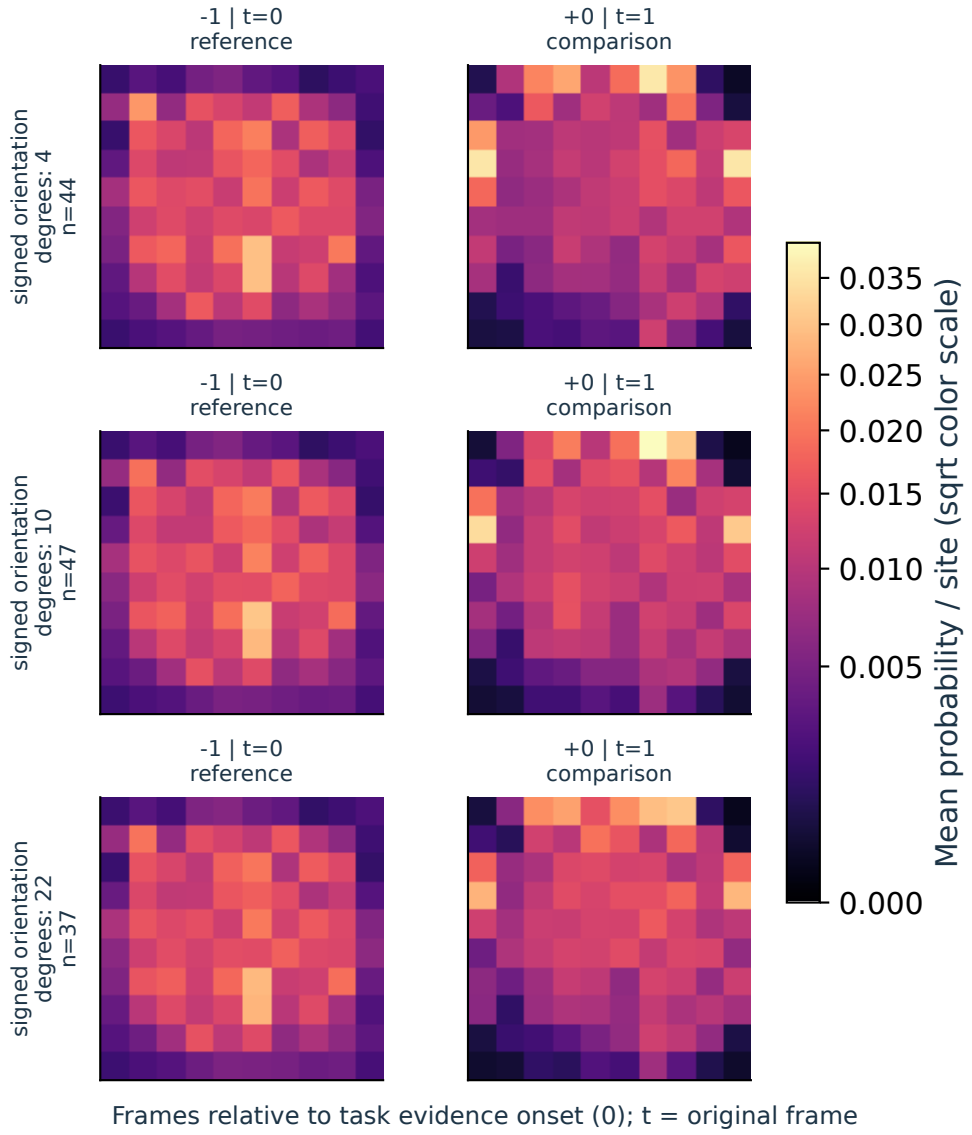


Figure 18: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Contrast comparison, spatial selection and recurrent state dependence

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized contrast in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 1 native conditions and 128 fresh trials, equal-condition accuracy was 100.0% (95% interval [97.1%, 100.0%]). For the representative condition mixed, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was spatial_gain0_t0, changing accuracy by -44.5 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

Contrast describes variation around an image brightness baseline. The task asks which of two grating intervals has the stronger contrast. Because the absolute pedestal varies, success requires comparing the two observations rather than applying one fixed intensity criterion.

The contrast increment is the primary difficulty variable. Separating sensitivity from interval preference is essential: a tendency to choose the second interval can change overall accuracy when samples are imbalanced. We therefore retain confusion matrices, class-balanced accuracy, and probability-based discrimination. Spatial priority interventions test whether the model can perform the comparison using a restricted part of the grating.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Campbell and Robson connected grating visibility to spatial-frequency content, while normalization models explain how stimulus drive and attentional allocation interact. This suggests analyzing both the physical contrast increment and the location from which evidence is selected. A priority manipulation could influence a contrast judgement by favoring reliable portions of the grating, by altering memory writes, or by changing the selected readout. Its behavioral effect should therefore be compared with the state at the same phase. The present recordings allow a distinction between a clear contrast-related representation and successful use of that representation by the decision pathway.

1.2 Biological experiments motivating this task

Experimental question. Does the model compare a retained contrast signal, and does its sensitivity follow the nonlinear contrast behavior seen in vision?

Legge and Foley (1980) Human grating discrimination showed facilitation at weak masking contrasts and increased thresholds at stronger masks. A nonlinear contrast transducer, spatial filtering and pooling explained the dependence on masker strength and frequency. [5]

Reynolds, Pasternak and Desimone (2000) Attention increased V4 responses more strongly at low than high stimulus contrast. The pattern was consistent with an increase in effective contrast, connecting attentional state with sensory sensitivity. [6]

Campbell and Robson (1968) Thresholds for several grating waveforms were explained by the visibility of their Fourier components. Higher harmonics became behaviorally useful when they exceeded their own thresholds, supporting frequency-selective visual mechanisms. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Which frame contains the higher-contrast grating?

Response labels are 0: frame 0; 1: frame 1. The native suite contains 1 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use mixed.

1.4 Stimulus sequence and spatial layout

For the representative native condition mixed, the sequence has 2 frames, indexed from zero: frame 0: reference; frame 1: comparison. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Contrast increment	0.025, 0.06, 0.13
Contrast pedestal	0.08, 0.18, 0.3
Stimulus family	contrast

The appearance pathway and local contrast statistics both provide signals relevant to this task. Their fusion is then routed through the shared attention map. The population analyses ask whether the answer is already readable in visual features or becomes more explicit after recurrent integration. A global additive priority shift is an exact control because softmax removes any common offset.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 100.0%, with macro AUC 1.000. The matched final checkpoint changed accuracy by -3.12 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was mixed at 100.0%; the highest was mixed at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (0.002) produced 60.9% correct responses, compared with 100.0% at the largest strength (0.13). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

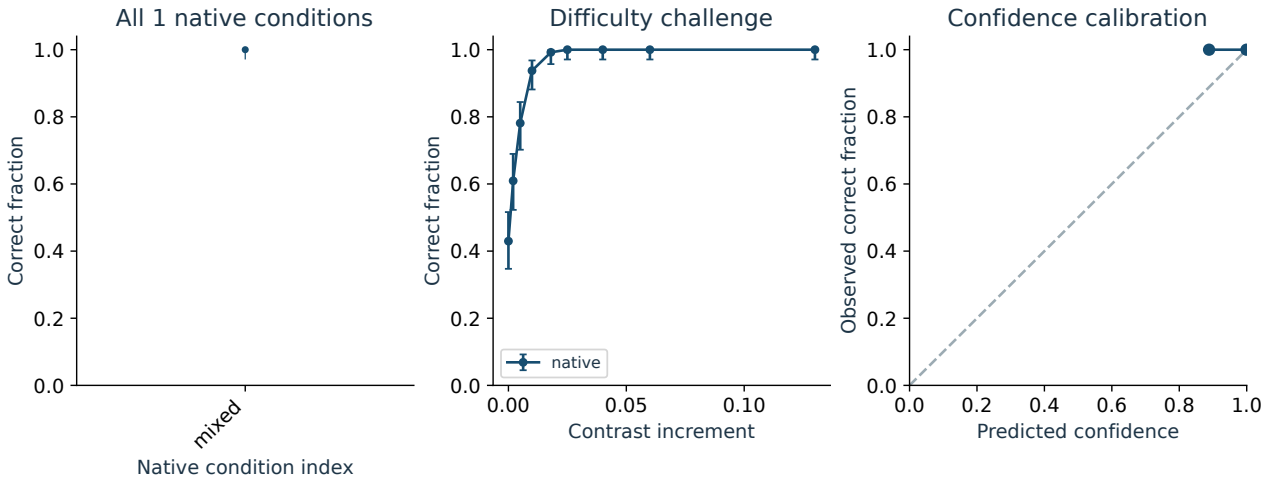


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)[1 - \exp(-(x/a)^b)]$. Here x is signal strength, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 0.004365 task units (bootstrap interval [0.003074, 0.005888]), with slope 51.85 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.070.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	100.0%	100.0%
0.7	100.0%	1.0	100.0%	97.7%
0.8	100.0%	1.0	100.0%	91.4%
0.9	98.4%	1.0	100.0%	85.2%

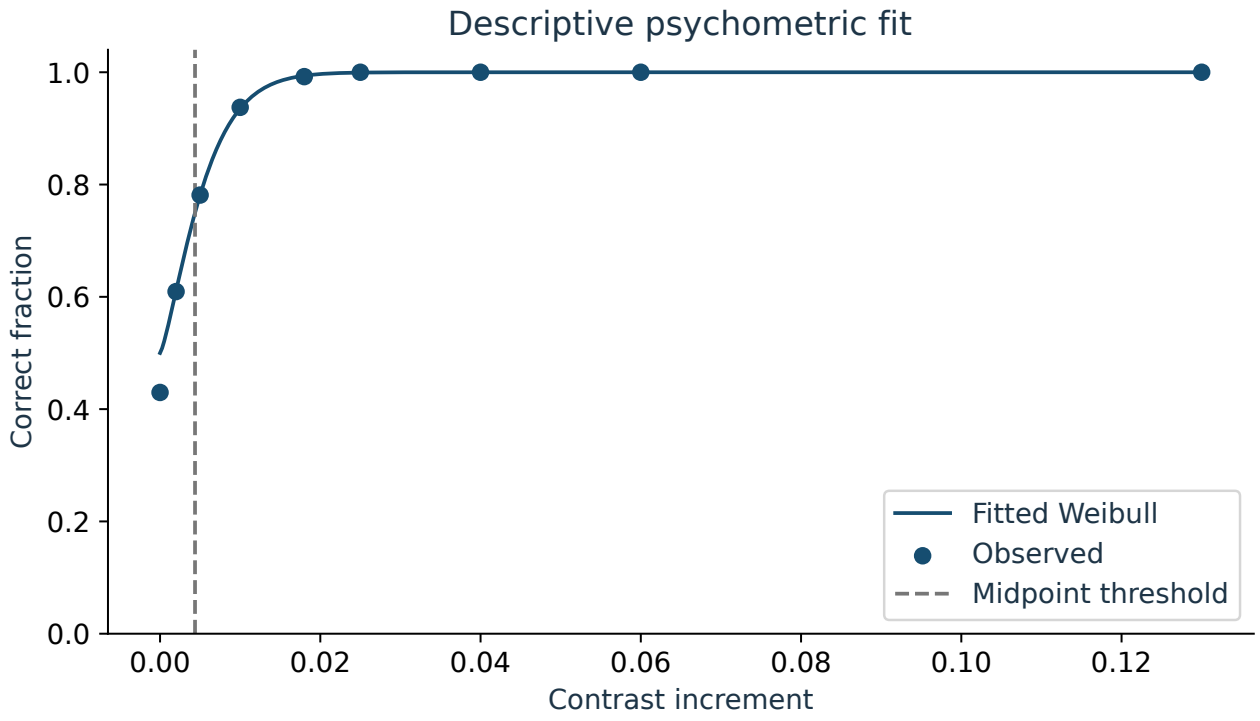


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

The representative condition mixed provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 128 of 128 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

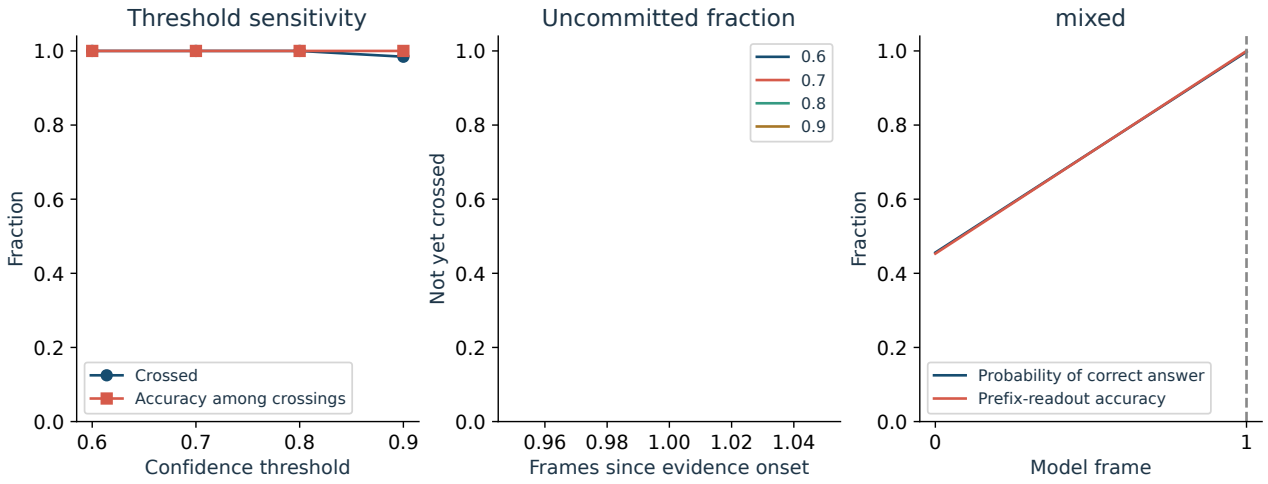


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

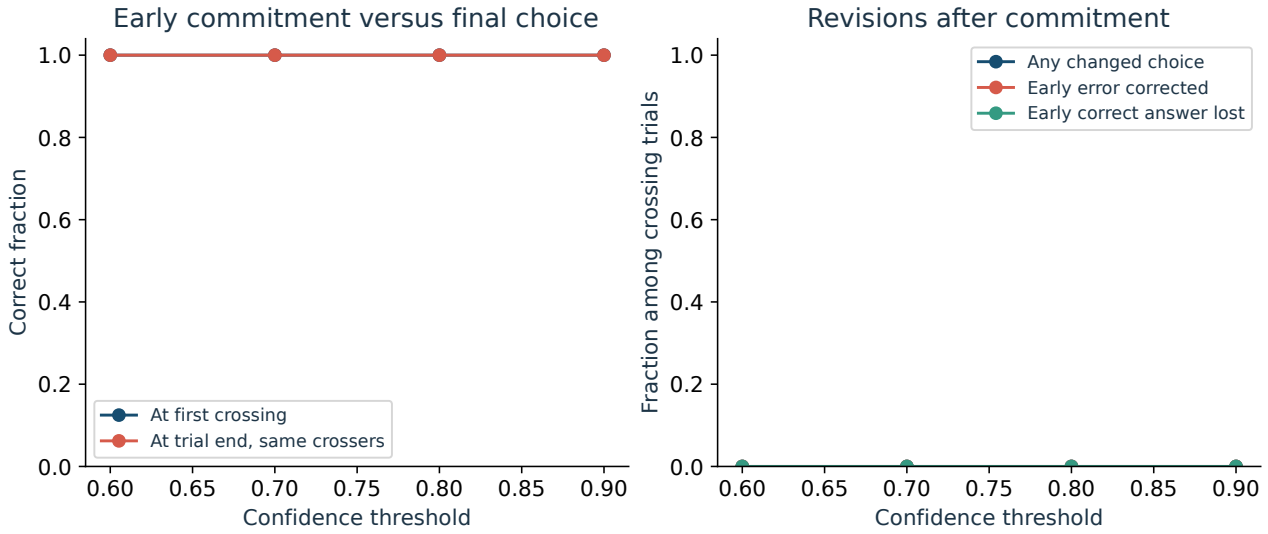


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

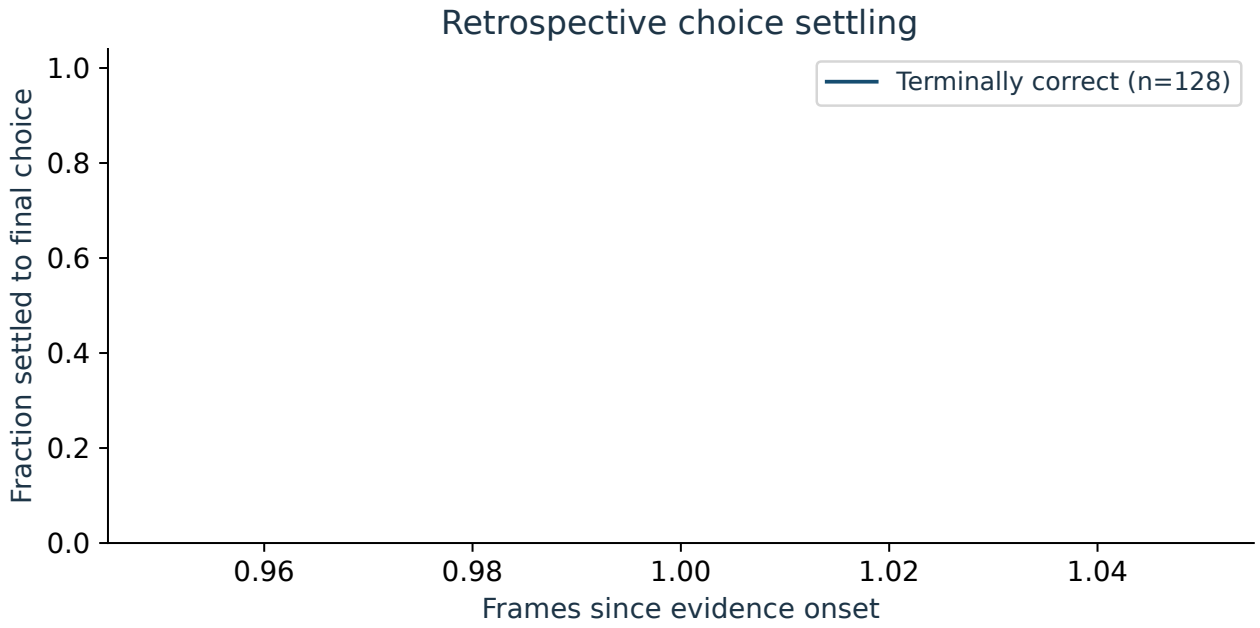


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

Rows separate contrast increments and pool pedestal levels and interval order. Traces refer to fixed image quadrants.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
contrast increment: 0.025	41	41
contrast increment: 0.06	47	47
contrast increment: 0.13	40	40

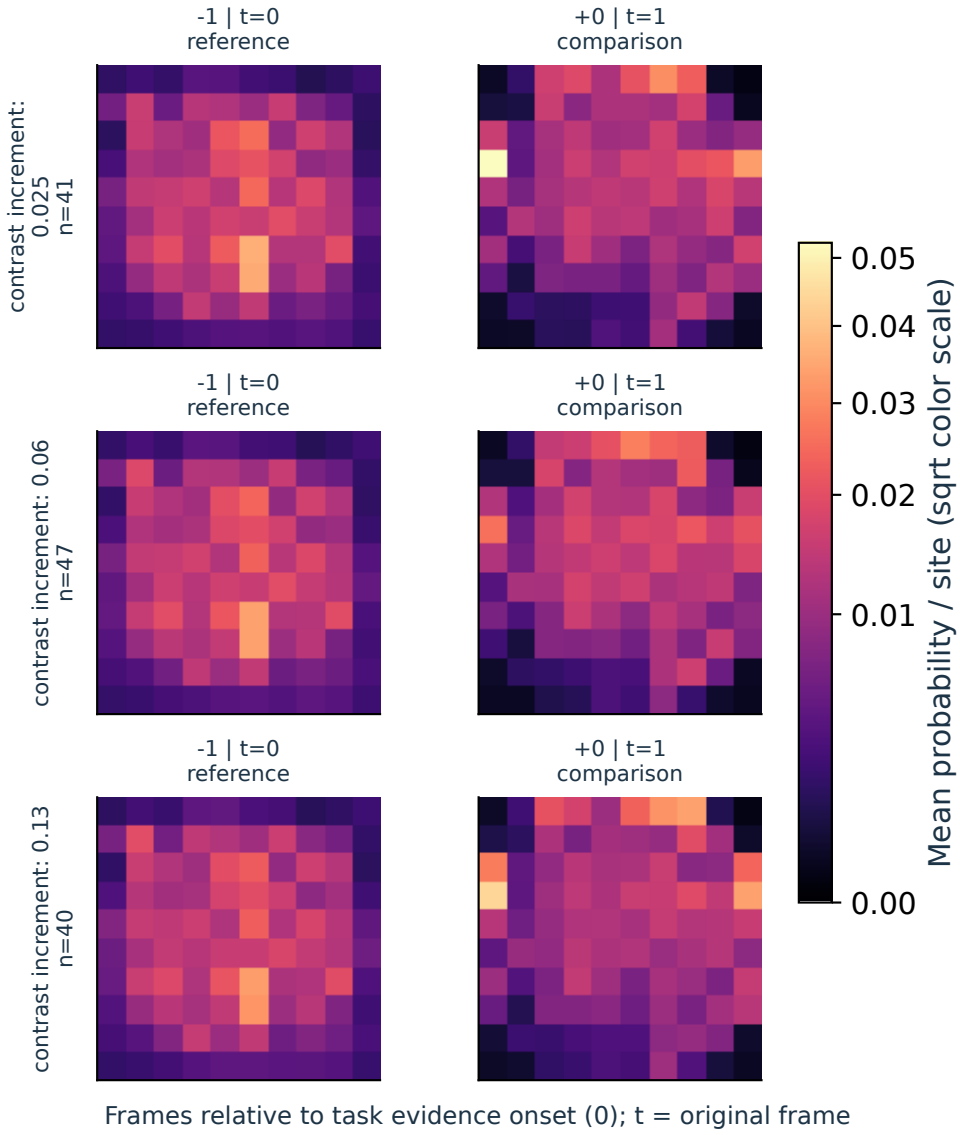


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

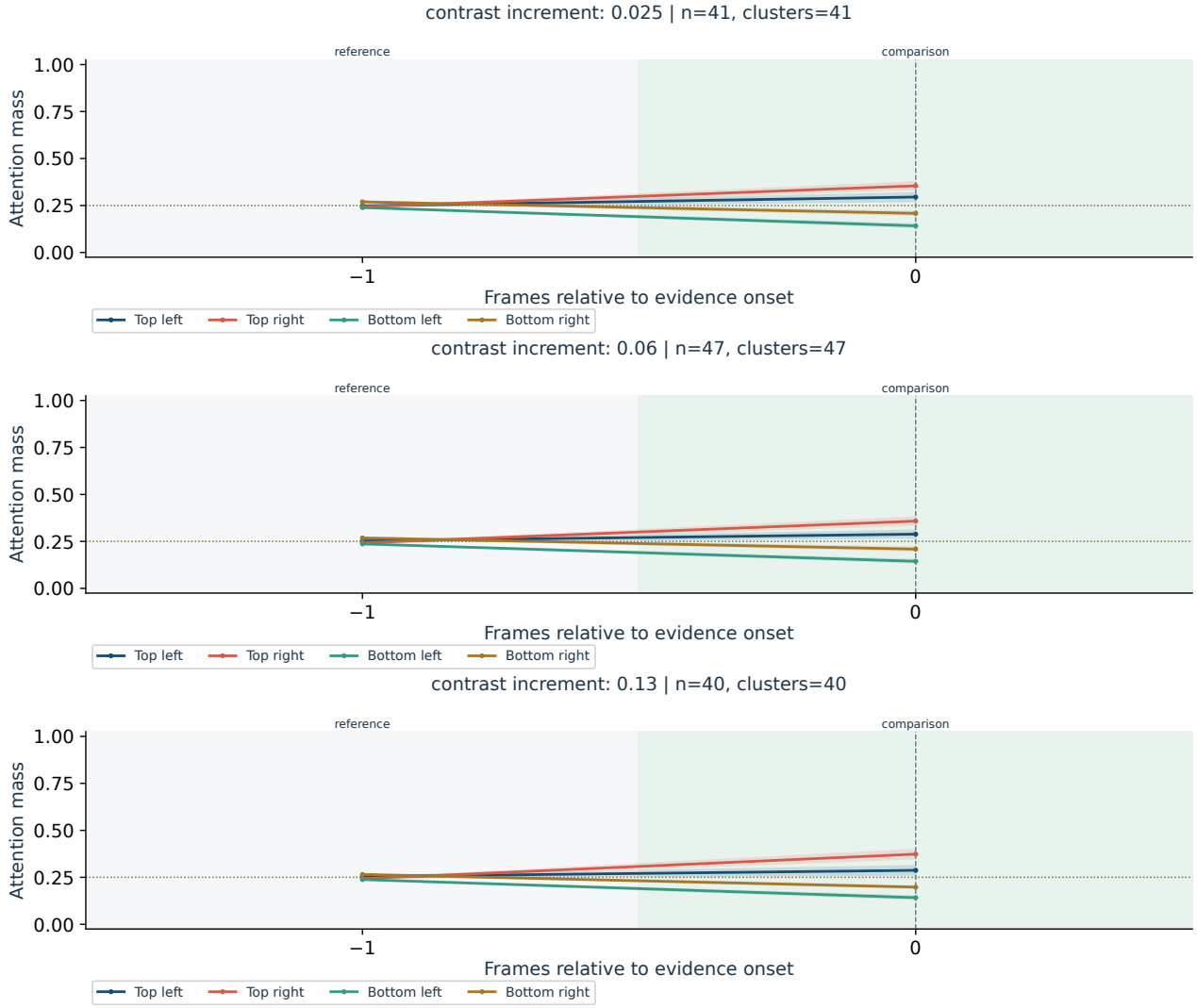


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

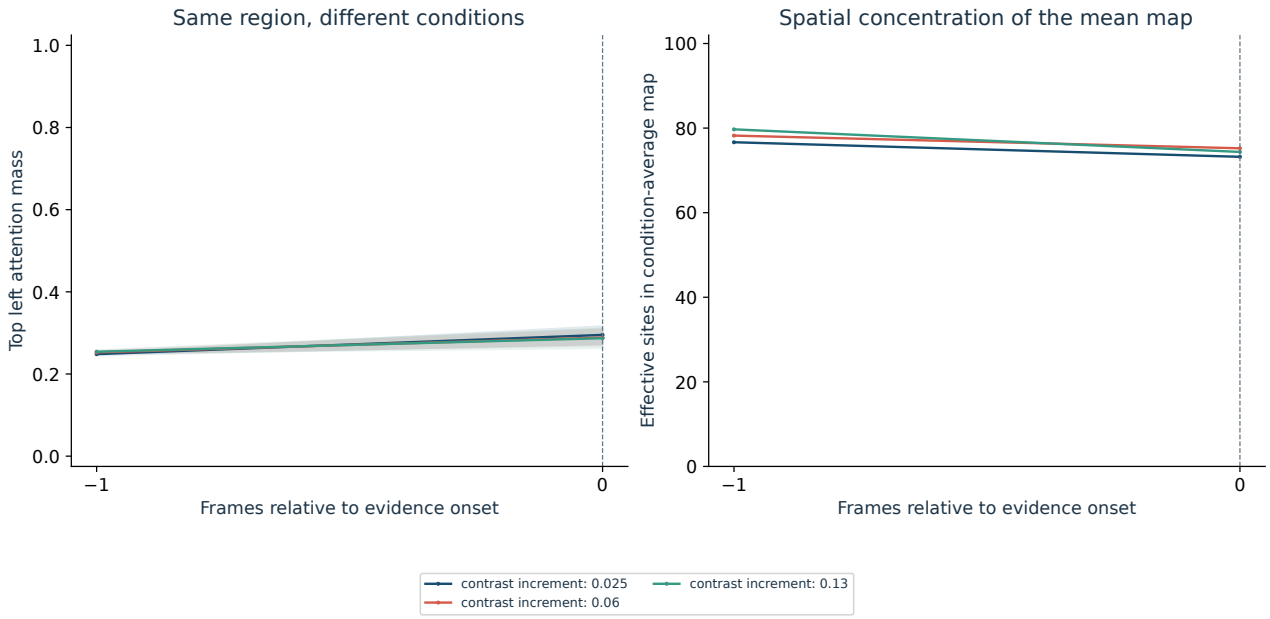


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. contrast increment: 0.025: top left mass peaks at 0.295 at original frame 1 (comparison), and is 0.295 on the final frame. contrast increment: 0.06: top left mass peaks at 0.289 at original frame 1 (comparison), and is 0.289 on the final frame. contrast increment: 0.13: top left mass peaks at 0.287 at original frame 1 (comparison), and is 0.287 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For spatial_gain0_t0, paired accuracy changed by -44.5 percentage points (paired 95% interval [-53.1%, -35.9%]; paired p=2.082e-16; within-task Holm p=6.453e-15). There were 1 corrected trials and 58 newly incorrect trials. For target_-8_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

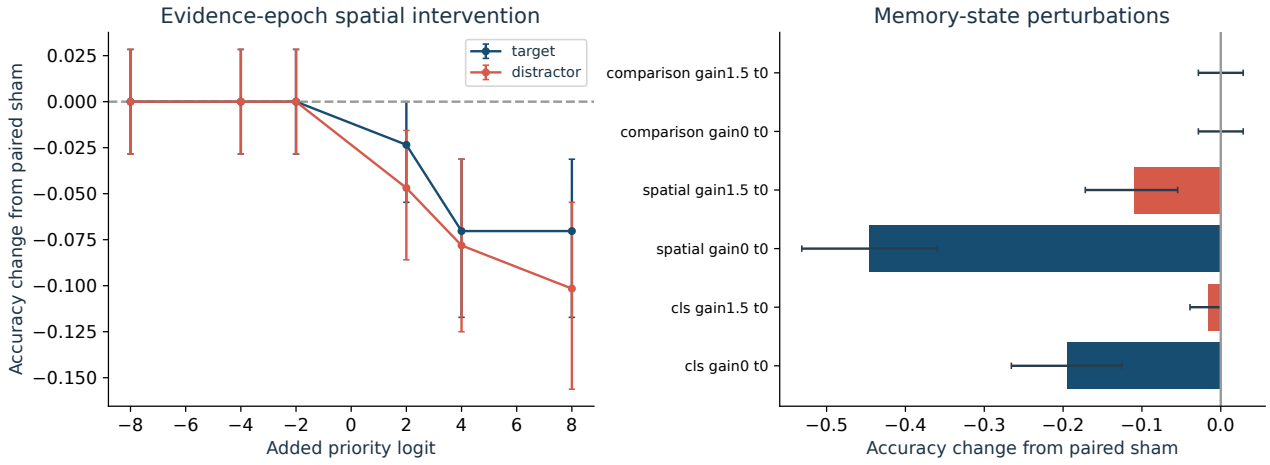


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
mixed	target_-4	+0.0	1
mixed	target_+4	-7.0	0.25
mixed	distractor_-4	+0.0	1
mixed	distractor_+4	-7.8	0.127

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.643	60.2%	0.060
fusion	0.880	76.6%	0.005
spatial	0.997	98.4%	0.005
cls	1.000	100.0%	0.005
comparison	Constant zero on this condition		
attention	0.914	81.2%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

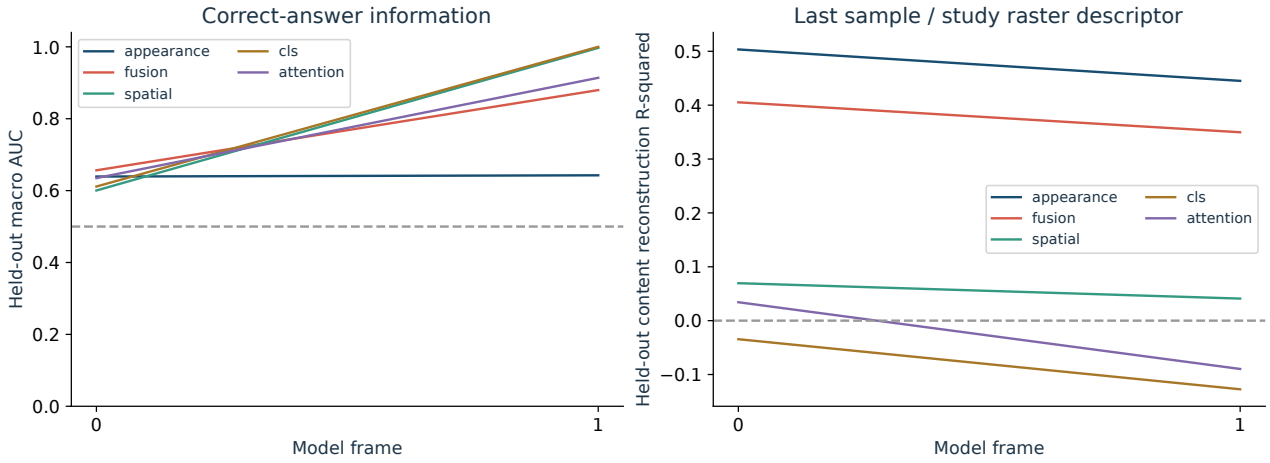


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 11.09 to 16.66 within the retained subspace. The final step displacement was 1.330 standardized RMS units. In spatial memory, the participation ratio ranged from 12.05 to 18.42 within the retained subspace. The final step displacement was 1.113 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 0, coarse sample-content reconstruction had R-squared -0.034 from CLS memory and 0.070 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -19.5 percentage points, spatial, -44.5 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.678	0.689	3200
fusion	0.945	0.692	6400
spatial	1.000	0.775	6400
cls	1.000	0.609	64
comparison	unavailable	unavailable	–
attention	0.931	0.668	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 43 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

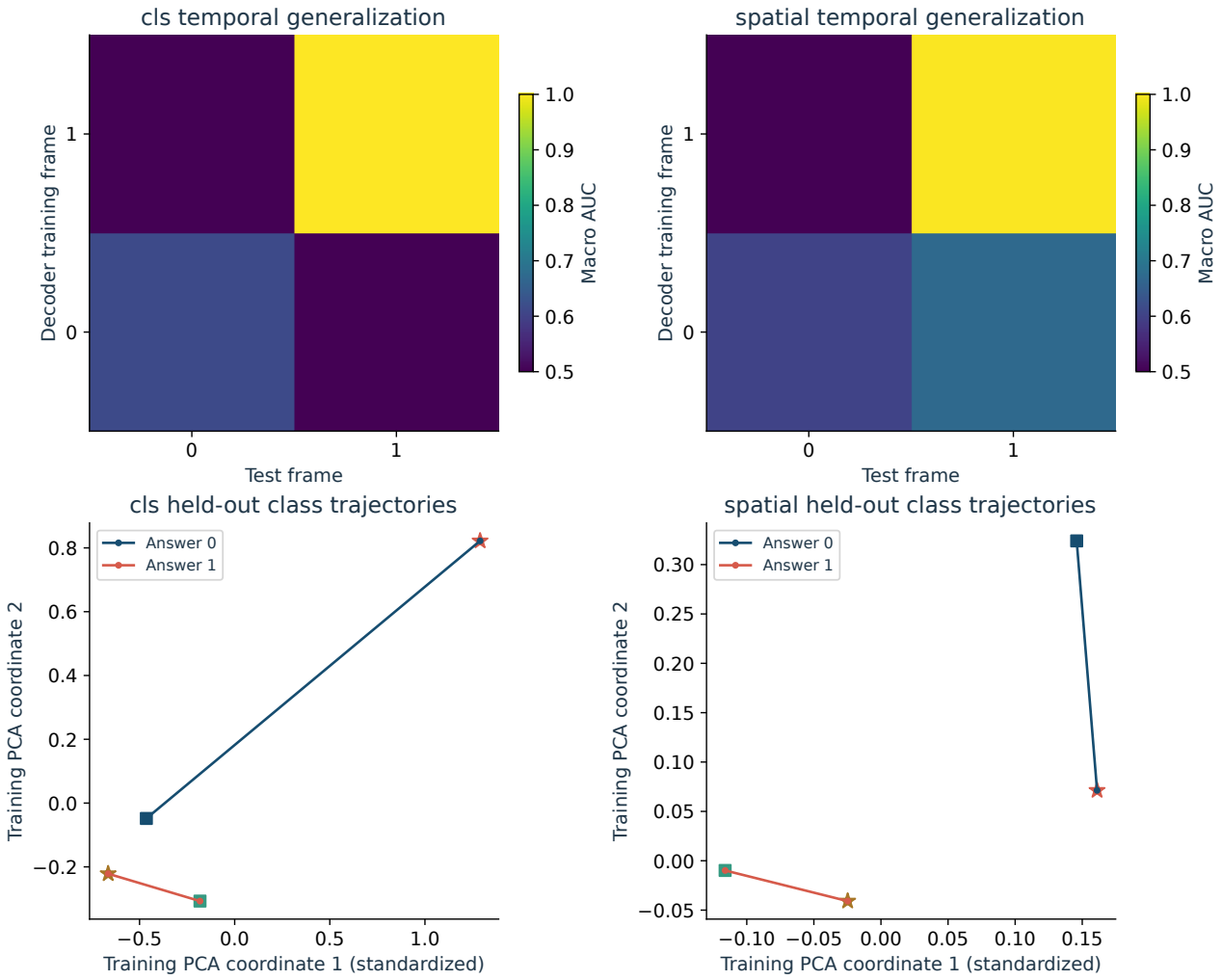


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

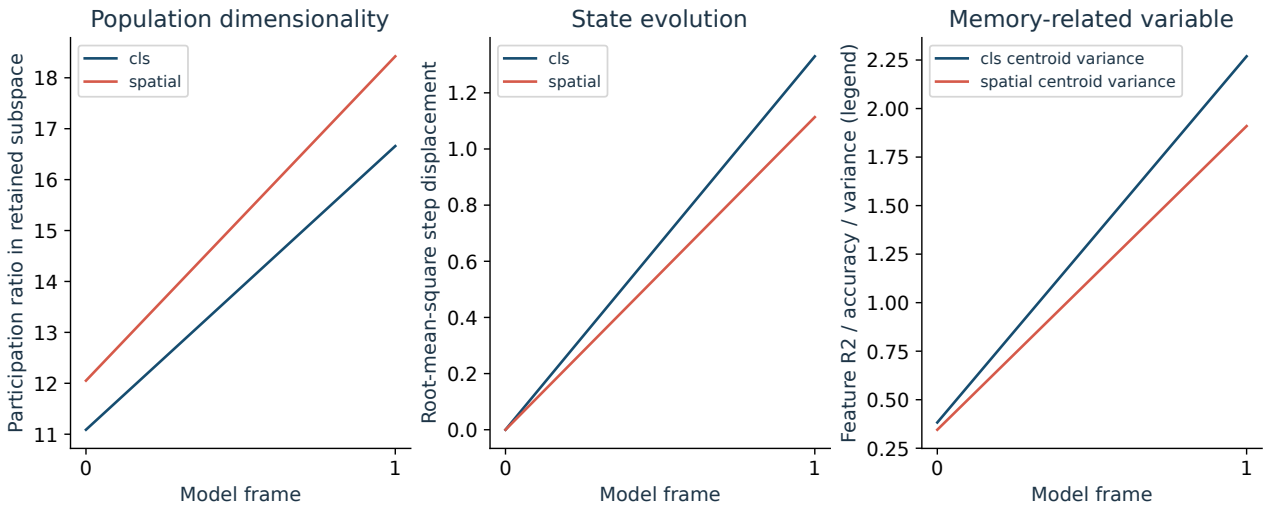


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

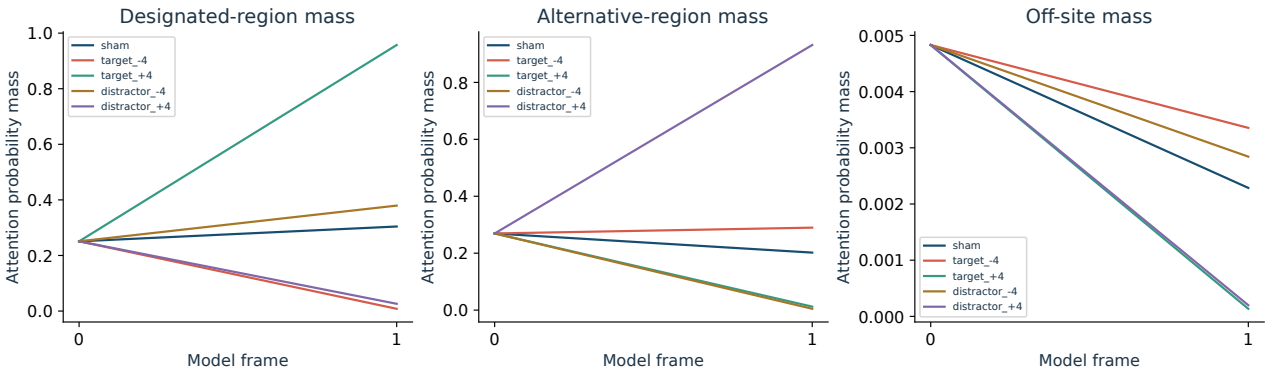


Figure 13: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

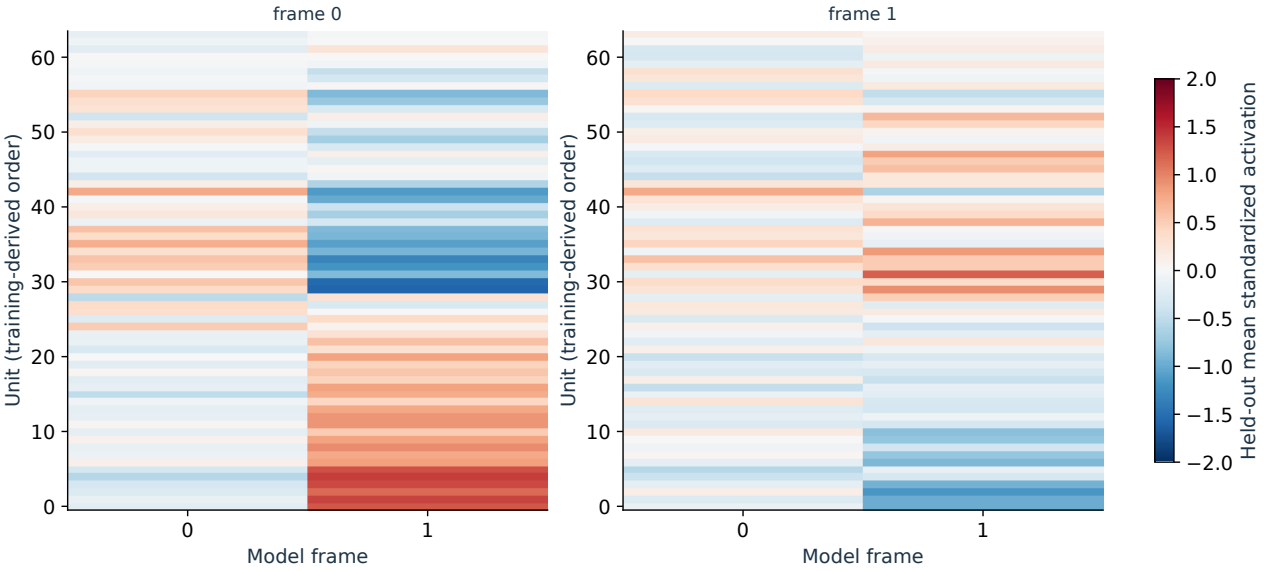


Figure 14: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	60.2%	60.2%	unavailable
fusion	76.6%	76.6%	unavailable
spatial	98.4%	98.4%	unavailable
cls	100.0%	100.0%	unavailable
attention	81.2%	81.2%	unavailable

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

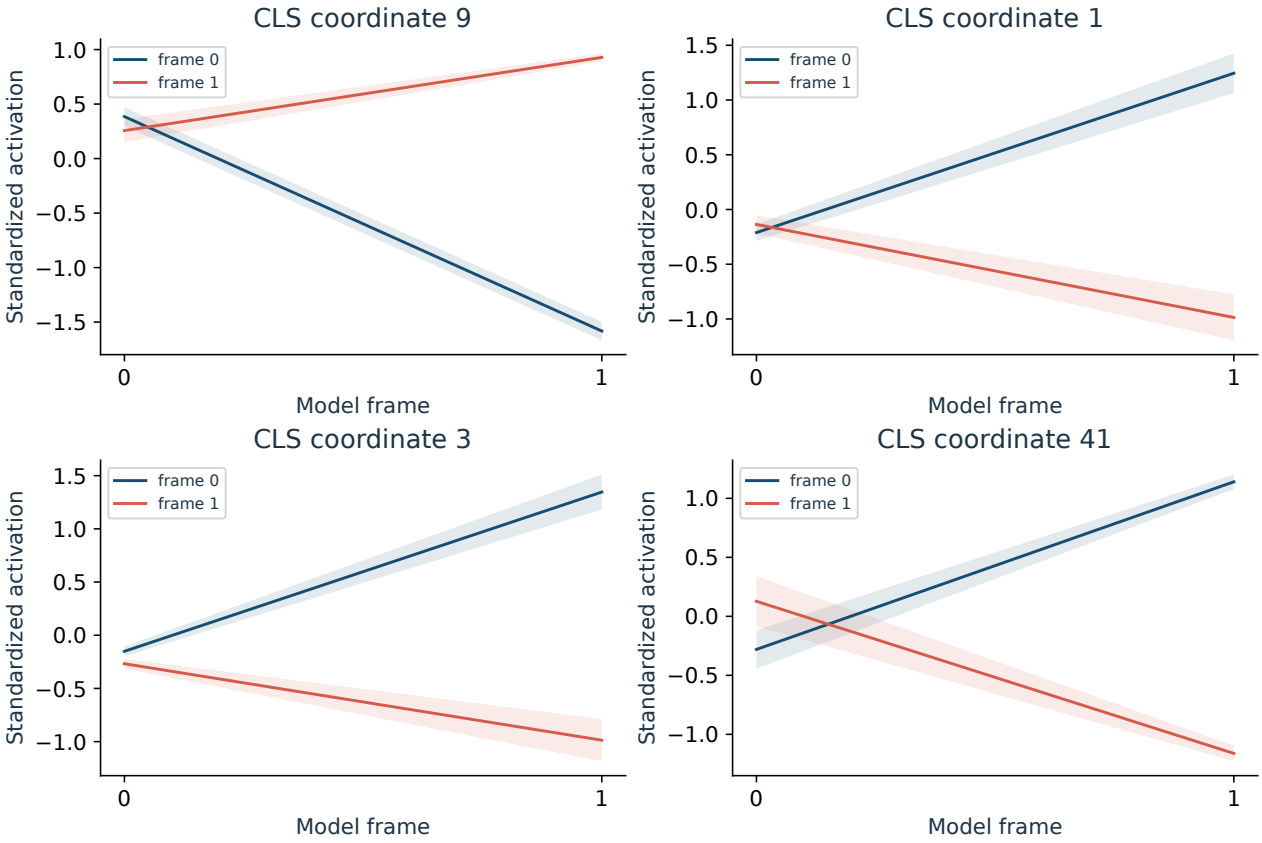


Figure 15: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

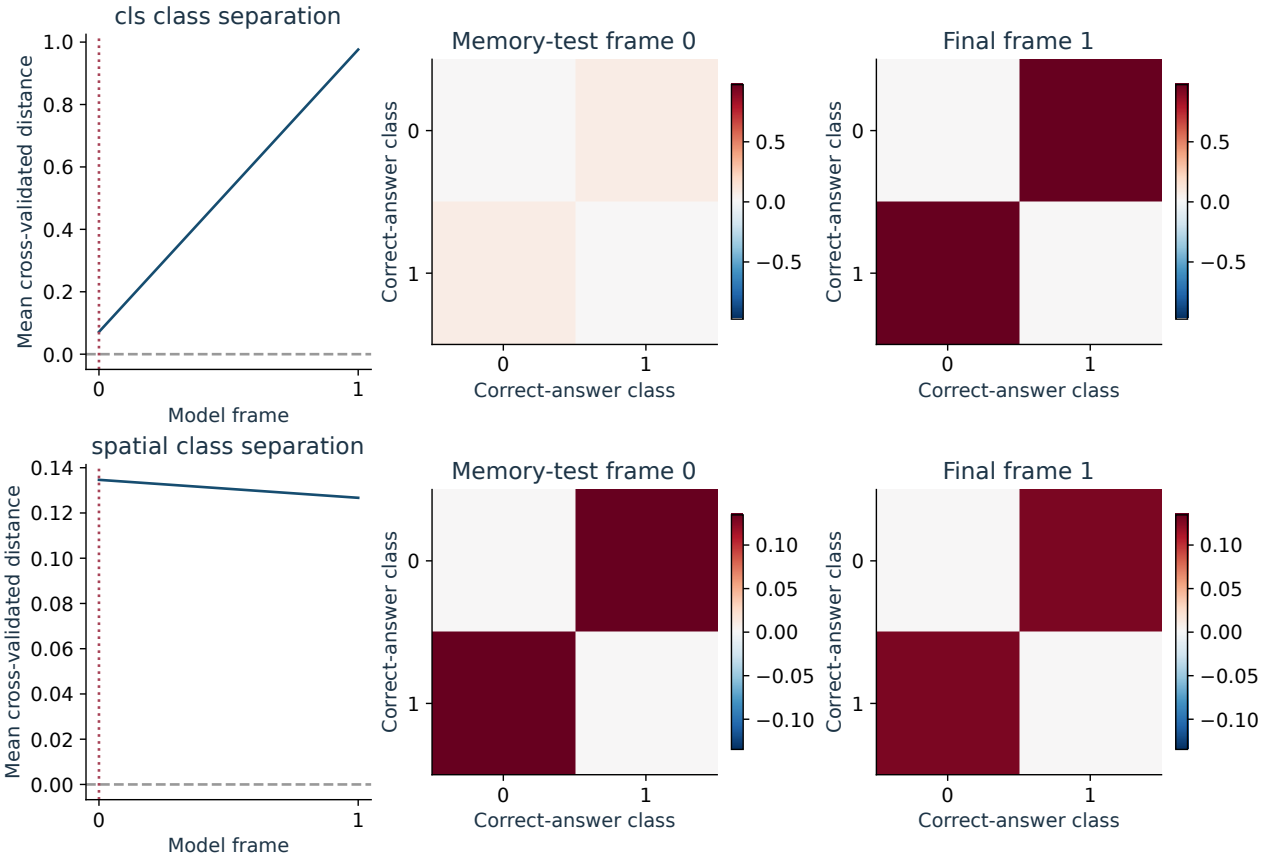


Figure 16: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

The fused representation already carries appreciable information about the correct interval, and recurrent populations make that information nearly fully readable. The largest reference-period memory-reset cost came from spatial memory. This connects the accurate final comparison to a retained spatial description of the first interval. Absolute contrast and the temporal image representation can each provide partial cues; the high terminal accuracy and reset effects reveal how those signals are combined in the trained observer. The subthreshold increment sweep measures the limit of that computation, while quadrant interventions test how concentrated sampling changes the reliability of the evidence being compared.

The behavioral result establishes 100.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies spatial_gain0_t0 as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

A pedestal-by-increment factorial would resolve whether thresholds follow a Weber-like relationship. Contrast polarity reversals and independently varied mean luminance would further separate contrast comparison from brightness cues.

4.1 Interpreting these results alongside biological experiments

4.1.1 Legge and Foley (1980)

Our increment challenge asks how small a contrast difference the observer can use. The human result asks how that threshold changes with the reference or masking contrast. Native ceiling accuracy and a single pooled increment curve cannot reveal the low-pedestal facilitation and high-pedestal masking pattern. The decisive comparison is a separate threshold curve at each pedestal, using the same spatial frequency and timing. [5]

4.1.2 Reynolds, Pasternak and Desimone (2000)

The monkey measurement is a contrast-response curve of a sensory population under attention. Our manipulation changes normalized spatial routing. To compare mechanisms, record the model's fused or recurrent response across contrast under the same attended and unattended geometry, then fit response amplitude and horizontal contrast shift separately. A behavioral improvement by itself would leave both possibilities open. [6]

4.1.3 Campbell and Robson (1968)

Contrast has meaning relative to a spatial pattern. Frequency-selective detection explains why an equal numerical contrast need not be equally visible for every grating. The model has a finite raster, fixed site spacing and learned image features. Varying frequency alongside pedestal would test whether its thresholds reflect a visual filter bank or a broad image-statistic comparison. [7]

4.2 A computational account of the observed behavior

Contrast discrimination combines a sensory transducer with comparison memory. A saturating transducer can make the same physical increment produce different internal differences at different pedestals. The spatial-memory reset cost in the present task shows that the reference image leaves a behaviorally useful spatial state. The final recurrent readout can then compare a new contrast-dependent pattern with that reference. Normalization theory adds a second operation: the response to one drive depends on competing drive. Layer normalization in the recurrent unit stabilizes channel scale, while attention softmax redistributes spatial weight. Their mathematical normalization does not by itself determine a physiological contrast-response law; that law must be measured by the pedestal experiment. [8]

4.3 Where this task agrees with the biological evidence

The model agrees with the broad requirement for retaining and comparing contrast-bearing evidence. The available data establish fine discrimination under the tested rendering. The characteristic pedestal dependence of human vision is the sharper biological test, and it is not determined by the present native ceiling result.

4.4 A follow-up experiment motivated by these results

Estimate thresholds over a two-dimensional pedestal-by-increment grid, with absolute frequency controlled. Repeat under broad and concentrated attention. Fit a saturating transducer and ask whether one parameter set predicts held-out pedestals and attention conditions.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Campbell FW, Robson JG (1968). Application of Fourier analysis to the visibility of gratings. Journal of Physiology 197:551-566. https://pubmed.ncbi.nlm.nih.gov/5666169/.

[2] Reynolds JH, Heeger DJ (2009). The normalization model of attention. Neuron 61:168-185. https://pubmed.ncbi.nlm.nih.gov/19186161/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] Legge and Foley (1980). Contrast masking in human vision. Primary source.

[6] Reynolds, Pasternak and Desimone (2000). Attention increases sensitivity of V4 neurons. Primary source.

[7] Campbell and Robson (1968). Application of Fourier analysis to the visibility of gratings. Primary source.

[8] Reynolds and Heeger (2009). The normalization model of attention. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
mixed	128	100.0%	1.000	0.000

B Every-frame attention atlas

Condition mixed. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 64 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

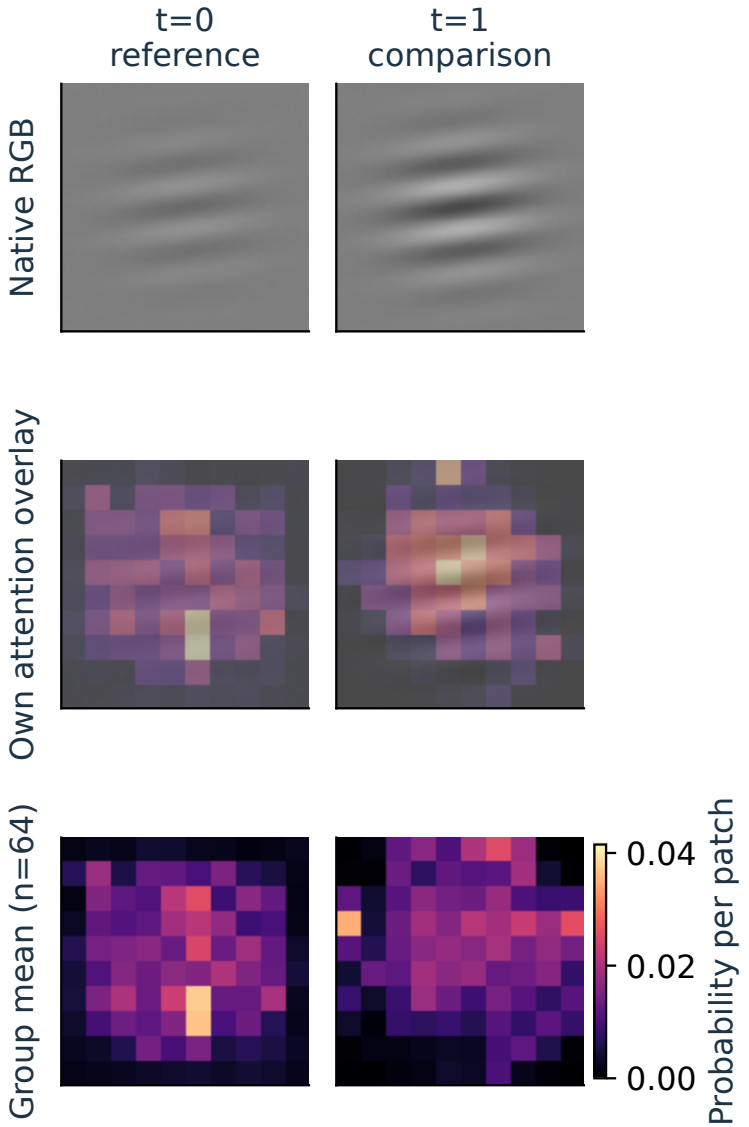


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

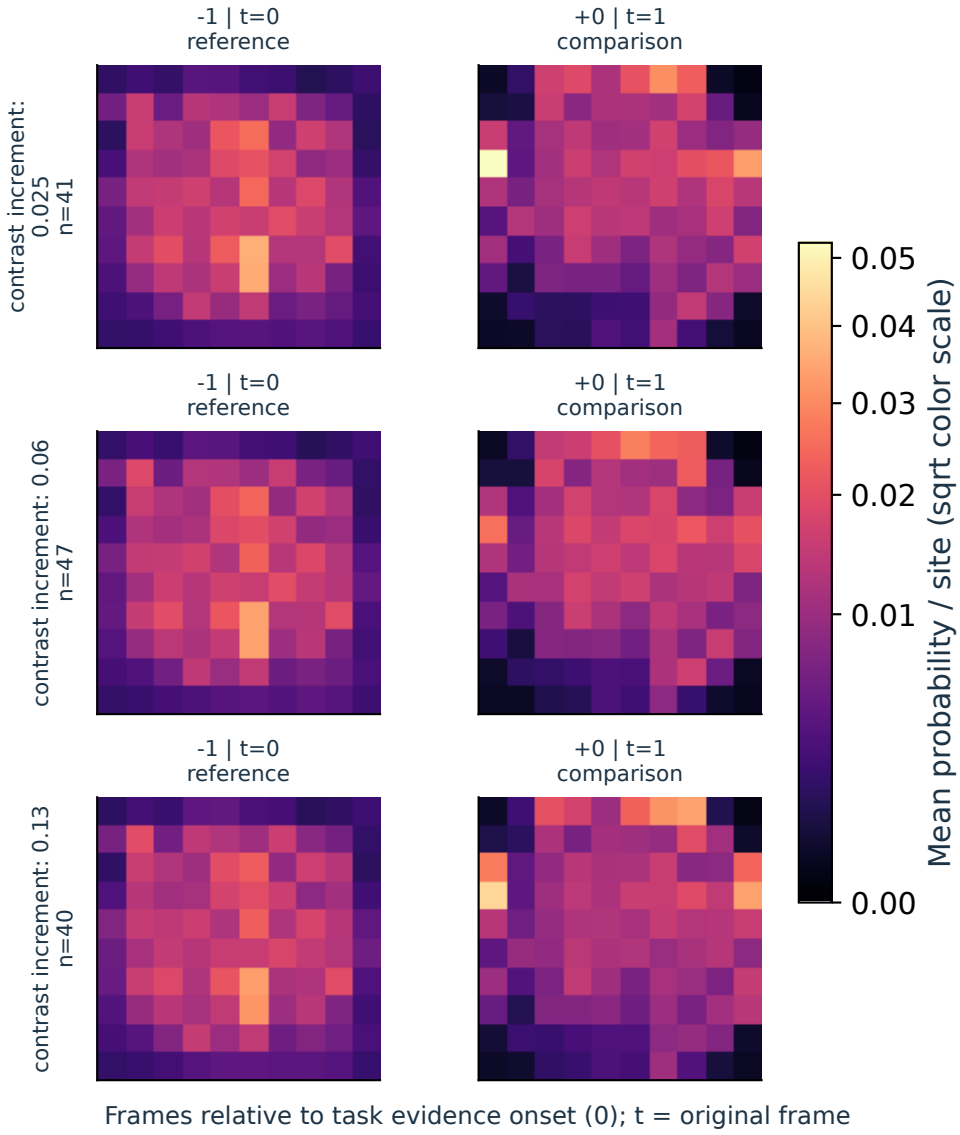


Figure 18: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Spatial-frequency discrimination in a shared visual architecture

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized spatial frequency in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 1 native conditions and 128 fresh trials, equal-condition accuracy was 100.0% (95% interval [97.1%, 100.0%]). For the representative condition mixed, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was cls_gain0_t0, changing accuracy by -43.8 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

Spatial frequency measures how rapidly a pattern changes across space. In this task the observer chooses the interval containing the finer grating. The required representation must remain useful across starting frequency, orientation and phase, while retaining the relative difference between intervals.

Difficulty is expressed as an octave difference, a logarithmic ratio of frequencies. Equal octave steps correspond to equal multiplicative changes. The psychometric function tests whether frequency discrimination degrades smoothly as the two patterns converge. Population decoding identifies where the interval ordering becomes accessible.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

De Valois and colleagues measured spatial-frequency selectivity in macaque visual cortex. Such selectivity provides a population basis for distinguishing coarse and fine patterns. For a two-interval judgement, however, the useful variable is their relation. The strongest mechanistic prediction is a progression from pattern-dependent activity to an interval-ordering code. Our regional decoders test that progression on held-out images, while memory perturbations test dependence on the earlier observation. A high terminal AUC in the global state can coexist with broad tuning in earlier visual features because the later state can combine many weak feature signals.

1.2 Biological experiments motivating this task

Experimental question. How can an observer report which of two gratings is finer, and what would establish a channel-based comparison?

Campbell and Robson (1968) Thresholds for several grating waveforms were explained by the visibility of their Fourier components. Higher harmonics became behaviorally useful when they exceeded their own thresholds, supporting frequency-selective visual mechanisms. [5]

De Valois, Albrecht and Thorell (1982) Measurements of striate neurons revealed many narrow spatial-frequency bandwidths and a wide range of preferred frequencies at a given eccentricity, supporting multiple spatial-frequency channels. [6]

Regan and Beverley (1983) Adaptation altered frequency-discrimination and grating-detection thresholds differently. The dissociation supported a distinction between detecting channel activity and comparing relative responses across overlapping channels. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Which frame contains the higher-spatial-frequency grating?

Response labels are 0: frame 0; 1: frame 1. The native suite contains 1 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use mixed.

1.4 Stimulus sequence and spatial layout

For the representative native condition mixed, the sequence has 2 frames, indexed from zero: frame 0: reference; frame 1: comparison. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Frequency increment (octaves)	0.08, 0.18, 0.35
Stimulus family	spatial_frequency

A response to local edges can be sensitive to both orientation and frequency. The task-specific head must select the relevant relation from their shared representation. Memory-state perturbations identify which recurrent component carries the comparison forward. Spatial quadrant manipulations assess how the representation tolerates restricting the evidence available for its decision.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 100.0%, with macro AUC 1.000. The matched final checkpoint changed accuracy by -3.12 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was mixed at 100.0%; the highest was mixed at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (0.005) produced 57.0% correct responses, compared with 100.0% at the largest strength (0.35). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

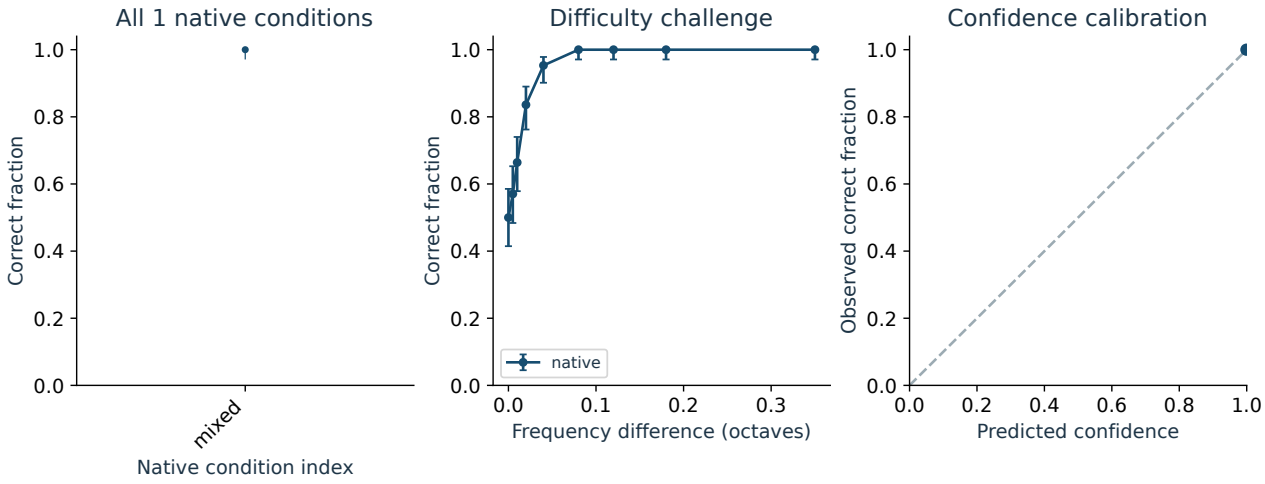


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)[1 - \exp(-(x/a)^b)]$. Here x is signal strength, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 0.01482 task units (bootstrap interval [0.01023, 0.02007]), with slope 15.29 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.015.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	100.0%	95.3%
0.7	100.0%	1.0	100.0%	89.8%
0.8	100.0%	1.0	100.0%	85.9%
0.9	100.0%	1.0	100.0%	74.2%

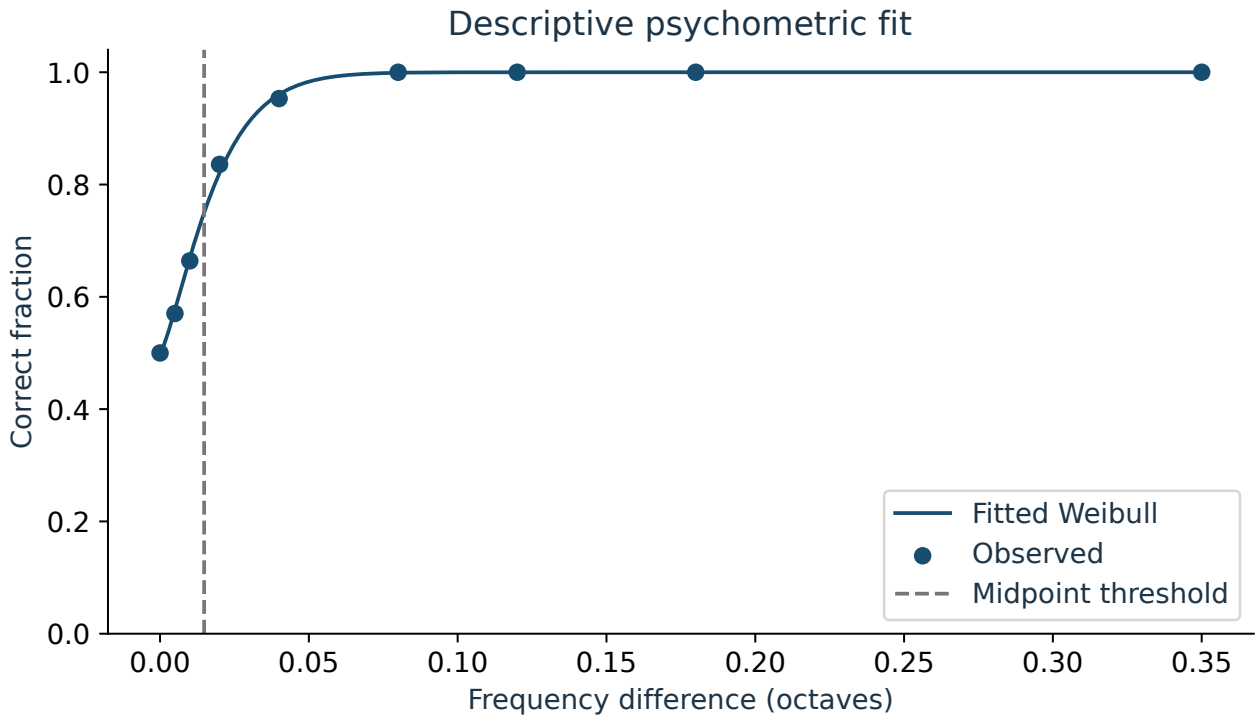


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

The representative condition mixed provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 128 of 128 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

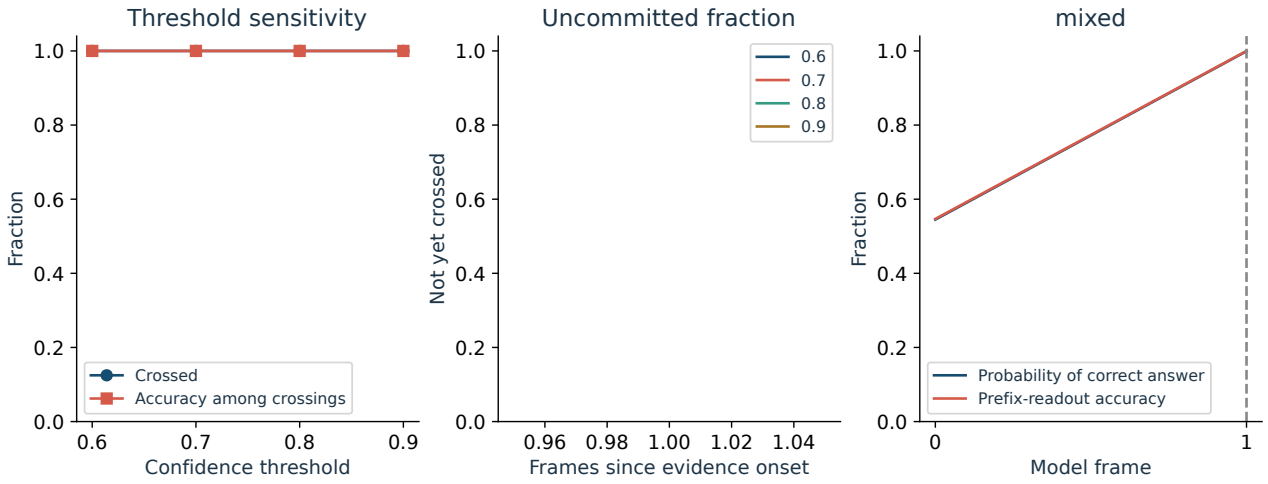


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

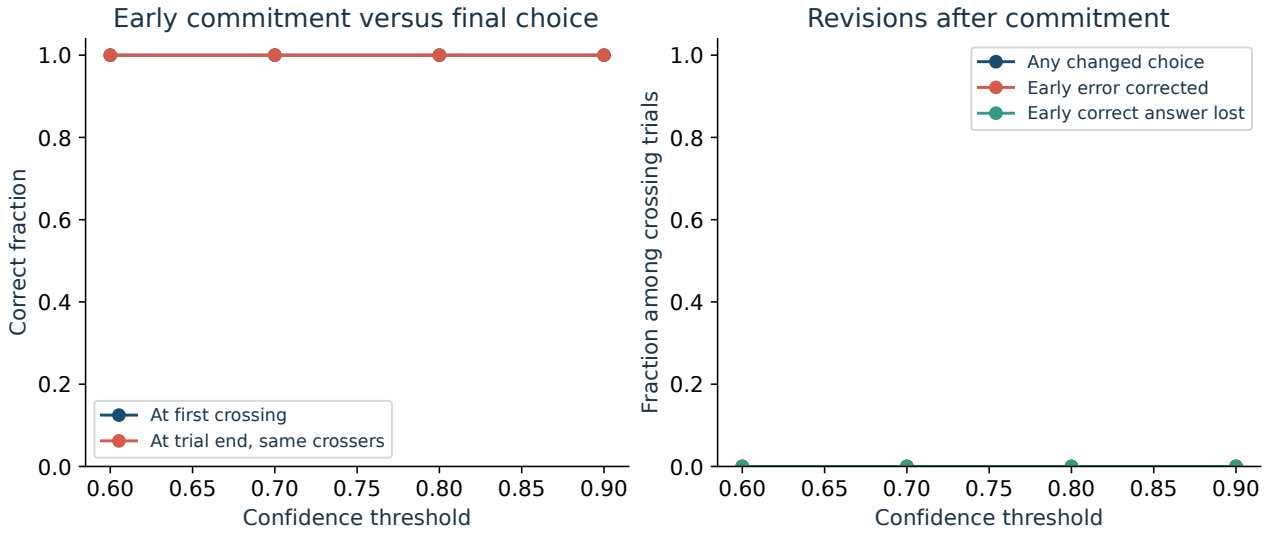


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

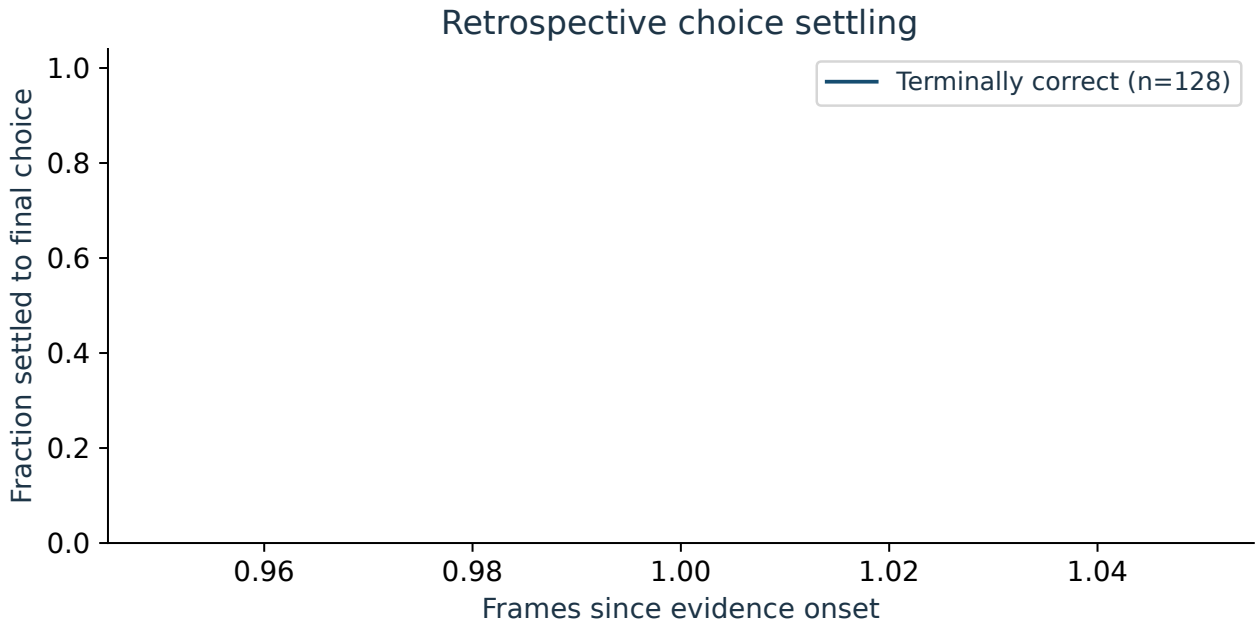


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

Rows separate octave increments and pool interval order and nuisance frequencies. Traces refer to fixed image quadrants.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
frequency octave increment: 0.08	45	45
frequency octave increment: 0.18	34	34
frequency octave increment: 0.35	49	49

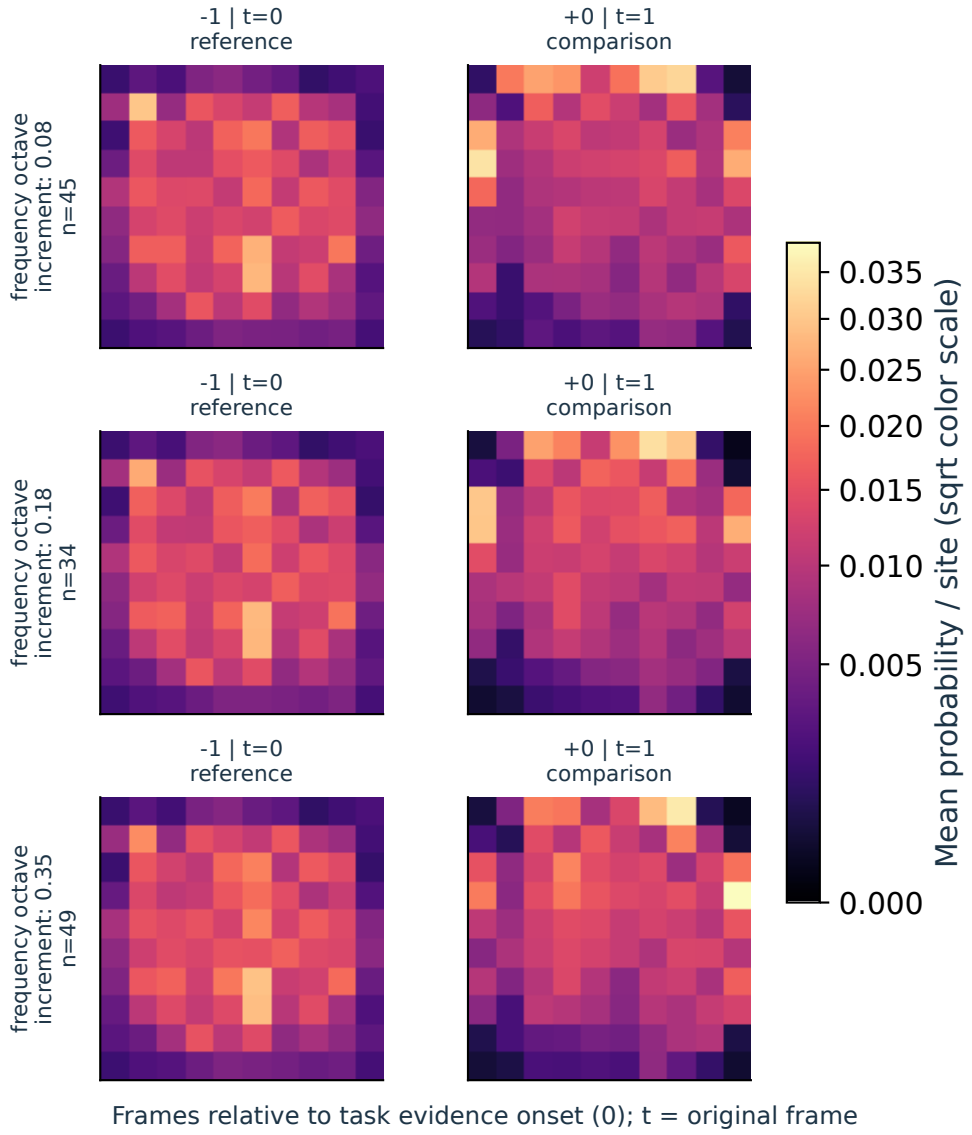


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

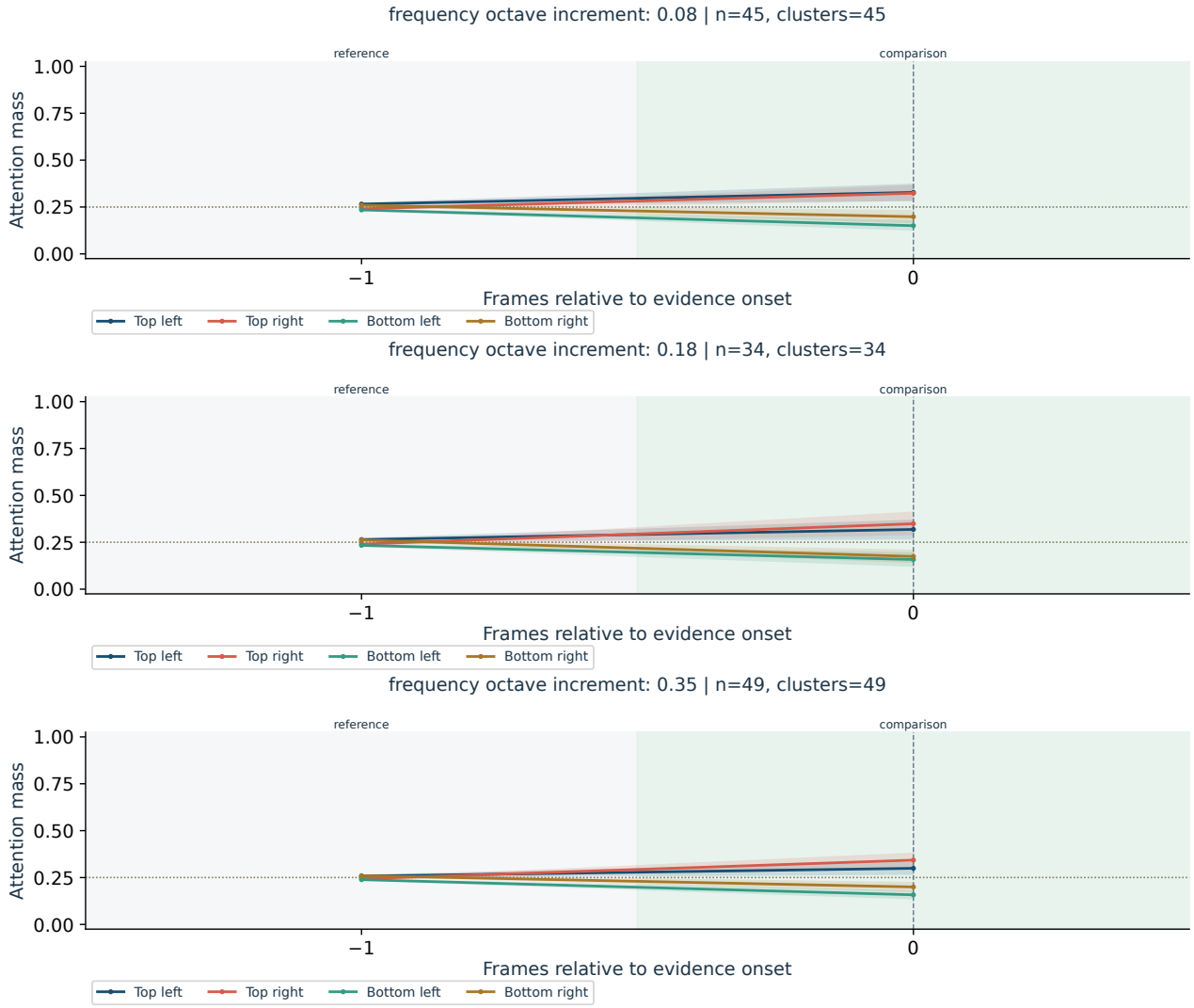


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

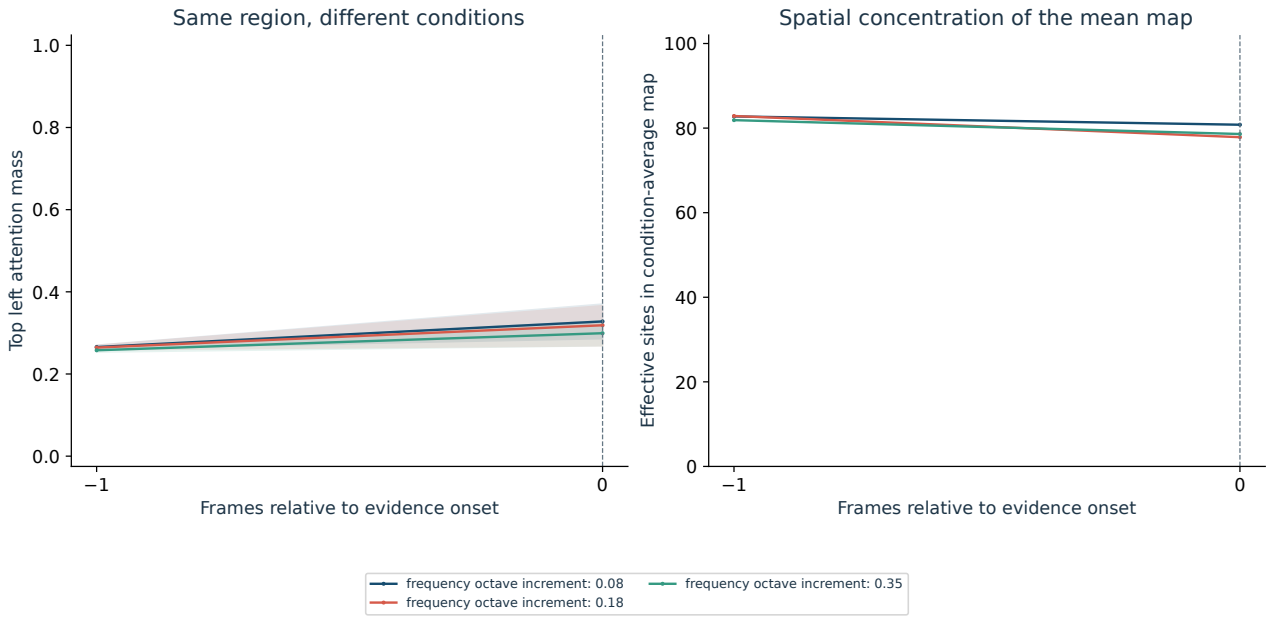


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. frequency octave increment: 0.08: top left mass peaks at 0.328 at original frame 1 (comparison), and is 0.328 on the final frame. frequency octave increment: 0.18: top left mass peaks at 0.319 at original frame 1 (comparison), and is 0.319 on the final frame. frequency octave increment: 0.35: top left mass peaks at 0.299 at original frame 1 (comparison), and is 0.299 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For cls_gain0_t0, paired accuracy changed by -43.8 percentage points (paired 95% interval [-52.3%, -35.2%]; paired p=2.776e-17; within-task Holm p=8.604e-16). There were 0 corrected trials and 56 newly incorrect trials. For target_-2_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

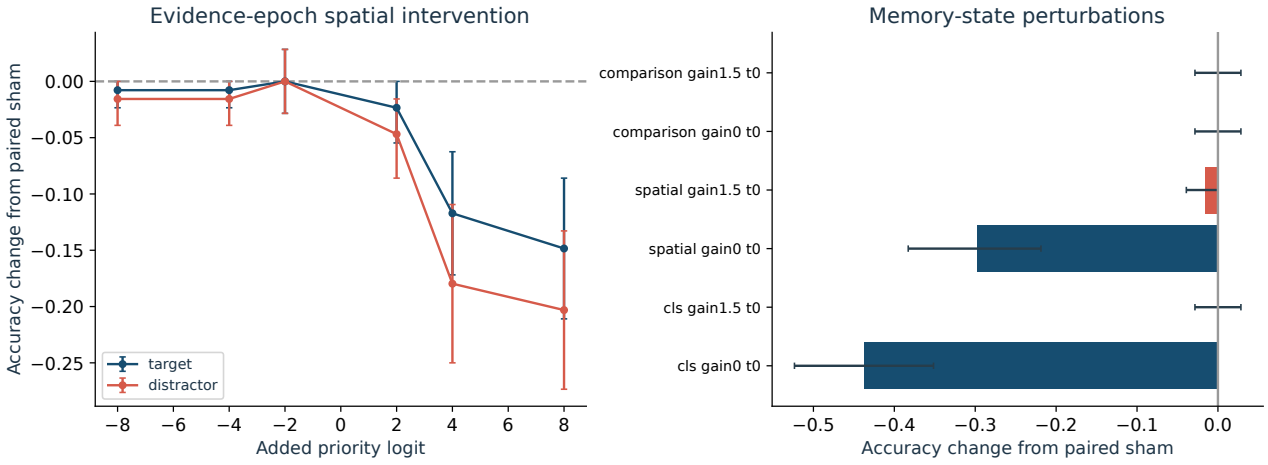


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
mixed	target_-4	-0.8	1
mixed	target_+4	-11.7	0.004517
mixed	distractor_-4	-1.6	1
mixed	distractor_+4	-18.0	1.955e-05

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.508	46.1%	0.800
fusion	0.569	50.0%	0.545
spatial	0.792	69.5%	0.005
cls	1.000	100.0%	0.005
comparison	Constant zero on this condition		
attention	0.576	56.2%	0.130

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

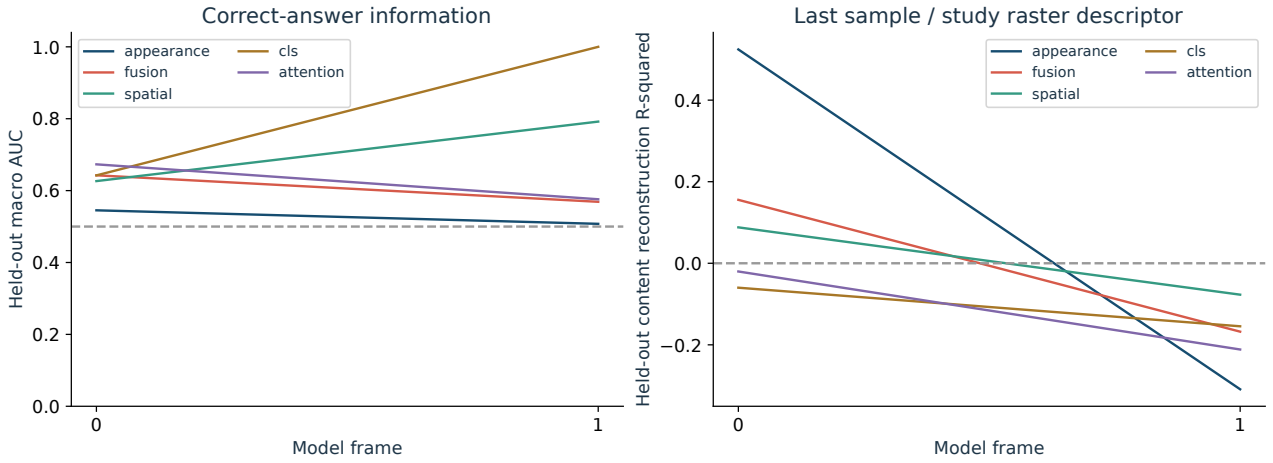


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 7.32 to 19.02 within the retained subspace. The final step displacement was 1.448 standardized RMS units. In spatial memory, the participation ratio ranged from 9.38 to 20.06 within the retained subspace. The final step displacement was 1.078 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 0, coarse sample-content reconstruction had R-squared -0.060 from CLS memory and 0.088 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -43.8 percentage points, spatial, -29.7 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.510	0.670	3200
fusion	0.496	0.562	6400
spatial	0.930	0.632	6400
cls	1.000	0.547	64
comparison	unavailable	unavailable	–
attention	0.570	0.592	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 31 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

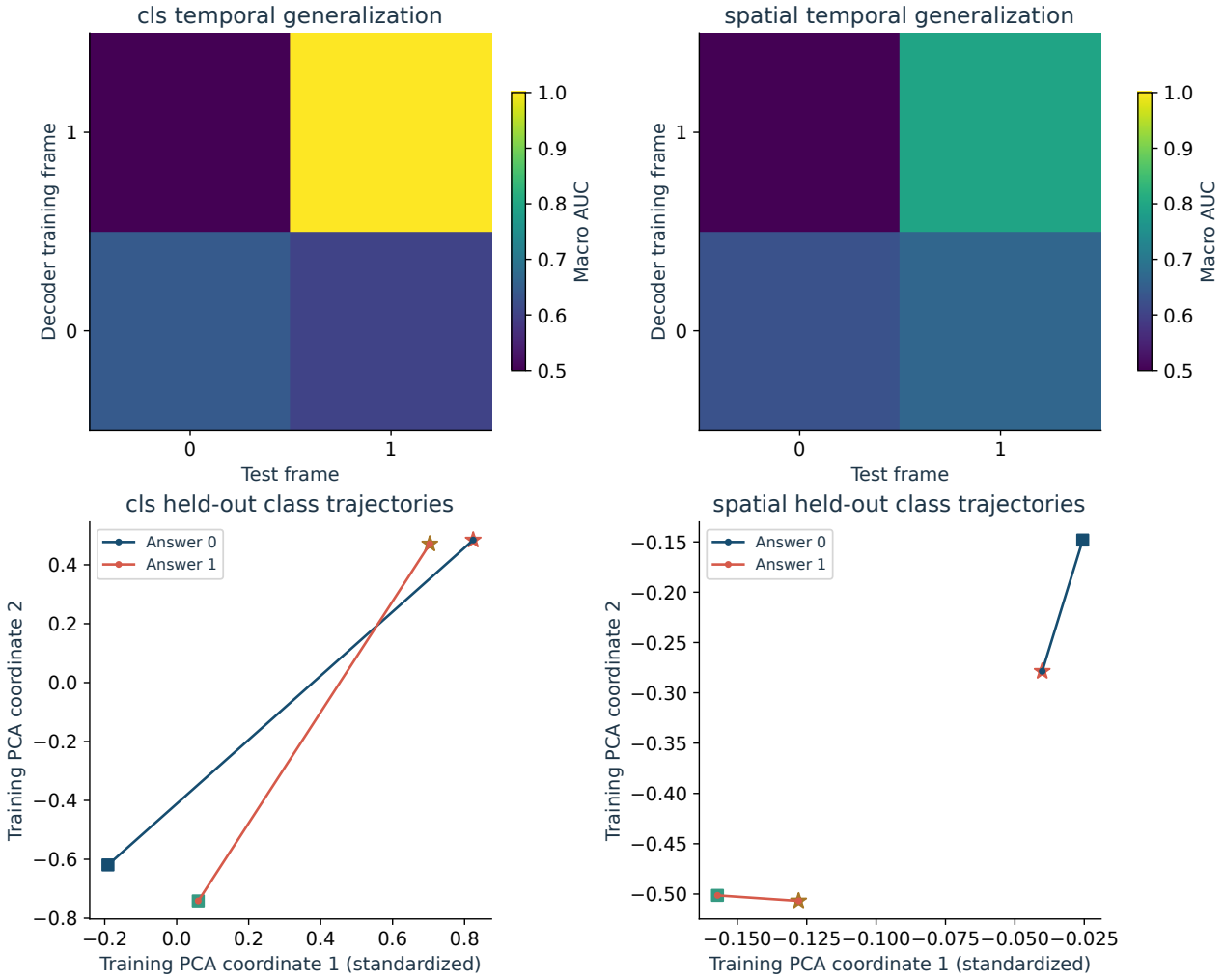


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

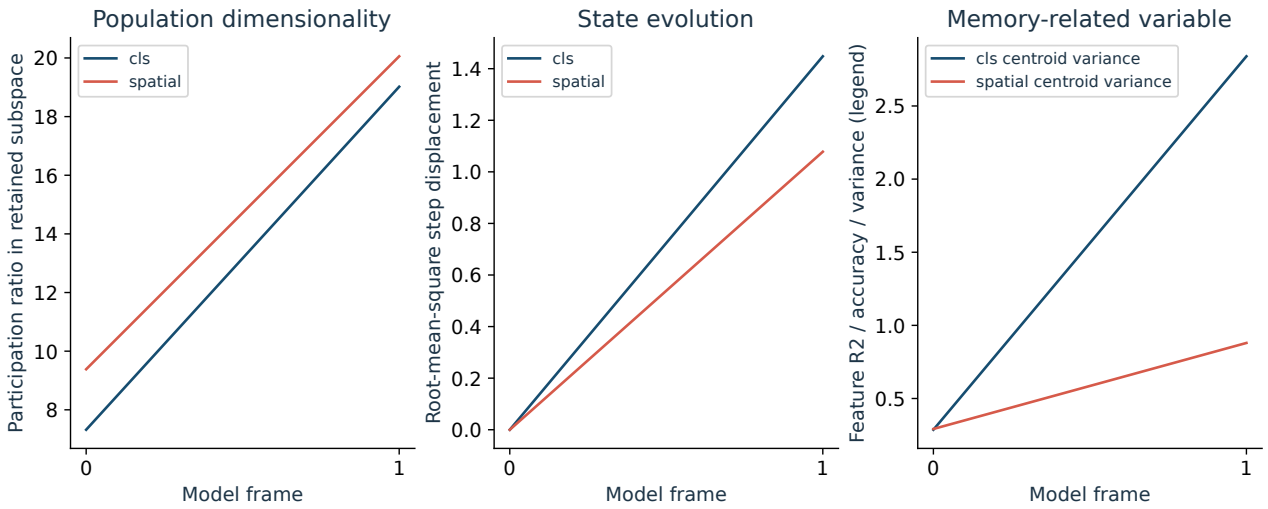


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

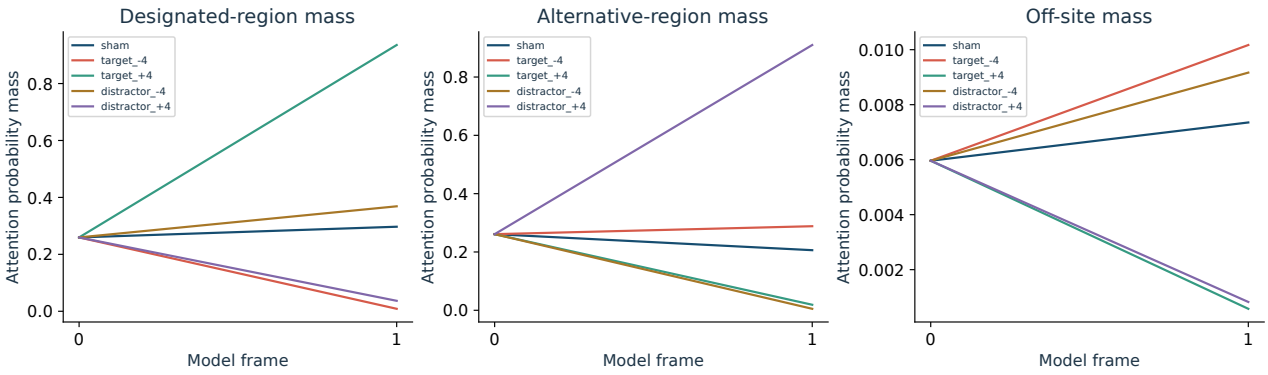


Figure 13: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

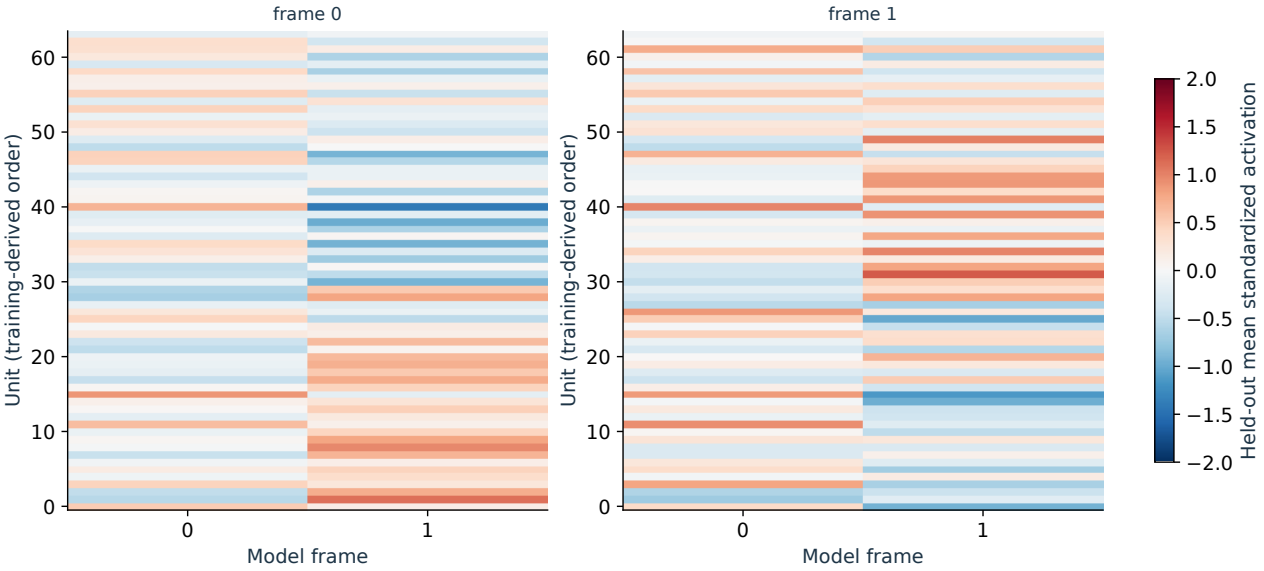


Figure 14: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	46.1%	46.1%	unavailable
fusion	50.0%	50.0%	unavailable
spatial	69.5%	69.5%	unavailable
cls	100.0%	100.0%	unavailable
attention	56.2%	56.2%	unavailable

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

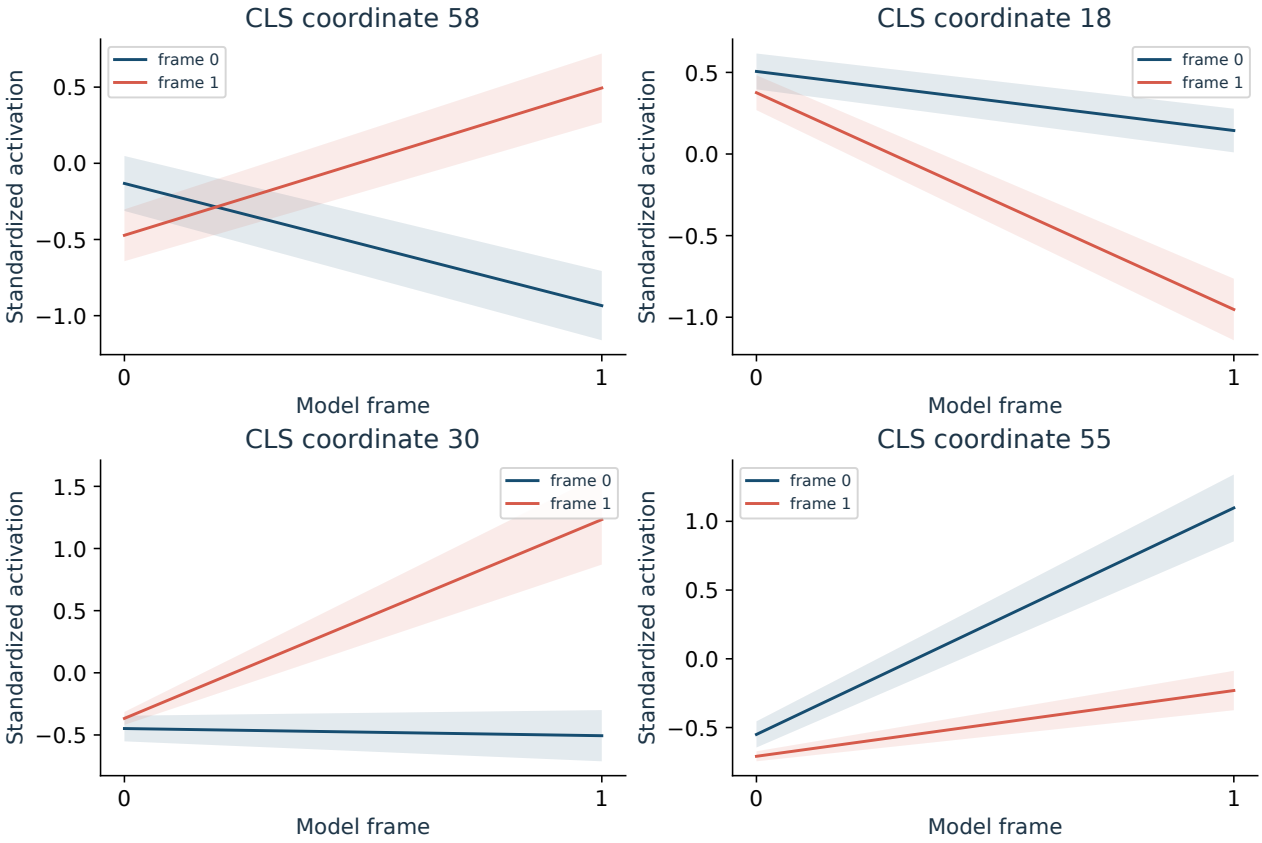


Figure 15: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

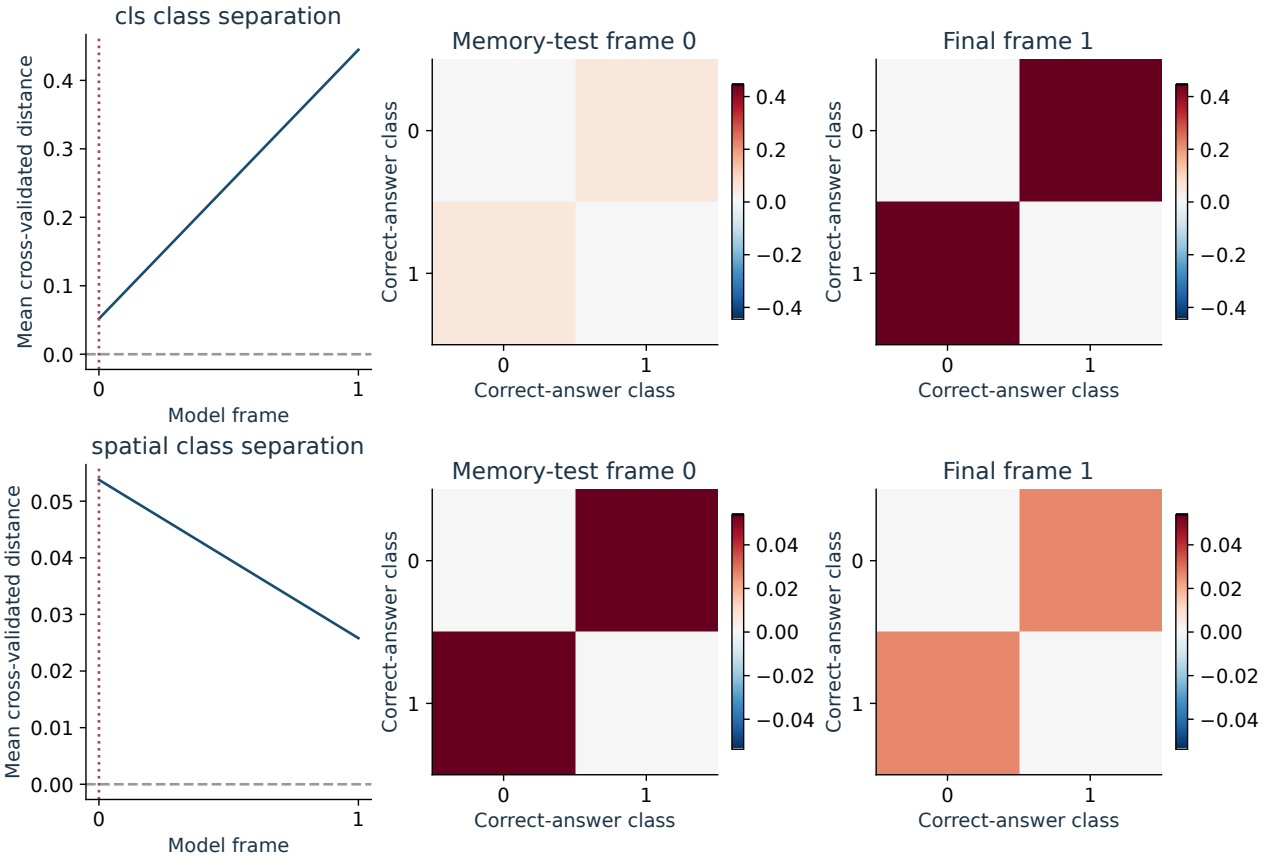


Figure 16: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

The answer is most clearly expressed in global recurrent memory, while both global and spatial reference-period resets impair the subsequent judgement. The results support a sequential representation of interval ordering rather than a direct interpretation of a single local edge detector. Spatial priority concentration can additionally reduce performance, consistent with the decision using evidence distributed over the grating. The octave axis makes the behavioral curve a test of relative frequency discrimination. Extending the pedestal range would determine whether the comparison remains accurate when the same relative difference is embedded in unfamiliar absolute frequencies.

The behavioral result establishes 100.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies cls_gain0_ t0 as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Testing novel pedestal frequencies and contrast levels would distinguish a general comparison rule from interpolation over familiar gratings. Aliasing should be measured explicitly when approaching the raster resolution limit.

4.1 Interpreting these results alongside biological experiments

4.1.1 Campbell and Robson (1968)

Our task compares frequencies above visibility threshold; the classic human experiment measured when components became visible. Both motivate describing image structure by scale, but successful frequency ordering does not establish the detector thresholds of separate Fourier components. Testing compound gratings would ask whether the model follows the fundamental or a more visible harmonic. [5]

4.1.2 De Valois, Albrecht and Thorell (1982)

A biological channel account predicts different preferred frequencies across a population. The model's appearance features and fixed crops offer candidate populations to inspect. Map each coordinate's response across frequency, phase and orientation, then test whether a decoder trained on some phases generalizes to new phases. This would identify a scale representation rather than a particular stripe pattern. [6]

4.1.3 Regan and Beverley (1983)

Detection and discrimination can change differently because the operations read different aspects of channel activity. Our recurrent reset costs identify a sequential comparison dependency. A model adaptation experiment would need an explicit adaptation process or a controlled preceding stimulus

sequence; merely lowering the contrast of a test image would address visibility instead of the historical adaptation manipulation. [7]

4.2 A computational account of the observed behavior

On a logarithmic frequency axis, a fixed octave increment means a fixed ratio between frequencies. A population of overlapping scale-sensitive channels can encode frequency by the relative distribution of its activity; a retained distribution from the first image can then be compared with the second. This is a plausible account of the strong final global-memory answer code and the costs of resetting either reference-period state. Concentrating attention can reduce the number of visible cycles sampled and make that estimate less stable. The 100-pixel image also creates a concrete sampling limit: the highest frequencies approach the raster's resolution and can change appearance sharply with phase. Separating scale sensitivity from aliasing is therefore essential to the interpretation.

4.3 Where this task agrees with the biological evidence

The agreement lies in scale-dependent visual evidence followed by a memory-dependent ordering judgment. The missing biological signature is a measured distribution of frequency tuning and its relation to thresholds across absolute frequencies. The present decoder results localize usable information without identifying the receptive-field basis that carries it.

4.4 A follow-up experiment motivated by these results

Sweep reference frequency over a wider range while holding octave differences fixed. Include independently randomized phase and compound gratings. Relate errors to measured channel tuning and to the number of cycles inside attended crops.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] De Valois RL, Albrecht DG, Thorell LG (1982). Spatial frequency selectivity of cells in macaque visual cortex. Vision Research 22:545-559. https://pubmed.ncbi.nlm.nih.gov/7112954/.

[2] Campbell FW, Robson JG (1968). Application of Fourier analysis to the visibility of gratings. Journal of Physiology 197:551-566. https://pubmed.ncbi.nlm.nih.gov/5666169/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] Campbell and Robson (1968). Application of Fourier analysis to the visibility of gratings. Primary source.

[6] De Valois, Albrecht and Thorell (1982). Spatial frequency selectivity of cells in macaque visual cortex. Primary source.

[7] Regan and Beverley (1983). Spatial-frequency discrimination and detection: comparison of postadaptation thresholds. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
mixed	128	100.0%	1.000	0.000

B Every-frame attention atlas

Condition mixed. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 64 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

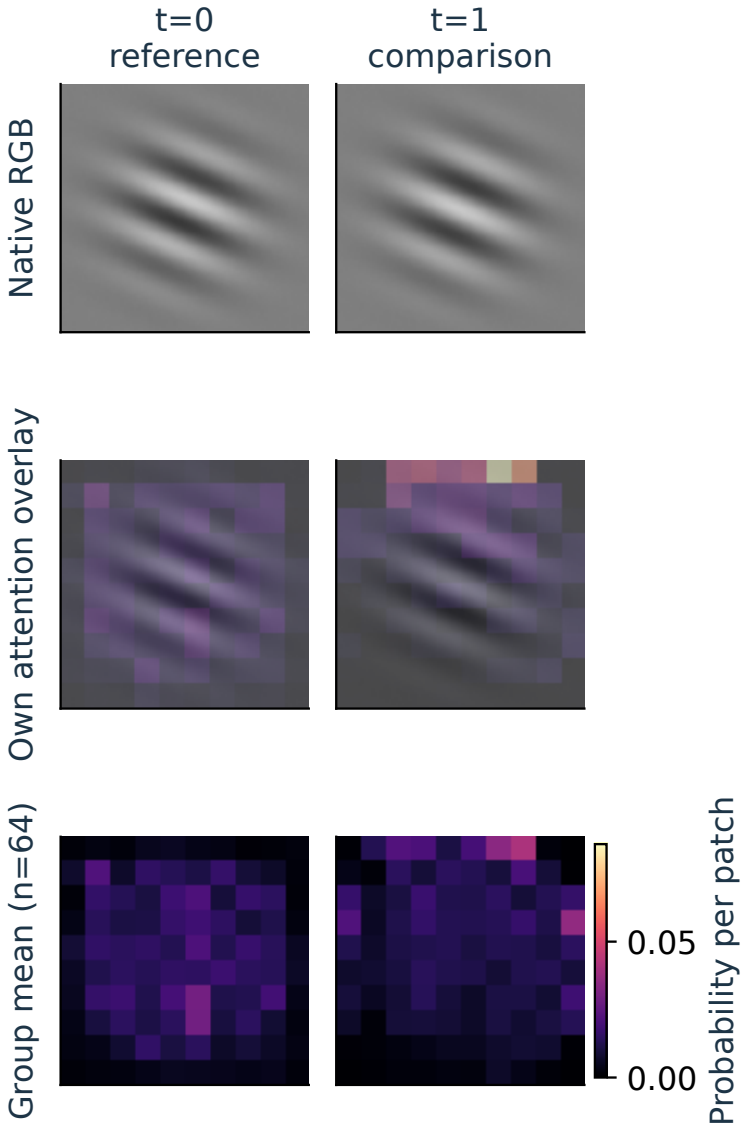


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

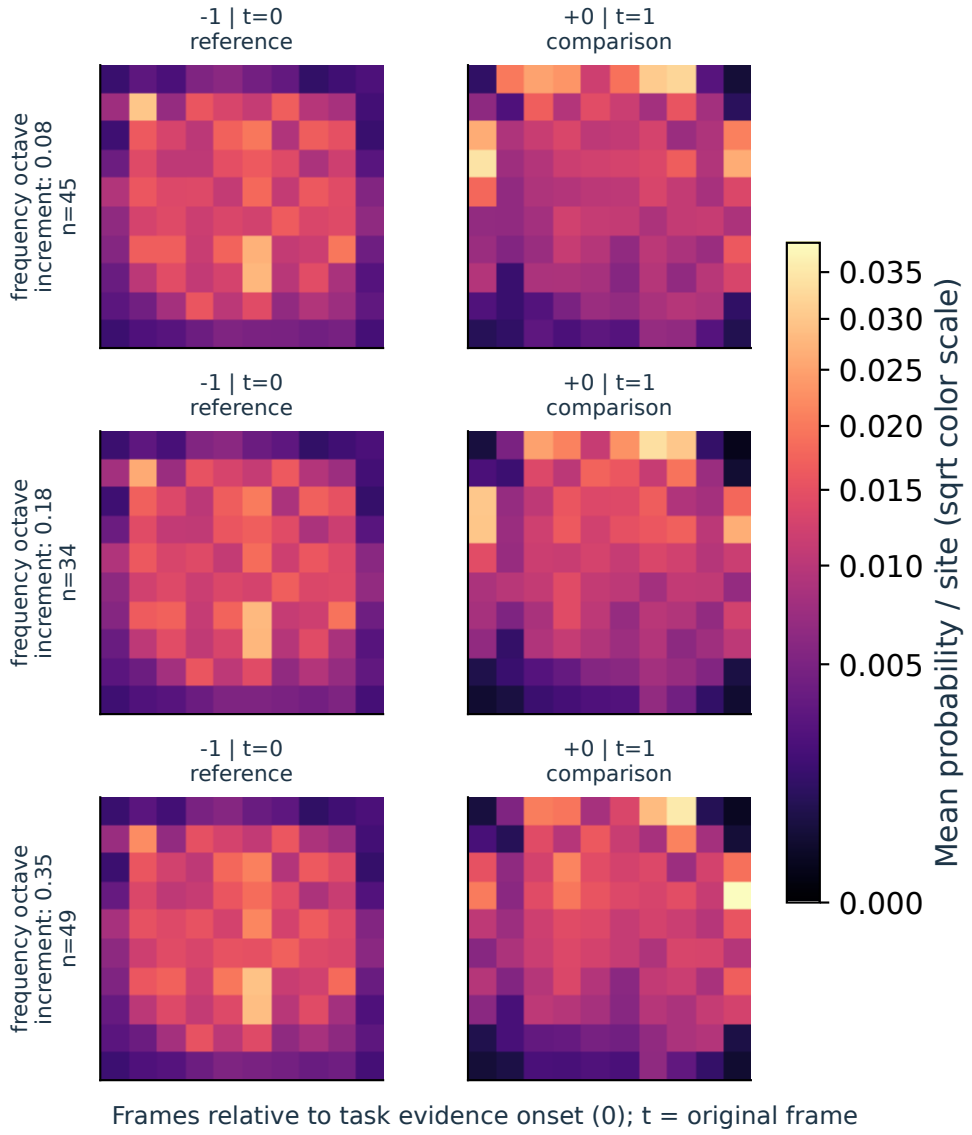


Figure 18: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Chromatic interval comparison under shared spatial priority

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized chromatic increment in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 1 native conditions and 128 fresh trials, equal-condition accuracy was 100.0% (95% interval [97.1%, 100.0%]). For the representative condition mixed, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was spatial_gain0_t0, changing accuracy by -39.8 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The observer identifies the interval with a positive increment along a defined color axis. The stimulus construction holds the intended luminance relation fixed while changing RGB components. The task therefore tests whether a representation shared with achromatic and motion tasks preserves useful chromatic information.

The psychometric variable is the magnitude of the linear-RGB increment. A color increment is a vector in the renderer color space, so its numerical magnitude depends on that definition. We measure discrimination across the renderer values and inspect whether the appearance and fused pathways support an independent linear readout of the answer.

The biological motivation is developed through [1]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Lennie and colleagues characterized chromatic mechanisms in macaque striate cortex. The relevant lesson for this experiment is that color can be represented across a population of responses with different sensitivities. The analysis therefore follows chromatic discrimination through a shared architecture rather than seeking a single dedicated color unit. Spatial priority acts on a site containing multiple kinds of visual evidence. A causal influence of that priority on a chromatic judgement would demonstrate a common selection operation even though the defining sensory signal differs from the moving-dot tasks.

1.2 Biological experiments motivating this task

Experimental question. How does a shared spatial-selection network use color information when its motion calculation is grayscale?

Derrington, Krauskopf and Lennie (1984) Color-space null measurements characterized how LGN neurons combine cone signals. The results exposed opponent chromatic mechanisms and separated chromatic from luminance-dependent response directions. [4]

Lennie, Krauskopf and Sclar (1990) V1 neurons responded to combinations of chromatic and luminance modulation. Their distribution of preferred color directions differed from the more distinctly grouped LGN mechanisms, showing a cortical transformation of cone-derived signals. [5]

Flanagan, Cavanagh and Favreau (1990) Opposite tilt aftereffects could be induced for chromatic and luminance gratings. The psychophysical dissociation supported orientation-selective mechanisms operating on multiple chromatic and achromatic signals. [6]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Which frame has the positive increment along the defined isoluminant linear-RGB axis?

Response labels are 0: frame 0; 1: frame 1. The native suite contains 1 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use mixed.

1.4 Stimulus sequence and spatial layout

For the representative native condition mixed, the sequence has 2 frames, indexed from zero: frame 0: reference; frame 1: comparison. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
RGB chromatic increment	0.018, 0.045, 0.1
Stimulus family	chromatic_increment

The model receives RGB images while the correspondence bank operates on grayscale averages. Chromatic information can enter through learned appearance features and local RGB statistics. The causal analyses ask whether spatial priority still regulates performance when the defining signal lies in those static channels. This is a direct architectural test of shared selection across visual features.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 100.0%, with macro AUC 1.000. The matched final checkpoint changed accuracy by +0.00 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was mixed at 100.0%; the highest was mixed at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (0.001) produced 60.2% correct responses, compared with 100.0% at the largest strength (0.1). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

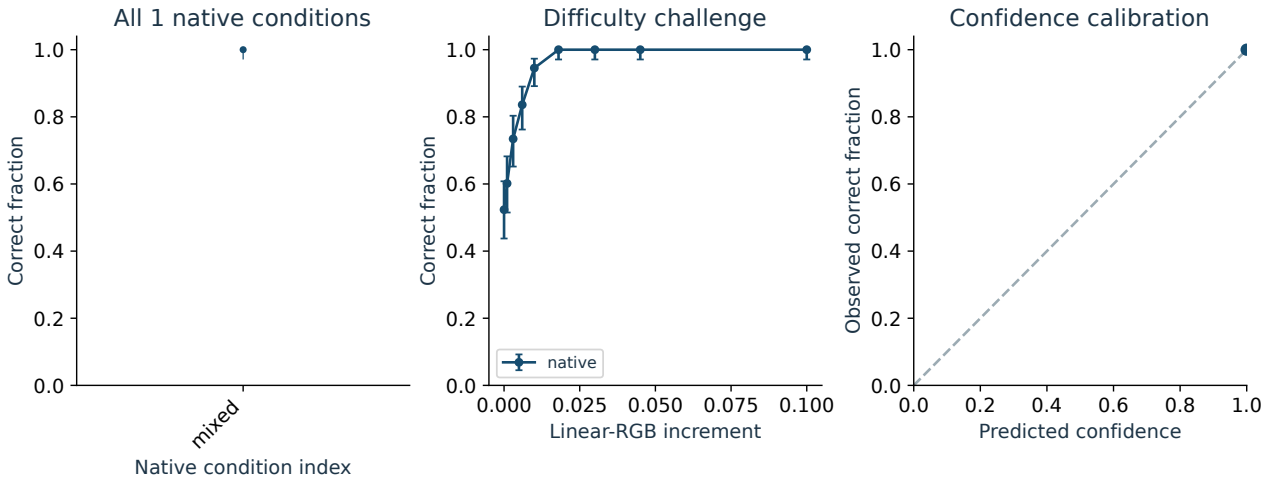


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)[1 - \exp(-(x/a)^b)]$. Here x is signal strength, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 0.003547 task units (bootstrap interval [0.002174, 0.005193]), with slope 56.52 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.028.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	100.0%	96.1%
0.7	100.0%	1.0	100.0%	89.1%
0.8	100.0%	1.0	100.0%	78.1%
0.9	100.0%	1.0	100.0%	68.8%

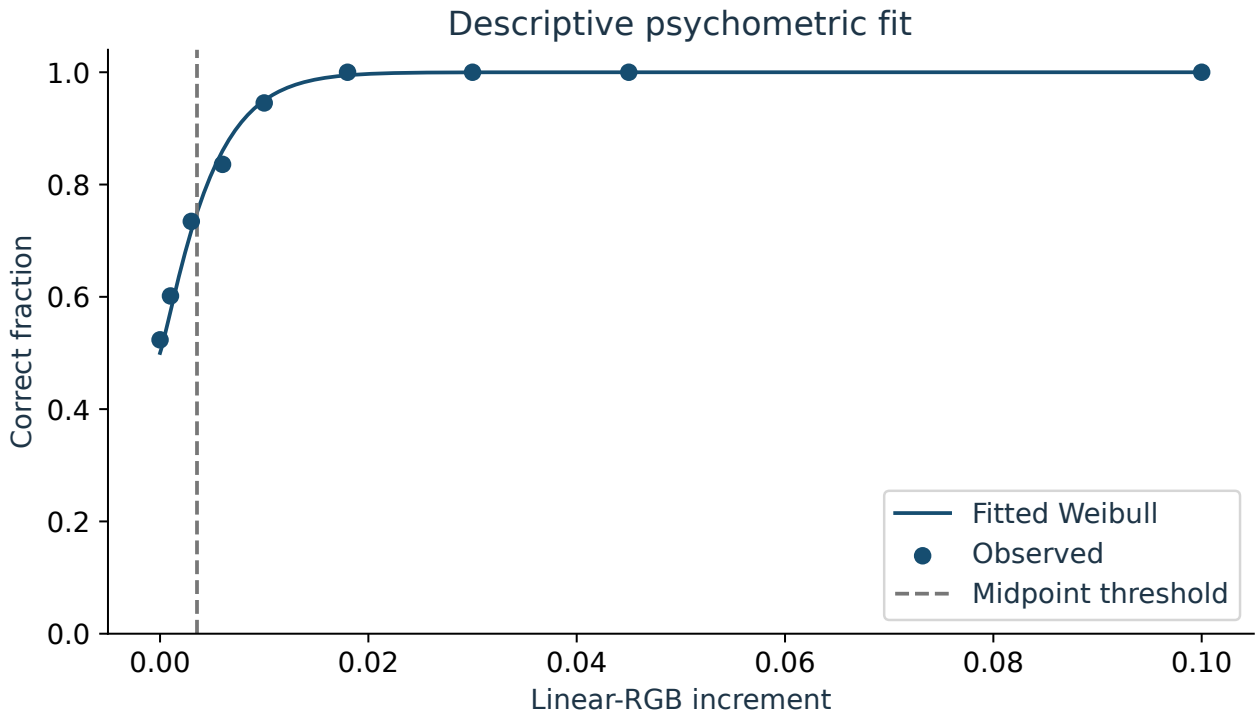


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

The representative condition mixed provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 128 of 128 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

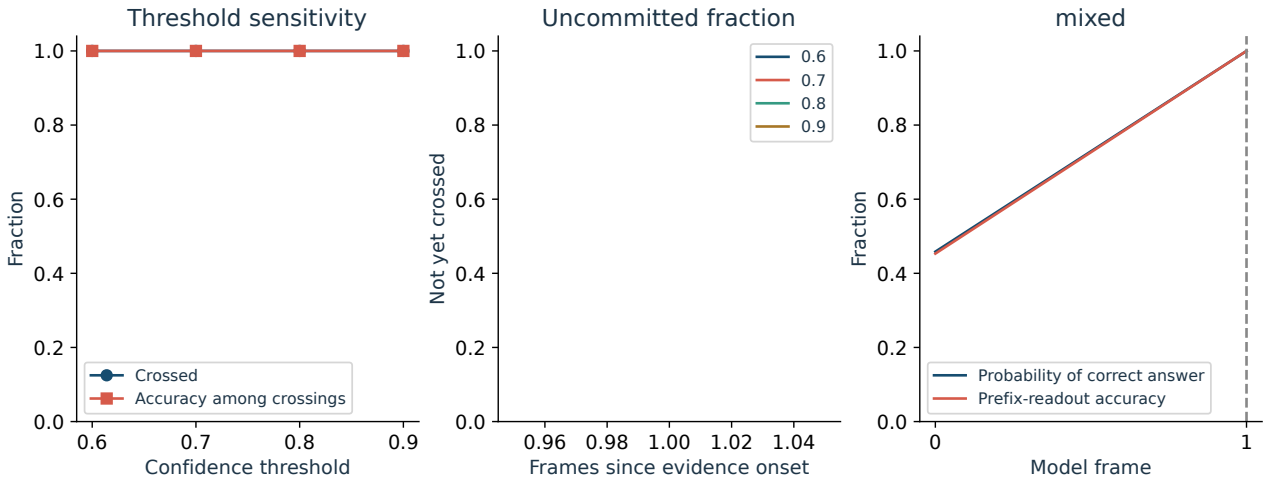


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

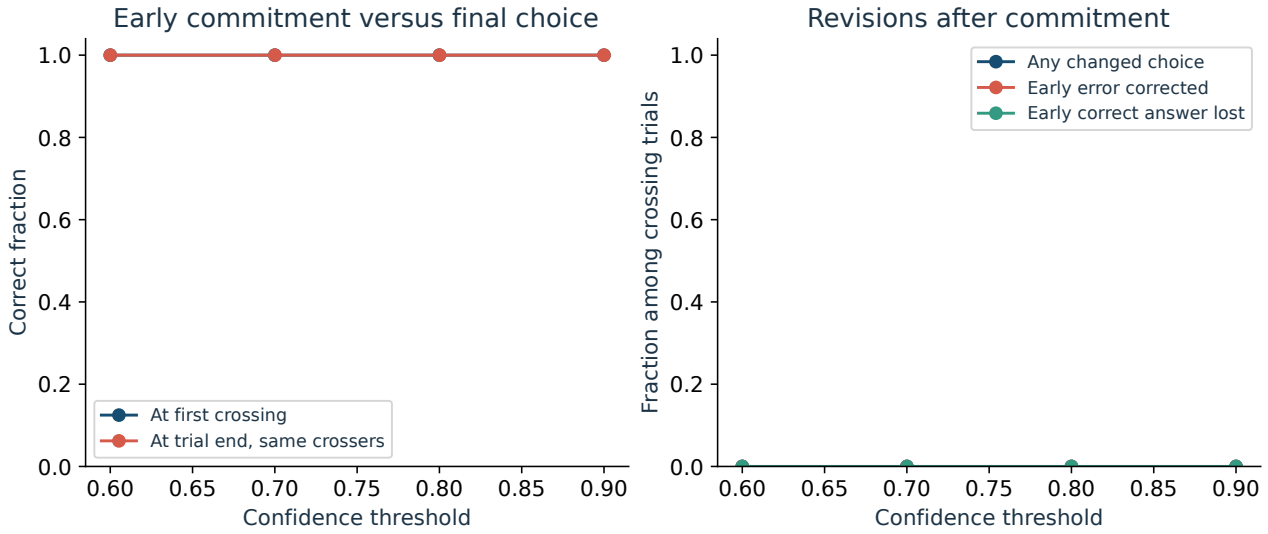


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

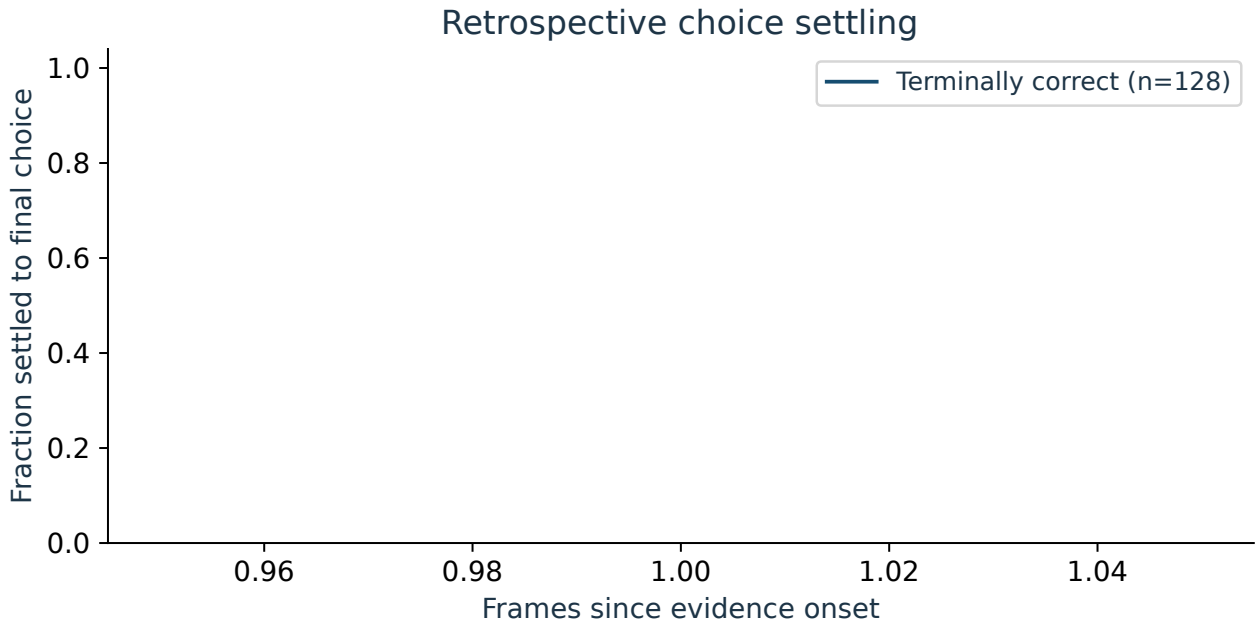


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

Rows separate chromatic increments and pool interval order. Traces refer to fixed image quadrants.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
chromatic increment: 0.018	43	43
chromatic increment: 0.045	46	46
chromatic increment: 0.1	39	39

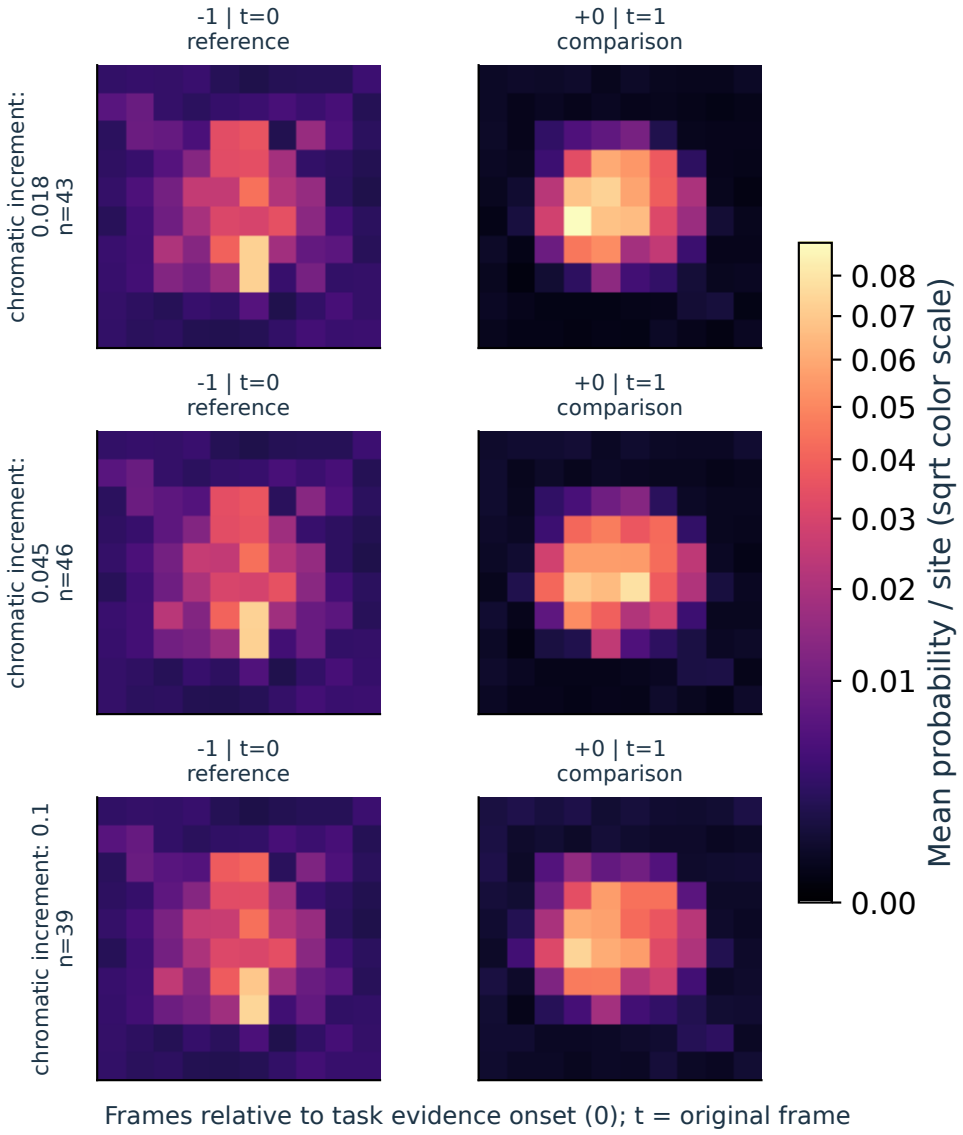


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

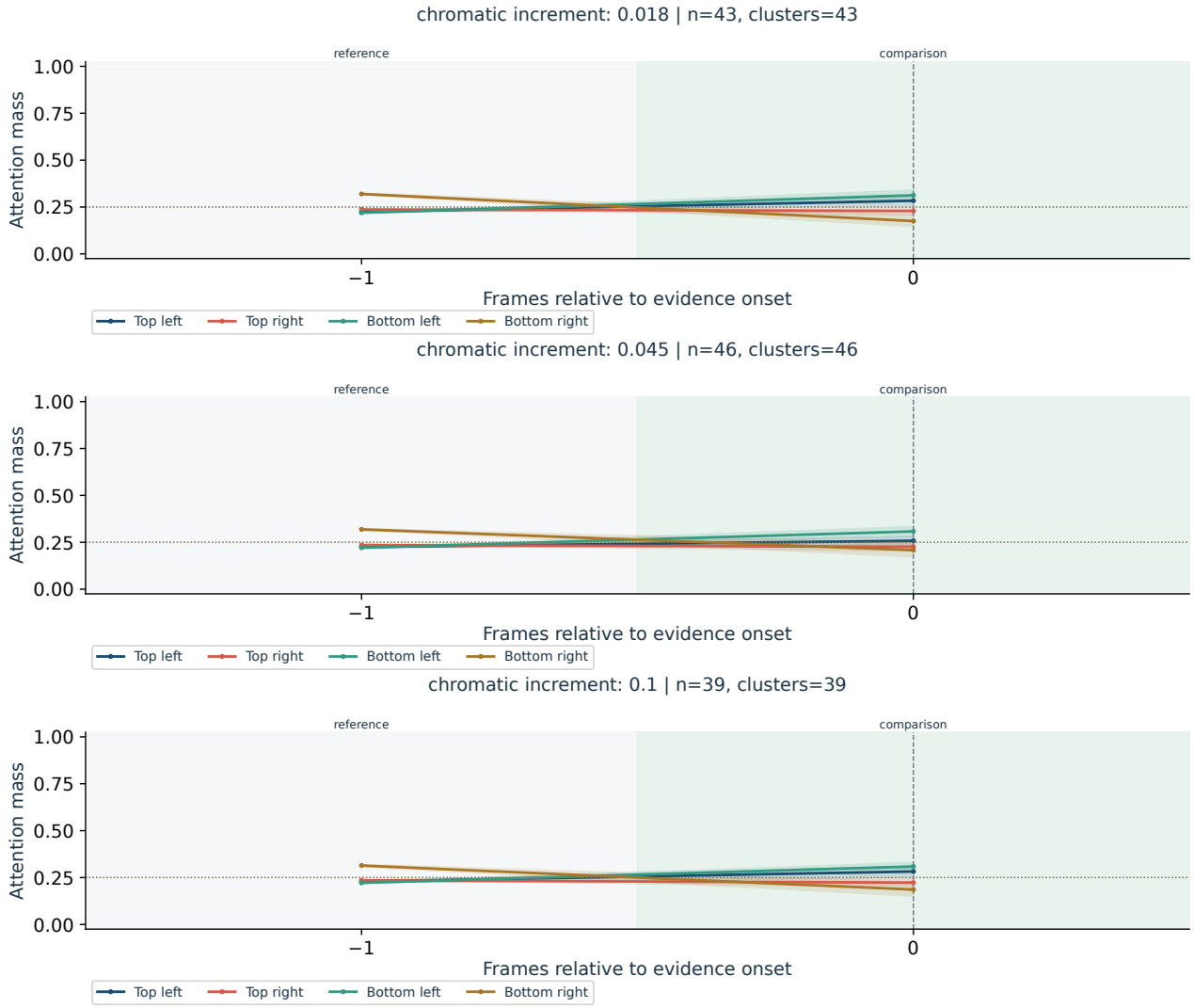


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

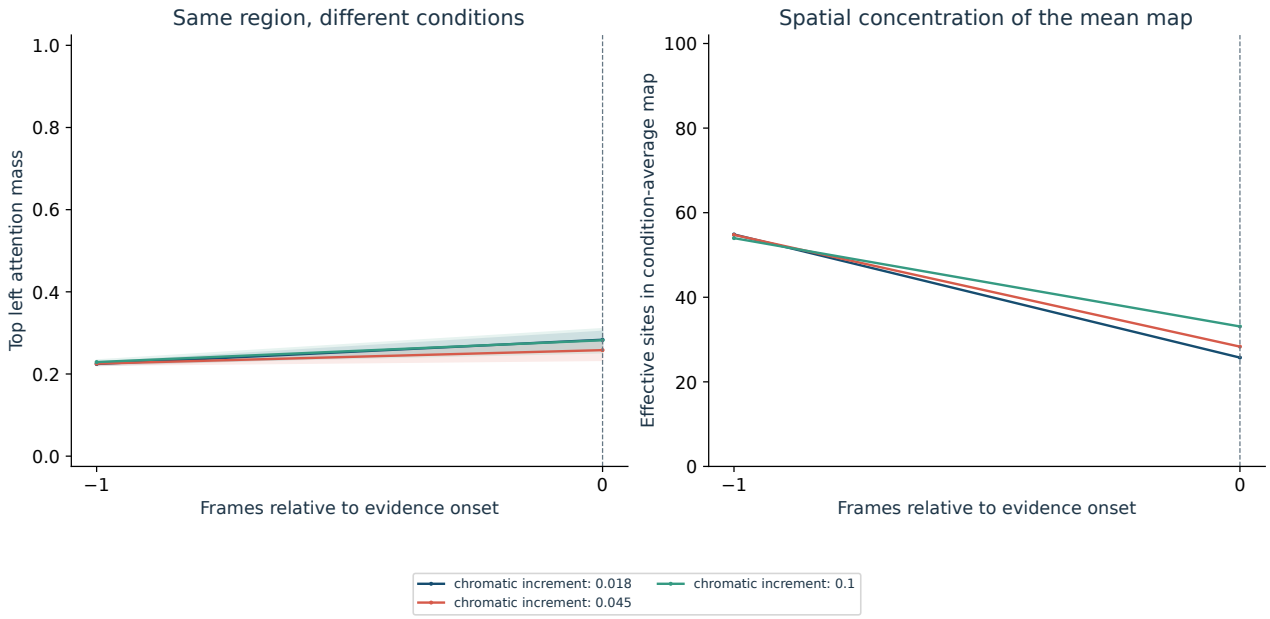


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. chromatic increment: 0.018: top left mass peaks at 0.283 at original frame 1 (comparison), and is 0.283 on the final frame. chromatic increment: 0.045: top left mass peaks at 0.258 at original frame 1 (comparison), and is 0.258 on the final frame. chromatic increment: 0.1: top left mass peaks at 0.282 at original frame 1 (comparison), and is 0.282 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For spatial_gain0_t0, paired accuracy changed by -39.8 percentage points (paired 95% interval [-48.4%, -32.0%]; paired p=8.882e-16; within-task Holm p=2.753e-14). There were 0 corrected trials and 51 newly incorrect trials. For target_-8_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

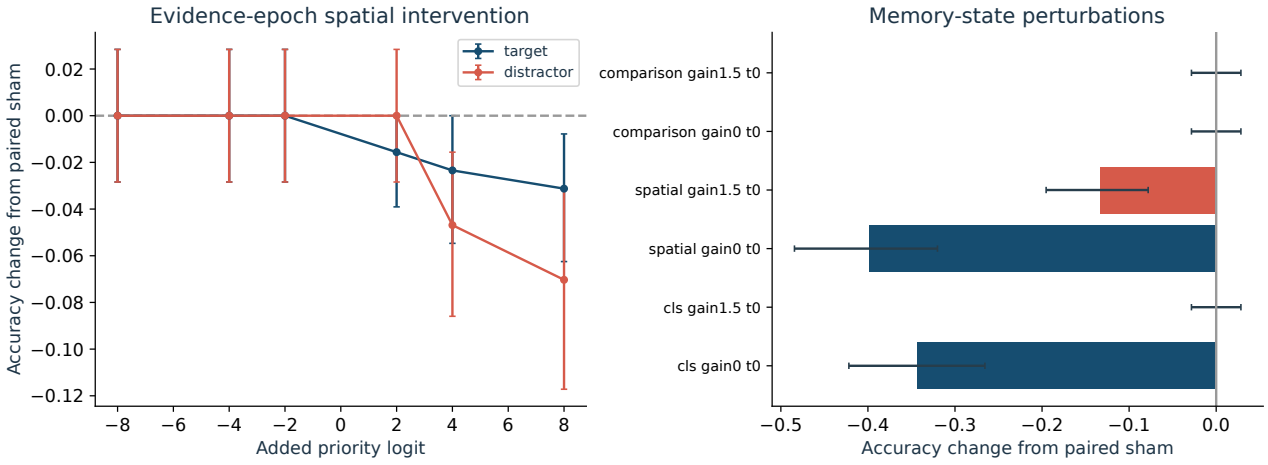


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
mixed	target_-4	+0.0	1
mixed	target_+4	-2.3	1
mixed	distractor_-4	+0.0	1
mixed	distractor_+4	-4.7	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.594	57.8%	0.080
fusion	0.588	57.8%	0.090
spatial	0.987	93.8%	0.005
cls	1.000	100.0%	0.005
comparison	Constant zero on this condition		
attention	0.726	64.8%	0.010

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

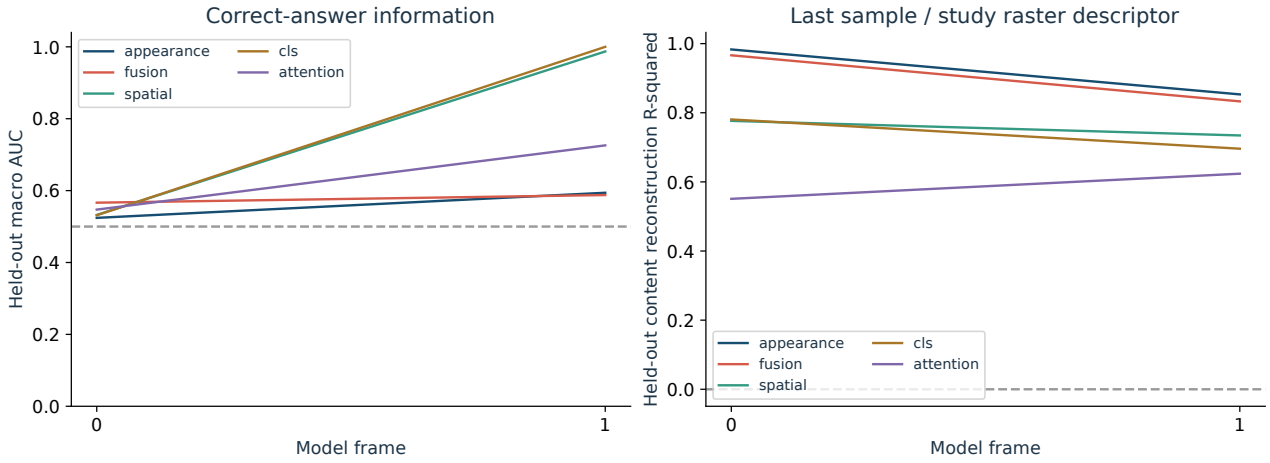


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 11.23 to 19.94 within the retained subspace. The final step displacement was 1.249 standardized RMS units. In spatial memory, the participation ratio ranged from 13.71 to 20.26 within the retained subspace. The final step displacement was 1.108 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 0, coarse sample-content reconstruction had R-squared 0.781 from CLS memory and 0.777 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -34.4 percentage points, spatial, -39.8 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.646	0.544	3200
fusion	0.637	0.583	6400
spatial	0.997	0.574	6400
cls	1.000	0.584	64
comparison	unavailable	unavailable	–
attention	0.702	0.564	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 41 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

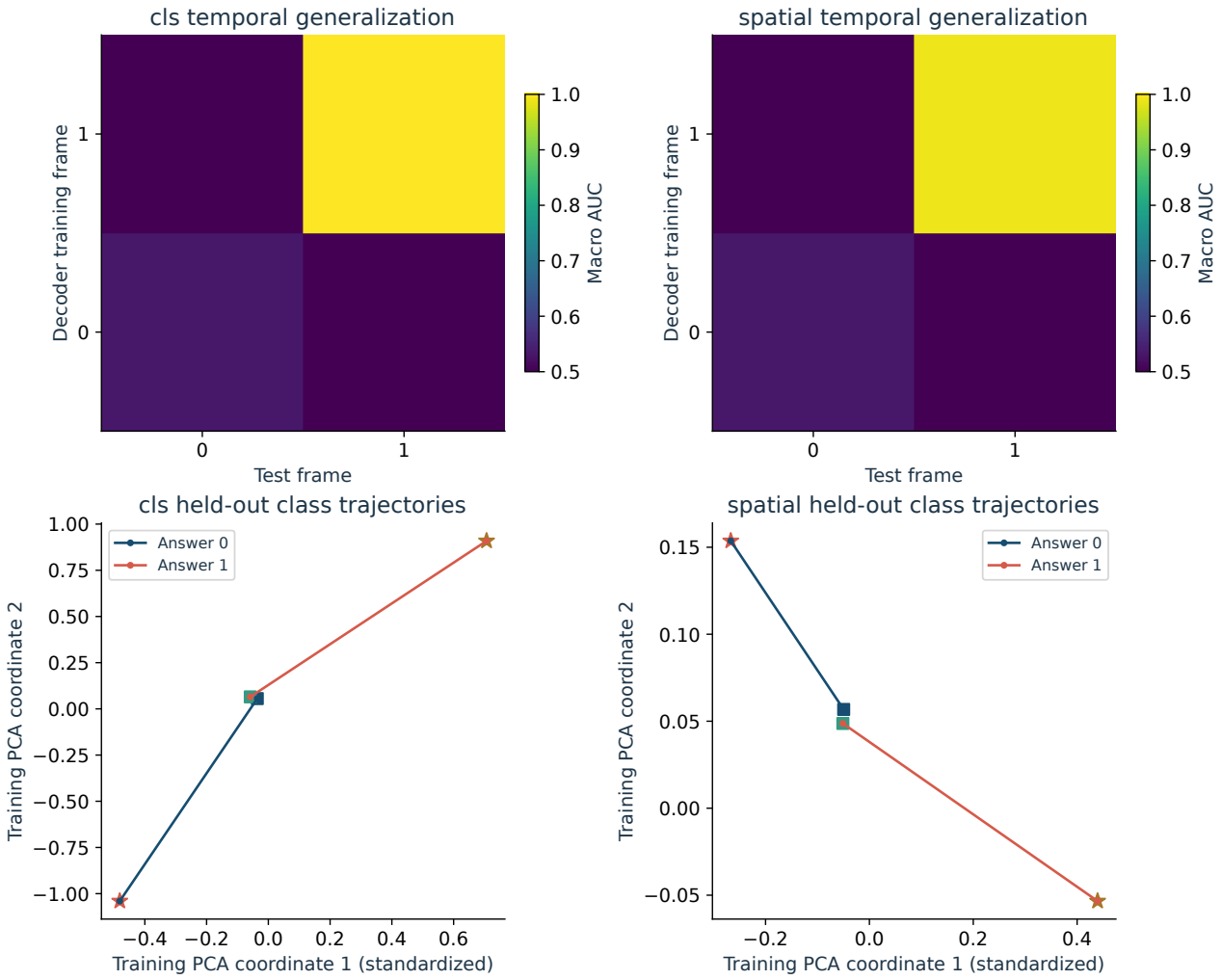


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

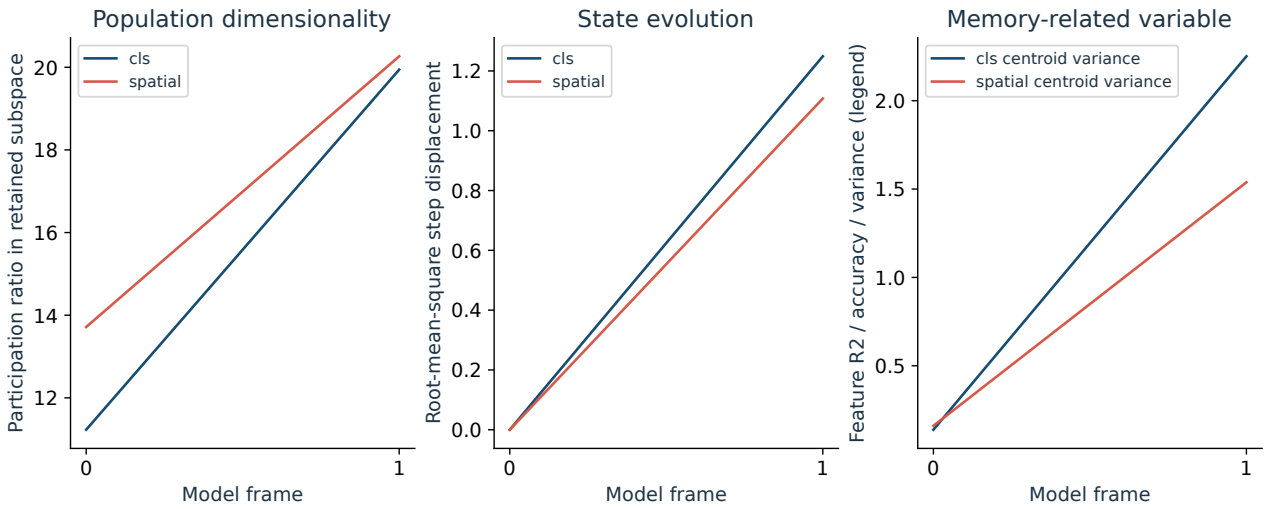


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

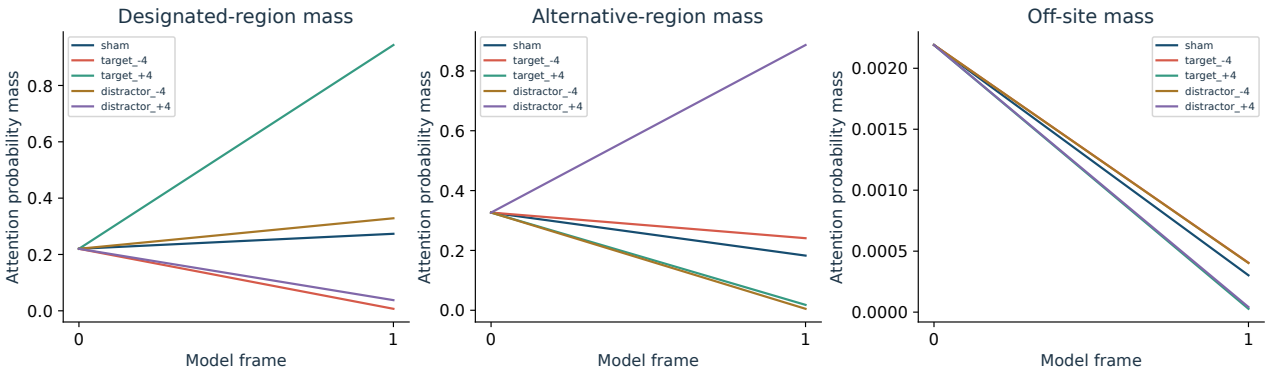


Figure 13: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

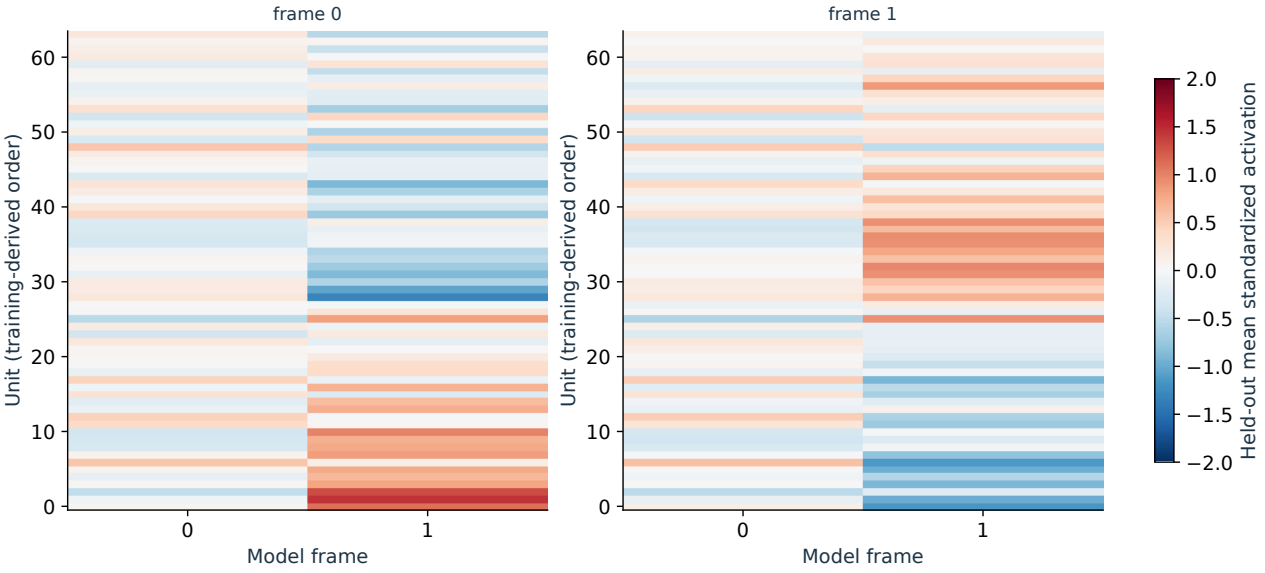


Figure 14: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	57.8%	57.0%	unavailable
fusion	57.8%	57.0%	unavailable
spatial	93.8%	93.0%	unavailable
cls	100.0%	99.2%	unavailable
attention	64.8%	65.6%	unavailable

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\text{fit}} d_j^{\text{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

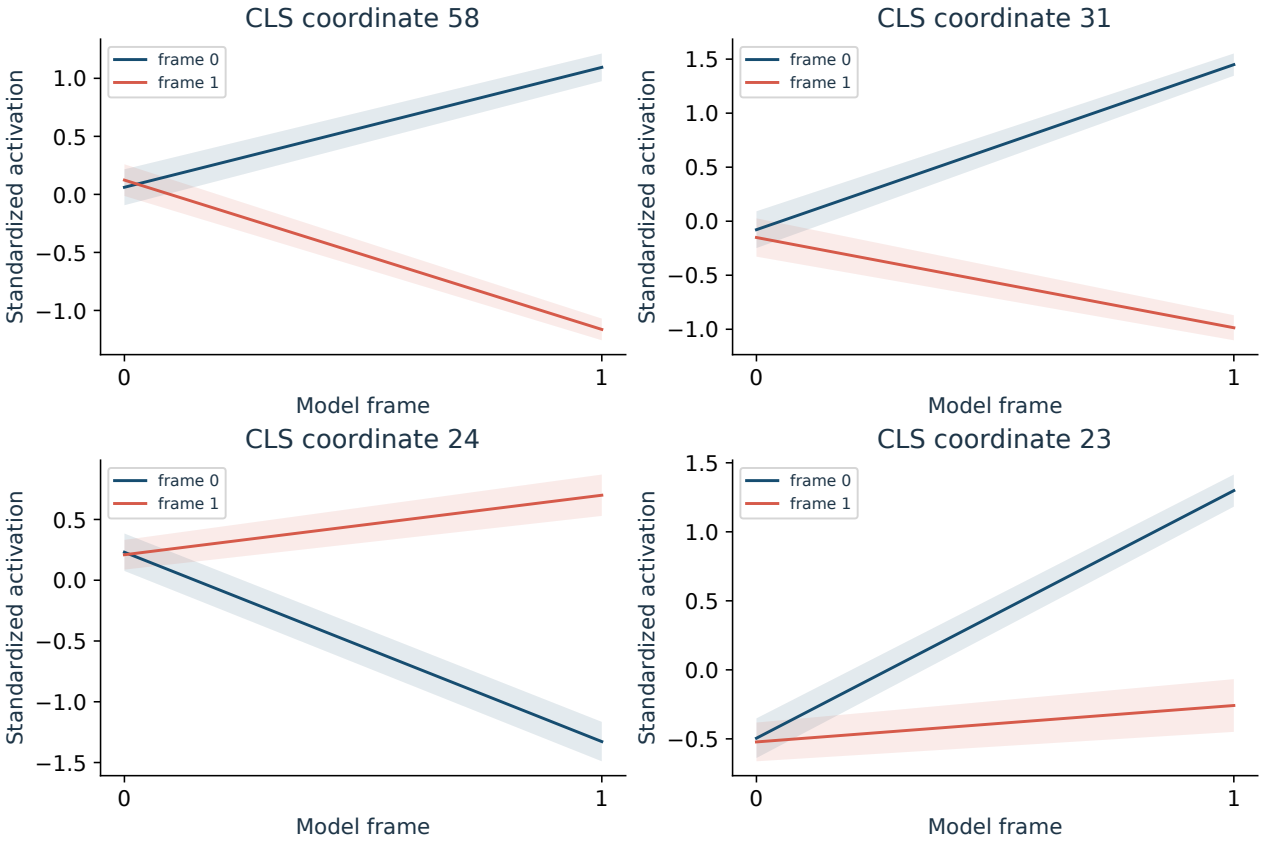


Figure 15: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

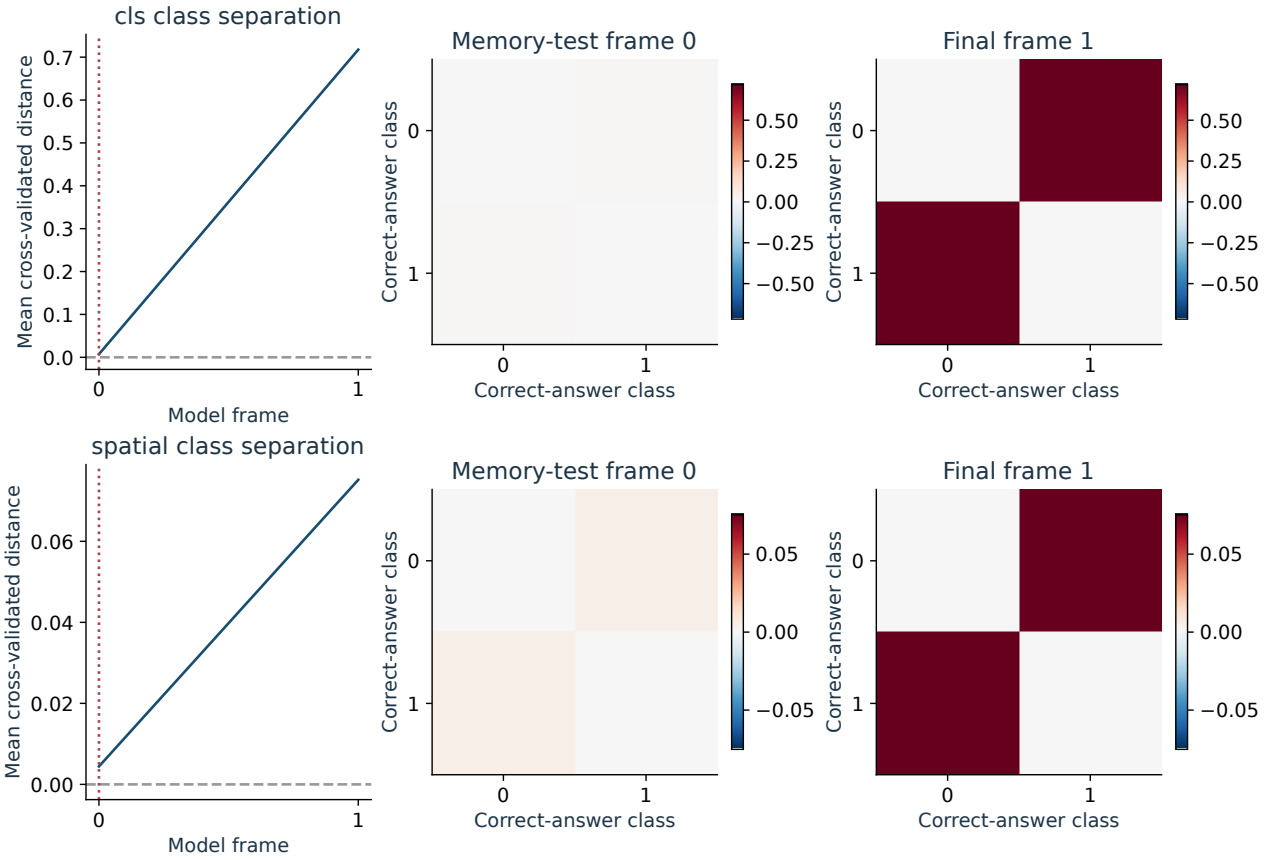


Figure 16: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

Accurate chromatic comparison survives in an architecture whose motion correspondence calculation uses grayscale input. The measured behavior is therefore supported by the color-bearing appearance and local-statistic channels together with recurrent processing. Both global and spatial reference-period resets have substantial effects, and the final answer is strongly readable in those recurrent populations. This provides a static-feature case for a shared spatial-selection architecture: the same priority distribution participates in a computation whose defining information is chromatic. A small effect of one quadrant manipulation can coexist with strong memory dependence when useful color evidence is present throughout the field.

The behavioral result establishes 100.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies spatial_gain0_t0 as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Additional axes and calibrated display-space transformations would test transfer beyond the trained color direction. Joint chromatic and motion conflicts would be especially informative about competition within the common attention map.

4.1 Interpreting these results alongside biological experiments

4.1.1 Derrington, Krauskopf and Lennie (1984)

The model receives RGB image coordinates, whereas the LGN study used cone-derived color coordinates. RGB increments have no direct physiological unit without a display spectrum and cone calibration. The useful comparison is structural: does the learned representation separate changes along independent color directions, and can those signals be selected and retained for a decision? [4]

4.1.2 Lennie, Krauskopf and Sclar (1990)

Cortical transformations can mix chromatic and luminance information into feature-sensitive responses. Our learned appearance channels and local color statistics provide the color-bearing route into fused tokens. The grayscale correspondence branch cannot alone preserve an isoluminant chromatic distinction. The model's successful task behavior therefore establishes use of additional visual information within the common attention architecture. [5]

4.1.3 Flanagan, Cavanagh and Favreau (1990)

The human aftereffect shows a way to test whether orientation processing can depend on chromatic signals separately from luminance. In the model, chromatic gratings with matched grayscale projection

would test whether color-bearing features support both color comparison and form. The current uniform chromatic increment task tests retention and comparison, while this extension would test feature interactions. [6]

4.2 A computational account of the observed behavior

A shared priority signal can act on a vector containing many visual attributes. At one site, the vector may include color, orientation and motion-related information; multiplying that vector by a spatial attention weight changes its contribution as a whole. This implements the user's shared-selection hypothesis at the routing stage. It also permits different feature transformations before that stage. The strong global and spatial reference-reset effects show that color decisions depend on retained state, despite small effects from some quadrant manipulations. For a broad uniform stimulus, many sites contain redundant evidence, so changing spatial allocation can preserve color information even while removing the reference state destroys the comparison.

4.3 Where this task agrees with the biological evidence

The model provides a concrete static-feature case of common spatial routing. Its biological difference is the input and feature basis: RGB channels and learned features replace calibrated cone responses and identified opponent circuits. The common selection computation can be tested independently of that sensory difference.

4.4 A follow-up experiment motivated by these results

Construct colors with identical model grayscale values, vary them along two independent RGB chromatic directions, and introduce localized color-versus-motion competition. At a fixed location, the same attention intervention should alter access to either feature while leaving feature identity encoded in distinct channel patterns.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Lennie P, Krauskopf J, Sclar G (1990). Chromatic mechanisms in striate cortex of macaque. Journal of Neuroscience 10:649-669. https://pubmed.ncbi.nlm.nih.gov/2303866/.

[2] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[3] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[4] Derrington, Krauskopf and Lennie (1984). Chromatic mechanisms in lateral geniculate nucleus of macaque. Primary source.

[5] Lennie, Krauskopf and Sclar (1990). Chromatic mechanisms in striate cortex of macaque. Primary source.

[6] Flanagan, Cavanagh and Favreau (1990). Independent orientation-selective mechanisms for the cardinal directions of colour space. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
mixed	128	100.0%	1.000	0.000

B Every-frame attention atlas

Condition mixed. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 64 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

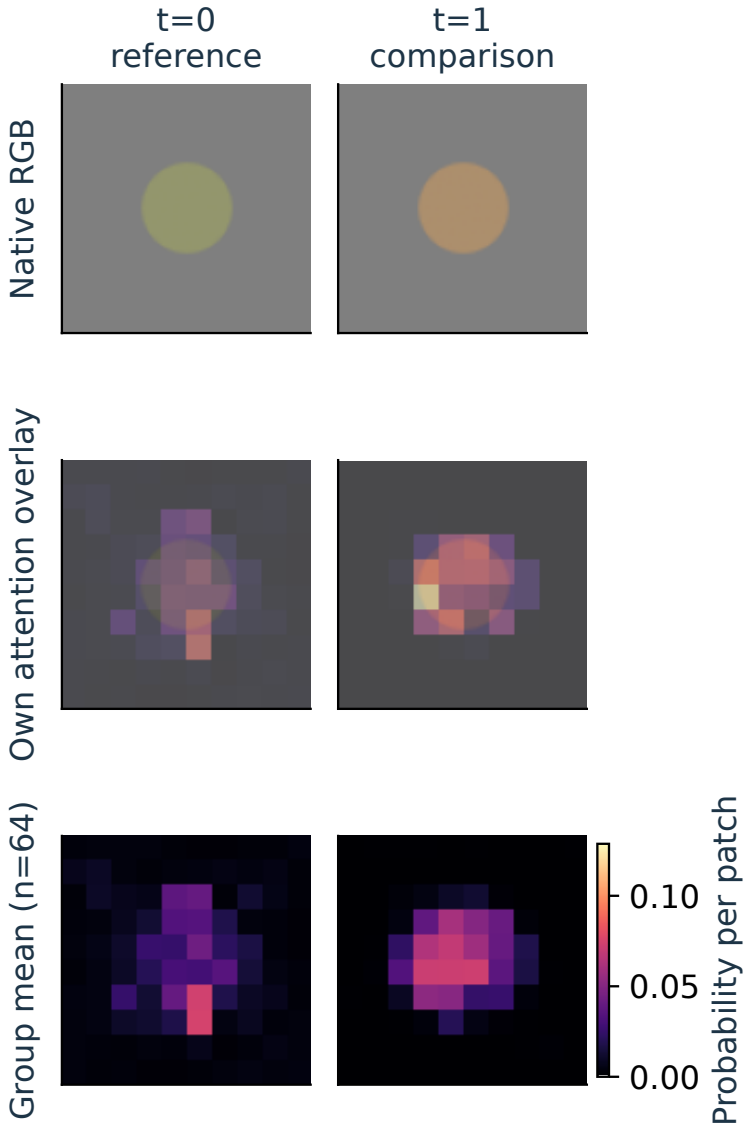


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

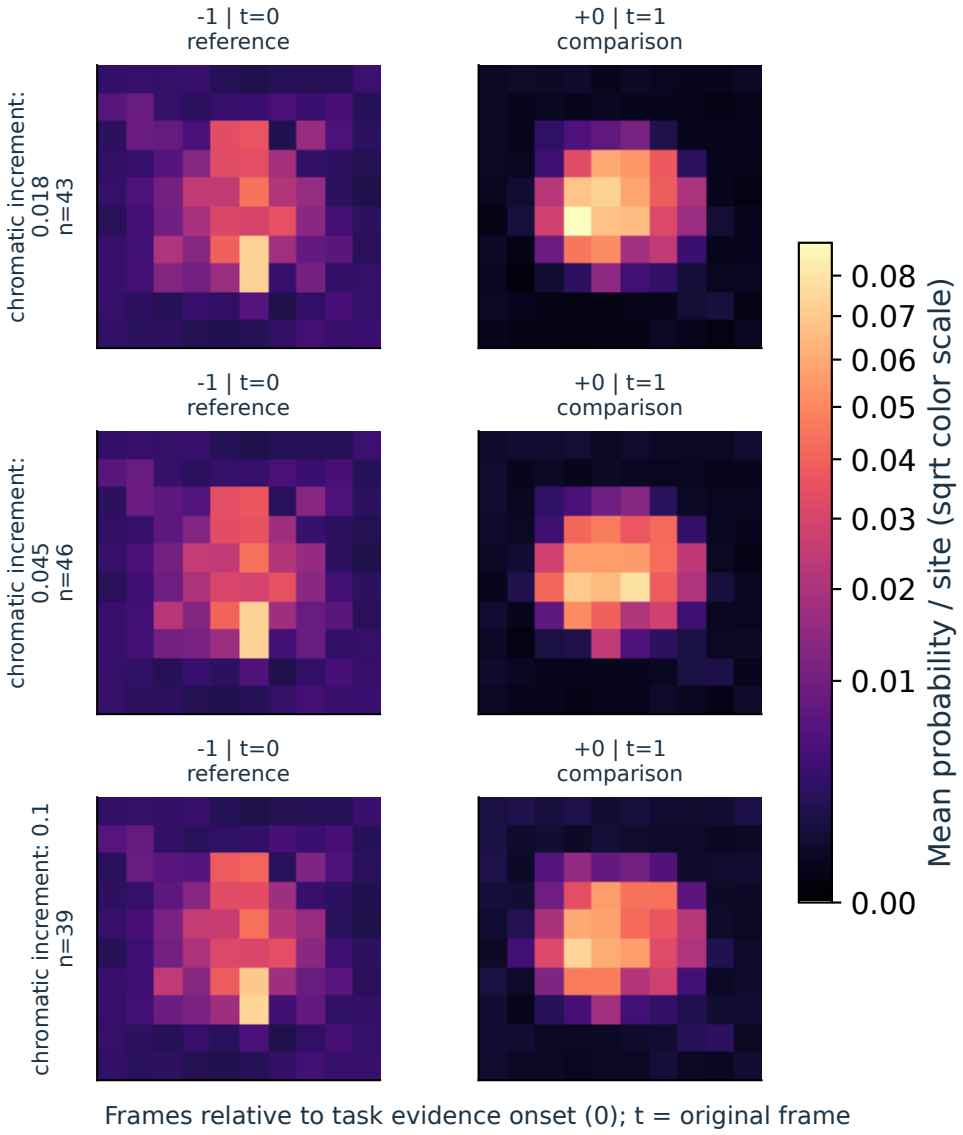


Figure 18: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Contour integration across local elements and recurrent populations

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized contour in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 1 native conditions and 128 fresh trials, equal-condition accuracy was 99.2% (95% interval [97.7%, 100.0%]). For the representative condition mixed, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was cls_gain0_t0, changing accuracy by -47.7 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

A contour is a spatial relationship among elements. The observer chooses which interval contains the aligned Gabor path among surrounding elements. Local orientation signals must be integrated into evidence for a coherent configuration, making this task a test of organization beyond a single isolated feature.

The perturbation axis is alignment jitter. Here increasing the numerical value makes the contour less orderly and typically makes detection harder. We preserve that direction in the plots. The analysis compares information in visual tokens with information in the spatial recurrent field and global state, then tests the causal cost of redistributing priority across the display.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Field, Hayes and Hess investigated contour integration using relationships between local oriented elements. This motivates treating the arrangement of features as an experimentally controlled variable. The model spatial recurrent field has an explicit neighborhood structure, which makes it a candidate substrate for integrating these relationships. The global state could then summarize the organized evidence. By measuring both population information and sensitivity to memory reset, the analysis can distinguish a contour representation that is present locally from one that has already been consolidated into a decision-relevant global signal.

1.2 Biological experiments motivating this task

Experimental question. Does the observer integrate relationships among local oriented elements, and how does selection affect that integration?

Field, Hayes and Hess (1993) People detected paths made of aligned Gabor elements within randomly oriented backgrounds. Detection depended strongly on the alignment and continuation of neighboring elements, motivating an association field linking compatible local orientations. [5]

Kapadia, Ito, Gilbert and Westheimer (1995) Collinear context improved human bar detection and facilitated responses in a subset of monkey V1 neurons. Relative position, orientation and intervening elements controlled the contextual effect. [6]

Li, Piech and Gilbert (2006) During contour detection, V1 responses tracked contour salience and were weaker when contours were missed or unattended. Neural discrimination connected contextual responses with perceptual performance. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Which frame contains the aligned Gabor contour?

Response labels are 0: frame 0; 1: frame 1. The native suite contains 1 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use mixed.

1.4 Stimulus sequence and spatial layout

For the representative native condition mixed, the sequence has 2 frames, indexed from zero: frame 0: reference; frame 1: comparison. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Contour orientation jitter (degrees)	2.0, 8.0, 16.0
Contour-path elements	7
Total elements	32
Stimulus family	contour

The spatial recurrent network can combine neighboring patch representations through its convolutional gates. The CLS state receives attention-weighted summaries of those representations. A population code that separates contour from distractor intervals need not consist of a single contour-selective unit; the decoding and state-space analyses are designed to expose distributed organization.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 99.2%, with macro AUC 1.000. The matched final checkpoint changed accuracy by -3.12 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was mixed at 99.2%; the highest was mixed at 99.2%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (2) produced 100.0% correct responses, compared with 53.9% at the largest strength (60). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

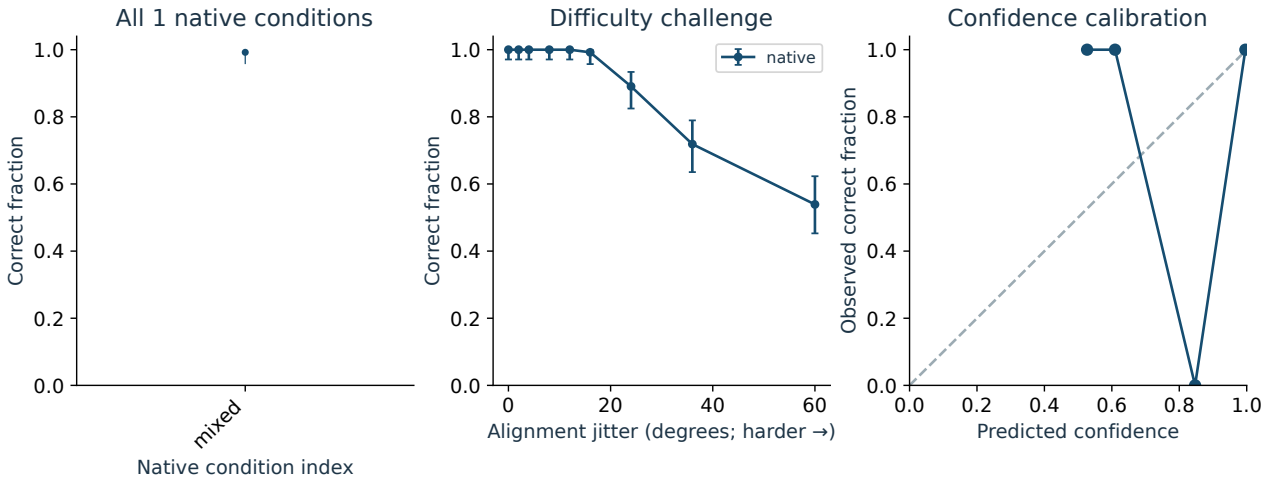


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)\exp(-(x/a)^b)$. Here x is contour orientation jitter in degrees, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 33.61 task units (bootstrap interval [30.32, 41.44]), with slope -0.02152 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.039.

Increasing jitter weakens contour alignment, so this fitted function decreases toward chance and its signed slope is negative.

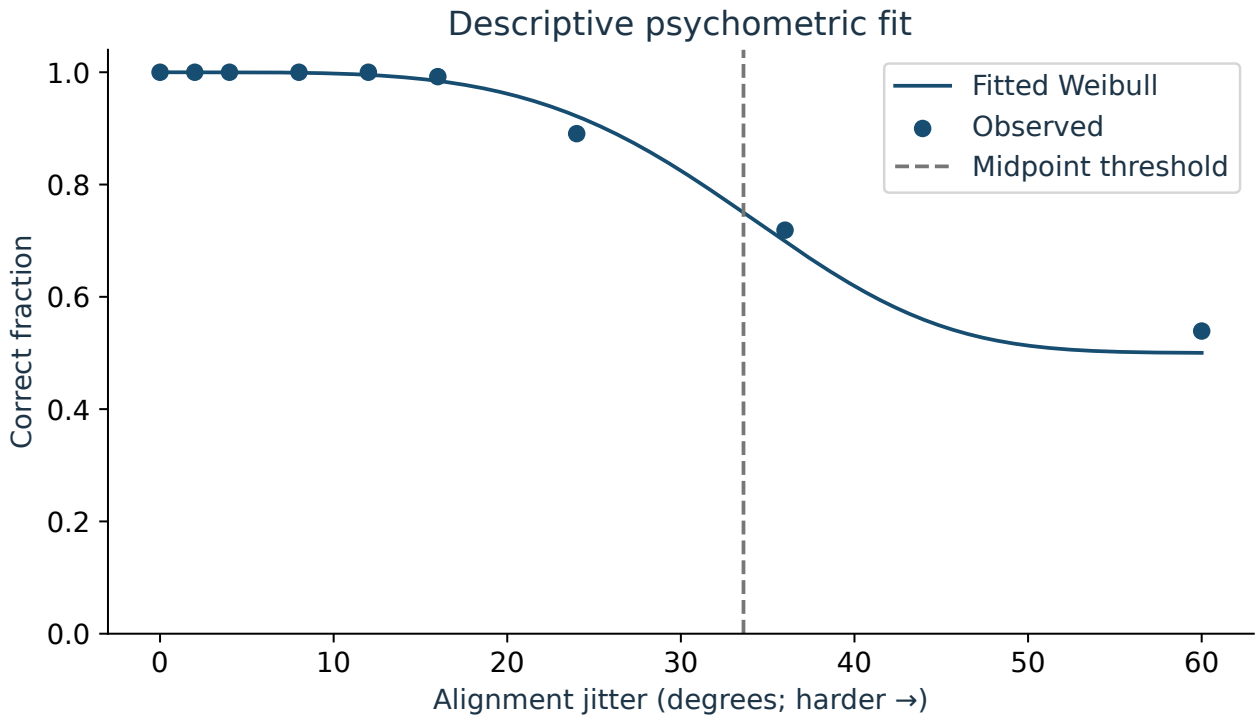


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	98.4%	1.0	99.2%	80.5%
0.7	97.7%	1.0	99.2%	45.3%
0.8	97.7%	1.0	99.2%	7.0%
0.9	96.9%	1.0	100.0%	0.0%

The representative condition mixed provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 125 of 128 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

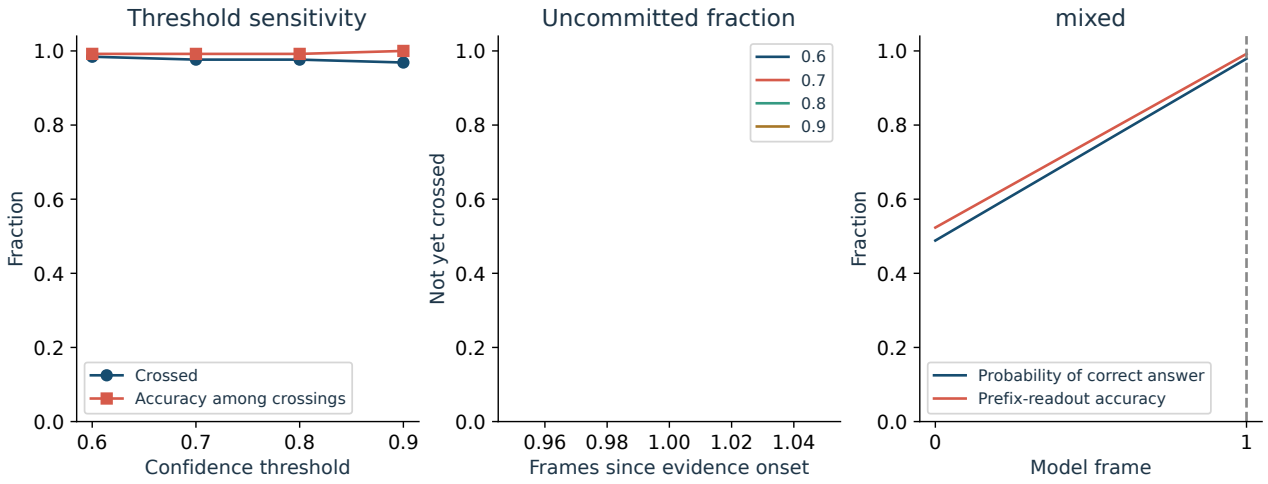


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

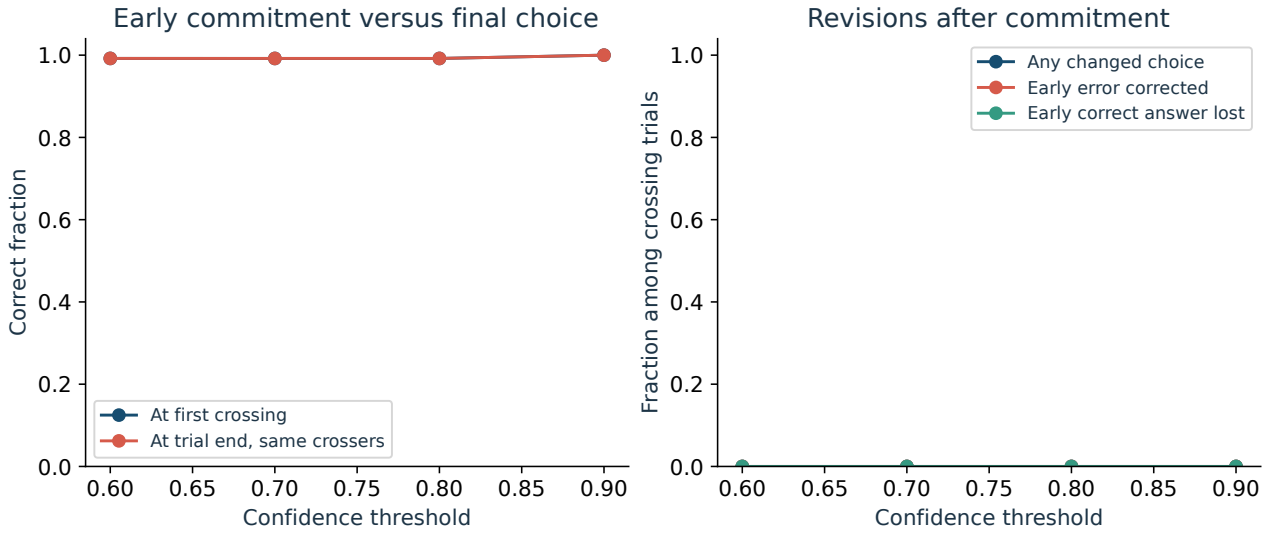


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

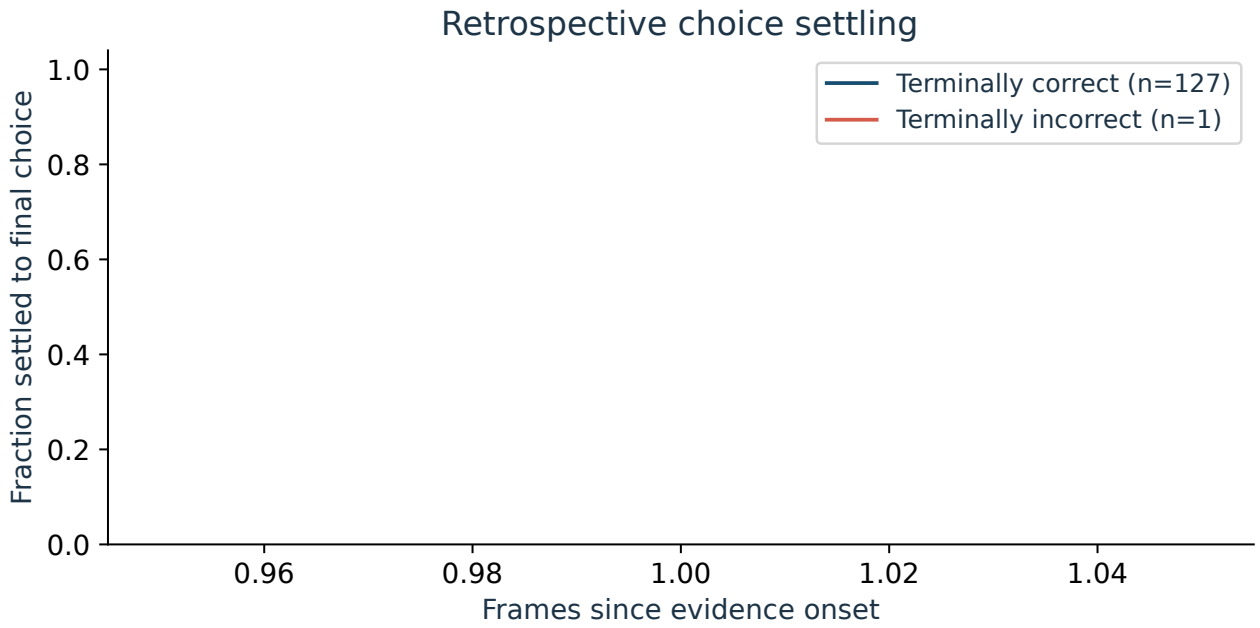


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

Rows separate contour-alignment jitter and pool interval order. Traces refer to fixed image quadrants.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
alignment jitter degrees: 2	41	41
alignment jitter degrees: 8	42	42
alignment jitter degrees: 16	45	45

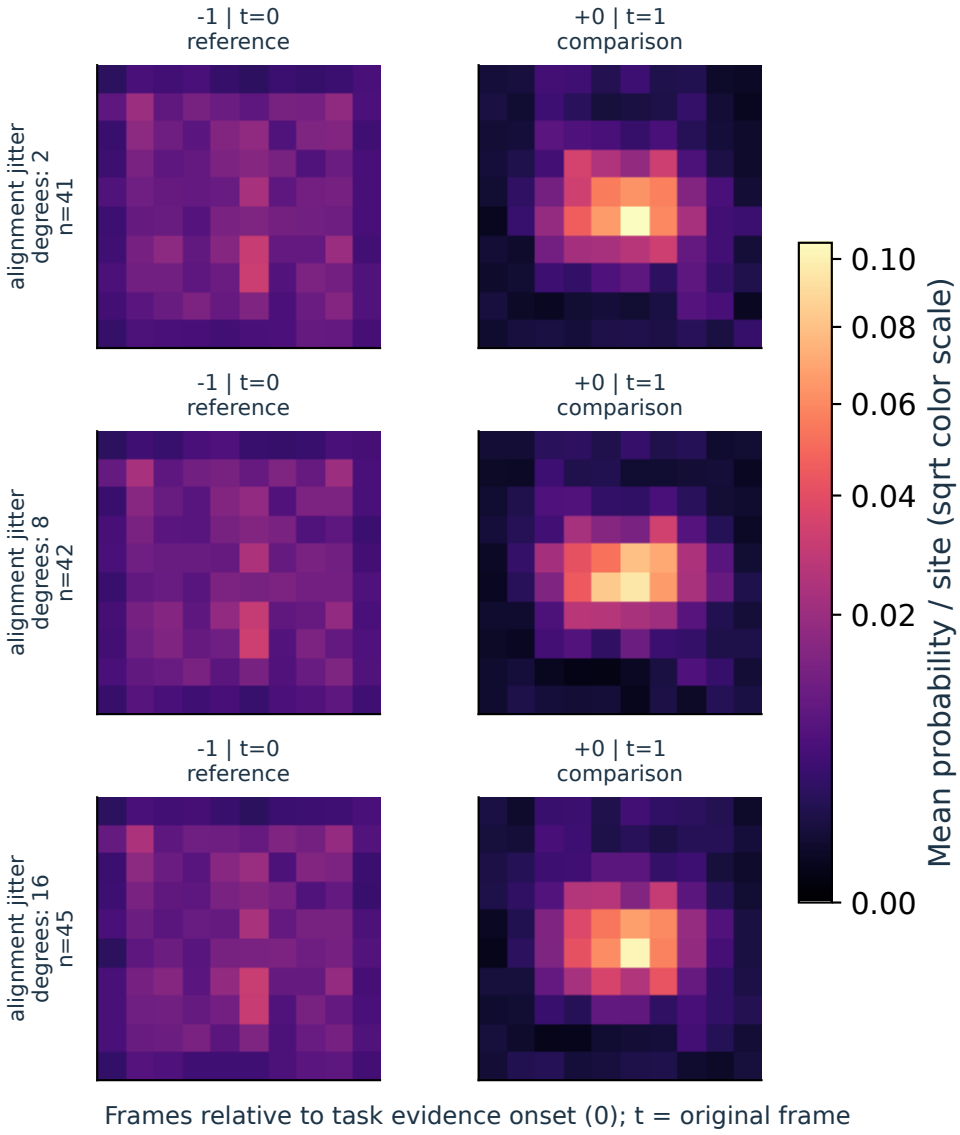


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

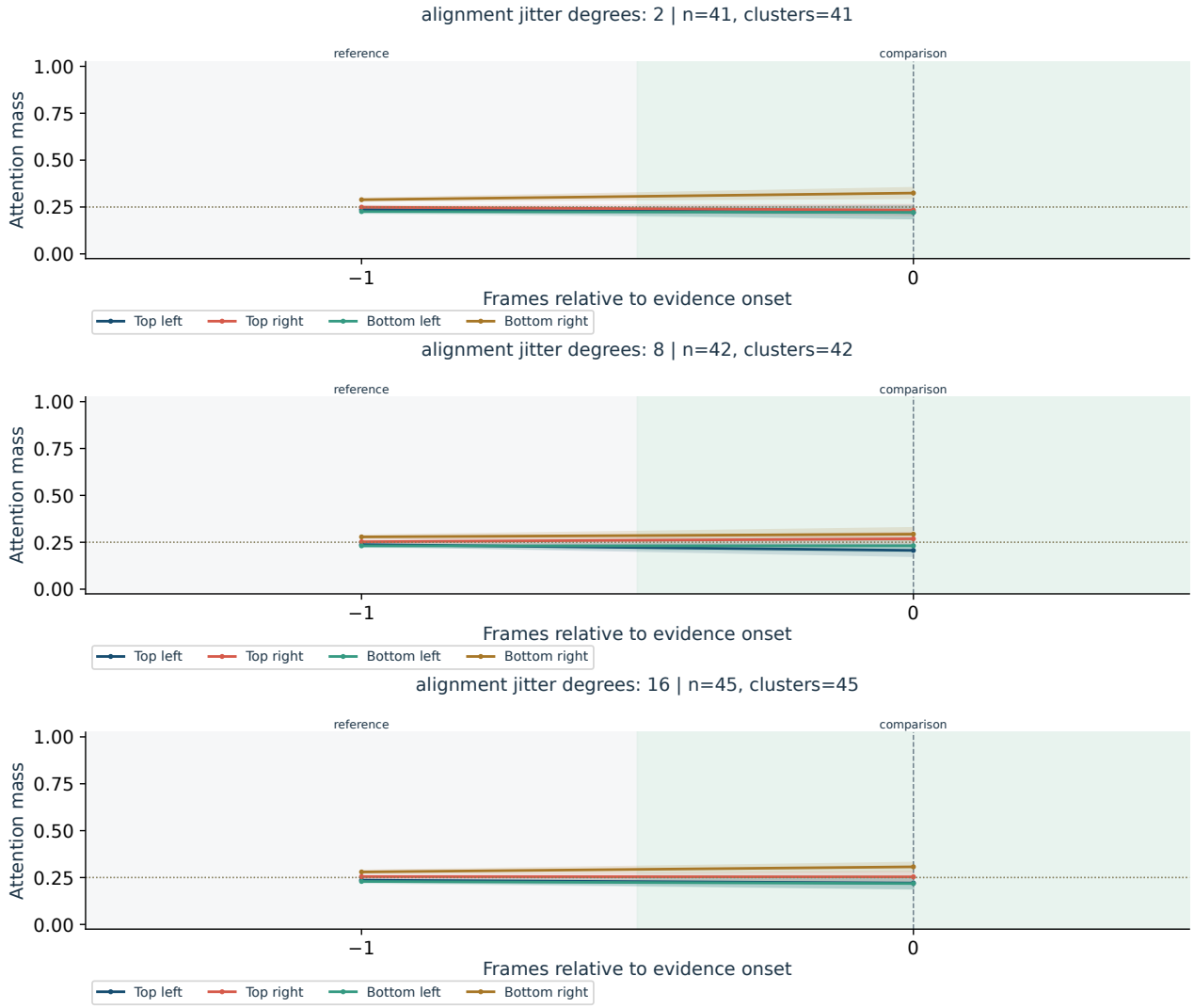


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

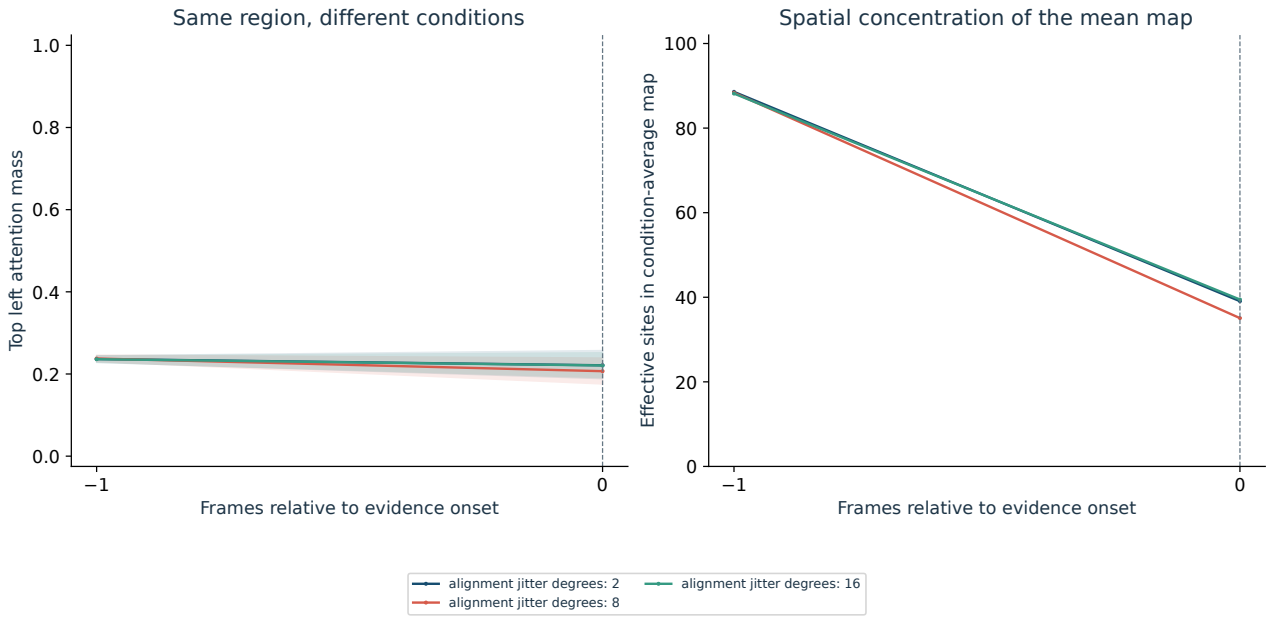


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. alignment jitter degrees: 2: top left mass peaks at 0.236 at original frame 0 (reference), and is 0.221 on the final frame. alignment jitter degrees: 8: top left mass peaks at 0.238 at original frame 0 (reference), and is 0.207 on the final frame. alignment jitter degrees: 16: top left mass peaks at 0.236 at original frame 0 (reference), and is 0.221 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For cls_gain0_t0, paired accuracy changed by -47.7 percentage points (paired 95% interval [-57.0%, -38.3%]; paired p=6.801e-16; within-task Holm p=2.108e-14). There were 3 corrected trials and 64 newly incorrect trials. For cls_gain1.5_t0, paired accuracy changed by +0.8 percentage points (paired 95% interval [0.0%, 2.3%]; paired p=1; within-task Holm p=1). There were 1 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

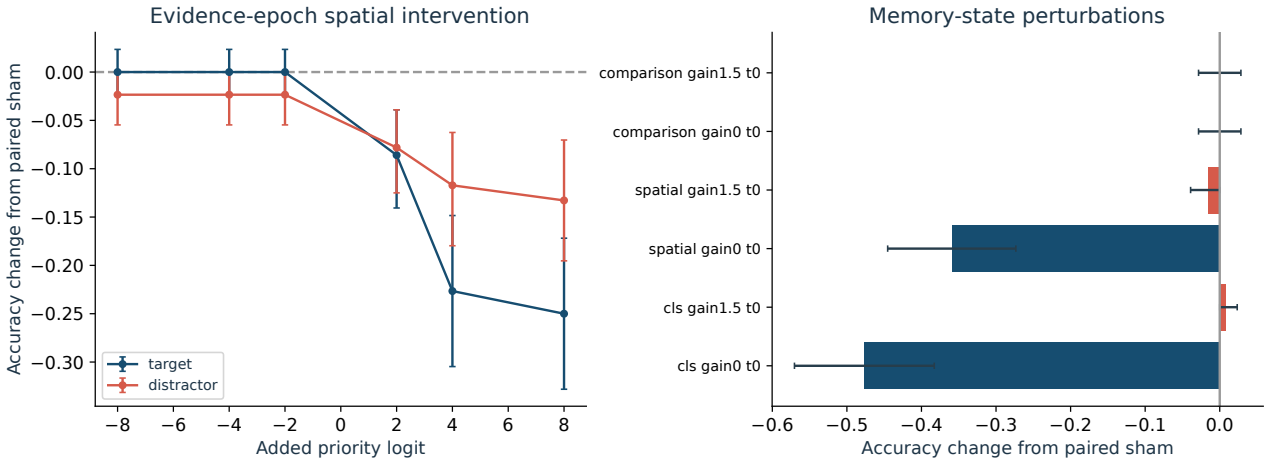


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
mixed	target_-4	+0.0	1
mixed	target_+4	-22.7	2.503e-06
mixed	distractor_-4	-2.3	1
mixed	distractor_+4	-11.7	0.01923

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.567	54.7%	0.180
fusion	0.580	57.8%	0.090
spatial	0.877	81.2%	0.005
cls	1.000	100.0%	0.005
comparison	Constant zero on this condition		
attention	0.607	60.9%	0.020

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

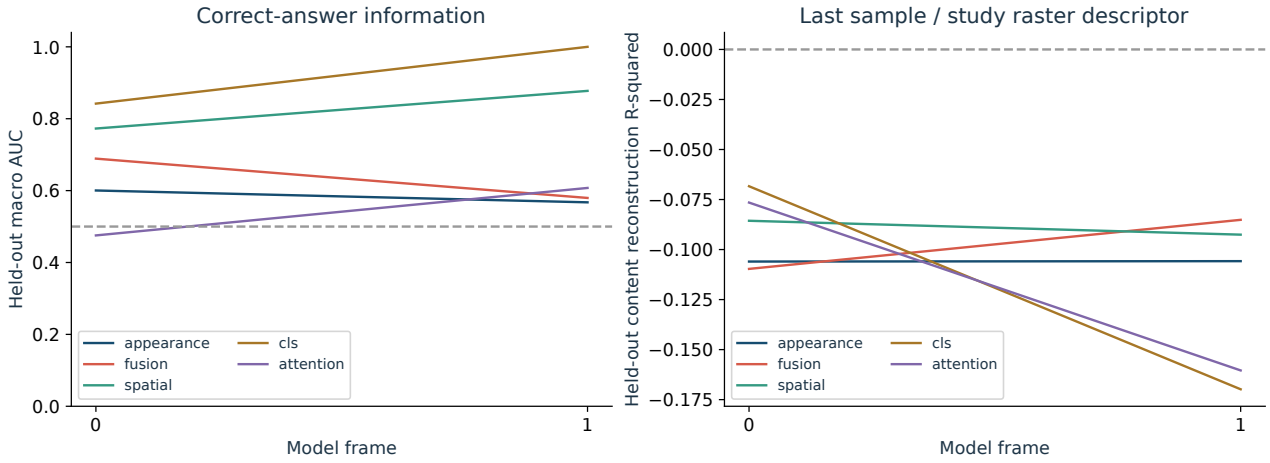


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 8.27 to 21.14 within the retained subspace. The final step displacement was 1.351 standardized RMS units. In spatial memory, the participation ratio ranged from 22.59 to 22.89 within the retained subspace. The final step displacement was 0.632 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 0, coarse sample-content reconstruction had R-squared -0.068 from CLS memory and -0.086 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -47.7 percentage points, spatial, -35.9 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.666	0.680	3200
fusion	0.631	0.807	6400
spatial	0.959	0.867	6400
cls	1.000	0.870	64
comparison	unavailable	unavailable	–
attention	0.591	0.546	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 47 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

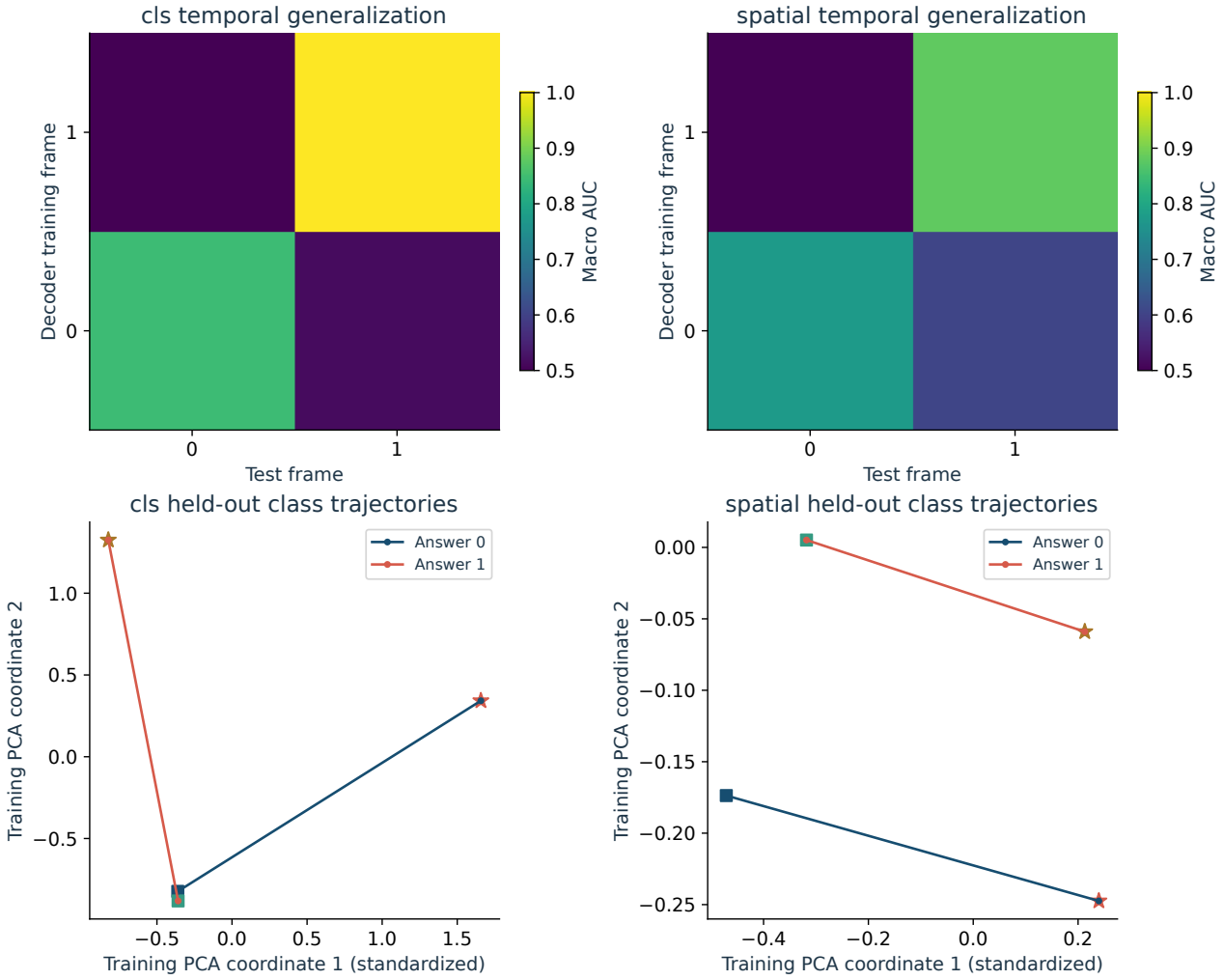


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

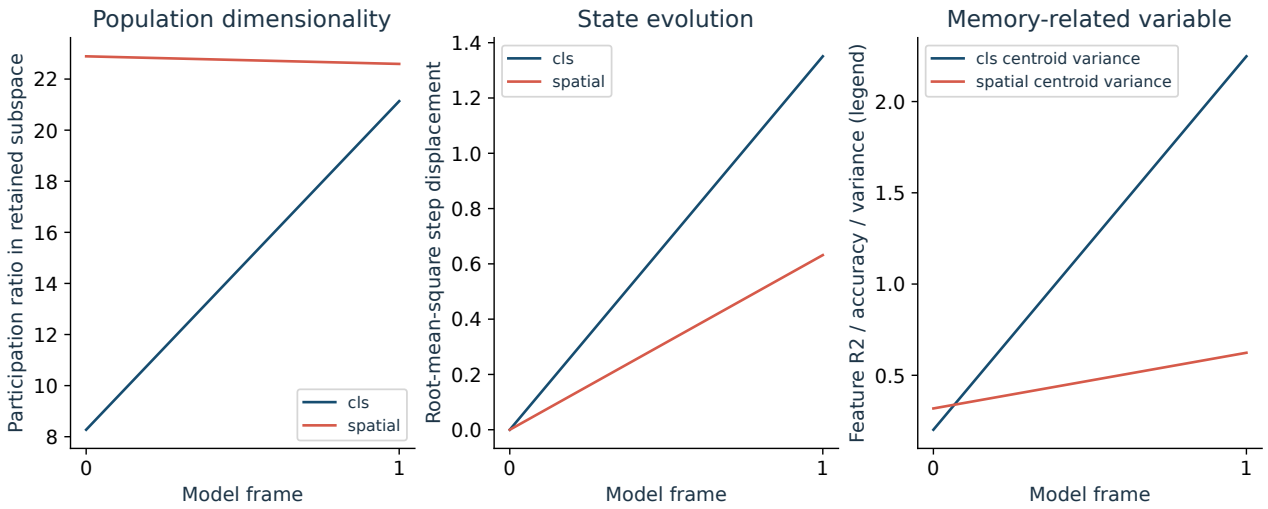


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

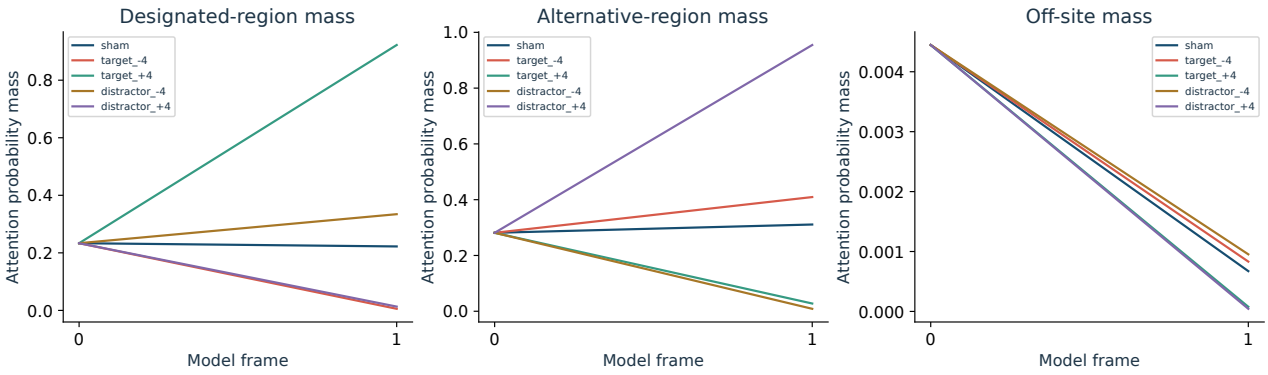


Figure 13: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

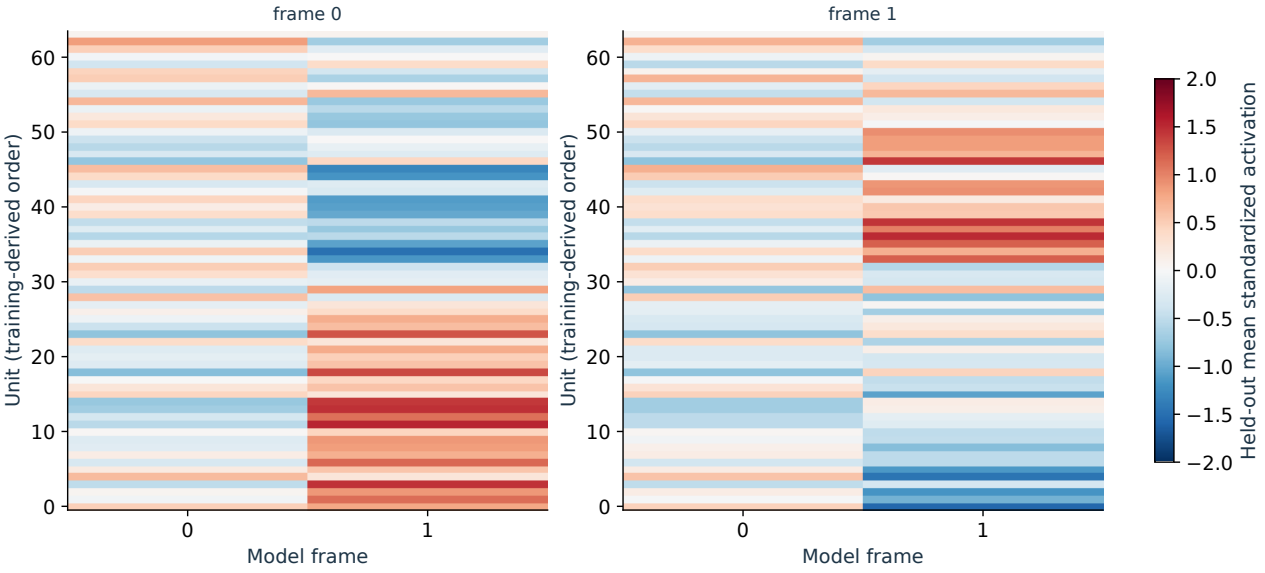


Figure 14: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	54.7%	56.2%	unavailable
fusion	57.8%	57.7%	unavailable
spatial	81.2%	79.8%	unavailable
cls	100.0%	98.5%	unavailable
attention	60.9%	60.3%	unavailable

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

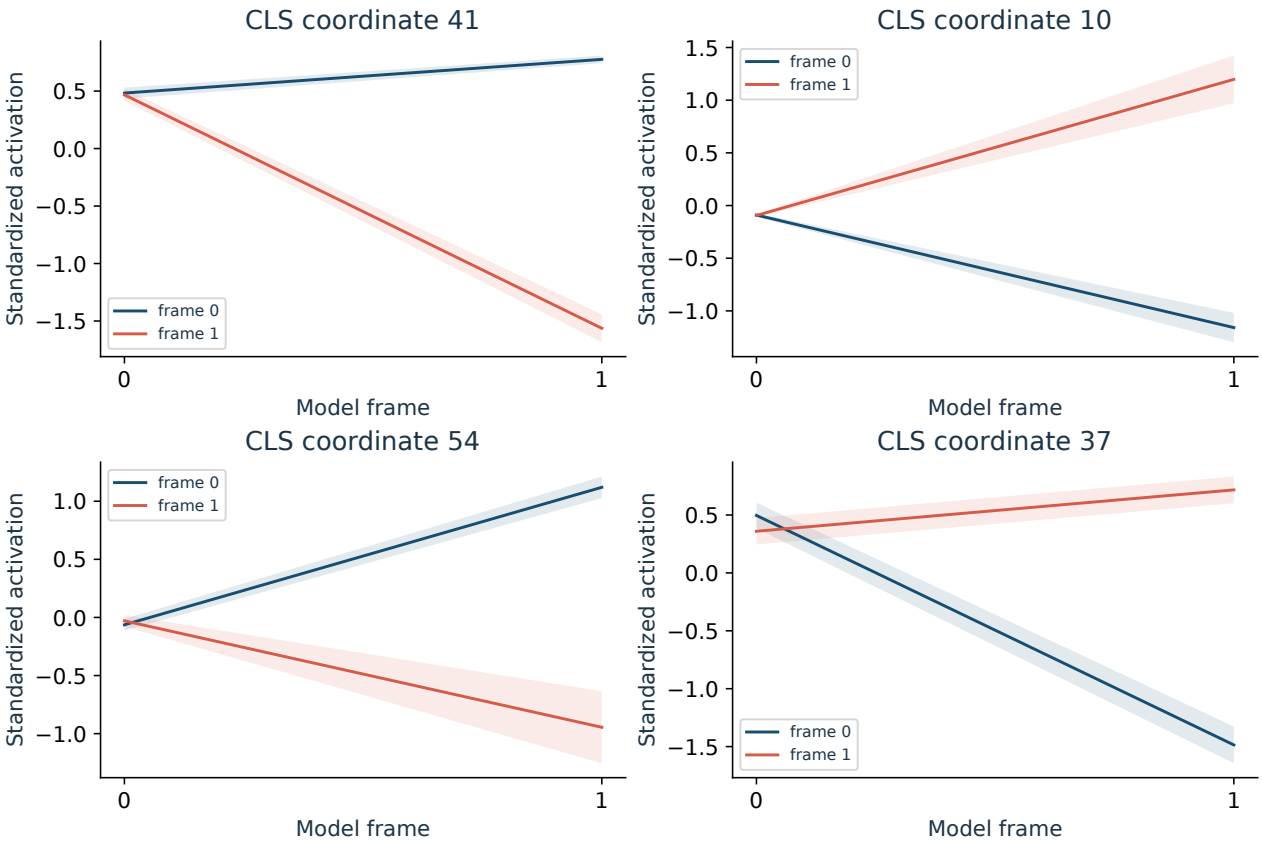


Figure 15: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

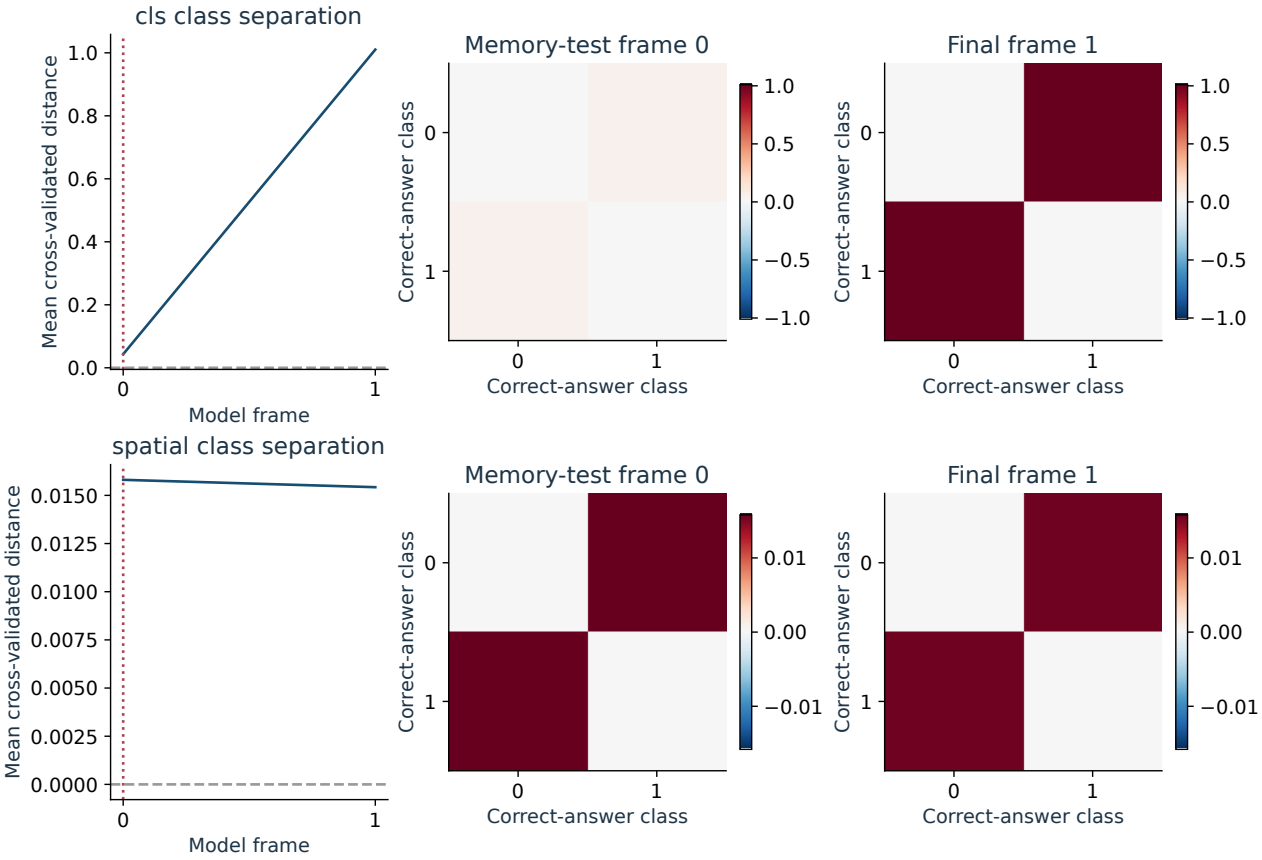


Figure 16: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

Contour decisions are almost perfect at the native settings, yet they are vulnerable to resetting either reference-period recurrent state and to concentrating priority in a fixed quadrant. The capacity-matched visual probes are weak compared with the final recurrent readout. This combination is consistent with a distributed configuration judgement assembled through the recurrent pathway. A contour spans multiple elements, so selecting a smaller region can remove useful relational structure even when the local orientation distribution remains informative. The current result establishes sensitivity to the native contour arrangement; independent manipulations of length and curvature would locate the spatial integration limit more precisely.

The behavioral result establishes 99.2% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies cls_gain0_t0 as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Changing contour length, spacing, curvature and local orientation jitter separately would test the spatial extent of integration. Occluding a contiguous path segment versus equally many scattered elements would isolate configuration sensitivity.

4.1 Interpreting these results alongside biological experiments

4.1.1 Field, Hayes and Hess (1993)

Our contour stimuli create a related need to distinguish an organized path from distractor structure. Near-perfect native performance and poorer performance after quadrant concentration are compatible with using relations across a spatially extended pattern. To identify an association-field-like rule, independently vary neighboring alignment, spacing and curvature while matching the distribution of local orientations. [5]

4.1.2 Kapadia, Ito, Gilbert and Westheimer (1995)

The biological comparison ties a local contextual effect to perception and V1 responses. The model's spatial recurrence can make one site depend on neighboring information, but its present task result does not measure the interaction kernel. Present one element alone, then add collinear or orthogonal neighbors, and inspect both local state change and behavioral sensitivity. [6]

4.1.3 Li, Piech and Gilbert (2006)

The animal result links contour-related responses with successful detection and attention. Our final recurrent populations carry stronger answer information than the capacity-matched early probes. Com-

paring correct and missed weak contours at matched stimulus strength would test whether model population separation tracks detection beyond physical salience. Existing native ceiling trials offer few misses, so challenge trials are the informative population. [7]

4.2 A computational account of the observed behavior

An association field is a proposed compatibility rule: nearby elements support one another when their orientations and positions fit a smooth path. In the model, recurrent processing could assemble such evidence through spatial interactions and shared global selection. A competing explanation is a simpler image statistic correlated with the native path. The current costs of reference-state reset and concentrated sampling establish dependence on recurrence and distributed evidence, but distinguishing these computations requires controls that preserve local statistics while rearranging relationships. The fixed crop size also defines an immediate spatial context; longer-range effects would have to emerge through repeated processing or global pooling.

4.3 Where this task agrees with the biological evidence

The measured behavior and perturbations agree with a distributed, recurrently supported configuration judgment. The central difference from the biological account is that the model's compatibility rule has not yet been mapped. Identifying the rule would connect the successful classifier to the geometric constraints of contour perception.

4.4 A follow-up experiment motivated by these results

Compare smooth paths, scrambled-position controls and orientation-matched nonpaths. Vary interelement spacing separately from element count. Measure the time at which path-versus-control separation appears locally and globally, then perturb the intermediate path locations.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Field DJ, Hayes A, Hess RF (1993). Contour integration by the human visual system: evidence for a local association field. Vision Research 33:173-193. https://pubmed.ncbi.nlm.nih.gov/8447091/.

[2] Treisman AM, Gelade G (1980). A feature-integration theory of attention. Cognitive Psychology 12:97-136. https://pubmed.ncbi.nlm.nih.gov/7351125/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] Field, Hayes and Hess (1993). Contour integration by the human visual system: evidence for a local association field. Primary source.

[6] Kapadia, Ito, Gilbert and Westheimer (1995). Improvement in visual sensitivity by changes in local context: parallel studies in human observers and in V1 of alert monkeys. Primary source.

[7] Li, Piech and Gilbert (2006). Contour saliency in primary visual cortex. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
mixed	128	99.2%	1.000	0.021

B Every-frame attention atlas

Condition mixed. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 64 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

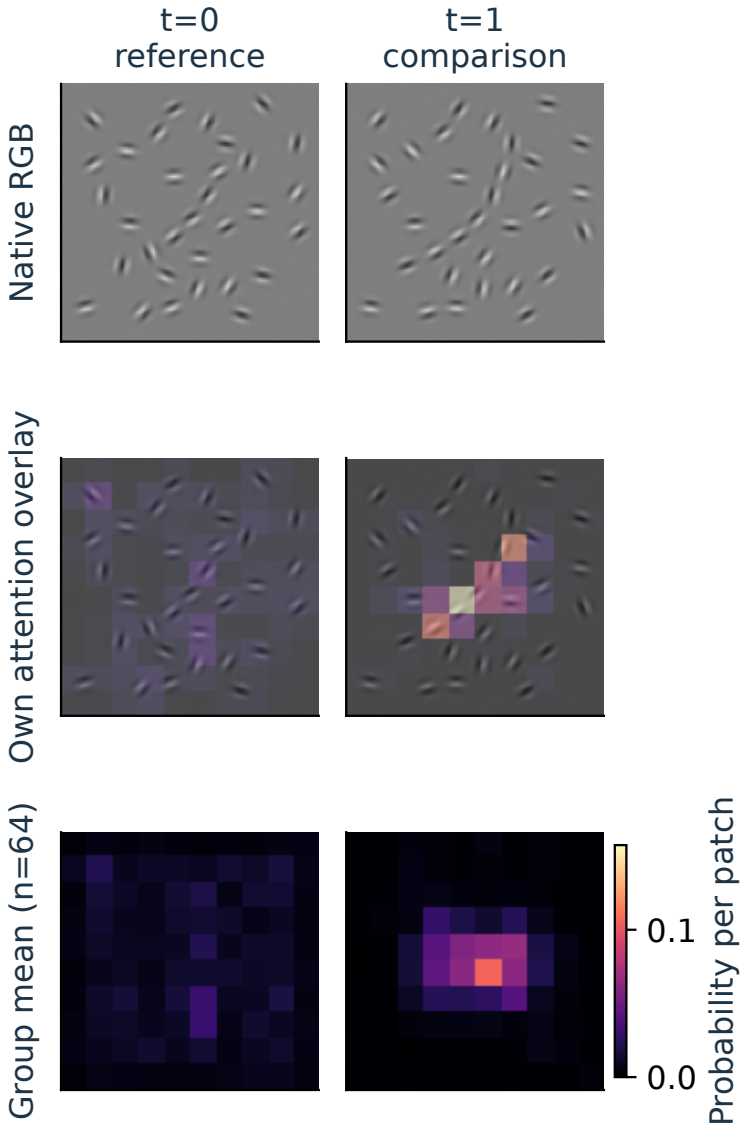


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

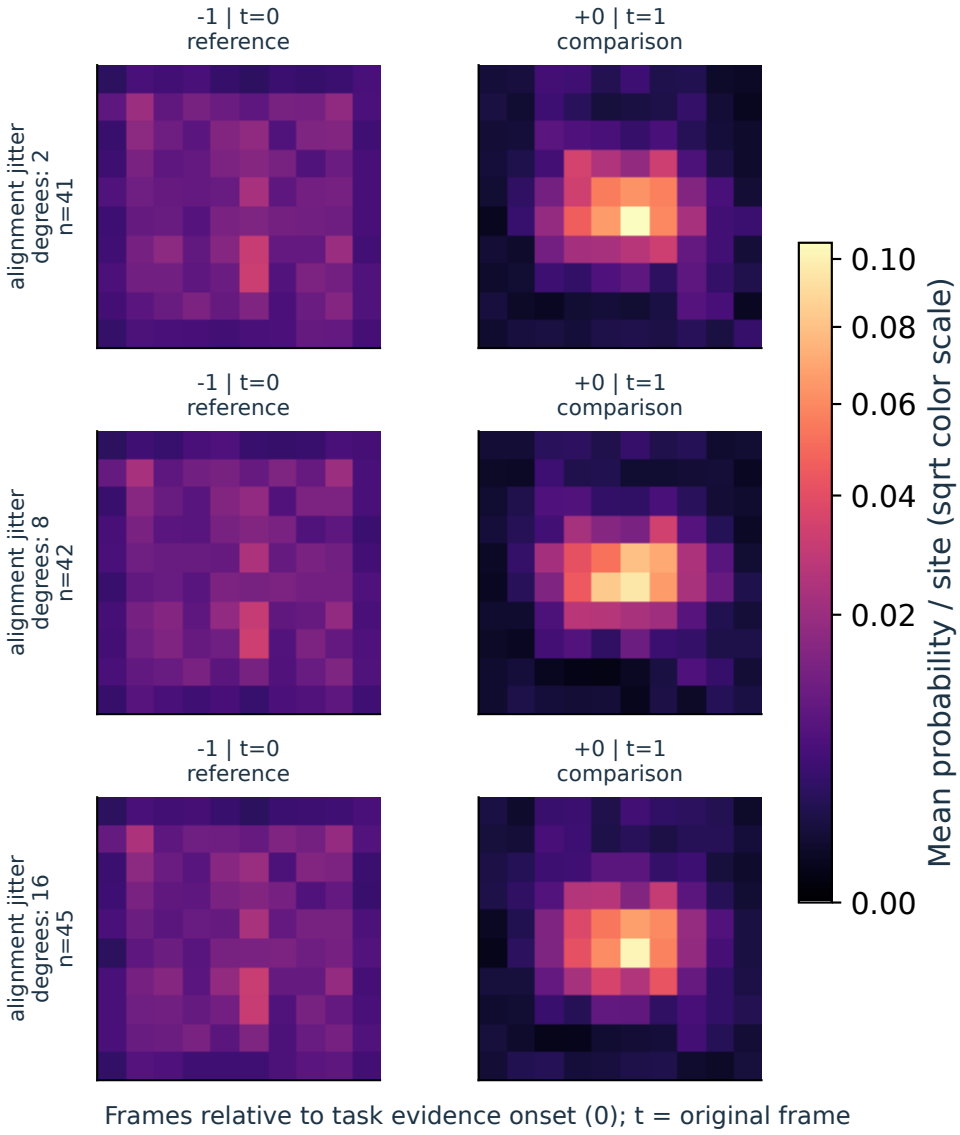


Figure 18: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Natural-image spectral comparison and the retention of texture statistics

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized natural spectrum in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 1 native conditions and 128 fresh trials, equal-condition accuracy was 100.0% (95% interval [95.9%, 100.0%]). For the representative condition mixed, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was spatial_gain0_t0, changing accuracy by -35.9 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

Natural images contain structure at many spatial scales. This task asks which interval has greater relative high-frequency detail after a controlled change to its spectral slope. The judgement concerns a statistical property of the image rather than object identity.

The slope difference sets difficulty. We retain the original natural-image split and evaluate new procedural comparisons. Readout from appearance, fused features and memory reveals whether the model carries a sufficient description of spectral content across intervals. Decoder training and testing use separate source-image splits.

The biological motivation is developed through [1]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Field related natural-image statistics to cortical response properties. Here that connection motivates a controlled spectral manipulation rather than an unrestricted image-category judgement. The same image can differ in the distribution of energy across scales while retaining recognizable structure. A useful population code should distinguish the relevant statistical change across different scenes. Keeping separate source-image splits for probe fitting and testing asks whether a linear readout generalizes across those scenes. Attention overlays then reveal which regions of a particular photograph contribute to the selected evidence.

1.2 Biological experiments motivating this task

Experimental question. Does successful comparison of image spectra reflect sensitivity to natural statistics, local contrast, or a learned spectral rule?

Tolhurst and Tadmor (1997) Sensitivity to spectral-slope changes could be explained by contrast changes within limited frequency bands. Which band was informative depended on the reference spectrum, linking global-looking judgments to selective spatial filtering. [4]

Tolhurst and Tadmor (2000) Experiments blending photograph spectra found enhanced sensitivity around natural spectral statistics after controlling image properties. The manipulation tested natural-image specialization more directly than successful discrimination of arbitrary slopes alone. [5]

Olshausen and Field (1996) Learning a sparse representation of natural image patches produced localized, oriented, bandpass basis functions. The theory connects environmental statistics to an efficient sensory representation. [6]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Which frame has greater relative high-frequency detail (smaller spectral beta)?

Response labels are 0: frame 0; 1: frame 1. The native suite contains 1 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use mixed.

1.4 Stimulus sequence and spatial layout

For the representative native condition mixed, the sequence has 2 frames, indexed from zero: frame 0: reference; frame 1: comparison. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Spectral-exponent difference	0.15, 0.3, 0.6
Stimulus family	natural_spectrum

A spatial priority map may emphasize textured regions that contain strong evidence about image detail. Averaging pixels across unrelated photographs would obscure that evidence, so the atlas pairs a single native image with its own attention and places the condition-average attention beside it. Those panels answer complementary questions about one trial and about systematic spatial preference.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 100.0%, with macro AUC 1.000. The matched final checkpoint changed accuracy by +0.00 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was mixed at 100.0%; the highest was mixed at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (0.01) produced 57.8% correct responses, compared with 100.0% at the largest strength (0.6). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

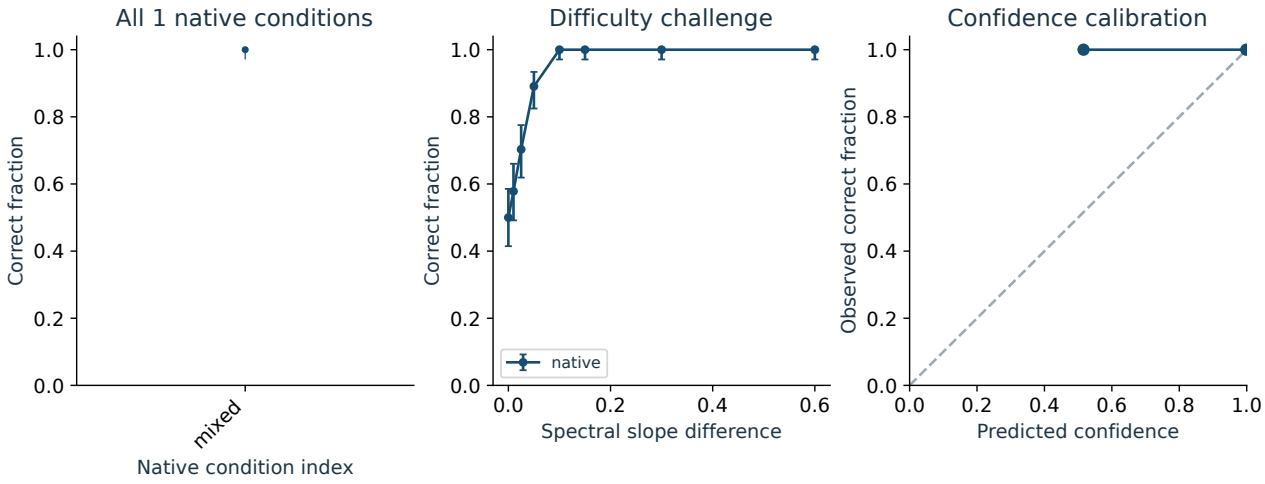


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)[1 - \exp(-(x/a)^b)]$. Here x is signal strength, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 0.02998 task units (bootstrap interval [0.0212, 0.03774]), with slope 9.699 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.026.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	99.2%	1.0	100.0%	94.5%
0.7	99.2%	1.0	100.0%	92.2%
0.8	99.2%	1.0	100.0%	88.3%
0.9	99.2%	1.0	100.0%	78.9%

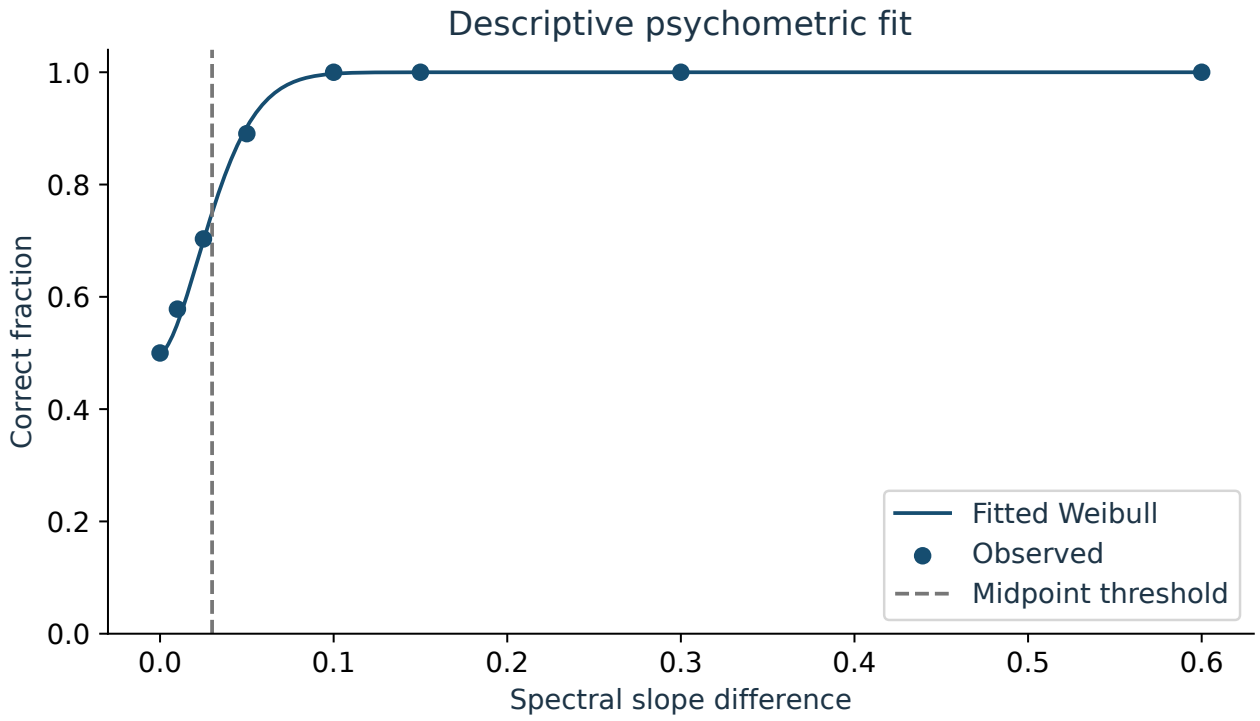


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

The representative condition mixed provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 127 of 128 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

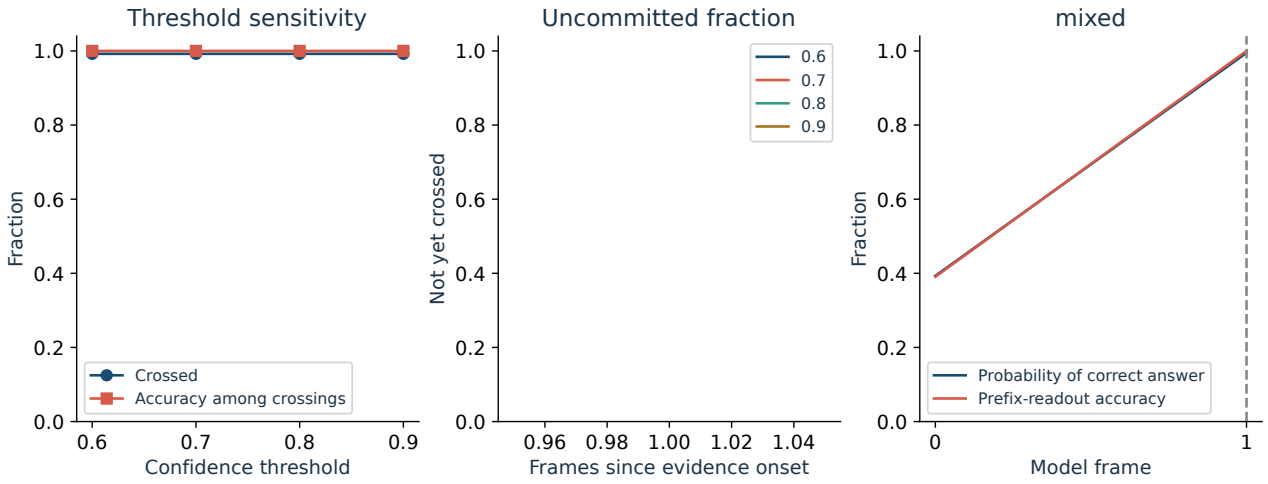


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

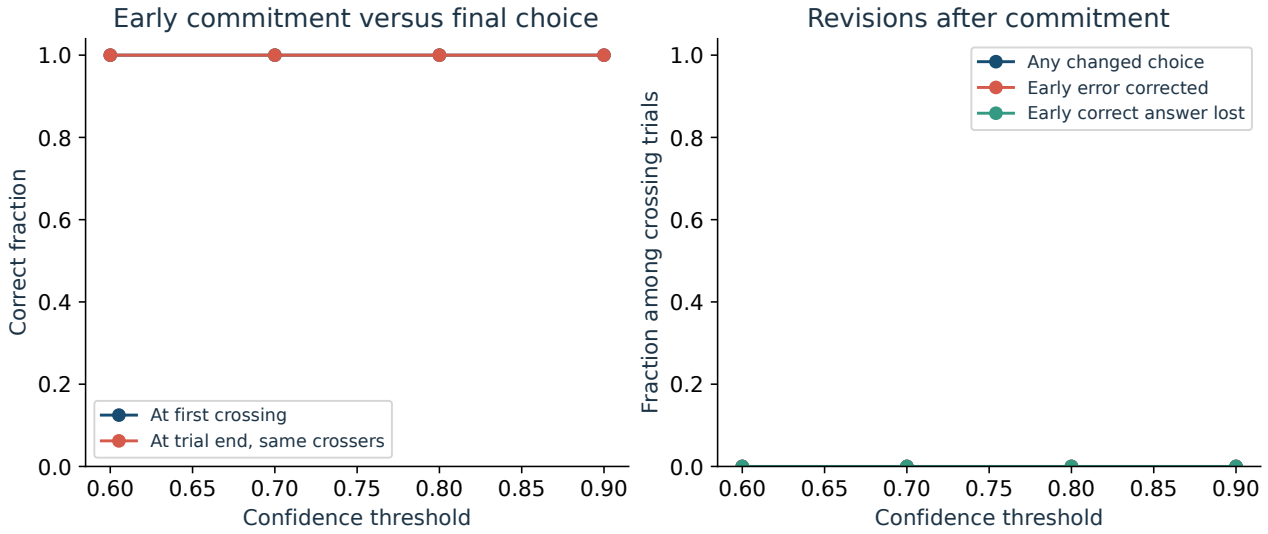


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

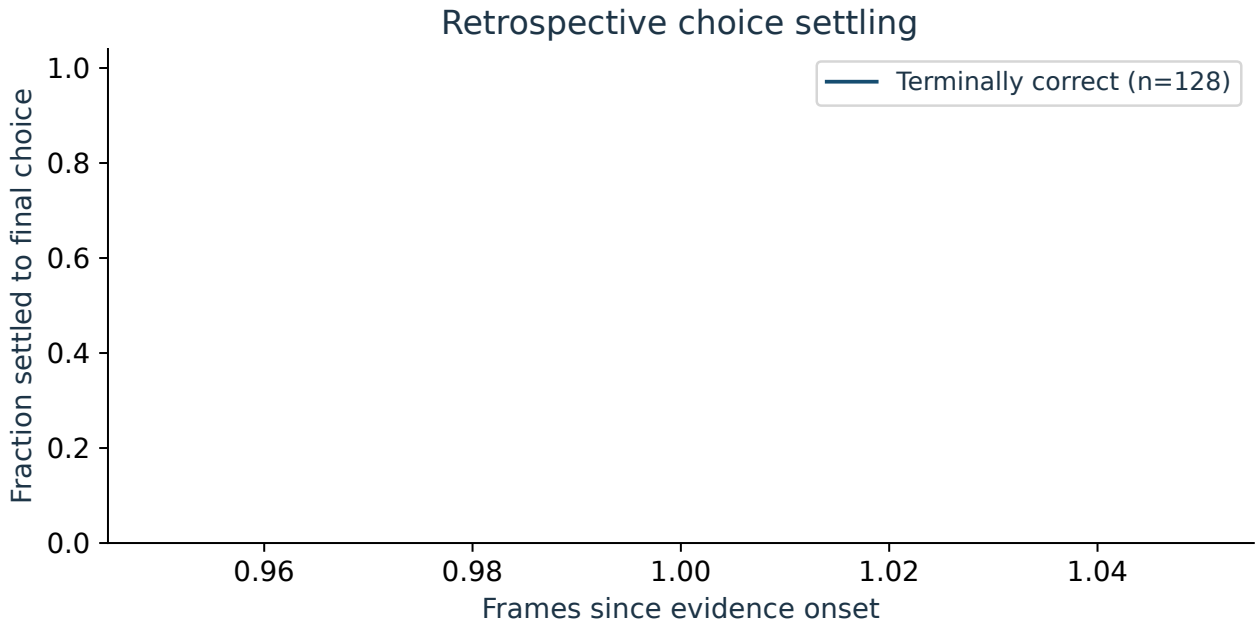


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

Rows separate spectral-exponent differences and pool interval order and photographs. The map average describes image-coordinate allocation across different photographs; each viewer overlay belongs to one replayed photograph.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
beta delta: 0.15	43	41
beta delta: 0.3	43	40
beta delta: 0.6	42	36

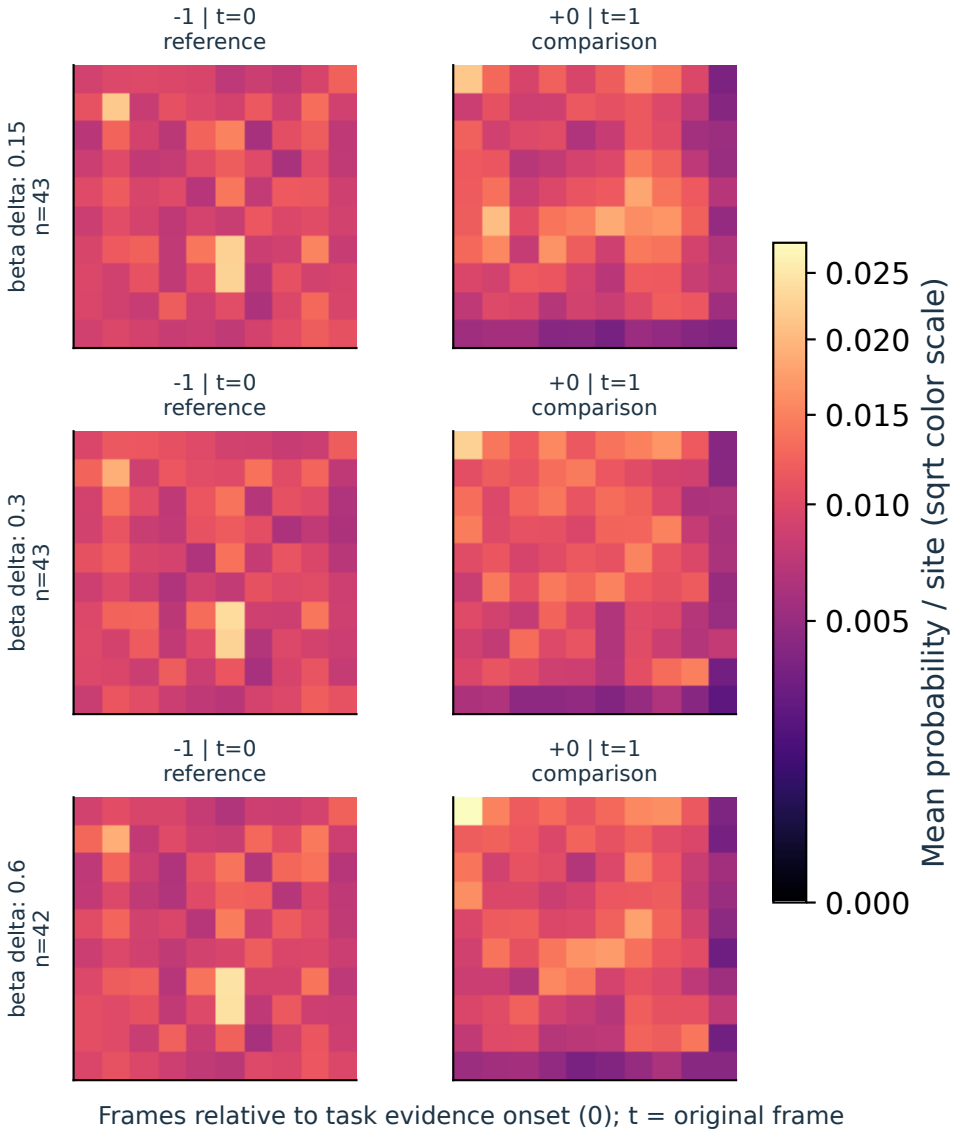


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

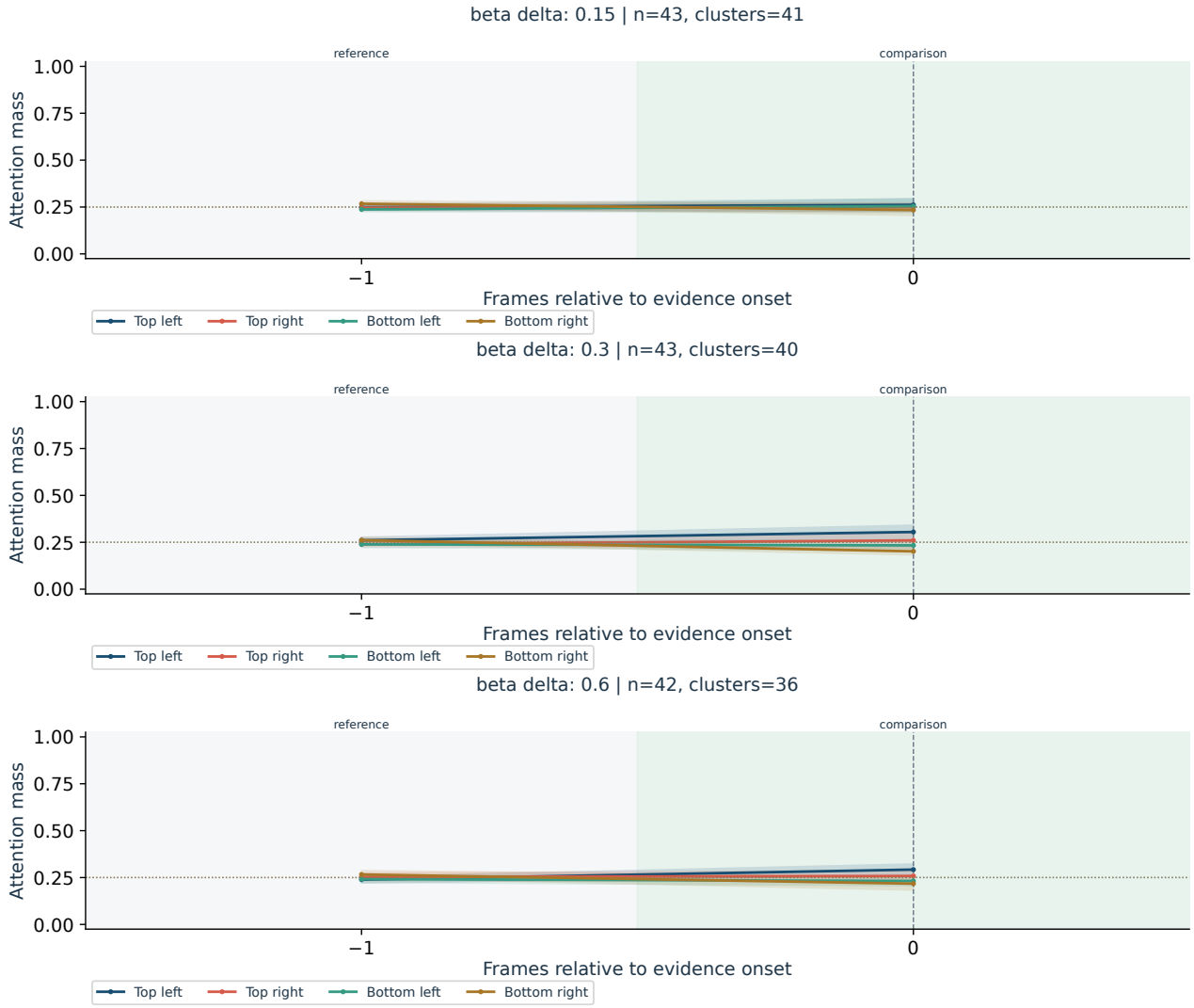


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

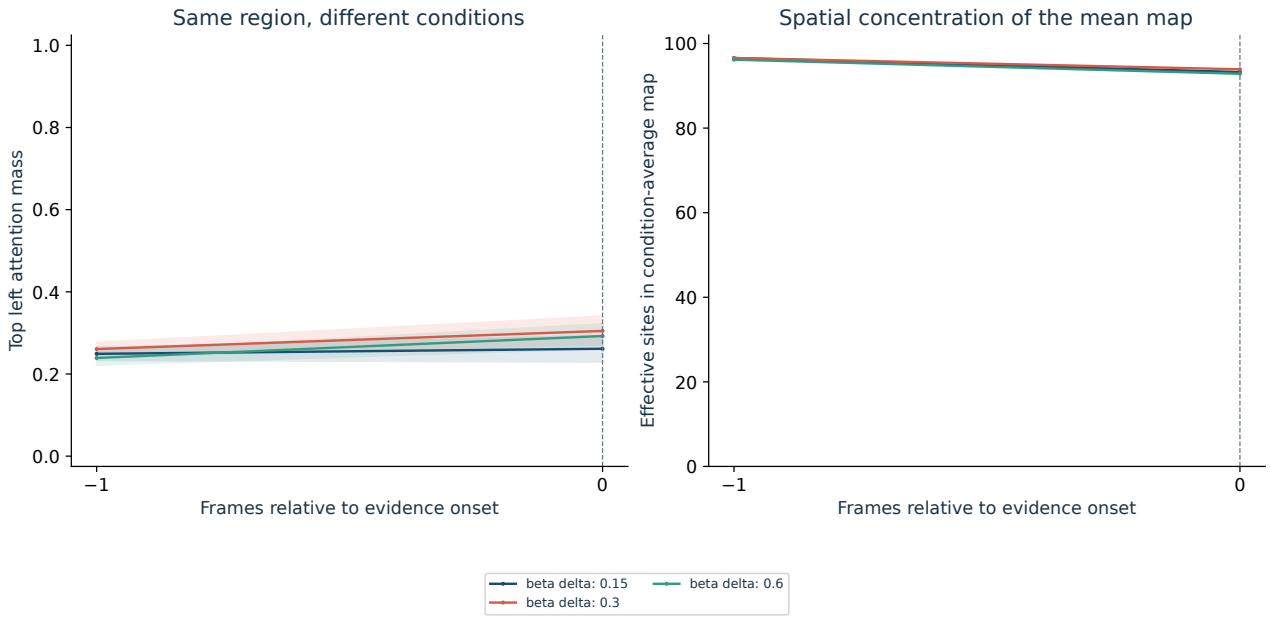


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. beta delta: 0.15: top left mass peaks at 0.261 at original frame 1 (comparison), and is 0.261 on the final frame. beta delta: 0.3: top left mass peaks at 0.305 at original frame 1 (comparison), and is 0.305 on the final frame. beta delta: 0.6: top left mass peaks at 0.292 at original frame 1 (comparison), and is 0.292 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For spatial_gain0_t0, paired accuracy changed by -35.9 percentage points (paired 95% interval [-44.5%, -27.2%]; paired p=0.0001; within-task Holm p=0.0031). There were 0 corrected trials and 46 newly incorrect trials. For target_-2_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-3.8%, 3.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

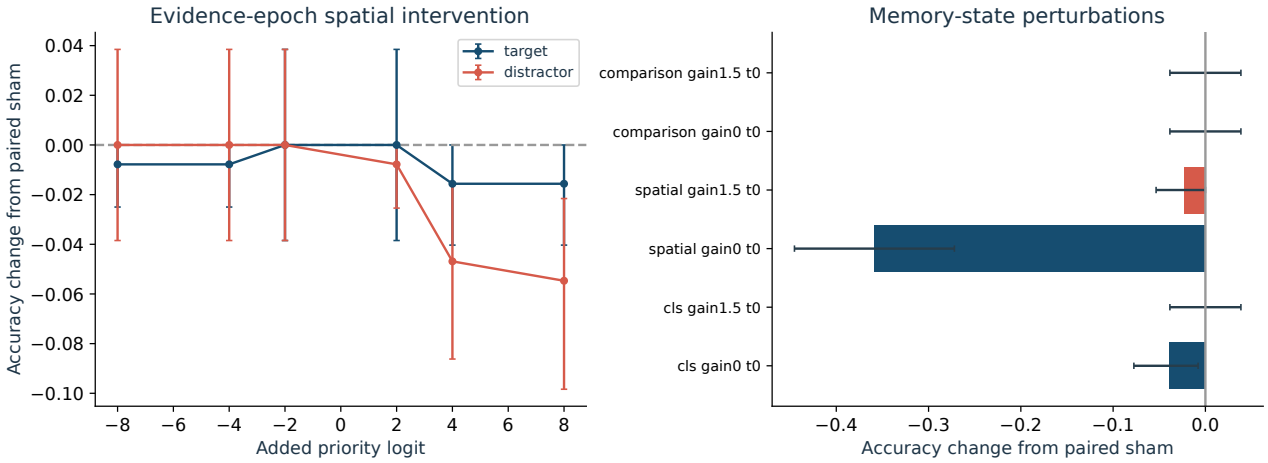


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
mixed	target_-4	-0.8	1
mixed	target_+4	-1.6	1
mixed	distractor_-4	+0.0	1
mixed	distractor_+4	-4.7	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.569	57.0%	0.095
fusion	0.751	71.1%	0.005
spatial	0.865	78.1%	0.005
cls	1.000	100.0%	0.005
comparison	Constant zero on this condition		
attention	0.616	59.4%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

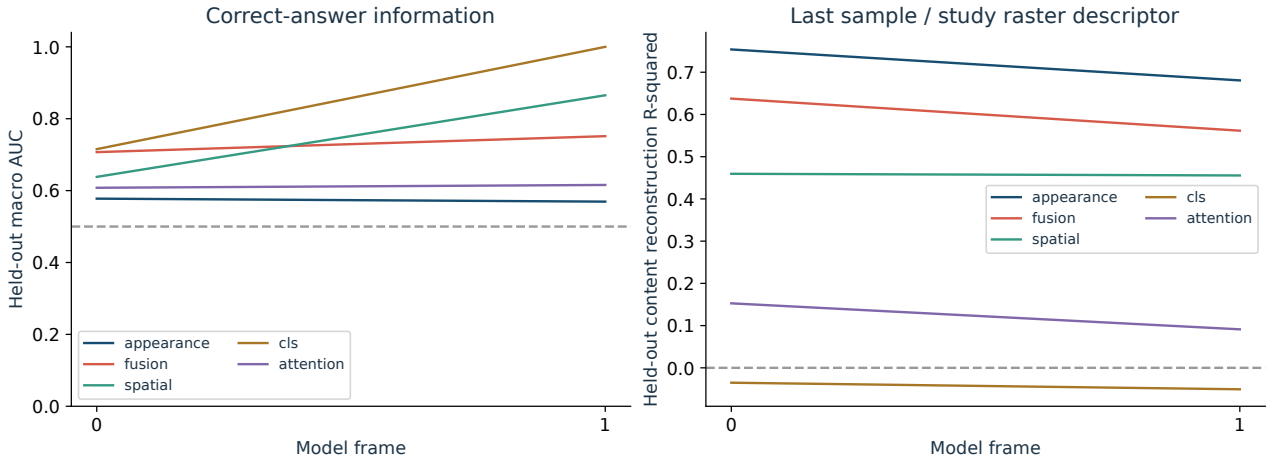


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 14.36 to 19.06 within the retained subspace. The final step displacement was 1.255 standardized RMS units. In spatial memory, the participation ratio ranged from 21.74 to 23.40 within the retained subspace. The final step displacement was 0.642 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 0, coarse sample-content reconstruction had R-squared -0.035 from CLS memory and 0.459 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -3.9 percentage points, spatial, -35.9 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.715	0.706	3200
fusion	0.819	0.732	6400
spatial	0.991	0.747	6400
cls	1.000	0.760	64
comparison	unavailable	unavailable	–
attention	0.650	0.576	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 43 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

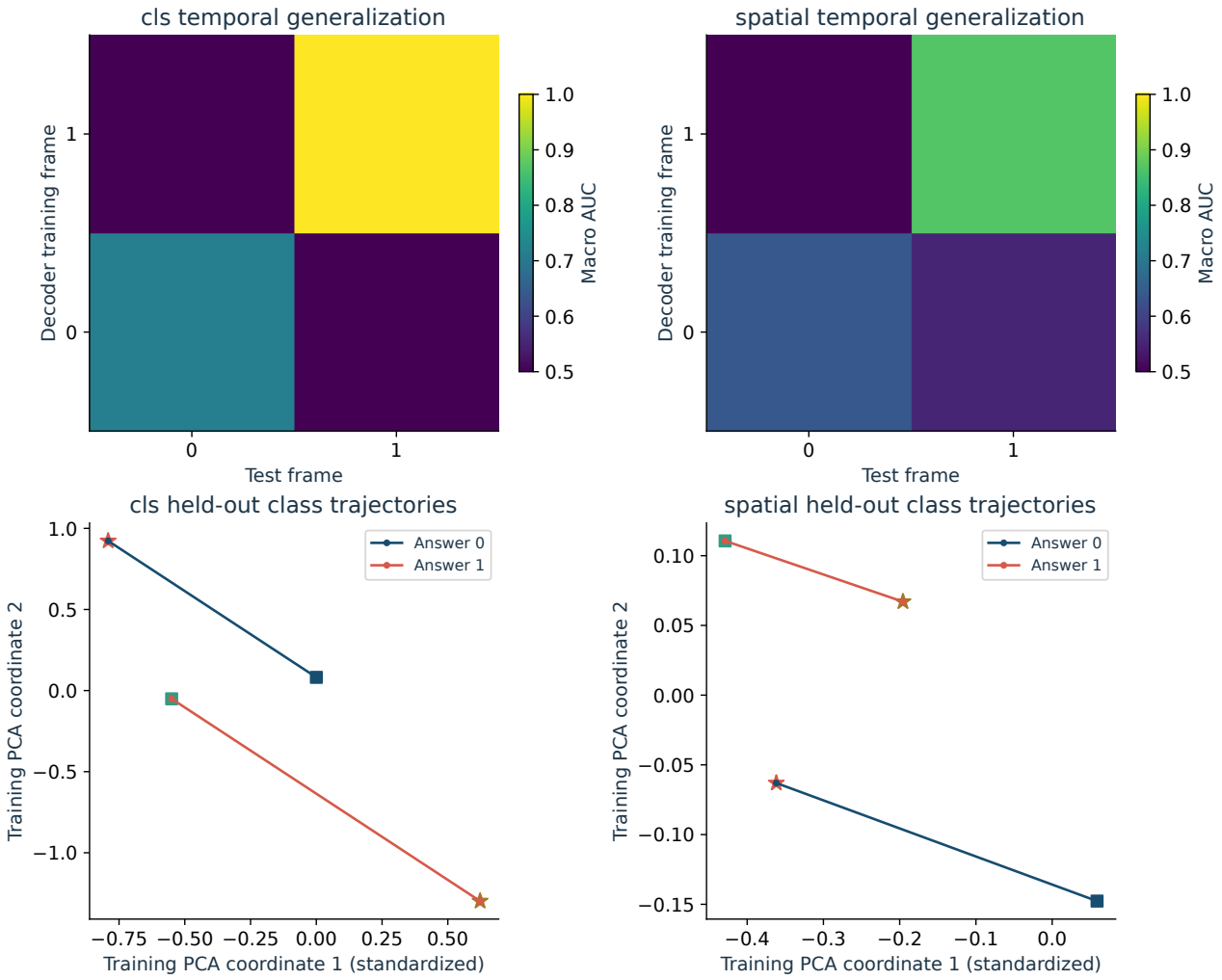


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

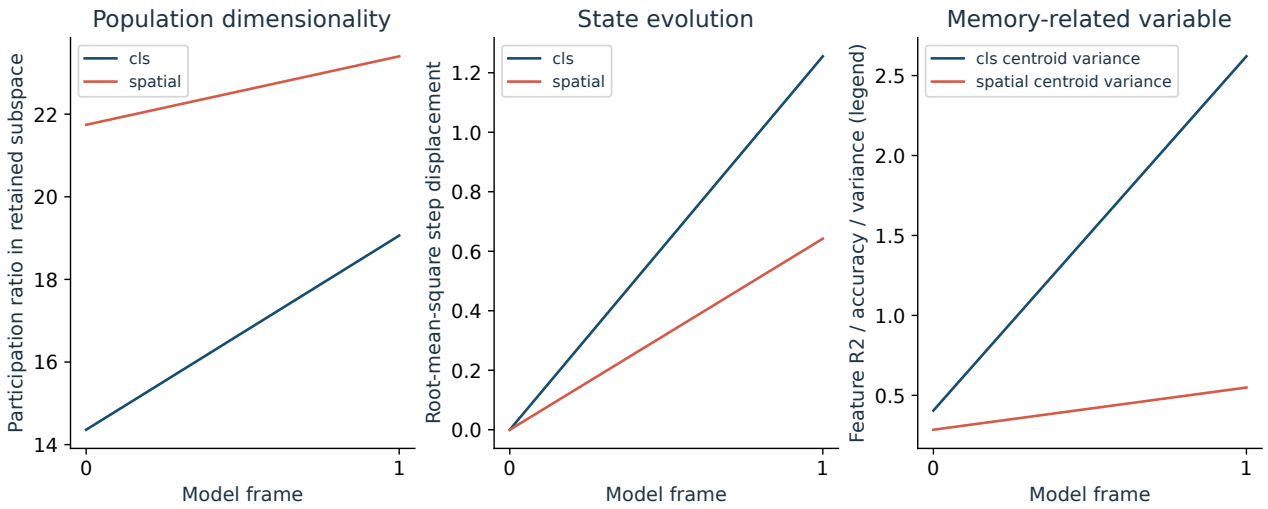


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

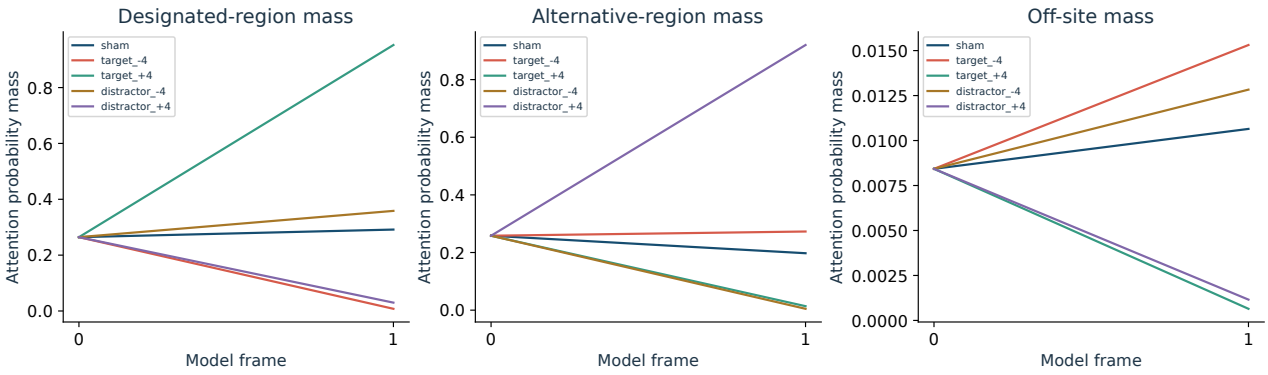


Figure 13: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

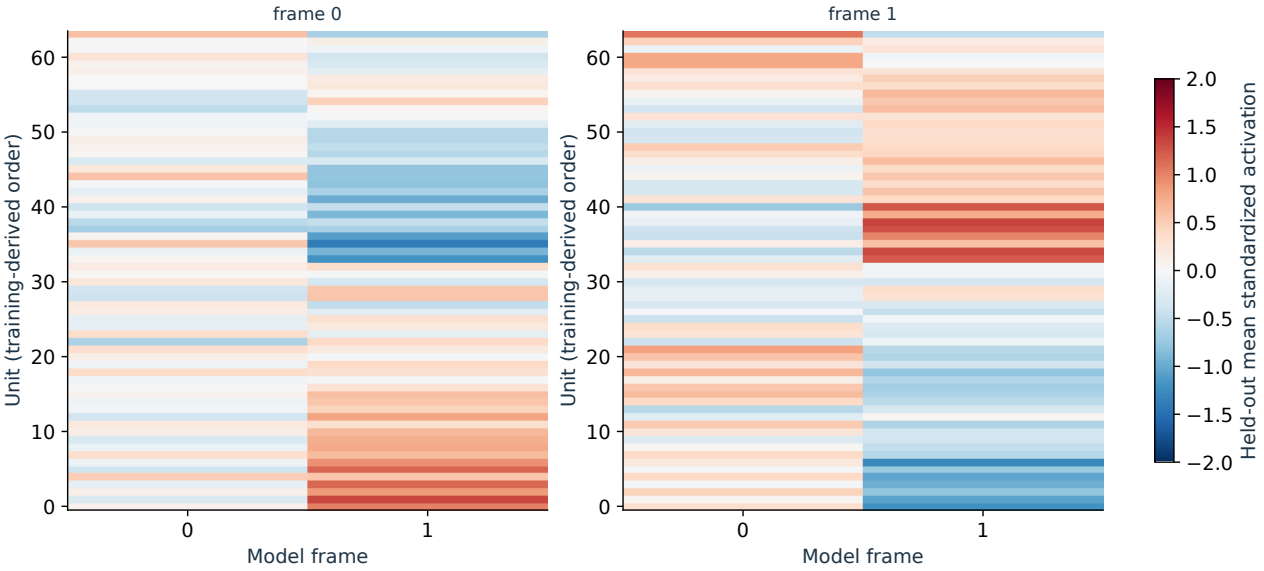


Figure 14: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	57.0%	57.0%	unavailable
fusion	71.1%	71.1%	unavailable
spatial	78.1%	78.1%	unavailable
cls	100.0%	100.0%	unavailable
attention	59.4%	59.4%	unavailable

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

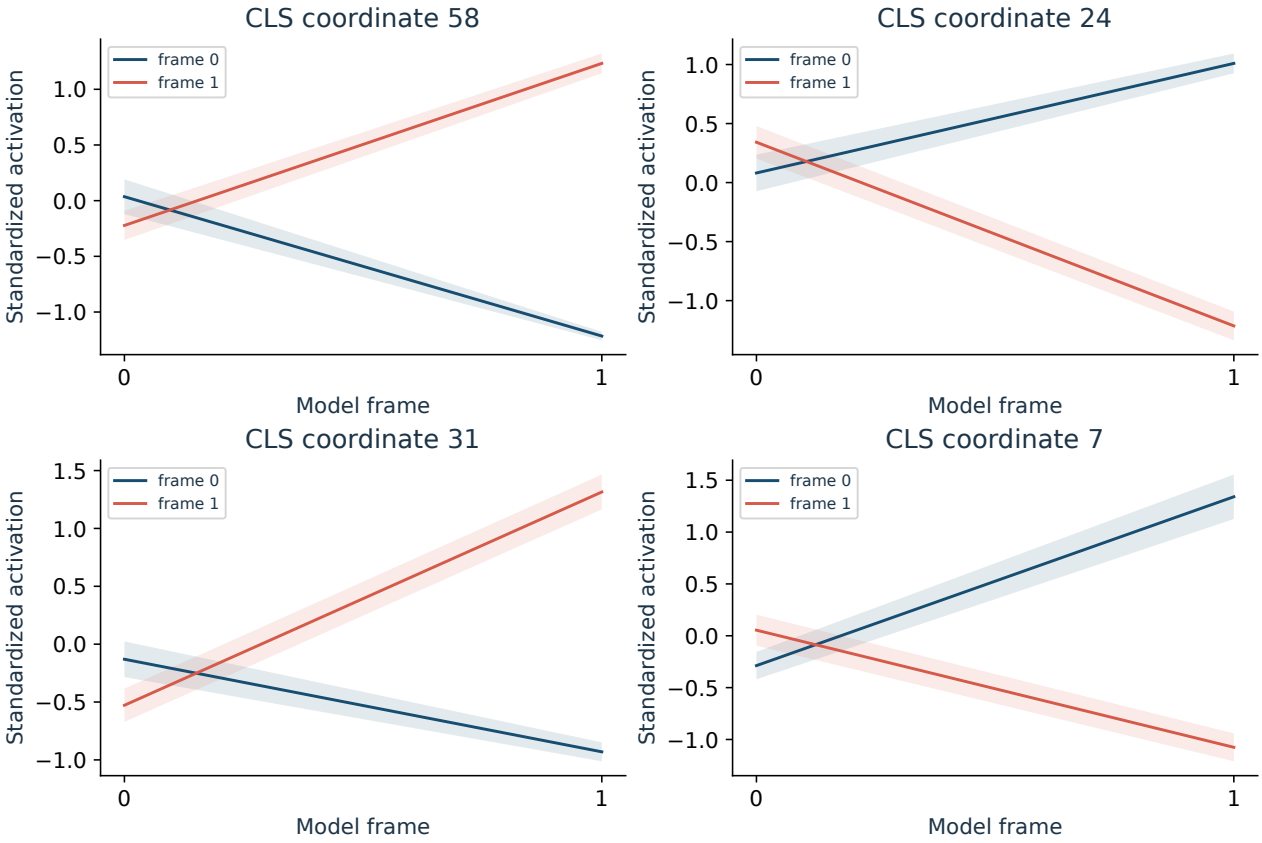


Figure 15: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

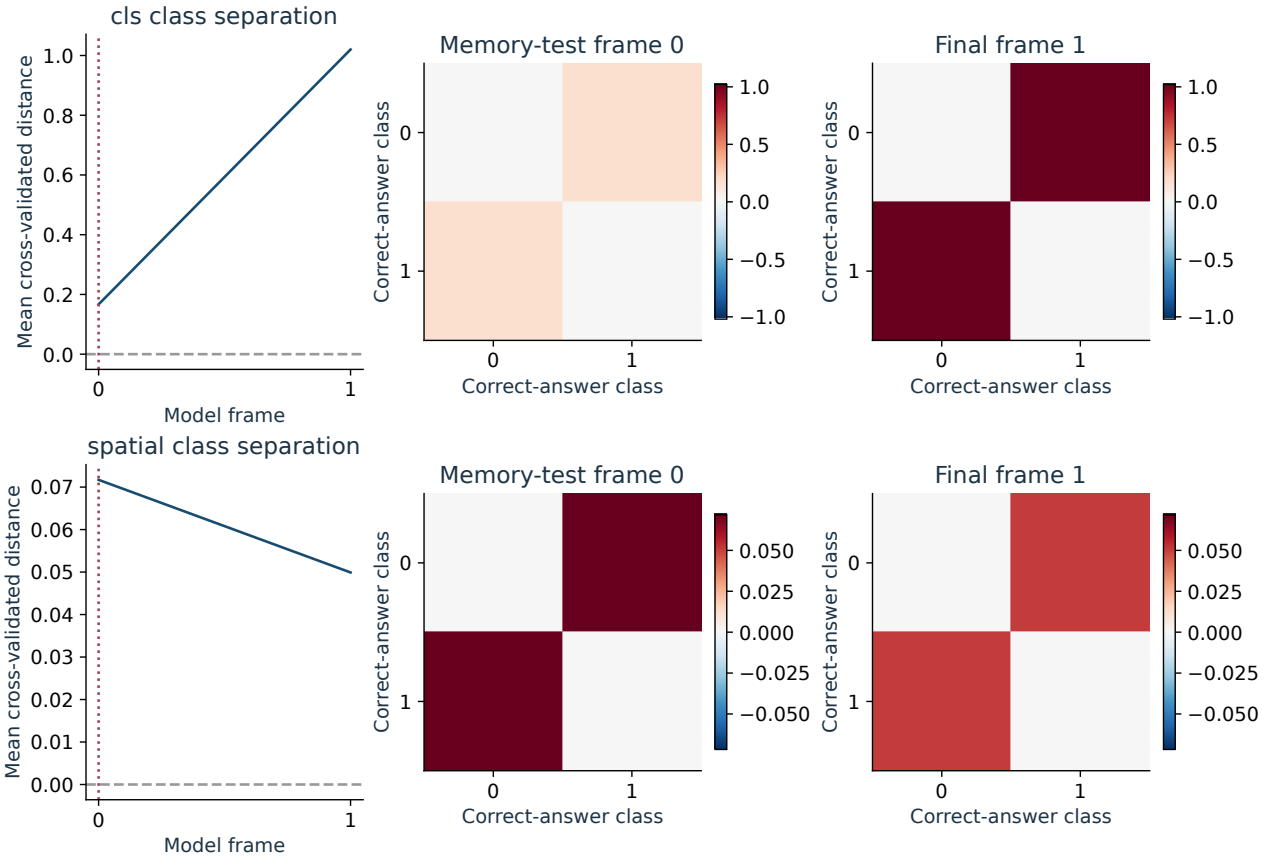


Figure 16: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

Spatial-reference memory is a stronger causal bottleneck than the global reference-period state in this experiment, although the final answer is most explicit in global memory. This illustrates a distinction between a representation used to preserve comparison information and the later population from which the decision is read. The natural-image pairs preserve a common mean and RMS contrast while altering spectral detail, so the measured discrimination concerns their structured image content. The example-specific attention overlays identify which image regions receive weight on individual trials, while the separately averaged maps reveal any repeatable spatial preference across photographs.

The behavioral result establishes 100.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies spatial_gain0_t0 as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Testing matched amplitude spectra with altered phase structure would separate sensitivity to spectral amplitude from sensitivity to recognizable content. Spatially localized filtering could reveal whether the selected region controls the judgement.

4.1 Interpreting these results alongside biological experiments

4.1.1 Tolhurst and Tadmor (1997)

Our pairs hold overall mean and RMS contrast fixed while changing spectral structure. A local frequency-band contrast can still change under those global controls. The human experiment makes this distinction useful: compare the agent with band-limited contrast predictors before attributing its response to an explicit estimate of a global spectral exponent. [4]

4.1.2 Tolhurst and Tadmor (2000)

The natural-statistics question asks where sensitivity is greatest along a family of reference spectra. Our present success shows that the trained distribution is discriminable. To test specialization for natural images, sweep the reference spectrum and ask whether equal perturbations become easier near natural statistics after controlling local and global contrast. [5]

4.1.3 Olshausen and Field (1996)

Sparse-coding theory concerns the basis used to represent image content. A sharply concentrated attention map instead describes which spatial sites contribute to processing. These can vary independently. Testing the theory in this model would involve receptive-field-like feature measurements,

response sparsity across channels and held-out natural-image reconstruction, in addition to the existing spatial allocation maps. [6]

4.2 A computational account of the observed behavior

A change in spectral slope redistributes image energy across scales. Even with equal total RMS contrast, finer details can weaken while coarser structure strengthens. Learned visual features can detect this redistribution locally, and spatial memory can retain a useful comparison pattern across the two observations. The large spatial-reference reset cost, together with the explicit final global answer, fits a division between retaining image information and constructing a response. Efficient-coding theory predicts that representation should reflect the statistics of experienced images. The present multi-task training objective rewards discrimination and pixel prediction; any efficient representation would emerge under those objectives and their particular image distribution.

4.3 Where this task agrees with the biological evidence

The agreement is sensitivity to structured scale information after global brightness and contrast controls. The scientific distinction is between discriminating a spectral manipulation and showing an advantage specifically tuned to natural statistics. Trial-specific maps identify selected image regions, while averages across unrelated photographs describe spatial preference rather than a common object.

4.4 A follow-up experiment motivated by these results

Compare natural photographs, phase-randomized versions and synthetic spectra at matched power. Fit local band-contrast predictors and test spectral-reference generalization. An advantage for natural structure that survives these controls would sharpen the efficient-coding interpretation.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Field DJ (1987). Relations between the statistics of natural images and the response properties of cortical cells. JOSA A 4:2379-2394. https://pubmed.ncbi.nlm.nih.gov/3430225/.

[2] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[3] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[4] Tolhurst and Tadmor (1997). Discrimination of changes in the slopes of the amplitude spectra of natural images: band-limited contrast and psychometric functions. Primary source.

[5] Tolhurst and Tadmor (2000). Discrimination of spectrally blended natural images: optimisation of the human visual system for encoding natural images. Primary source.

[6] Olshausen and Field (1996). Emergence of simple-cell receptive field properties by learning a sparse code for natural images. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
mixed	128	100.0%	1.000	0.004

B Every-frame attention atlas

Condition mixed. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 64 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

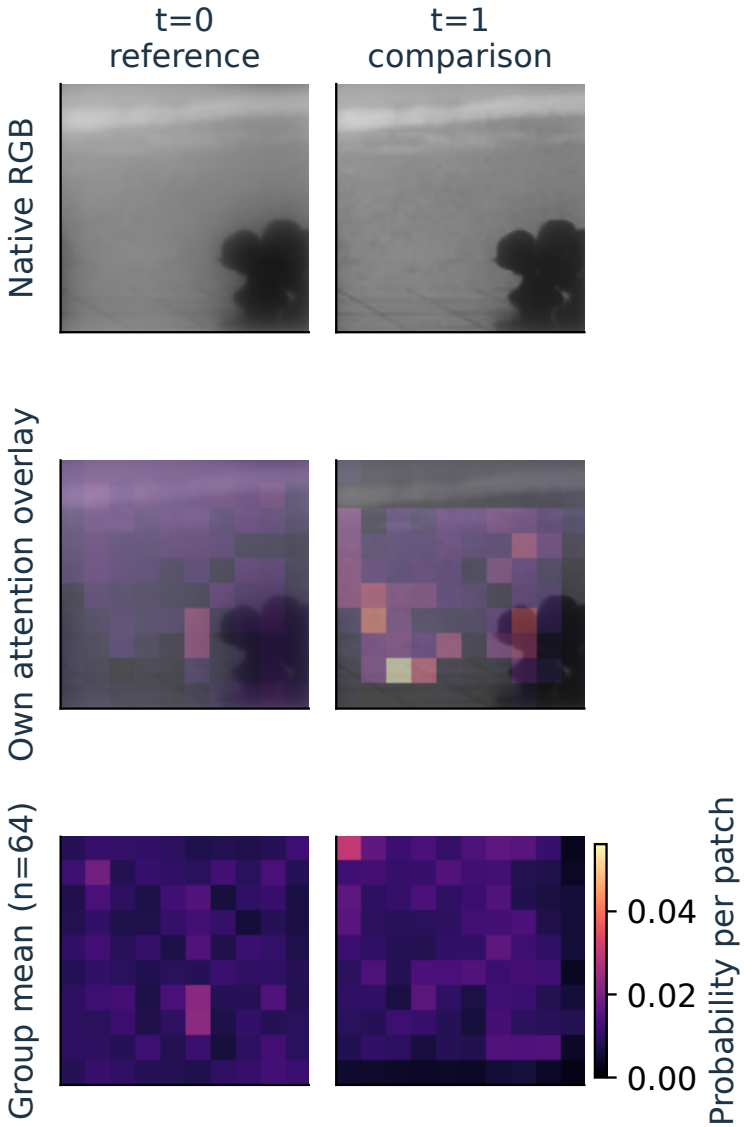


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

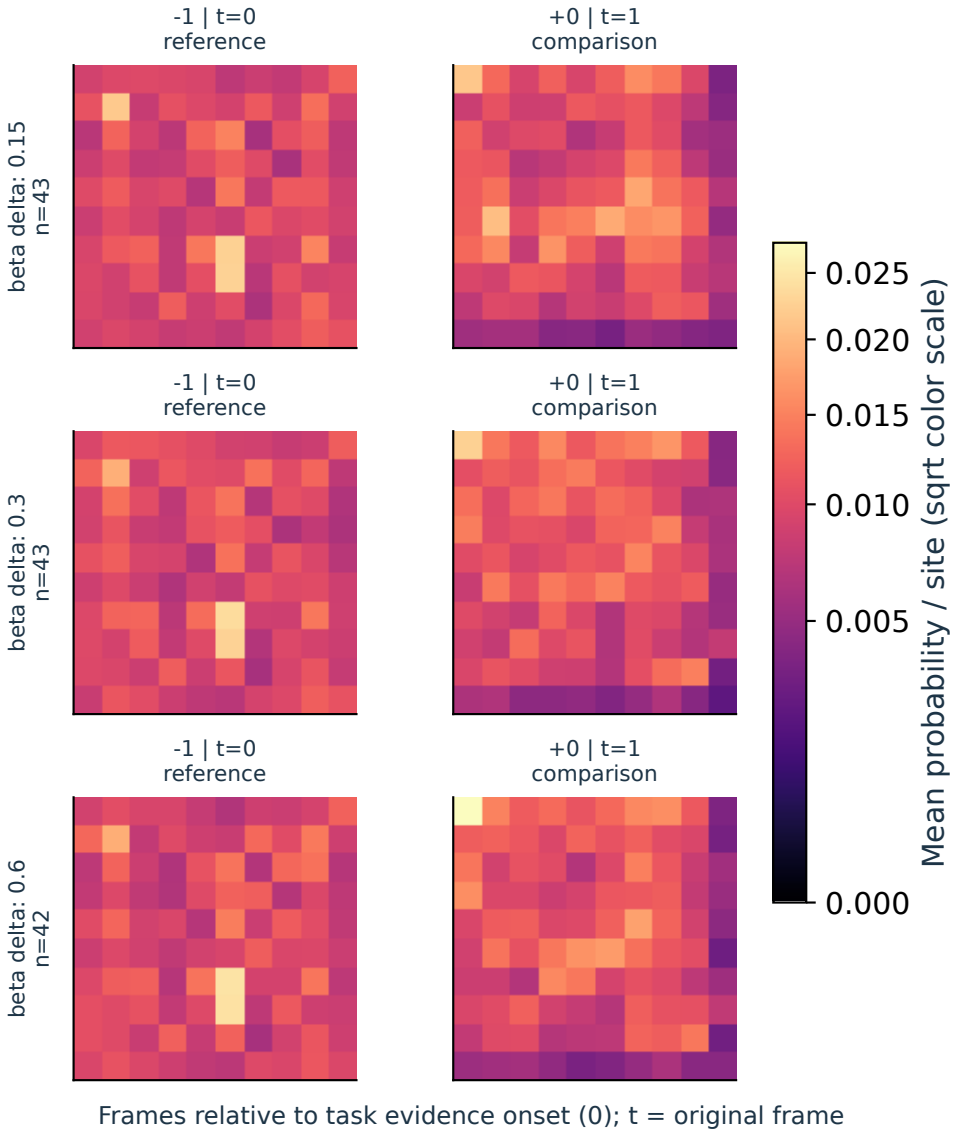


Figure 18: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Location-cued orientation comparison with a persistent ring cue

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized orientation ring in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 1 native conditions and 128 fresh trials, equal-condition accuracy was 100.0% (95% interval [97.1%, 100.0%]). For the representative condition D0, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was distractor_+8_evidence, changing accuracy by -44.5 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The observer reports the sign of rotation at a ring-cued Gabor location. Distractor locations also change, so a successful comparison must preserve which location is relevant. The task links a spatial instruction to a feature-specific report.

Rotation magnitude controls sensory difficulty. We decode the cued location independently of the response label, since remembering where to inspect and determining how the target rotated are distinct computations. The attention atlas follows the cue, sample and comparison in their native order.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Funahashi and colleagues described mnemonic spatial coding in monkey prefrontal cortex. A location instruction can therefore be analyzed as information that persists beyond its first presentation and constrains subsequent processing. In this task, cue and rotation variables can be decoded separately. Their temporal ordering matters: a location code during the sample establishes access to the instruction, whereas an answer code after the probe reflects the instructed comparison. The intervention epochs exploit this distinction to test whether priority is most consequential during encoding or during final selection.

1.2 Biological experiments motivating this task

Experimental question. How does a spatial instruction select the orientation comparison that should control the response?

McAdams and Maunsell (1999) During an orientation attention task, V4 responses increased with attention largely through scaling of tuning curves, without a systematic change in tuning width. V1 modulation was smaller. [5]

Griffin and Nobre (2003) Cues presented before or after a visual array improved recognition of relevant locations. Behavioral and electrophysiological results linked orienting in perception and memory while revealing differences between precue and retrocue processing. [6]

Reynolds, Chelazzi and Desimone (1999) For two stimuli within a receptive field, attention shifted the response toward that evoked by the attended item alone. The findings supported biased competition between simultaneously represented stimuli. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

What is the sign of rotation at the ring-cued Gabor location?

Response labels are 0: negative axial angle change; 1: positive axial angle change. The native suite contains 1 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use D0.

1.4 Stimulus sequence and spatial layout

For the representative native condition D0, the sequence has 4 frames, indexed from zero: frame 0: cue; frames 1–2: sample; frame 3: probe. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Rotation magnitude (degrees)	15, 30, 45

The shared recurrent map can carry location-specific evidence while the CLS state summarizes selected information. Spatial priority manipulations at the target and a designated distractor test competition between those locations. Resetting memory immediately before the probe assesses whether pre-probe processing contributes to the comparison.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 100.0%, with macro AUC 1.000. The matched final checkpoint changed accuracy by +0.00 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was D0 at 100.0%; the highest was D0 at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (2) produced 64.1% correct responses, compared with 100.0% at the largest strength (45). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

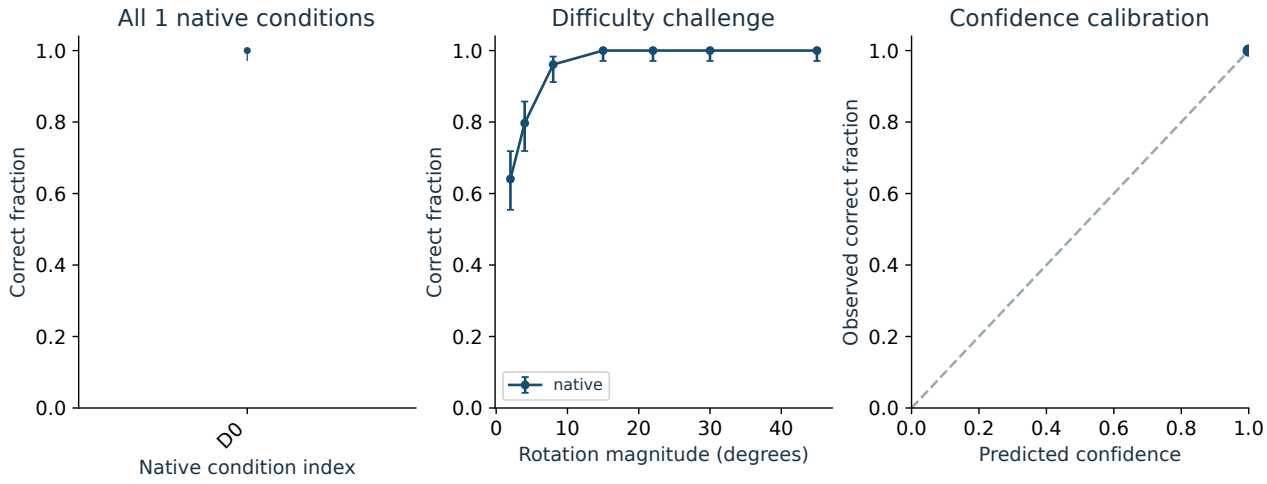


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)[1 - \exp(-(x/a)^b)]$. Here x is signal strength, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 3.352 task units (bootstrap interval [2.397, 4.376]), with slope 0.07854 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.005.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	100.0%	100.0%
0.7	100.0%	1.0	100.0%	100.0%
0.8	100.0%	1.0	100.0%	100.0%
0.9	100.0%	1.0	100.0%	100.0%

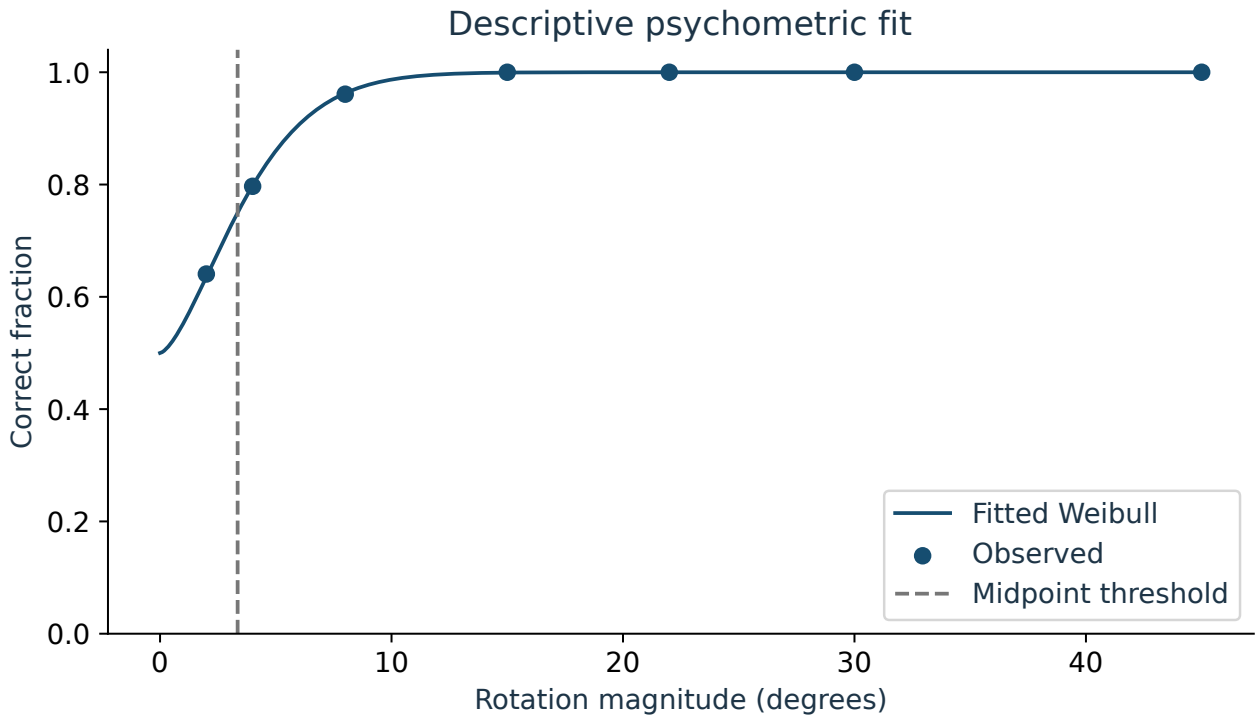


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

The representative condition D0 provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 128 of 128 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

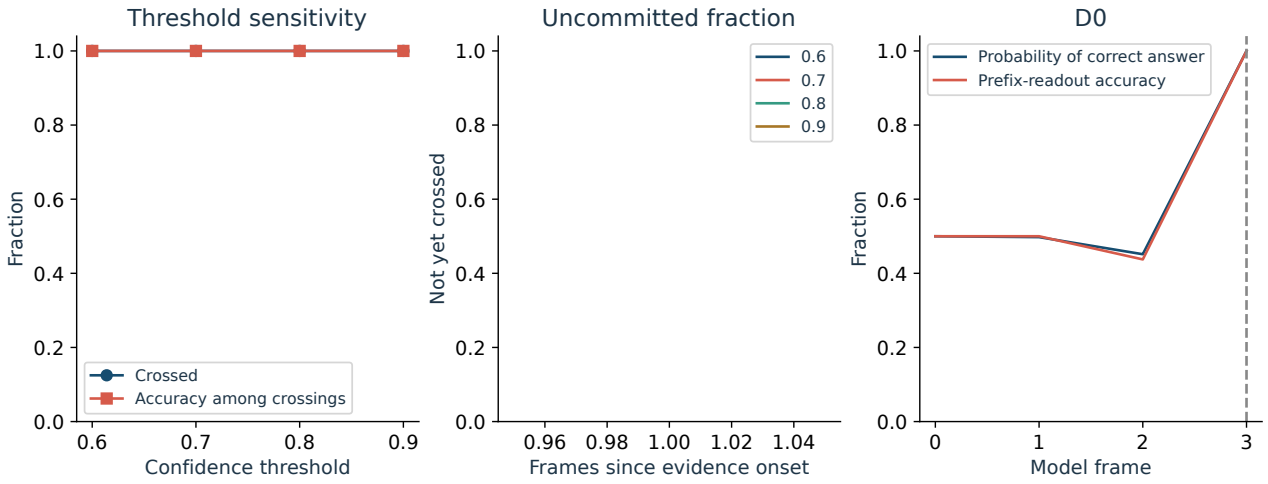


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

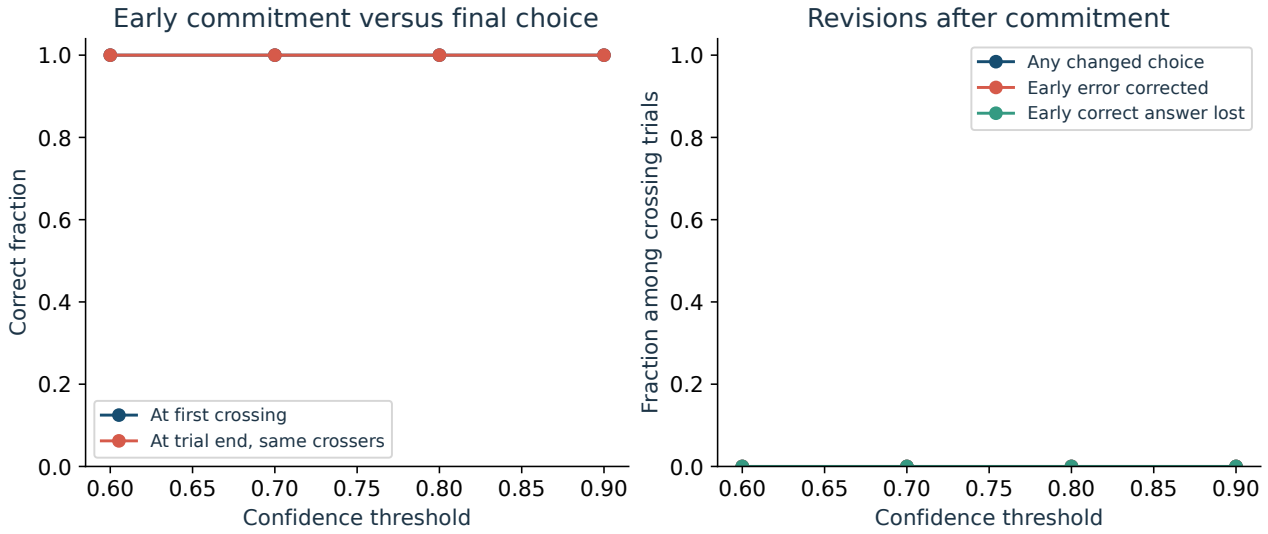


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

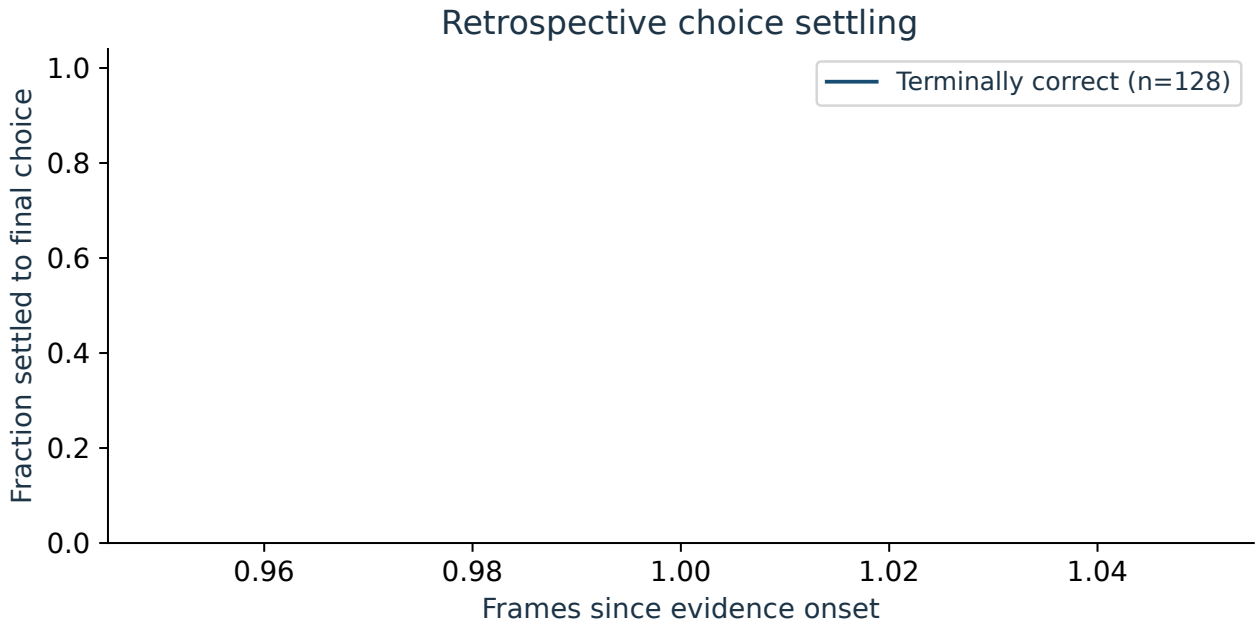


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The target is fixed at top left. Rows compare clockwise and counterclockwise changes in the zero-delay native condition.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Clockwise	16	16
Counterclockwise	16	16

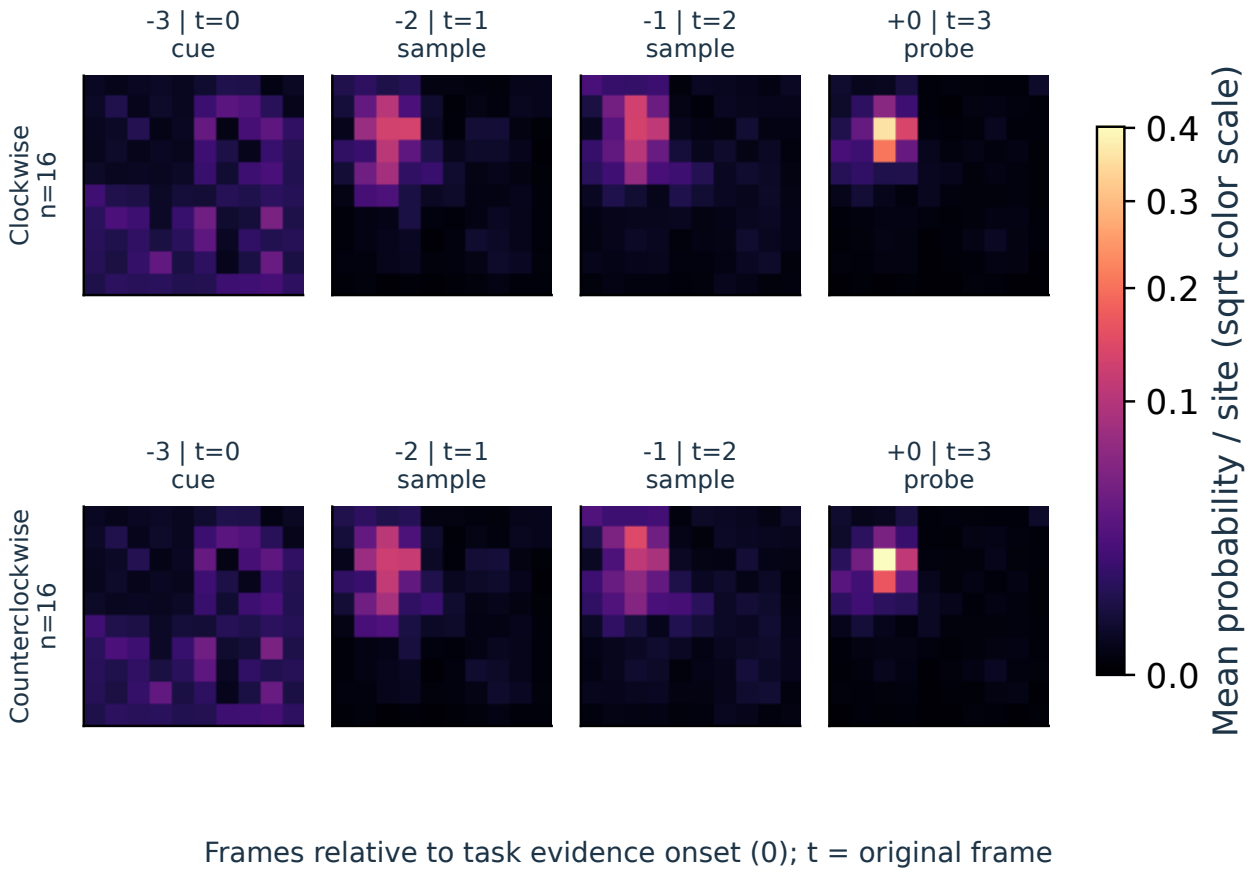


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

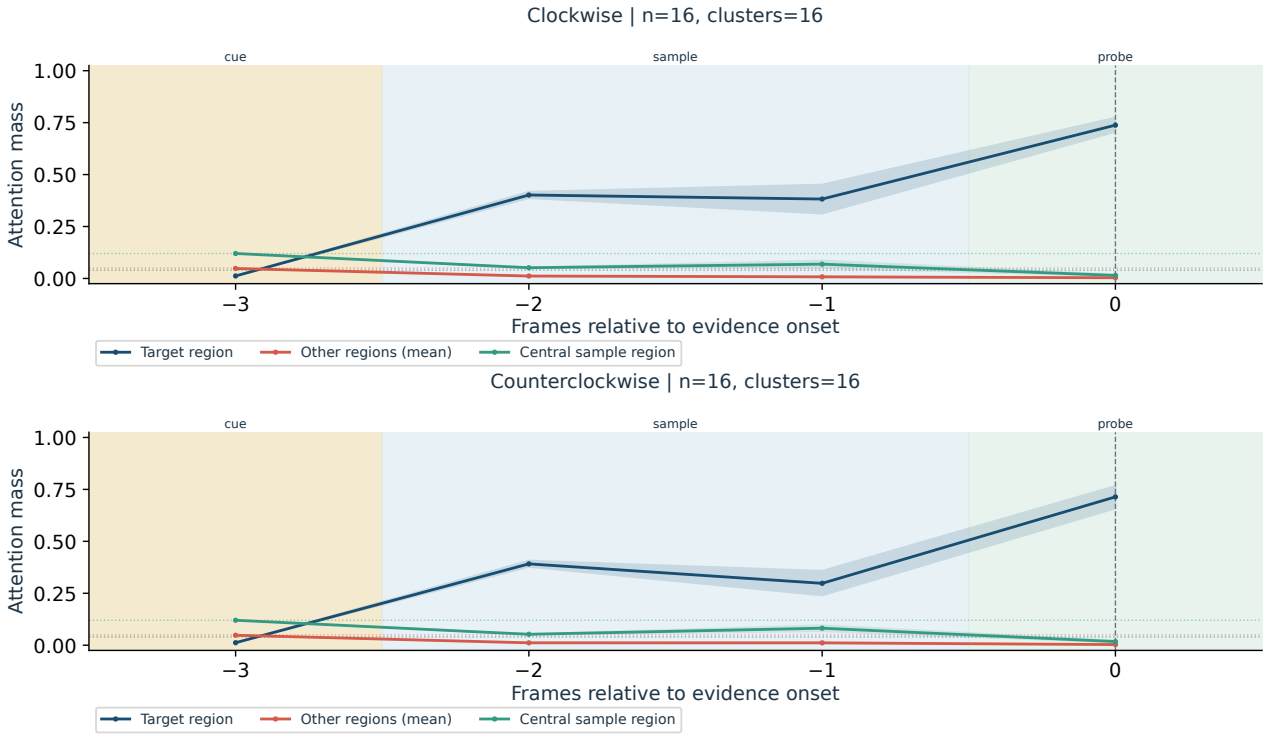


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

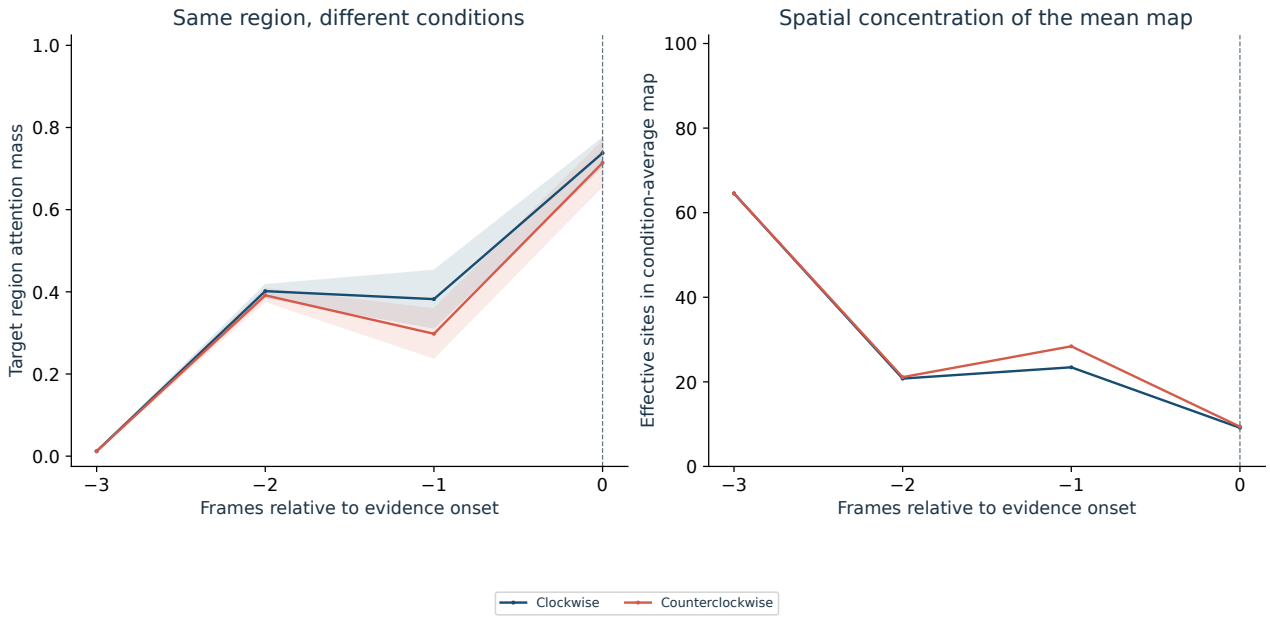


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Clockwise: target region mass peaks at 0.738 at original frame 3 (probe), and is 0.738 on the final frame. Counterclockwise: target region mass peaks at 0.714 at original frame 3 (probe), and is 0.714 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For distractor_+8_evidence, paired accuracy changed by -44.5 percentage points (paired 95% interval [-53.1%, -35.9%]; paired p=1.388e-17; within-task Holm p=4.302e-16). There were 0 corrected trials and 57 newly incorrect trials. For target_+2_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

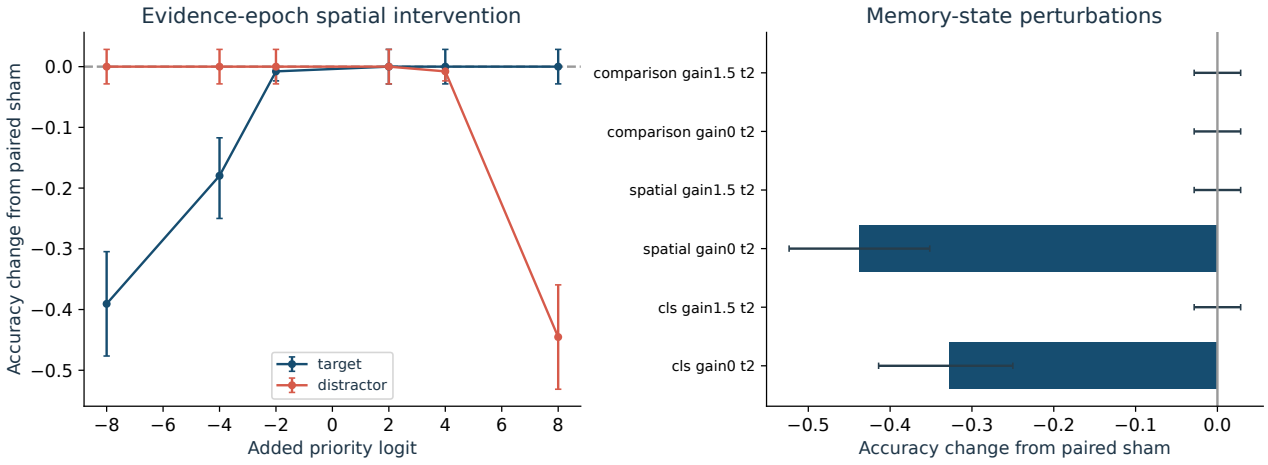


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
D0	target_-4	-18.0	1.955e-05
D0	target_+4	+0.0	1
D0	distractor_-4	+0.0	1
D0	distractor_+4	-0.8	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.450	49.2%	0.550
fusion	0.457	44.5%	0.900
spatial	0.394	42.2%	0.960
cls	1.000	100.0%	0.005
comparison	Constant zero on this condition		
attention	0.572	54.7%	0.145

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

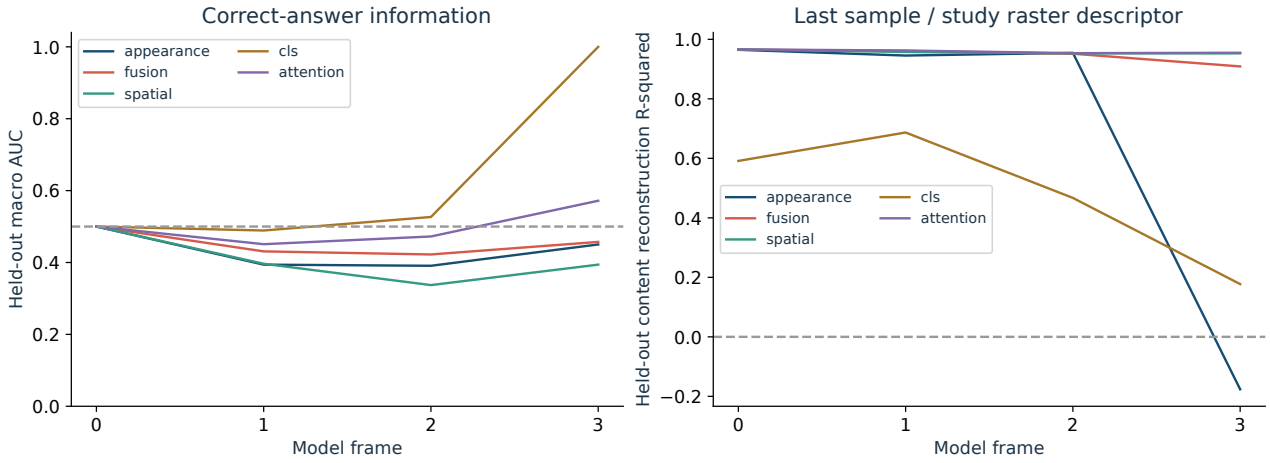


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 1.51 to 16.97 within the retained subspace. The final step displacement was 1.577 standardized RMS units. In spatial memory, the participation ratio ranged from 3.00 to 15.13 within the retained subspace. The final step displacement was 1.066 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 2, reconstruction of the sample orientation at the instructed location, represented by the cosine and sine of twice its angle had held-out R-squared 0.986 in CLS memory and 0.950 in spatial memory. This continuous target describes stimulus content, while the final-answer decoder tests the response required after the subsequent evidence.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -32.8 percentage points, spatial, -43.8 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.396	0.480	3200
fusion	0.476	0.479	6400
spatial	0.986	0.410	6400
cls	1.000	0.490	64
comparison	unavailable	unavailable	–
attention	0.565	0.469	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 20 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

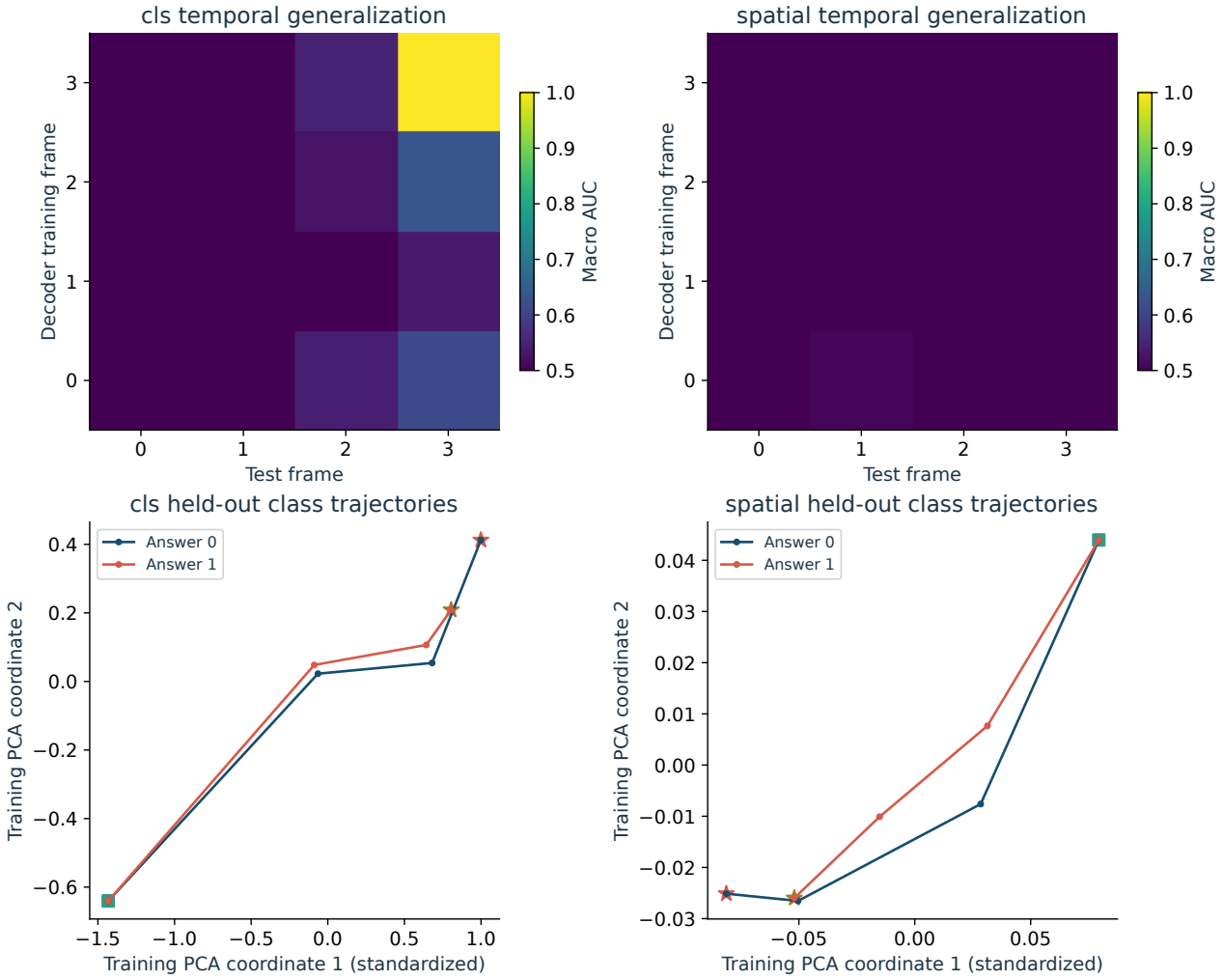


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

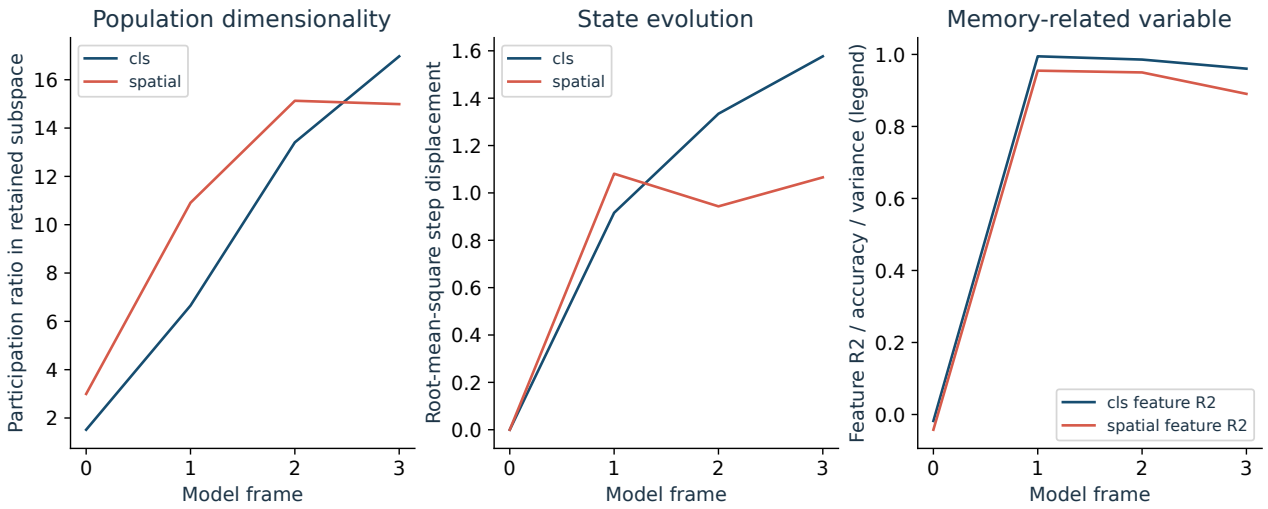


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

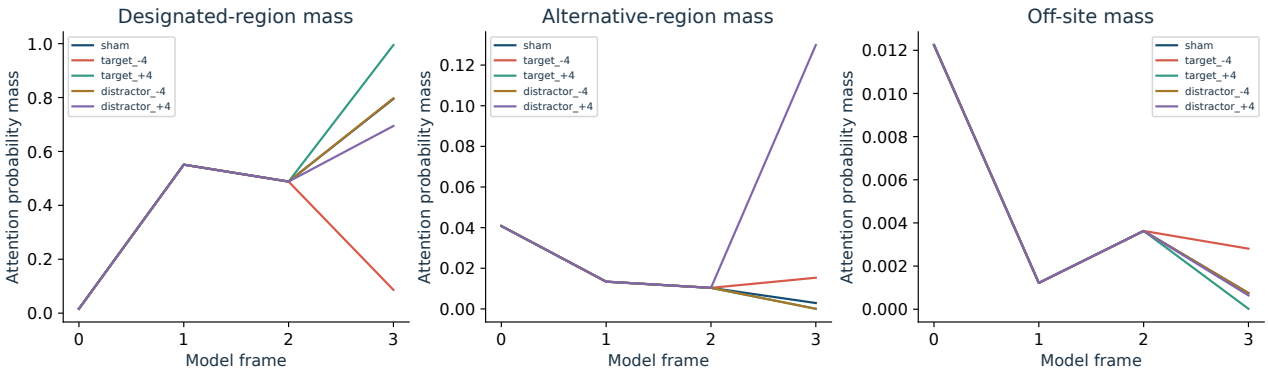


Figure 13: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

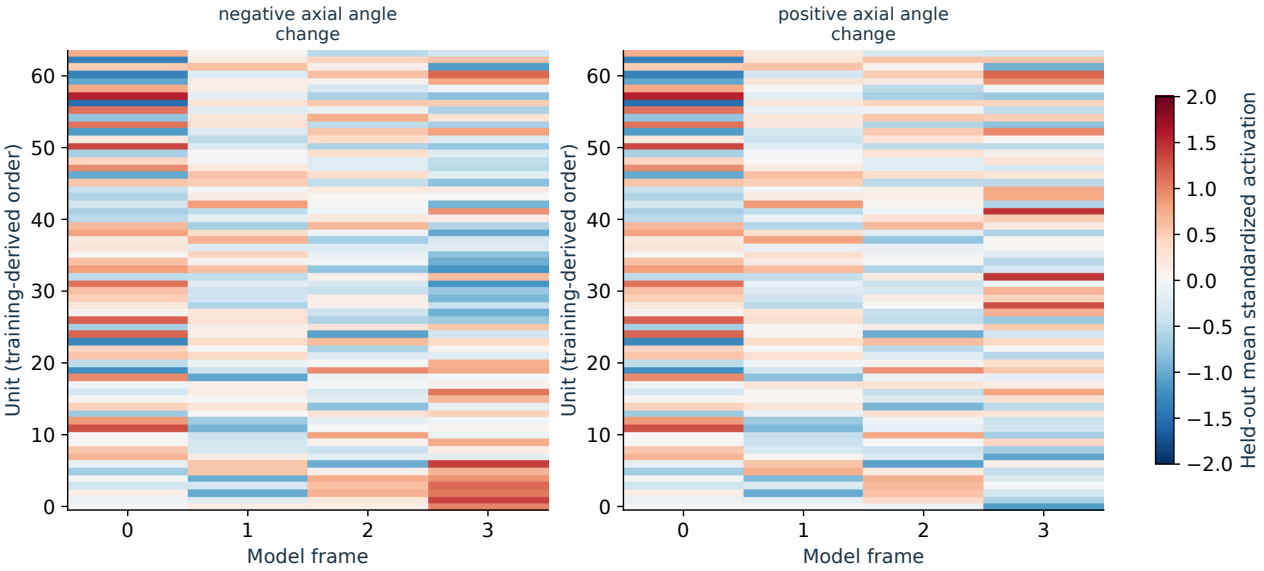


Figure 14: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	49.2%	49.2%	22.7%
fusion	44.5%	44.5%	100.0%
spatial	42.2%	42.2%	100.0%
cls	100.0%	100.0%	57.0%
attention	54.7%	54.7%	100.0%

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\text{fit}} d_j^{\text{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

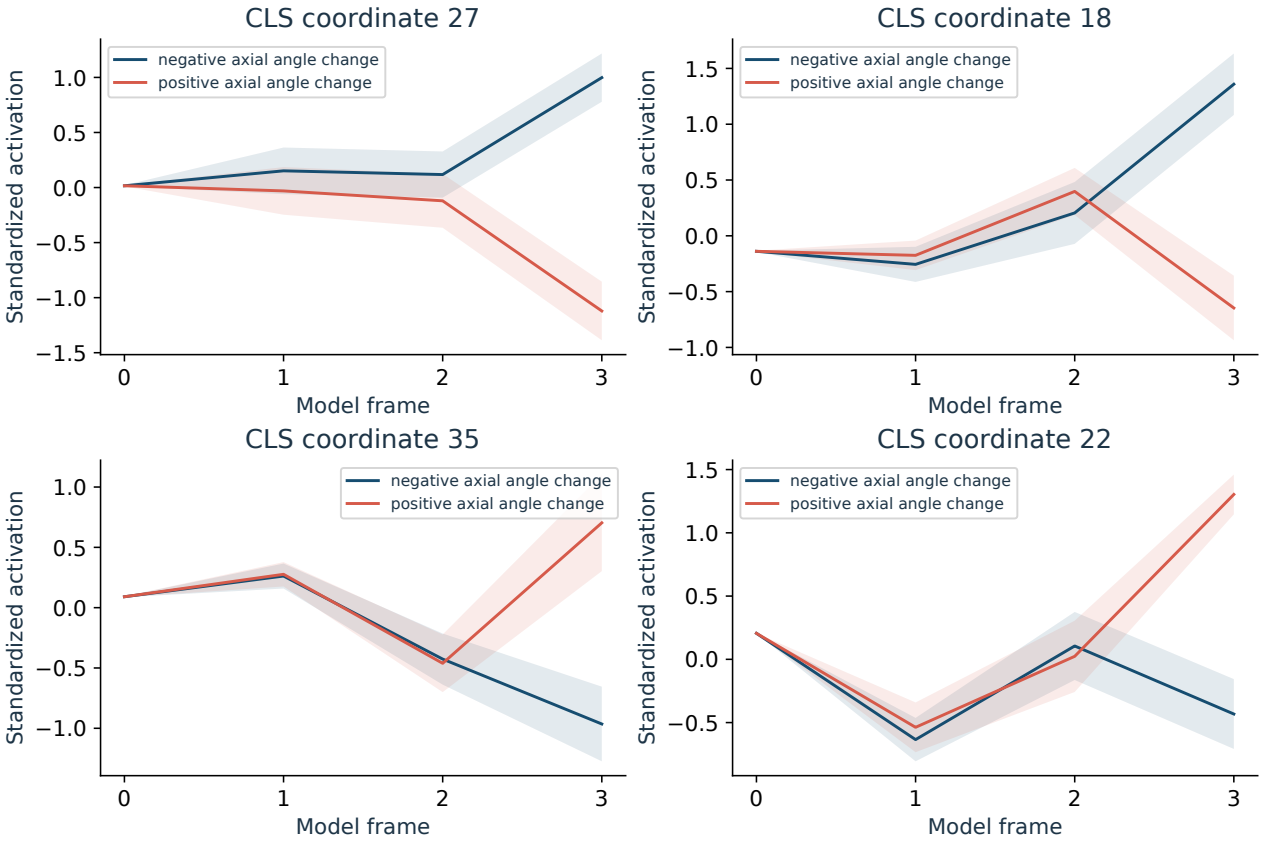


Figure 15: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

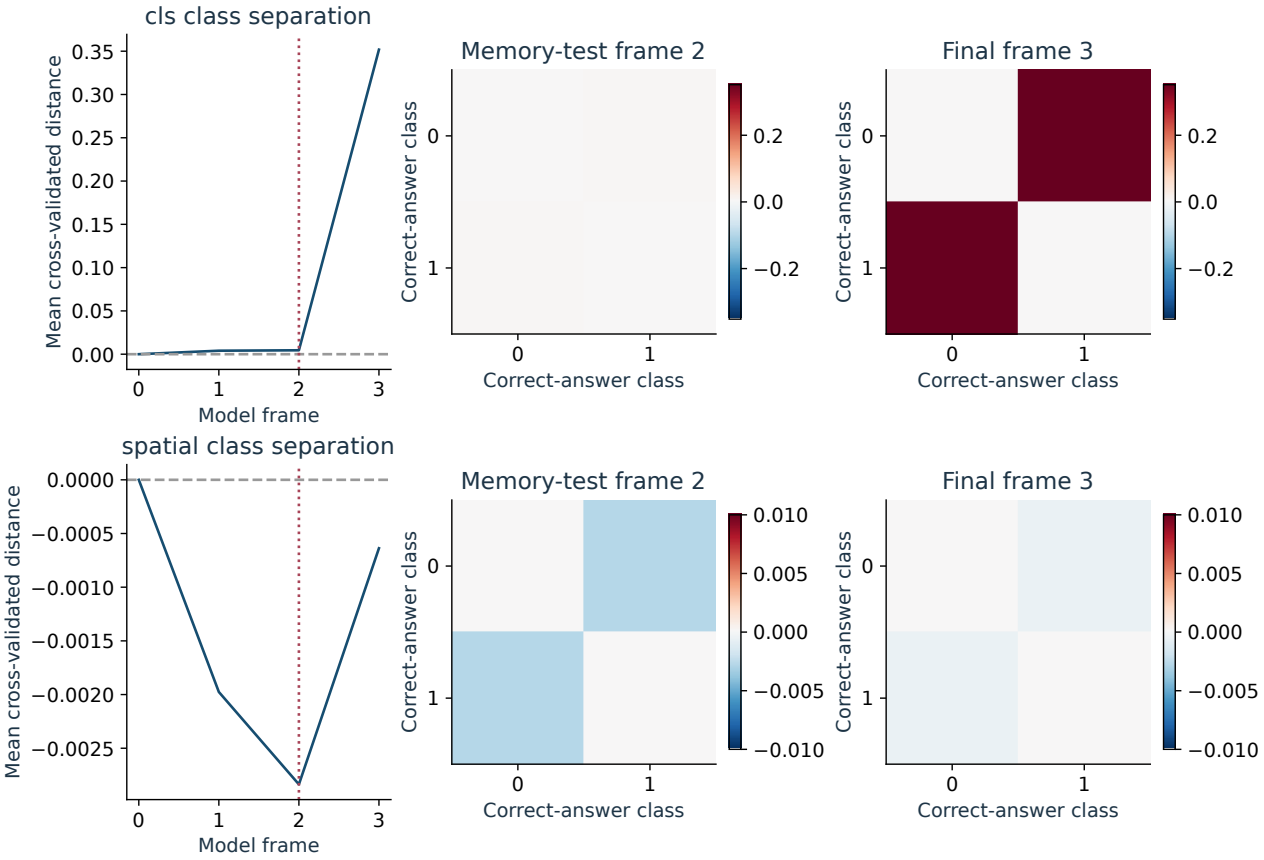


Figure 16: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

The visible ring supplies a location instruction, and target inhibition at the comparison stage impairs the sign judgement. Resetting either recurrent state immediately before the probe also causes a substantial loss. Final answer information is especially explicit in global memory, while a fixed-capacity spatial probe reads the answer poorly. These findings are compatible with spatial memory retaining task-relevant content that is transformed into an explicit response only when it is selected and compared. The full-state probe control is important here because a small projected subspace need not preserve every feature that the original nonlinear readout can use.

The behavioral result establishes 100.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies distractor_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Removing the ring before the sample and adding longer retention intervals would separate cue visibility from remembered priority. Independent target and distractor rotations would support a full distractor-congruence analysis.

4.1 Interpreting these results alongside biological experiments

4.1.1 McAdams and Maunsell (1999)

The model's ring identifies which location matters, and target inhibition during comparison harms performance. The V4 experiment offers a sensory-gain comparison: measure the response to the same oriented item when its location is selected or ignored. Our attention mass is the routing coefficient; a physiological comparison additionally requires the orientation-dependent value or state response after that routing. [5]

4.1.2 Griffin and Nobre (2003)

A visible location cue can improve processing of current input, while a later cue can select information already stored. The ring condition should be interpreted according to its actual timing in the trial atlas. Comparing ring-before-sample and ring-after-sample versions with identical arrays would separate encoding selection from retrieval selection rather than treating every cue effect as one process. [6]

4.1.3 Reynolds, Chelazzi and Desimone (1999)

Several represented locations create competition over which comparison contributes to the report. The strong target intervention and recurrent reset costs support that joint requirement. A decisive analogue of the biological experiment would compare the attended item alone with the same item among

competitors, asking whether attention drives the model's selected representation toward the item-alone state. [7]

4.2 A computational account of the observed behavior

A location instruction solves a correspondence problem: the first and second orientation must be compared at the same relevant position. Spatial memory can preserve separate local contents, while the global state constructs a categorical response after selecting one position. This explains why the final answer can be easy to decode globally yet difficult to read from a compressed spatial probe. The spatial state may store several orientations rather than the answer itself, and its useful coordinates may be lost by a fixed low-dimensional projection. The causal reset and the full-state control therefore complement the ordinary answer decoder. The mechanism is best described as content retention followed by selective comparison.

4.3 Where this task agrees with the biological evidence

The model agrees with location-dependent selection and with a behavioral dependence on retained information. Its softmax competition supplies one explicit implementation. Biological gain, spatial memory and readout selection are distinct measurements, so agreement becomes stronger when the same attended-item contrast appears in all three.

4.4 A follow-up experiment motivated by these results

Move cue timing across sample, delay and probe while preserving the movie content. Compare identical target orientations with and without distractors. Predict that a late cue requires storage of multiple locations, whereas an early cue can reduce that storage requirement.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Funahashi S, Bruce CJ, Goldman-Rakic PS (1989). Mnemonic coding of visual space in the monkey dorsolateral prefrontal cortex. Journal of Neurophysiology 61:331-349. https://pubmed.ncbi.nlm.nih.gov/2918358/.

[2] Reynolds JH, Heeger DJ (2009). The normalization model of attention. Neuron 61:168-185. https://pubmed.ncbi.nlm.nih.gov/19186161/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] McAdams and Maunsell (1999). Effects of attention on orientation-tuning functions of single neurons in macaque cortical area V4. Primary source.

[6] Griffin and Nobre (2003). Orienting attention to locations in internal representations. Primary source.

[7] Reynolds, Chelazzi and Desimone (1999). Competitive mechanisms subserve attention in macaque areas V2 and V4. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
D0	128	100.0%	1.000	0.000

B Every-frame attention atlas

Condition D0. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 16 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

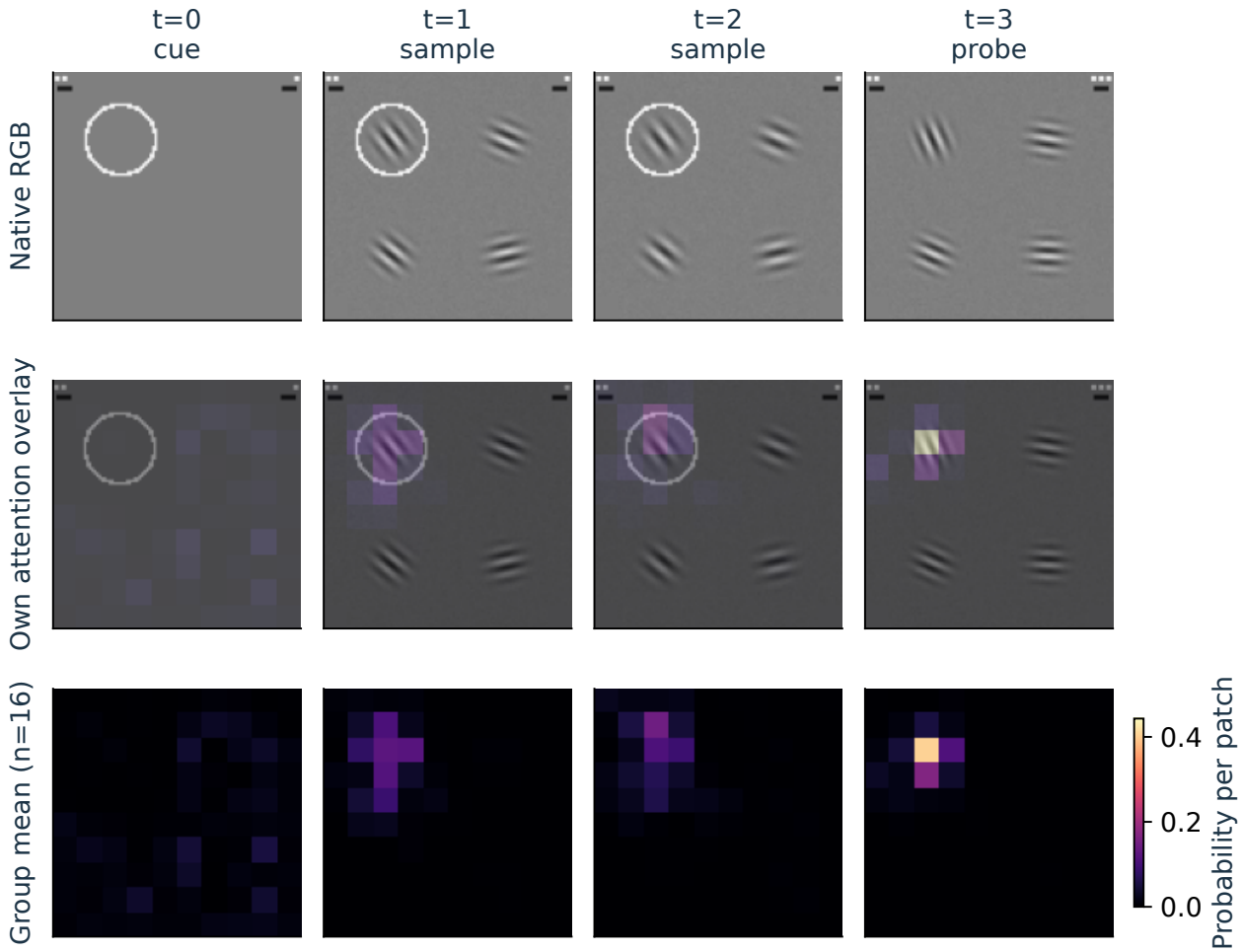


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

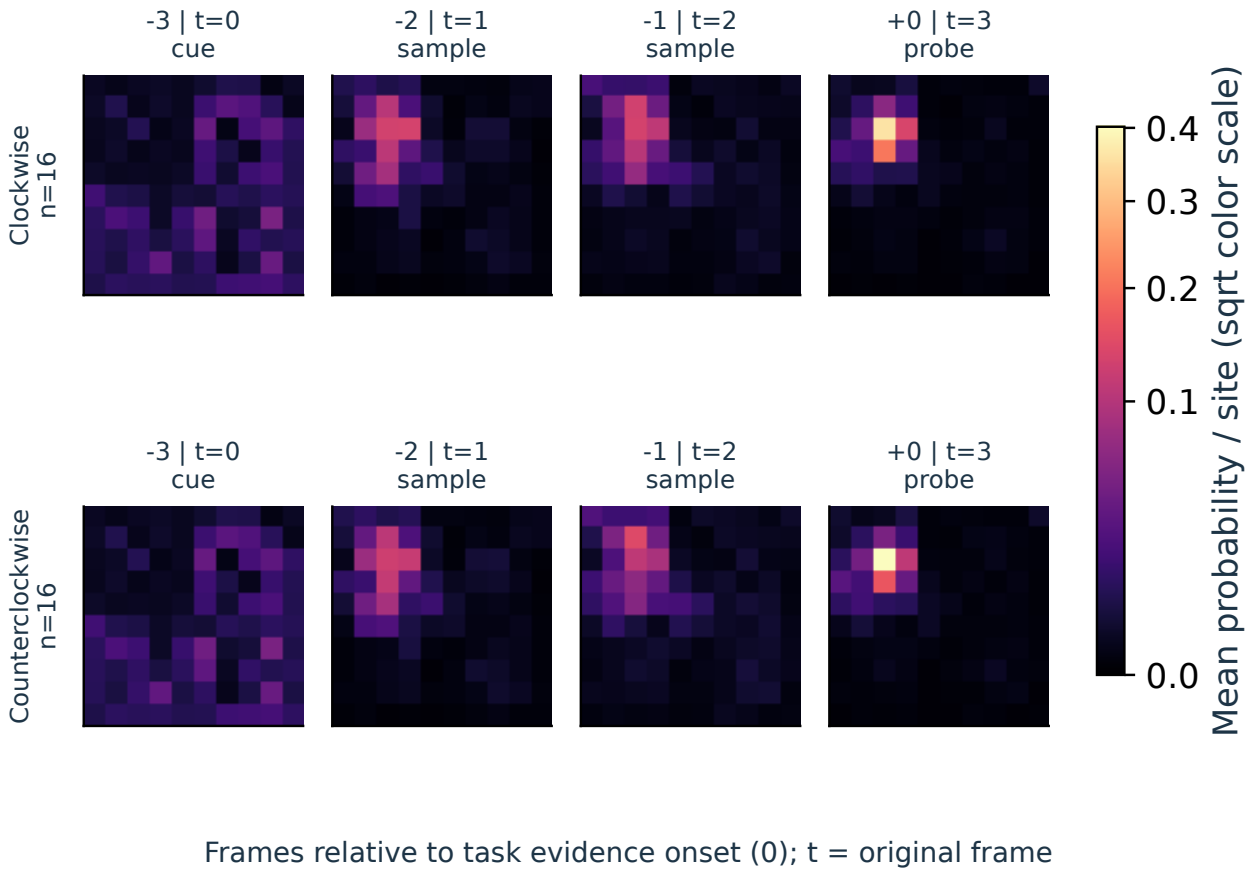


Figure 18: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Rule-dependent orientation judgement across a retention interval

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized orientation cued in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 4 native conditions and 512 fresh trials, equal-condition accuracy was 99.8% (95% interval [99.4%, 100.0%]). For the representative condition D24, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was target_ -8_evidence, changing accuracy by -56.2 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

This task combines a location instruction with a requested rotation sign. A positive report requires that the target rotate in the instructed direction. Target preservation, opposite rotation and distracting changes create competing sources of evidence, while blank intervals require the instruction and sample to persist.

We examine rotation magnitude, delay duration, cue-location decoding, remembered orientation and the final instructed response. The key mechanistic distinction is between preserving the visual sample and preserving the rule that determines how to interpret it. Time-generalization matrices ask whether readable information maintains a common population code across the delay.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Studies of dynamic prefrontal coding motivate tracking representations across the entire sequence rather than inspecting only a terminal state. The location and sign instruction can remain functionally available even while the pattern of recurrent activity changes. A decoder trained during one phase may therefore transfer well, poorly, or only after the relevant cue reappears. The time-generalization matrix tests this geometry explicitly. Resetting a state during the blank interval adds a causal test of maintenance: a subsequent performance loss identifies dependence on that state at a time when the task-relevant sample is absent.

1.2 Biological experiments motivating this task

Experimental question. How does a remembered orientation become a rule-dependent decision after a delay?

Harrison and Tong (2009) Human fMRI patterns in V1-V4 identified the remembered grating orientation during a blank delay, including cases with little sustained average activation. Information in a population pattern could persist despite low mean activity. [4]

Griffin and Nobre (2003) Cues presented before or after a visual array improved recognition of relevant locations. Behavioral and electrophysiological results linked orienting in perception and memory while revealing differences between precue and retrocue processing. [5]

Stokes and colleagues (2013) Population analyses examined how prefrontal coding changed while establishing, maintaining and using task context. The study motivates measuring representational transformations through time rather than only mean persistent activation. [6]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Did the target Gabor rotate in the direction requested by its location/sign glyph?

Response labels are 0: unchanged or opposite to cue sign; 1: rotation in cue sign. The native suite contains 4 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use D24.

1.4 Stimulus sequence and spatial layout

For the representative native condition D24, the sequence has 28 frames, indexed from zero: frame 0: cue; frames 1–2: sample; frames 3–26: delay; frame 27: probe. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Rotation magnitude (degrees)	15, 30, 45

Stimulus centers in the 100 by 100 input are [[27.0, 27.0], [73.0, 27.0], [27.0, 73.0], [73.0, 73.0]]. Coordinate x increases rightward and y downward.

Attention manipulations are separated by encoding, delay and evidence epochs. Encoding interventions can alter what enters memory; delay interventions can change ongoing selection and memory writes; probe interventions can alter which comparison influences the decision. A reciprocal target cost and distractor benefit would support competitive spatial selection.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 99.8%, with macro AUC 1.000. The matched final checkpoint changed accuracy by +0.78 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was D24 at 99.2%; the highest was D0 at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (2) produced 50.0% correct responses, compared with 100.0% at the largest strength (45). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

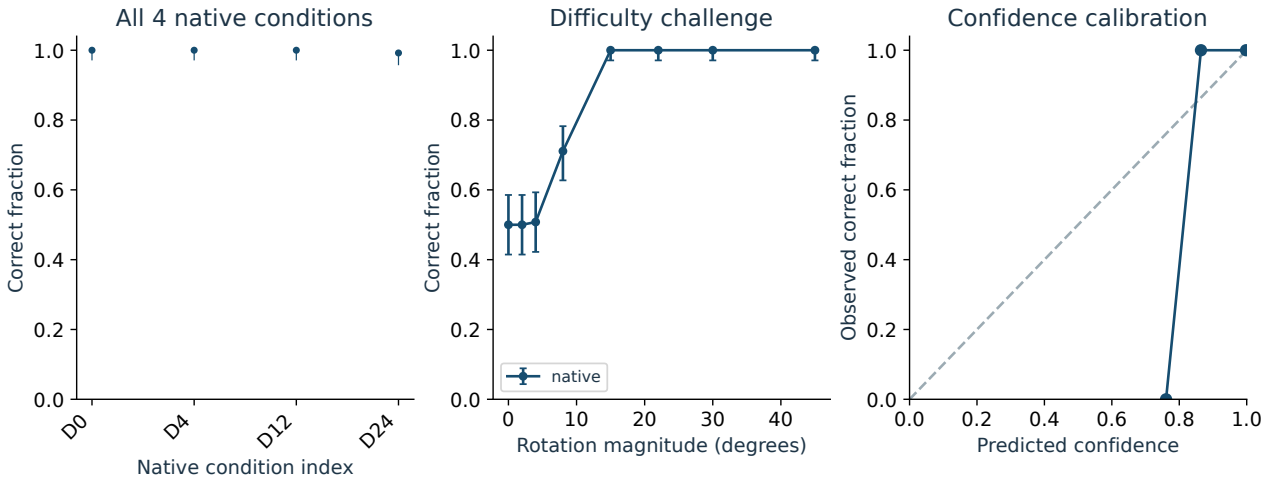


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Exploratory psychometric threshold fit

The native-label accuracy sweep is summarized by $P(x) = \gamma + (1 - \gamma - \lambda)[1 - \exp(-(x/a)^b)]$. Here x is signal strength, γ is reciprocal class count, $a > 0$ sets scale, $b > 0$ sets shape, and $\lambda \in [0, 0.2]$ is the fitted lapse fraction. Binomial likelihood is minimized over the empirical levels. The midpoint threshold $x_{50} = a(\log 2)^{1/b}$ lies halfway between the guessing floor and fitted upper asymptote. This is an exploratory descriptive addition after observing the empirical curves. Three hundred bootstrap resamples retain the same nuisance-trial index across signal levels. Event-conditioned detection reports remain empirical. The approach follows the fitting and uncertainty principles discussed by Wichmann and Hill (2001; https://pubmed.ncbi.nlm.nih.gov/11800458/, https://pubmed.ncbi.nlm.nih.gov/11800459/).

The fitted midpoint was 8.373 task units (bootstrap interval [7.511, 8.841]), with slope 0.1067 probability per task unit and lapse 0.0%. The largest absolute discrepancy from an observed point was 0.000.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	99.2%	100.0%
0.7	100.0%	1.0	99.2%	100.0%
0.8	99.2%	1.0	100.0%	100.0%
0.9	99.2%	1.0	100.0%	100.0%

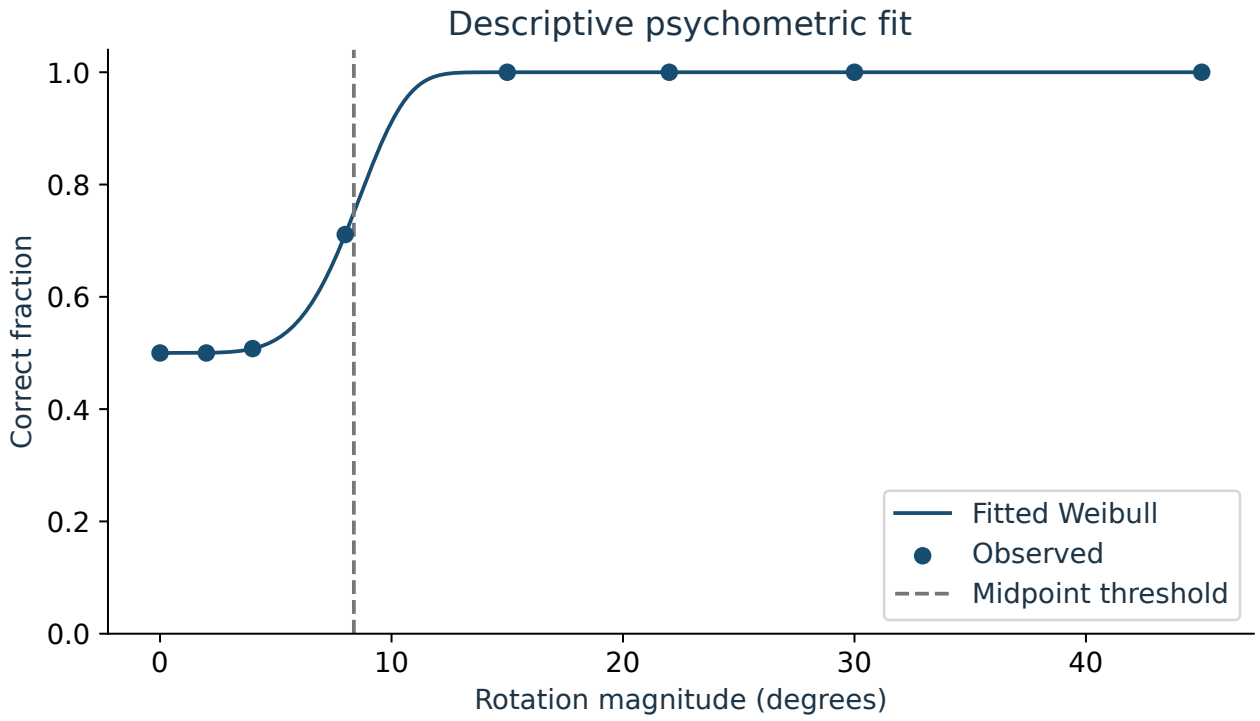


Figure 2: Empirical points and descriptive Weibull function. The vertical line marks the midpoint between guessing and the fitted upper asymptote. Threshold units match the stimulus-strength axis.

The representative condition D24 provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 511 of 512 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

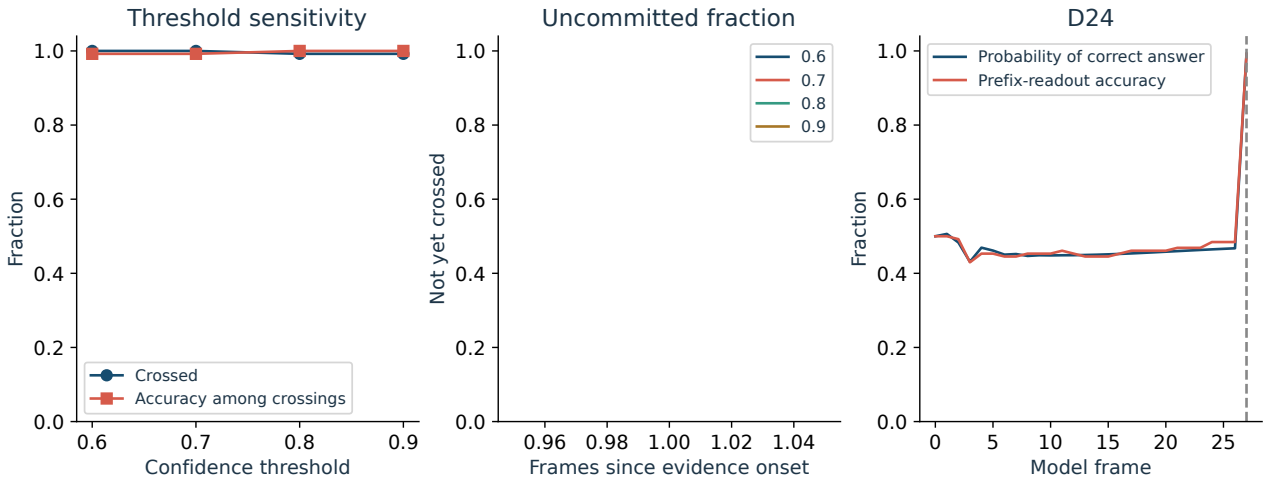


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

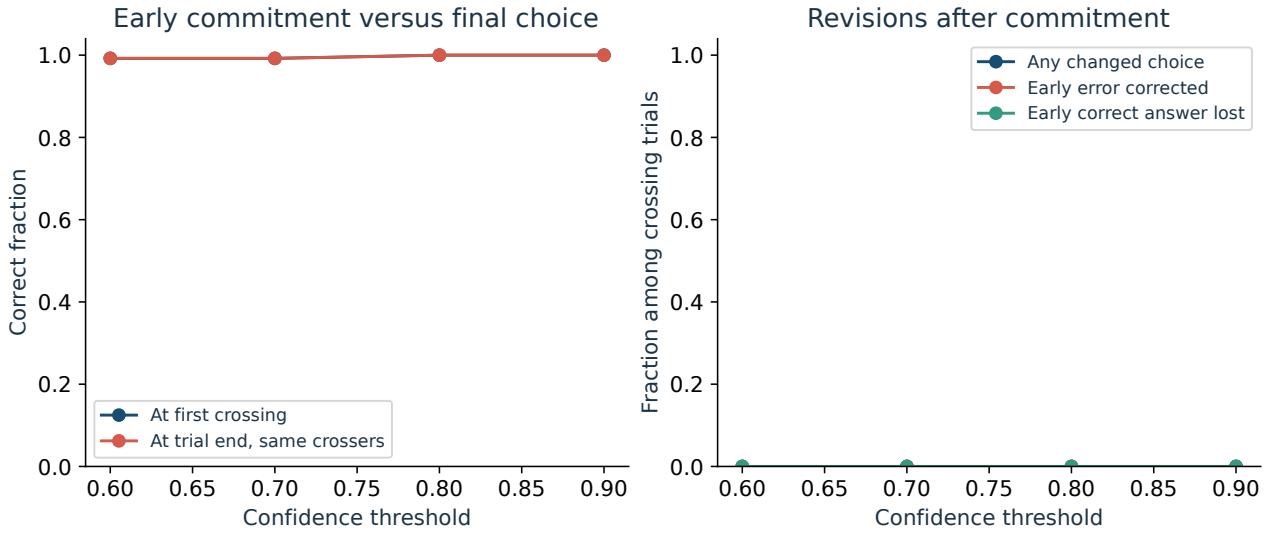


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

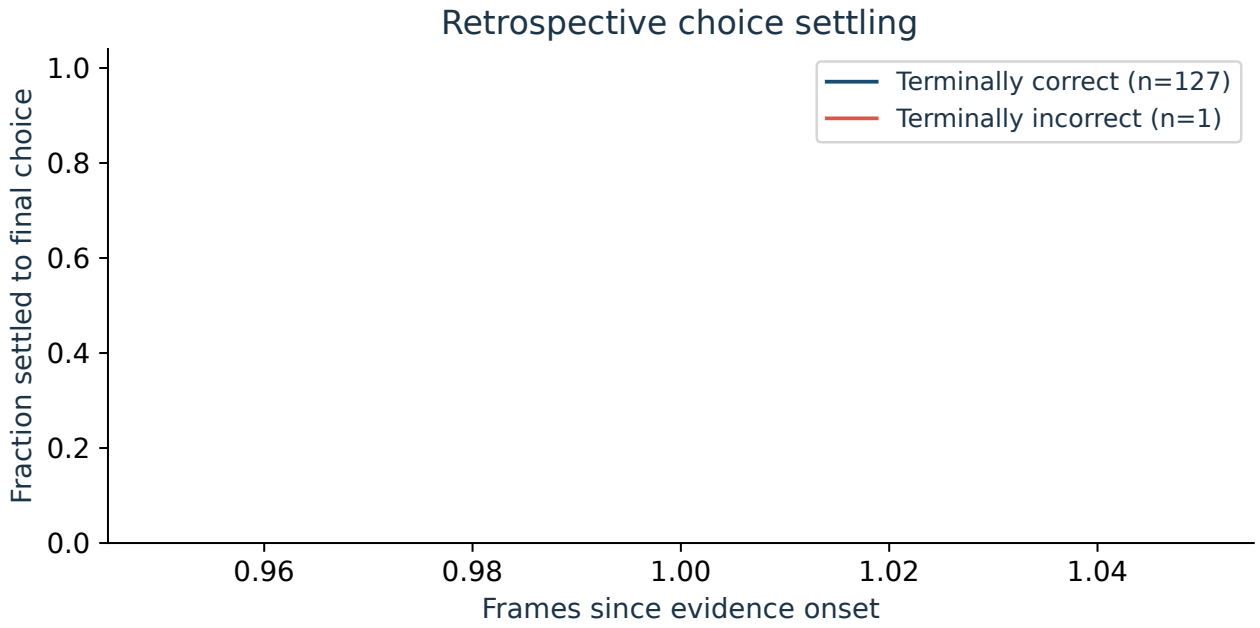


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The target is fixed at top left with a 24-frame delay. Rows separate target changes matching the instruction, unchanged targets, and target changes opposing the instruction. Clockwise and counterclockwise instructions are pooled.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Matches instructed rotation	16	16
Target unchanged	11	11
Opposes instructed rotation	5	5

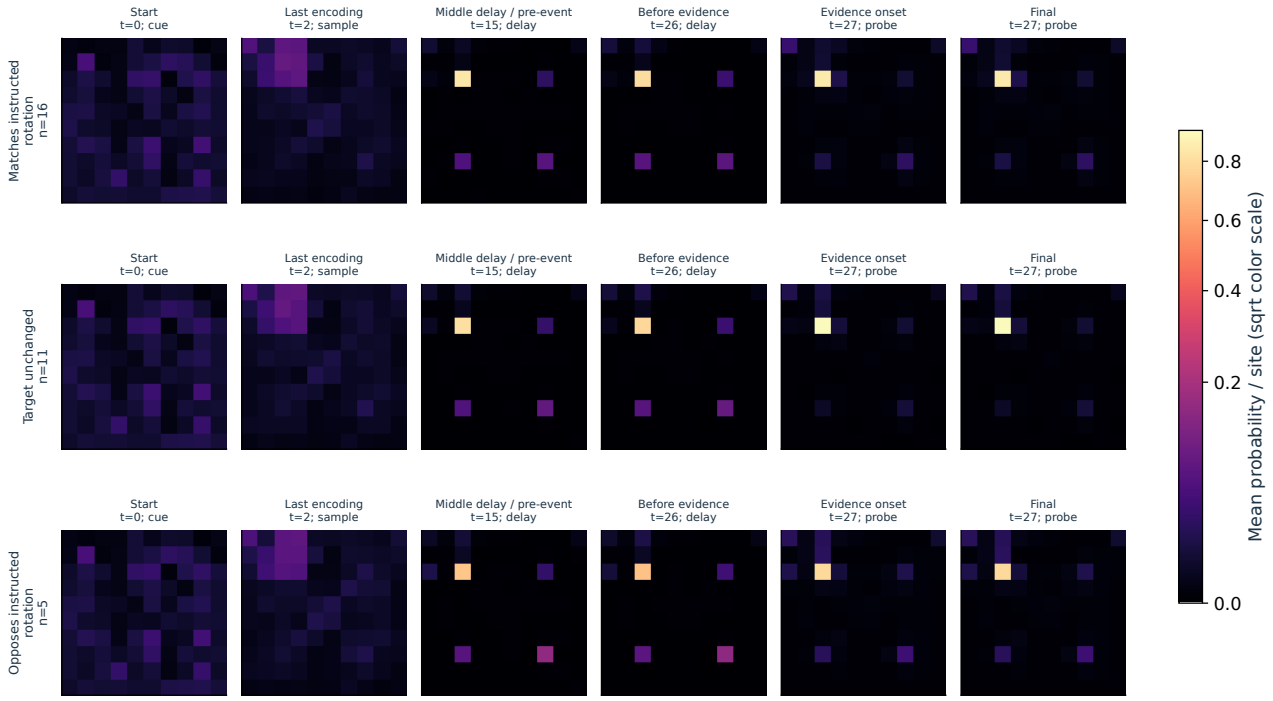


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

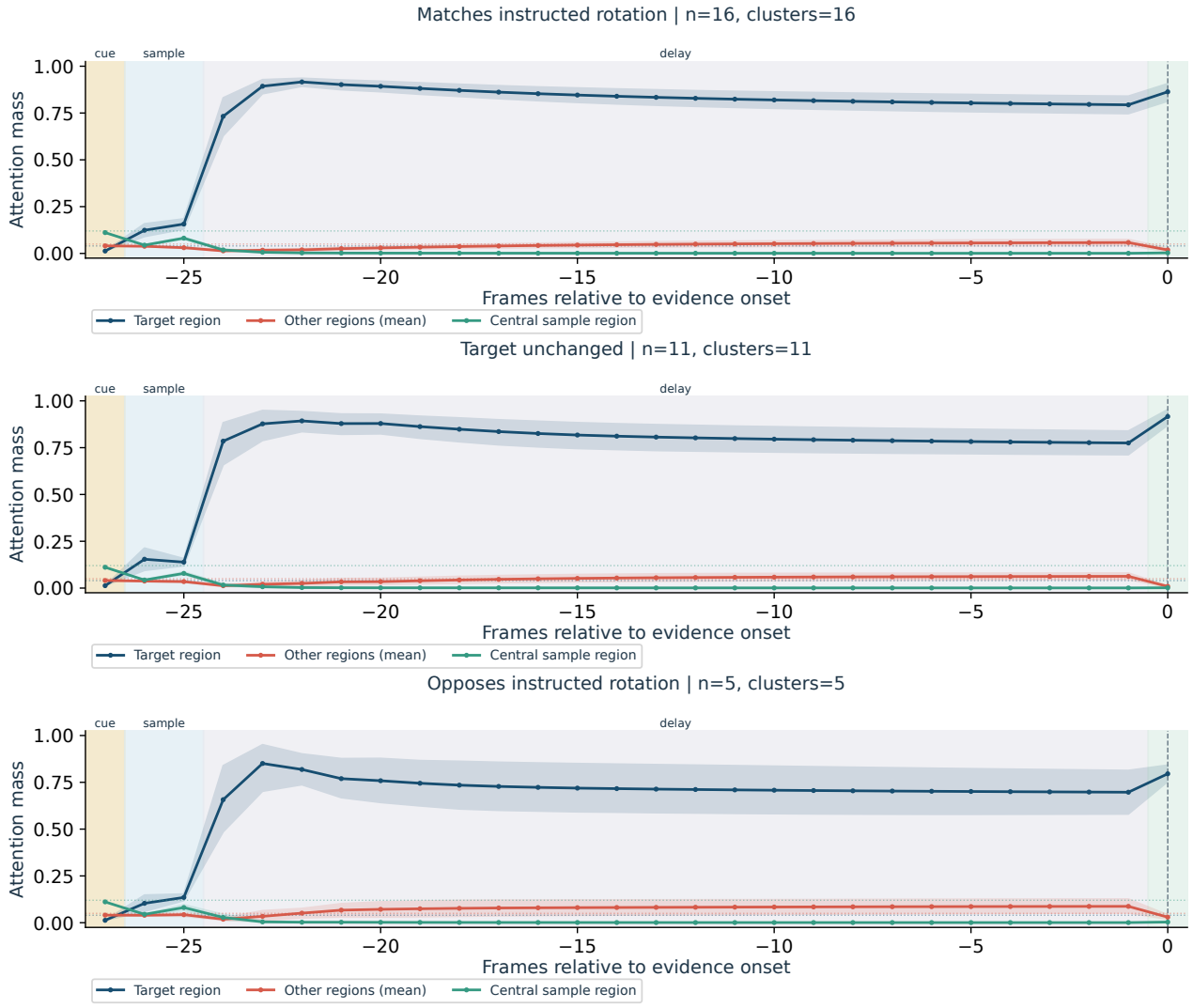


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

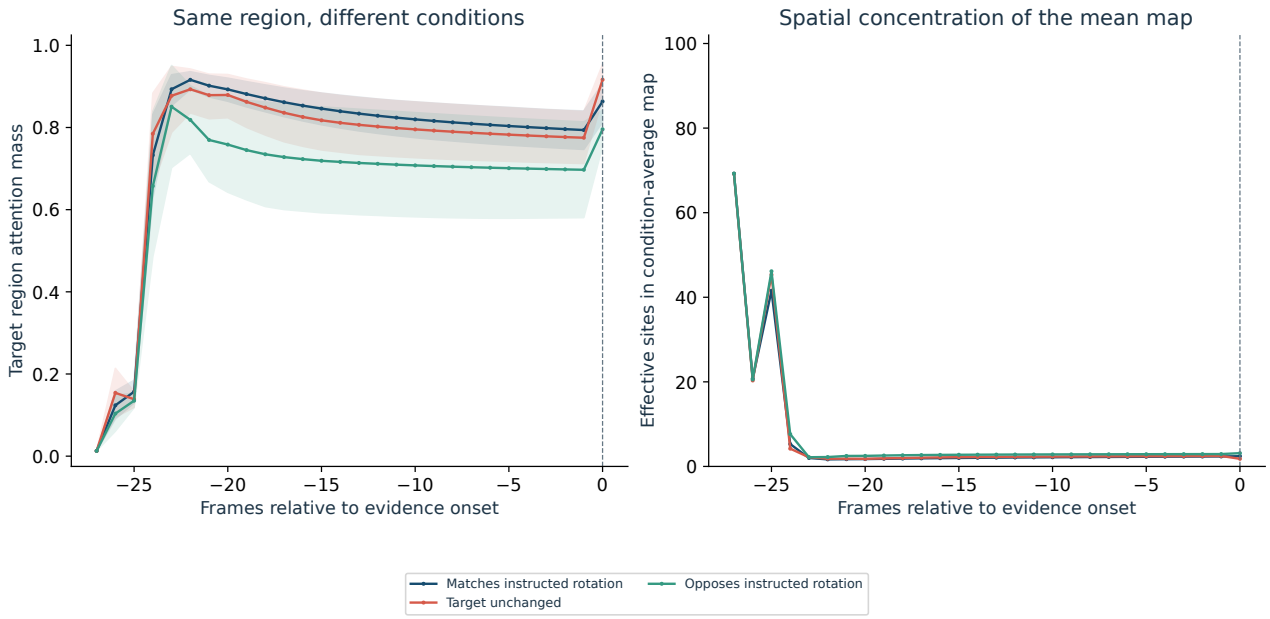


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Matches instructed rotation: target region mass peaks at 0.916 at original frame 5 (delay), and is 0.863 on the final frame. Target unchanged: target region mass peaks at 0.916 at original frame 27 (probe), and is 0.916 on the final frame. Opposes instructed rotation: target region mass peaks at 0.851 at original frame 4 (delay), and is 0.796 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For target_-8_evidence, paired accuracy changed by -56.2 percentage points (paired 95% interval [-64.8%, -47.7%]; paired p=4.235e-22; within-task Holm p=1.313e-20). There were 0 corrected trials and 72 newly incorrect trials. For target_-2_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

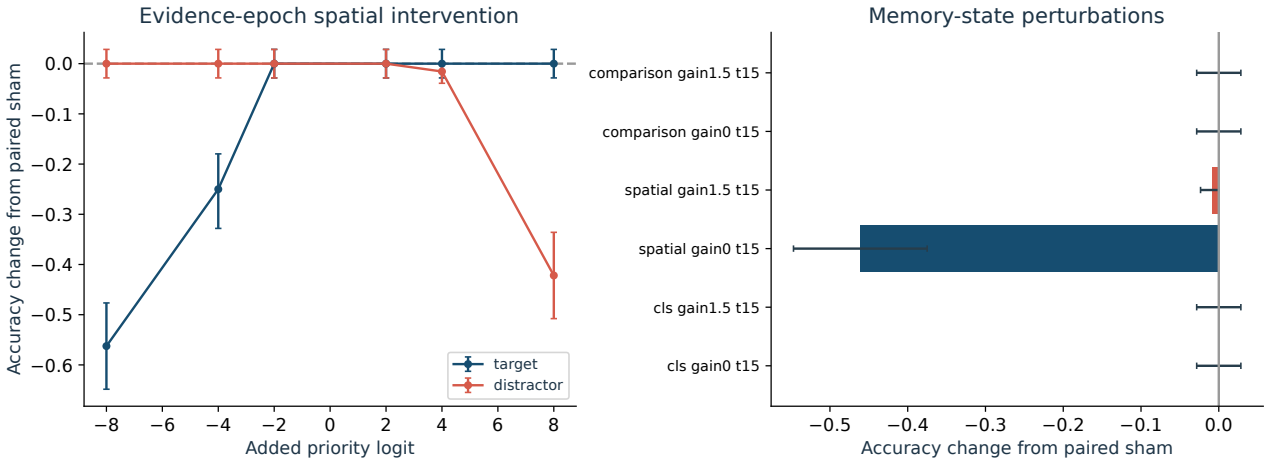


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
D24	target_-4	-25.0	4.051e-08
D24	target_+4	+0.0	1
D24	distractor_-4	+0.0	1
D24	distractor_+4	-1.6	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.509	50.8%	0.525
fusion	0.519	52.3%	0.350
spatial	0.469	50.0%	0.555
cls	1.000	100.0%	0.005
comparison	0.455	46.1%	0.940
attention	0.589	57.0%	0.020

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

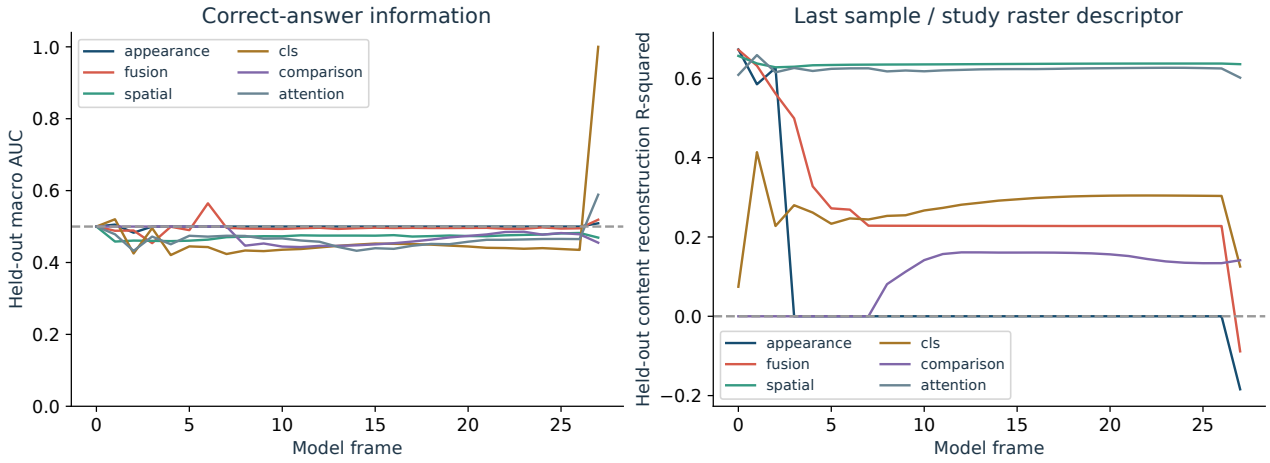


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 2.17 to 17.87 within the retained subspace. The final step displacement was 1.469 standardized RMS units. In spatial memory, the participation ratio ranged from 2.97 to 21.68 within the retained subspace. The final step displacement was 0.216 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 15, reconstruction of the sample orientation at the instructed location, represented by the cosine and sine of twice its angle had held-out R-squared 0.953 in CLS memory and 0.772 in spatial memory. This continuous target describes stimulus content, while the final-answer decoder tests the response required after the subsequent evidence.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, +0.0 percentage points, spatial, -46.1 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

3.11 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.544	unavailable	3200
fusion	0.479	0.483	6400
spatial	0.617	0.434	6400
cls	1.000	0.418	64
comparison	0.487	0.499	735
attention	0.590	0.470	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 26 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.12 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

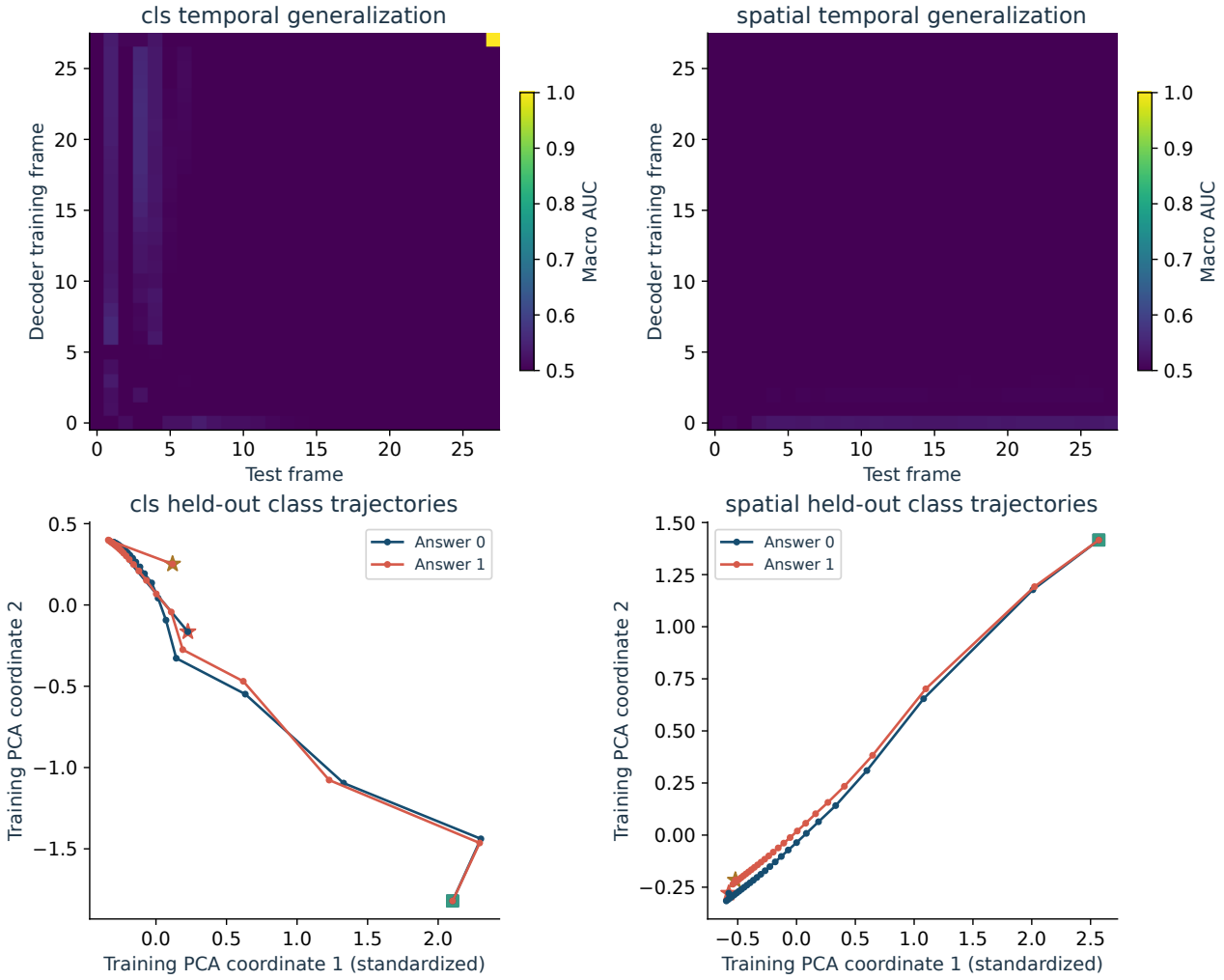


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

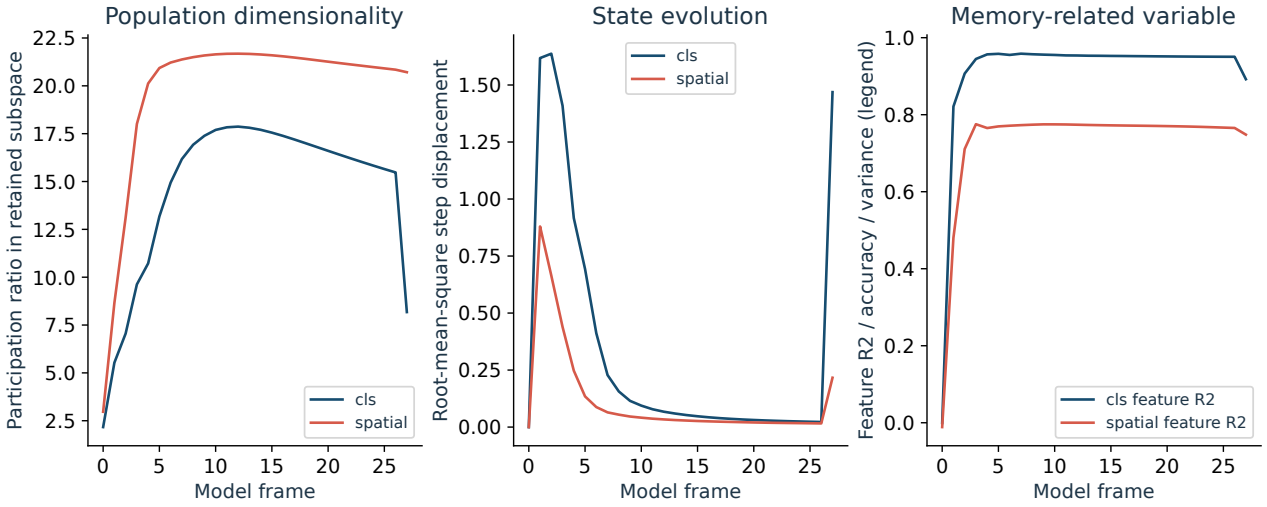


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

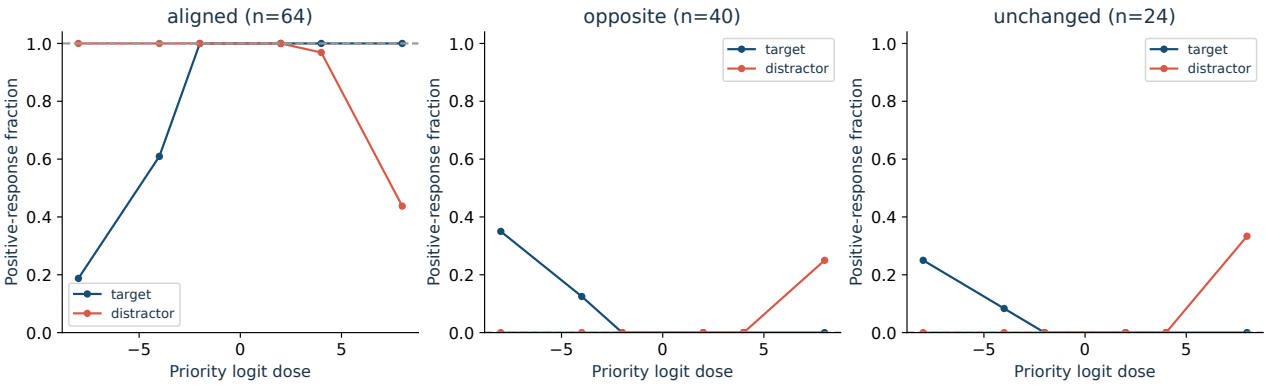


Figure 13: Event-specific causal response functions. Curves show positive reports after evidence-epoch target or alternative-site intervention; the horizontal line is paired sham. Event counts are stated above each panel. A positive report has the task-specific meaning defined in the text.

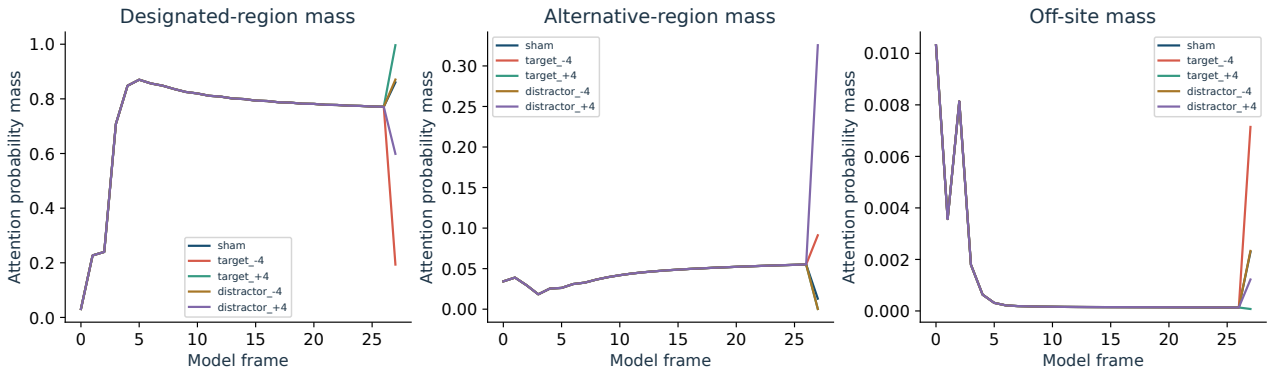


Figure 14: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

3.13 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

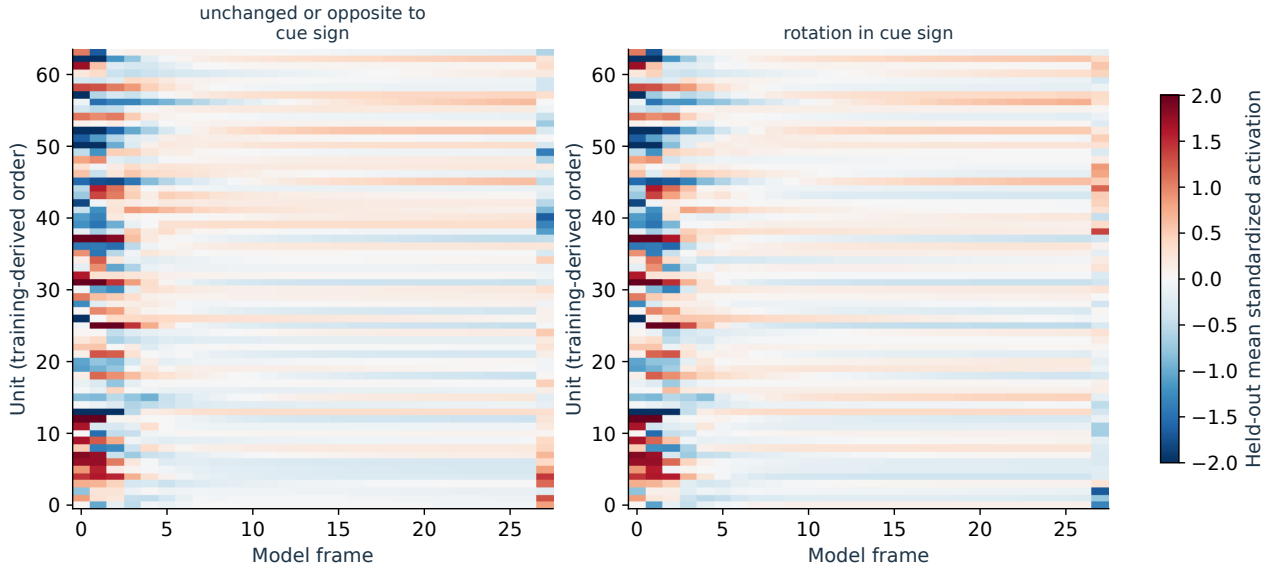


Figure 15: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.14 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional

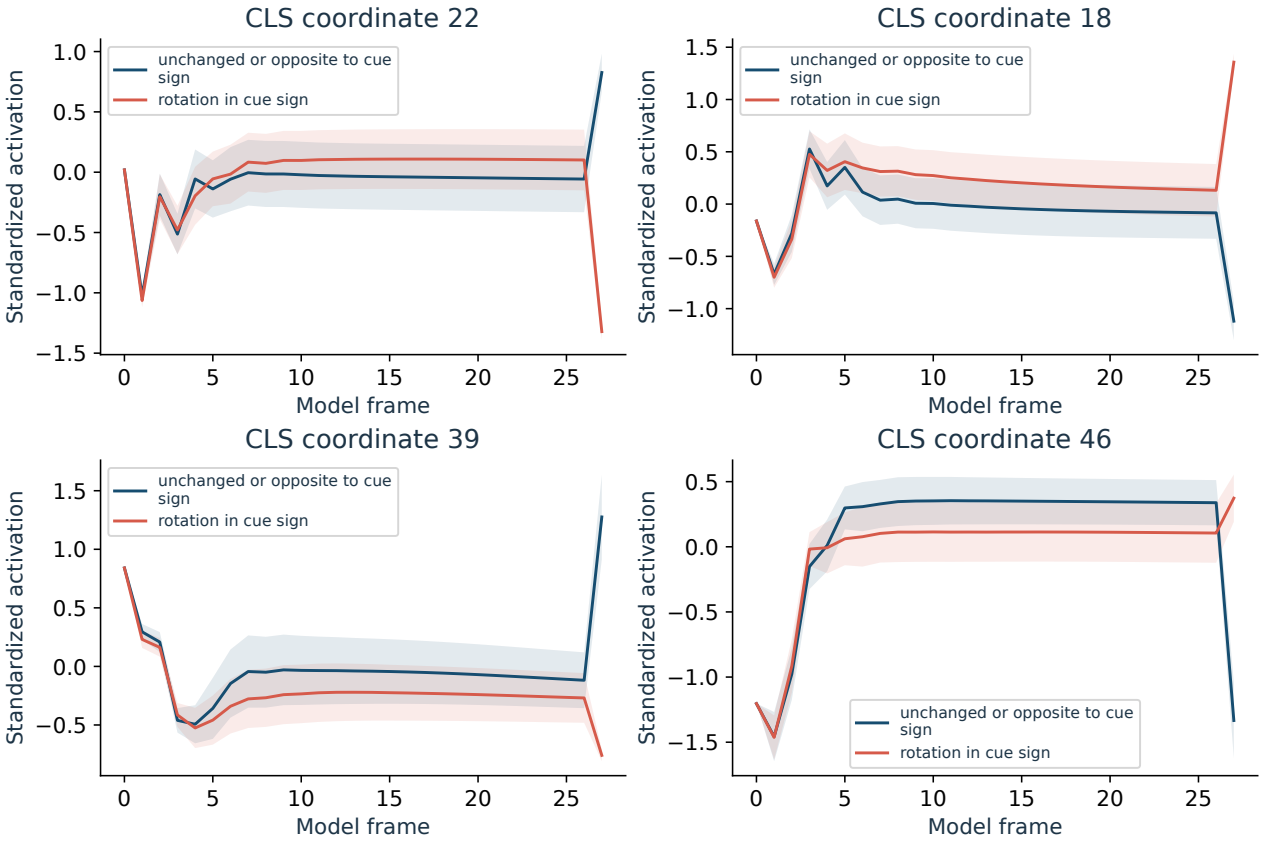


Figure 16: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	50.8%	50.8%	25.8%
fusion	52.3%	52.3%	28.1%
spatial	50.0%	50.0%	100.0%
cls	100.0%	100.0%	66.4%
comparison	46.1%	46.1%	53.9%
attention	57.0%	57.0%	100.0%

3.15 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

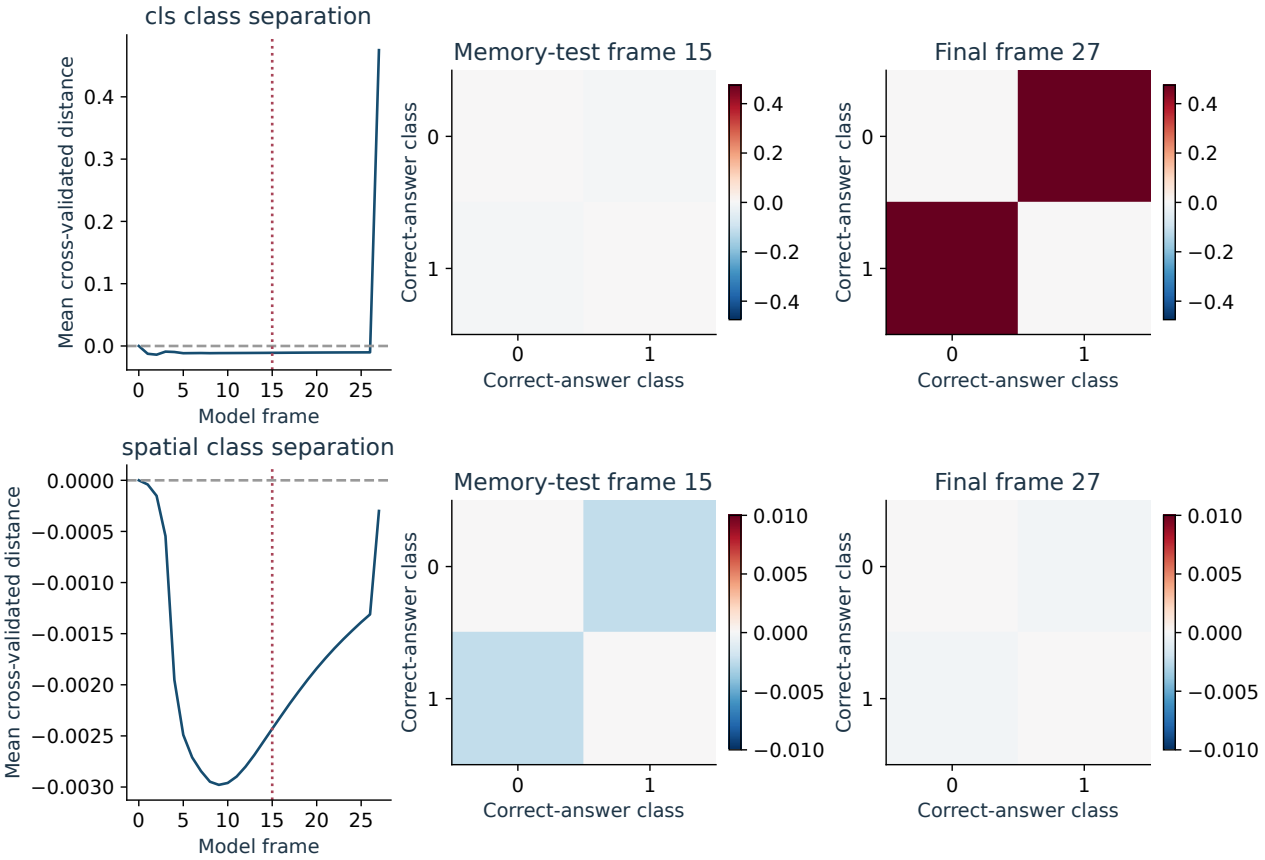


Figure 17: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

3.16 Remembered orientation inventory at all four locations

A separate content analysis reconstructs all four original sample orientations at every frame. Each orientation angle is encoded as its doubled-angle cosine and sine, respecting the 180-degree periodicity of a grating. The same training-only 32-component representation and ridge penalty used in the main analysis predict the eight continuous target values. Each location has a separate held-out R-squared relative to its fitting-mean predictor. This directly measures location-bound sample content. In the retrospective-cue task, the identity of the eventual target is revealed only after the delay, so all-location reconstruction is the appropriate pre-cue storage measurement.

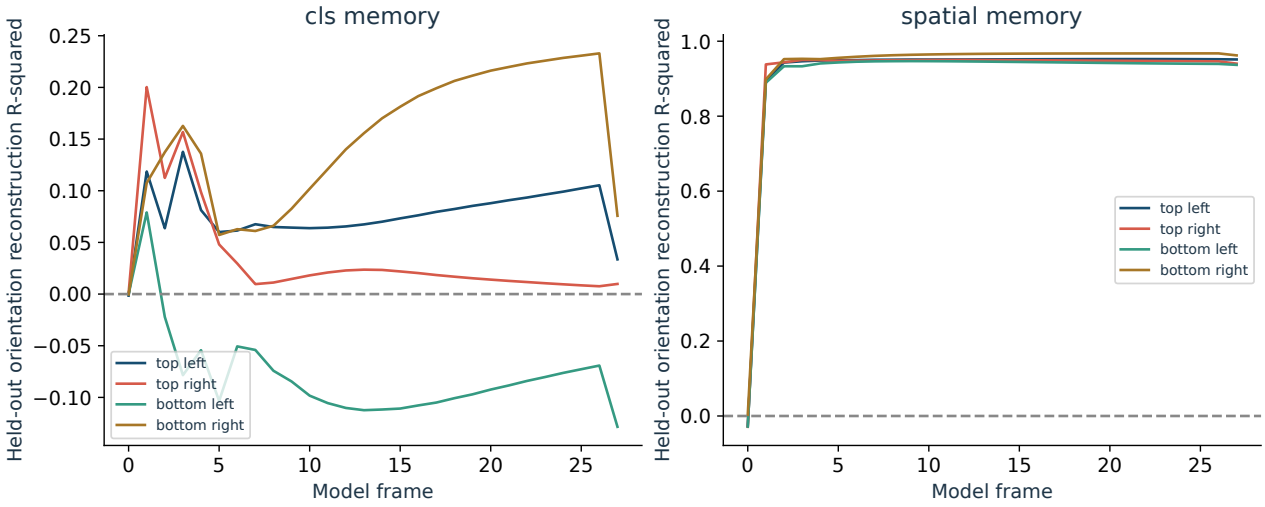


Figure 18: The original orientation at each of the four sample locations decoded from global and spatial memory. Curves track continuous sample content rather than the later correct-answer label. Zero is the fitting-mean baseline.

4 Discussion

The continuous-feature analysis and the response-label analysis describe different stages of the task. Sample orientation remains readable during the delay, whereas the final answer depends on the later rotation and cue rule. A mid-delay spatial reset causes a large subsequent accuracy loss, while the same-time global reset produces no observed loss in this sample. The latter state can be reconstructed during the remaining frames from information still available elsewhere. Target inhibition at the probe separately demonstrates a causal selection requirement. The combined result supports a sequence of retained spatial content, later selection, and rule-dependent response construction.

The behavioral result establishes 99.8% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies target_-8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

A full factorial of cue sign, target rotation and foil congruence would distinguish mnemonic degradation from rule confusion. Applying a late countercue would test how rapidly the stored representation can be read under a different instruction.

4.1 Interpreting these results alongside biological experiments

4.1.1 Harrison and Tong (2009)

Sample orientation is readable during the model's delay, matching the key computational observation that absent visual content can remain available in a population pattern. The model adds a causal intervention: resetting spatial memory during retention causes a large later cost. This links information availability to behavior more directly than a decoder alone. [4]

4.1.2 Griffin and Nobre (2003)

The cue determines which retained content or location matters for a subsequent comparison. The human cue experiments motivate separating a benefit at encoding from selection within memory. In this model, probe-period target inhibition provides a later selection test, while the delay reset tests retention. Their separate effects support distinct stages of the computation. [5]

4.1.3 Stokes and colleagues (2013)

The model's answer becomes meaningful only after the probe and rule define it. A population can therefore transform from an orientation code into a decision code. Cross-temporal decoding tests whether one linear readout remains useful across those phases. A changing geometry can support stable task performance when the downstream computation reads the appropriate phase-dependent state. [6]

4.2 A computational account of the observed behavior

The crucial theoretical distinction is between storing a stimulus and storing a response. Before the probe, the network can retain sample orientation and spatial context; after the probe it can compute a signed rotation and apply the required rule. The large spatial reset cost localizes a necessary retained contribution. The absence of a measured cost from a global reset at the same instant has a narrower interpretation: global state can be restored or bypassed using other remaining state during the subsequent frames. It does not imply that global memory never participates. The time-resolved maps and decoder matrices make these stages visible: encoding, retention, renewed sensory selection and final response construction.

4.3 Where this task agrees with the biological evidence

The model agrees with retained sensory content and subsequent attentional selection. Its division of labor favors spatial memory at the tested delay time. The biological theory of dynamic coding becomes a testable account when content decoders transfer differently across phases while the causal dependence on the stored information remains.

4.4 A follow-up experiment motivated by these results

Apply brief resets at multiple delay times and immediately before the probe. Decode the sample angle after each reset, then measure whether behavior recovers. Separate rapid reconstruction from true tolerance to loss of the represented orientation.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Funahashi S, Bruce CJ, Goldman-Rakic PS (1989). Mnemonic coding of visual space in the monkey dorsolateral prefrontal cortex. Journal of Neurophysiology 61:331-349. https://pubmed.ncbi.nlm.nih.gov/2918358/.

[2] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Harrison and Tong (2009). Decoding reveals the contents of visual working memory in early visual areas. Primary source.

[5] Griffin and Nobre (2003). Orienting attention to locations in internal representations. Primary source.

[6] Stokes and colleagues (2013). Dynamic coding for cognitive control in prefrontal cortex. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
D0	128	100.0%	1.000	0.001
D4	128	100.0%	1.000	0.000
D12	128	100.0%	1.000	0.000
D24	128	99.2%	1.000	0.009

B Every-frame attention atlas

Condition D24. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 16 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

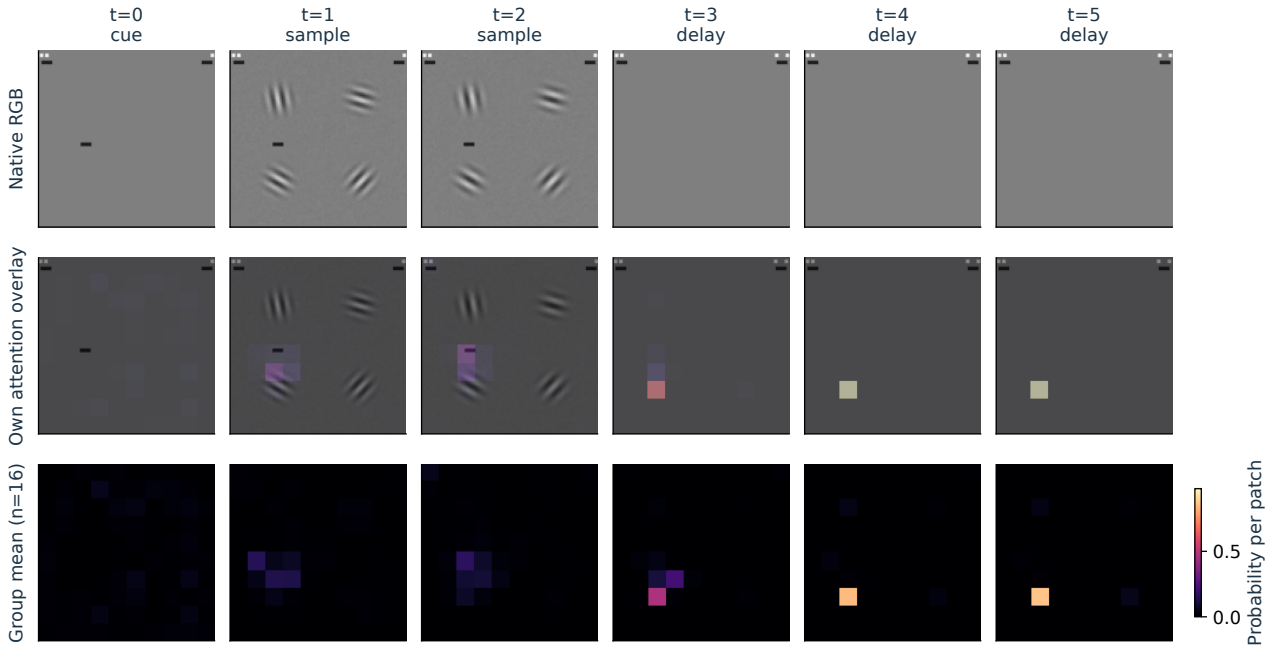


Figure 19: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

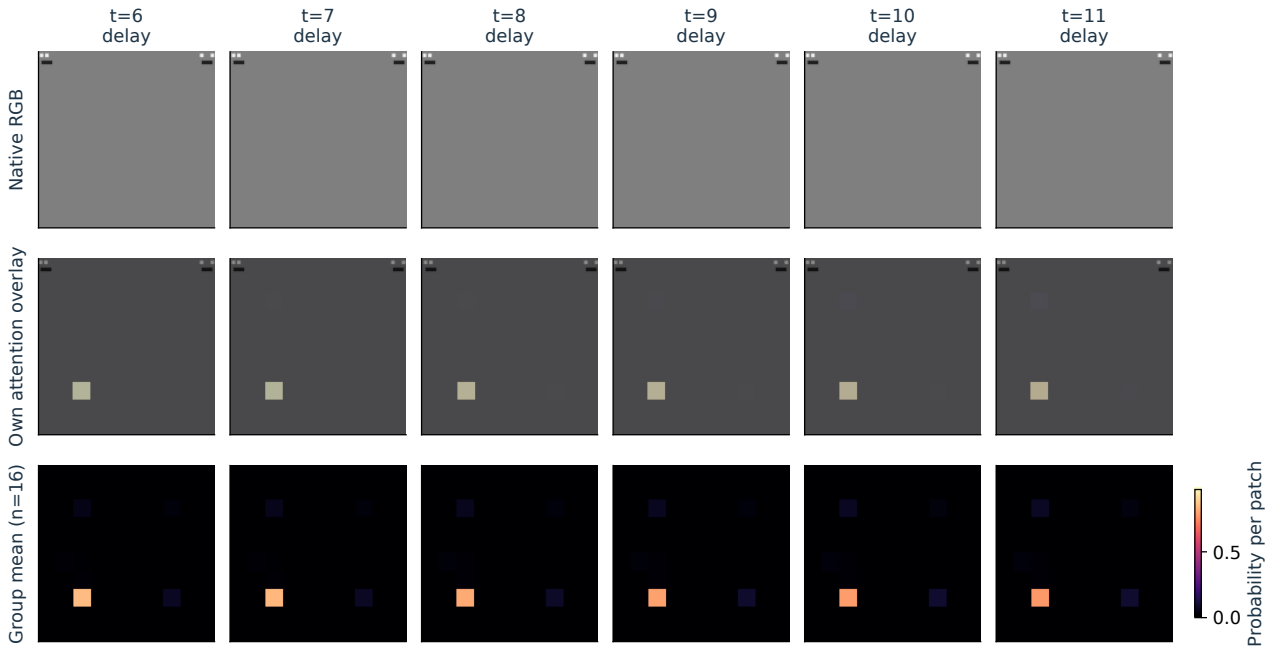


Figure 20: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

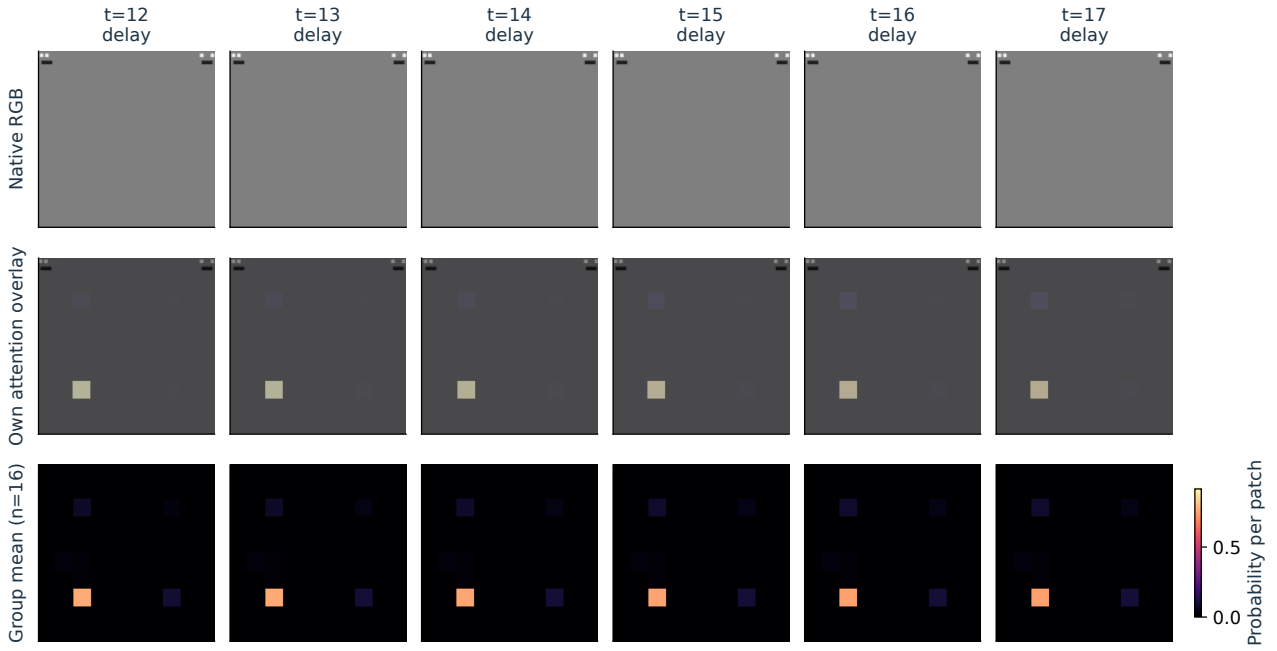


Figure 21: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

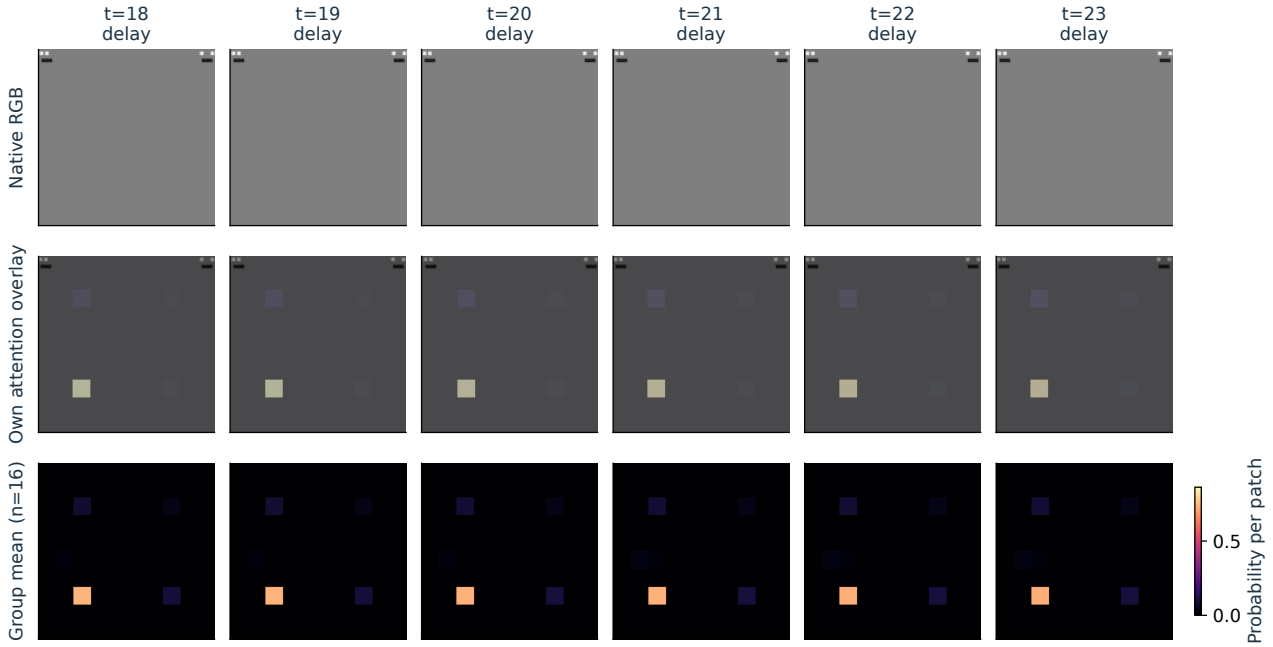


Figure 22: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

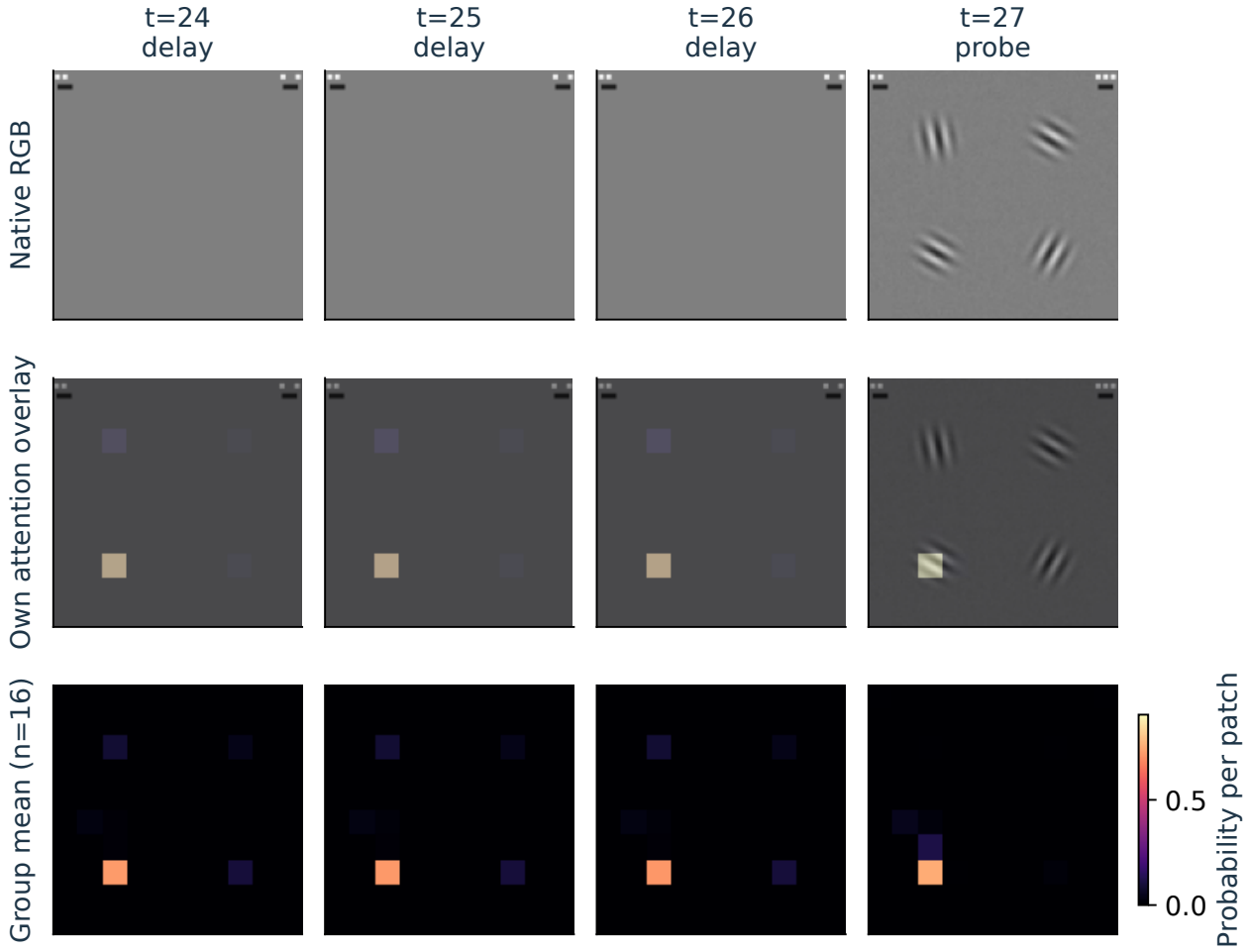


Figure 23: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

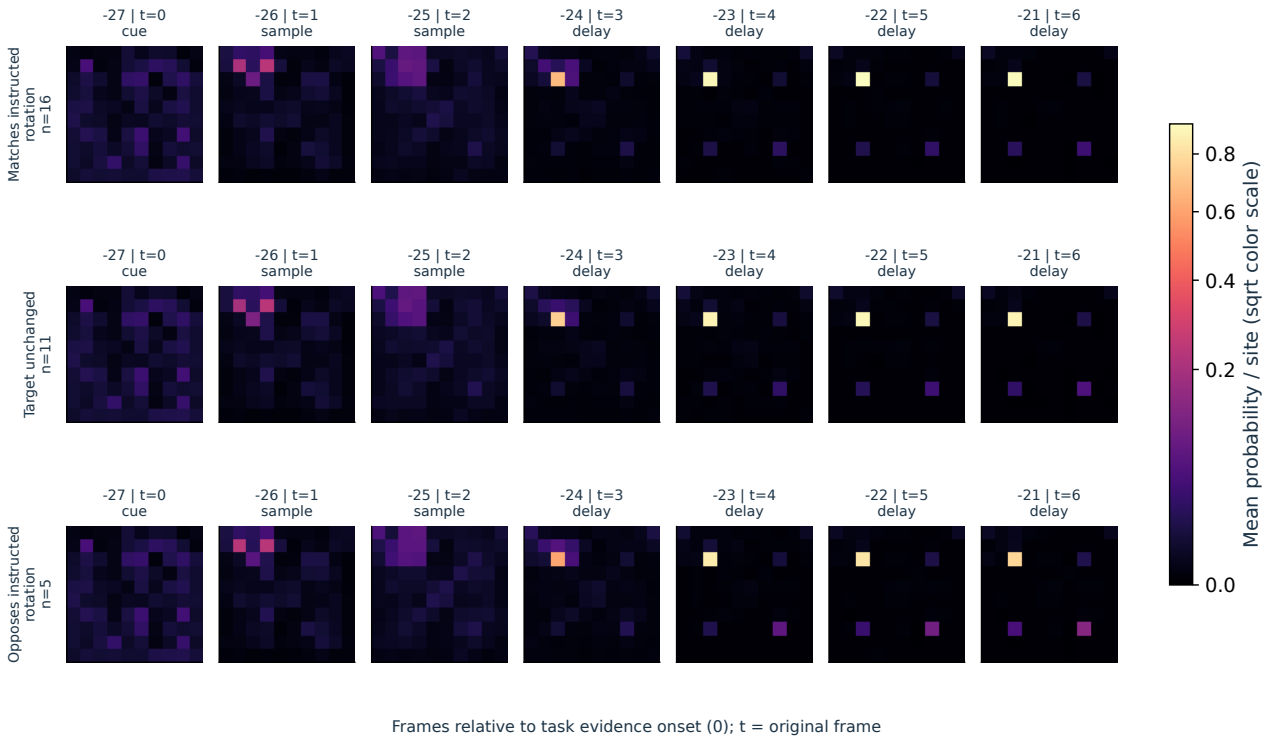


Figure 24: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

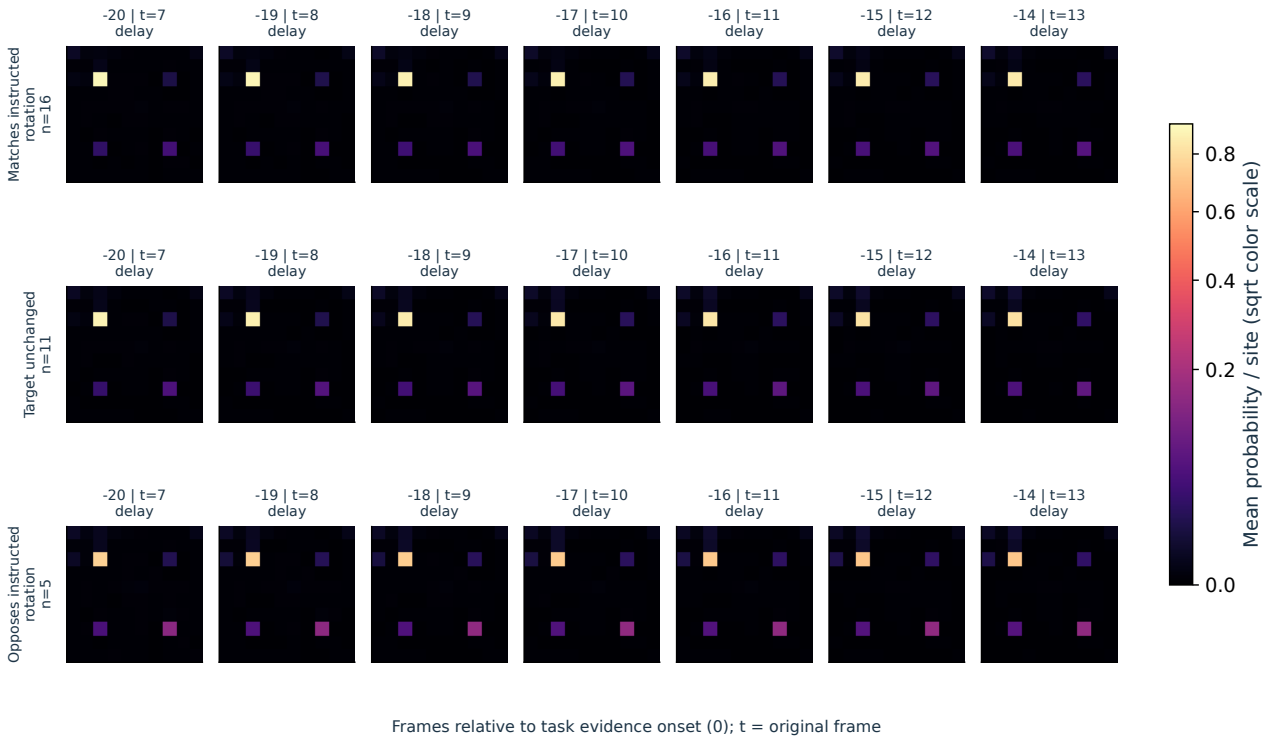


Figure 25: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

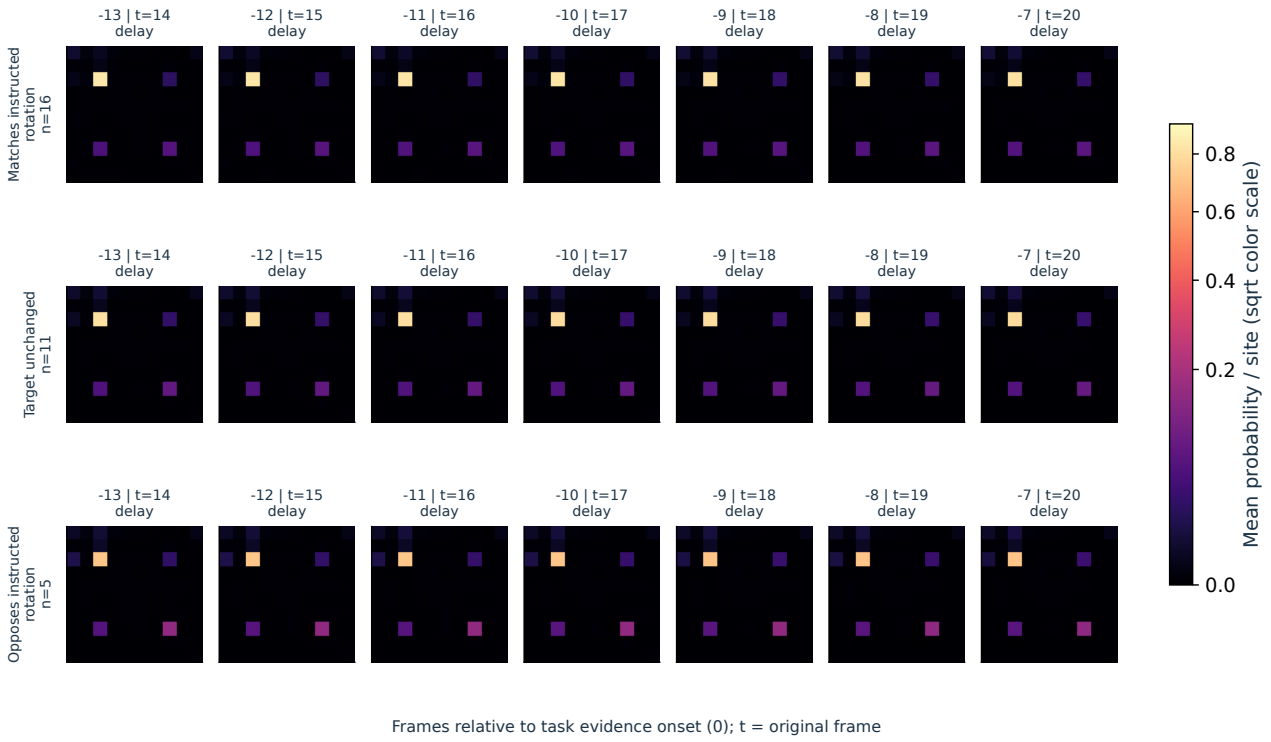


Figure 26: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

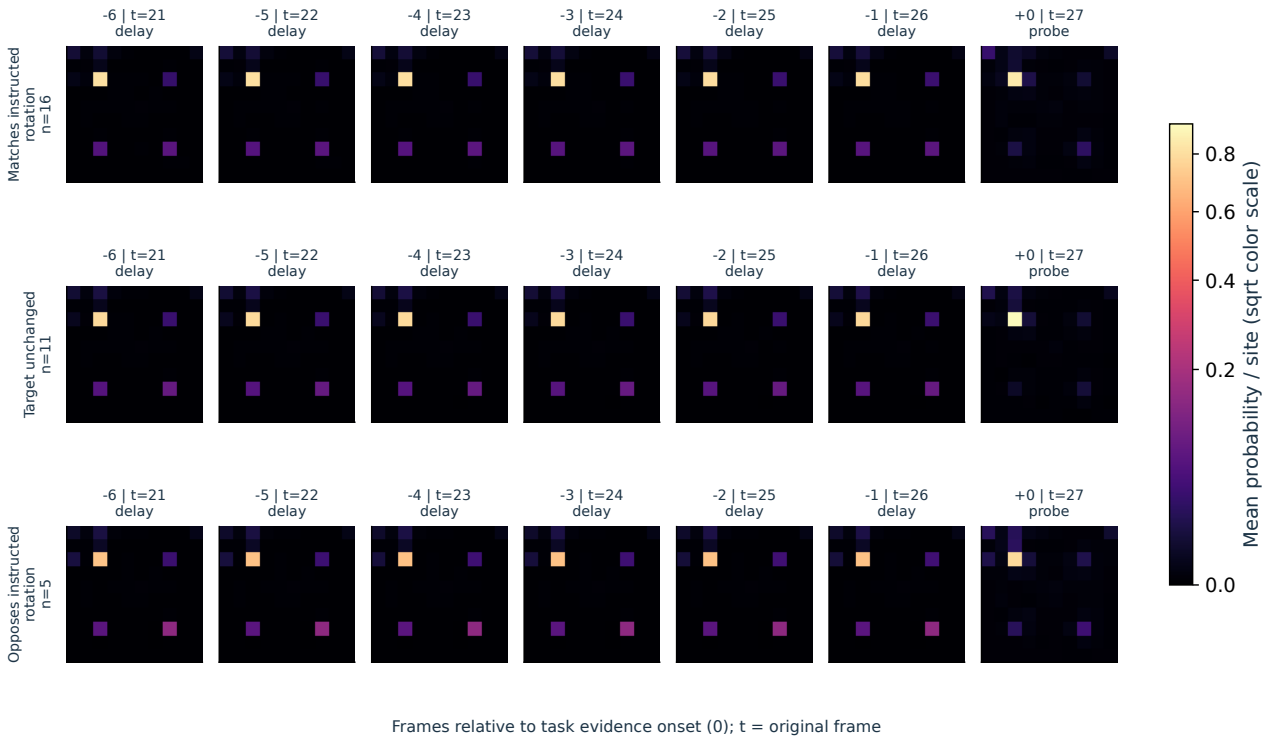


Figure 27: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Temporal voting over cued motion and maintenance of the winning direction

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized motion duration cued in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 4 native conditions and 512 fresh trials, equal-condition accuracy was 97.5% (95% interval [96.1%, 98.6%]). For the representative condition D24, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was distractor_+8_evidence, changing accuracy by -71.1 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The cued motion patch changes cardinal direction across eight transitions. The observer must report the direction occupying the greatest number of transitions. A single transition is insufficient; the task requires accumulating evidence over time and preserving the winning direction through a later blank interval.

Native conditions vary retention delay. The temporal decoder shows when the eventual winner becomes readable, while the confidence analysis asks when the unchanged terminal readout first commits to an alternative. Because later transitions can overturn an early lead, premature confidence is an informative property of the computation.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

The task combines a sensory motion judgement with a memory-guided spatial instruction. A single direction-selective response is insufficient to identify the most frequent transition unless the evidence is integrated across the movie. This motivates examining the trajectory of the eventual winning class and the point at which that winner becomes decodable. During a blank delay, a stable answer code would be compatible with maintaining a summarized decision; a changing code with successful cross-temporal decoding would support maintenance in a dynamically evolving population. Mid-delay state perturbation tests whether that maintained variable is behaviorally necessary.

1.2 Biological experiments motivating this task

Experimental question. Does the model retain a temporally accumulated summary of the selected motion stream?

Kiani, Hanks and Shadlen (2008) Monkeys could commit to a motion choice before externally controlled viewing ended. LIP activity supported accumulation to a bound, demonstrating that late available evidence need not influence the decision. [5]

Stine, Zylberberg, Ditterich and Shadlen (2020) Task manipulations distinguished prolonged evidence integration from nonintegration strategies that can reproduce common behavioral signatures. Most tested observers showed evidence for prolonged integration under the diagnostic design. [6]

Mante, Sussillo, Shenoy and Newsome (2013) Monkeys selected motion or color evidence according to context. Prefrontal population activity and trained recurrent-network dynamics linked context-dependent selection with evidence integration. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Which direction occupied the most of eight transitions in the cued patch?

Response labels are 0: right; 1: up; 2: left; 3: down. The native suite contains 4 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use D24.

1.4 Stimulus sequence and spatial layout

For the representative native condition D24, the sequence has 35 frames, indexed from zero: frame 0: cue; frame 1: reference; frames 2–9: motion; frames 10–33: delay; frame 34: report. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Motion transitions	8
Dots per aperture	32
Displacement per transition (pixels)	0.8, 1.2, 1.6
Aperture radius (pixels)	11.5

Stimulus centers in the 100 by 100 input are [[27.0, 27.0], [73.0, 27.0], [27.0, 73.0], [73.0, 73.0]]. Coordinate x increases rightward and y downward.

Spatial selection determines which patch contributes evidence, whereas temporal integration determines how successive directions affect the state. The comparison-memory branch and recurrent states can

make different contributions. Reset and gain perturbations at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists) test the stored result after the visible evidence has ended.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding

global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression $k^{\mathsf{T}}q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^\delta A/(1 - A + e^\delta A)$. Thus its odds against the remaining image are multiplied by e^δ; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance

probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ±4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated

class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 97.5%, with macro AUC 0.999. The matched final checkpoint changed accuracy by +2.34 percentage points on average across conditions, using 32 same-movie comparisons per condition. The lowest native cell was D0 at 93.8%; the highest was D12 at 99.2%. These comparisons retain the native sampling design rather than selecting trials after observing success.

Marginal accuracy by delay was 0: 93.8%; 4: 98.4%; 12: 99.2%; 24: 98.4%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

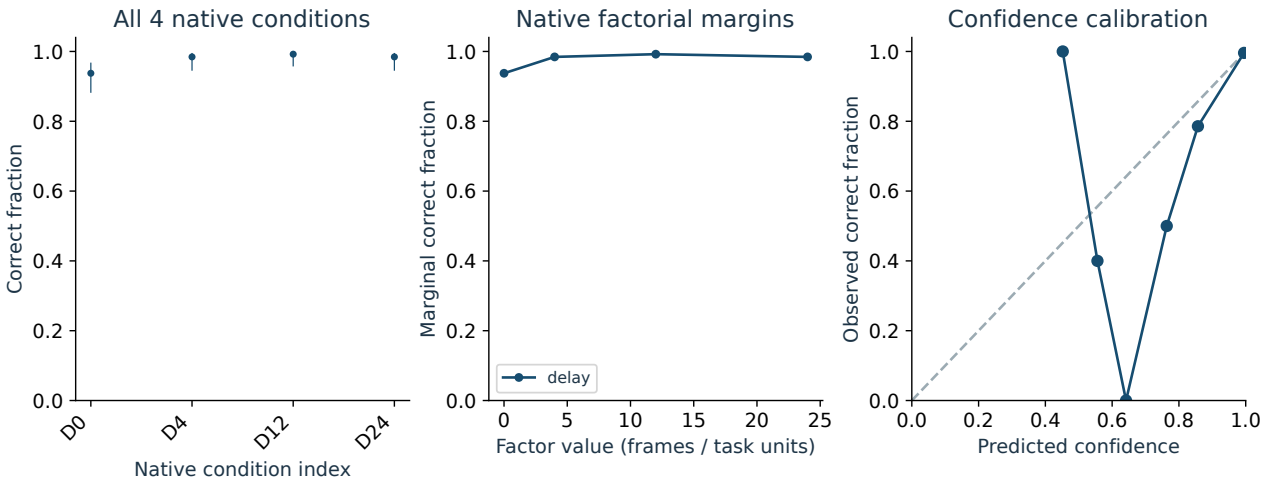


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	25.8%	100.0%
0.7	100.0%	1.0	24.2%	100.0%
0.8	99.2%	1.0	26.8%	99.2%
0.9	98.4%	2.0	48.4%	80.5%

The representative condition D24 provides 33 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

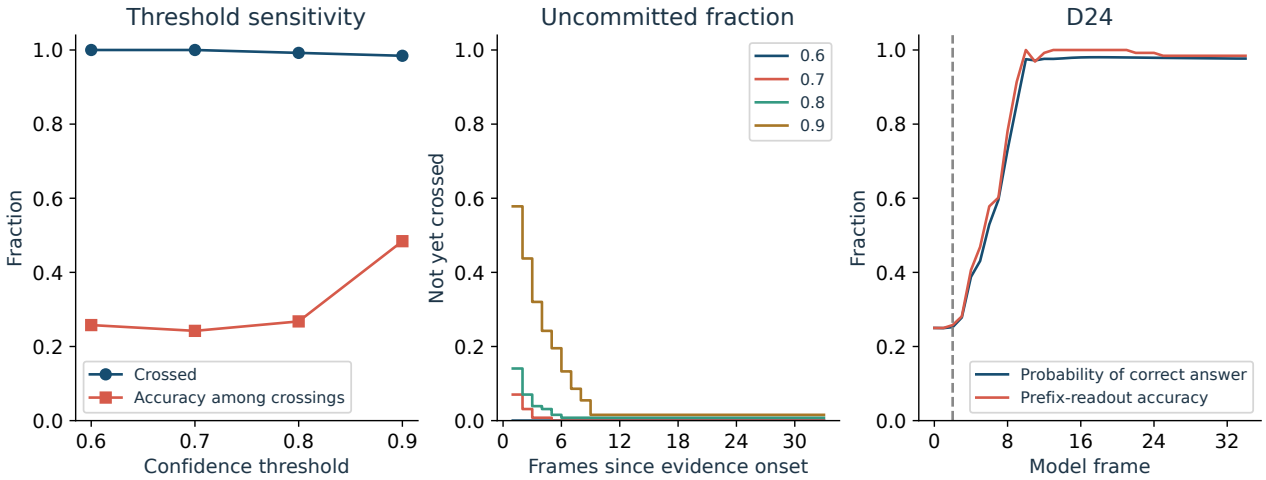


Figure 2: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

3.3 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 511 of 512 trials crossed. Of those, 364 changed their answer later, 354 corrected an early error, and 2 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

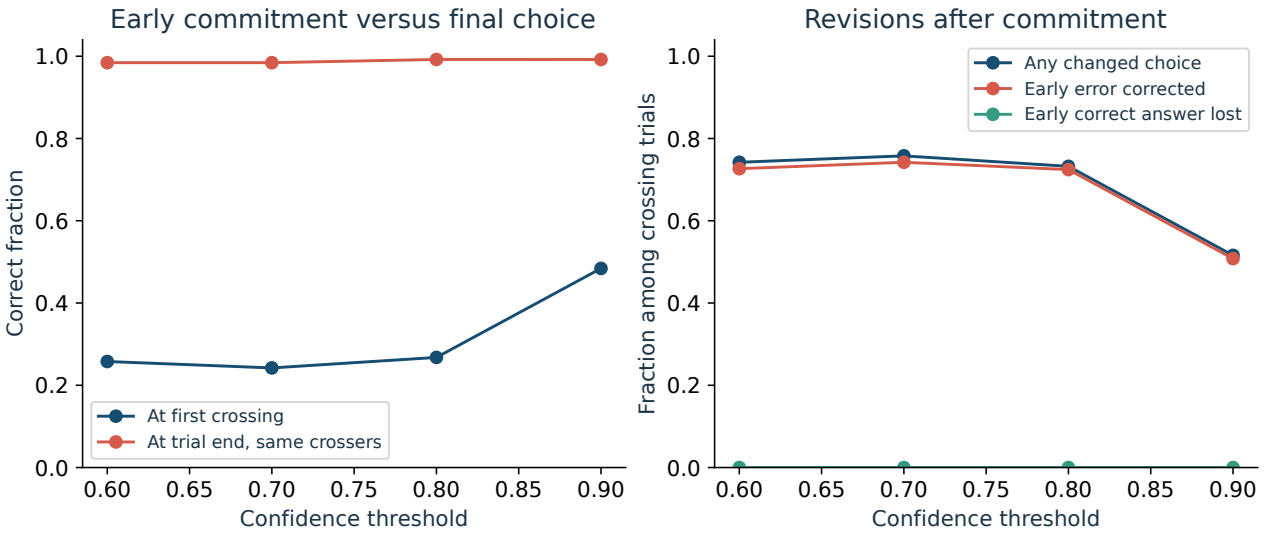


Figure 3: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

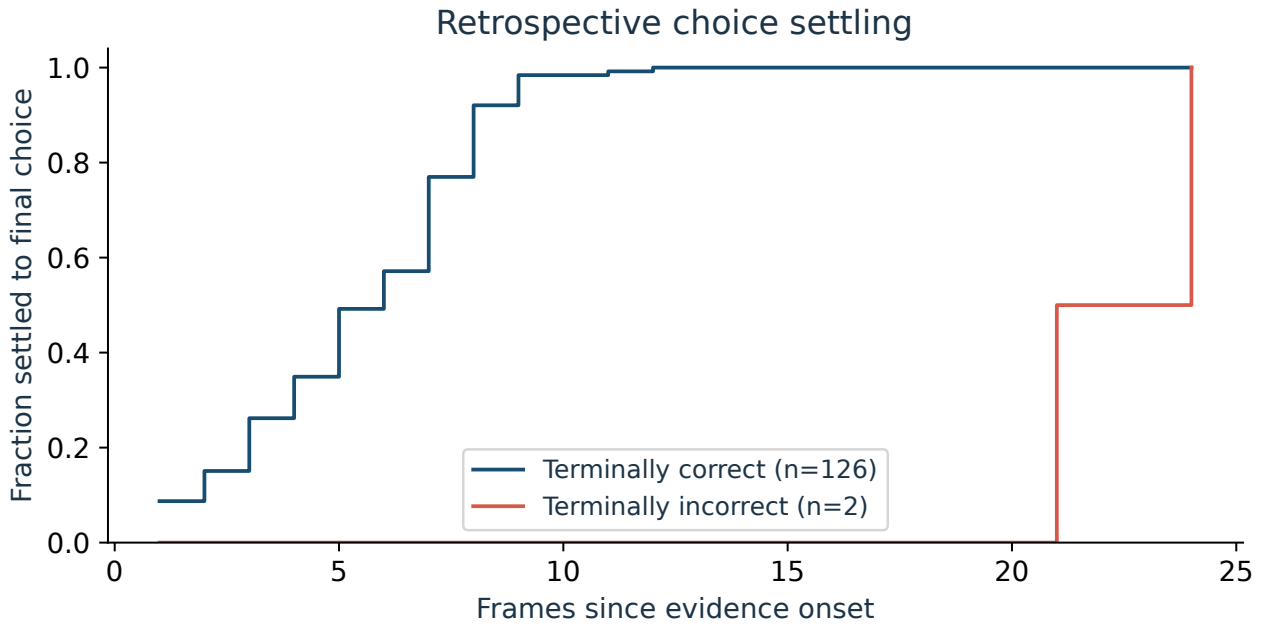


Figure 4: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.4 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The cued patch is fixed at top left. Rows compare zero-, twelve- and twenty-four-frame retention delays. Motion onset defines frame zero; the later retention interval and report remain on the same time axis.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Delay: 0 frames	32	32
Delay: 12 frames	32	32
Delay: 24 frames	32	32

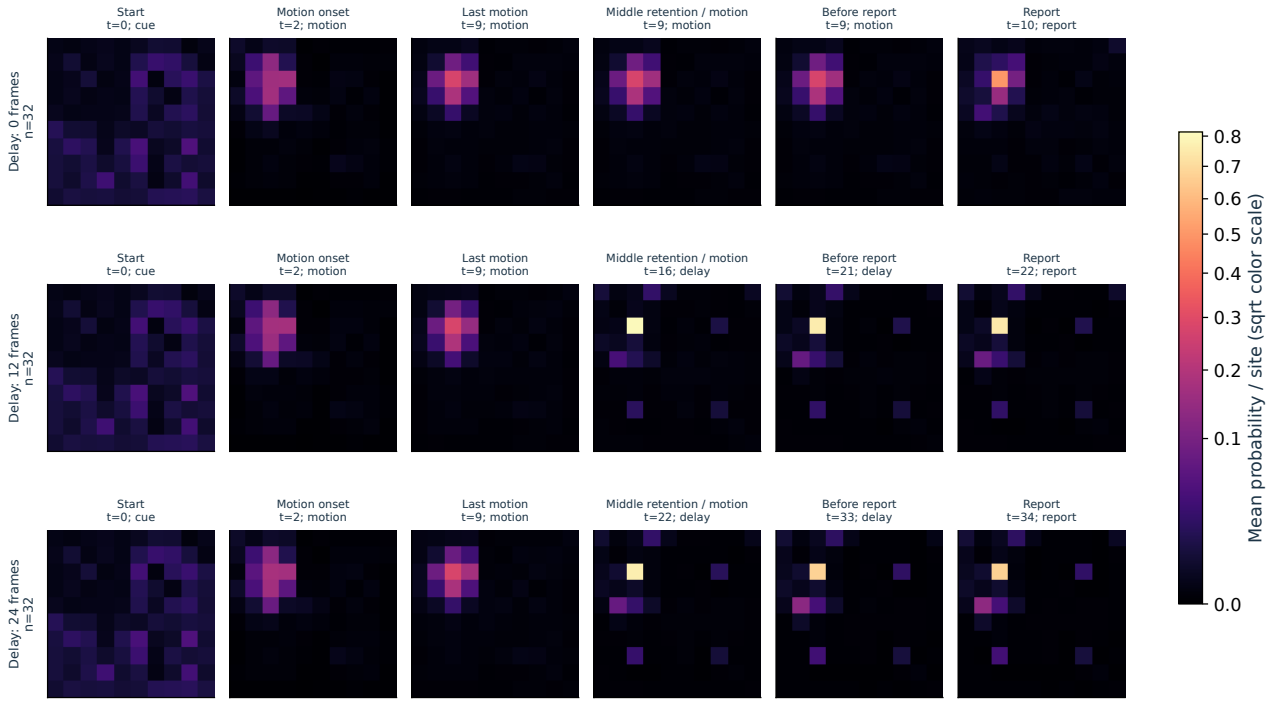


Figure 5: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

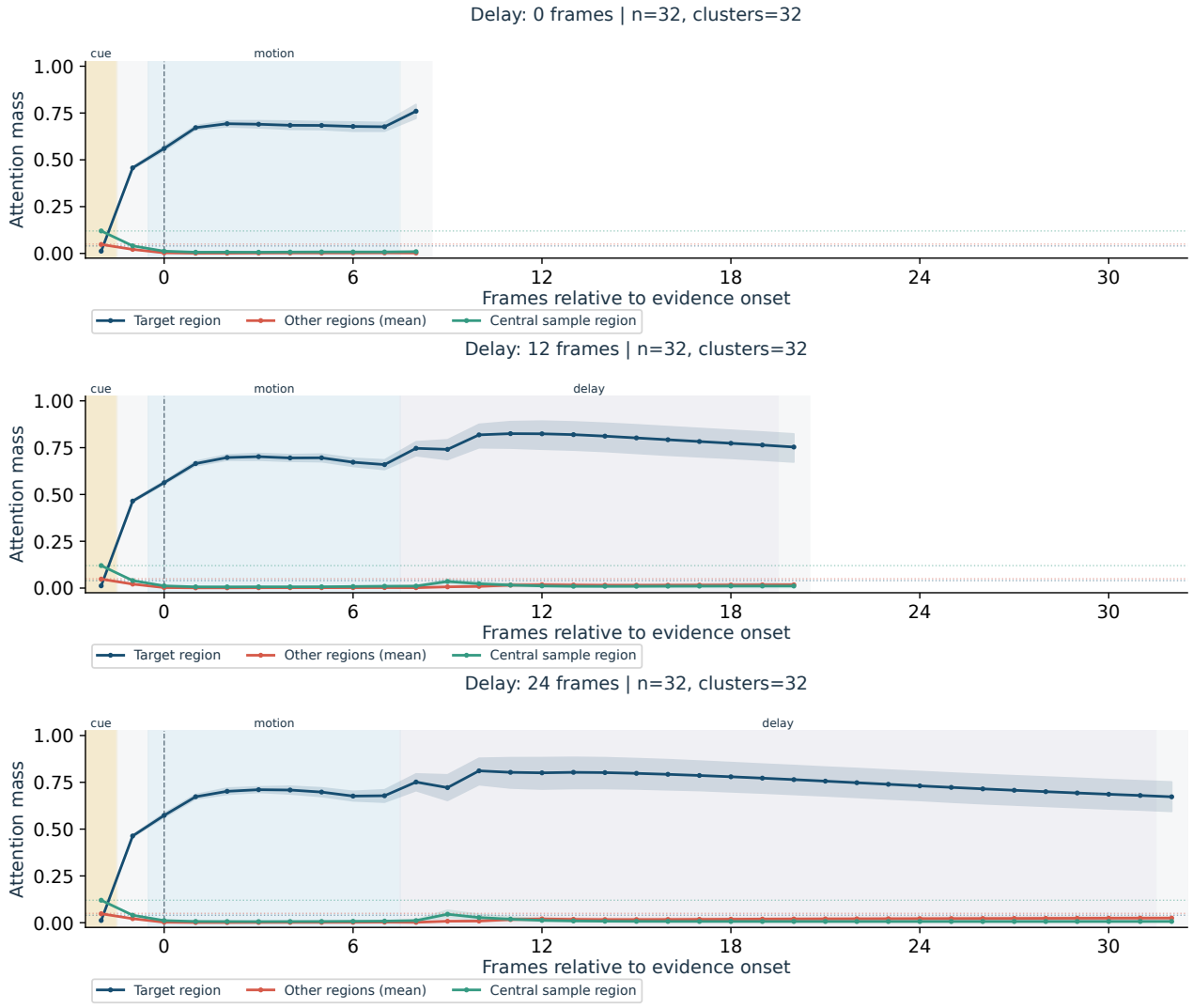


Figure 6: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

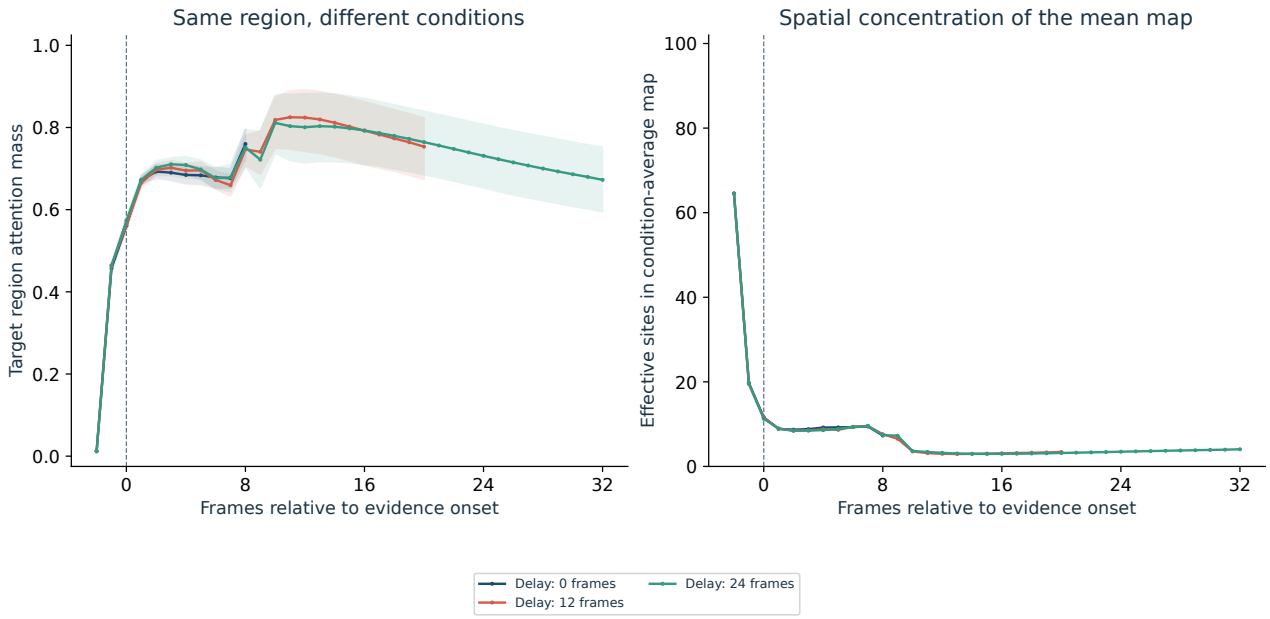


Figure 7: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Delay: 0 frames: target region mass peaks at 0.760 at original frame 10 (report), and is 0.760 on the final frame. Delay: 12 frames: target region mass peaks at 0.825 at original frame 13 (delay), and is 0.753 on the final frame. Delay: 24 frames: target region mass peaks at 0.811 at original frame 12 (delay), and is 0.673 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.5 Causal sensitivity to spatial and mnemonic interventions

For distractor_+8_evidence, paired accuracy changed by -71.1 percentage points (paired 95% interval [-78.9%, -63.3%]; paired p=8.078e-28; within-task Holm p=2.504e-26). There were 0 corrected trials and 91 newly incorrect trials. For distractor_-8_evidence, paired accuracy changed by +0.8 percentage points (paired 95% interval [0.0%, 2.3%]; paired p=1; within-task Holm p=1). There were 1 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

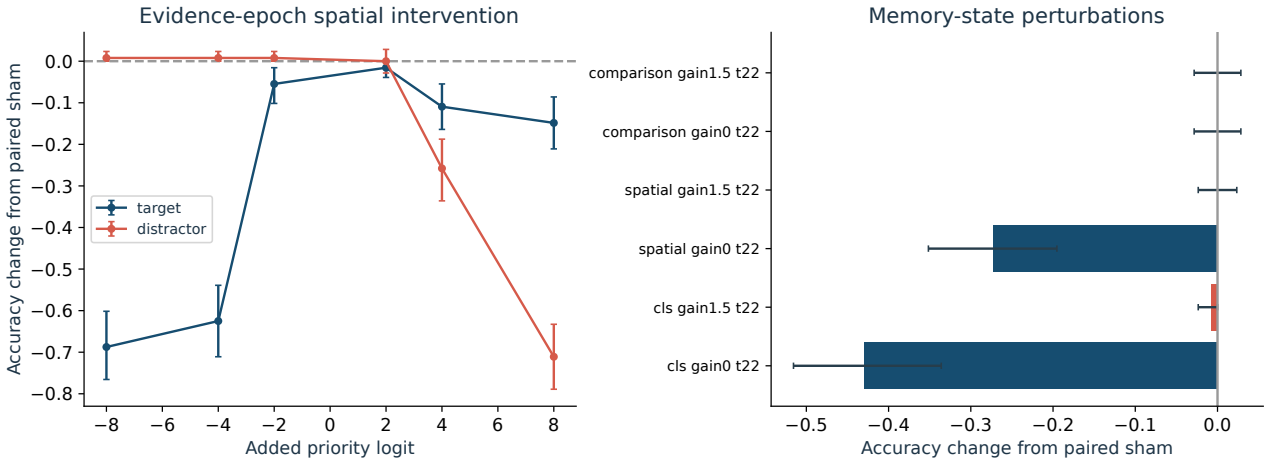


Figure 8: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.6 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
D24	target_-4	-62.5	1.654e-22
D24	target_+4	-10.9	0.008789
D24	distractor_-4	+0.8	1
D24	distractor_+4	-25.8	2.049e-08

3.7 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.500	25.0%	1.000
fusion	0.477	22.7%	0.980
spatial	0.996	94.5%	0.005
cls	1.000	99.2%	0.005
comparison	0.830	48.4%	0.010
attention	0.935	82.0%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

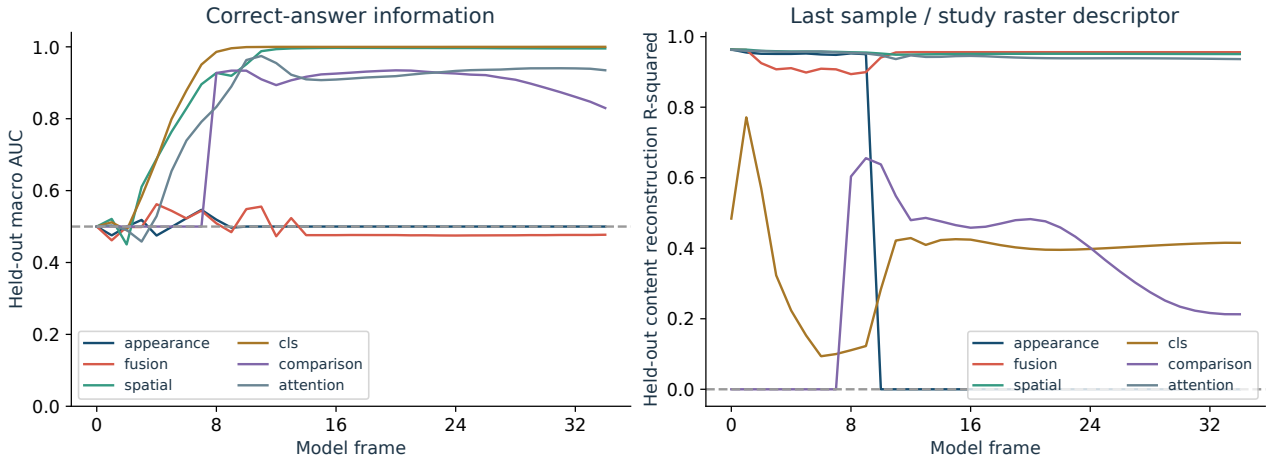


Figure 9: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.8 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 1.80 to 14.35 within the retained subspace. The final step displacement was 0.053 standardized RMS units. In spatial memory, the participation ratio ranged from 2.84 to 19.17 within the retained subspace. The final step displacement was 0.037 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 22, coarse sample-content reconstruction had R-squared 0.395 from CLS memory and 0.952 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -43.0 percentage points, spatial, -27.3 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.9 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	unavailable	unavailable	–
fusion	0.530	0.529	6400
spatial	0.999	0.999	6400
cls	1.000	1.000	64
comparison	0.914	0.957	735
attention	0.960	0.951	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 63 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.10 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

3.11 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting

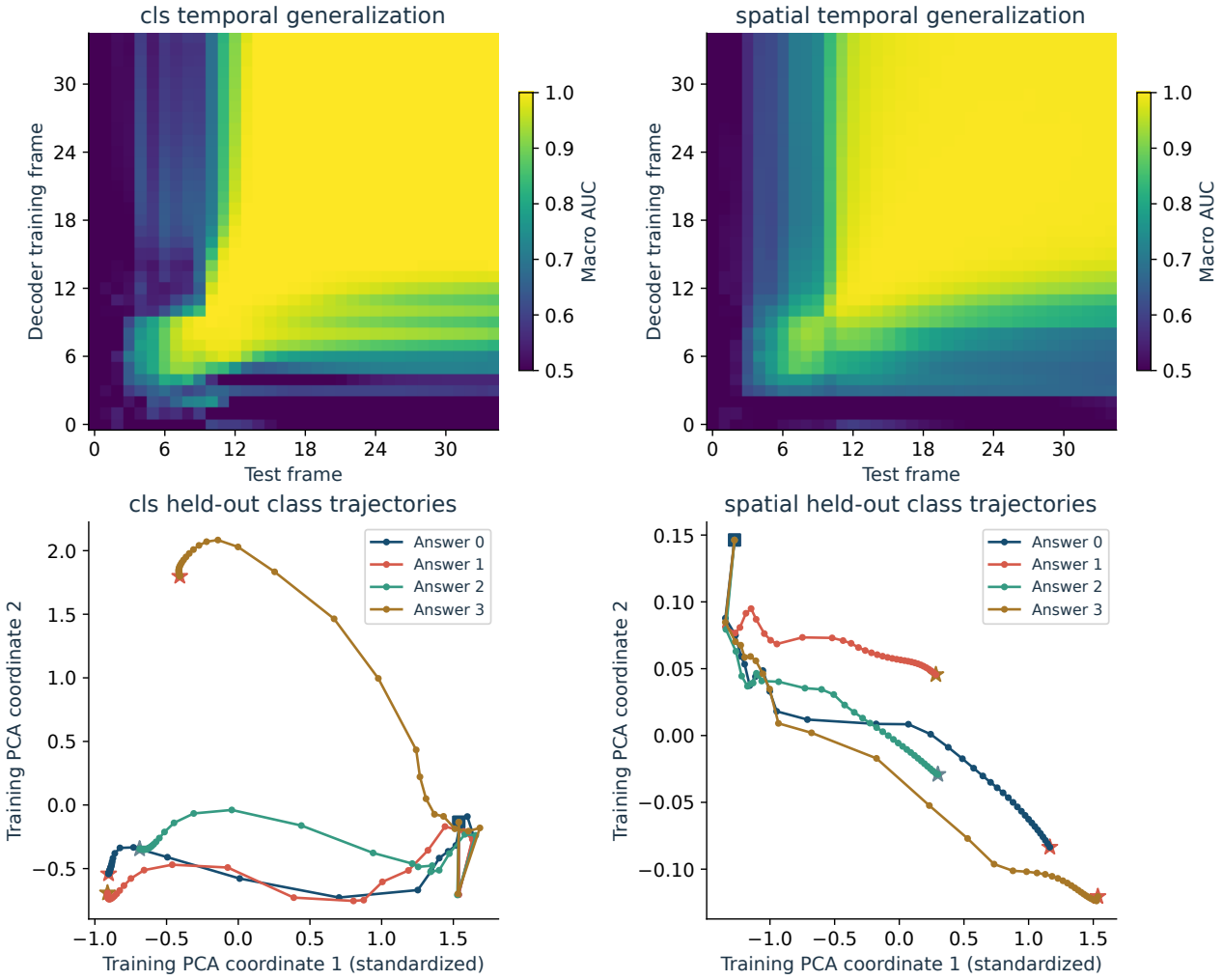


Figure 10: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

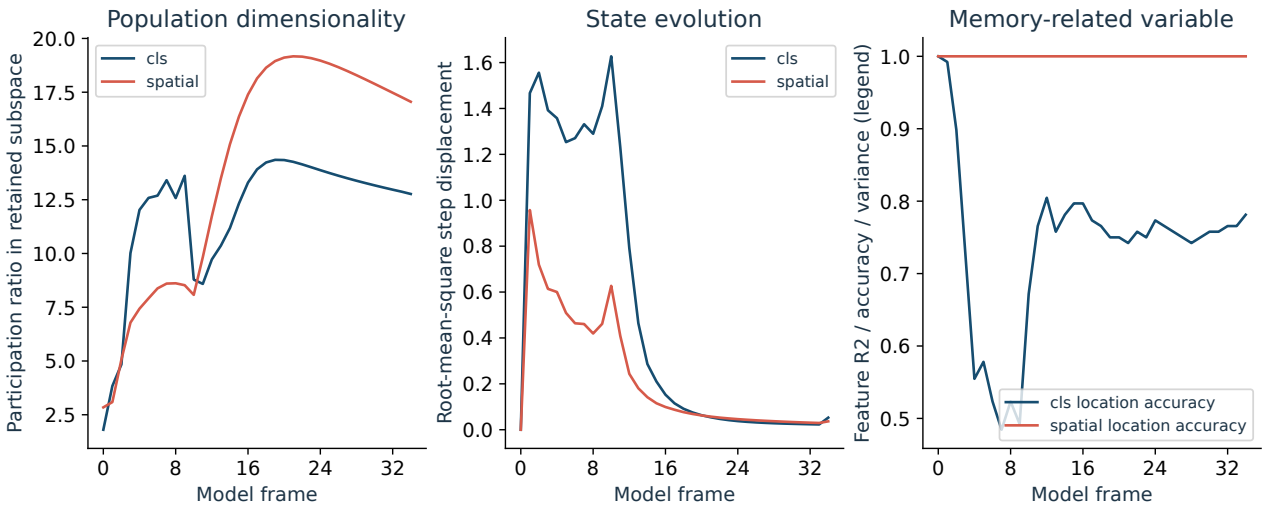


Figure 11: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

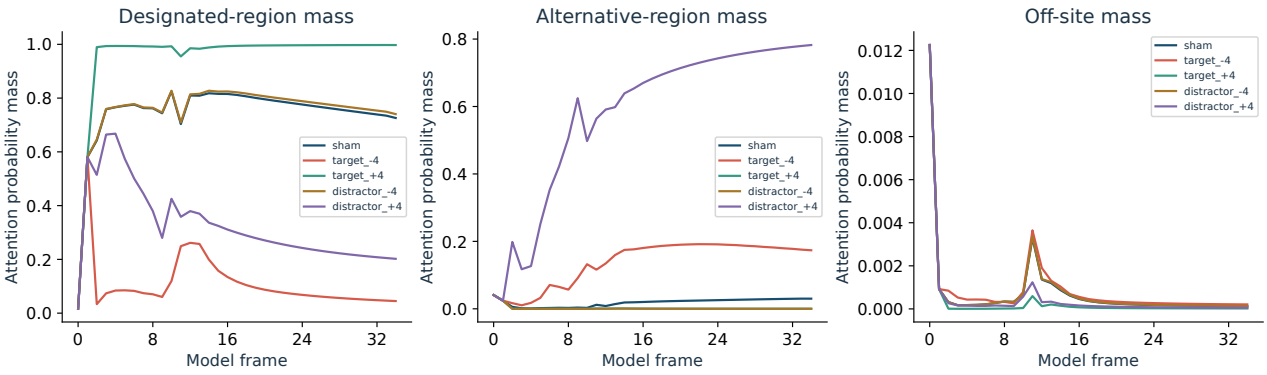


Figure 12: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

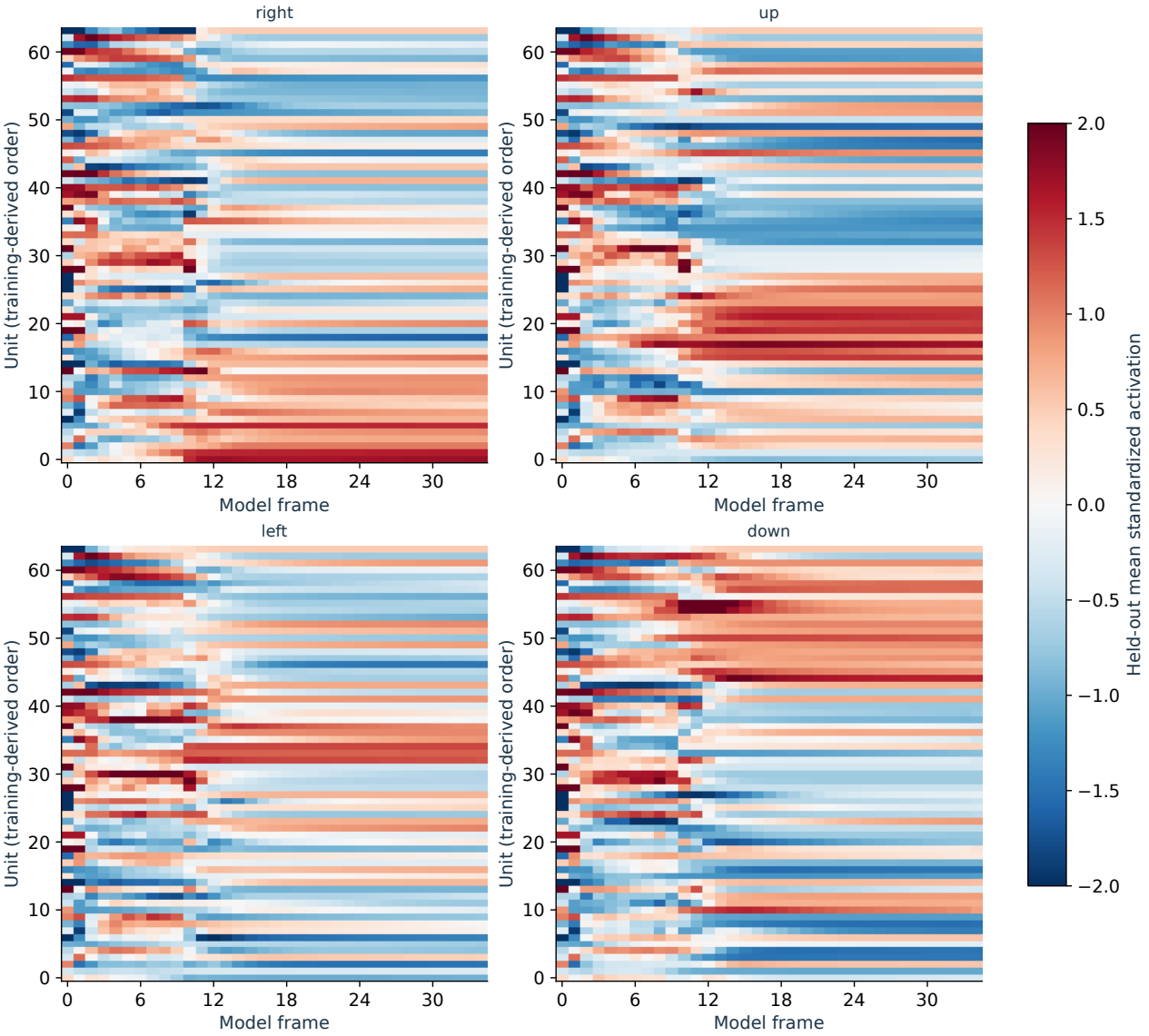


Figure 13: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.12 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

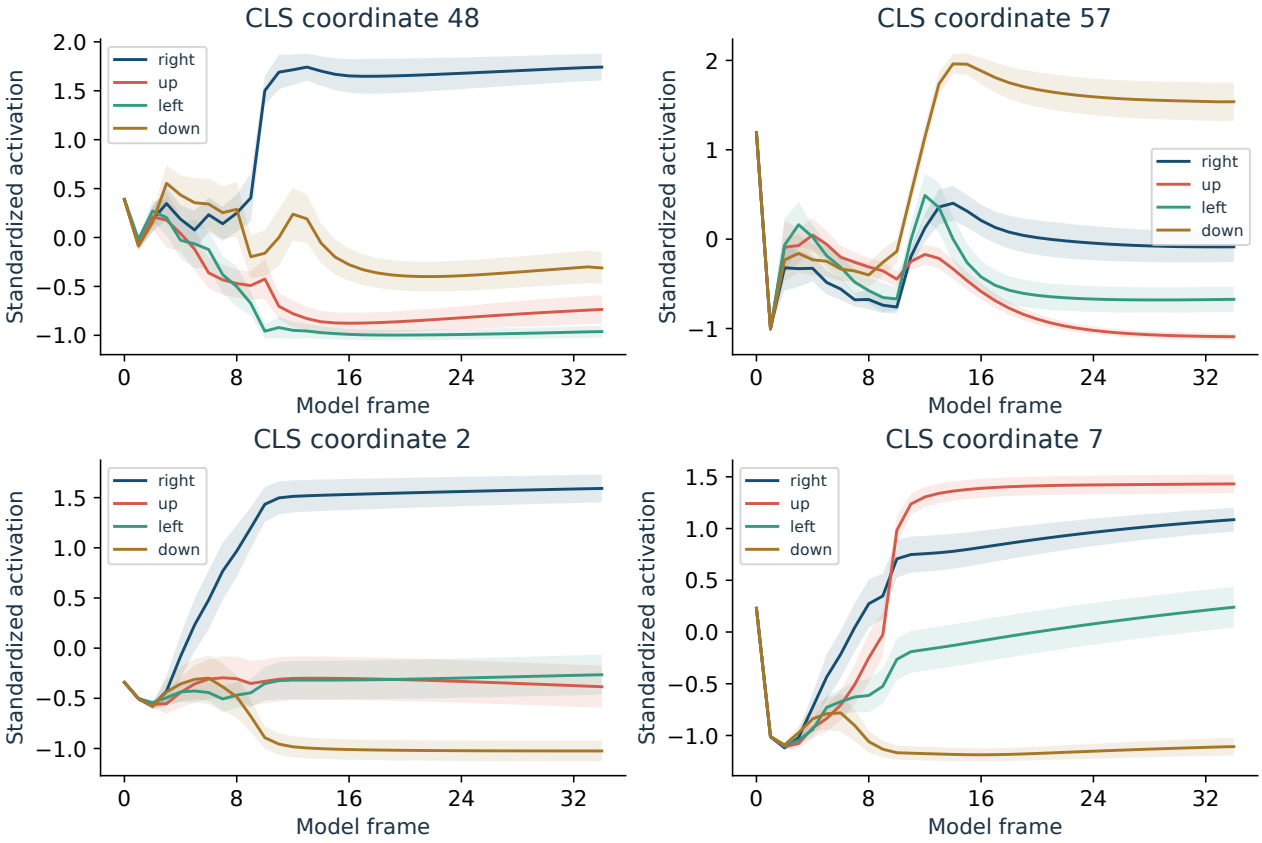


Figure 14: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

Region	Correct answer	Actual choice	Cue/target location
appearance	25.0%	25.0%	25.0%
fusion	22.7%	23.4%	100.0%
spatial	94.5%	95.3%	100.0%
cls	99.2%	99.2%	78.1%
comparison	48.4%	48.4%	55.5%
attention	82.0%	77.3%	100.0%

3.13 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

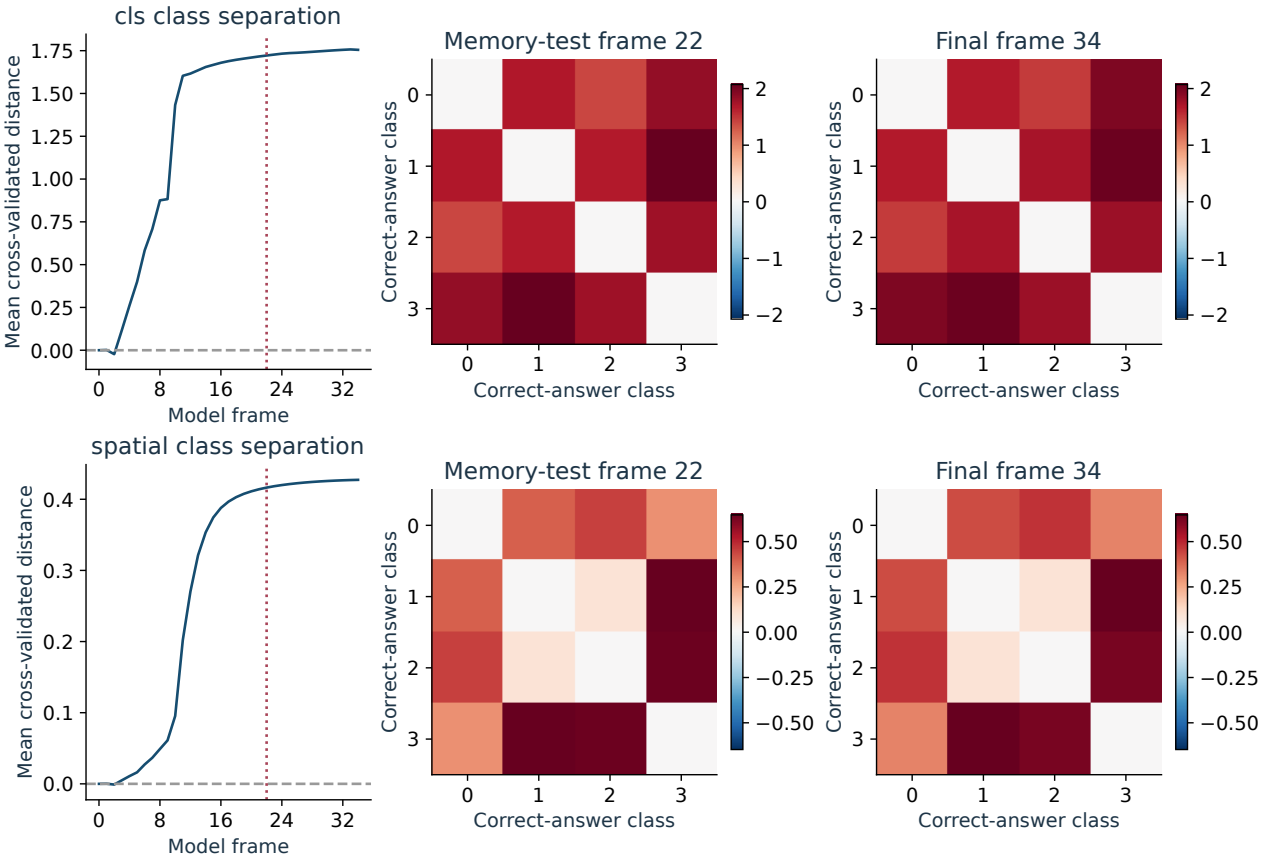


Figure 15: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

3.14 Additional analysis prompted by the biological comparison

The full eight-transition count winner is compared with three fixed shortcuts: report the first direction, report the last direction, or report the most frequent direction in the last three transitions (ties favor the most recent tied direction). These rules use the cued stream and receive perfect access to its native metadata. The trained model receives pixels. Thus this comparison gives each shortcut an especially favorable sensory input.

Within each native condition, there are 128 independent procedural draws. The conflict subset consists of trials on which the shortcut disagrees with the correct count winner. Reported intervals are Wilson 95 percent intervals within that subset. Conditions remain separate; no additional model forwards or fitted parameters are used.

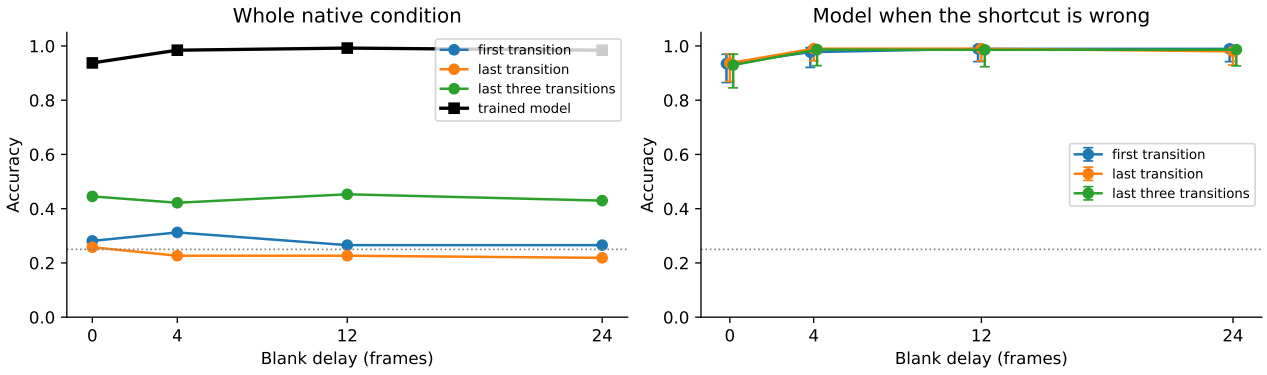


Figure 16: Saved-trial temporal-strategy comparison. Left: trained-model and shortcut accuracies. Right: model accuracy restricted to trials where each shortcut is wrong. Error bars are within-condition Wilson intervals. Dotted line marks four-choice chance. The plotted delay affects retention after the same number of motion transitions.

Delay	n	Shortcut	Rule correct	Conflict n	Model on conflict
0	128	First	28.1%	92	93.5%
0	128	Last three	44.5%	71	93.0%
0	128	Last	25.8%	95	93.7%
4	128	First	31.2%	88	97.7%
4	128	Last three	42.2%	74	98.6%
4	128	Last	22.7%	99	99.0%
12	128	First	26.6%	94	98.9%
12	128	Last three	45.3%	70	98.6%
12	128	Last	22.7%	99	99.0%
24	128	First	26.6%	94	98.9%
24	128	Last three	43.0%	73	98.6%
24	128	Last	21.9%	100	98.0%

The model remains accurate when these short-history rules fail. This directly rejects the tested shortcut descriptions for these movies. It establishes use of a richer temporal summary; order-controlled evidence is the next test for distinguishing exact counts from other recurrent summaries.

4 Discussion

The final visual input need not display the accumulated winning direction, yet that direction is strongly readable in recurrent memory. Resetting global or spatial memory during retention impairs the report, and reducing target priority during the moving evidence produces a particularly large cost. This connects performance to both selecting the correct stream and preserving a summary of its temporal history. The comparison-state decoder also contains answer information, but a single reset of that state had little effect at the selected instant. The measured dependencies therefore favor recurrent integration as a major contributor to the duration judgement, while leaving the temporal front end available throughout the movie.

The behavioral result establishes 97.5% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies distractor_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Varying the winner margin, order of transitions and total sequence length independently would reveal whether the model counts evidence, averages motion populations, or disproportionately weights recent observations.

4.1 Interpreting these results alongside biological experiments

4.1.1 Kiani, Hanks and Shadlen (2008)

Our task asks which direction occupied the greatest number of transitions, whereas the monkey task accumulated noisy evidence for a direction. Both require a temporal decision variable, but a winning count and a likelihood ratio are different summaries. Early confident model outputs can also be revised because training supervises the terminal report. Threshold crossing should therefore be compared with actual persistence of the choice. [5]

4.1.2 Stine, Zylberberg, Ditterich and Shadlen (2020)

High accuracy by itself can sometimes be explained by a shortcut using only a small part of the sequence. The additional reanalysis below directly compares the trained model with first-direction, last-direction and final-three-transition rules on the same saved trials. Performance when a shortcut disagrees with the true count winner is the diagnostic subset. [6]

4.1.3 Mante, Sussillo, Shenoy and Newsome (2013)

The cued patch identifies which stream should contribute to the temporal computation. Target-priority inhibition during motion is especially harmful, and resetting recurrent memory during retention also

harms the report. This combines a selection requirement with a storage requirement. The relation to context-dependent recurrent computation is strongest at that functional level, with the task's count rule specifying the information to integrate. [7]

4.2 A computational account of the observed behavior

A useful reference computation maintains four counts, one for each direction, and increments only the count corresponding to the cued patch's current direction. At report, the largest count wins. The trained network is free to represent equivalent differences between counts rather than literal counters. The final image need not contain the winning direction, which makes retained history necessary on diagnostic trials. A leaky summary would overweight late transitions; an early commitment rule would underweight evidence after commitment. The saved sequence comparison tests simple versions of those alternatives. Order-controlled movies would further distinguish exact counting from an approximate recurrent summary that happens to agree on the native sequences.

4.3 Where this task agrees with the biological evidence

The existing retention and attention interventions support selected temporal integration. The new shortcut comparison quantifies how much the behavior exceeds single-observation explanations. A full temporal-weighting account additionally requires count-matched sequences with systematically rearranged early and late evidence.

4.4 A follow-up experiment motivated by these results

Keep direction counts fixed and reverse or permute their order. Then exchange early and late evidence while preserving the count margin. Fit temporal weights on held-out sequences and test whether a learned bound, leak or uniform count best predicts errors.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Salzman CD, Britten KH, Newsome WT (1990). Cortical microstimulation influences perceptual judgements of motion direction. Nature 346:174-177. https://pubmed.ncbi.nlm.nih.gov/2366872/.

[2] Funahashi S, Bruce CJ, Goldman-Rakic PS (1989). Mnemonic coding of visual space in the monkey dorsolateral prefrontal cortex. Journal of Neurophysiology 61:331-349. https://pubmed.ncbi.nlm.nih.gov/2918358/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] Kiani, Hanks and Shadlen (2008). Bounded integration in parietal cortex underlies decisions even when viewing duration is dictated by the environment. Primary source.

[6] Stine, Zylberberg, Ditterich and Shadlen (2020). Differentiating between integration and non-integration strategies in perceptual decision making. Primary source.

[7] Mante, Sussillo, Shenoy and Newsome (2013). Context-dependent computation by recurrent dynamics in prefrontal cortex. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
D0	128	93.8%	0.998	0.088
D4	128	98.4%	1.000	0.021
D12	128	99.2%	1.000	0.016
D24	128	98.4%	1.000	0.015

B Every-frame attention atlas

Condition D24. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 8 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

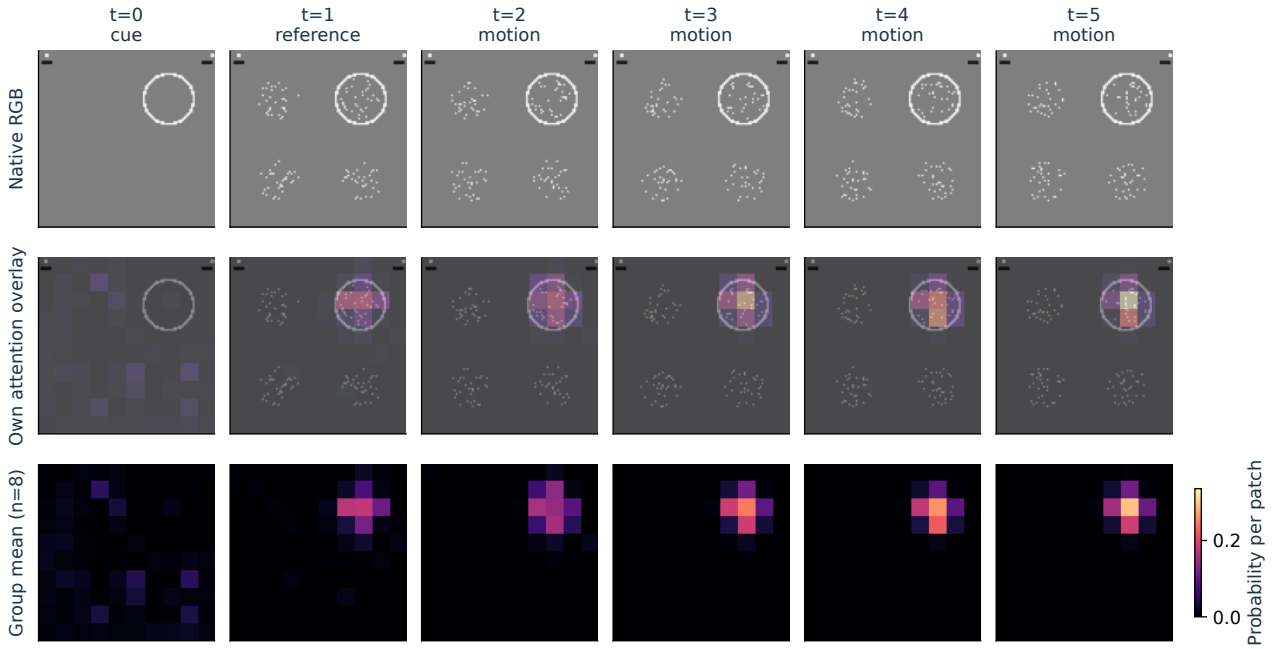


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

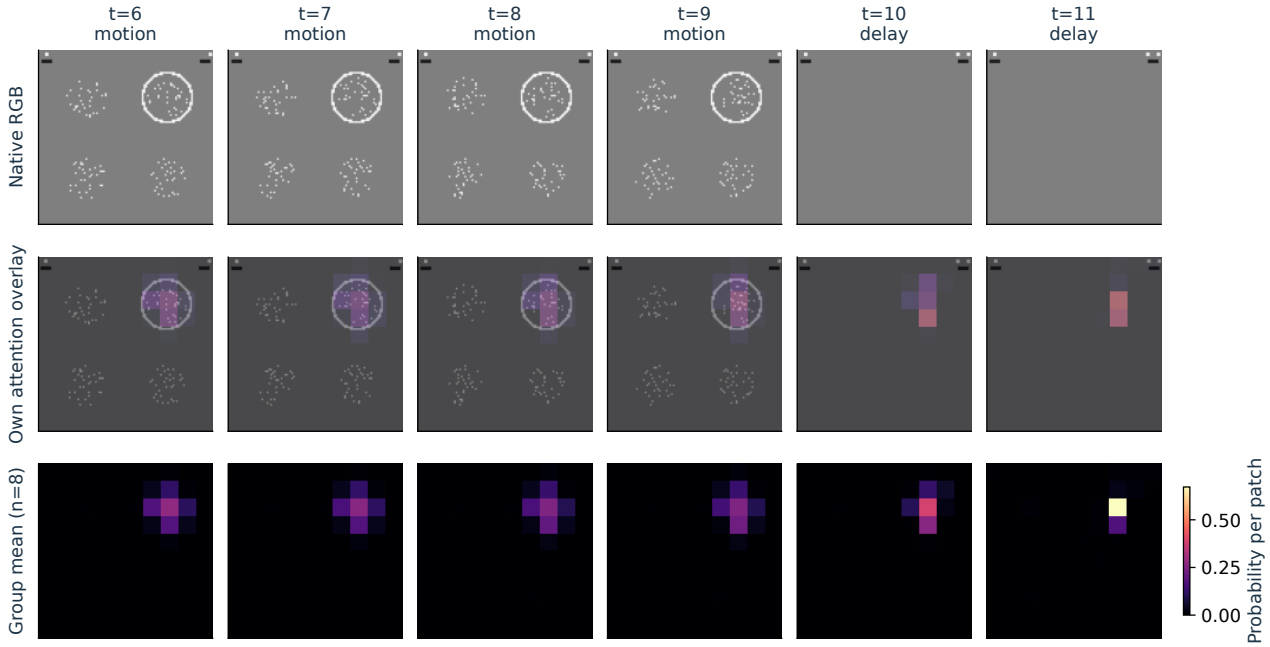


Figure 18: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

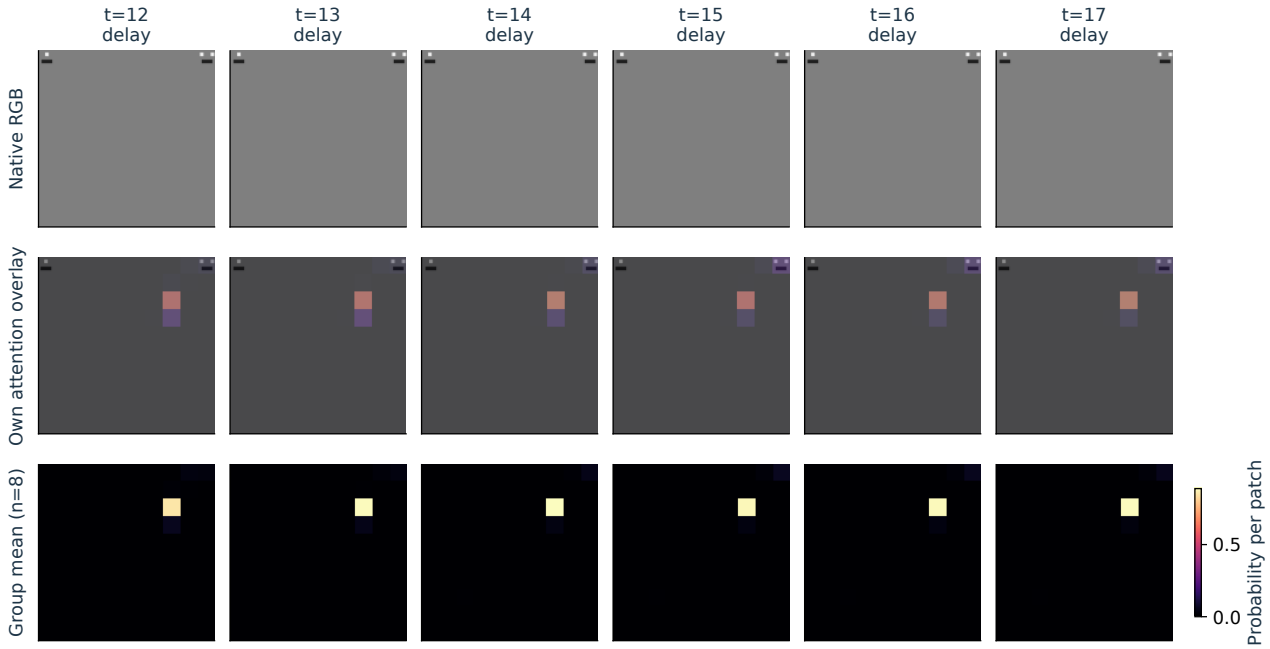


Figure 19: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

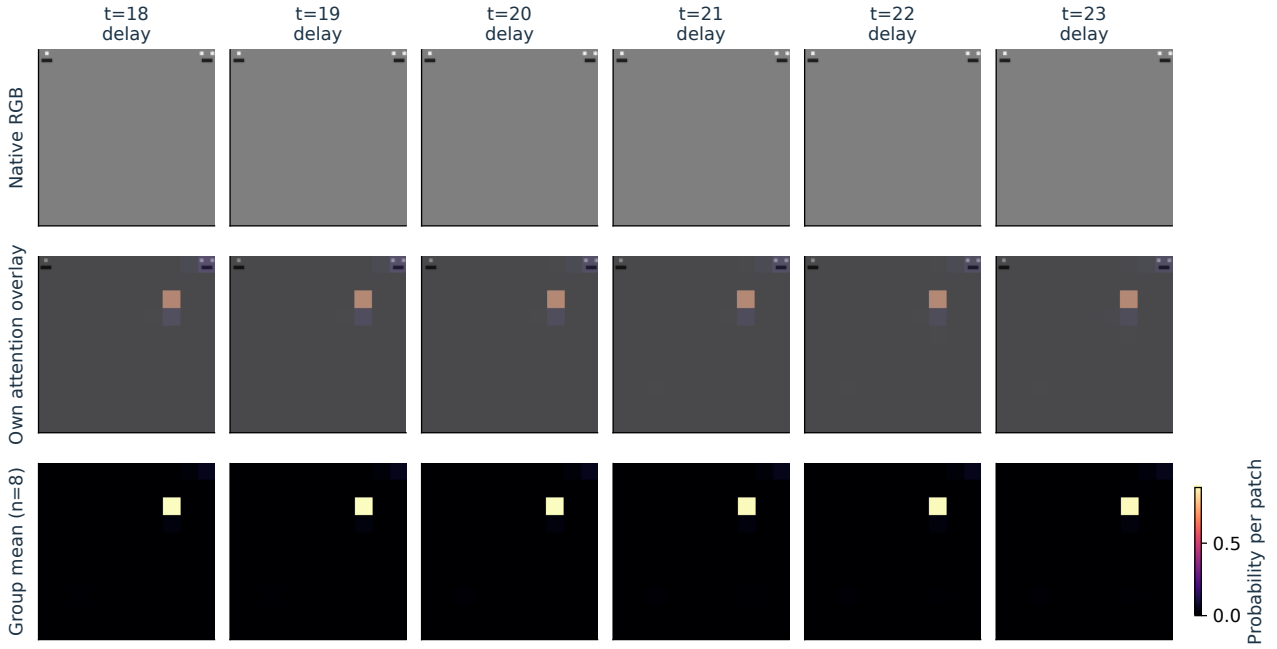


Figure 20: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

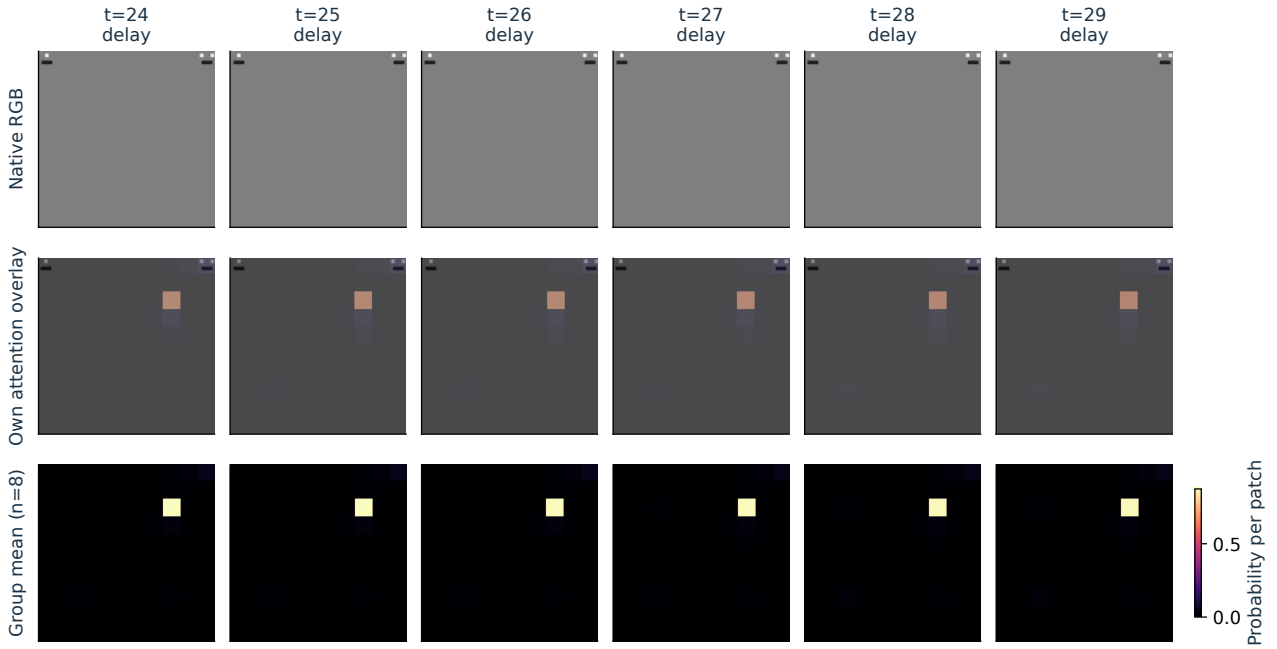


Figure 21: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

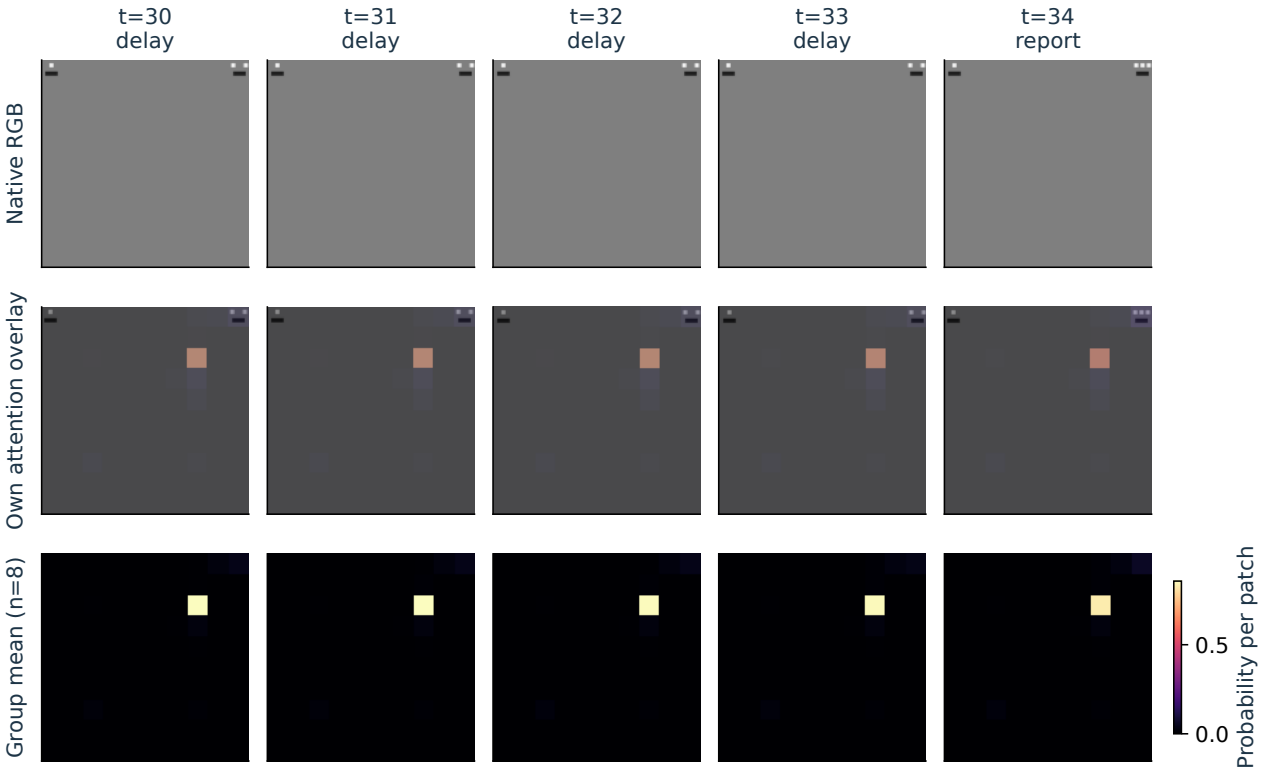


Figure 22: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

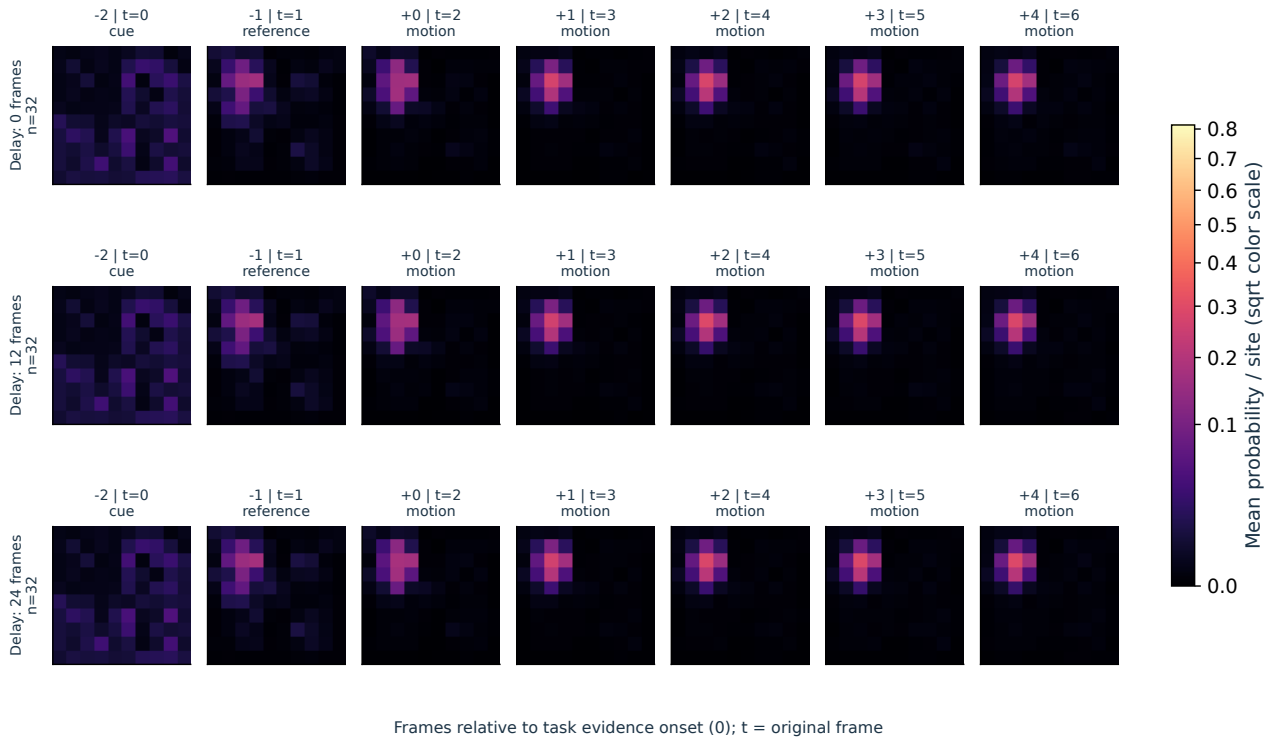


Figure 23: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

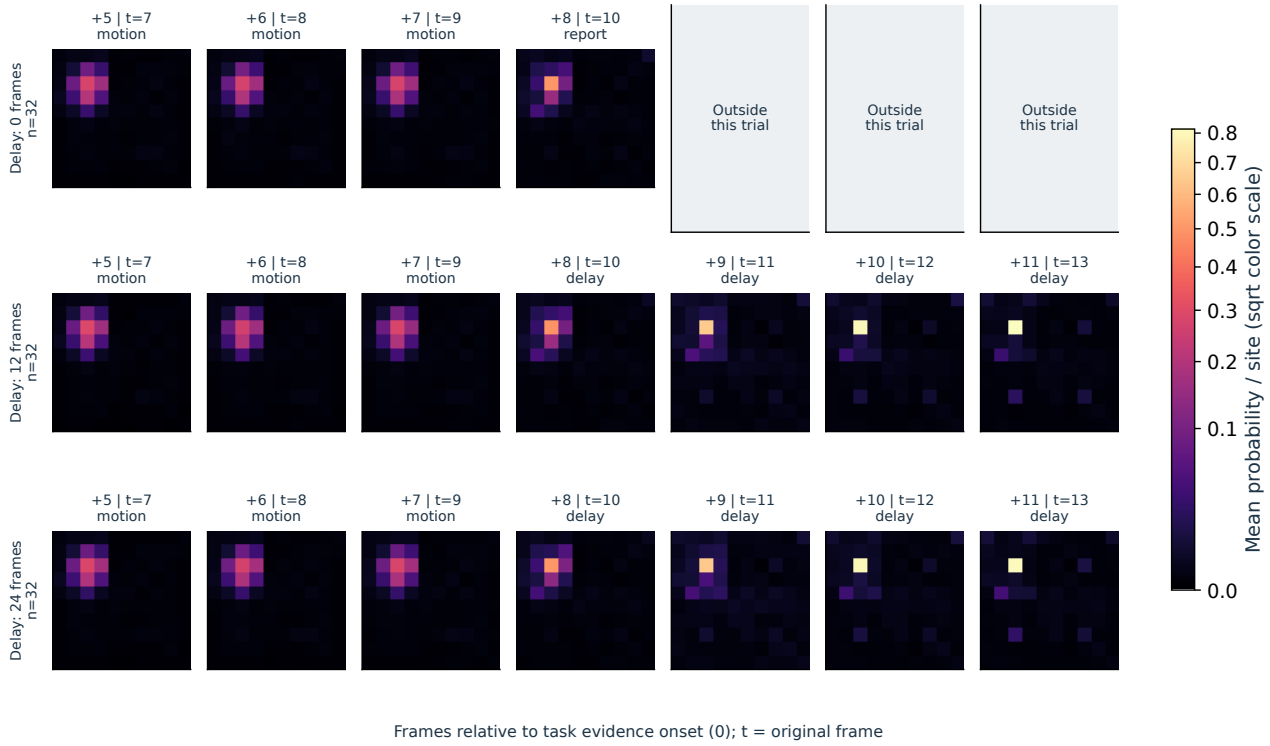


Figure 24: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

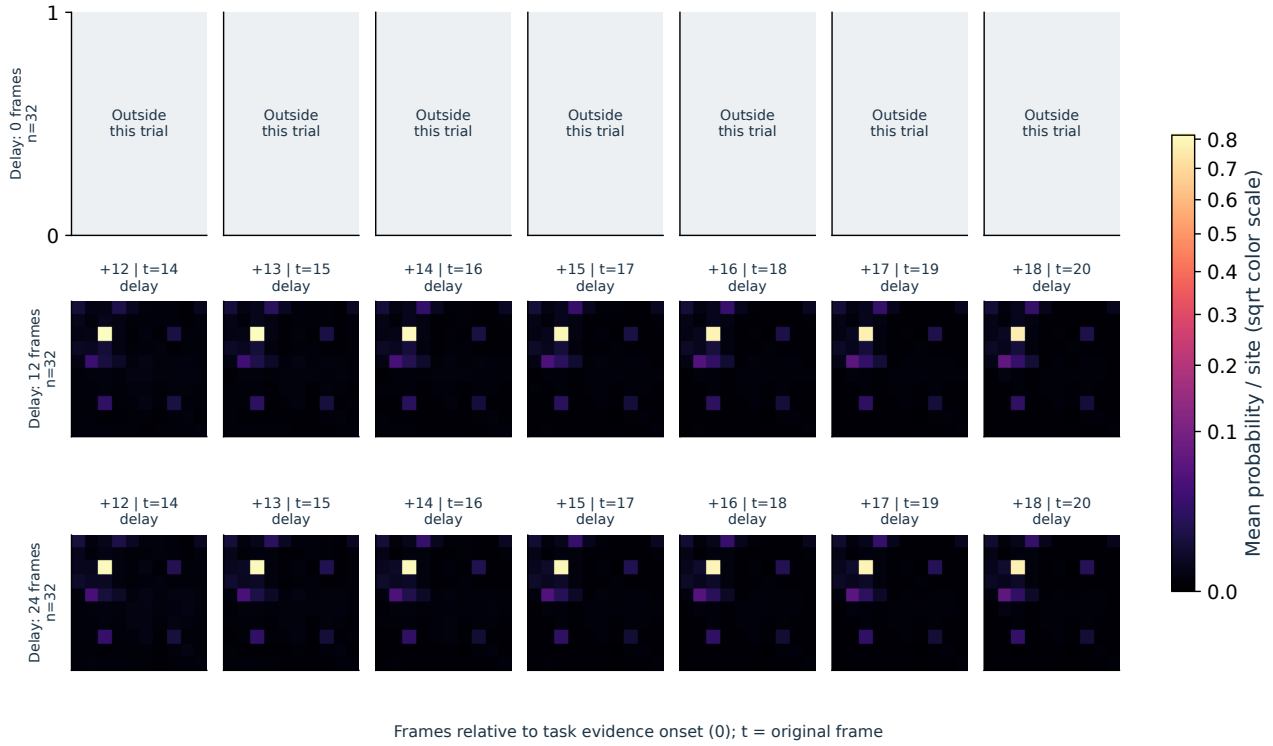


Figure 25: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

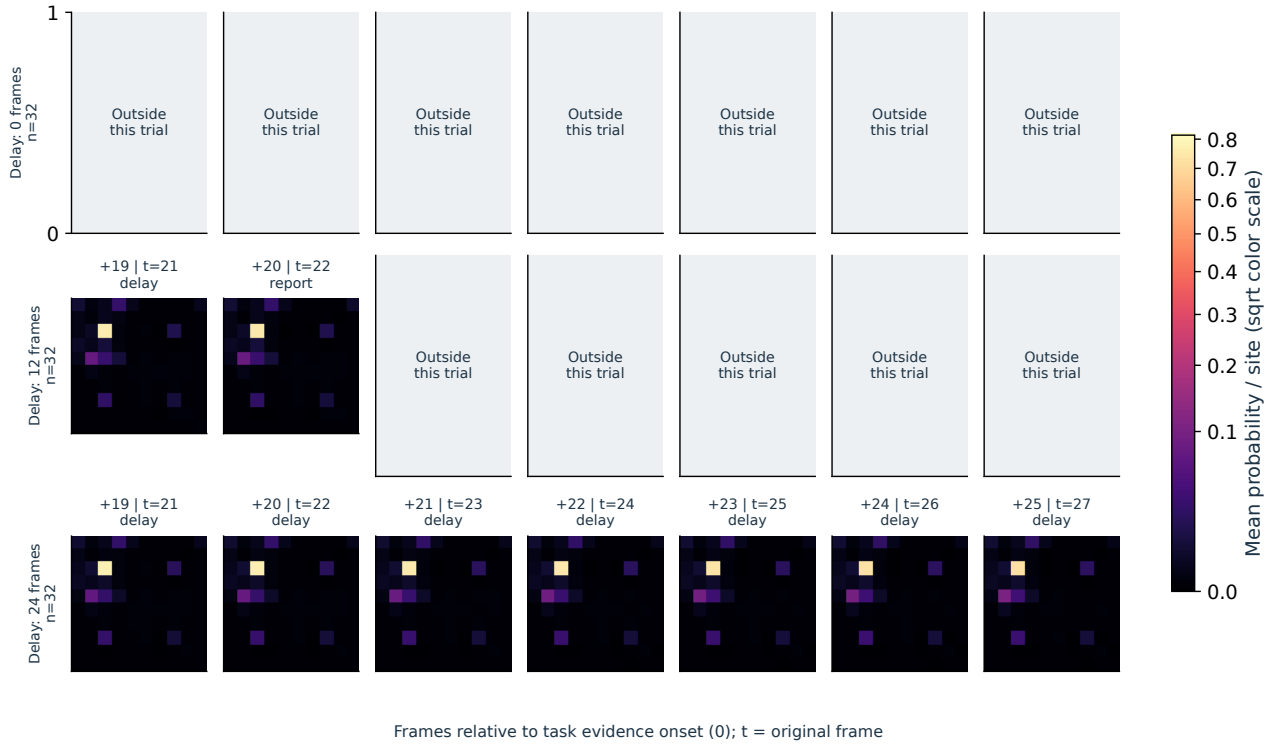


Figure 26: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

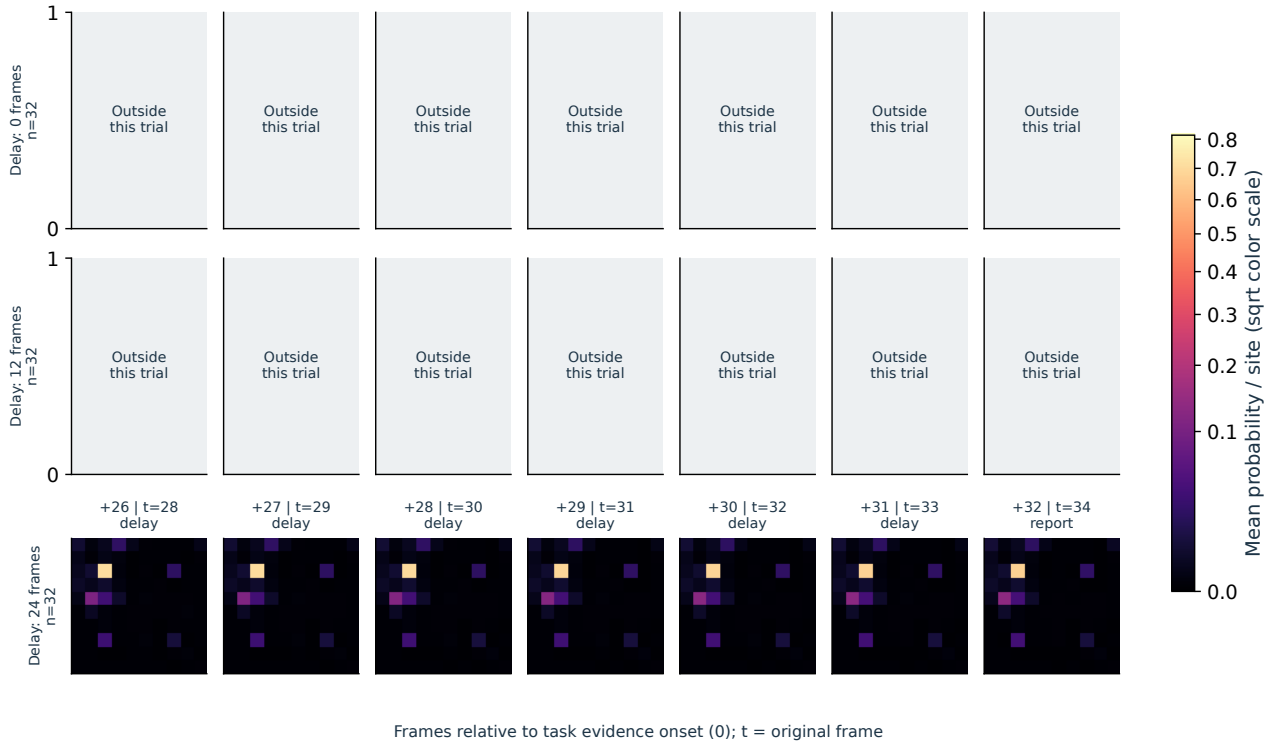


Figure 27: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Target selection and distractor rejection in cued motion-change detection

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized krauzlis cued motion in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 3 native conditions and 384 fresh trials, equal-condition accuracy was 95.8% (95% interval [93.8%, 97.7%]). For the representative condition B28, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 0.910. The largest observed impairment in that condition was distractor_+8_evidence, changing accuracy by -63.3 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

A precue identifies one of two moving-dot patches. The observer reports a change in that patch and withholds a change report for foil-only changes and catches. This makes selection behaviorally essential: strong evidence that a change occurred somewhere is not sufficient to answer correctly.

We vary motion-change magnitude and preserve target, foil and catch events as separate categories. Hits measure detection of relevant changes; reports on foil trials measure distractor capture; catch reports measure spontaneous false alarms. Their separation permits a precise test of the proposed dual effect of spatial perturbation.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Arcizet and Krauzlis studied covert spatial selection in a cued motion-change paradigm, and reversible inactivation studies establish causal roles for the superior colliculus and frontal eye fields. These experiments motivate a distinction between detecting a physical change and selecting the behaviorally relevant change. A model can contain clear foil evidence and still correctly withhold a positive report. The most diagnostic intervention therefore measures hits, foil-driven errors and catches separately. Inhibiting the target and inhibiting the foil need not have symmetric behavioral consequences, because the instruction assigns different meanings to the same physical event at the two sites.

1.2 Biological experiments motivating this task

Experimental question. Which moving signal is allowed to influence a covert change report?

Arcizet and Krauzlis (2018) Monkeys detected cued motion changes with a joystick while maintaining fixation. Caudate neurons encoded relevant location and task epoch, extending spatial-selection signals beyond overt orienting movements. [5]

Bollimunta, Bogadhi and Krauzlis (2018) Reversible FEF and SC inactivation impaired covert motion-change detection. Comparing the two interventions in a shared behavioral framework tested their contributions to selection during maintained fixation. [6]

Lovejoy and Krauzlis (2010) SC inactivation shifted motion judgments away from signals in the affected field toward competing information elsewhere. Deficits depended strongly on informative distractors; affecting the foil location favored judgments based on the cued signal. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Did mean motion direction change in the precued patch rather than the foil?

Response labels are 0: foil-only change or catch; 1: target change. The native suite contains 3 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use B28.

1.4 Stimulus sequence and spatial layout

For the representative native condition B28, the sequence has 45 frames, indexed from zero: frames 0–1: cue; frames 2–6: fixation; frames 7–35: baseline; frames 36–43: change; frame 44: report. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Dots per aperture	16
Displacement per transition (pixels)	0.375
Aperture radius (pixels)	8.125

Stimulus centers in the 100 by 100 input are [[20.0, 50.0], [80.0, 50.0]]. Coordinate x increases rightward and y downward.

The long baseline makes delayed local motion comparisons available. One shared spatial distribution gates both learned visual values and explicit comparison features. Inhibiting a selected site can reduce its contribution while increasing the relative contribution of the other site. The causal test is whether these competing changes in contribution produce reciprocal behavioral costs and benefits.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 95.8%, with macro AUC 0.995. The matched final checkpoint changed accuracy by +2.08 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was B20 at 95.3%; the highest was B12 at 96.9%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (4) produced 3.1% correct responses, compared with 100.0% at the largest strength (60). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

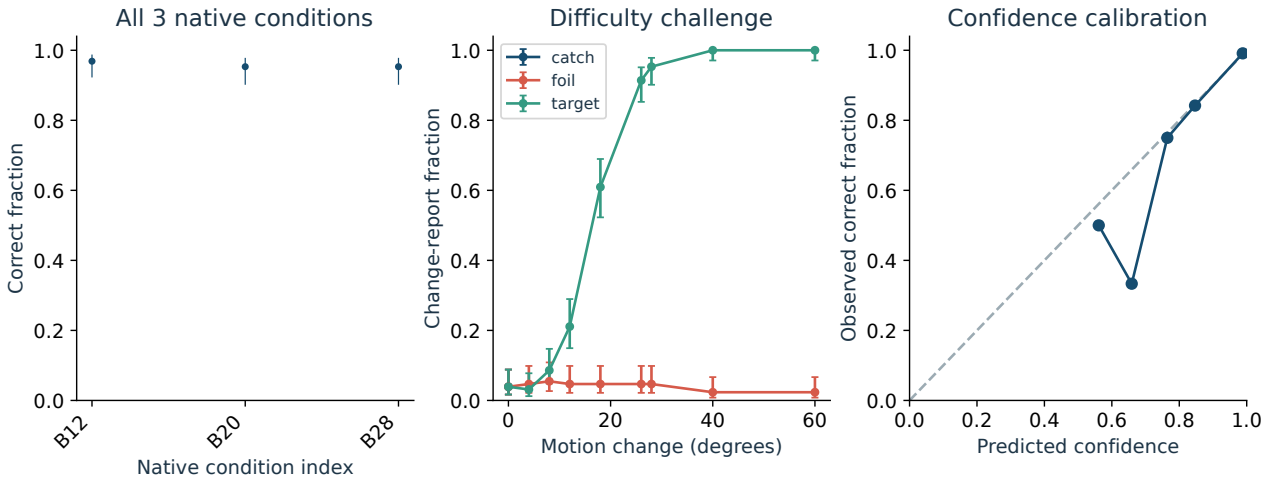


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	44.5%	100.0%
0.7	100.0%	1.0	43.8%	100.0%
0.8	100.0%	1.0	46.1%	100.0%
0.9	100.0%	1.0	50.0%	100.0%

The representative condition B28 provides 9 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.3 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

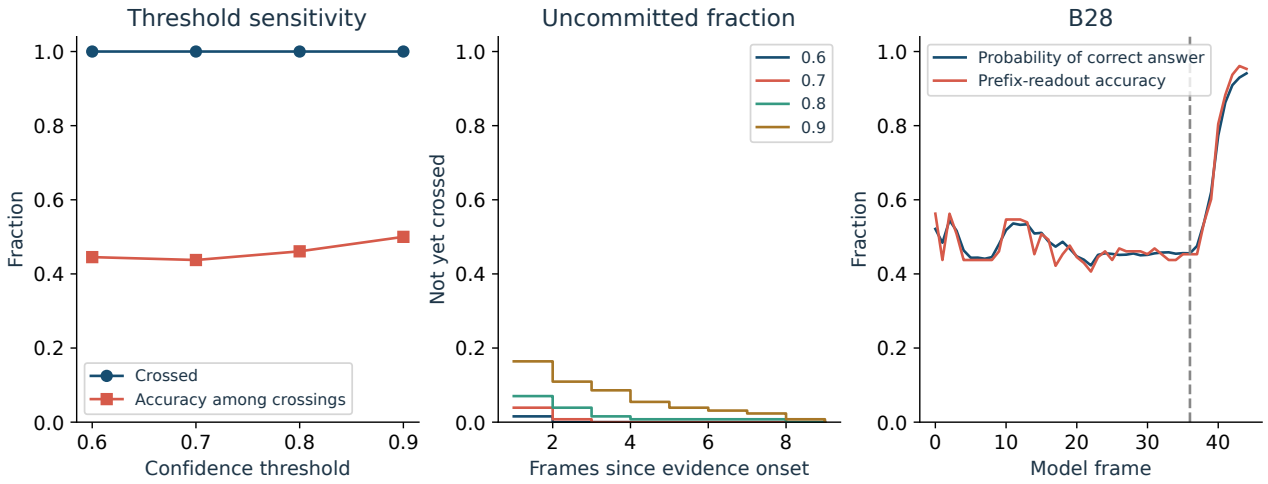


Figure 2: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

At threshold 0.8 across all native presentations, 384 of 384 trials crossed. Of those, 201 changed their answer later, 197 corrected an early error, and 4 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

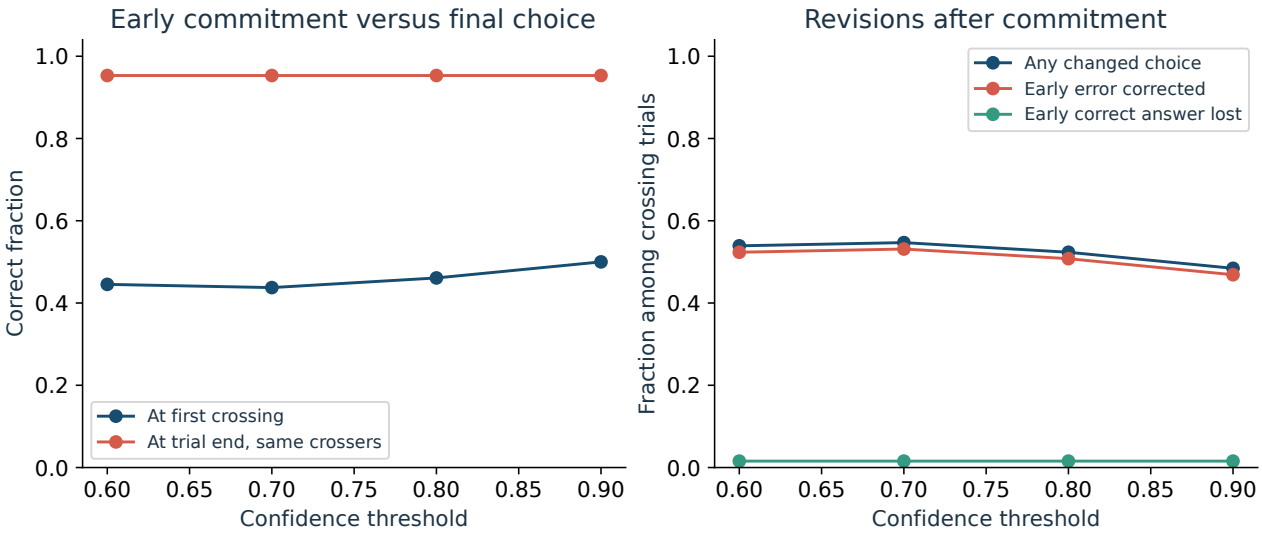


Figure 3: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

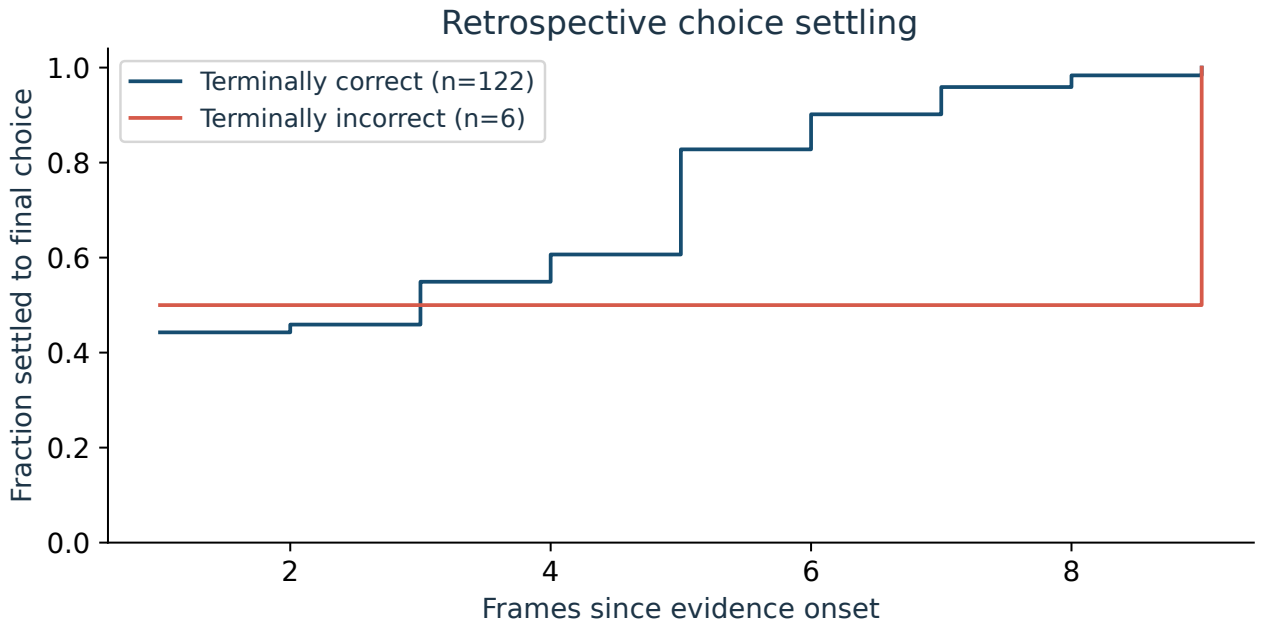


Figure 4: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.4 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The cue is fixed at the right patch, with 28 baseline transitions. Rows separate a target change, a left-foil change, and a catch trial. The shared time origin is the first postchange frame, including its scheduled counterpart on catches.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Change at right cue	37	37
Change at left foil	21	21
No change	9	9

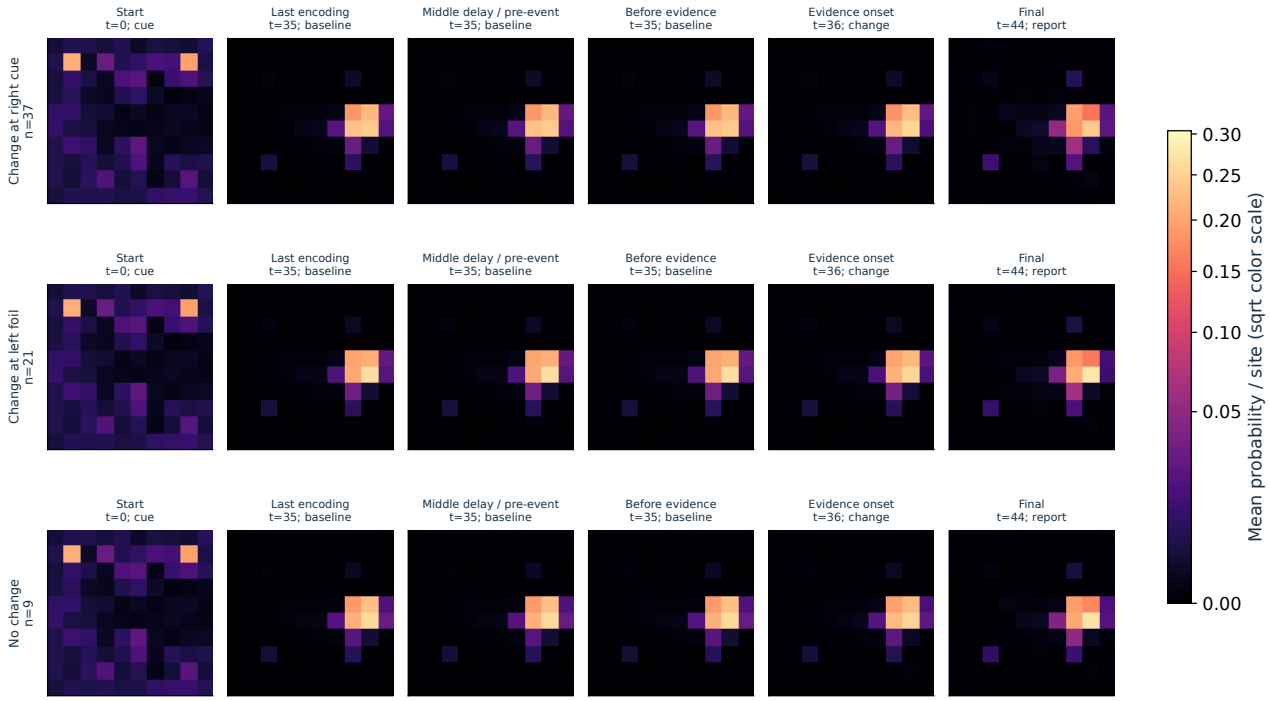


Figure 5: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

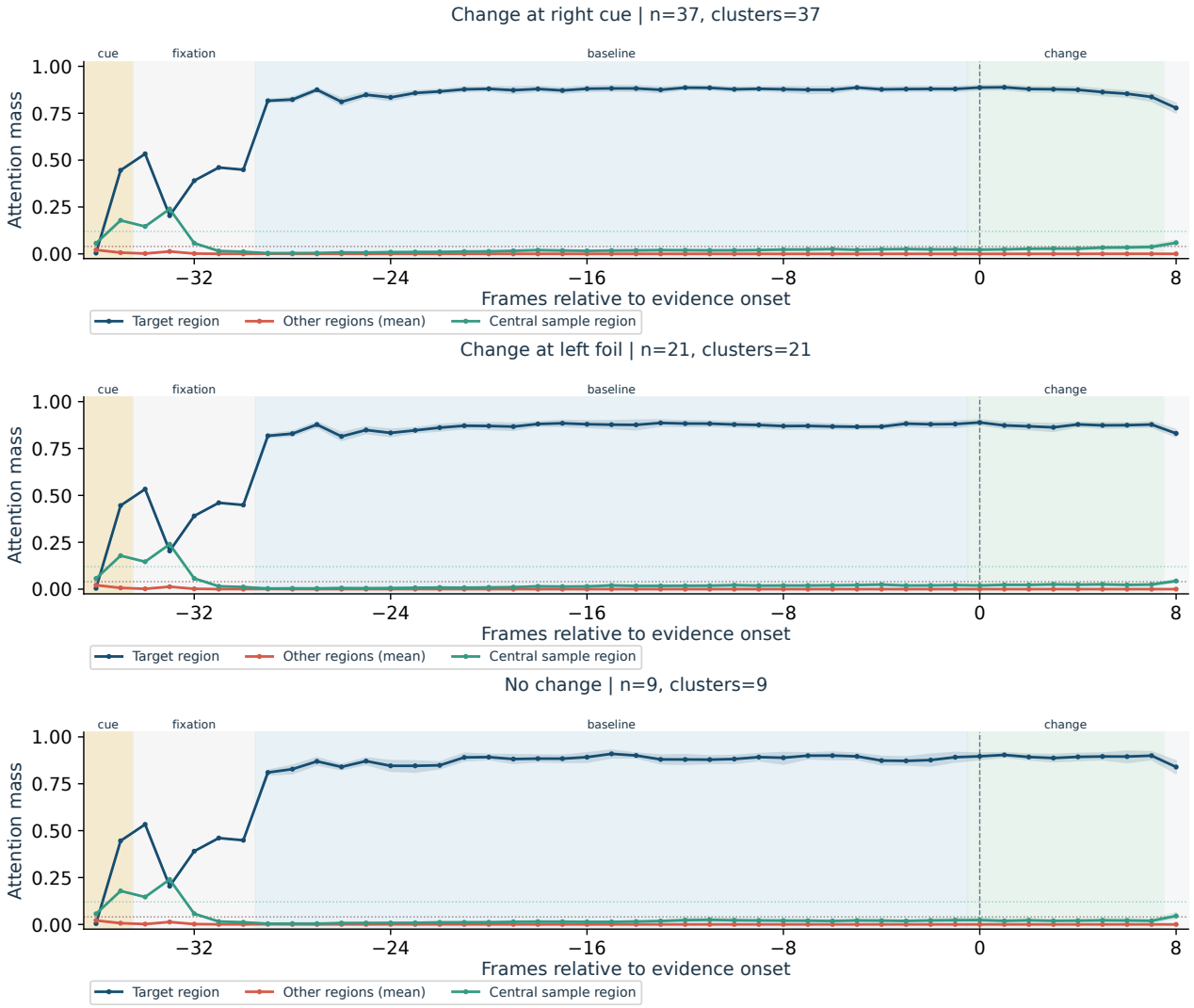


Figure 6: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

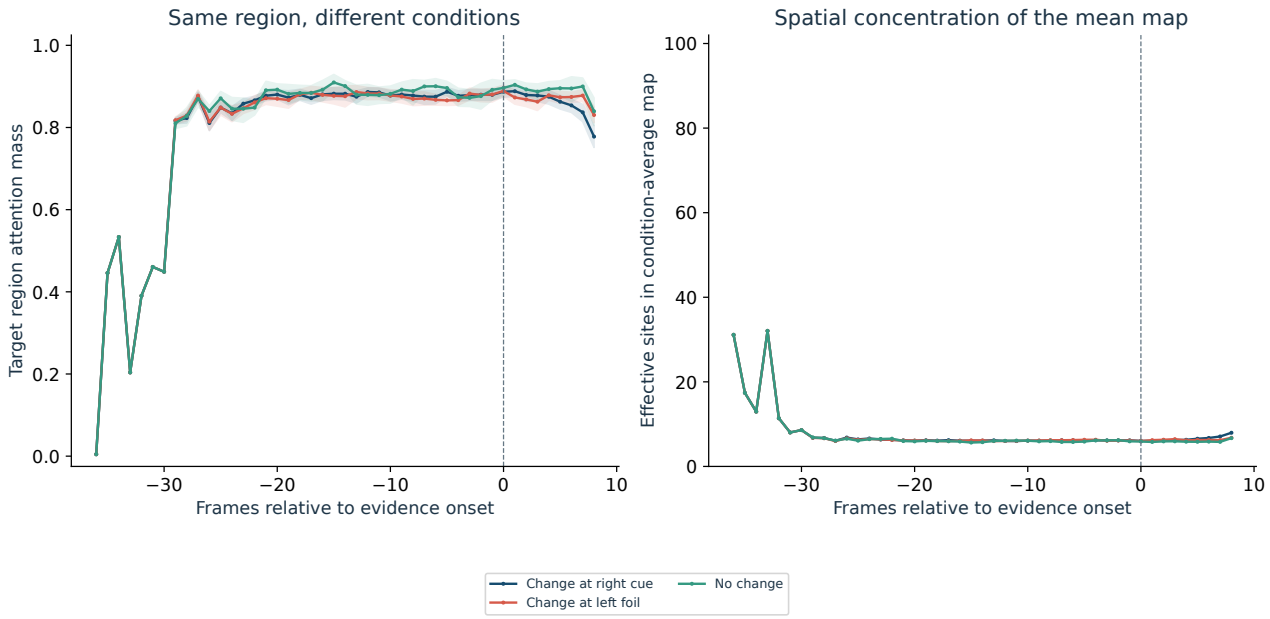


Figure 7: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Change at right cue: target region mass peaks at 0.888 at original frame 37 (change), and is 0.778 on the final frame. Change at left foil: target region mass peaks at 0.889 at original frame 36 (change), and is 0.830 on the final frame. No change: target region mass peaks at 0.910 at original frame 21 (baseline), and is 0.839 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.5 Causal sensitivity to spatial and mnemonic interventions

For distractor_+8_evidence, paired accuracy changed by -63.3 percentage points (paired 95% interval [-71.9%, -53.9%]; paired p=1.89e-22; within-task Holm p=5.859e-21). There were 2 corrected trials and 83 newly incorrect trials. For spatial_gain1.5_t35, paired accuracy changed by +0.8 percentage points (paired 95% interval [0.0%, 2.3%]; paired p=1; within-task Holm p=1). There were 1 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

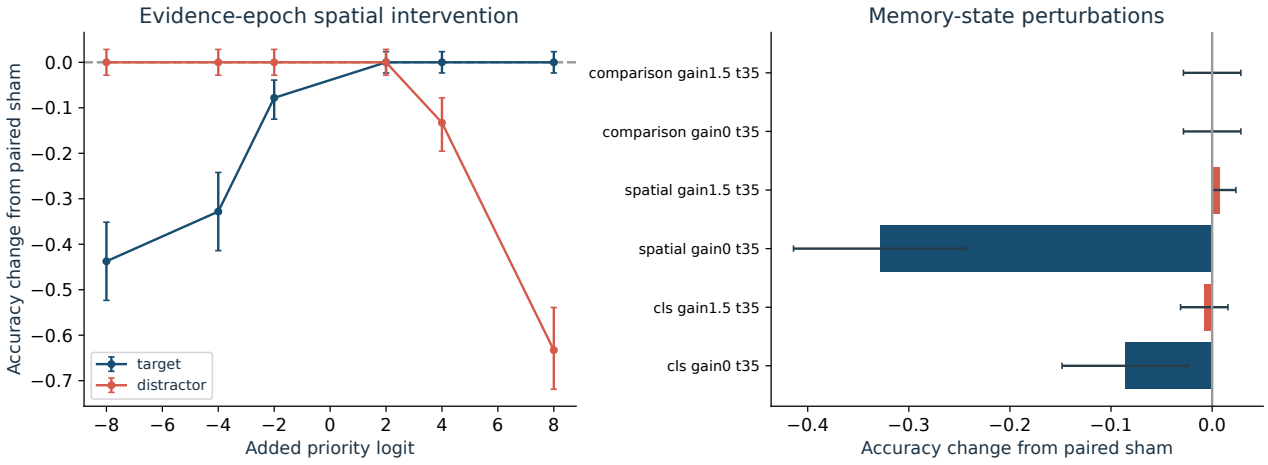


Figure 8: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.6 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
B28	target_-4	-32.8	4.553e-10
B28	target_+4	+0.0	1
B28	distractor_-4	+0.0	1
B28	distractor_+4	-13.3	0.001205

3.7 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.500	50.0%	1.000
fusion	0.508	50.0%	0.815
spatial	0.608	56.5%	0.070
cls	0.910	84.9%	0.005
comparison	0.886	76.8%	0.005
attention	0.739	71.1%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.



Figure 9: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.8 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 1.00 to 17.58 within the retained subspace. The final step displacement was 1.589 standardized RMS units. In spatial memory, the participation ratio ranged from 1.00 to 17.47 within the retained subspace. The final step displacement was 1.343 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 35, coarse sample-content reconstruction had R-squared 0.233 from CLS memory and 0.994 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -8.6 percentage points, spatial, -32.8 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.9 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

3.10 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	unavailable	0.494	–
fusion	0.518	0.496	6400
spatial	0.727	0.415	6400
cls	0.924	0.399	64
comparison	0.869	0.428	735
attention	0.720	0.467	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 13 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.11 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

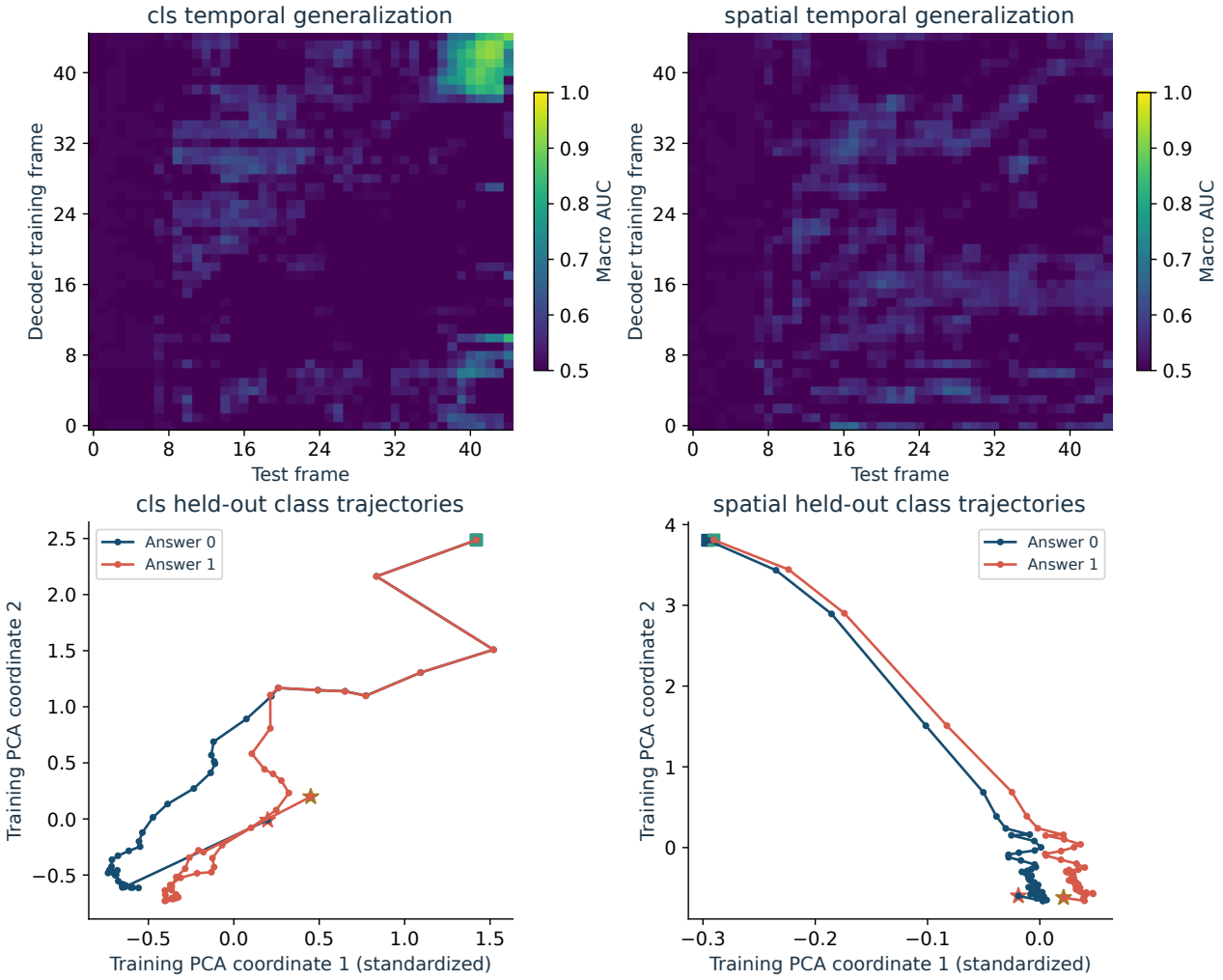


Figure 10: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

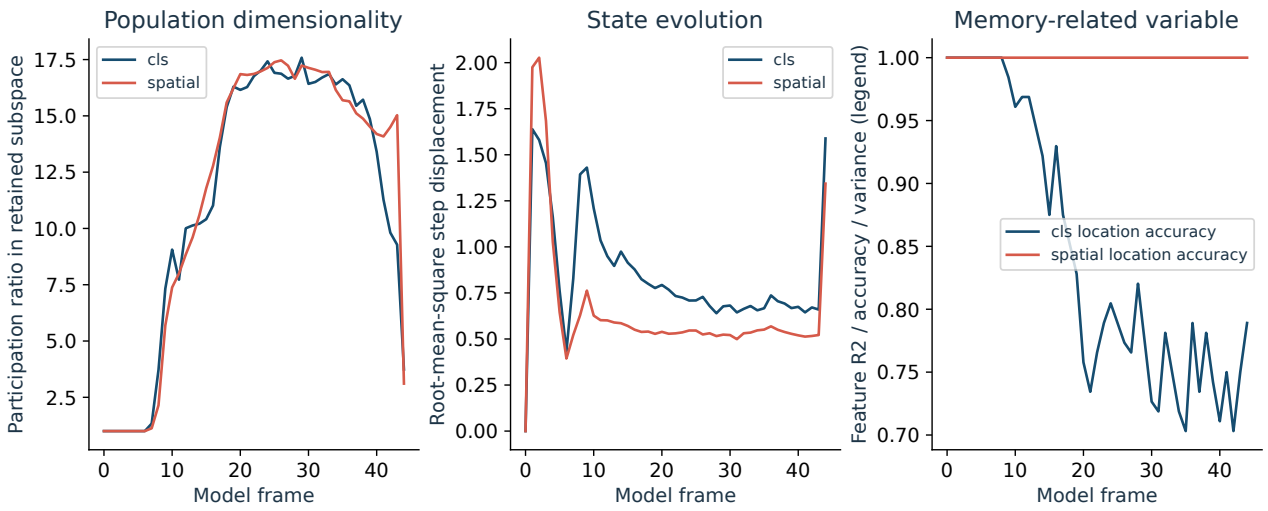


Figure 11: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

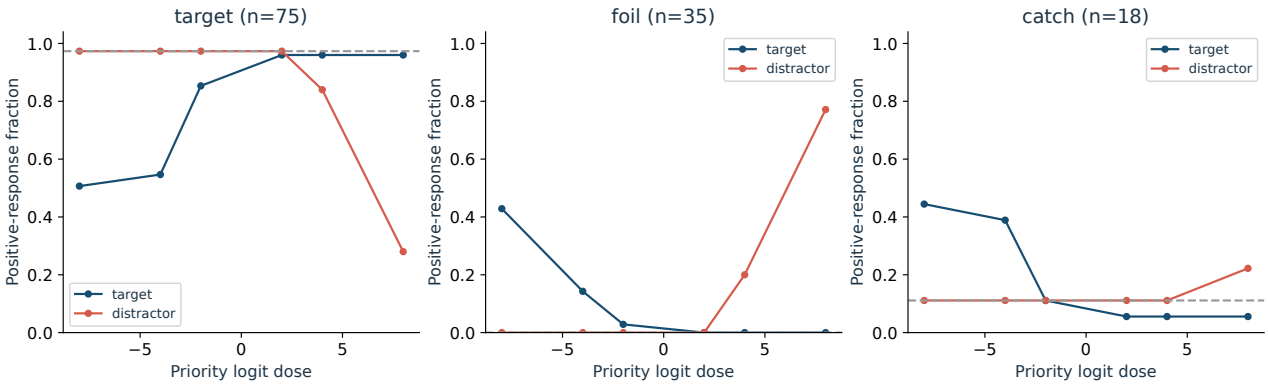


Figure 12: Event-specific causal response functions. Curves show positive reports after evidence-epoch target or alternative-site intervention; the horizontal line is paired sham. Event counts are stated above each panel. A positive report has the task-specific meaning defined in the text.

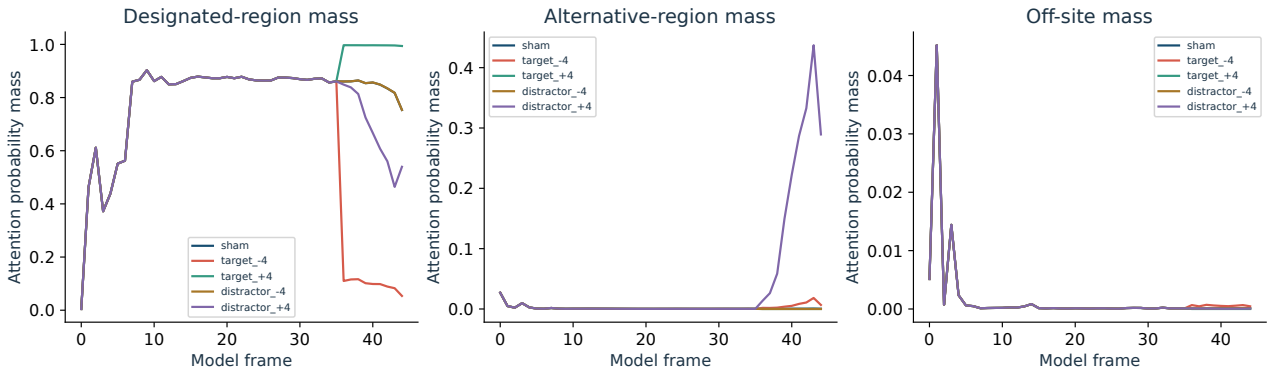


Figure 13: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

3.12 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

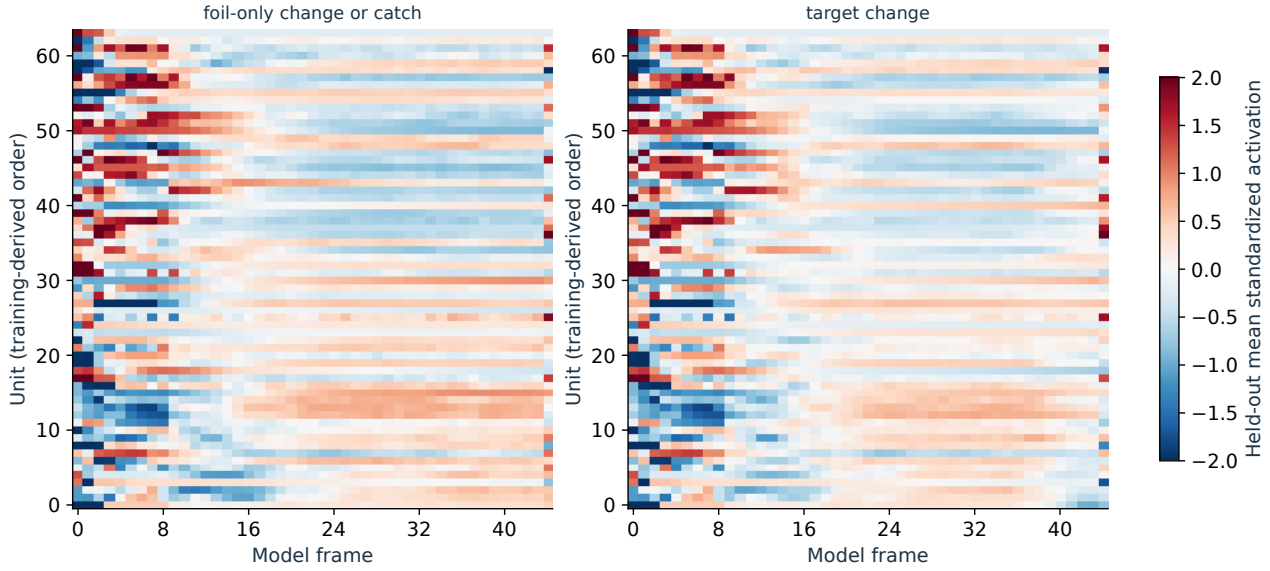


Figure 14: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.13 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional

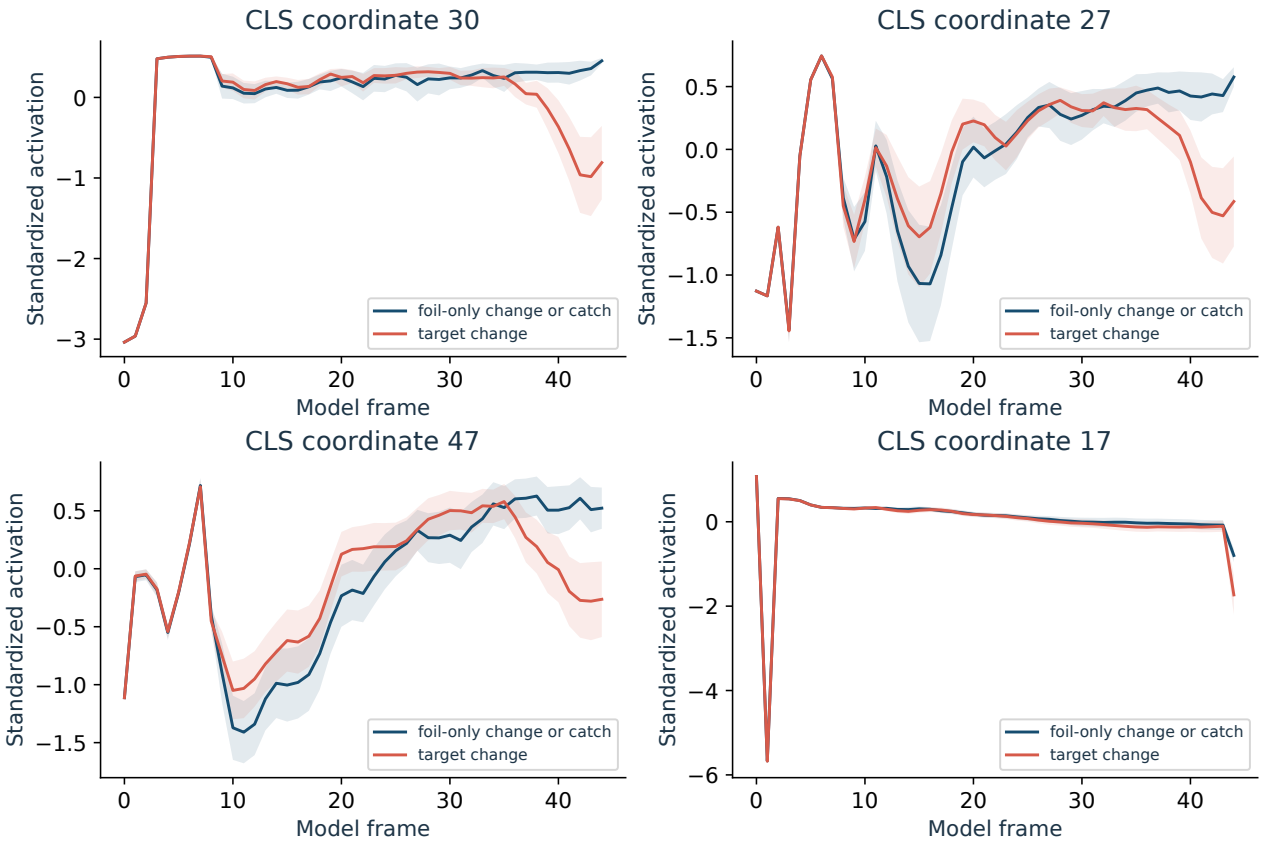


Figure 15: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	50.0%	50.0%	50.0%
fusion	50.0%	52.1%	95.3%
spatial	56.5%	57.8%	100.0%
cls	84.9%	84.7%	78.9%
comparison	76.8%	74.7%	53.1%
attention	71.1%	72.4%	100.0%

3.14 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

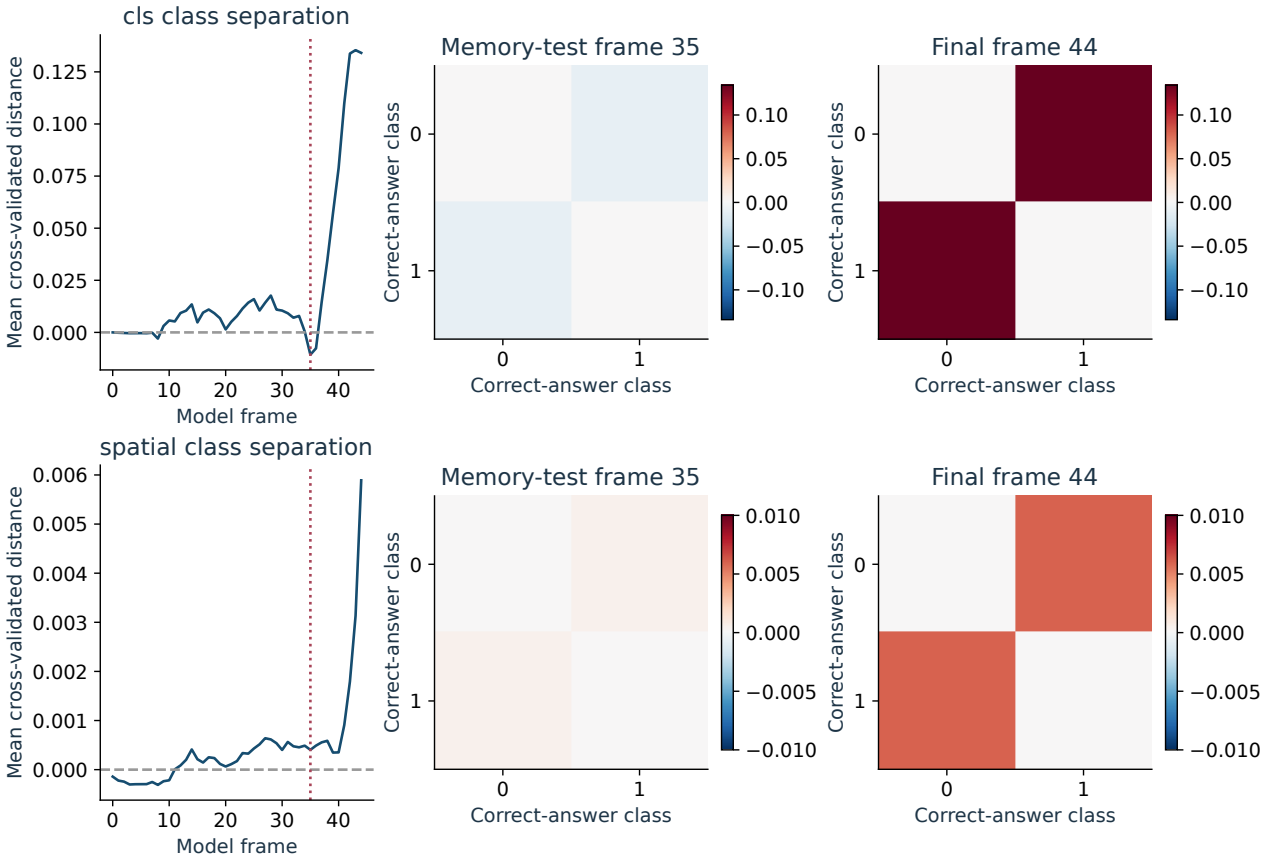


Figure 16: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

4 Discussion

The target and foil interventions reveal a selection-dependent detector. Suppressing target priority reduces correct reports, and increasing foil priority also reduces performance. The baseline experiment provides little evidence for an additional benefit from foil suppression at the tested native signal strength. Spatial-memory disruption immediately before the change is especially costly, linking the event decision to the preceding cued state. Both global and comparison memories contain readable event information at the end of the trial. A pre-change comparison-memory reset tests dependence at that moment; subsequent visual comparisons remain available and can rebuild later evidence. Event-separated response curves distinguish missed relevant changes from inappropriate foil reports.

The behavioral result establishes 95.8% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies distractor_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

A decisive follow-up would compare matched short and long baselines while perturbing the comparison branch independently of attention. That design would separate temporal evidence availability from the allocation of shared spatial priority.

4.1 Interpreting these results alongside biological experiments

4.1.1 Arcizet and Krauzlis (2018)

This is the closest task-family relation: maintain fixation, monitor cued motion and ignore an uncued event. The model's spatial state can carry location and phase information, and its head reports a discrete event decision. Epoch and location decoders therefore parallel the questions asked of caudate activity. The model has a distributed recurrent implementation rather than an anatomically identified caudate circuit. [5]

4.1.2 Bollimunta, Bogadhi and Krauzlis (2018)

Target suppression and foil enhancement impair the model's detection, matching the functional importance of spatial selection. A logit dose changes normalized routing immediately and can alter later recurrent states. Biological inactivation changes a region's activity and network interactions over a different spatial and temporal extent. The useful common comparison is the pattern of target hits, foil reports and catches under matched stimulus strength. [6]

4.1.3 Lovejoy and Krauzlis (2010)

The animal result demonstrates competition-dependent use of motion evidence and a favorable shift when the foil is affected. Our native experiment shows little additional benefit from foil suppression at its tested strength. This is a substantive incomplete agreement: the model shows selection costs, while the predicted benefit from removing competition needs more favorable measurement conditions or a different computation. [7]

4.2 A computational account of the observed behavior

A shared-priority account says the decision depends on selected evidence, not simply on whether a visual change exists. Spatial memory retains the cue and prechange context; visual comparison provides event evidence; the readout must use the relevant location. Suppressing the target can remove useful evidence, and enhancing the foil can admit misleading evidence. The complementary improvement after foil suppression requires that the foil was limiting performance and that the remaining target evidence is usable. Native target changes here are relatively large, so a high baseline can hide an improvement. However, a ceiling explanation should be tested by lowering target strength, rather than assumed from the direction of attention redistribution.

4.3 Where this task agrees with the biological evidence

The model agrees strongly with a causal selection requirement and with retained spatial context. It only partly reproduces the proposed reciprocal benefit-and-cost pattern. The next comparison should isolate informative competition, because that was central to the animal result and is more specific than a generic attention lesion.

4.4 A follow-up experiment motivated by these results

Cross target strength with absent, uninformative and counterinformative foil events. Apply target and foil suppression separately, retain matched catches, and fit sensitivity and criterion. Predict a selective benefit of foil suppression only where foil information actually degrades the untreated decision.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Arcizet F, Krauzlis RJ (2018). Covert spatial selection in primate basal ganglia. PLOS Biology 16:e2005930. https://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.2005930.

[2] Bollimunta A, Bogadhi AR, Krauzlis RJ (2018). Comparing frontal eye field and superior colliculus contributions to covert spatial attention. Nature Communications 9:3553. https://www.nature.com/articles/s41467-018-06042-2.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] Arcizet and Krauzlis (2018). Covert spatial selection in primate basal ganglia. Primary source.

[6] Bollimunta, Bogadhi and Krauzlis (2018). Comparing frontal eye field and superior colliculus contributions to covert spatial attention. Primary source.

[7] Lovejoy and Krauzlis (2010). Inactivation of primate superior colliculus impairs covert selection of signals for perceptual judgments. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
B12	128	96.9%	0.995	0.047
B20	128	95.3%	0.995	0.061
B28	128	95.3%	0.995	0.062

B Every-frame attention atlas

Condition B28. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 30 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

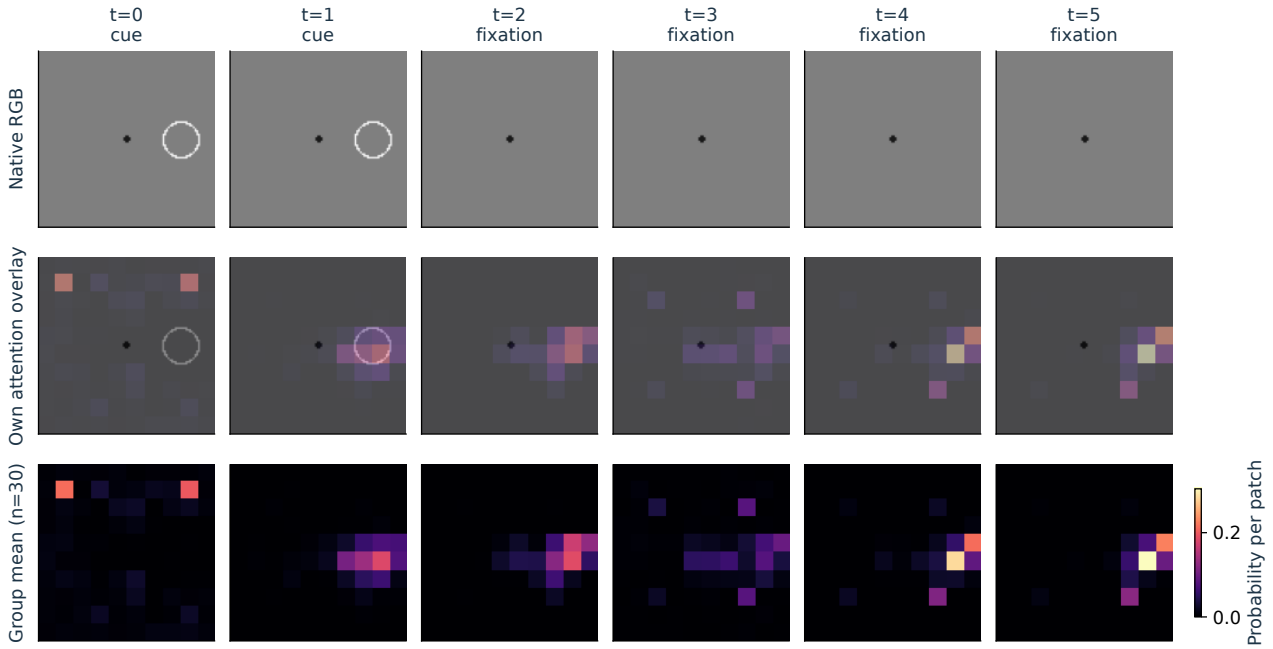


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

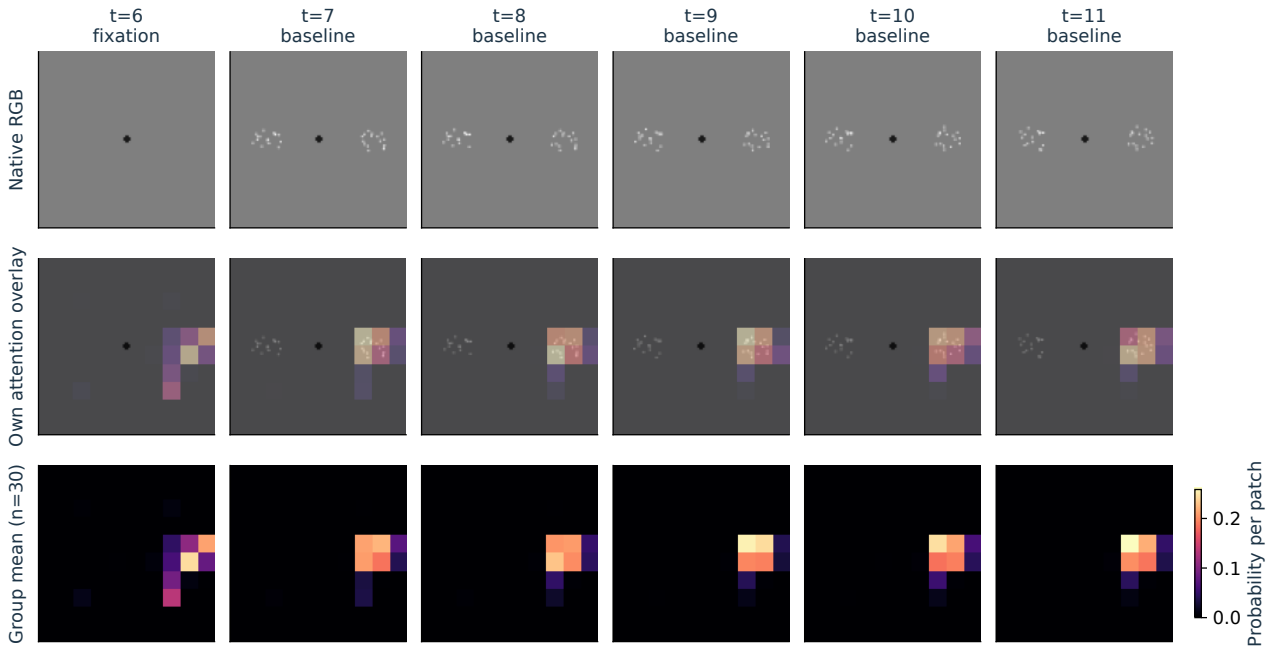


Figure 18: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

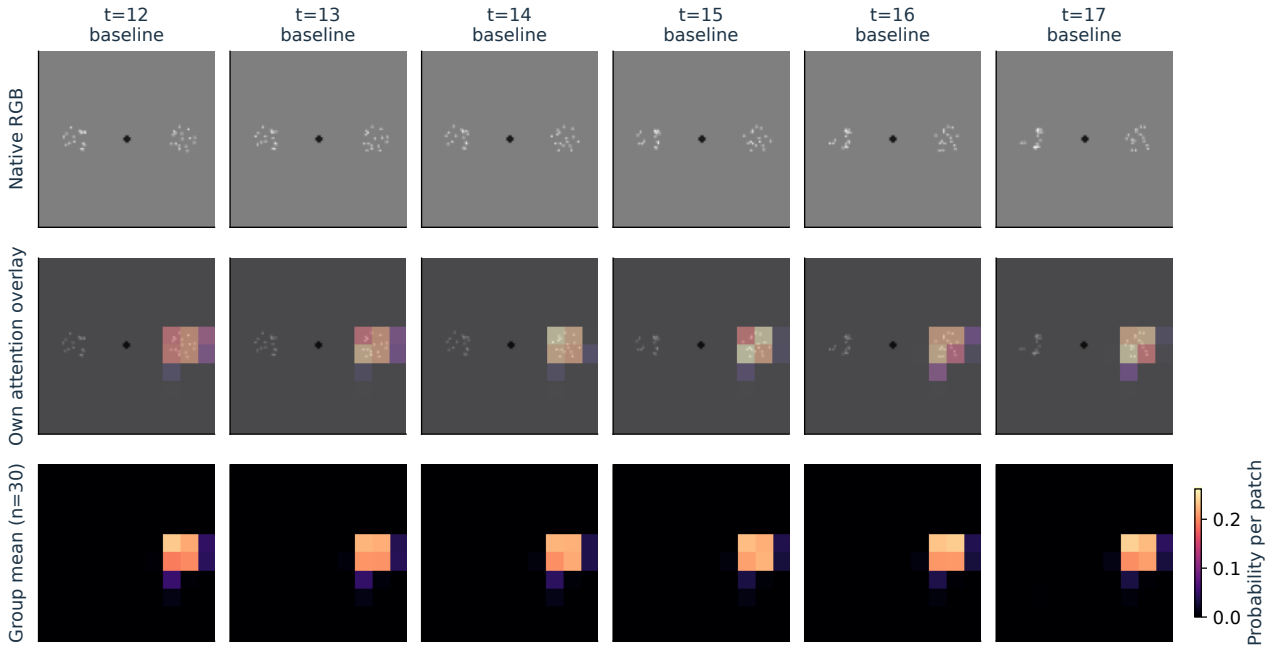


Figure 19: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

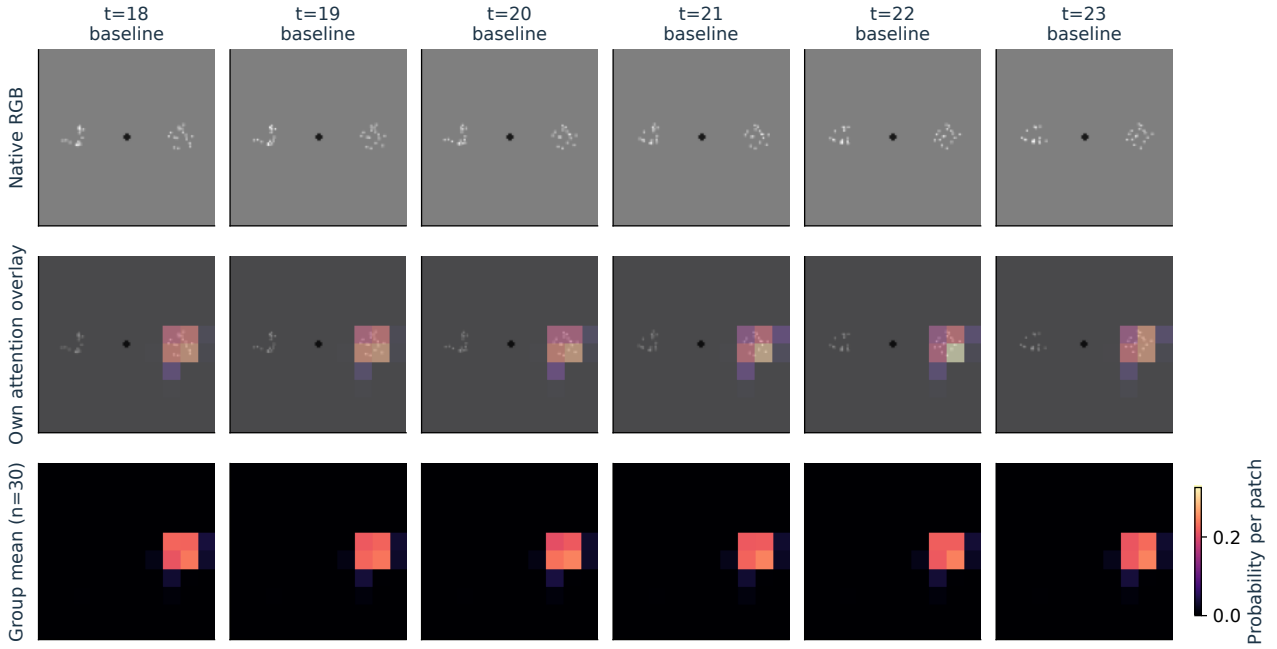


Figure 20: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

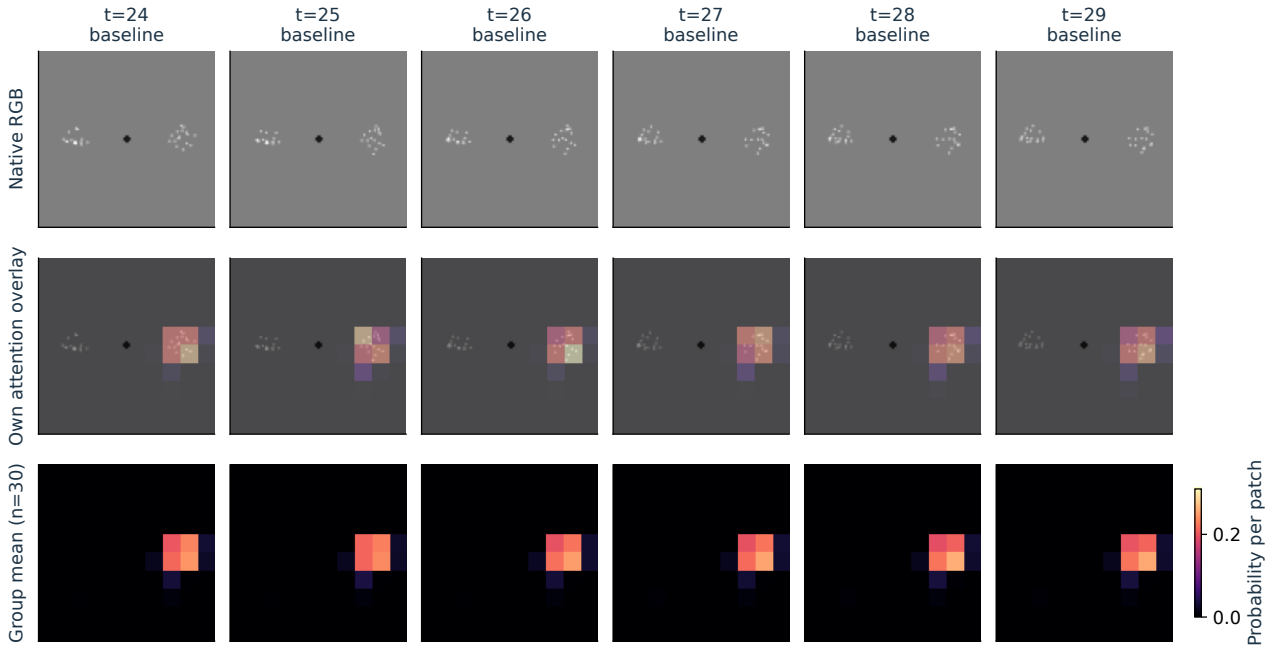


Figure 21: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

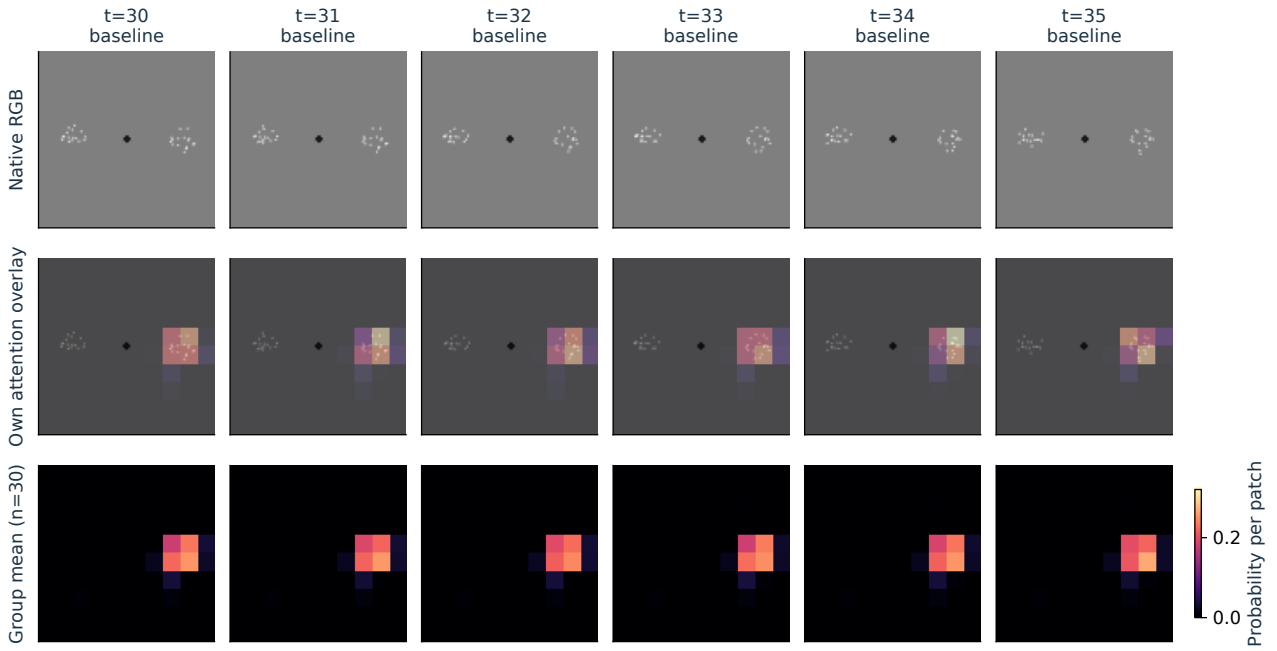


Figure 22: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

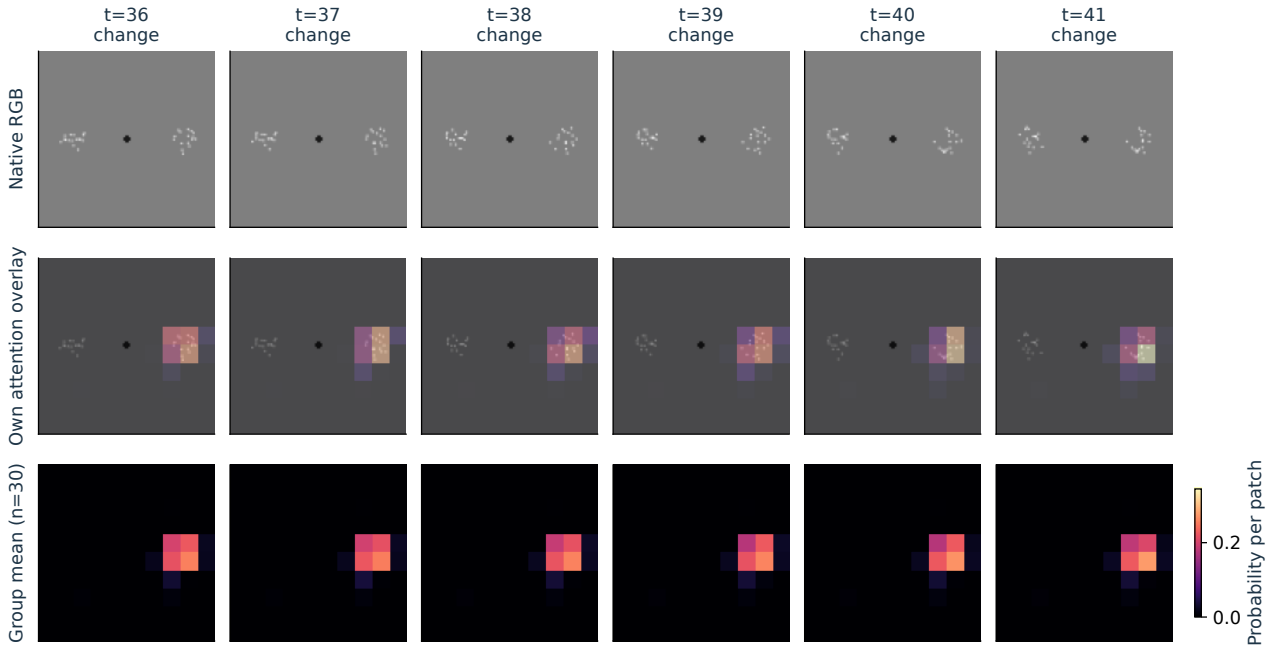


Figure 23: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

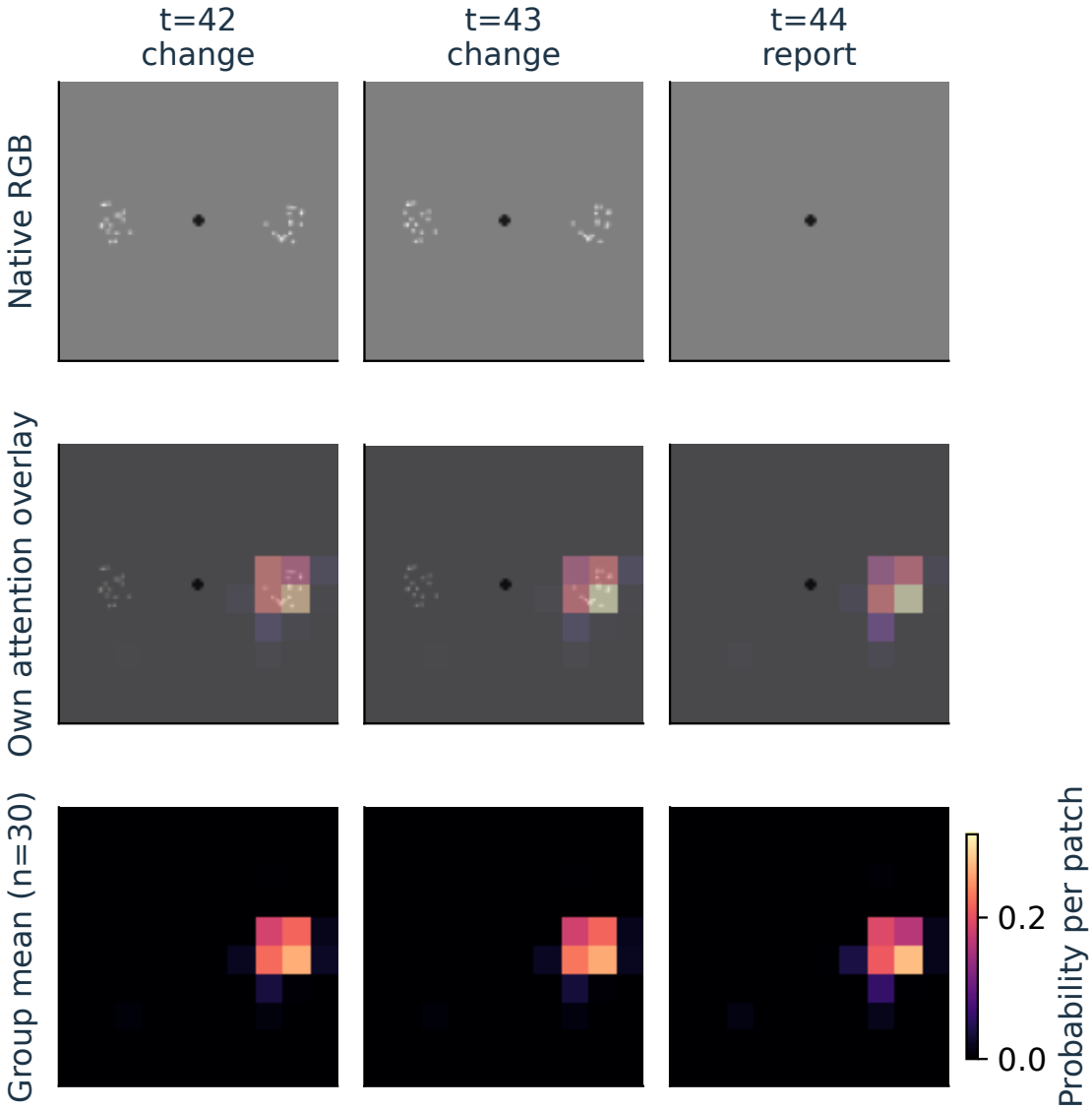


Figure 24: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

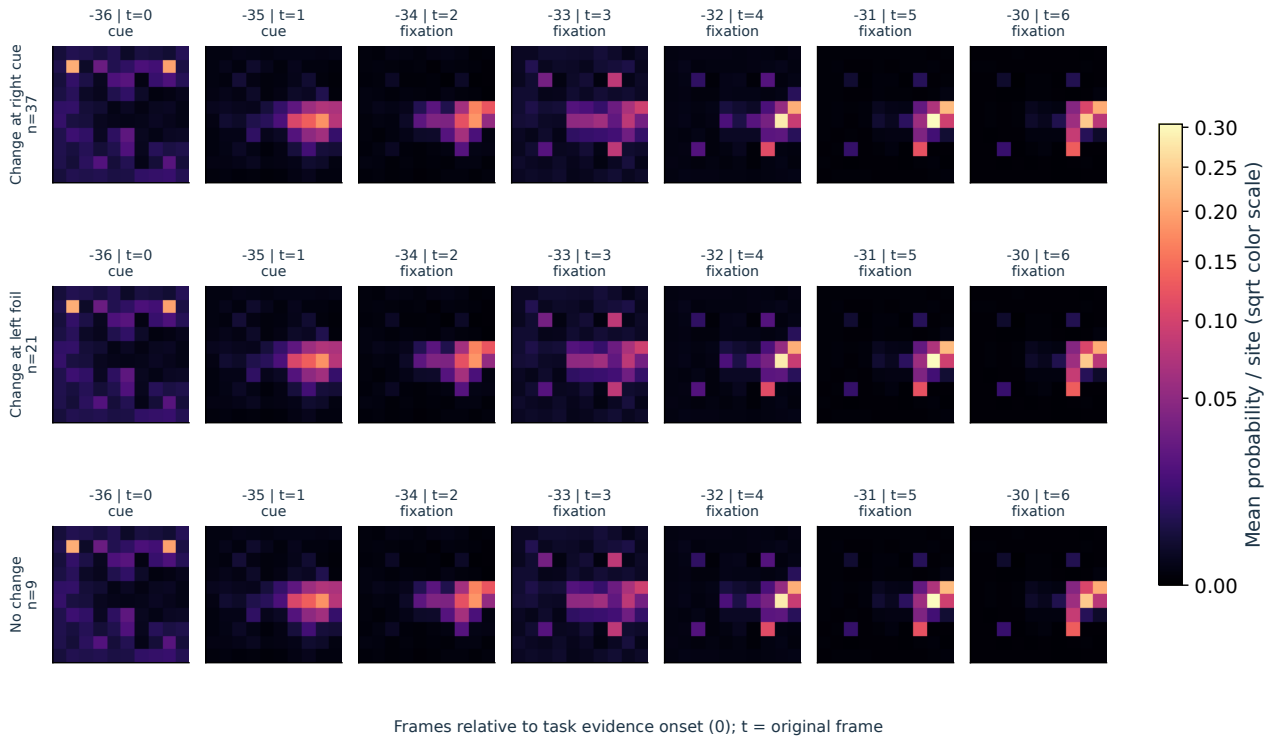


Figure 25: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

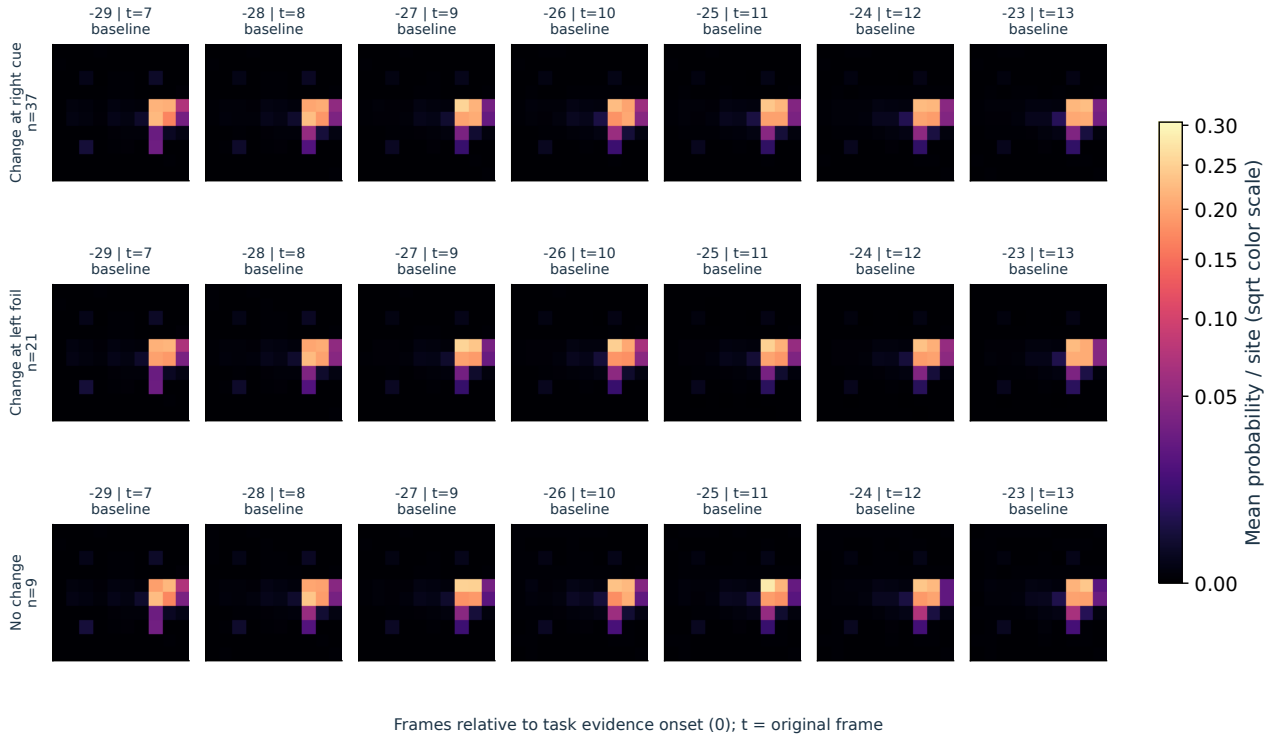


Figure 26: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

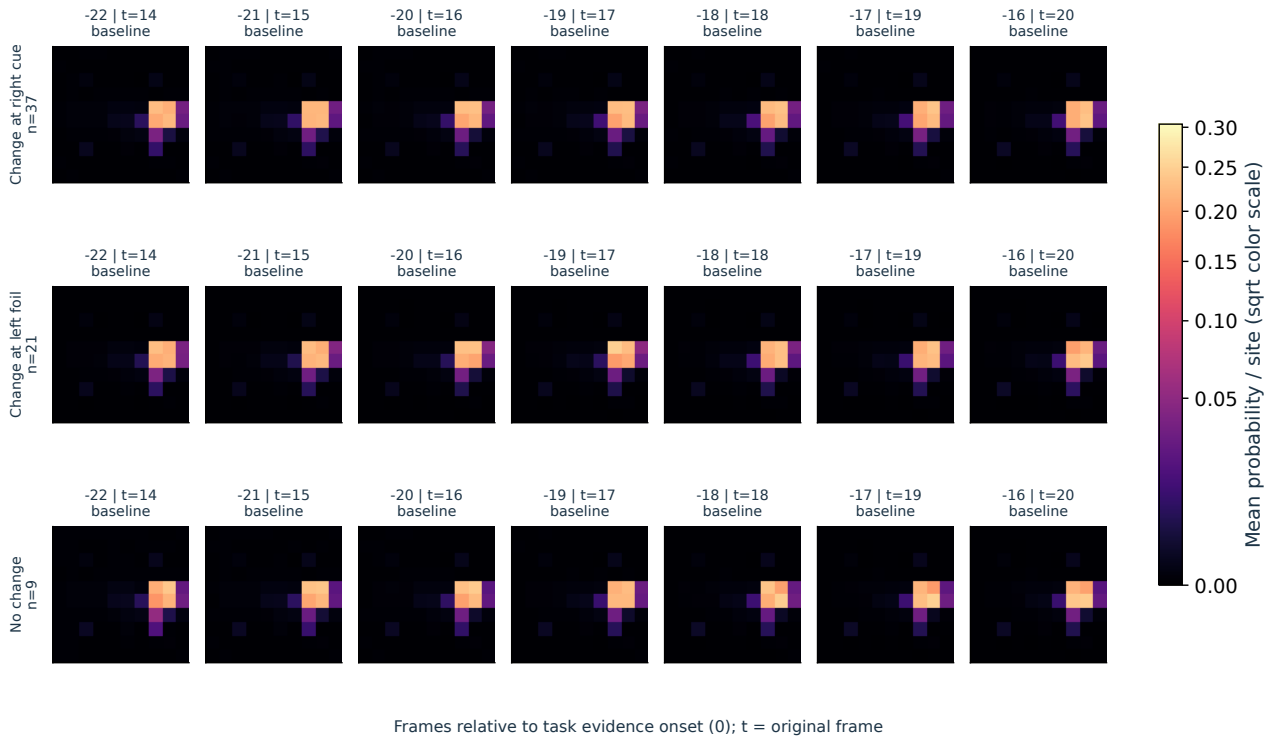


Figure 27: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

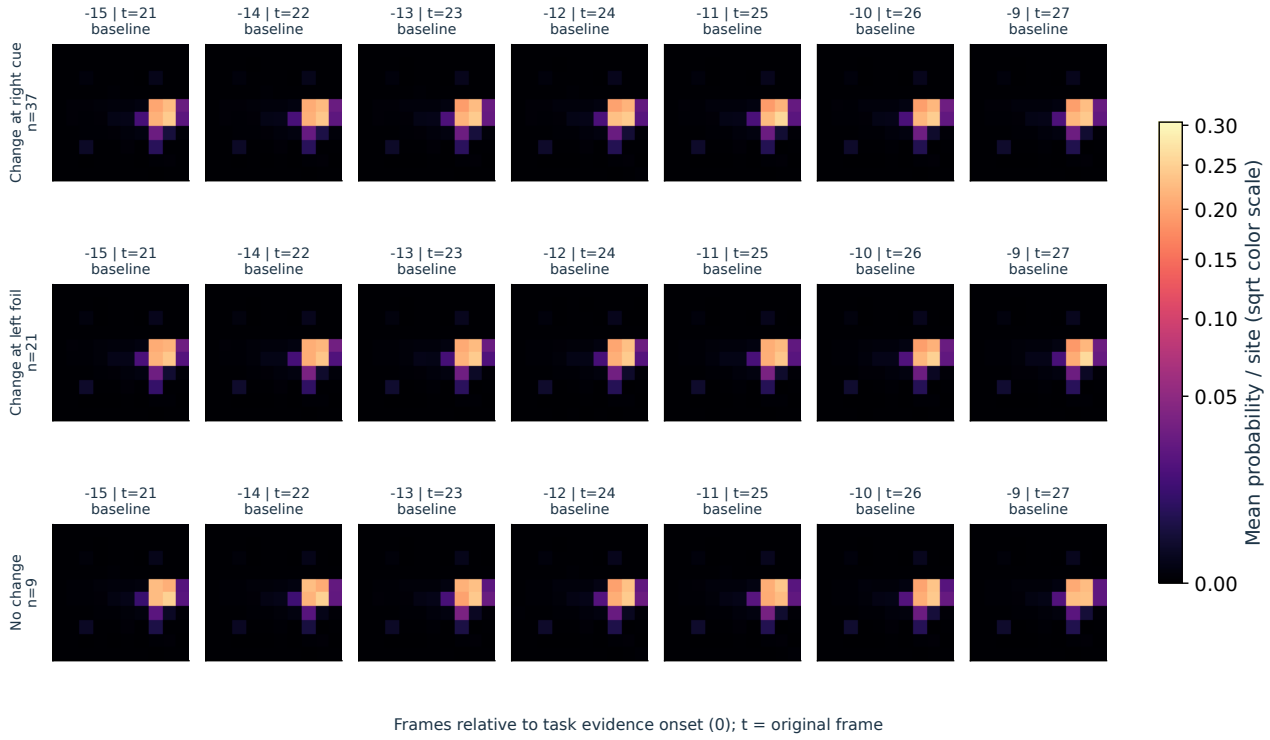


Figure 28: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

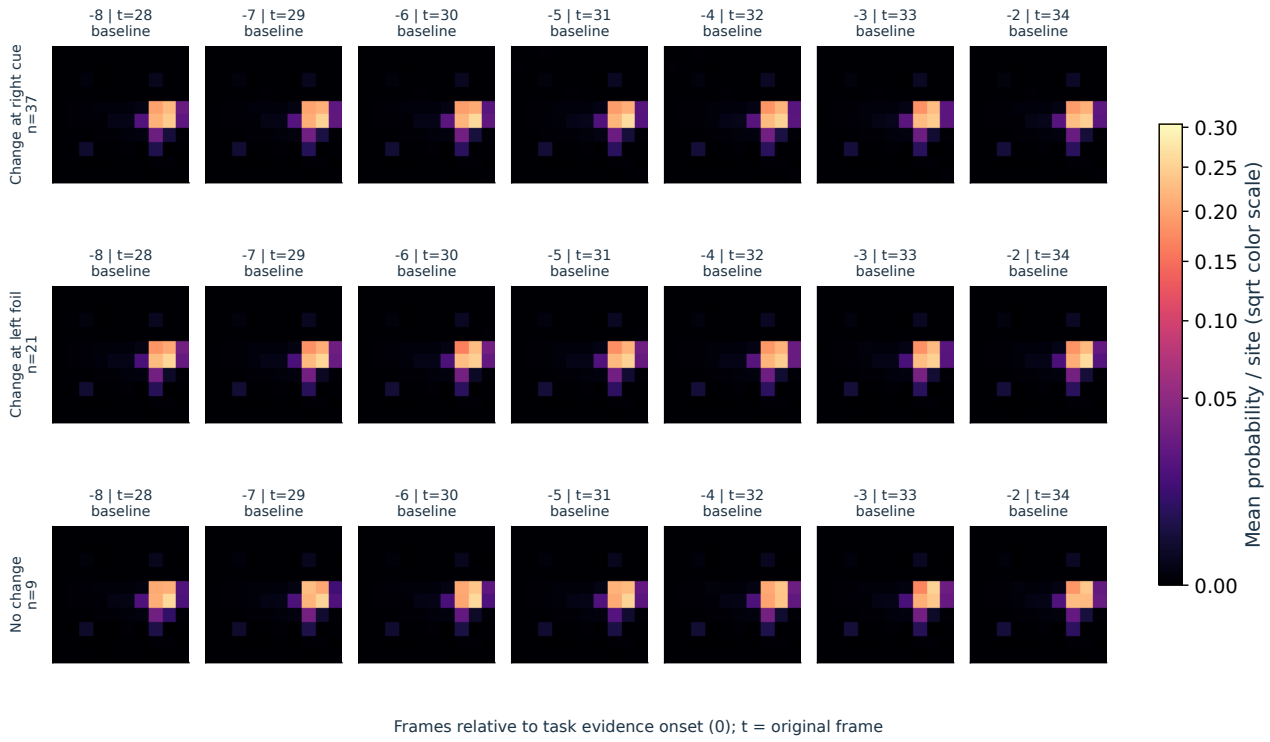


Figure 29: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

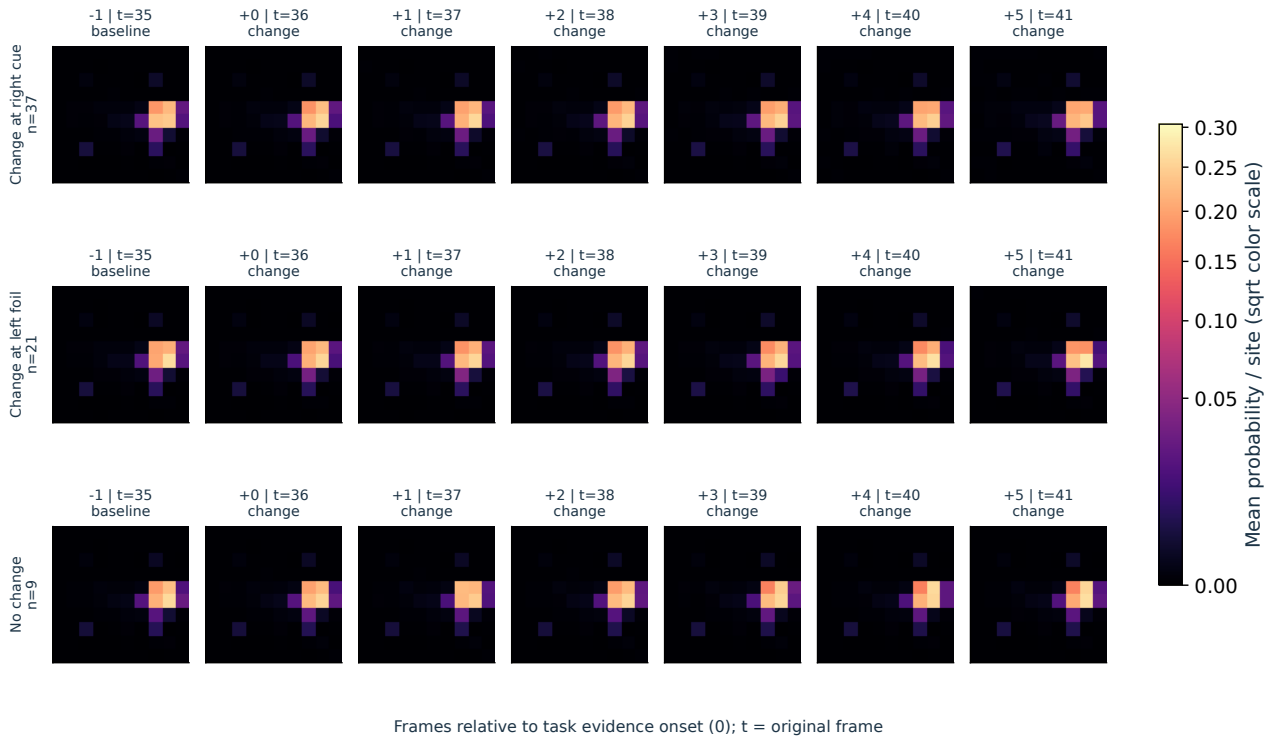


Figure 30: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

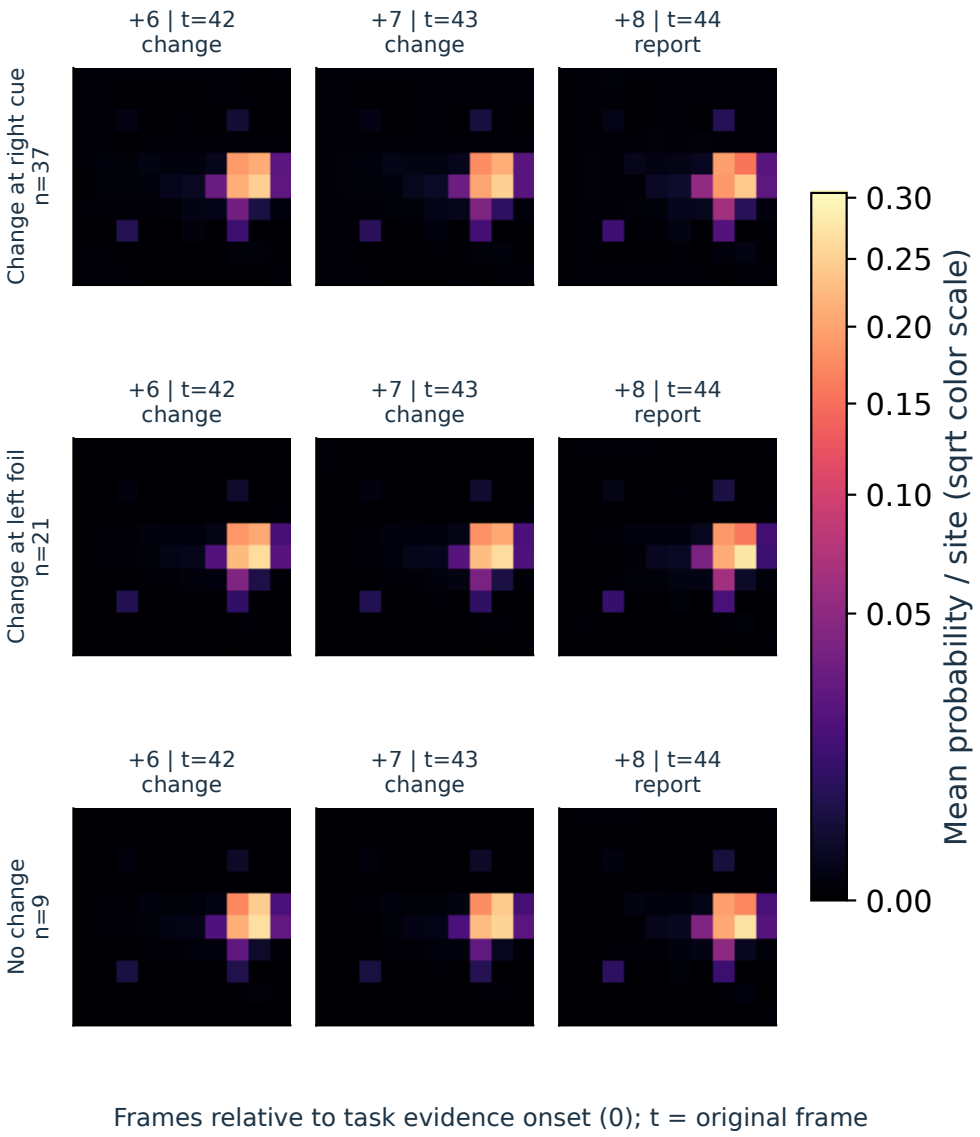


Figure 31: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Retrospective selection of an orientation-location binding

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized spatial binding in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 4 native conditions and 512 fresh trials, equal-condition accuracy was 100.0% (95% interval [99.3%, 100.0%]). For the representative condition D24, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was target_-8_evidence, changing accuracy by -75.0 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The observer sees an array of orientations, retains it through a blank interval, then receives a retrospective cue and a probe. The report concerns whether the orientation at the queried location was exchanged with another location. The orientation inventory is preserved, so recognizing the set of features alone cannot solve the task.

The critical memory is a conjunction of what and where. All locations are initially eligible because the query arrives after retention. We examine delay-dependent accuracy and the availability of location-specific sample information in spatial memory and CLS state. The retrospective cue offers a natural landmark for changes in population geometry.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Feature-integration and spatial-memory work motivate representing an object together with its location. A swap changes the assignment of features to positions while preserving the inventory of features. The retrospective query makes this a useful test of distributed storage: before the cue, all locations must remain available. A model that stores only an average orientation or a histogram of orientations would lose the binding needed for the response. The spatial-memory recordings and mid-delay perturbations address whether a location-preserving state supports the later query.

1.2 Biological experiments motivating this task

Experimental question. Can the observer retain which feature belonged where, then retrieve the relevant assignment after a cue?

Wheeler and Treisman (2002) Change-detection experiments separated memory for features from memory for their combinations. Binding was vulnerable to interference, motivating a distinction between retained feature information and attention-dependent integration. [5]

Bays, Catalao and Husain (2009) Continuous-report errors reflected both feature and location memory. Accounting for their joint uncertainty supported a shared-resource description in which precision changes with the number of remembered objects. [6]

Griffin and Nobre (2003) Cues presented before or after a visual array improved recognition of relevant locations. Behavioral and electrophysiological results linked orienting in perception and memory while revealing differences between precue and retrocue processing. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Was the orientation at the retrospectively queried location exchanged with another location?

Response labels are 0: foil-only exchange; 1: target participates in exchange. The native suite contains 4 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use D24.

1.4 Stimulus sequence and spatial layout

For the representative native condition D24, the sequence has 29 frames, indexed from zero: frame 0: observation; frames 1–2: sample; frames 3–26: delay; frame 27: cue; frame 28: probe. The complete condition table specifies the other tested timing and difficulty settings.

Stimulus centers in the 100 by 100 input are [[27.0, 27.0], [73.0, 27.0], [27.0, 73.0], [73.0, 73.0]]. Coordinate x increases rightward and y downward.

The native orientation inventory consists of four angles spaced 45 degrees apart around one random base angle, randomly assigned to locations. The probe exchanges two locations while preserving that inventory.

A distributed spatial memory can maintain several local descriptions before the query selects one. Delay lesions therefore probe storage, while probe-period priority changes probe selection of the stored binding. If target localization becomes decodable only after the cue, that is appropriate to the trial information structure rather than evidence of delayed perception.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 100.0%, with macro AUC 1.000. The matched final checkpoint changed accuracy by +0.00 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was D0 at 100.0%; the highest was D0 at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

Marginal accuracy by delay was 0: 100.0%; 4: 100.0%; 12: 100.0%; 24: 100.0%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

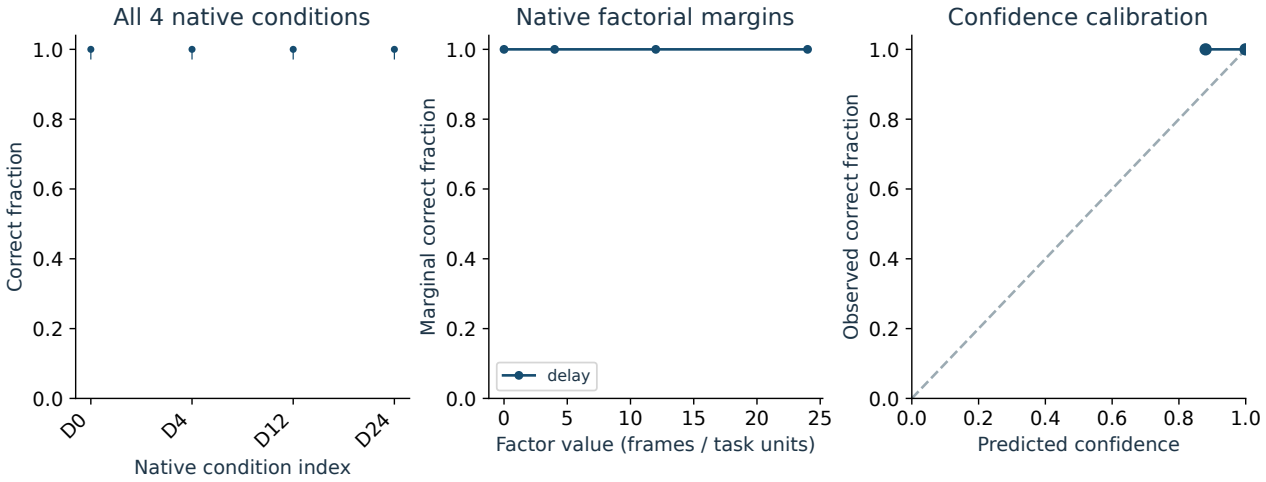


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	100.0%	100.0%
0.7	100.0%	1.0	100.0%	100.0%
0.8	100.0%	1.0	100.0%	100.0%
0.9	100.0%	1.0	100.0%	100.0%

The representative condition D24 provides 1 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.3 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 512 of 512 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are

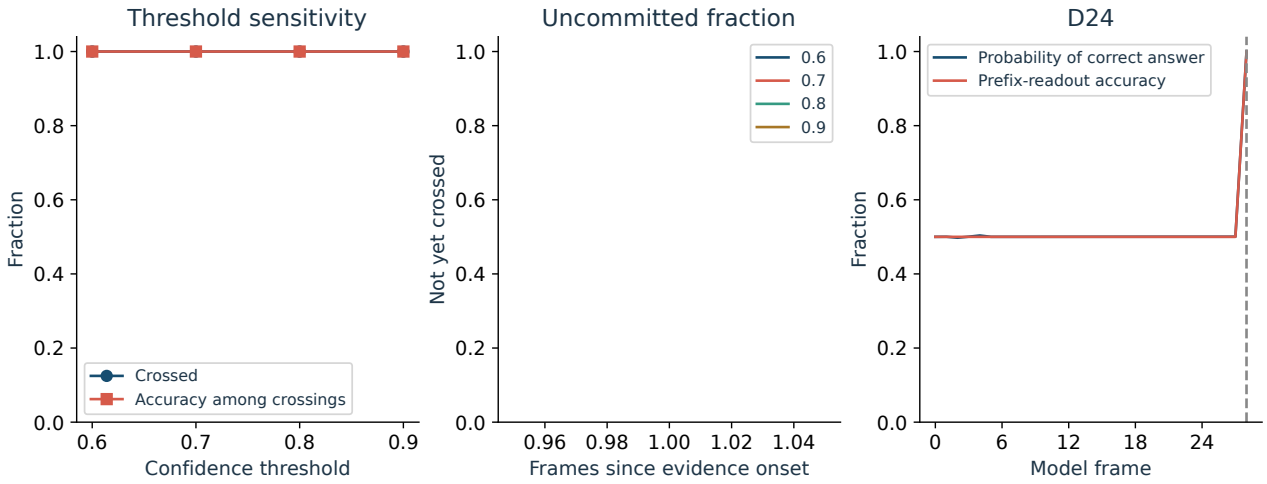


Figure 2: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

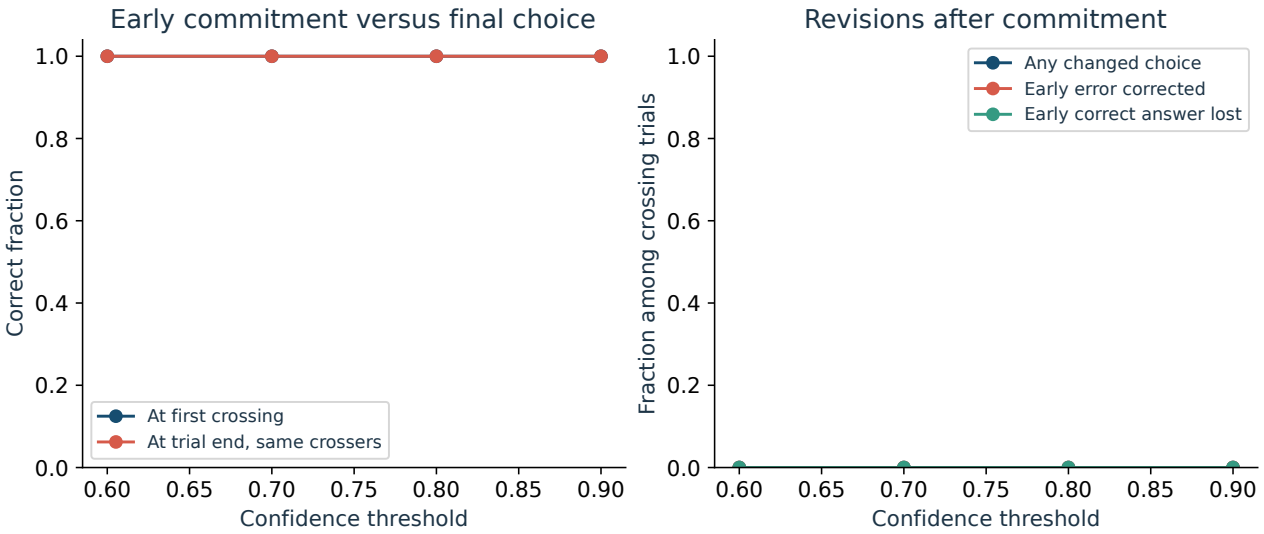


Figure 3: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

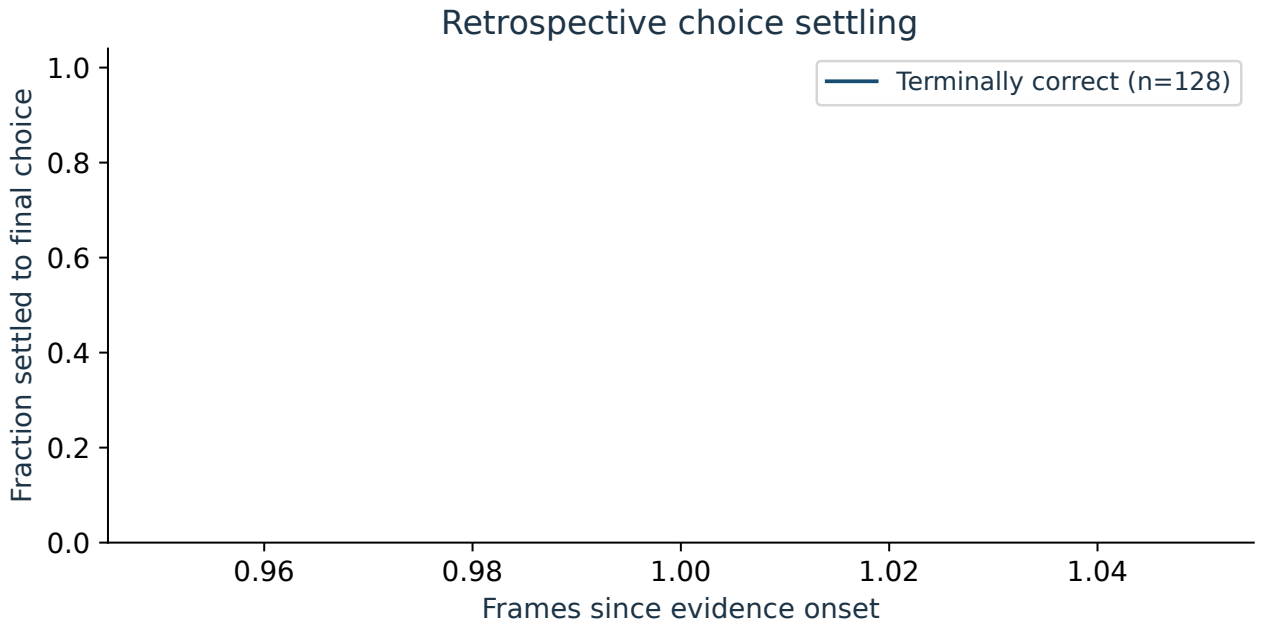


Figure 4: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.4 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The retrocued target is fixed at top left with a 24-frame retention interval. Rows separate swaps involving that target from swaps confined to other locations.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Foil-only swap	16	16
Target-involving swap	16	16

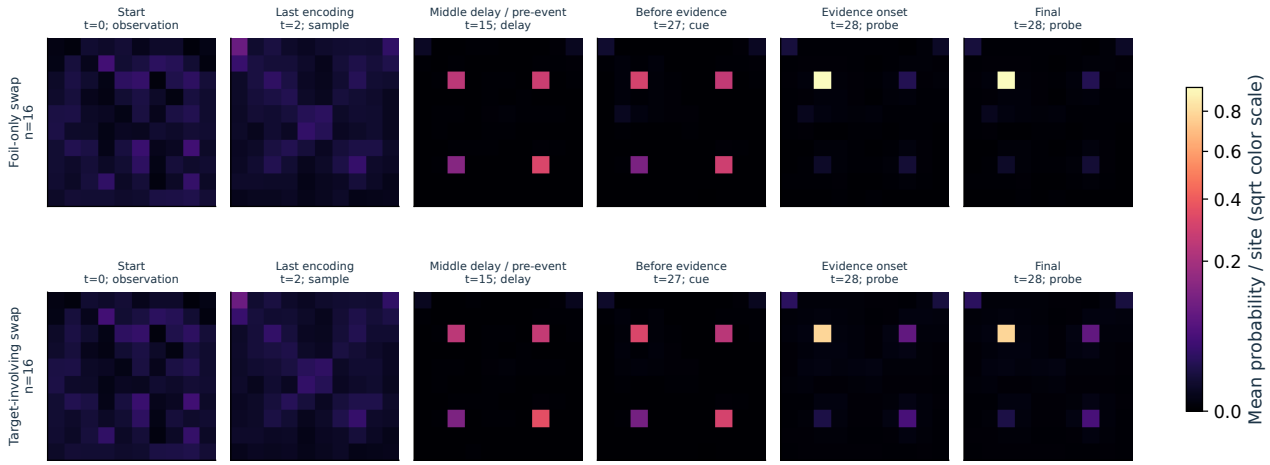


Figure 5: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

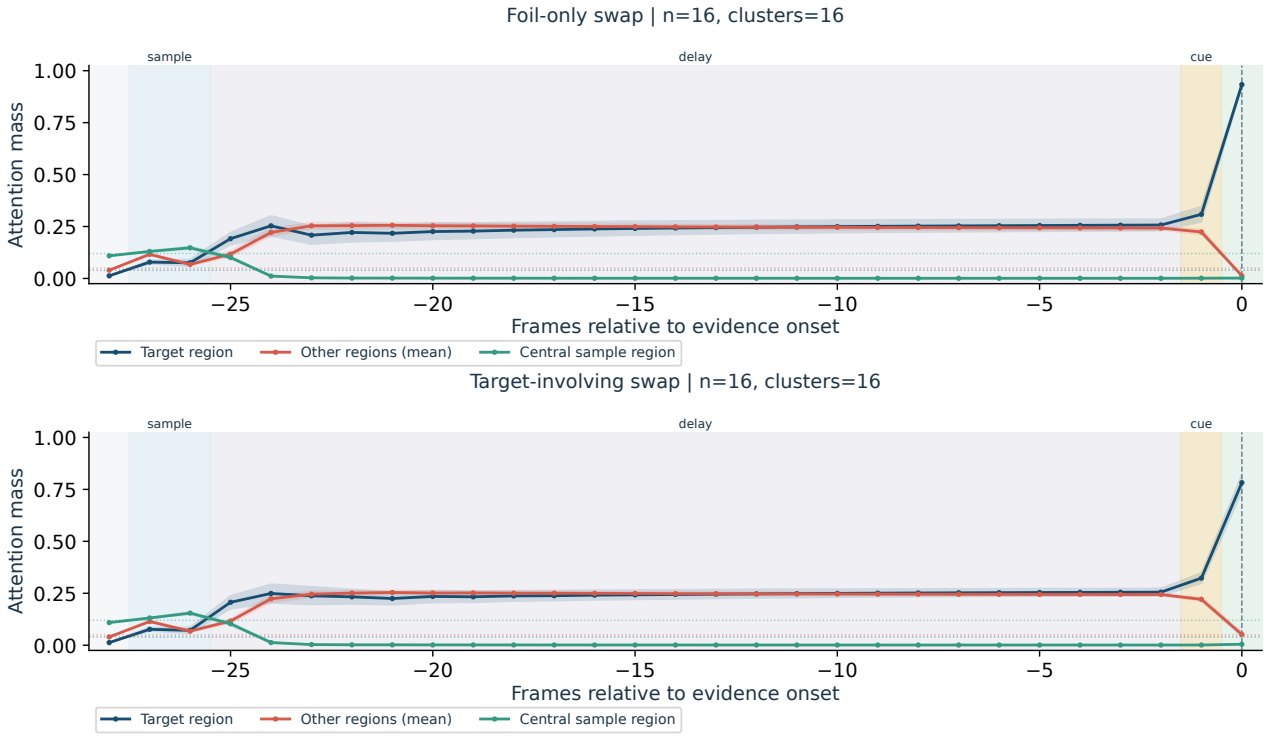


Figure 6: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

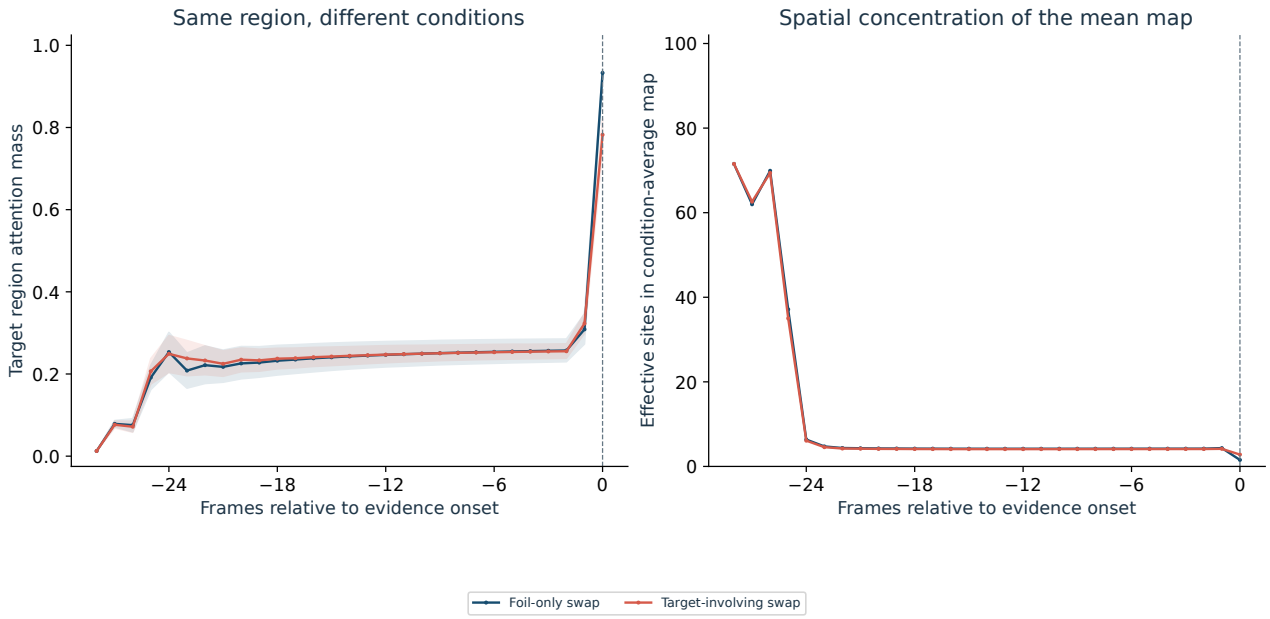


Figure 7: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Foil-only swap: target region mass peaks at 0.933 at original frame 28 (probe), and is 0.933 on the final frame. Target-involving swap: target region mass peaks at 0.782 at original frame 28 (probe), and is 0.782 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.5 Causal sensitivity to spatial and mnemonic interventions

For target_-8_evidence, paired accuracy changed by -75.0 percentage points (paired 95% interval [-82.0%, -67.2%]; paired p=2.524e-29; within-task Holm p=7.826e-28). There were 0 corrected trials and 96 newly incorrect trials. For target_+2_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

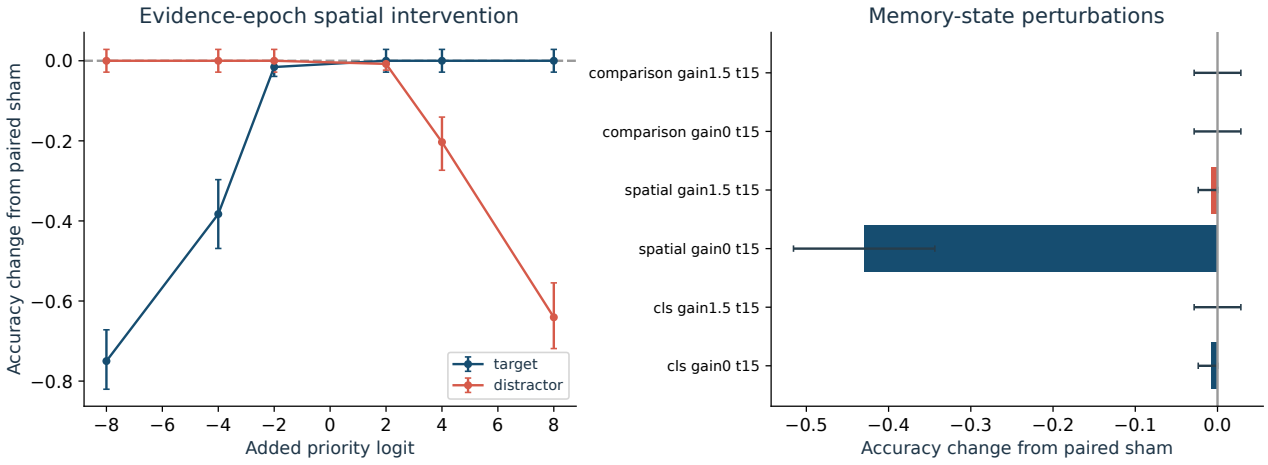


Figure 8: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.6 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
D24	target_-4	-38.3	3.233e-13
D24	target_+4	+0.0	1
D24	distractor_-4	+0.0	1
D24	distractor_+4	-20.3	2.503e-06

3.7 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.436	44.5%	0.920
fusion	0.496	47.7%	0.690
spatial	0.511	48.4%	0.730
cls	1.000	100.0%	0.005
comparison	0.844	72.7%	0.030
attention	0.905	78.9%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

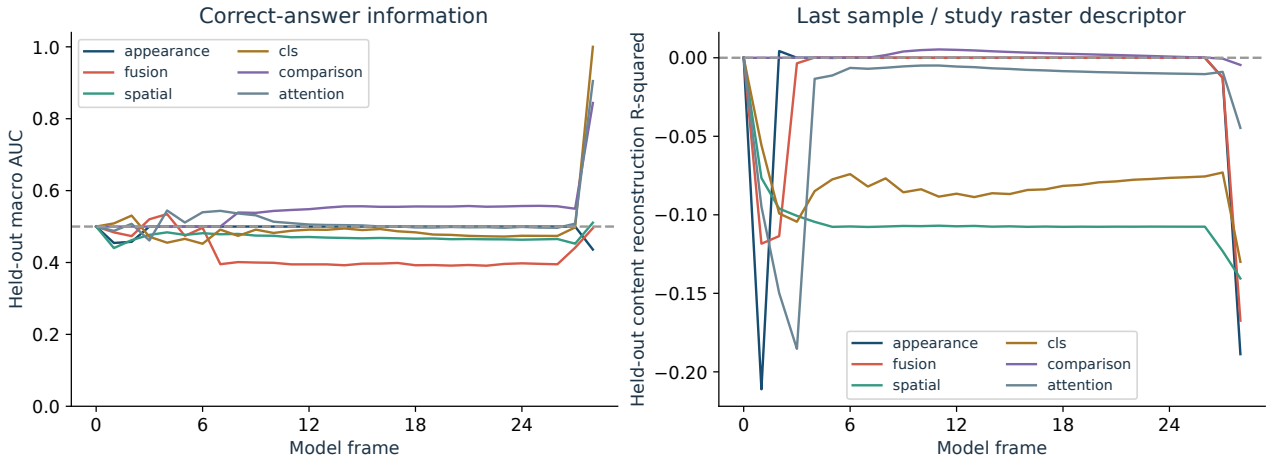


Figure 9: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.8 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 0.00 to 13.08 within the retained subspace. The final step displacement was 2.980 standardized RMS units. In spatial memory, the participation ratio ranged from 0.00 to 19.36 within the retained subspace. The final step displacement was 0.435 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 15, reconstruction of the original orientation at the location that will be cued after retention had held-out R-squared -0.149 in CLS memory and -0.076 in spatial memory. This continuous target describes stimulus content, while the final-answer decoder tests the response required after the subsequent evidence.

Before the retrospective cue, the model has not been told which location this single-target regression will query. The all-four-location inventory analysis below directly tests the stored orientation-location bindings without requiring advance knowledge of the future cue.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -0.8 percentage points, spatial, -43.0 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.9 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.496	unavailable	3200
fusion	0.477	0.472	6400
spatial	0.652	0.463	6400
cls	1.000	0.475	64
comparison	0.736	0.548	735
attention	0.832	0.477	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 35 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.10 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

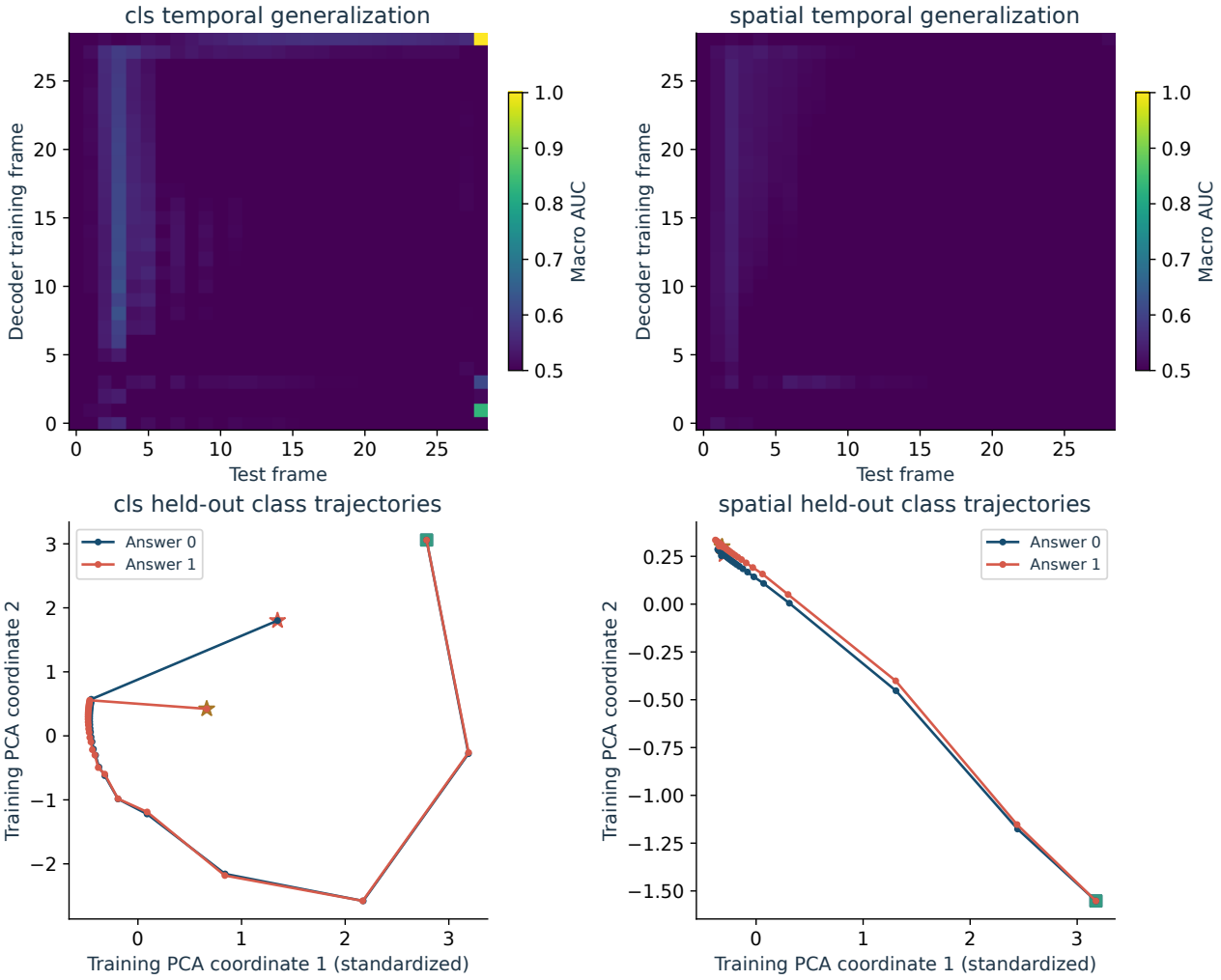


Figure 10: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

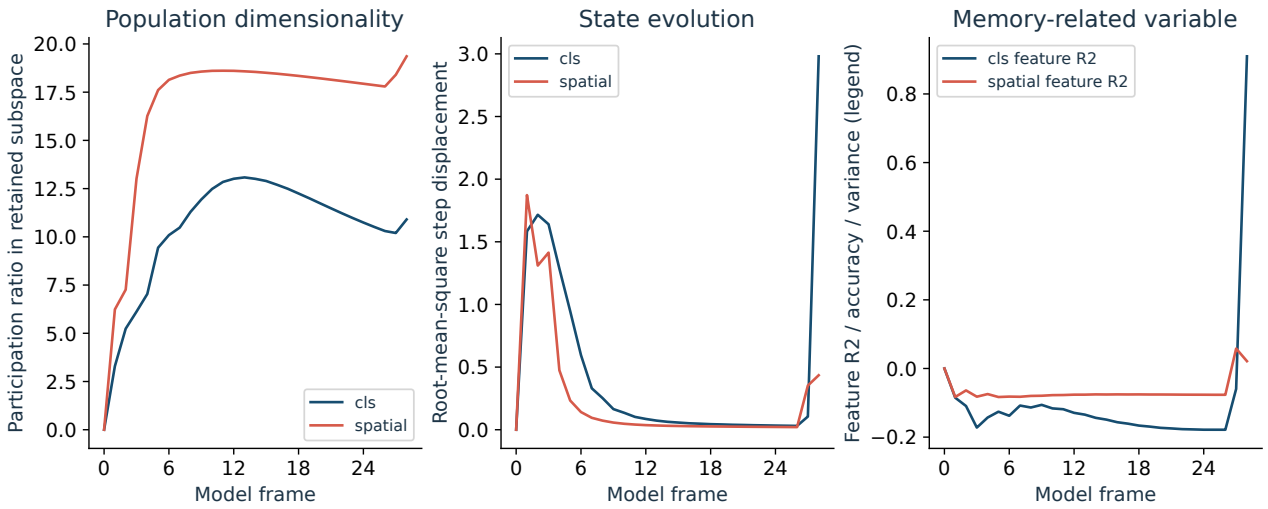


Figure 11: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

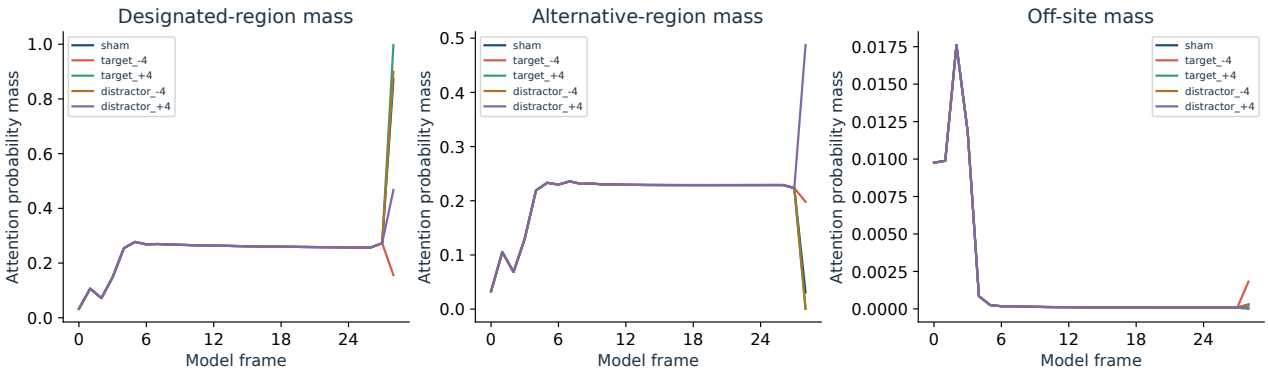


Figure 12: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

3.11 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

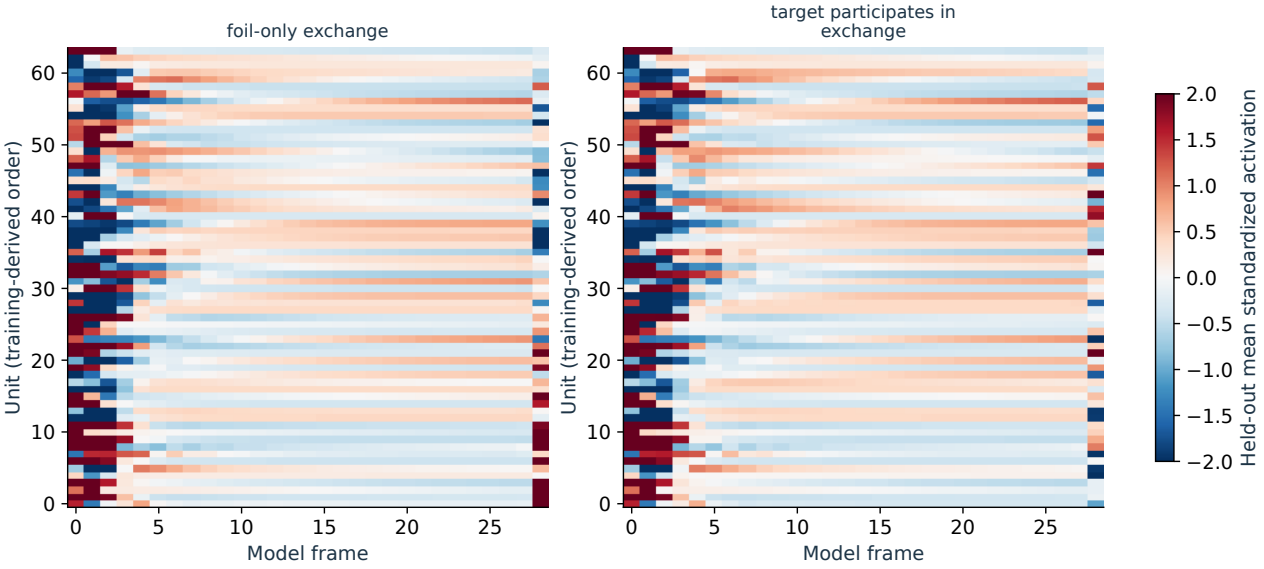


Figure 13: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.12 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	44.5%	44.5%	25.8%
fusion	47.7%	47.7%	100.0%
spatial	48.4%	48.4%	58.6%
cls	100.0%	100.0%	45.3%
comparison	72.7%	72.7%	38.3%
attention	78.9%	78.9%	100.0%

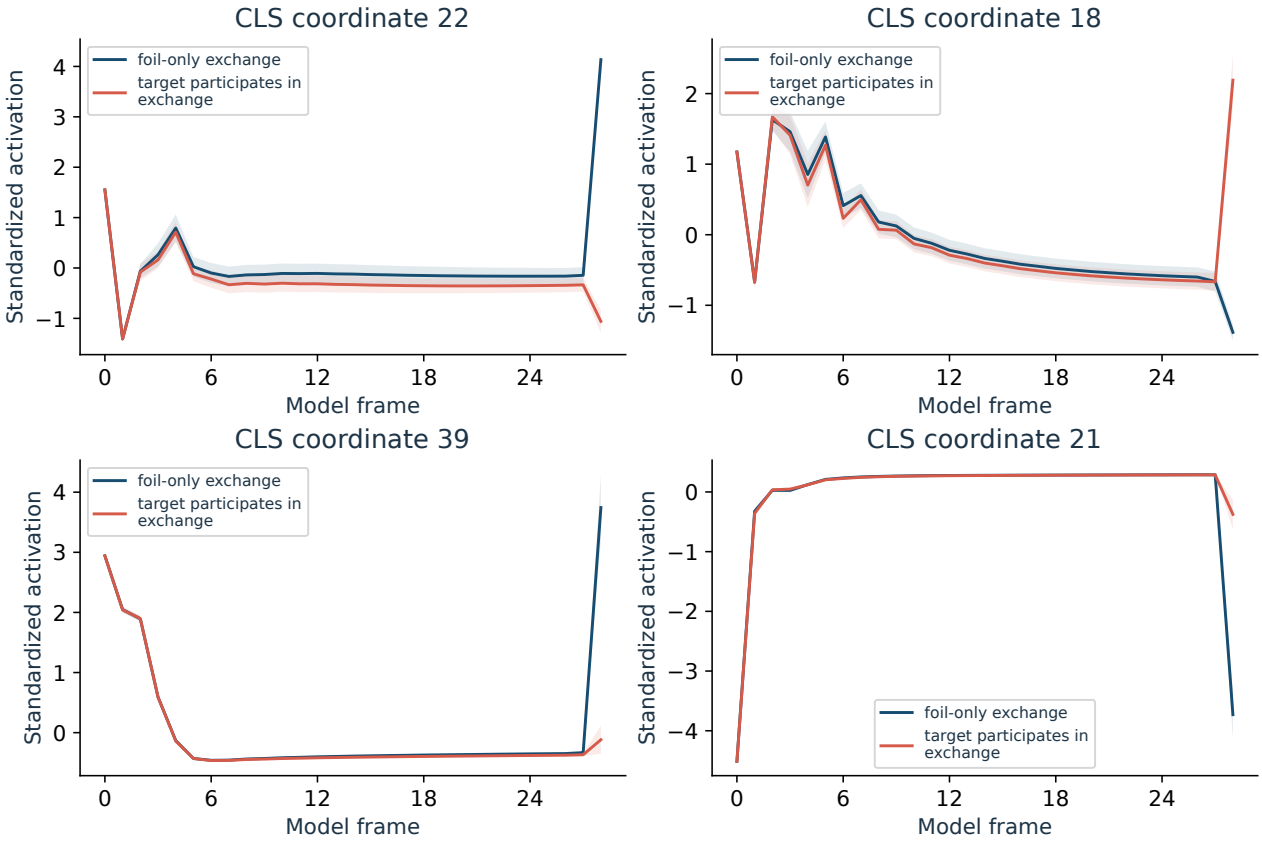


Figure 14: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

3.13 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

3.14 Remembered orientation inventory at all four locations

A separate content analysis reconstructs all four original sample orientations at every frame. Each orientation angle is encoded as its doubled-angle cosine and sine, respecting the 180-degree periodicity of a grating. The same training-only 32-component representation and ridge penalty used in the main analysis predict the eight continuous target values. Each location has a separate held-out R-squared relative to its fitting-mean predictor. This directly measures location-bound sample content. In the retrospective-cue task, the identity of the eventual target is revealed only after the delay, so all-location reconstruction is the appropriate pre-cue storage measurement.

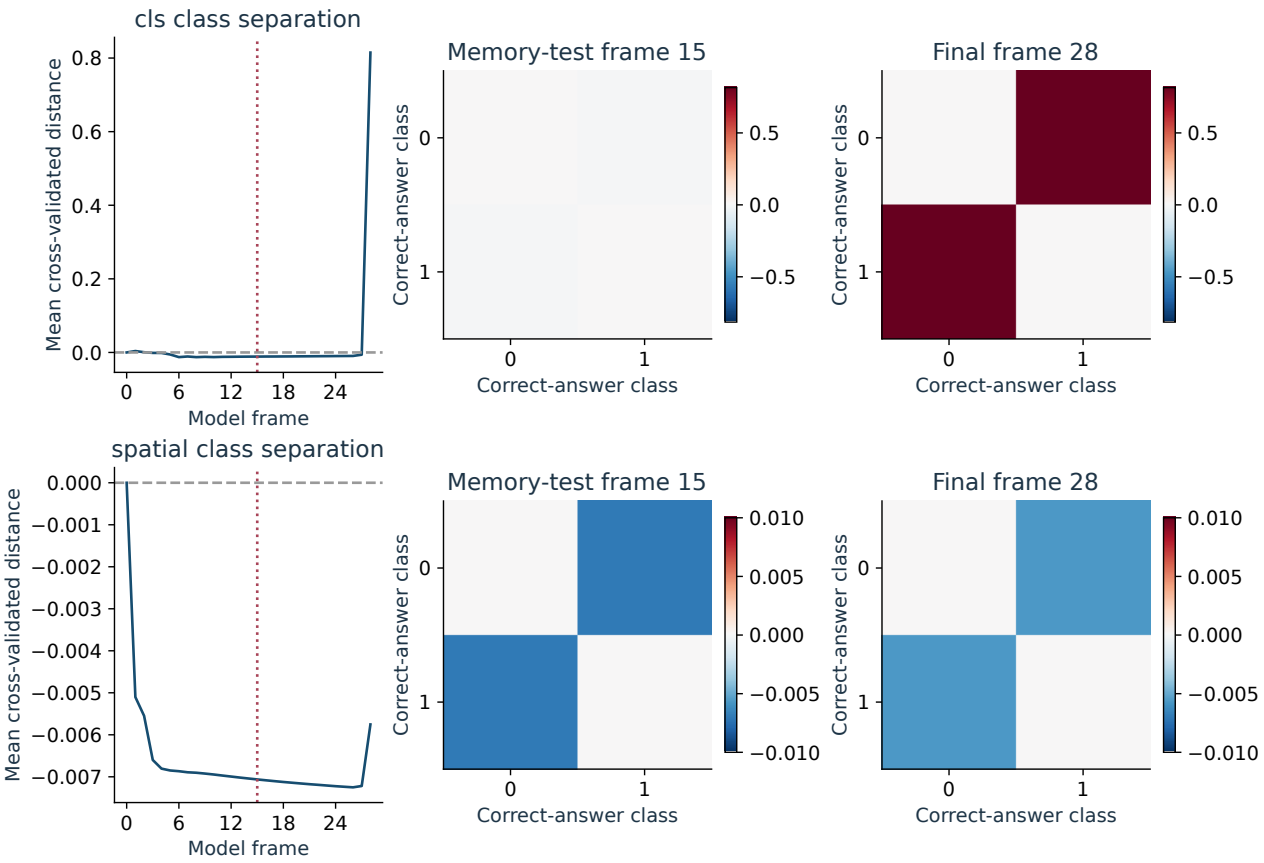


Figure 15: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

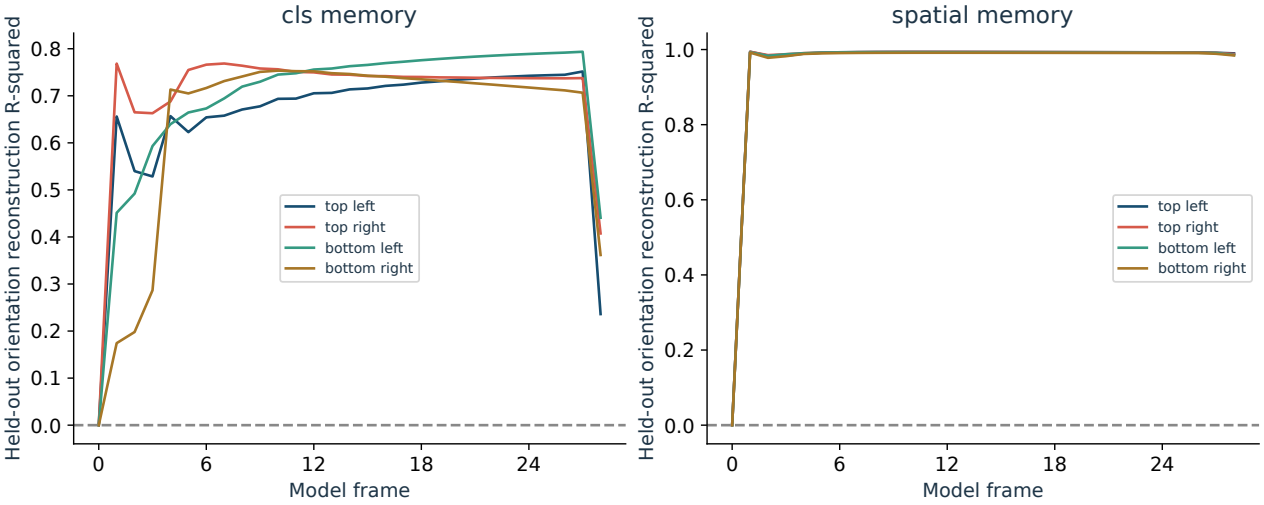


Figure 16: The original orientation at each of the four sample locations decoded from global and spatial memory. Curves track continuous sample content rather than the later correct-answer label. Zero is the fitting-mean baseline.

4 Discussion

The all-location reconstruction provides a direct storage result: the original orientation inventory remains readable during the blank interval, before the retrospective cue identifies a target. Spatial-memory reconstruction is particularly strong. A mid-delay spatial reset and a later target-priority suppression each impair the final binding judgement, identifying storage and retrieval selection as separate causal stages. The final answer becomes explicit in global memory after the cue and probe supply the information needed to define it. The preserved inventory has native structureâ€"four equally spaced orientations assigned to locationsâ€"so the result concerns retention of that structured array and its spatial assignment.

The behavioral result establishes 100.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies target_-8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Set size and swap distance should be varied independently. The present native generator draws a rotated inventory of four orientations spaced 45 degrees apart and permutes their locations. A follow-up with independently sampled orientations would test memory precision across a richer array distribution.

4.1 Interpreting these results alongside biological experiments

4.1.1 Wheeler and Treisman (2002)

Our probe can exchange orientations between locations while preserving the overall inventory. This directly tests assignment rather than the mere presence of features. Delay reconstruction shows the inventory is readable before the retrospective cue, while spatial reset and later target inhibition separately impair storage and retrieval. Those measurements make the binding interpretation more concrete than terminal accuracy alone. [5]

4.1.2 Bays, Catalao and Husain (2009)

A wrong-location report can reflect an assignment error even when feature content survives. Our binary swap decision compresses those possibilities into correct or incorrect. Reconstructing every location's orientation permits a more informative error analysis: compare decoded angles with the true local angle and with the other stored angles. The equally spaced native inventory makes this especially important because it introduces predictable structure. [6]

4.1.3 Griffin and Nobre (2003)

The late cue selects an internal representation rather than a visible sample. The model's before-cue reconstruction and after-cue attention sequence provide a natural parallel. A successful retrocue computation should first preserve several assignments, then prioritize the relevant one. Probe-period inhibition tests the second operation without erasing the original movie. [7]

4.2 A computational account of the observed behavior

Binding can be understood as retaining a joint relation between a location and a feature. Knowing all four orientations is insufficient if their positions are forgotten. A spatial recurrent field provides an explicit coordinate system for such relations, while global memory can later express whether the cued assignment changed. The current all-location reconstruction supports retained structured content; causal spatial disruption shows that this state matters. A resource account predicts gradual loss of precision or increasing assignment errors as load and similarity increase. The native task fixes a structured set of four orientations, so a load-dependent resource law must be tested with independent feature draws and additional set sizes.

4.3 Where this task agrees with the biological evidence

The strongest agreement is the separation of retained content, spatial assignment and retrospective selection. The main protocol difference is the constrained orientation inventory. That structure can make inference easier than remembering arbitrary independent feature-location pairs and should be varied before interpreting native perfection as unlimited binding capacity.

4.4 A follow-up experiment motivated by these results

Use independent orientations, then vary set size and minimum angular separation. Include feature substitutions and pure swaps. Measure local-angle precision, other-item confusions and binary behavior together, with delay and retrieval interventions separated in time.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Treisman AM, Gelade G (1980). A feature-integration theory of attention. Cognitive Psychology 12:97-136. https://pubmed.ncbi.nlm.nih.gov/7351125/.

[2] Funahashi S, Bruce CJ, Goldman-Rakic PS (1989). Mnemonic coding of visual space in the monkey dorsolateral prefrontal cortex. Journal of Neurophysiology 61:331-349. https://pubmed.ncbi.nlm.nih.gov/2918358/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] Wheeler and Treisman (2002). Binding in short-term visual memory. Primary source.

[6] Bays, Catalao and Husain (2009). The precision of visual working memory is set by allocation of a shared resource. Primary source.

[7] Griffin and Nobre (2003). Orienting attention to locations in internal representations. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
D0	128	100.0%	1.000	0.000
D4	128	100.0%	1.000	0.000
D12	128	100.0%	1.000	0.000
D24	128	100.0%	1.000	0.000

B Every-frame attention atlas

Condition D24. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 16 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

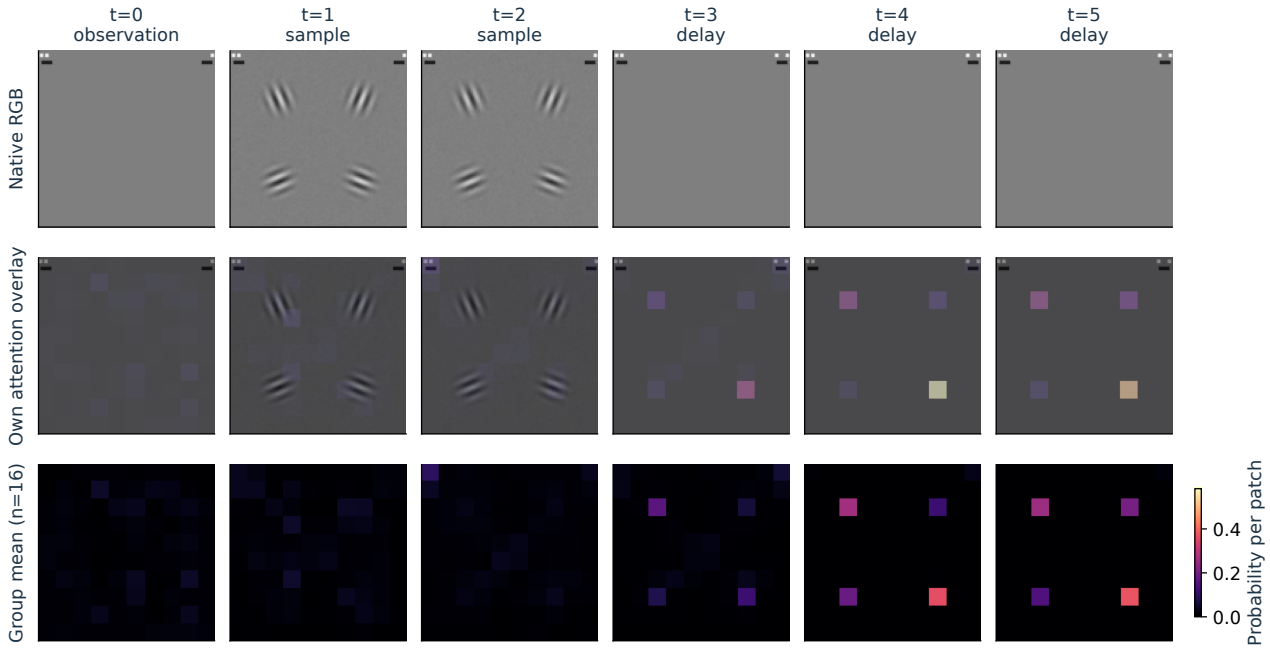


Figure 17: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

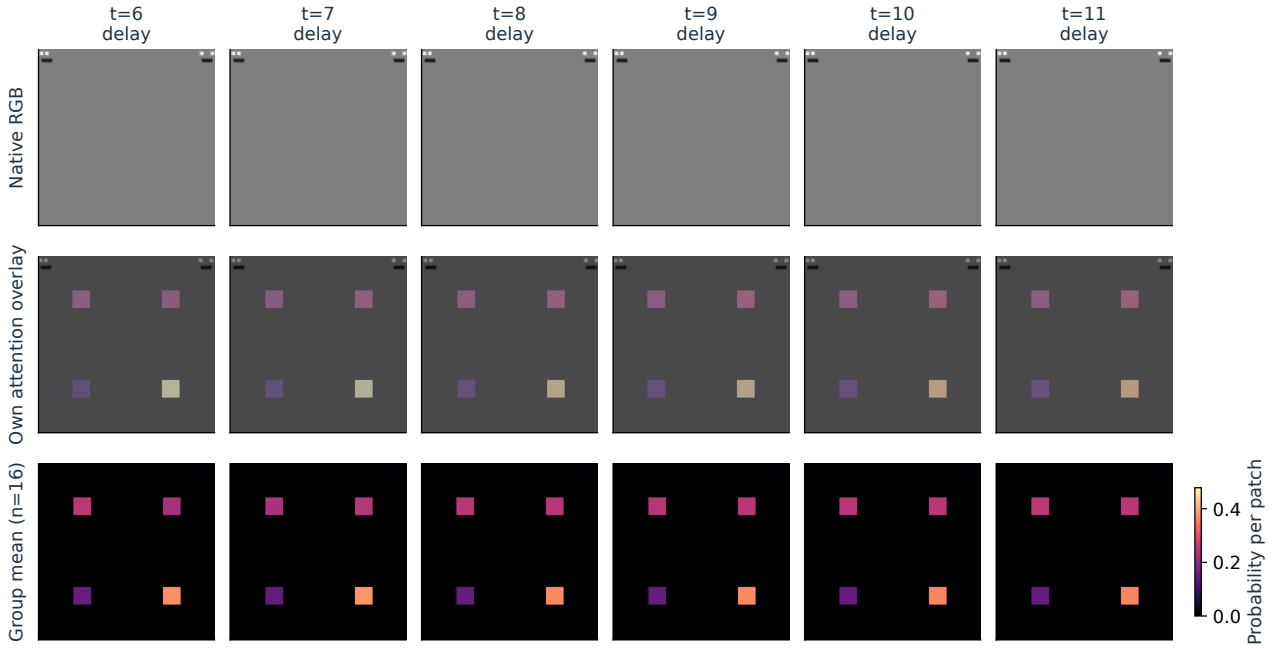


Figure 18: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

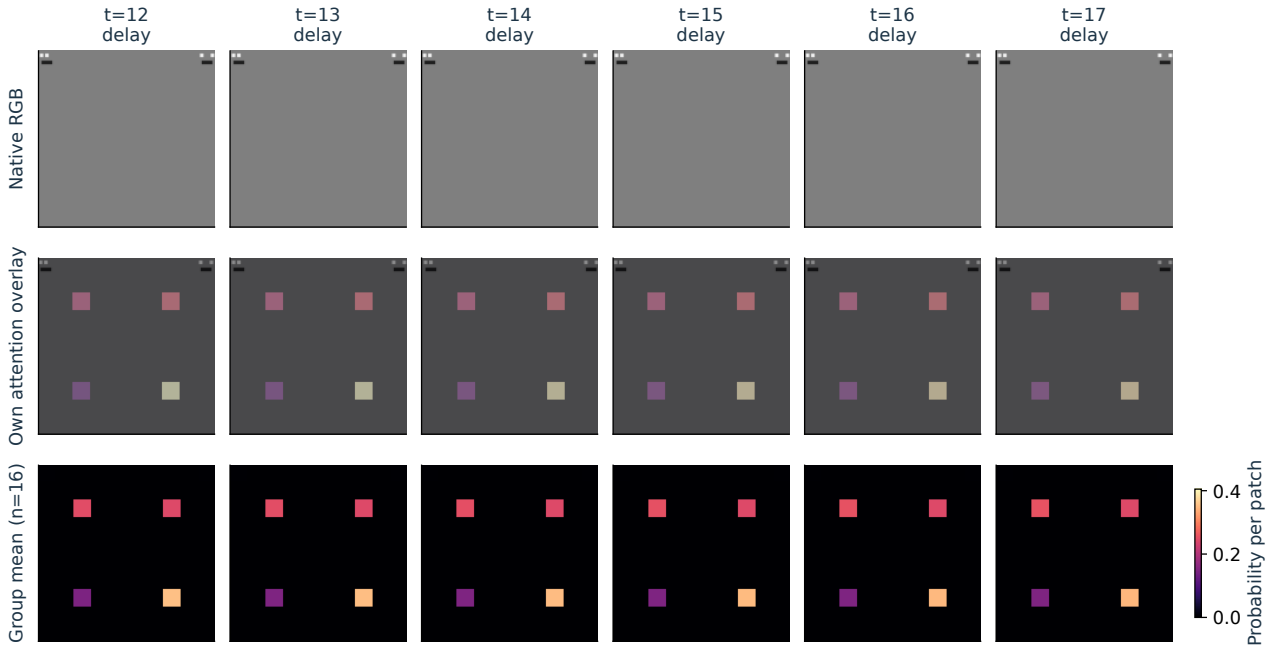


Figure 19: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

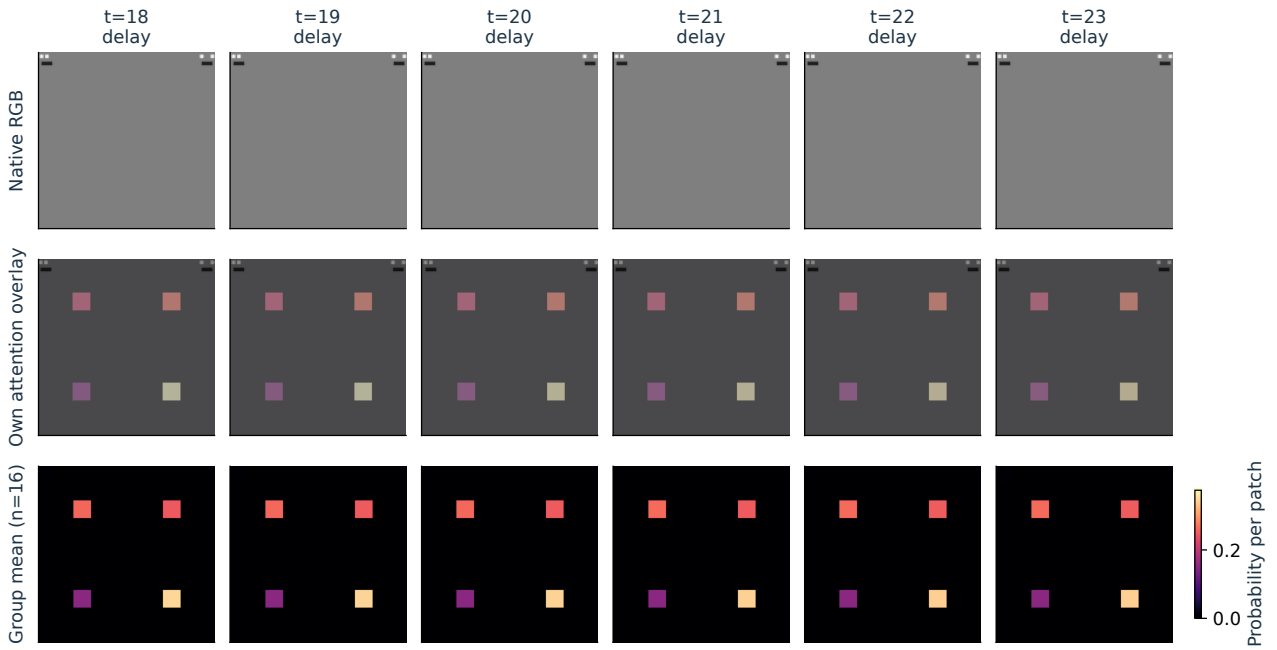


Figure 20: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

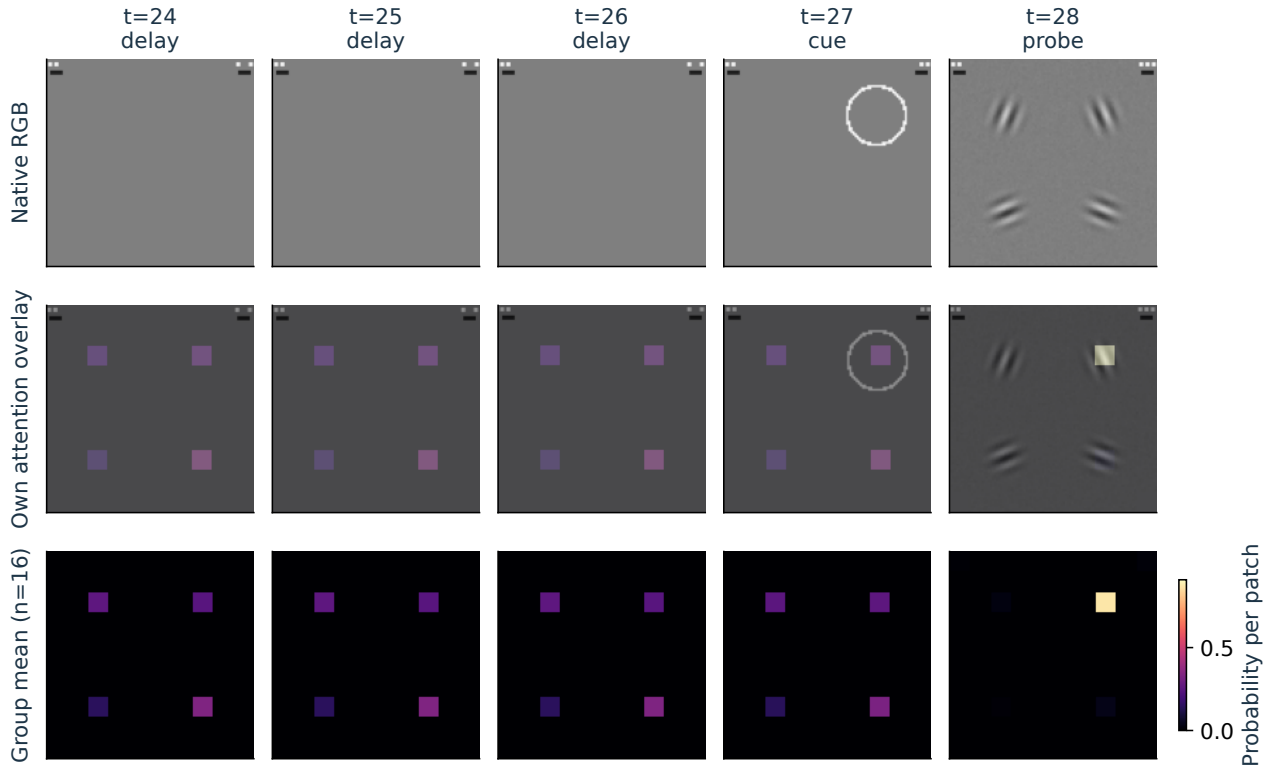


Figure 21: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

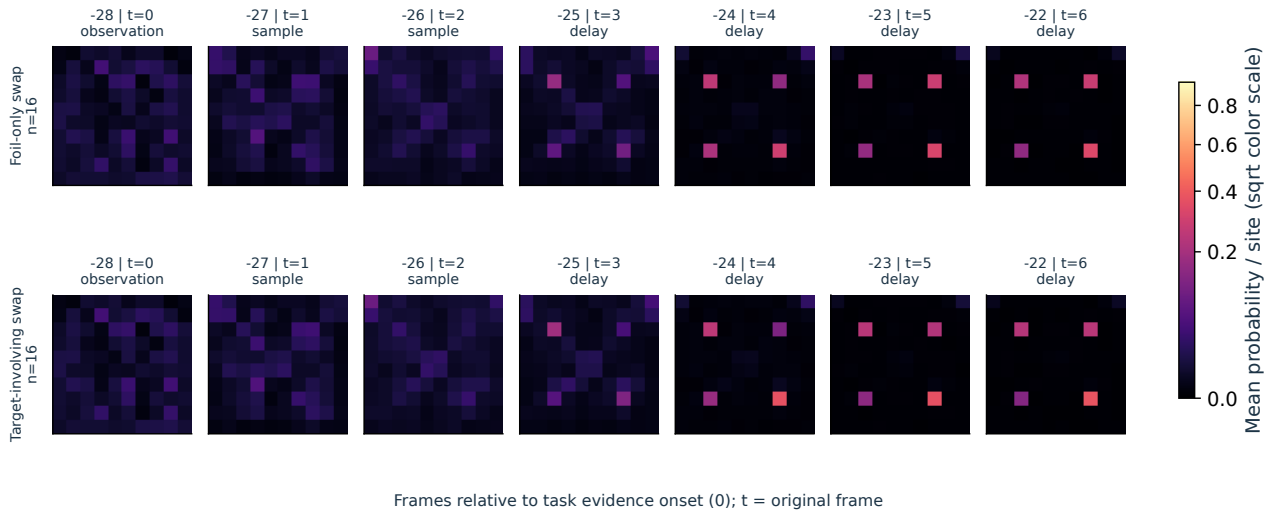


Figure 22: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

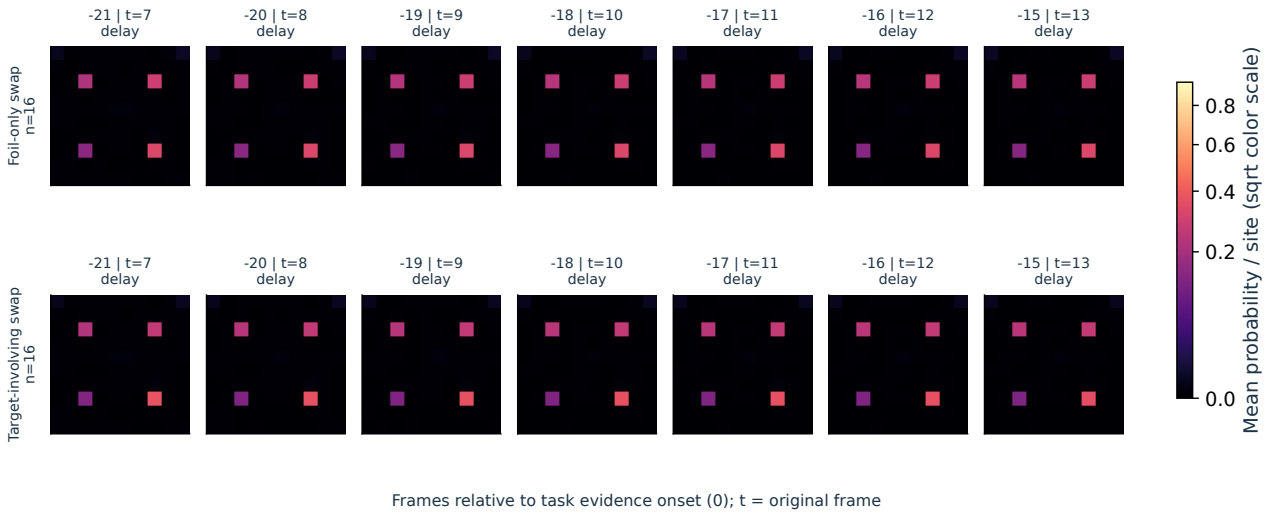


Figure 23: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

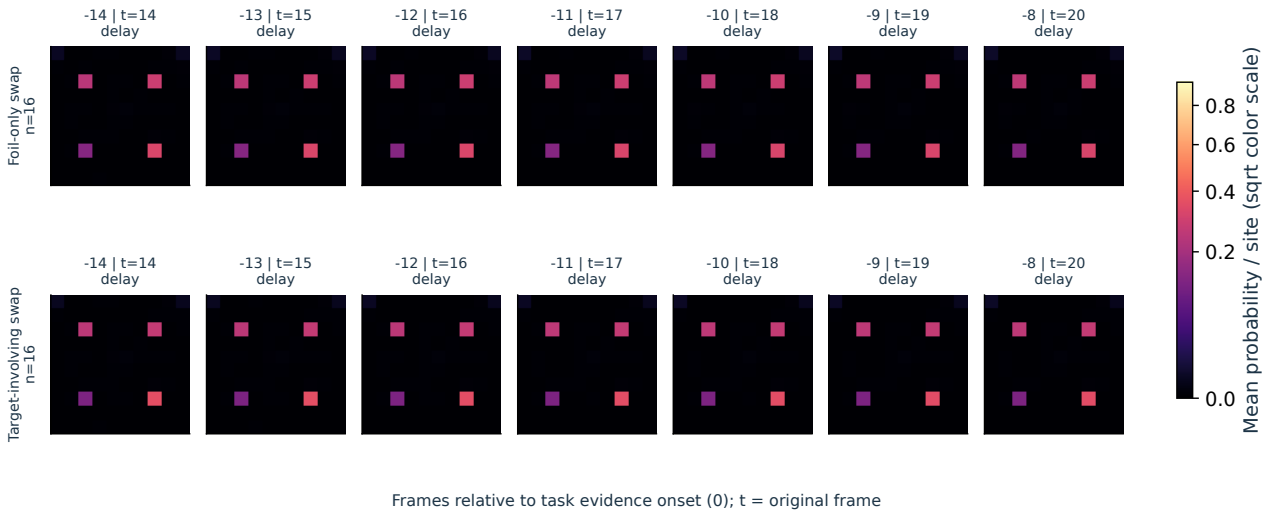


Figure 24: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

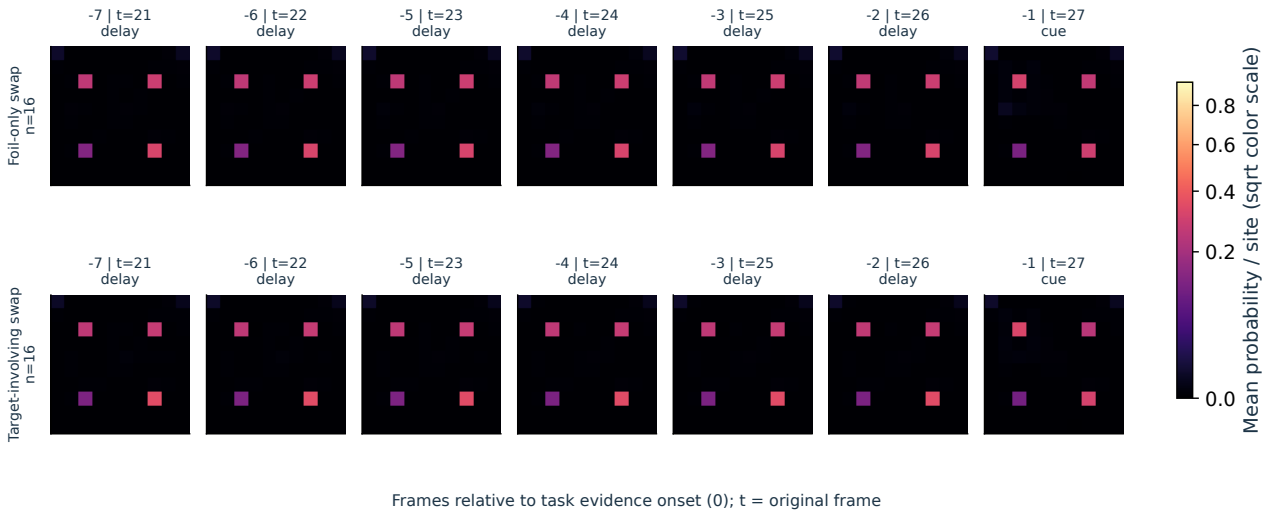


Figure 25: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

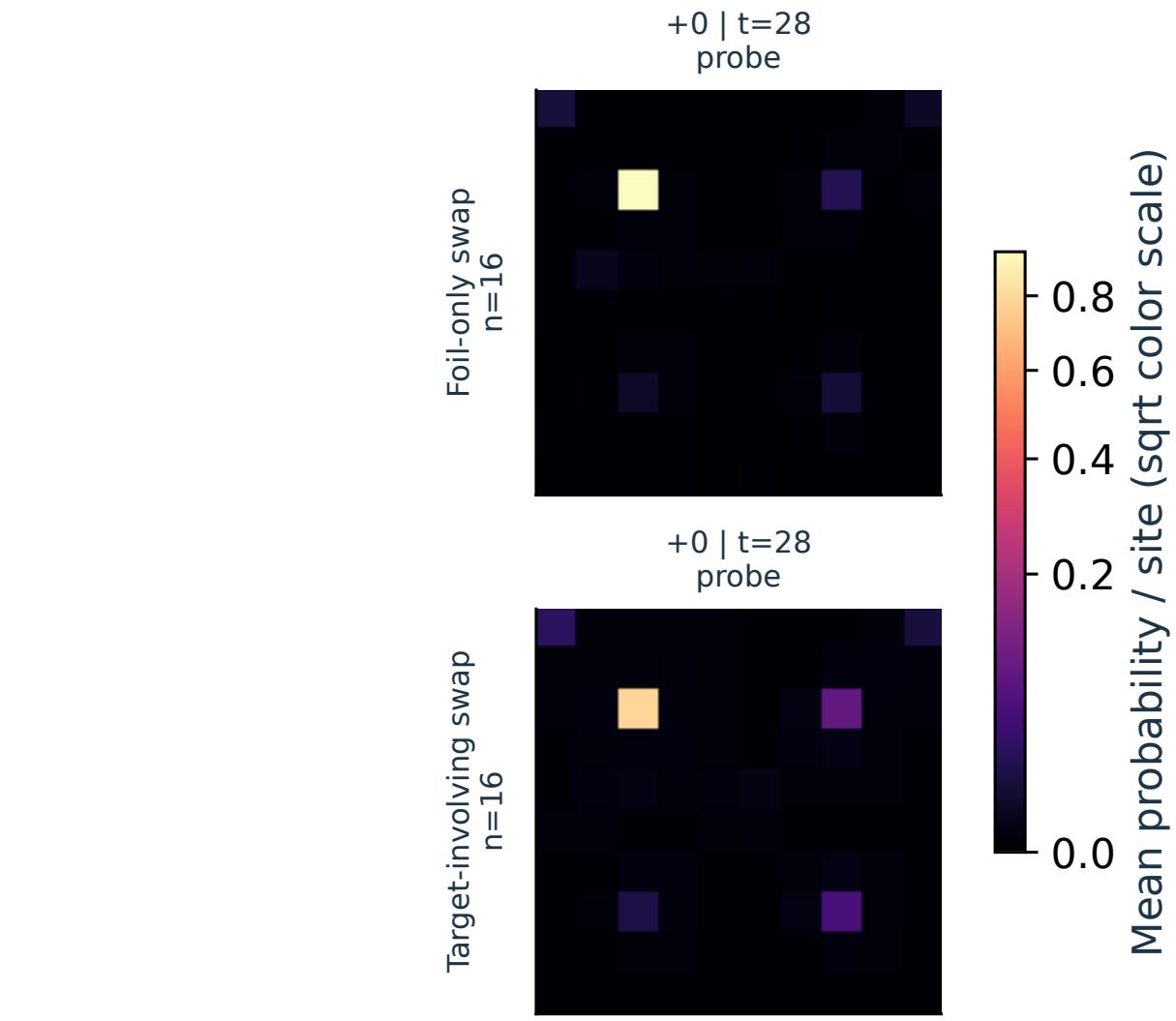


Figure 26: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Recognition of an exact natural-image sample among intervening images

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized image recognition in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 12 native conditions and 1,536 fresh trials, equal-condition accuracy was 94.3% (95% interval [93.1%, 95.4%]). For the representative condition N4_H5, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 0.992. The largest observed impairment in that condition was spatial_gain0_t6, changing accuracy by -52.3 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The observer studies a sequence of natural images and then judges whether the repeated probe raster appeared in that sequence. Study load changes the number of images that must remain available. Probe duration changes the time allowed to evaluate the retrieval signal.

Load and probe hold are analyzed separately. The zero-load condition is an always-novel control; recognition sensitivity requires both seen and novel trials. Neural analyses distinguish readable information about the last study image from readable evidence that the later probe belongs to the study set. These are different targets for a memory decoder.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Miller and colleagues studied visual working memory using delayed matching in macaque prefrontal cortex. That tradition motivates separating sample-related activity, retention, and a match-dependent response to the probe. In a multi-image study sequence, each new image can displace, transform or coexist with earlier information. Population trajectories track that evolution, and load curves quantify its behavioral consequences. The recognition decision concerns membership of the entire study set, while the continuous decoder asks the narrower question of whether a coarse description of the last study image remains recoverable. Reporting both prevents those two memory questions from being conflated.

1.2 Biological experiments motivating this task

Experimental question. Why does recognition remain good for repeats while novel images become increasingly likely to be accepted?

Miller, Li and Desimone (1993) IT responses carried match-related information during delayed matching. Stimulus-selective delay activity could be disrupted by intervening stimuli, separating visible delay activity from successful later recognition. [5]

Miller, Erickson and Desimone (1996) A delayed matching task with intervening test objects examined prefrontal signals for remembered samples and matching decisions. It provided a comparison with temporal-cortex mechanisms under interference. [6]

Bays, Catalao and Husain (2009) Continuous-report errors reflected both feature and location memory. Accounting for their joint uncertainty supported a shared-resource description in which precision changes with the number of remembered objects. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Is the repeated probe raster an exact member of the preceding study-image set?

Response labels are 0: not in study set; 1: in study set. The native suite contains 12 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use N4_H5, N24_H5.

1.4 Stimulus sequence and spatial layout

For the representative native condition N4_H5, the sequence has 13 frames, indexed from zero: frame 0: observation; frames 1–4: study; frames 5–7: delay; frames 8–12: probe. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Study images	4
Probe frames	5

A trajectory that changes with each study image may still support recognition if old information is retained in a distributed population. The 8-by-8 content reconstruction is a coarse image descriptor, whereas the behavioral label requires exact membership in the original raster sequence. Attention maps are therefore shown on individual photographs, with separate location-averaged panels.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 94.3%, with macro AUC 0.963. The matched final checkpoint changed accuracy by -0.26 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was N24_H5 at 79.7%; the highest was N0_H3 at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

Marginal accuracy by load was 0: 100.0%; 4: 99.5%; 12: 94.0%; 24: 83.6%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

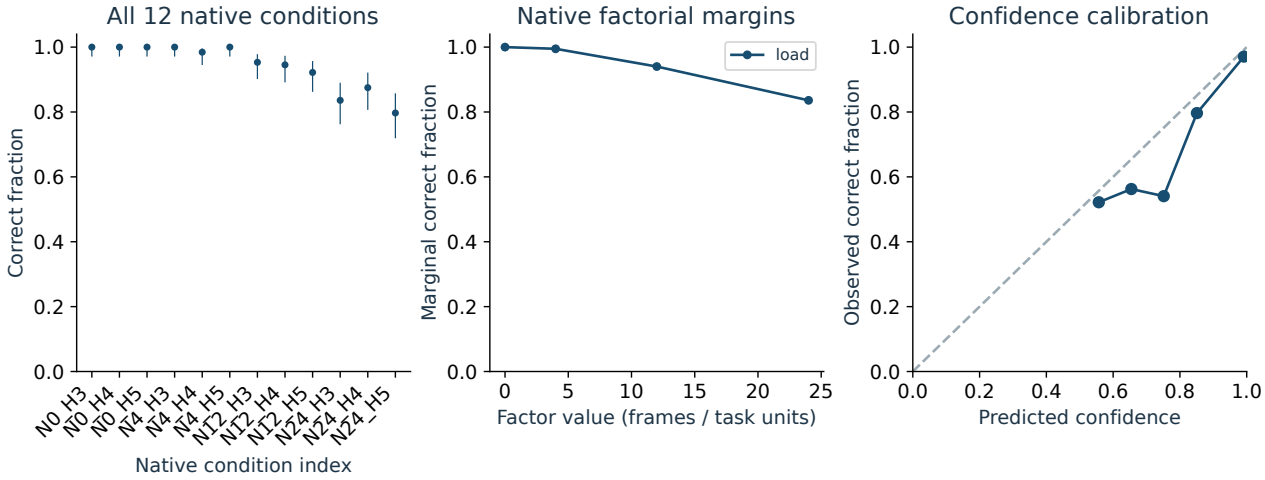


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Load, false recognition and serial position

Zero-study trials are negative controls: every probe is novel. The nonzero study loads test recognition after four, twelve or twenty-four images. Hit and false-alarm curves separate missed stored items from false familiarity. Serial-position quartiles run from the earliest to latest study items, pooled across probe-hold durations. Intervals resample probe-image identities across those conditions.

At load 4, the hit rate was 100.0% and the false-alarm rate was 1.0%. At load 12, the hit rate was 98.4% and the false-alarm rate was 10.4%. At load 24, the hit rate was 94.8% and the false-alarm rate was 27.6%. The load-related error pattern can therefore be assessed as a change in recognition sensitivity and report tendency, alongside the population-memory measurements.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	57.0%	100.0%
0.7	100.0%	1.0	59.4%	100.0%
0.8	100.0%	1.0	62.5%	100.0%
0.9	98.4%	1.0	73.0%	100.0%

The representative condition N4_H5 provides 5 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–

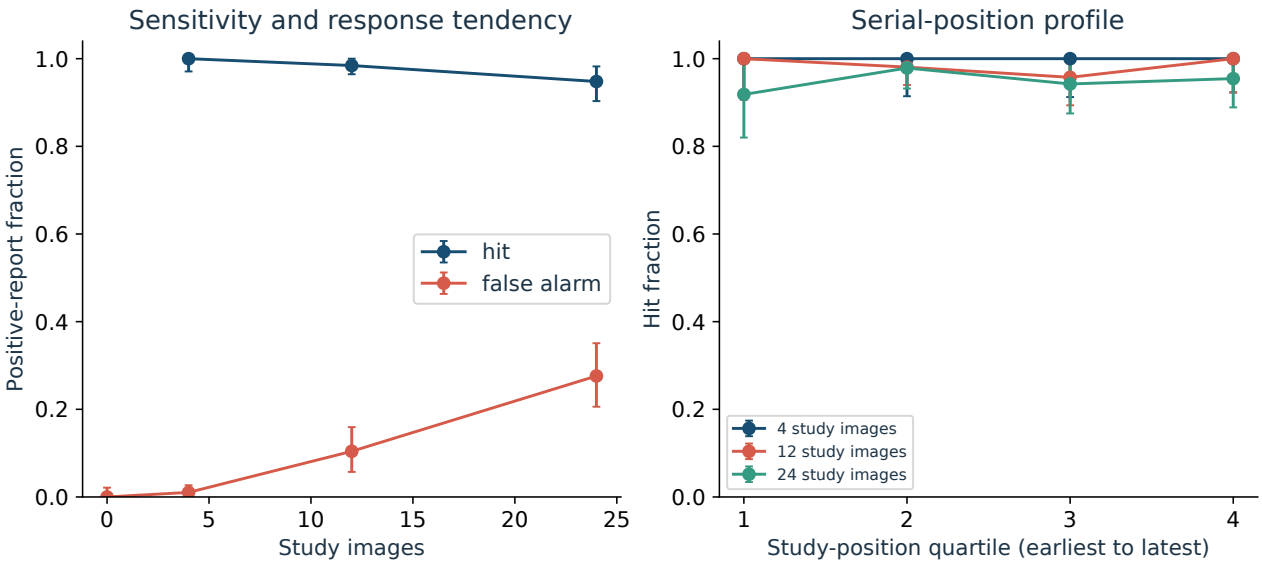


Figure 2: Recognition error structure. Left: hit and false-alarm fractions by study load. Right: hit fractions by study-position quartile. Intervals preserve repeated probe photographs across probe-hold conditions.

coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 1,515 of 1,536 trials crossed. Of those, 438 changed their answer later, 428 corrected an early error, and 10 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

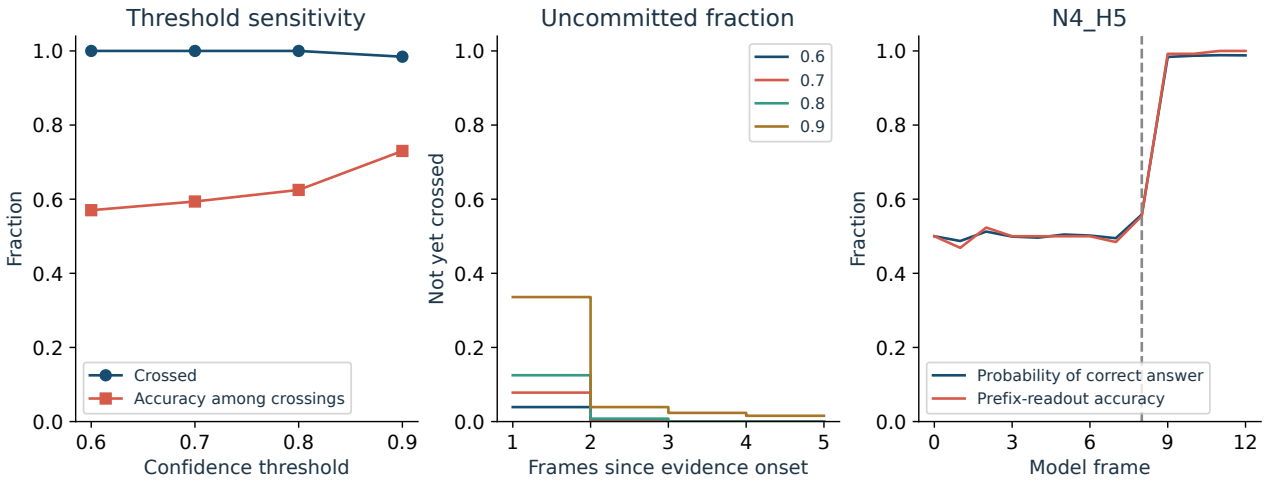


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

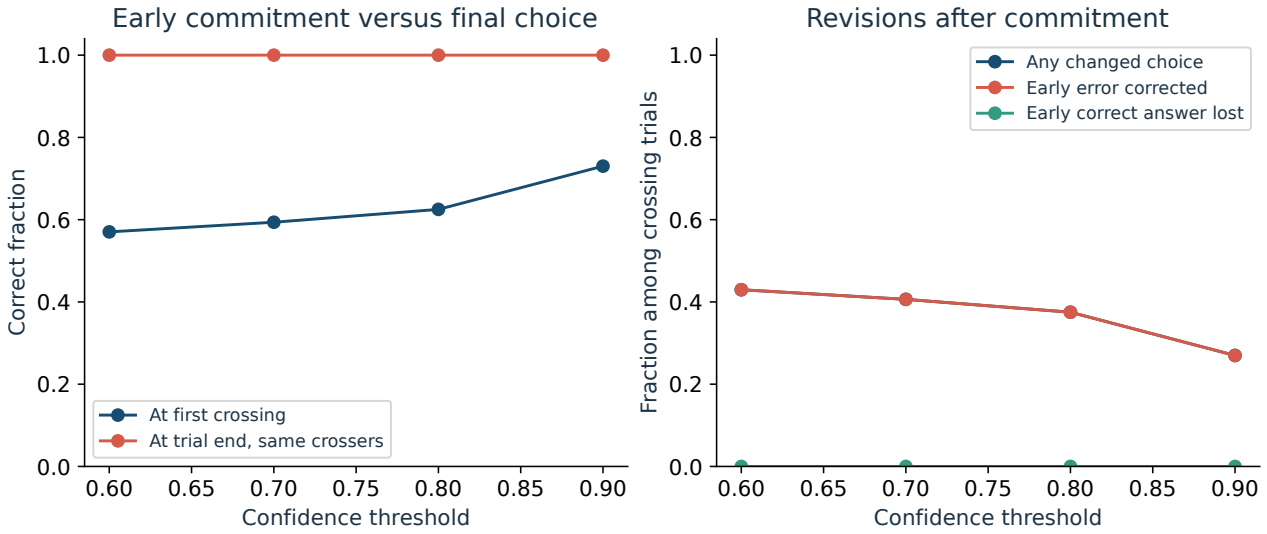


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

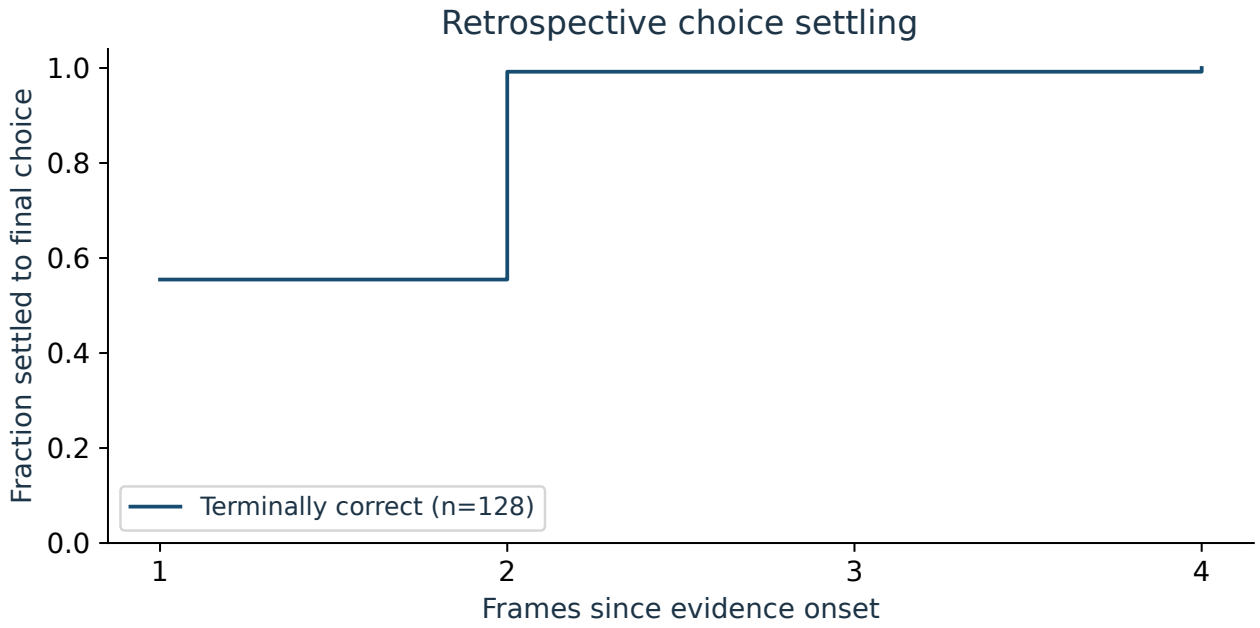


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

Rows cross four versus twenty-four study images with seen versus novel probes, using five probe frames. Probe onset defines the common time origin. The central region is the middle 4-by-4 patch square; the second trace sums the remaining 84 sites. An average over different photographs measures a consistent spatial preference; the viewer separately preserves the exact photograph and its own attention.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
4 study images; novel probe	64	54
4 study images; seen probe	64	59
24 study images; novel probe	64	56
24 study images; seen probe	64	59

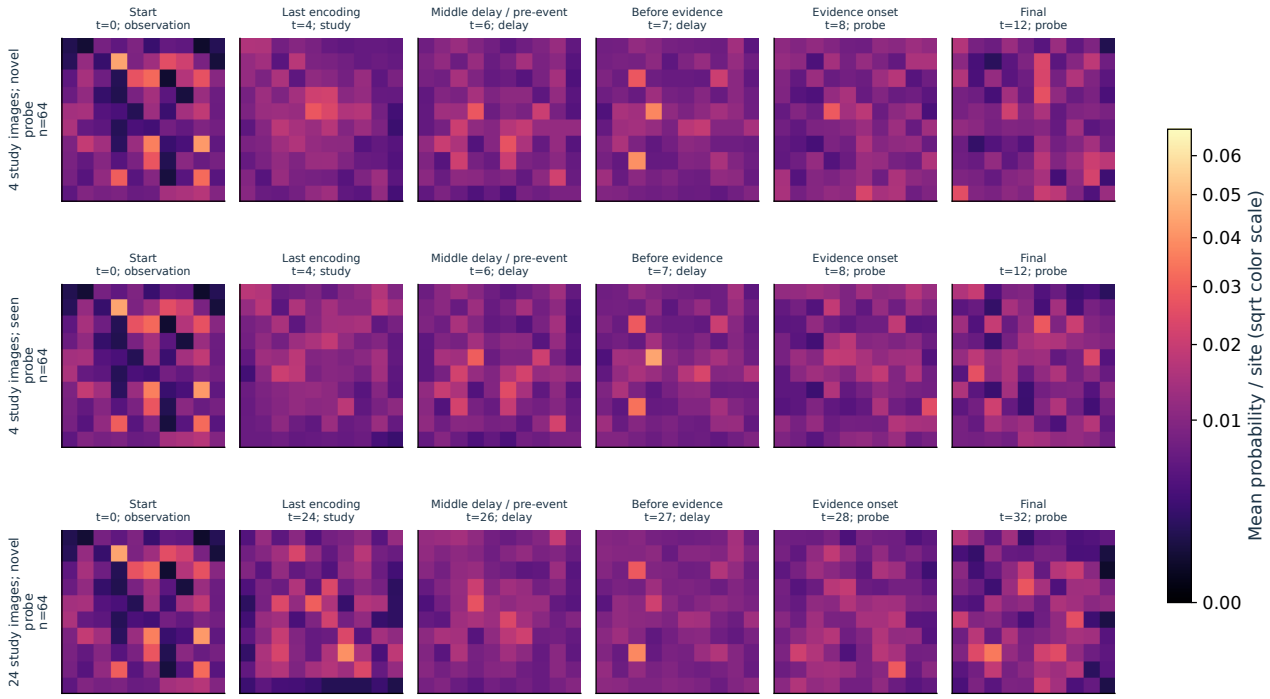


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

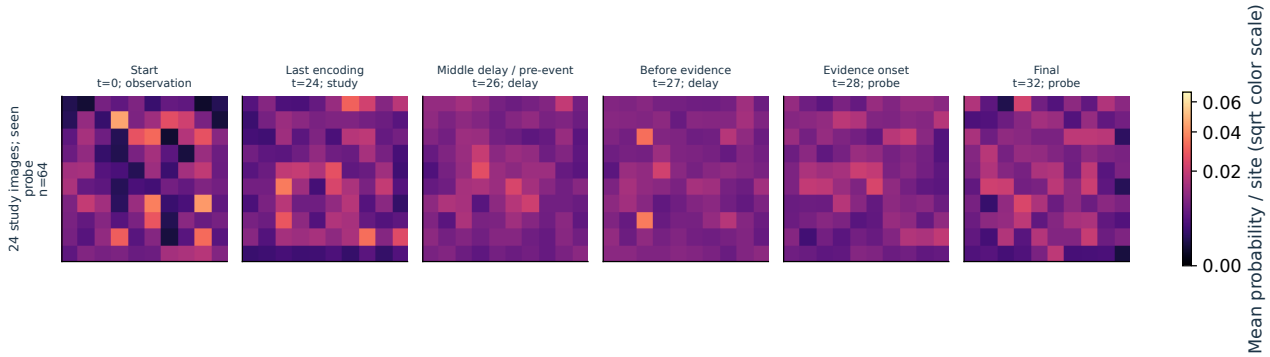


Figure 7: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

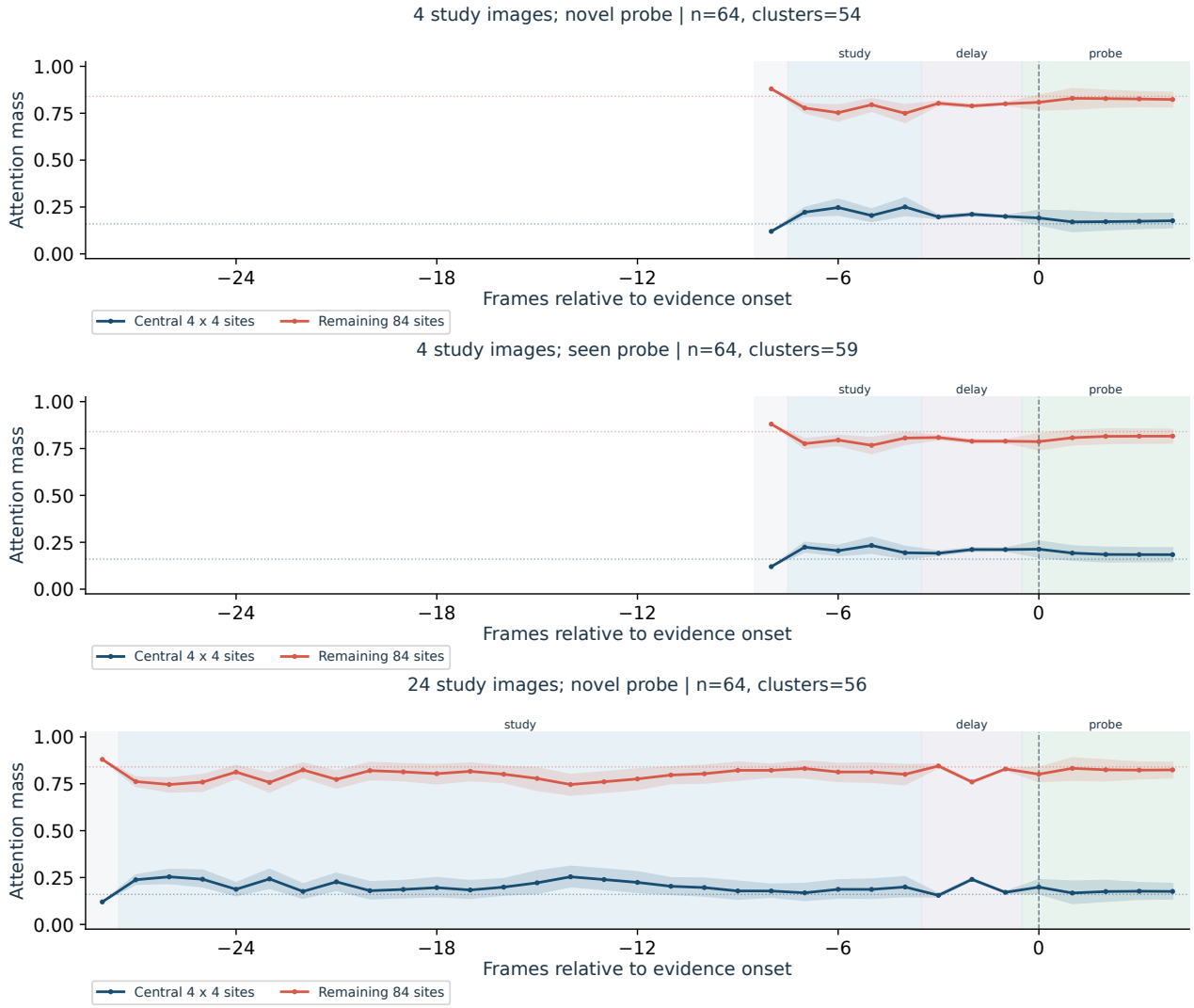


Figure 8: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

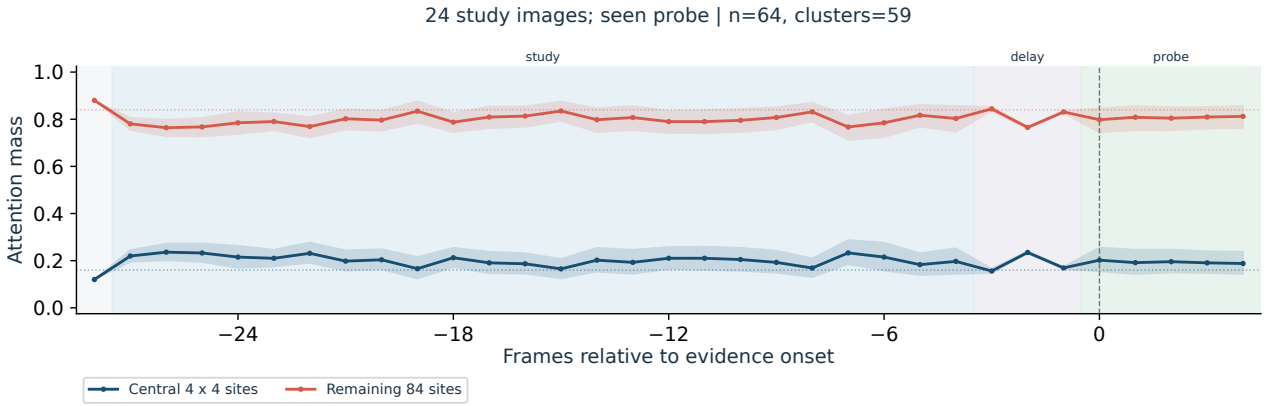


Figure 9: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

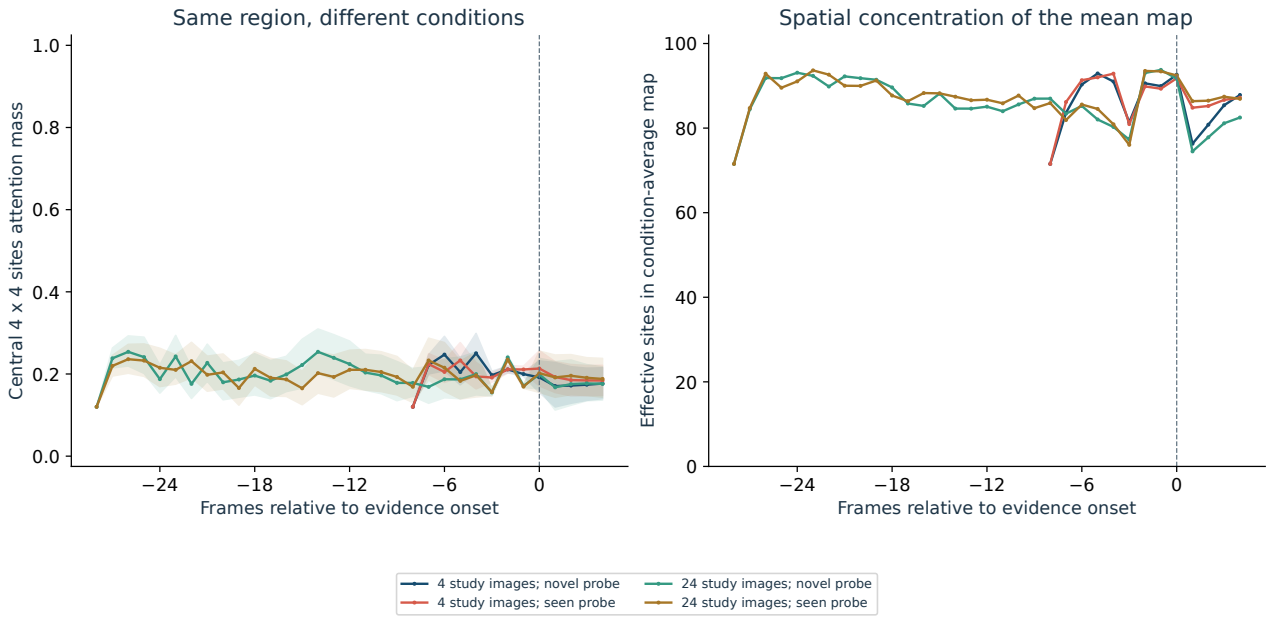


Figure 10: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. 4 study images; novel probe: central 4 x 4 sites mass peaks at 0.250 at original frame 4 (study), and is 0.177 on the final frame. 4 study images; seen probe: central 4 x 4 sites mass peaks at 0.233 at original frame 3 (study), and is 0.184 on the final frame. 24 study images; novel probe: central 4 x 4 sites mass peaks at 0.254 at original frame 2 (study), and is 0.176 on the final frame. 24 study images; seen probe: central 4 x 4 sites mass peaks at 0.236 at original frame 2 (study), and is 0.188 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For spatial_gain0_t6, paired accuracy changed by -52.3 percentage points (paired 95% interval [-60.6%, -43.3%]; paired p=0.0001; within-task Holm p=0.0062). There were 2 corrected trials and 69 newly incorrect trials. For target_-4_delay, paired accuracy changed by +0.8 percentage points (paired 95% interval [-1.6%, 3.8%]; paired p=1; within-task Holm p=1). There were 2 corrected trials and 1 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

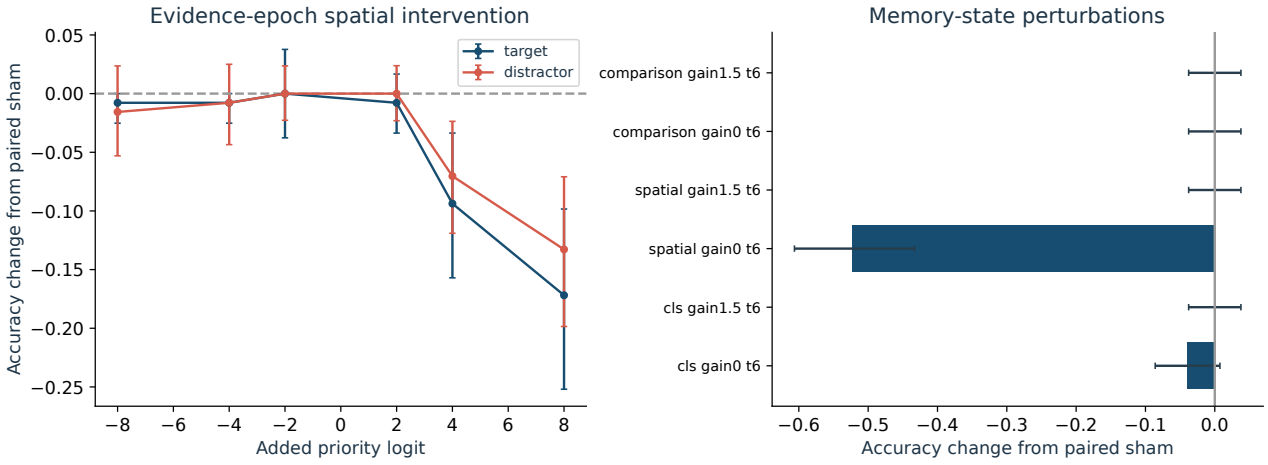


Figure 11: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
N24_H5	target_-4	-3.9	1
N24_H5	target_+4	-18.0	0.0276
N24_H5	distractor_-4	-3.1	1
N24_H5	distractor_+4	-11.7	0.0276
N4_H5	target_-4	-0.8	1
N4_H5	target_+4	-9.4	0.4851
N4_H5	distractor_-4	-0.8	1
N4_H5	distractor_+4	-7.0	0.708

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.511	50.8%	0.485
fusion	0.538	55.5%	0.155
spatial	0.788	75.8%	0.005
cls	0.992	97.7%	0.005
comparison	0.644	60.9%	0.020
attention	0.543	53.9%	0.155

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 0.00 to 13.08 within the retained subspace. The final step displacement was 0.459 standardized RMS units. In spatial memory, the participation ratio ranged from 0.00 to 19.09 within the retained subspace. The final step displacement was 0.118 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 6, coarse sample-content reconstruction had R-squared 0.053 from CLS memory and 0.099 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -3.9 percentage points, spatial, -52.3 percentage points, comparison, +0.0 percentage points. Comparing

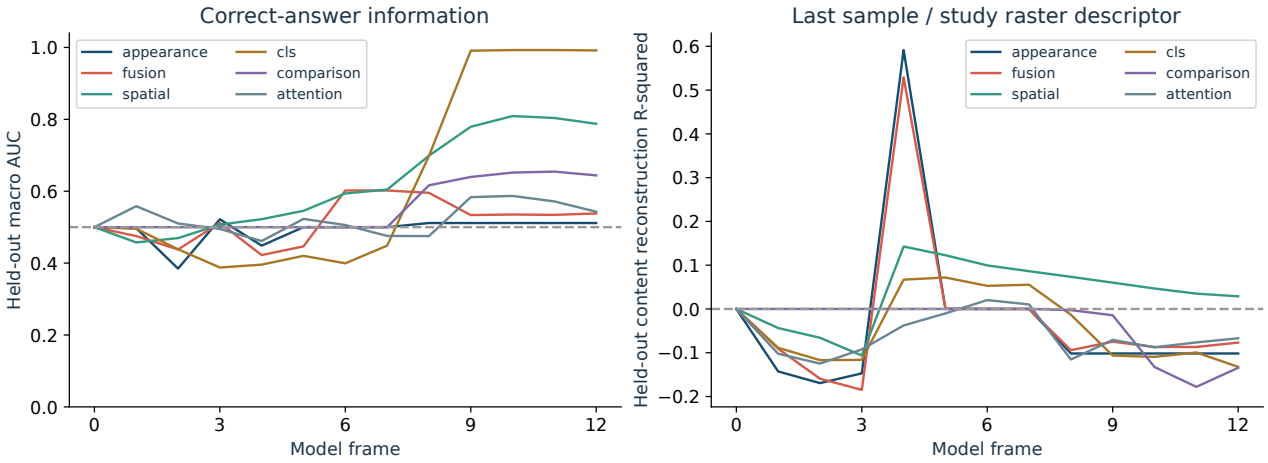


Figure 12: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

3.11 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.565	unavailable	3200
fusion	0.561	0.600	6400
spatial	0.822	0.510	6400
cls	0.995	0.514	64
comparison	0.630	unavailable	735
attention	0.514	0.493	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 43 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

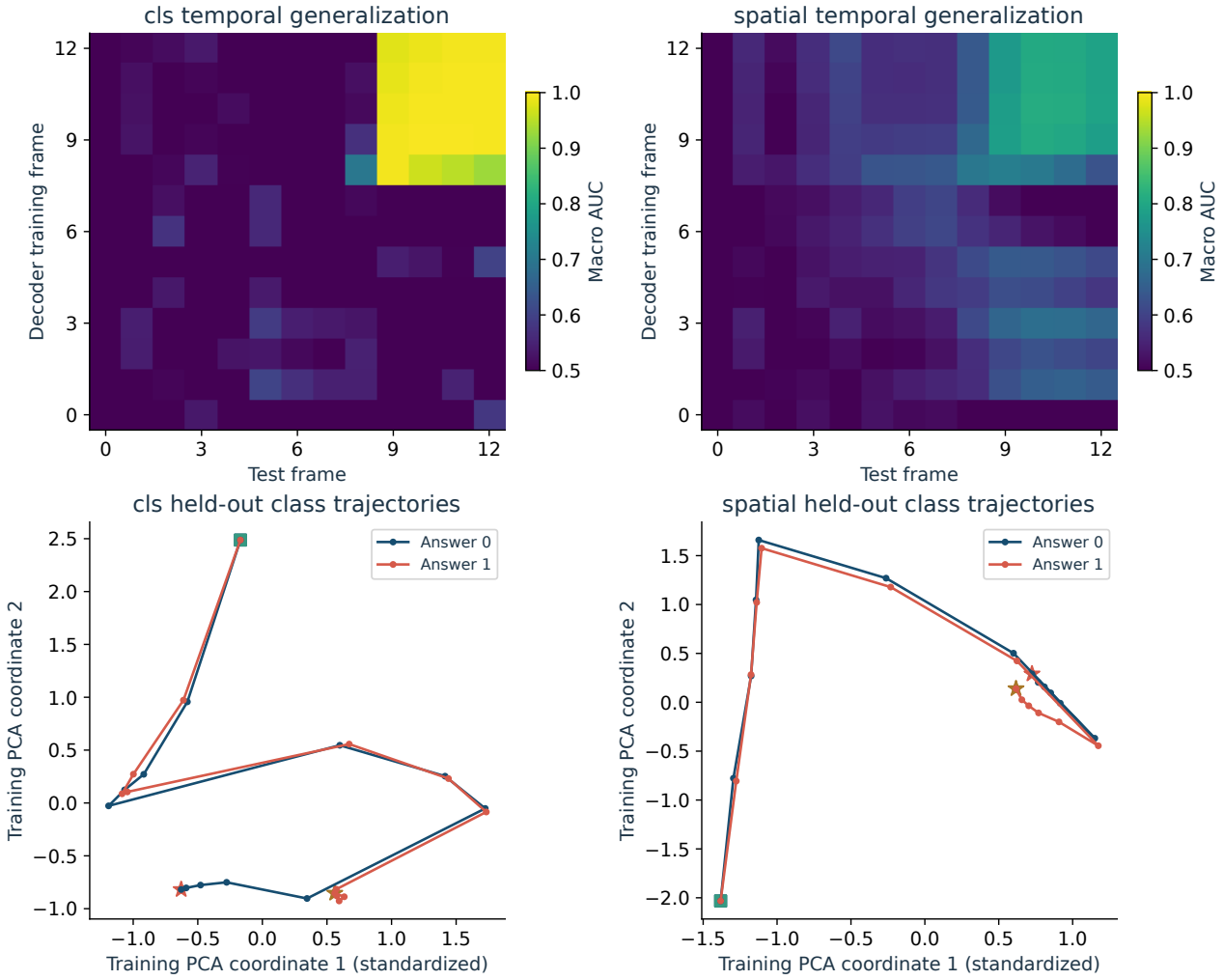


Figure 13: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

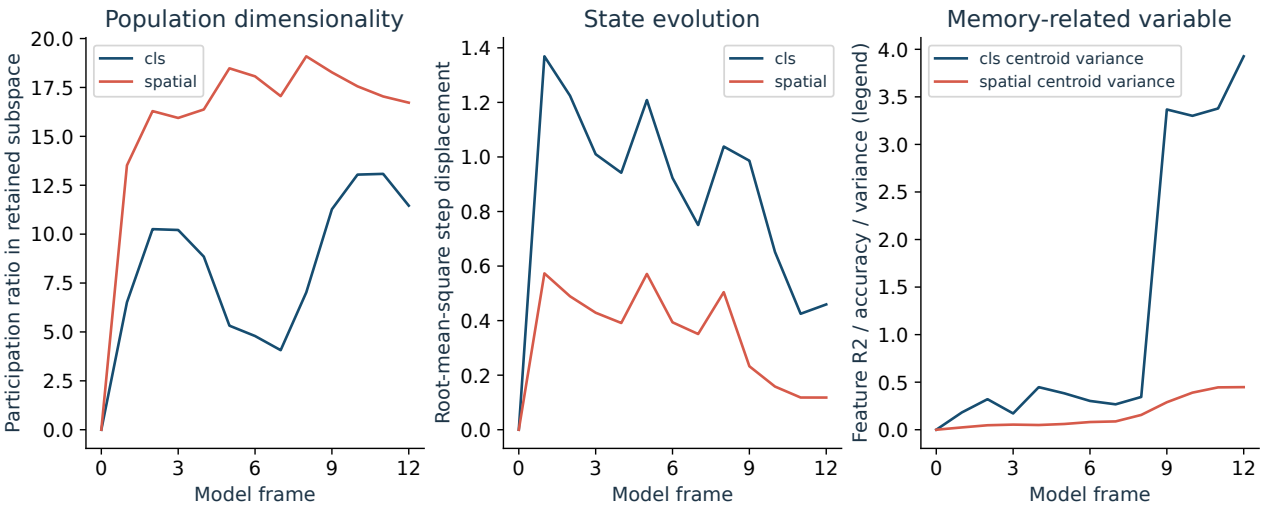


Figure 14: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

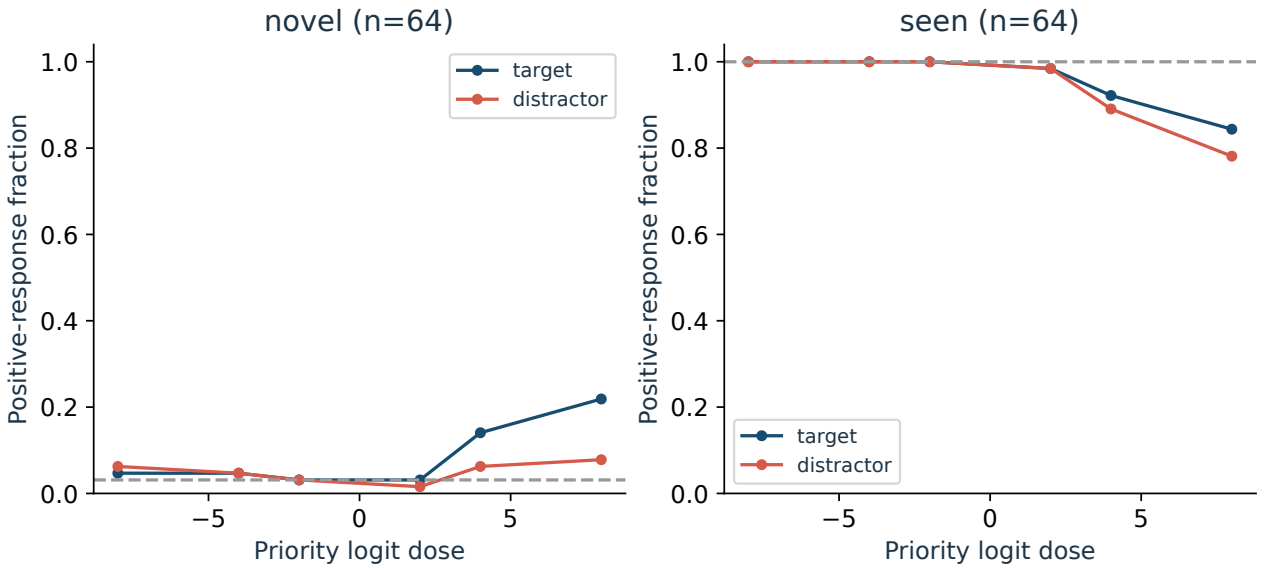


Figure 15: Event-specific causal response functions. Curves show positive reports after evidence-epoch target or alternative-site intervention; the horizontal line is paired sham. Event counts are stated above each panel. A positive report has the task-specific meaning defined in the text.

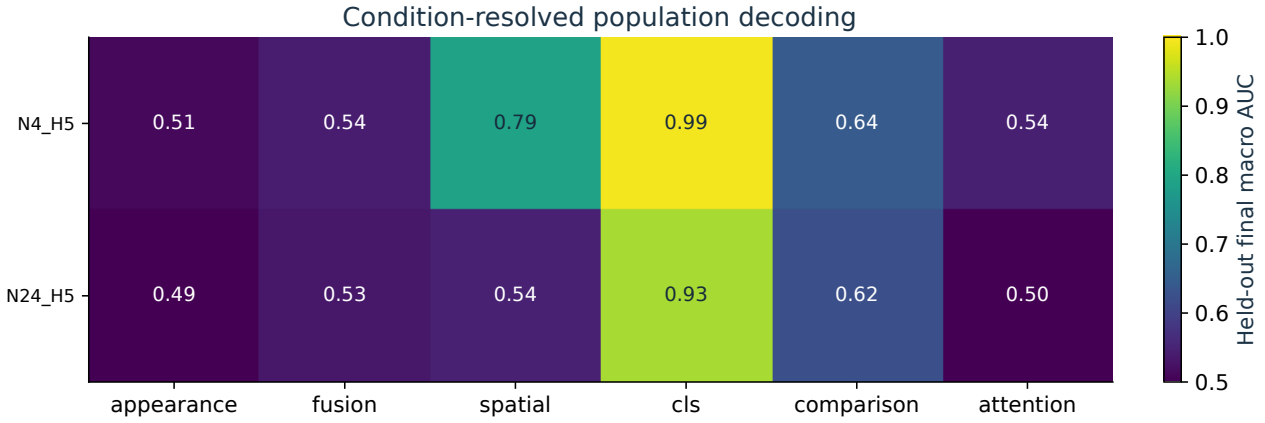


Figure 16: Held-out final answer decoding across selected conditions and recording sites. A constant-zero comparison representation is displayed at chance AUC.

3.12 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

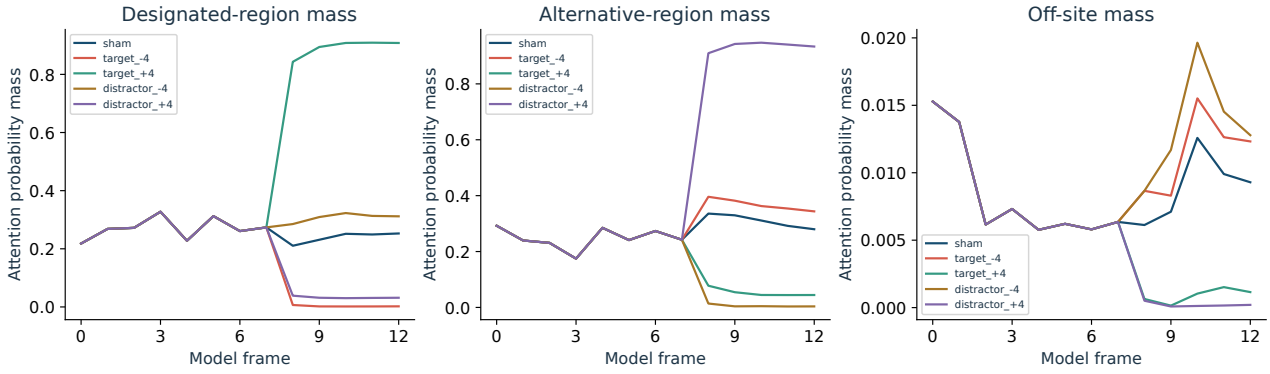


Figure 17: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

3.13 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

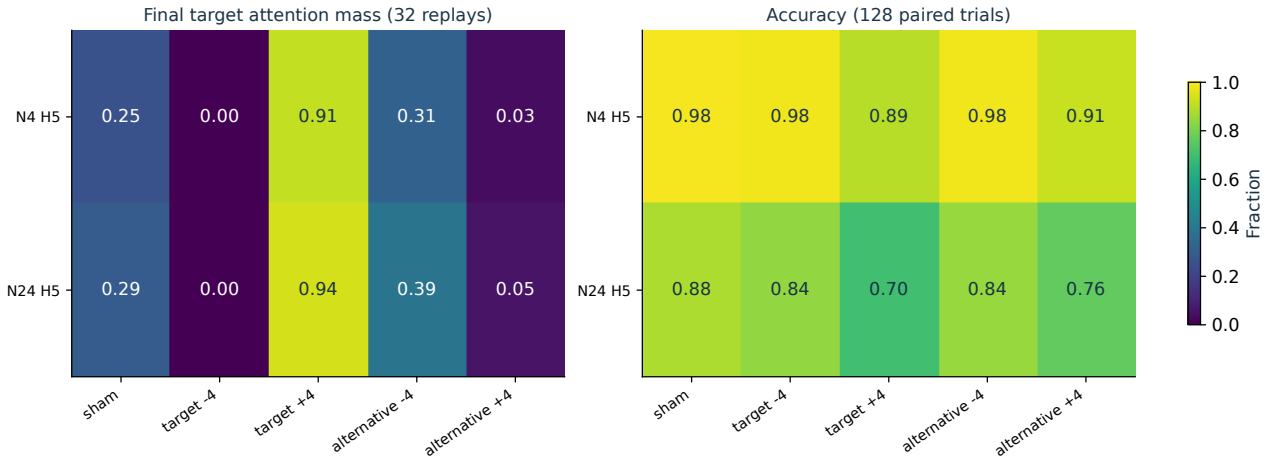


Figure 18: Condition-resolved manipulation checks. The left panel measures final target-region allocation on 32 exact replays; the right measures accuracy on all 128 paired intervention trials. All displayed spatial doses have magnitude four and act during evidence frames. Rows preserve each selected condition, making redistribution and behavioral rescue separately visible.

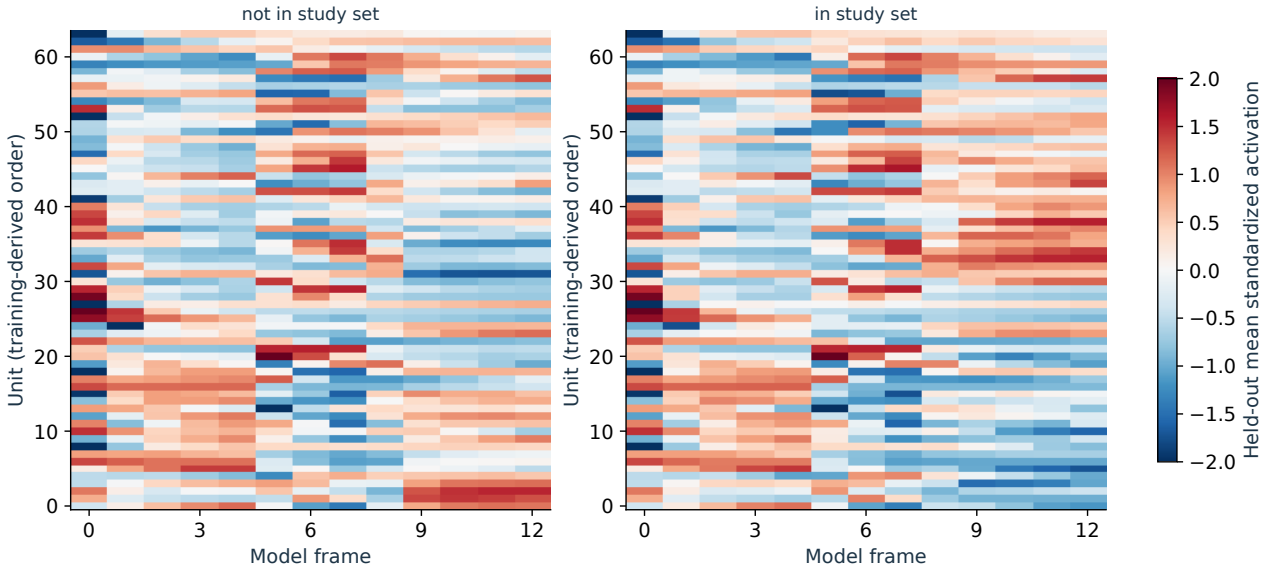


Figure 19: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

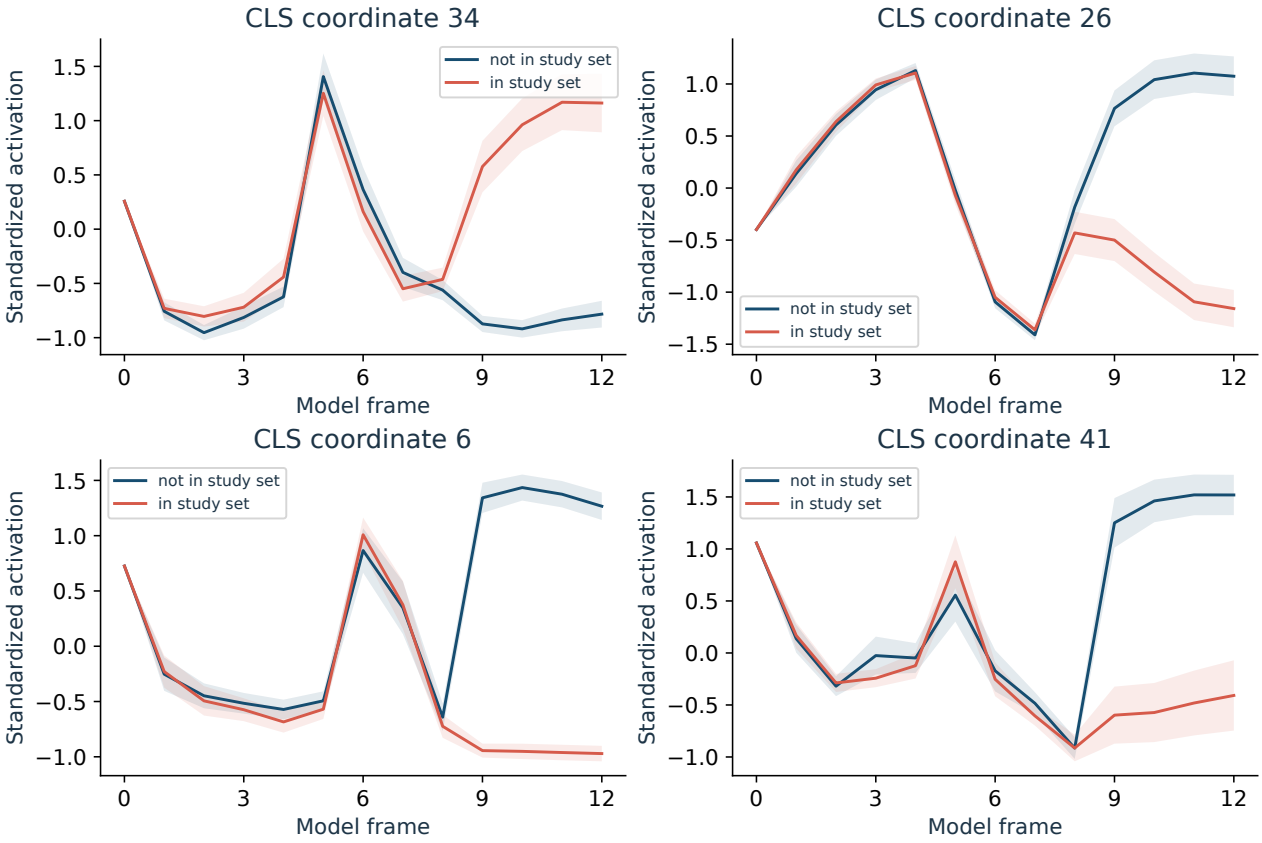


Figure 20: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

3.14 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	50.8%	52.1%	unavailable
fusion	55.5%	55.1%	unavailable
spatial	75.8%	74.4%	unavailable
cls	97.7%	99.2%	unavailable
comparison	60.9%	58.4%	unavailable
attention	53.9%	54.5%	unavailable

3.15 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that

frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\text{fit}} d_j^{\text{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

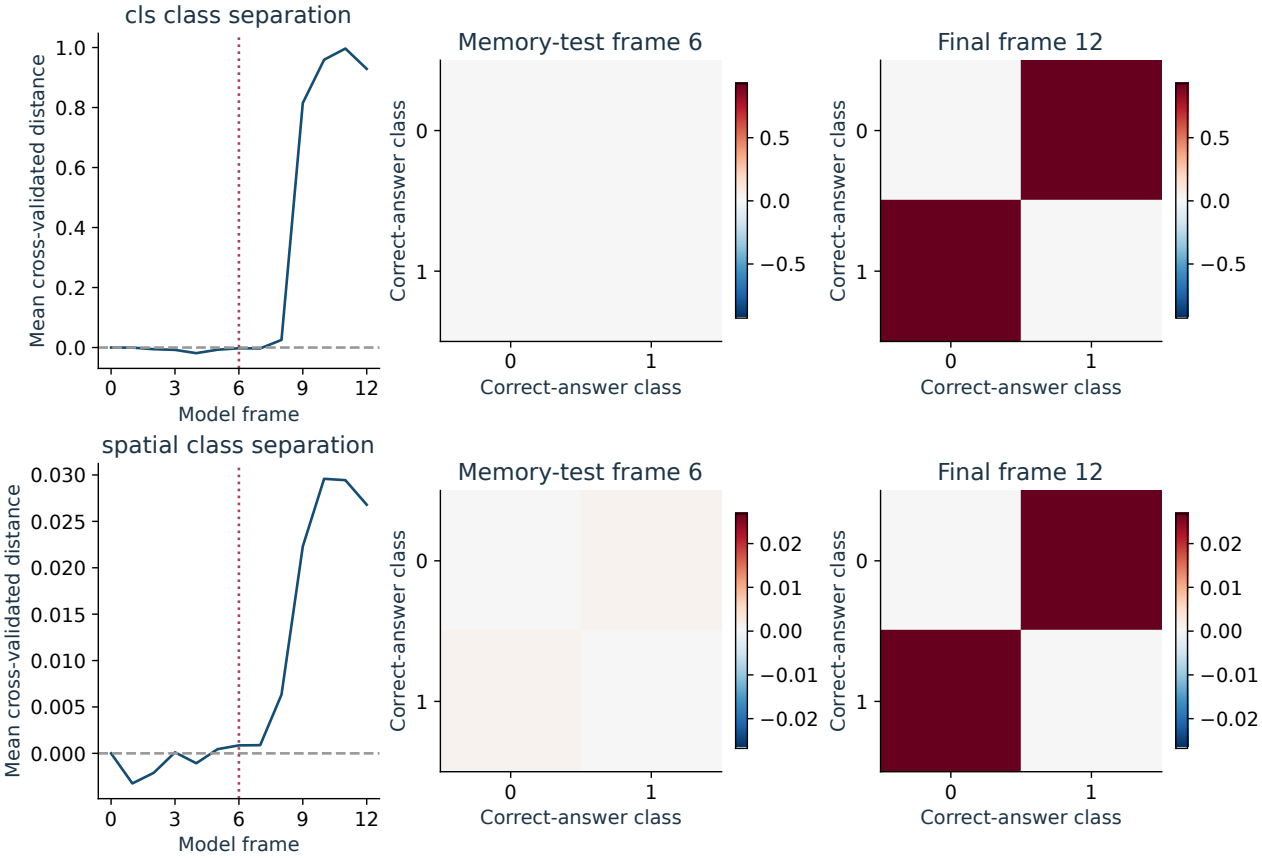


Figure 21: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

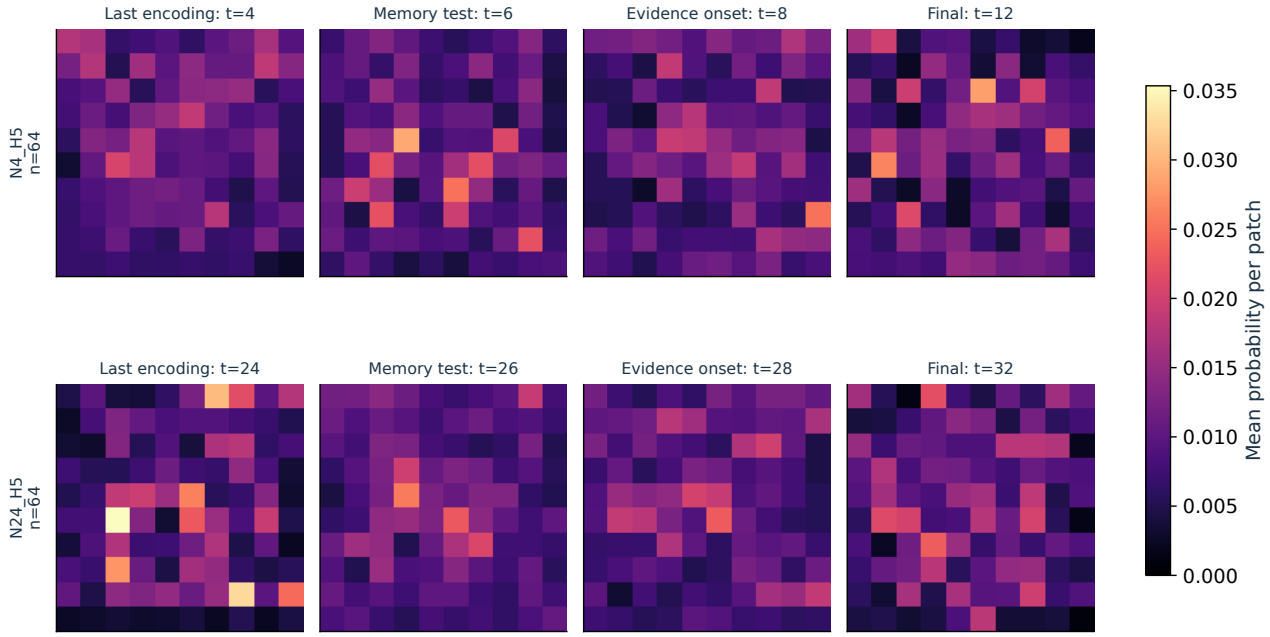


Figure 22: Condition-resolved attention at four temporal landmarks. Memory-search and pop-out rows average trials with top-left targets. Image-recognition rows average seen-probe trials. Each map is shown in the original image layout with a common probability scale across conditions; the row label gives the number of averaged trials. The memory-test frame is the middle delay, or the frame before evidence when a blank delay is absent.

3.16 Additional analysis prompted by the biological comparison

The table re-expresses the existing native recognition decisions as repeat hits, novel-image false alarms, sensitivity and criterion. Positive means "previously studied. Sensitivity is $d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F})$ and criterion is $c = -[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]/2$. Here Φ^{-1} converts a cumulative probability to a standard-normal quantile; $\widetilde{H} = (h + 0.5)/(n_1 + 1)$ and $\widetilde{F} = (f + 0.5)/(n_0 + 1)$ use counts of hits h, false alarms f, repeat opportunities n_1 and novel opportunities n_0. The correction keeps perfect observed rates finite. Negative criterion indicates a tendency toward positive reports in this descriptive signal-detection coordinate system.

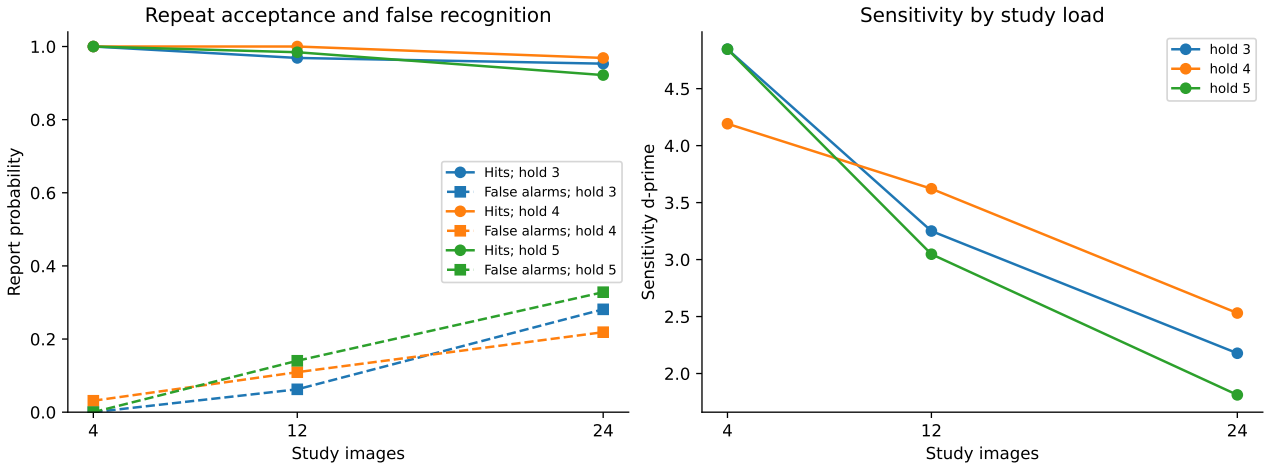


Figure 23: Recognition reanalysis of the saved native trials. Curves retain the three probe-hold conditions. Hits stay high while false recognition grows with study load; sensitivity generally decreases. These are descriptive condition estimates. There are 64 repeats and 64 novel probes per nonempty-list condition. Source-image clustering used elsewhere in the paper remains necessary for inferential comparisons across repeated photographs.

Load	Hold	Hits/repeats	False/novel	d'	c
0	3	0/0	0/128	–	–
0	4	0/0	0/128	–	–
0	5	0/0	0/128	–	–
4	3	64/64	0/64	4.85	-0.00
4	4	64/64	2/64	4.19	-0.33
4	5	64/64	0/64	4.85	-0.00
12	3	62/64	4/64	3.25	-0.14
12	4	64/64	7/64	3.62	-0.61
12	5	63/64	9/64	3.05	-0.47
24	3	61/64	18/64	2.18	-0.52
24	4	62/64	14/64	2.53	-0.50
24	5	59/64	21/64	1.81	-0.47

The empty-list condition has no repeat opportunities, so sensitivity and criterion are undefined there. At higher loads, reduced sensitivity and a positive-report tendency coexist. A criterion-only explanation would fail to capture the observed decrease in separation under this convention; an item-similarity analysis can identify what makes the novel probes seem familiar.

4 Discussion

Increasing the study load primarily increases false recognition of novel probes, while genuine repeats retain a high hit rate. This separates the dominant error pattern from simple disappearance of every stored image. The spatial-memory reset produces a large cost in the lower-load neural condition, and the final global state carries a strong recognition signal. The serial-position curves describe which stored items are most likely to be accepted; the last-image reconstruction describes only one component of the retained content. Together, these measures support studying familiarity accumulation and interference, alongside the precision with which individual study items can be recovered.

The behavioral result establishes 94.3% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies spatial_gain0_t6 as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Matched lure similarity and controlled intervening-image interference would extend the measured serial-position and false-alarm profiles. Reconstructing every study item would test how multiple stored images contribute to the recognition decision beyond the last-item probe reported here.

4.1 Interpreting these results alongside biological experiments

4.1.1 Miller, Li and Desimone (1993)

The model's growing false-recognition rate with study load makes match-related population signals especially relevant. The biological result distinguishes a later recognition signal from uninterrupted activity for a sample. Our last-image reconstruction describes one retained item, while the recognition decision may depend on a broader aggregate of studied content. These are different readouts of memory. [5]

4.1.2 Miller, Erickson and Desimone (1996)

Intervening images are a direct interference manipulation in both task families. The model must determine membership in a list, whereas delayed matching can designate one sample as the relevant item. That rule difference changes which intervening images should be stored or ignored. Comparing the same image sequence under sample-only and any-studied-item rules would reveal whether the network can protect a selected memory. [6]

4.1.3 Bays, Catalao and Husain (2009)

A distributed-resource account predicts declining precision as more content is retained. Recognition errors, however, also depend on the threshold for declaring a match. The signal-detection reanalysis

below separates sensitivity from positive-response tendency across load. A high hit rate with rising false alarms can arise from overlapping familiarity distributions, a criterion shift, or both; accuracy alone does not decide among them. [7]

4.2 A computational account of the observed behavior

A familiarity-style computation compares the probe with an aggregate trace of previous images. As more images contribute, a novel probe can accidentally resemble some part of that trace. A more item-specific computation stores separable representations and compares the probe against each. The model's spatial-reset cost establishes that retained state supports the judgment, while its high repeat acceptance and increasing novel acceptance identify a characteristic interference pattern. The added sensitivity-and-criterion table expresses that pattern in a common recognition framework. It uses finite-sample correction for perfect rates and keeps probe-hold conditions separate. An empty study list contains no genuine repeats, so sensitivity is undefined there even when correct rejection is perfect.

4.3 Where this task agrees with the biological evidence

The model agrees with interference-sensitive recognition and a population-level memory signal. It differs from a simple all-or-none forgetting account because errors are concentrated in accepting novel probes. The biological comparison becomes sharper by measuring image similarity and list membership separately rather than treating a single average recognition score as memory capacity.

4.4 A follow-up experiment motivated by these results

Stratify novel probes by similarity to the study list and compare them with repeated items at different serial positions. Fit a held-out familiarity predictor and an itemwise nearest-memory predictor. Use matched images across list lengths to separate interference from differences in picture difficulty.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Miller EK, Erickson CA, Desimone R (1996). Neural mechanisms of visual working memory in prefrontal cortex of the macaque. Journal of Neuroscience 16:5154-5167. https://pubmed.ncbi.nlm.nih.gov/8756444/.

[2] Chelazzi L, Miller EK, Duncan J, Desimone R (1993). A neural basis for visual search in inferior temporal cortex. Nature 363:345-347. https://pubmed.ncbi.nlm.nih.gov/8497317/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] Miller, Li and Desimone (1993). Activity of neurons in anterior inferior temporal cortex during a short-term memory task. Primary source.

[6] Miller, Erickson and Desimone (1996). Neural mechanisms of visual working memory in prefrontal cortex of the macaque. Primary source.

[7] Bays, Catalao and Husain (2009). The precision of visual working memory is set by allocation of a shared resource. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
N0_H3	128	100.0%	unavailable	0.000
N0_H4	128	100.0%	unavailable	0.000
N0_H5	128	100.0%	unavailable	0.000
N4_H3	128	100.0%	1.000	0.003
N4_H4	128	98.4%	0.997	0.038
N4_H5	128	100.0%	1.000	0.005
N12_H3	128	95.3%	0.986	0.085
N12_H4	128	94.5%	0.979	0.094
N12_H5	128	92.2%	0.974	0.114
N24_H3	128	83.6%	0.936	0.232
N24_H4	128	87.5%	0.927	0.203
N24_H5	128	79.7%	0.867	0.300

B Every-frame attention atlas

Condition N4_H5. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 64 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.



Figure 24: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

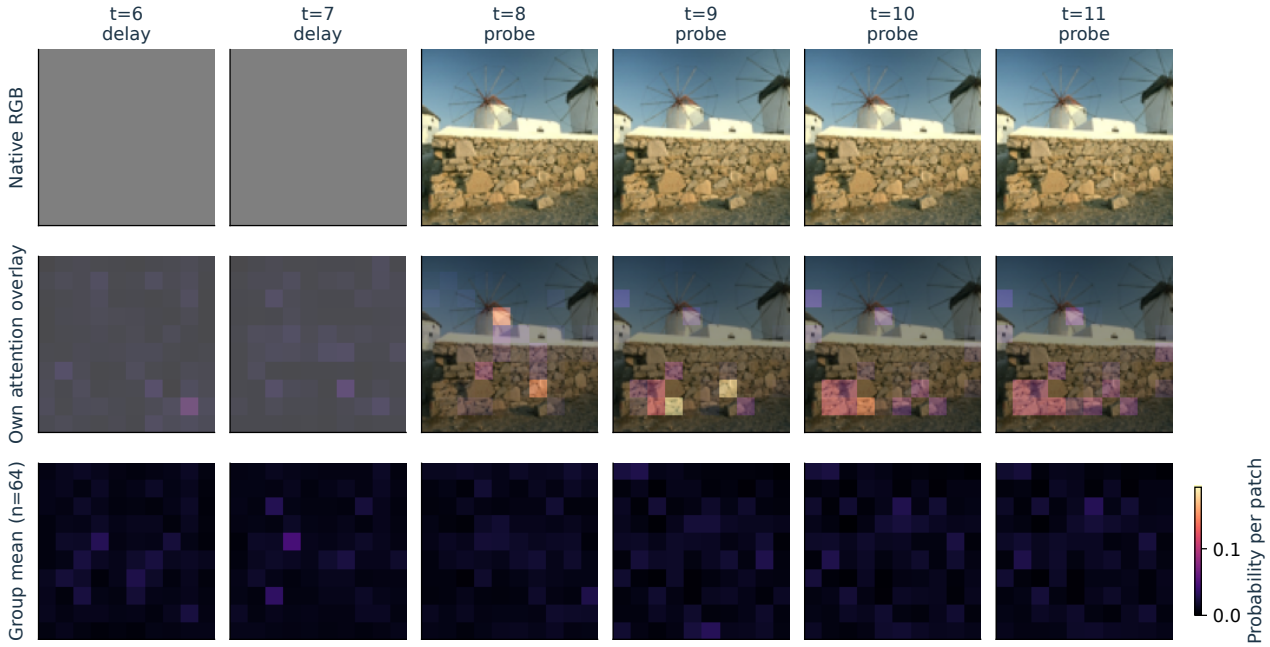


Figure 25: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

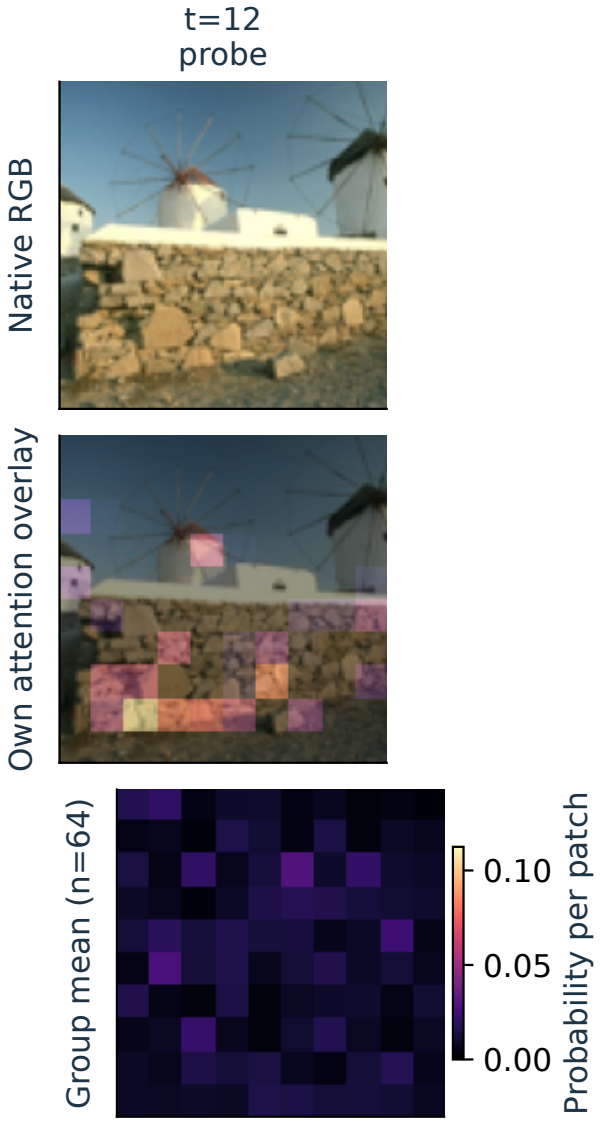


Figure 26: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.



Figure 27: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

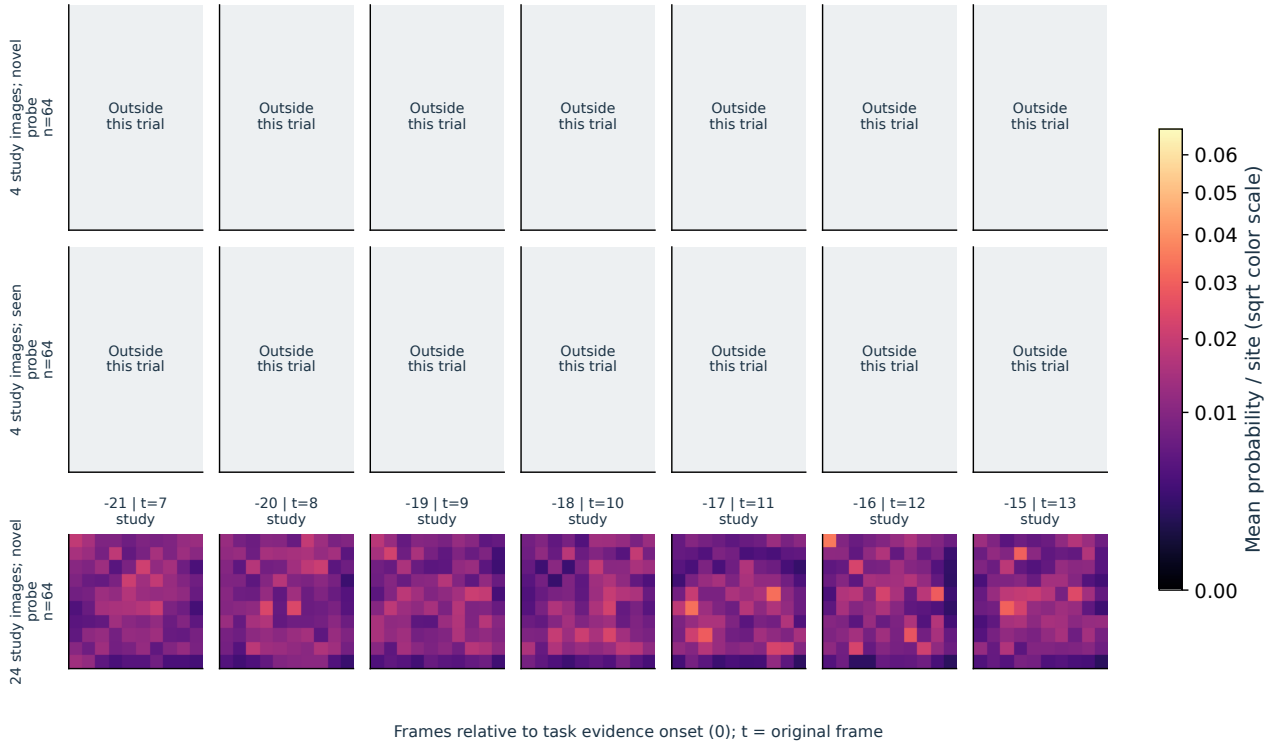


Figure 28: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.



Figure 29: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

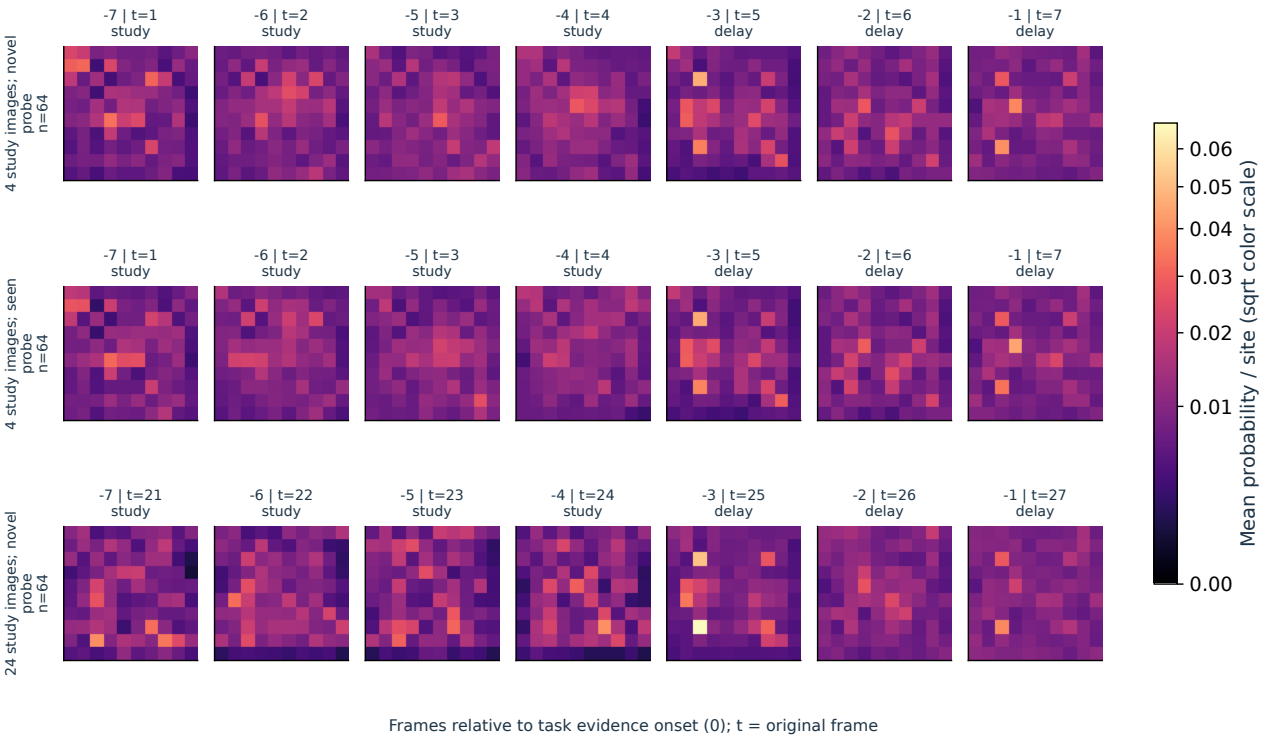


Figure 30: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

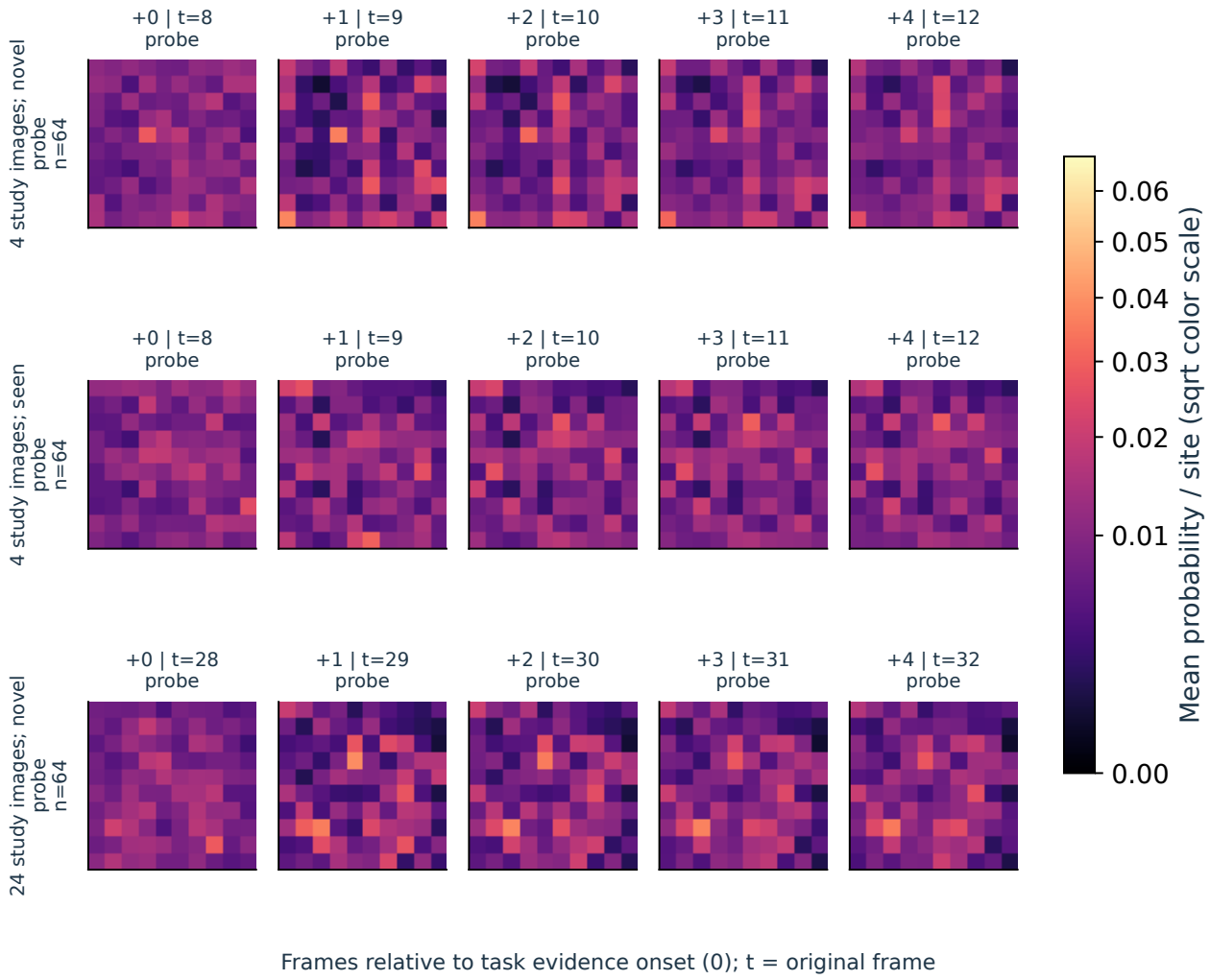


Figure 31: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.



Figure 32: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.



Figure 33: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.



Figure 34: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.



Figure 35: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

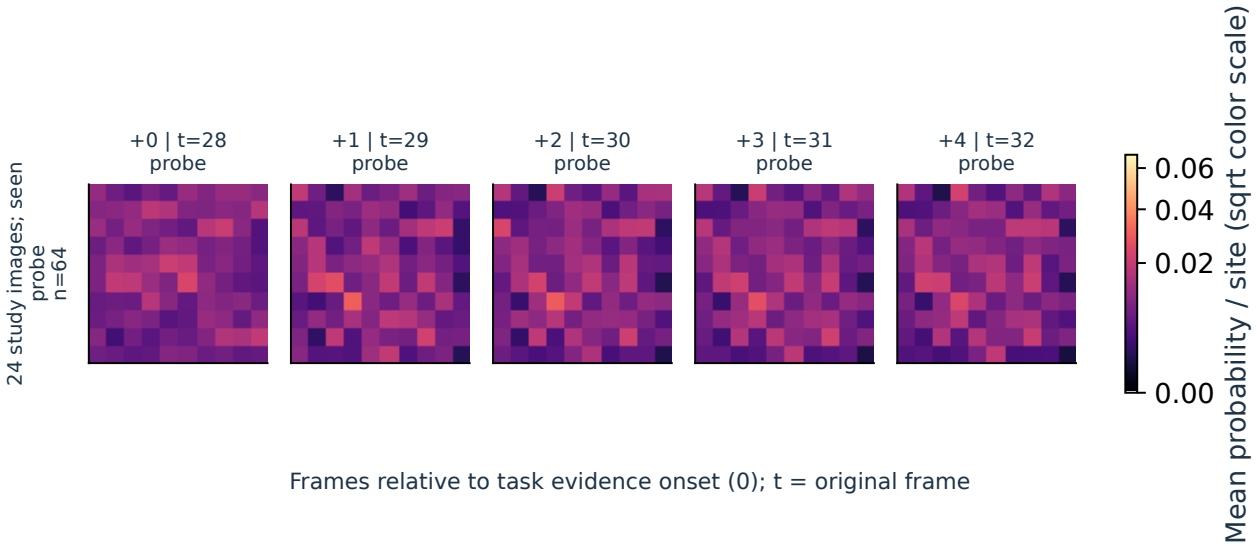


Figure 36: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Value-cued priority and causal competition in a four-location change task

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized vda4 in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 3 native conditions and 384 fresh trials, equal-condition accuracy was 68.0% (95% interval [63.5%, 72.7%]). For the representative condition TH65, the strongest final answer readout among the tested regions was spatial, with held-out macro AUC 0.927. The largest observed impairment in that condition was distractor_+8_evidence, changing accuracy by -32.0 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

Four Gabor locations compete for processing under a cue that carries probability and value information. The report concerns a change anywhere in the array. A cue can therefore improve the allocation of resources while also making an unexpected change elsewhere easier to miss.

Native angle conditions quantify performance under the trained cue distribution. The central counterfactual fixes a 100 percent bottom-right cue and places a change at top left. Inhibiting bottom-right priority during the two change frames tests whether releasing resources from the expected location improves unexpected-change reports.

The biological motivation is developed through [1], [2], [3], [4]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Superior-colliculus attention experiments motivate perturbing a spatially organized priority signal while measuring perception. The important computational prediction is competitive: lowering priority at one location can change the influence of evidence elsewhere. The four-location layout makes the corner-to-corner test especially clear. A cue establishes an expectation, a controlled event violates or satisfies it, and the intervention acts only when new evidence appears. Reciprocal effects on cued and uncued changes would support redistribution. The catch condition distinguishes such redistribution from a general increase in positive reports.

1.2 Biological experiments motivating this task

Experimental question. Does suppressing priority at a highly probable location harm detection there while helping a change elsewhere?

Lovejoy and Krauzlis (2010) SC inactivation shifted motion judgments away from signals in the affected field toward competing information elsewhere. Deficits depended strongly on informative distractors; affecting the foil location favored judgments based on the cued signal. [7]

Basso and Wurtz (1998) SC buildup activity varied with target probability, including when probability changed while the display remained fixed. More possible targets reduced preparatory activity, linking selection state with uncertainty and saccade latency. [8]

Zenon and Krauzlis (2012) SC inactivation impaired attentive behavior while established attentional modulation in visual cortex persisted. Behavioral selection and cortical response enhancement could therefore dissociate. [9]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Did any Gabor change orientation?

Response labels are 0: no change; 1: change anywhere. The native suite contains 3 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use TH65.

1.4 Stimulus sequence and spatial layout

For the representative native condition TH65, the sequence has 7 frames, indexed from zero: frame 0: blank; frame 1: cue; frame 2: blank; frames 3–4: sample; frames 5–6: test. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Native rotation parameter (degrees)	65.0
Native angular noise parameter (degrees)	5.0
Displayed cue validity	0.25, 0.5, 0.75, 1.0
Native trial reward value	1.0, 3.0, 5.0
Original raster width (pixels)	50

The counterfactual includes cued-change, uncued-change and catch trials on matched nuisance seeds. A benefit for uncued changes accompanied by a cost for cued changes is a redistribution signature. Catch reports show whether the intervention also shifts the response criterion. These seven-frame movies

keep the long-window comparison branch at zero, locating any effect in the recurrent visual-attention computation.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding

global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression $k^{\mathsf{T}}q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^\delta A/(1 - A + e^\delta A)$. Thus its odds against the remaining image are multiplied by e^δ; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance

probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ±4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated

class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 68.0%, with macro AUC 0.755. The matched final checkpoint changed accuracy by +6.25 percentage points on average across conditions, using 32 same-movie comparisons per condition. The lowest native cell was TH8 at 52.3%; the highest was TH65 at 76.6%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (2) produced 33.6% correct responses, compared with 100.0% at the largest strength (65). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

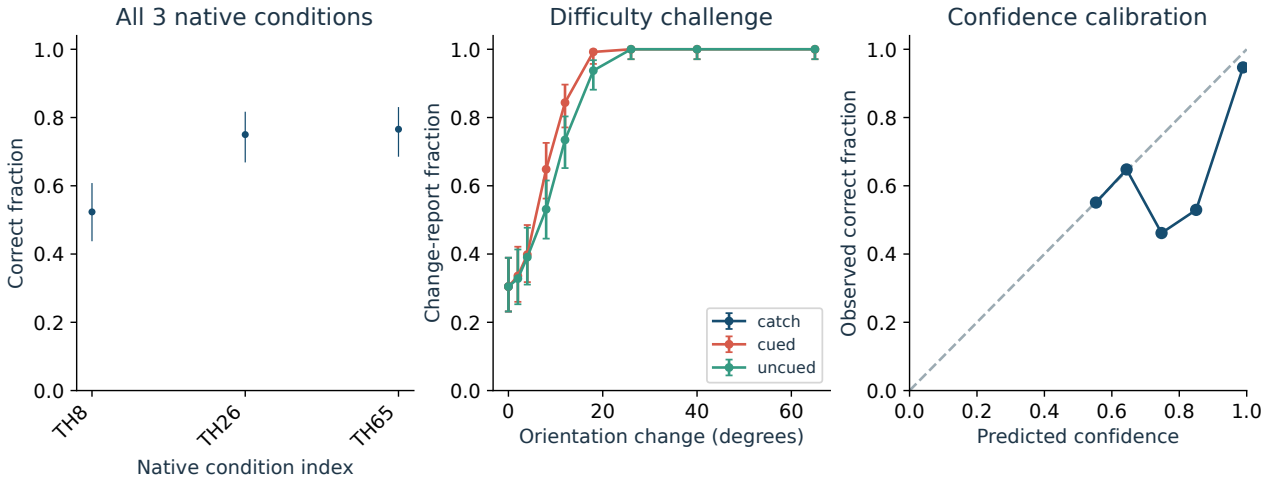


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	92.2%	1.0	77.1%	100.0%
0.7	50.0%	1.0	85.9%	100.0%
0.8	43.8%	1.0	94.6%	100.0%
0.9	39.1%	2.0	98.0%	100.0%

The representative condition TH65 provides 2 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

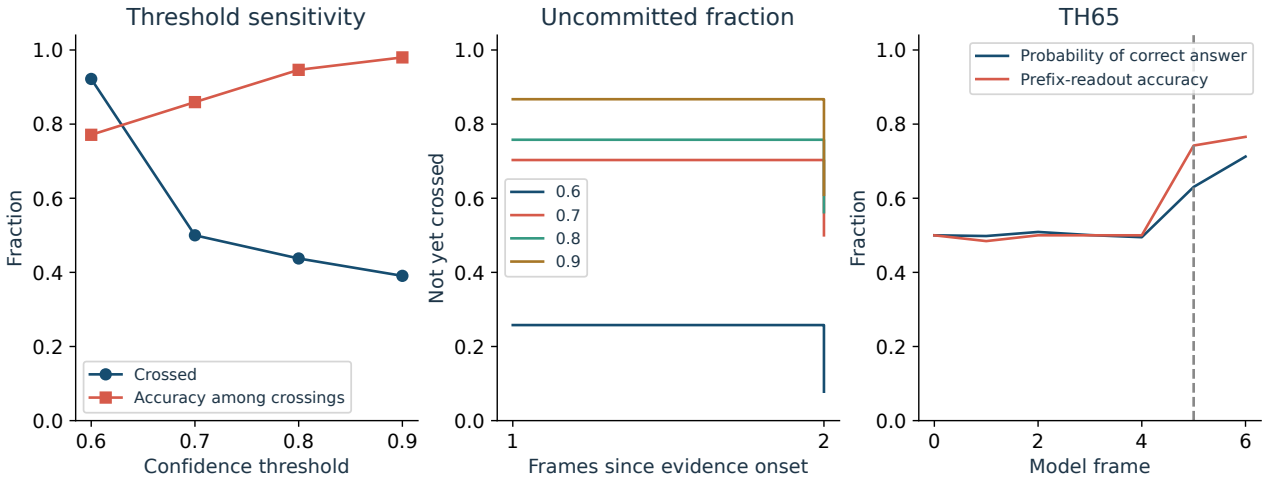


Figure 2: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

3.3 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 116 of 384 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

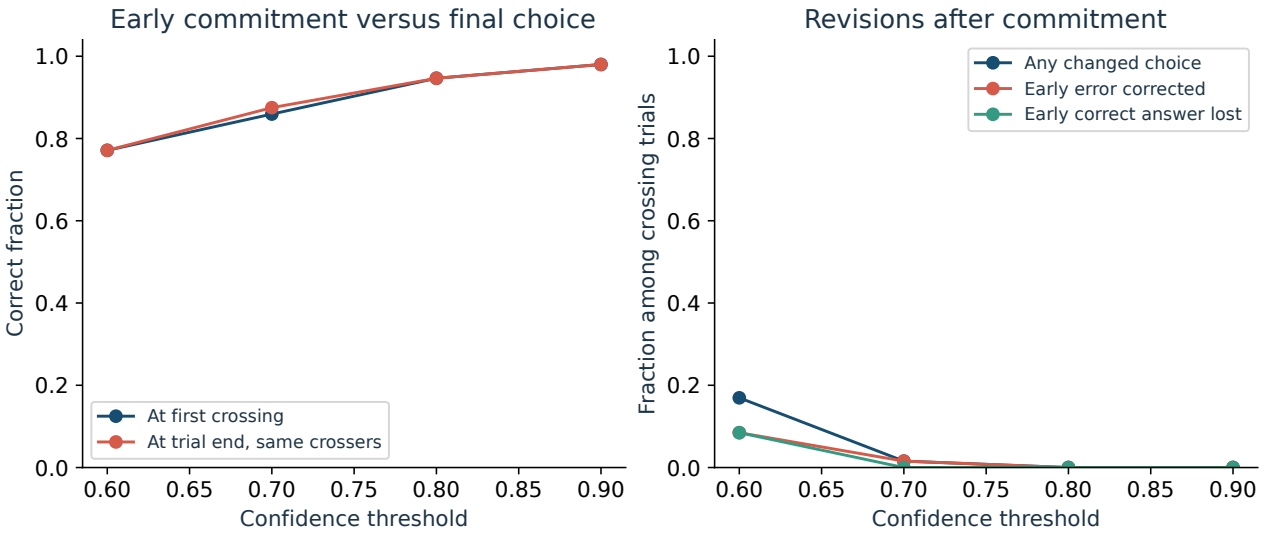


Figure 3: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

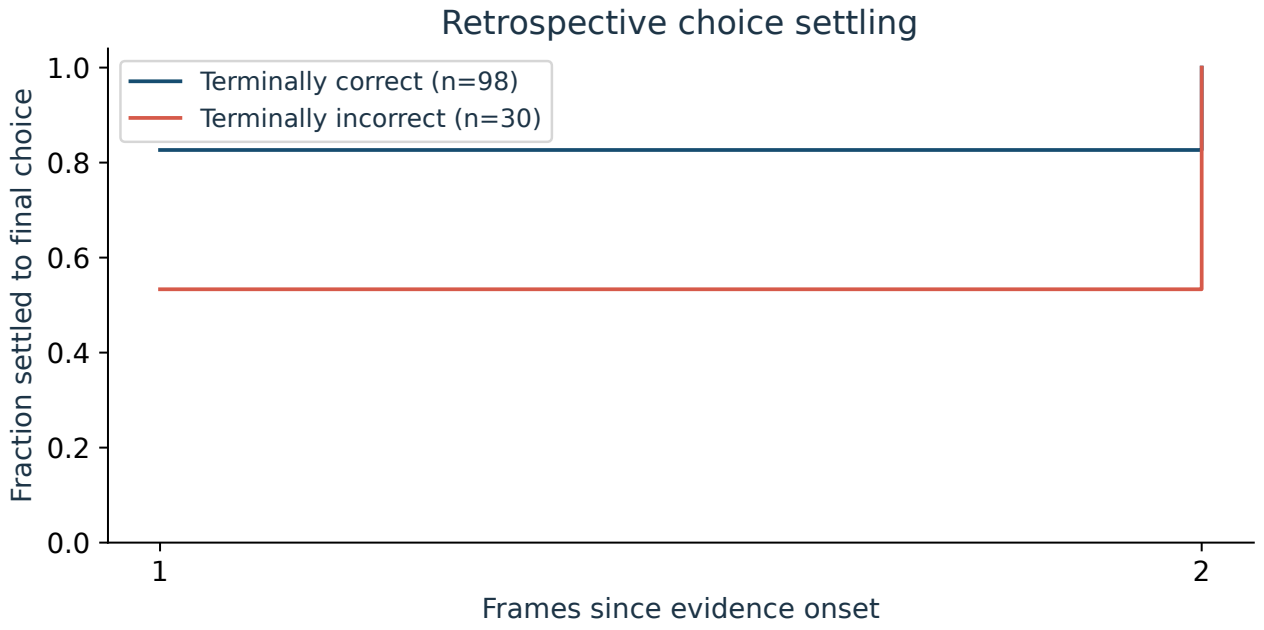


Figure 4: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.4 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

All rows are no-change trials with the cue fixed at bottom right. Rows compare the four displayed cue validities, pooling the three native difficulty cells and reward colors. This most directly follows the condition-by-time organization of Figure 4 in Morgan, Albanna and Herman.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Cue validity 25%; no change	11	11
Cue validity 50%; no change	10	10
Cue validity 75%; no change	13	13
Cue validity 100%; no change	14	14

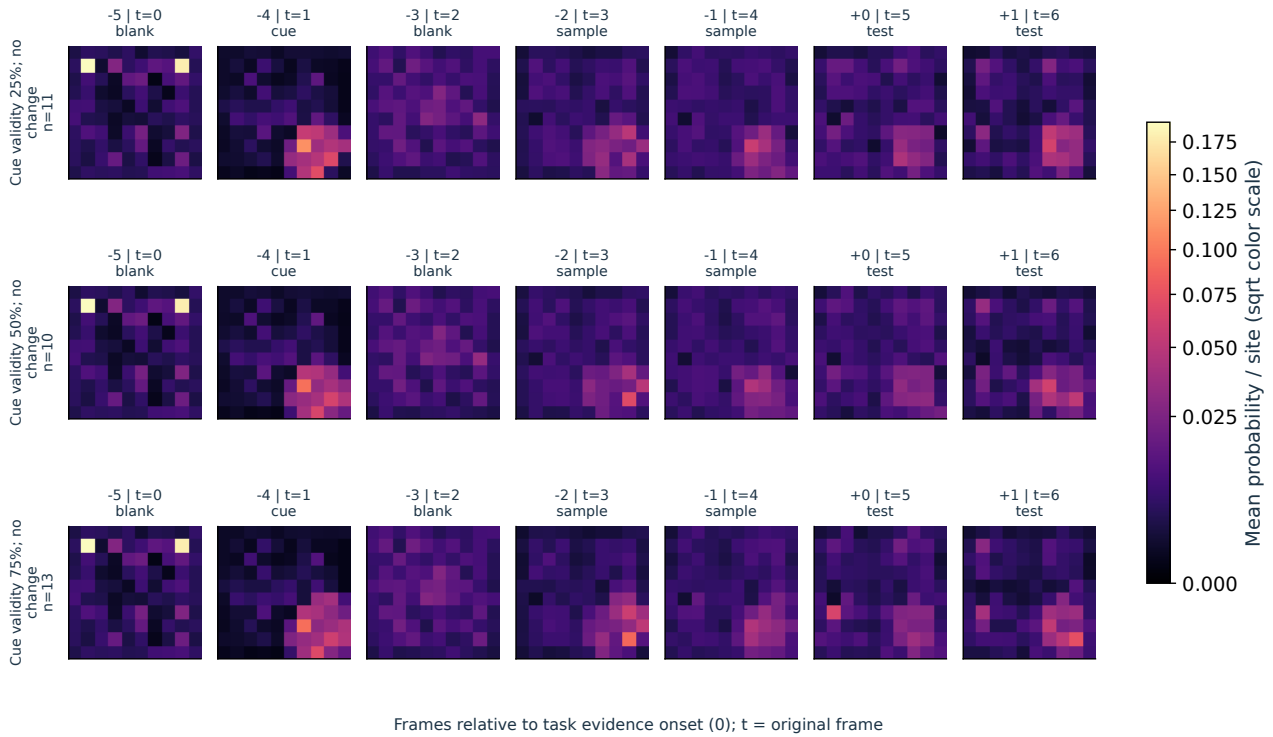


Figure 5: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

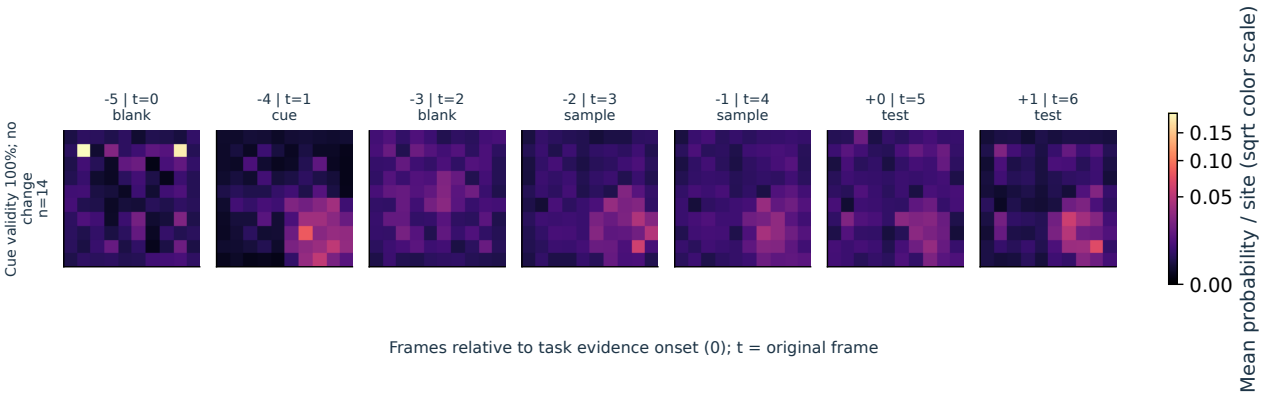


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

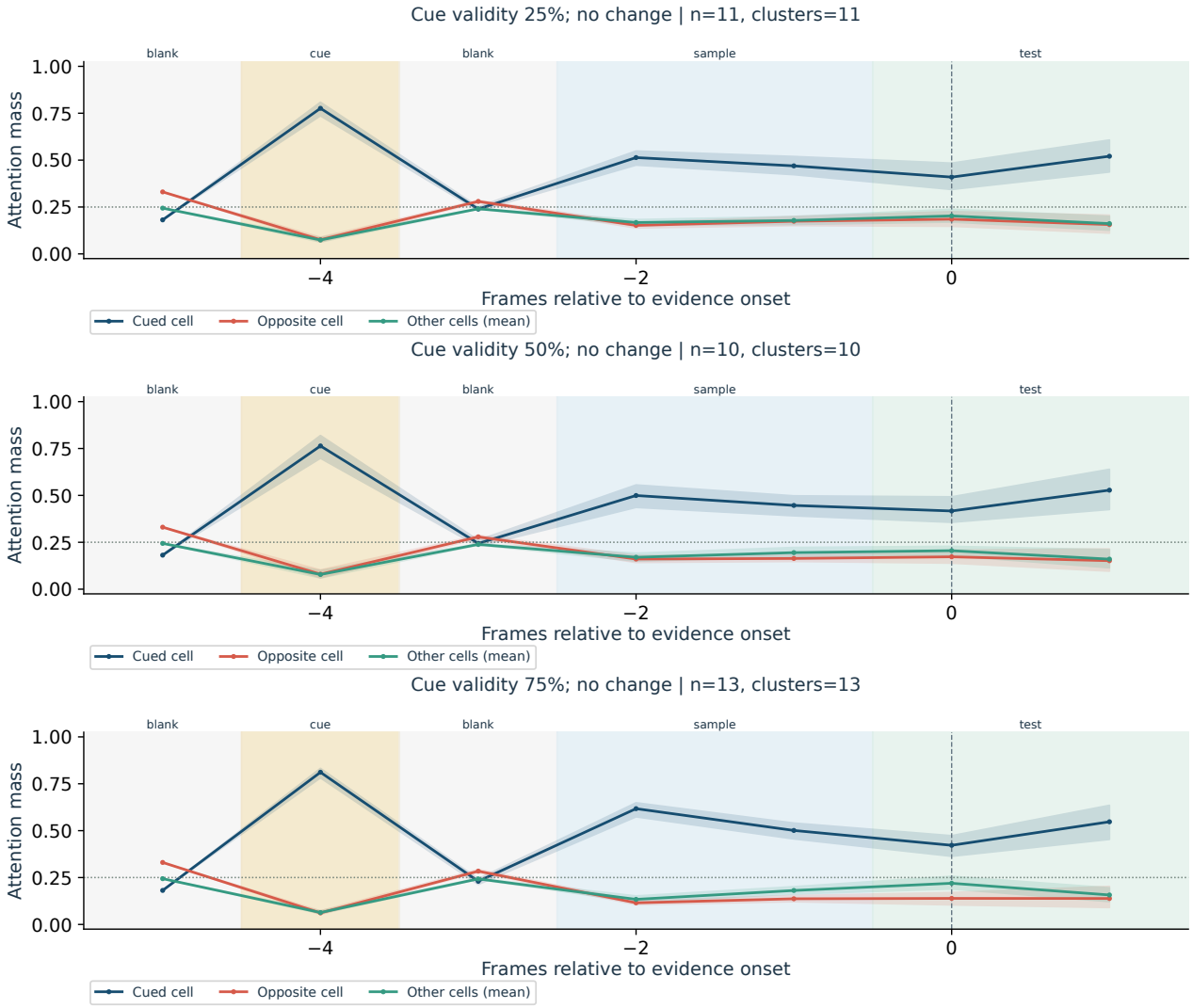


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

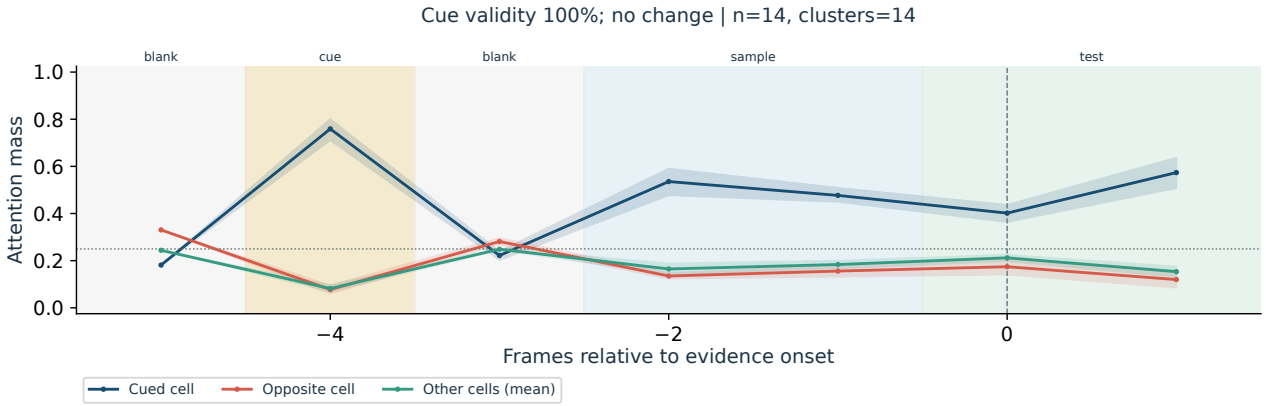


Figure 8: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

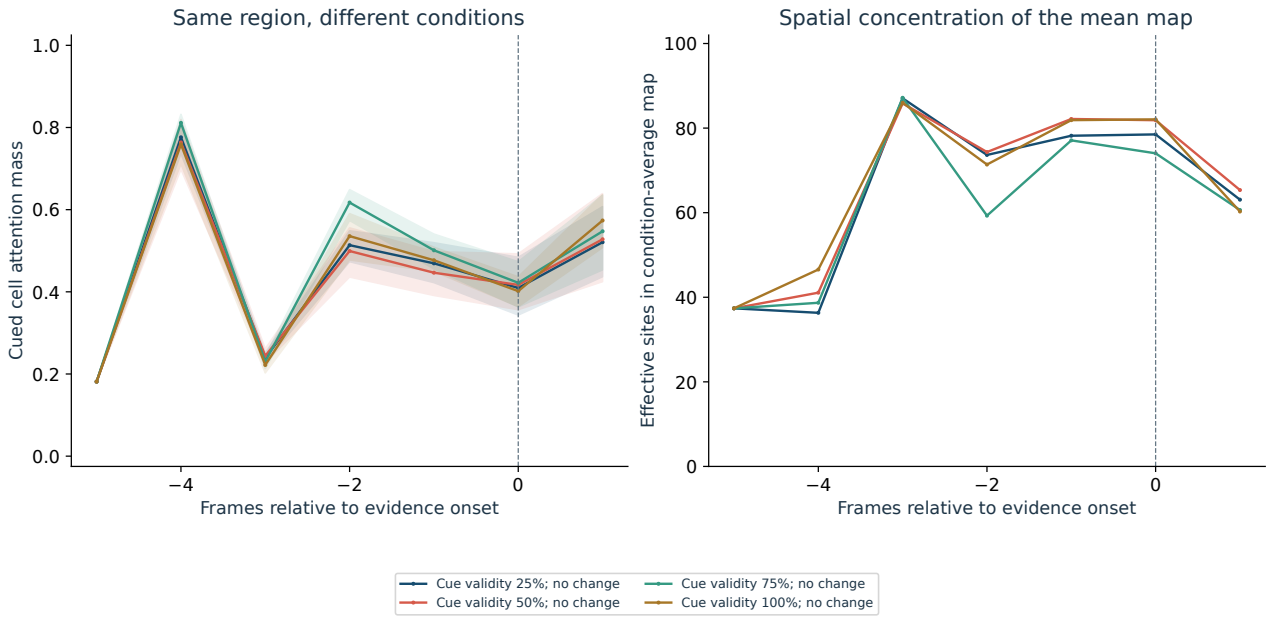


Figure 9: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Cue validity 25%; no change: cued cell mass peaks at 0.776 at original frame 1 (cue), and is 0.521 on the final frame. Cue validity 50%; no change: cued cell mass peaks at 0.764 at original frame 1 (cue), and is 0.528 on the final frame. Cue validity 75%; no change: cued cell mass peaks at 0.811 at original frame 1 (cue), and is 0.547 on the final frame. Cue validity 100%; no change: cued cell mass peaks at 0.759 at original frame 1 (cue), and is 0.574 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.5 Causal sensitivity to spatial and mnemonic interventions

For distractor_+8_evidence, paired accuracy changed by -32.0 percentage points (paired 95% interval [-43.0%, -21.9%]; paired p=9.624e-08; within-task Holm p=4.523e-06). There were 10 corrected trials and 51 newly incorrect trials. For target_-2_evidence, paired accuracy changed by +3.1 percentage points (paired 95% interval [-2.3%, 8.6%]; paired p=0.424; within-task Holm p=1). There were 9 corrected trials and 5 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

3.6 Bottom-right cue and top-left counterfactual change

For catch events, bottom-right inhibition at dose -4 changed correct-report frequency by +9.0 percentage points, with paired interval [3.1%, 14.8%]. For cued events, bottom-right inhibition at dose -4 changed correct-report frequency by -53.9 percentage points, with paired interval [-60.2%, -48.0%]. For uncued

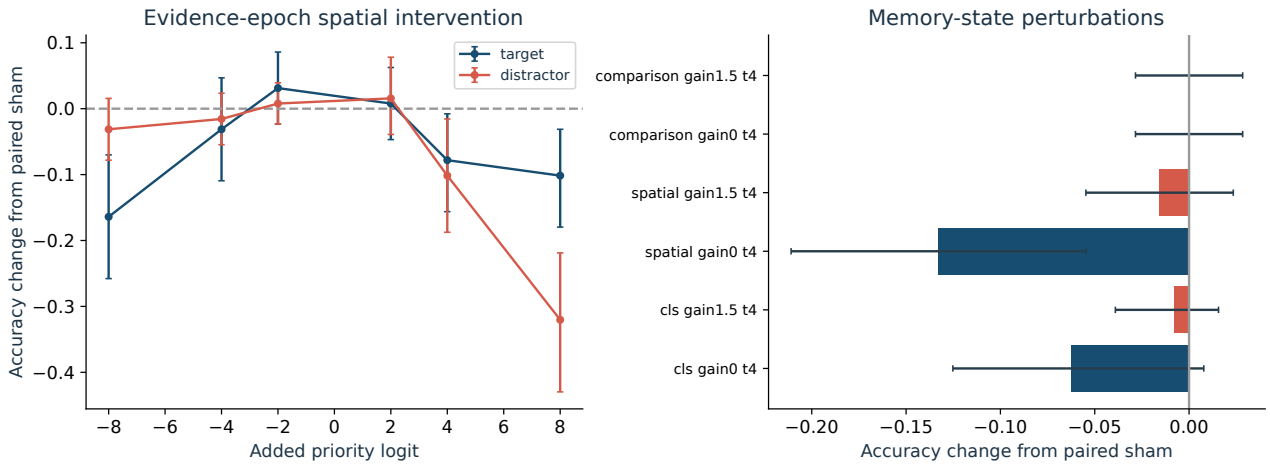


Figure 10: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

events, bottom-right inhibition at dose -4 changed correct-report frequency by +0.8 percentage points, with paired interval [-1.2%, 3.1%]. The 100 percent cue and uncued-change combination is deliberately outside the native cue-validity rule. A change at top left still has a correct positive label because VDA requires a report for a change anywhere. Catch trials have no physical change and provide the false-alarm control.

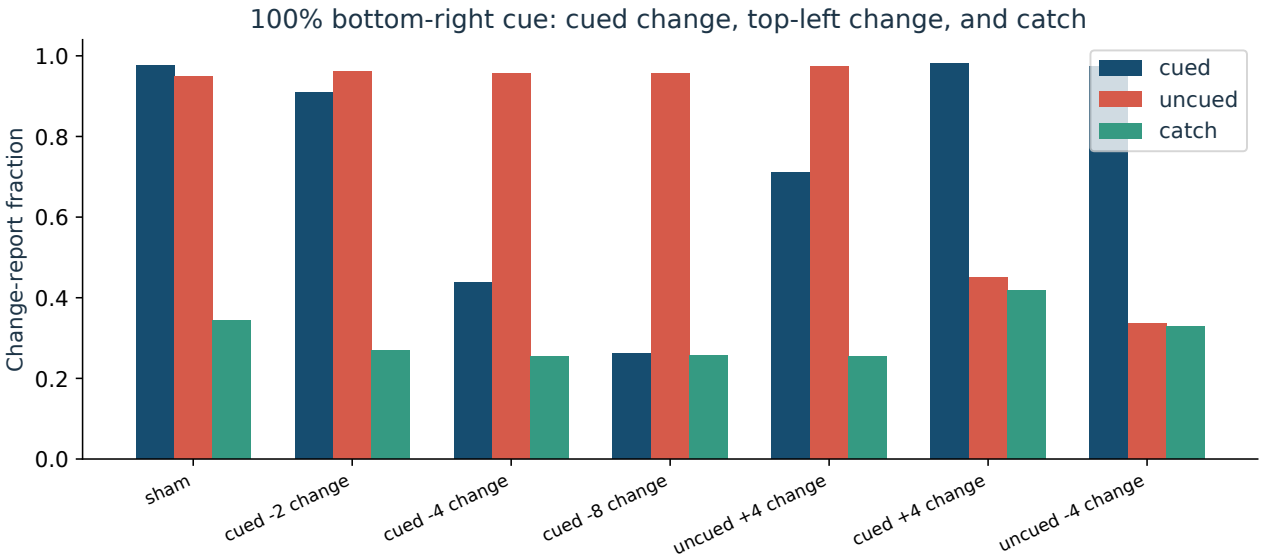


Figure 11: Requested counterfactual experiment. Every movie has a 100 percent bottom-right cue. Change reports are plotted separately for changes at the cue, changes at top left and catches. Priority perturbations occur only on the two change frames. Each event type contains 256 matched nuisance draws per intervention arm.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
TH65	target_-4	-3.1	1
TH65	target_+4	-7.8	1
TH65	distractor_-4	-1.6	1
TH65	distractor_+4	-10.2	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.542	53.4%	0.170
fusion	0.780	72.9%	0.005
spatial	0.927	86.2%	0.005
cls	0.916	85.5%	0.005
comparison	Constant zero on this condition		
attention	0.830	75.2%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.



Figure 12: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 0.00 to 13.80 within the retained subspace. The final step displacement was 1.362 standardized RMS units. In spatial memory, the participation ratio ranged from 0.00 to 19.02 within the retained subspace. The final step displacement was 0.470 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 4, coarse sample-content reconstruction had R-squared 0.056 from CLS memory and 0.427 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -6.2 percentage points, spatial, -13.3 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

3.11 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.583	0.560	3200
fusion	0.741	0.531	6400
spatial	0.902	0.534	6400
cls	0.877	0.625	64
comparison	unavailable	unavailable	–
attention	0.850	0.645	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 39 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.12 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

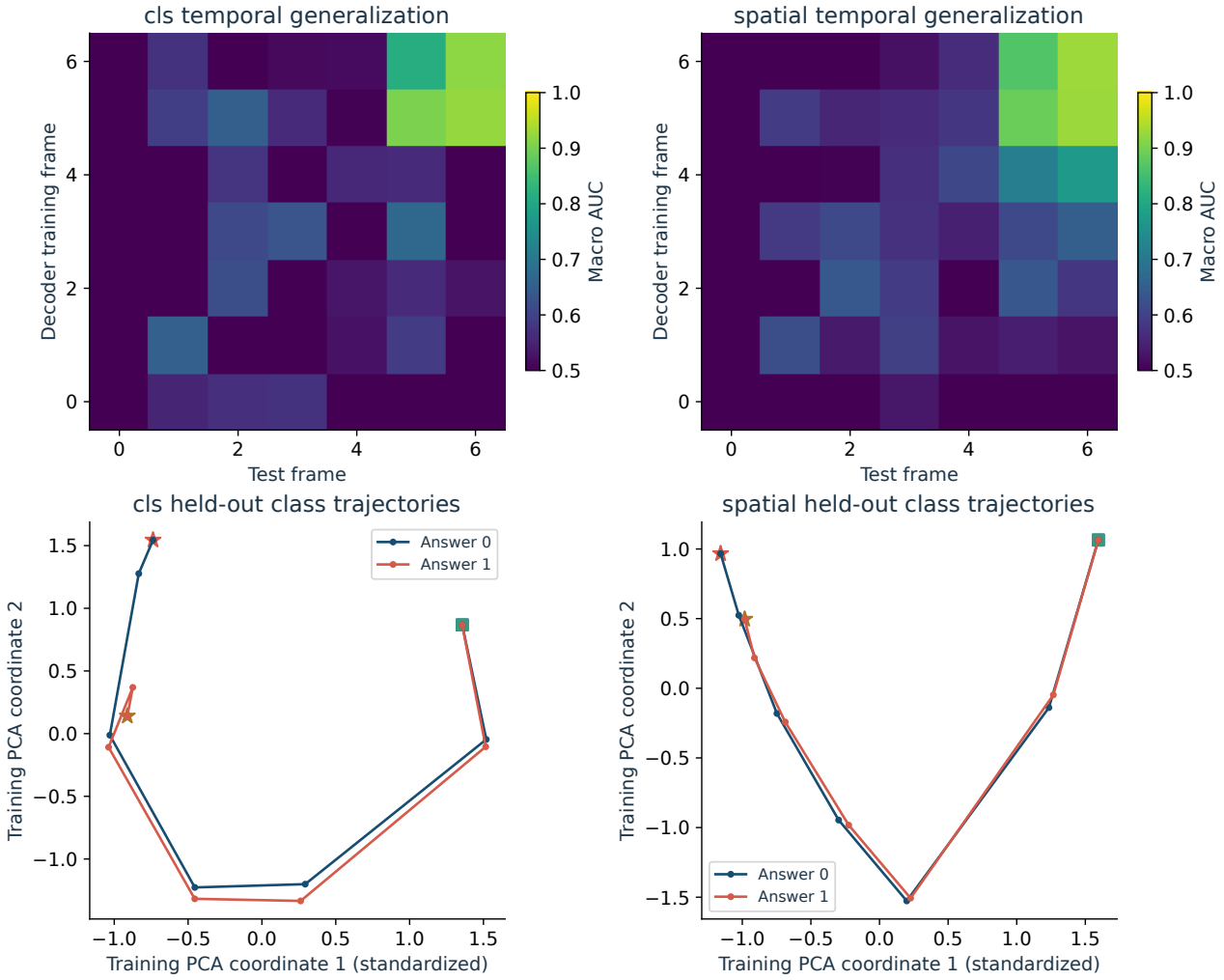


Figure 13: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

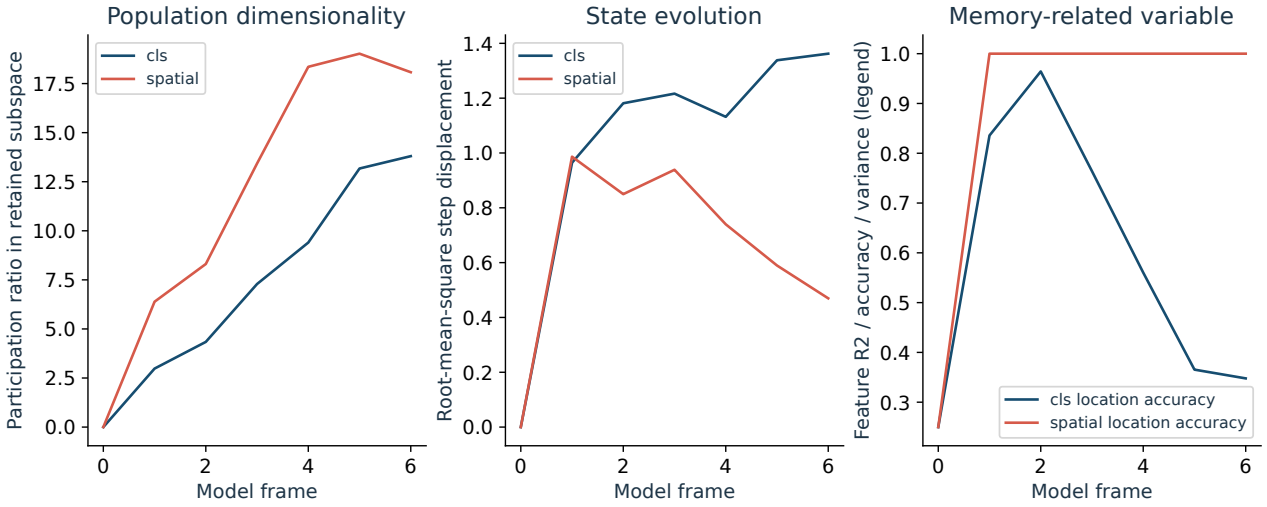


Figure 14: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

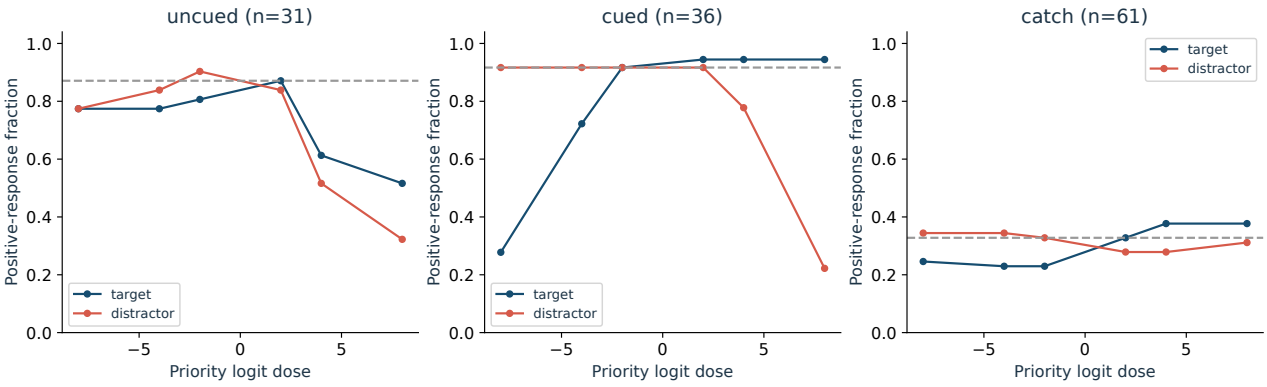


Figure 15: Event-specific causal response functions. Curves show positive reports after evidence-epoch target or alternative-site intervention; the horizontal line is paired sham. Event counts are stated above each panel. A positive report has the task-specific meaning defined in the text.

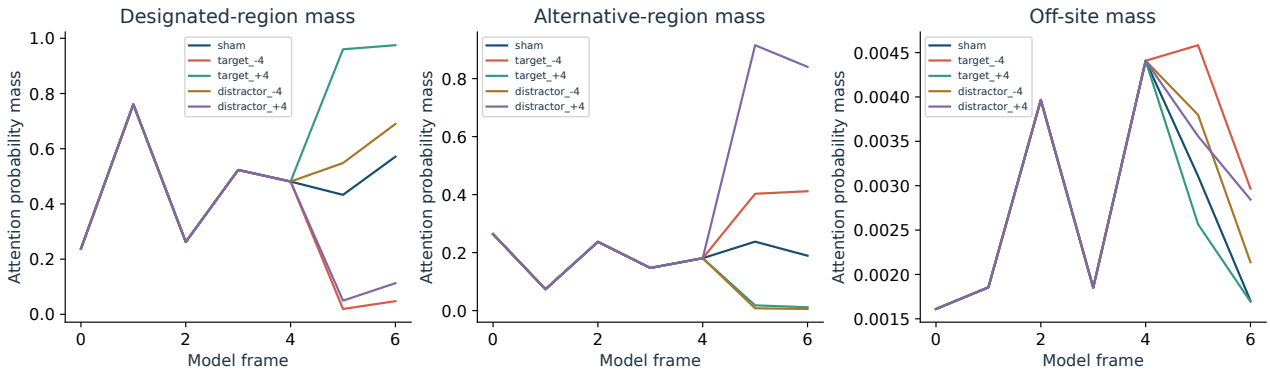


Figure 16: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

3.13 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

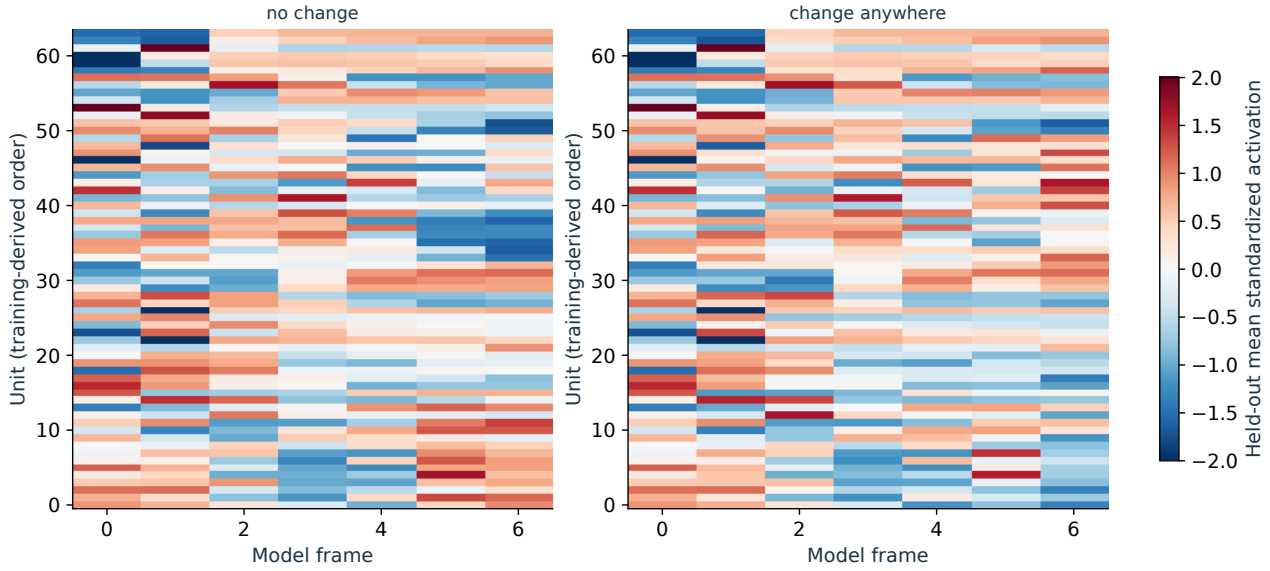


Figure 17: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.14 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional

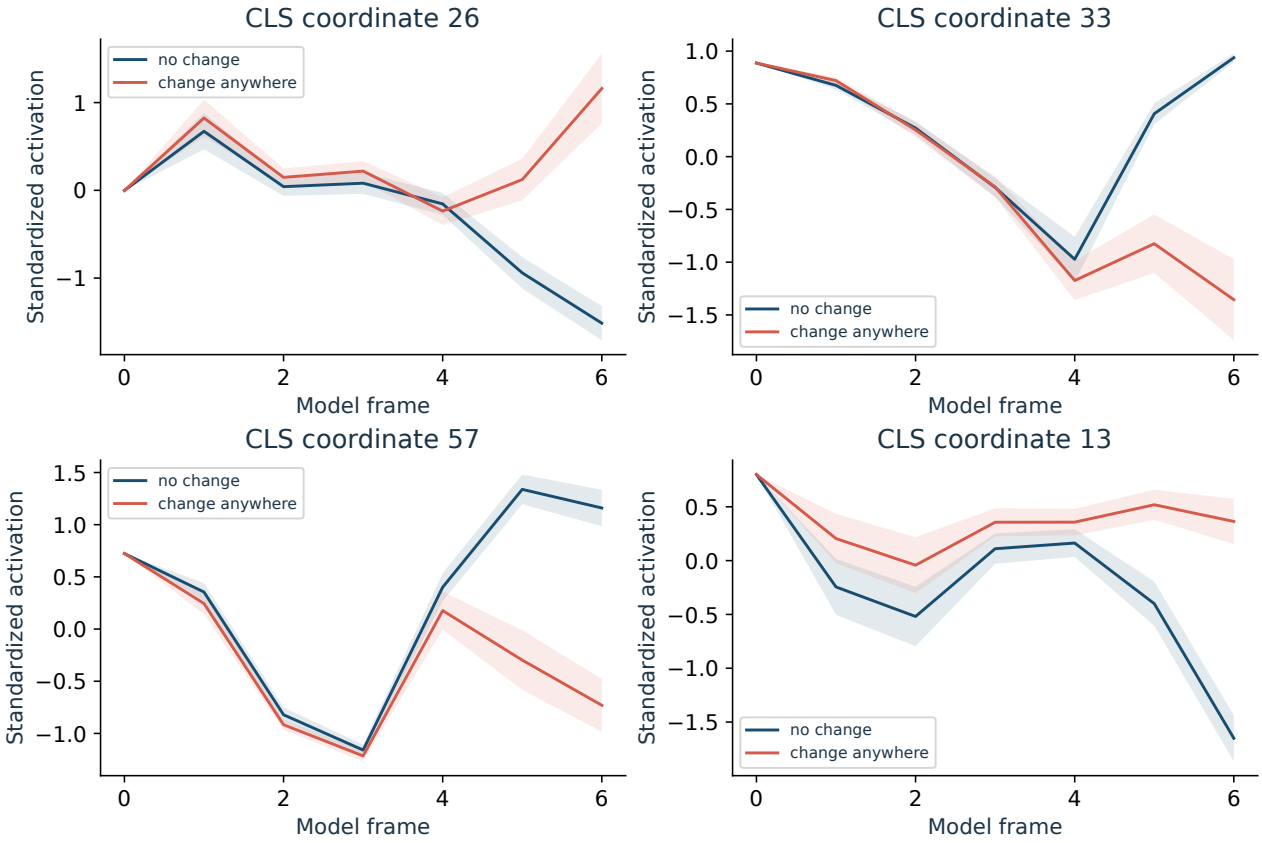


Figure 18: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	53.4%	50.8%	22.9%
fusion	72.9%	69.5%	27.0%
spatial	86.2%	81.6%	100.0%
cls	85.5%	95.0%	34.8%
attention	75.2%	73.6%	82.5%

3.15 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

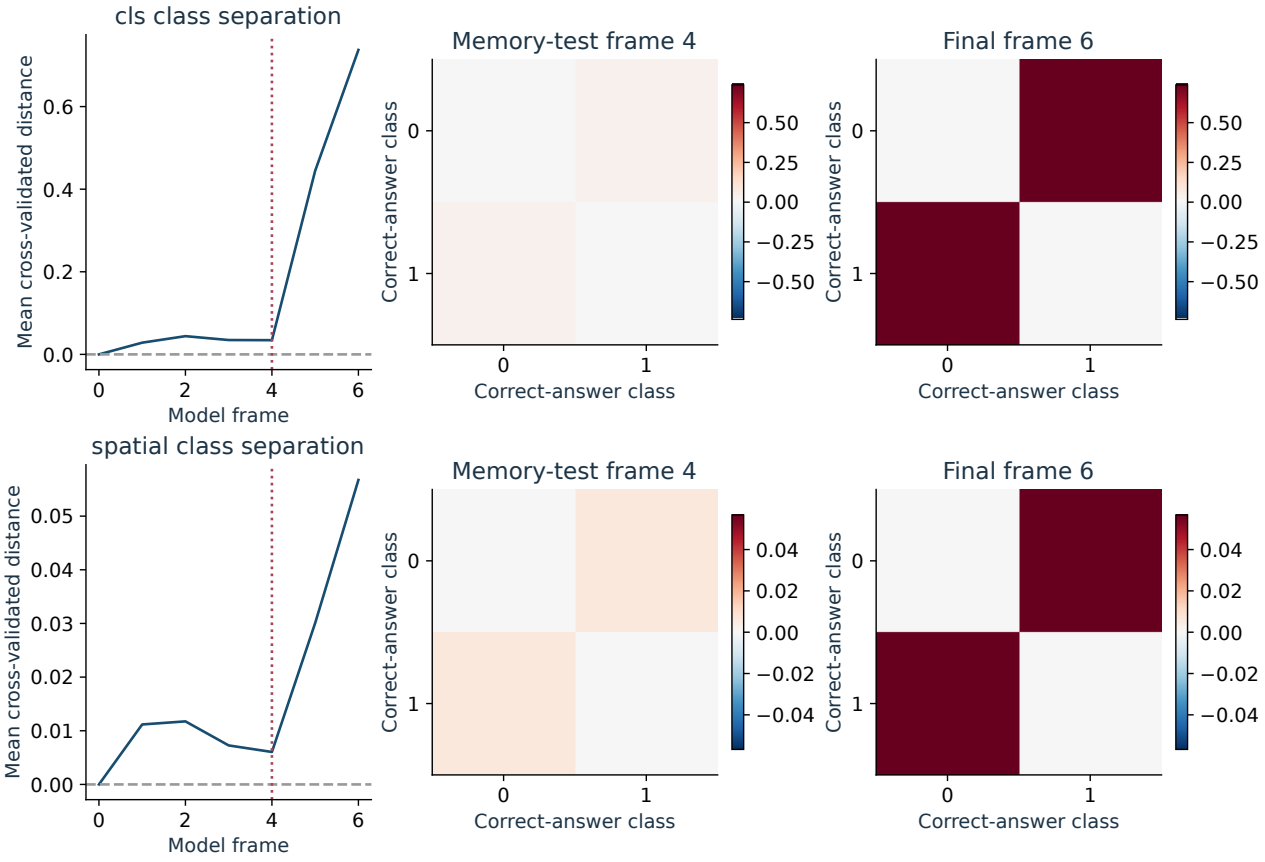


Figure 19: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

3.16 Weaker-signal follow-up to test the improvement prediction

After the 18-degree VDA4 counterfactual showed high uncued baseline detection, an exploratory follow-up tested 2-, 4- and 8-degree changes, plus catches, in all three VDA sizes. Each event-strength cell contains 128 new matched nuisance draws per arm. The bottom-right cue remains certain and perturbations occur during the change frames. Holm adjustment below covers the entire follow-up family within this task. These tests address whether the apparent absence of an uncued benefit at strong signals reflects limited headroom.

Event	Degrees	Sham accuracy	Cue-inhibition change	Holm p
catch	0	72.7%	-0.8 pp	1
cued	2	28.9%	-0.8 pp	1
uncued	2	29.7%	-1.6 pp	1
cued	4	35.2%	-7.0 pp	1
uncued	4	32.0%	+0.0 pp	1
cued	8	57.8%	-28.9 pp	1.16e-05
uncued	8	47.7%	+0.0 pp	1

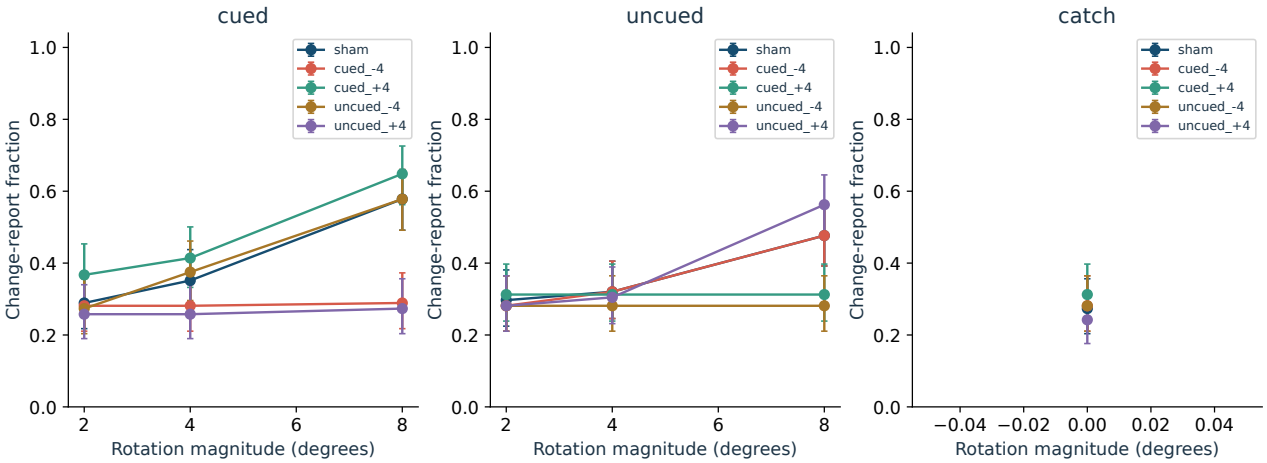


Figure 20: Exploratory weak-change counterfactual. Cued and uncued hit fractions are shown across three magnitudes; the catch panel shows false reports without a physical change. Error bars are Wilson 95 percent intervals. All interventions reuse identical movies within each cell.

3.17 Probability cues, reward value and prechange allocation

The native task varies displayed change-location probability and cue color, which specifies trial value. We stratify the original 384 trials by displayed probability, event location and value. Cued and uncued change reports are hits; reports on catch trials are false alarms. Prechange cue mass is measured at frame 4, before the two test frames, and is expressed relative to a uniform distribution over the 100 sites. Probability intervals use Wilson bounds; attention intervals resample native trials 3,000 times. These native stratifications characterize how the learned observer treats the task instructions, with the paired interventions supplying the causal test.

The model obtained 67.6% of the available native value, calculated as the sum of value on correctly answered trials divided by total offered value. At value 1, accuracy was 68.8% across 128 trials. At value 3, accuracy was 68.1% across 116 trials. At value 5, accuracy was 67.1% across 140 trials.

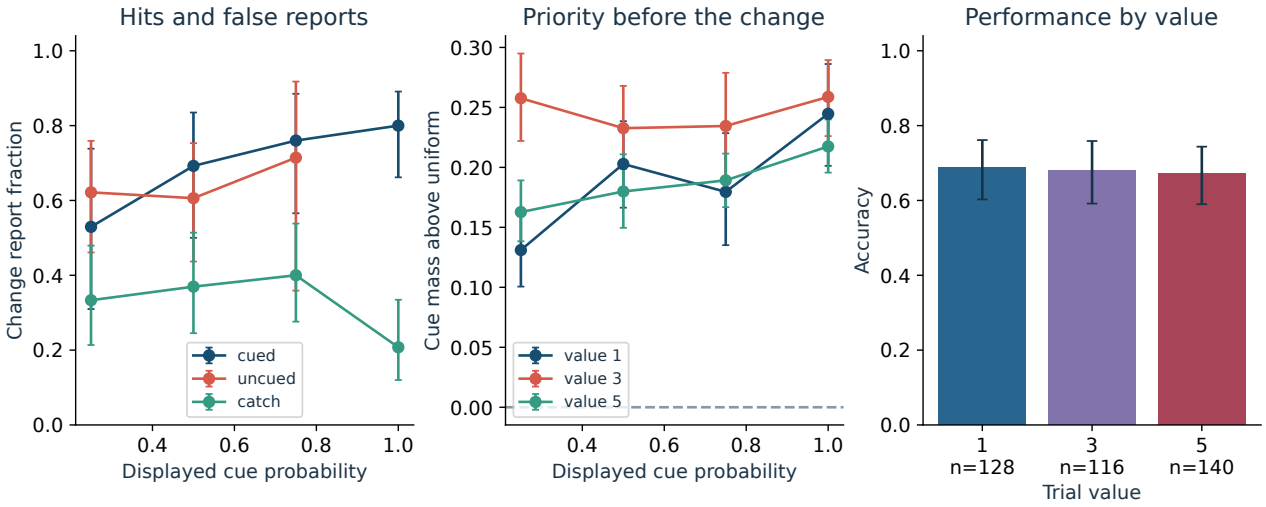


Figure 21: Native probability and value stratification. Left: cued hits, uncued hits and catch false reports. Middle: cue-region attention above uniform at the last prechange sample frame, separated by trial value. Right: accuracy by offered value. Sparse event-probability groups have wide sampling intervals; their sample counts are supplied in the source table.

4 Discussion

The strong-change counterfactual reveals a pronounced cost of removing priority from the cued location and a large cost of increasing that priority when the change is elsewhere. Uncued detection is already high in the 18-degree sham condition. The exploratory two-, four- and eight-degree follow-up removes that ceiling, but still finds no uncued improvement under cue inhibition; at eight degrees, cued detection falls by about 29 percentage points while uncued detection is unchanged. Thus the requested combination of cued cost and uncued benefit is not recovered for this checkpoint in these experiments. Catch trials show how priority also changes positive-response tendency. Native performance at the smallest angle is substantially poorer than at larger angles, making a single pooled score an incomplete description. Because comparison memory remains zero throughout these seven frames, the measured competition is implemented by the shared visual-attention and recurrent pathways.

The behavioral result establishes 68.0% native accuracy for this trained subject across the tested condition distribution. The information result identifies spatial as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies distractor_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Variable trial length would allow the same cue manipulation to recruit the delayed comparison branch. Independently changing cue validity and reward value would separate probability-guided allocation from reward-guided allocation.

4.1 Interpreting these results alongside biological experiments

4.1.1 Lovejoy and Krauzlis (2010)

The biological experiment motivates testing both sides of competition. In VDA, a change anywhere receives a positive label; in the cued motion discrimination experiment, the instructed signal determines the correct direction. Our counterfactual bottom-right cue with a top-left change deliberately tests behavior outside the native 100-percent-validity contingency. It probes how strongly the cue governs the detector despite physically informative evidence elsewhere. [7]

4.1.2 Basso and Wurtz (1998)

Cue validity is a probability instruction. The SC study shows why probability should be manipulated independently of the visible array. Our condition-resolved maps can ask whether the same four Gabors receive different allocation under different displayed validity. The biological endpoint was saccade preparation, while our endpoint is a change report; both expose how expected relevance changes a spatial selection state. [8]

4.1.3 Zenon and Krauzlis (2012)

The dissociation between cortical enhancement and behavior makes it important to inspect both the allocation map and the final report. In this model, the direct manipulation checks show that the intervention changes routing. A changed map alone does not imply improved use of the newly favored evidence. The weak-change follow-up supplies the corresponding behavioral test. [9]

4.2 A computational account of the observed behavior

The immediate softmax effect of inhibiting one region is mathematically guaranteed: the remaining regions receive a larger share of total attention. Behavioral improvement is a separate prediction. It requires useful change information to be represented, enough weight to reach it, and a readout that interprets it correctly. The strong-change sham already detects many uncued events. Lowering the angle to two, four and eight degrees removes that easy explanation for a missing benefit. At eight degrees, inhibition costs about 29 percentage points for cued events while uncued detection remains unchanged. The seven-frame trials keep the long-window comparison memory at zero, so these effects arise through the shared visual and recurrent pathways. This focuses the mechanism on routing, local temporal evidence and recurrent readout.

4.3 Where this task agrees with the biological evidence

The model shows the cued impairment side of the proposed dual effect. It does not recover a reliable uncued improvement in the tested weak-change follow-up. This updates the earlier promising strong-signal interpretation: the evidence supports asymmetric behavioral consequences of redistribution, even though spatial competition itself is present.

4.4 A follow-up experiment motivated by these results

Use one matched change-strength grid for cued, uncued and catch events, with independent cue-validity levels. Plot sensitivity and criterion alongside allocation. The crucial prediction is a positive uncued sensitivity change paired with a negative cued change, rather than merely more positive reports after inhibition.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Morgan J, Albanna B, Herman JP (2025). A recurrent vision transformer shows signatures of primate visual attention. arXiv:2502.10955. https://arxiv.org/abs/2502.10955.

[2] Bollimunta A, Bogadhi AR, Krauzlis RJ (2018). Comparing frontal eye field and superior colliculus contributions to covert spatial attention. Nature Communications 9:3553. https://www.nature.com/articles/s41467-018-06042-2.

[3] Reynolds JH, Heeger DJ (2009). The normalization model of attention. Neuron 61:168-185. https://pubmed.ncbi.nlm.nih.gov/19186161/.

[4] Wolfe JM (2021). Guided Search 6.0: An updated model of visual search. Psychonomic Bulletin and Review 28:1060-1092. https://pubmed.ncbi.nlm.nih.gov/33547630/.

[5] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[6] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[7] Lovejoy and Krauzlis (2010). Inactivation of primate superior colliculus impairs covert selection of signals for perceptual judgments. Primary source.

[8] Basso and Wurtz (1998). Modulation of neuronal activity in superior colliculus by changes in target probability. Primary source.

[9] Zenon and Krauzlis (2012). Attention deficits without cortical neuronal deficits. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
TH8	128	52.3%	0.521	0.548
TH26	128	75.0%	0.819	0.348
TH65	128	76.6%	0.926	0.293

B Every-frame attention atlas

Condition TH65. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 16 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

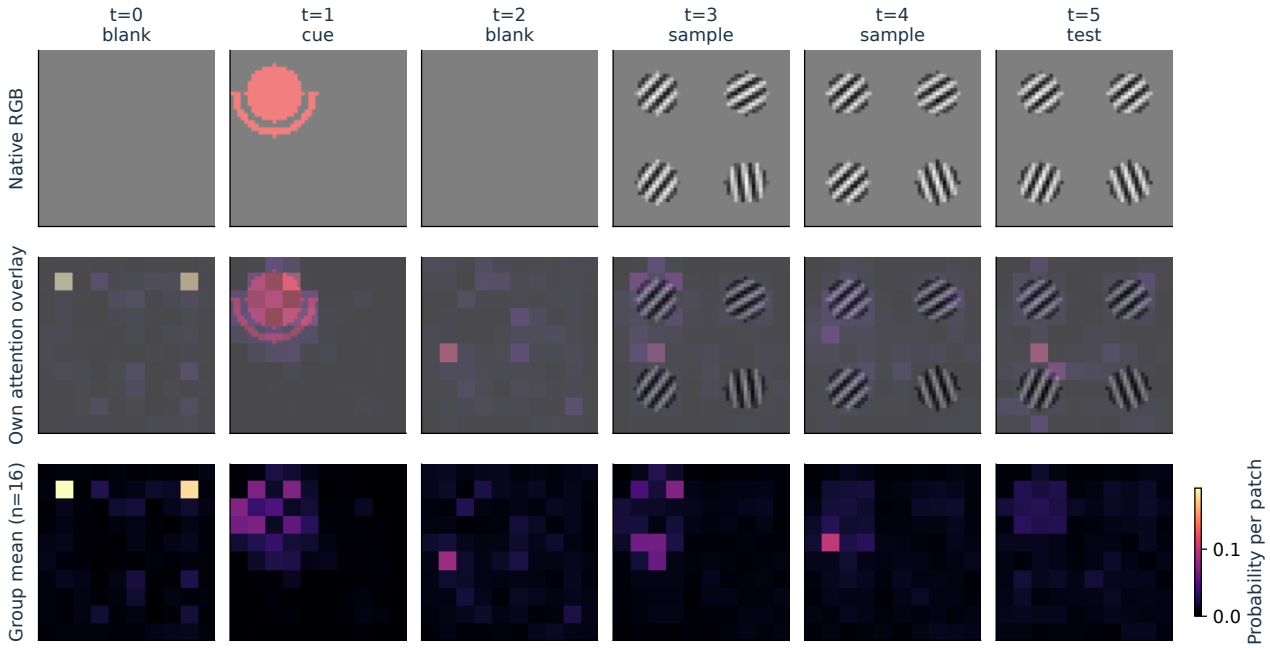


Figure 22: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

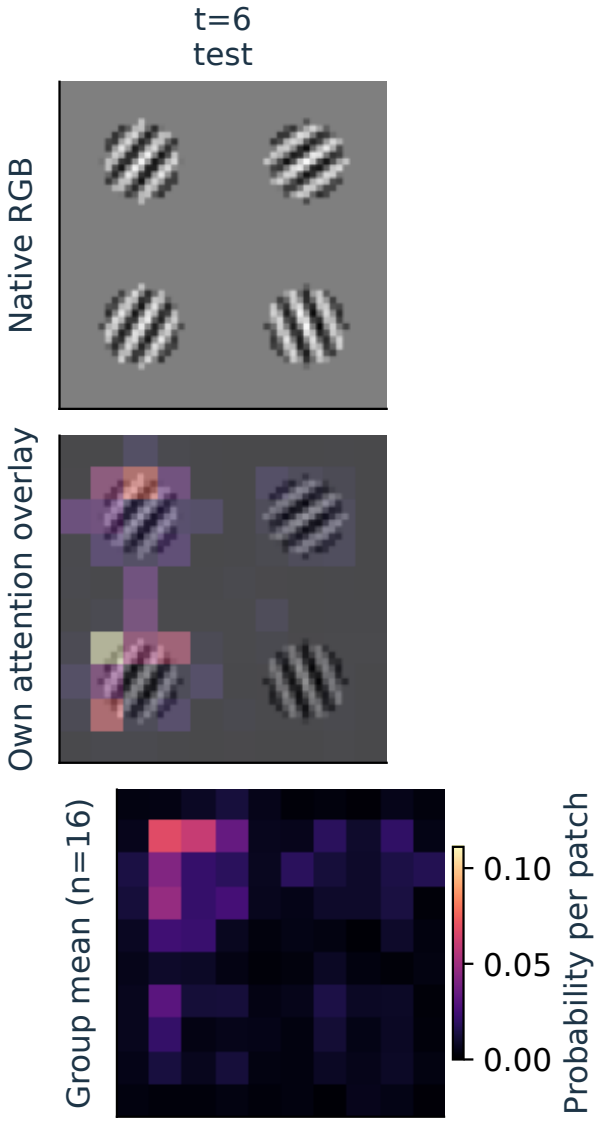


Figure 23: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

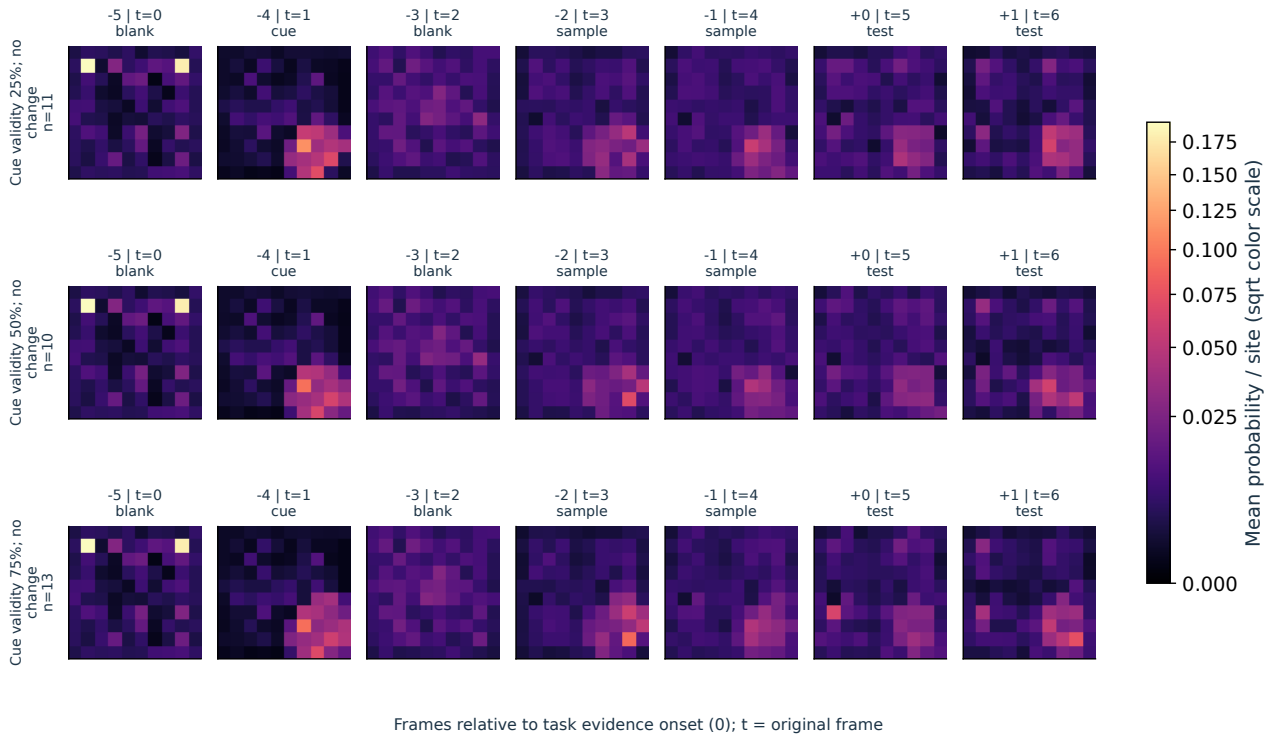


Figure 24: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

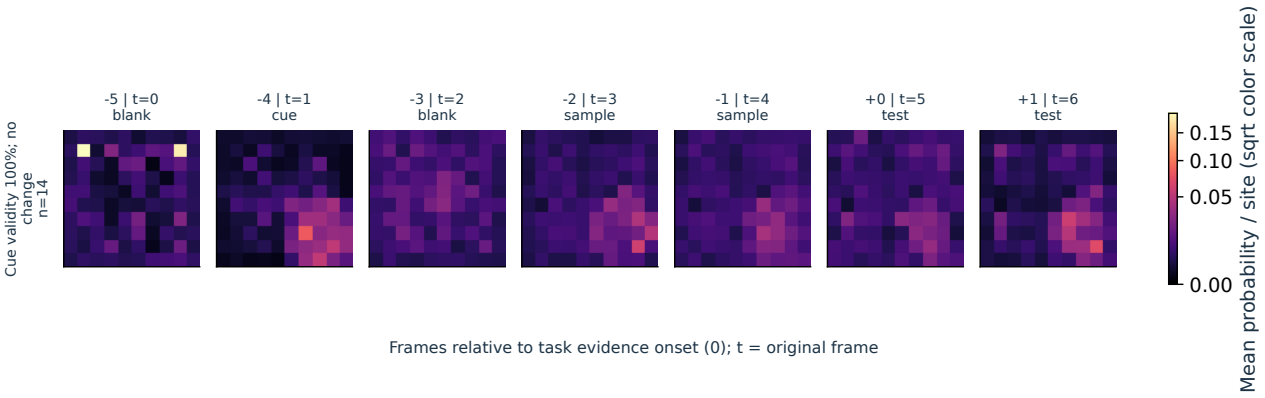


Figure 25: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Priority competition and unexpected changes across nine visual locations

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized vda9 in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 3 native conditions and 384 fresh trials, equal-condition accuracy was 68.5% (95% interval [64.1%, 72.9%]). For the representative condition TH65, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 0.961. The largest observed impairment in that condition was distractor_+8_evidence, changing accuracy by -23.4 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The nine-location task expands the number of candidate change sites while preserving a probabilistic and value-bearing cue. The shared priority distribution must allocate a fixed total mass across a larger display, creating a stronger test of spatial competition than a four-site array.

We measure native discrimination over angle conditions and probe the same corner-to-corner counterfactual used in the four-location task. The exact mapping from native image cells to model patches is retained. Since nine locations do not divide the ten-by-ten model grid evenly, affected patch counts are part of the intervention geometry.

The biological motivation is developed through [1], [2], [3], [4]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

The larger array asks whether the same spatial-priority computation scales with the number of competing sites. Neuroscientific causal experiments often compare effects inside and outside the affected field; our corner-to-corner design makes that comparison explicit. The grid also imposes an important geometric constraint: the nine stimulus cells occupy different numbers of the model ten-by-ten sites. Measuring probability mass and documenting the masks therefore makes the intervention reproducible. The behavioral analysis then determines whether a geometrically specified redistribution benefits an unexpected location.

1.2 Biological experiments motivating this task

Experimental question. How does cue-dependent competition scale when nine visual locations share the same spatial representation?

Reynolds, Chelazzi and Desimone (1999) For two stimuli within a receptive field, attention shifted the response toward that evoked by the attended item alone. The findings supported biased competition between simultaneously represented stimuli. [7]

Basso and Wurtz (1998) SC buildup activity varied with target probability, including when probability changed while the display remained fixed. More possible targets reduced preparatory activity, linking selection state with uncertainty and saccade latency. [8]

Bollimunta, Bogadhi and Krauzlis (2018) Reversible FEF and SC inactivation impaired covert motion-change detection. Comparing the two interventions in a shared behavioral framework tested their contributions to selection during maintained fixation. [9]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Did any Gabor change orientation?

Response labels are 0: no change; 1: change anywhere. The native suite contains 3 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use TH65.

1.4 Stimulus sequence and spatial layout

For the representative native condition TH65, the sequence has 7 frames, indexed from zero: frame 0: blank; frame 1: cue; frame 2: blank; frames 3–4: sample; frames 5–6: test. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Native rotation parameter (degrees)	65.0
Native angular noise parameter (degrees)	5.0
Displayed cue validity	0.0, 0.25, 0.5, 0.75, 1.0
Native trial reward value	1.0, 3.0, 5.0
Original raster width (pixels)	75

Spatial allocation should be read relative to the area occupied by each stimulus cell. An equal additive bias per patch is a precise computational dose, while total priority mass can differ across locations with different patch counts. Cue-only, sample, delay and change-epoch manipulations distinguish instruction processing from selection of sensory evidence.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 68.5%, with macro AUC 0.760. The matched final checkpoint changed accuracy by +9.38 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was TH8 at 50.8%; the highest was TH65 at 81.2%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (2) produced 32.0% correct responses, compared with 100.0% at the largest strength (65). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

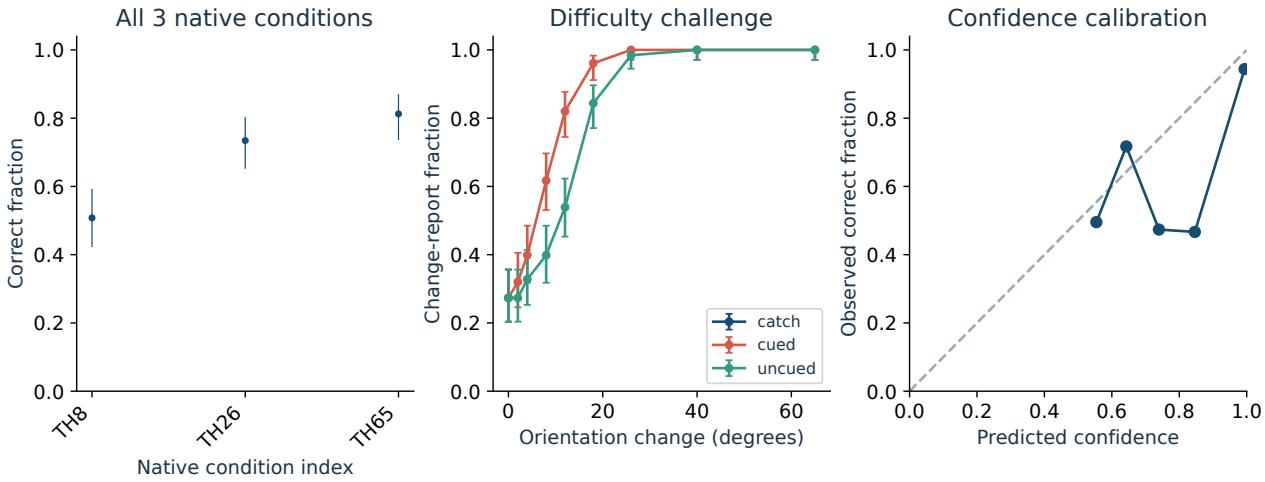


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	88.3%	1.0	81.4%	100.0%
0.7	48.4%	1.0	85.5%	100.0%
0.8	44.5%	1.0	91.2%	100.0%
0.9	39.8%	2.0	98.0%	100.0%

The representative condition TH65 provides 2 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.3 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

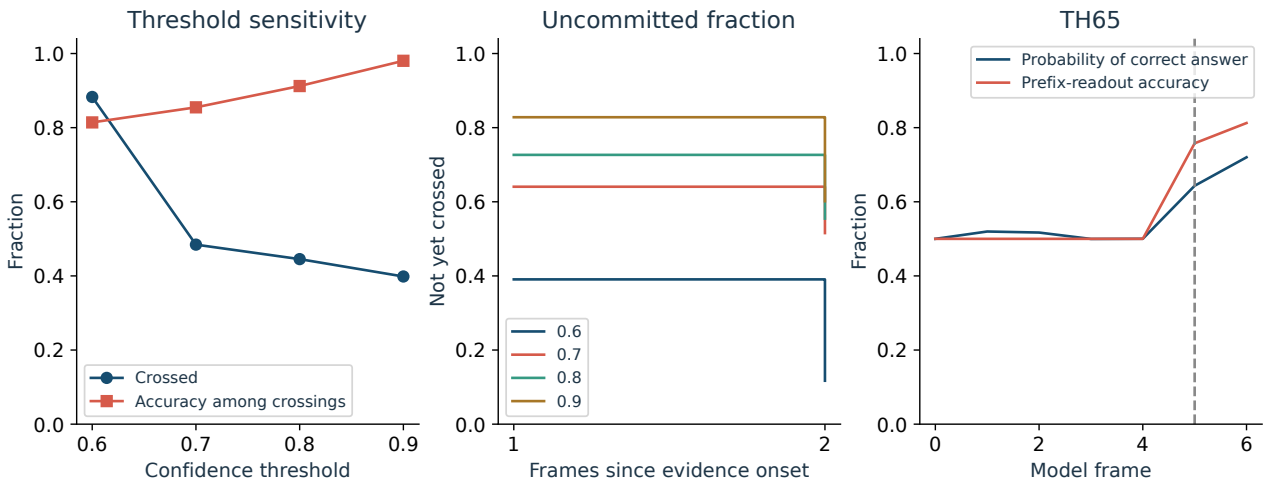


Figure 2: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

At threshold 0.8 across all native presentations, 106 of 384 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

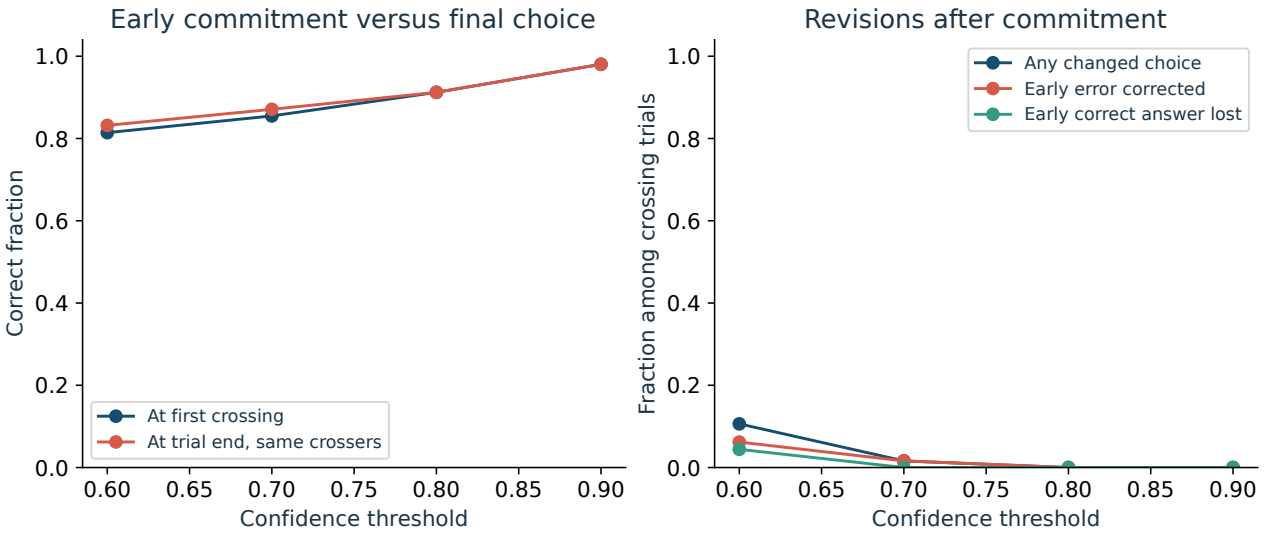


Figure 3: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

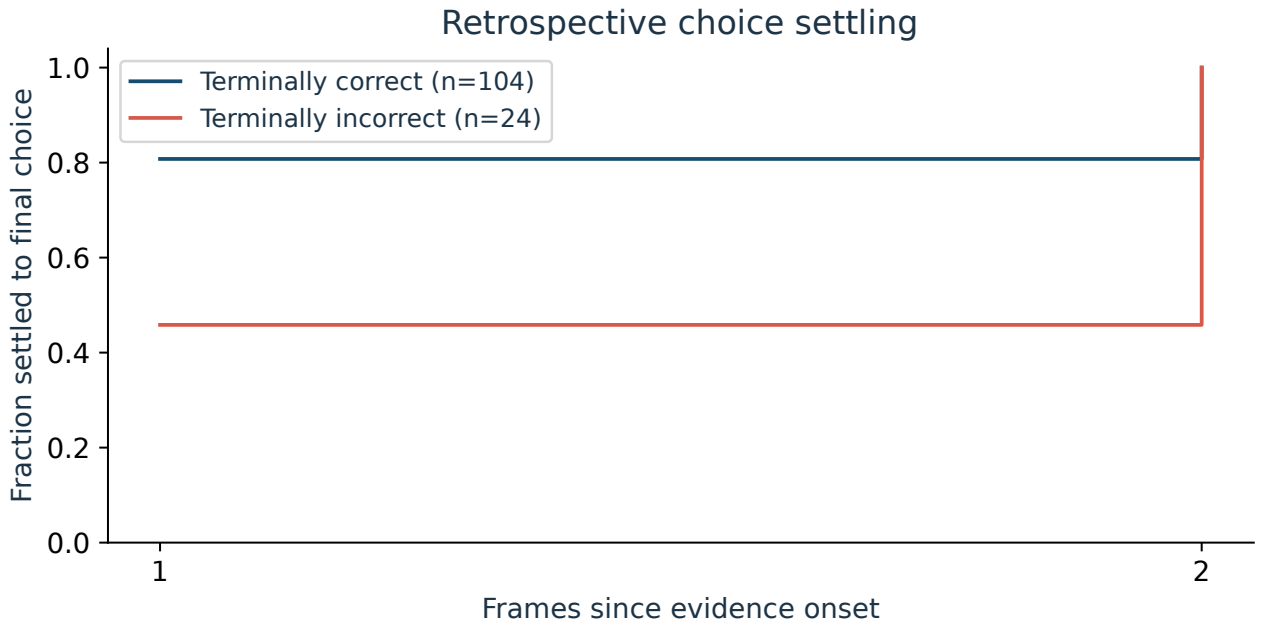


Figure 4: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.4 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

All rows are no-change trials with the cue fixed at bottom right. Rows compare zero, 25, 50, 75 and 100 percent displayed cue validity, pooling the three native difficulty cells and reward colors.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Cue validity 0%; no change	19	19
Cue validity 25%; no change	20	20
Cue validity 50%; no change	14	14
Cue validity 75%; no change	27	27
Cue validity 100%; no change	18	18

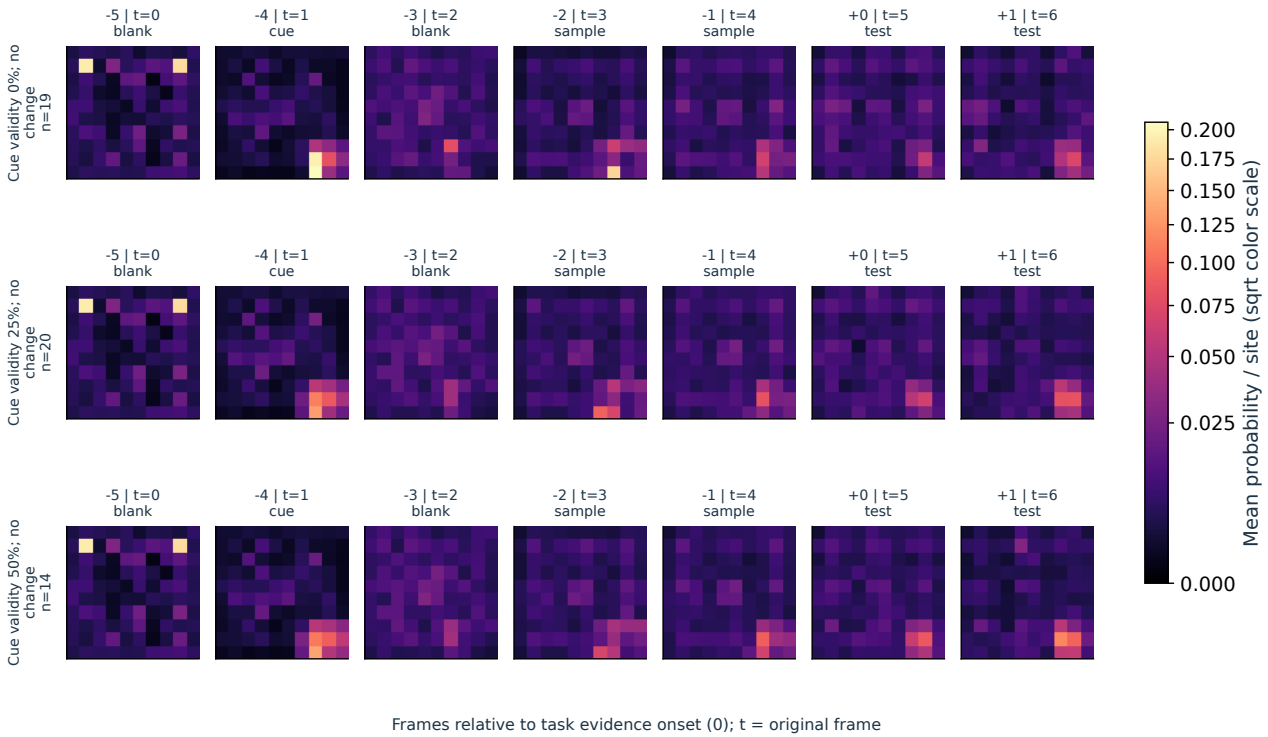


Figure 5: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

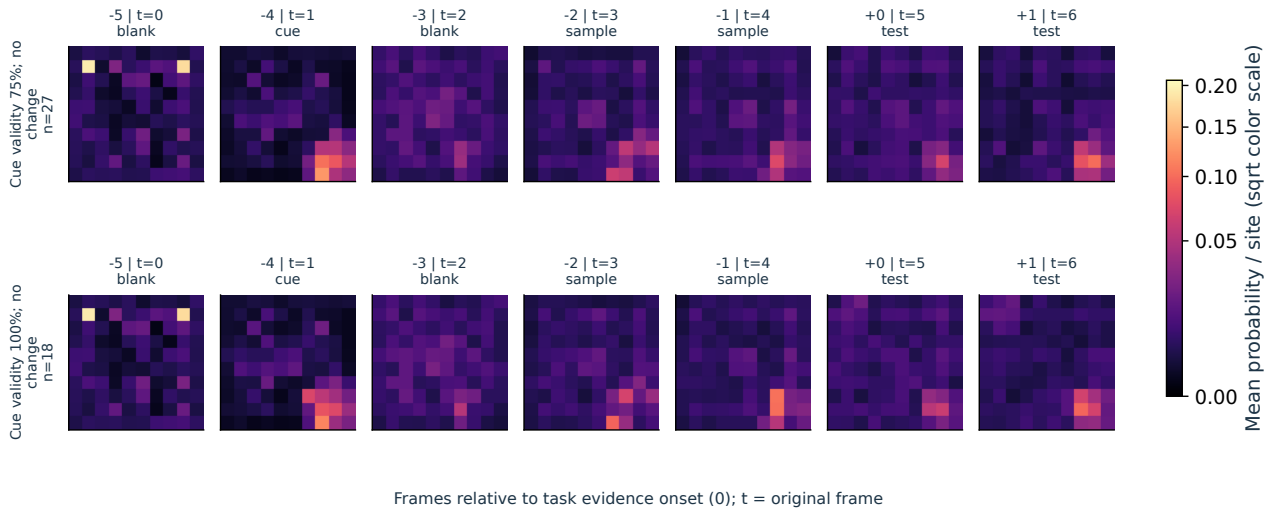


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

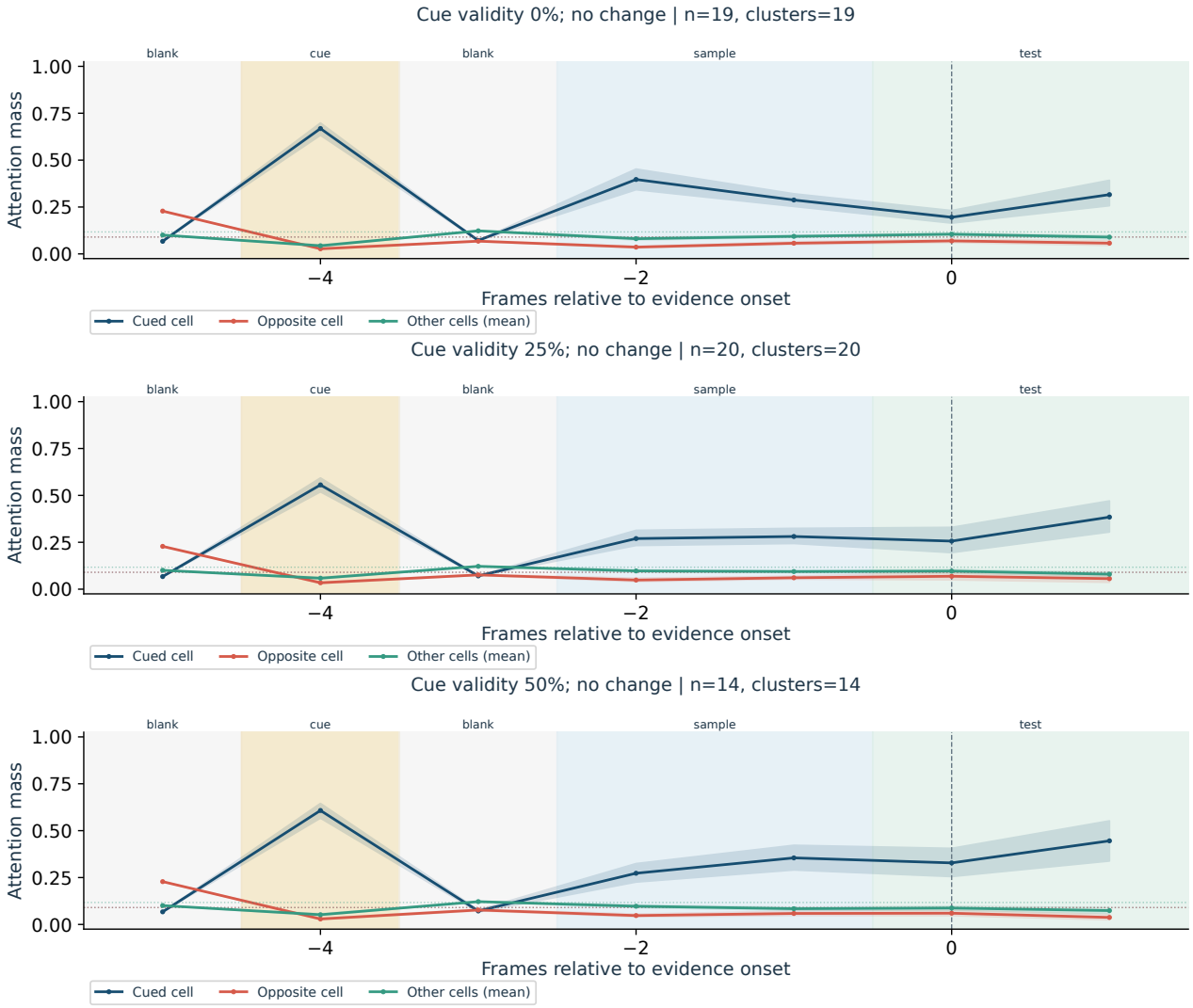


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.



Figure 8: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

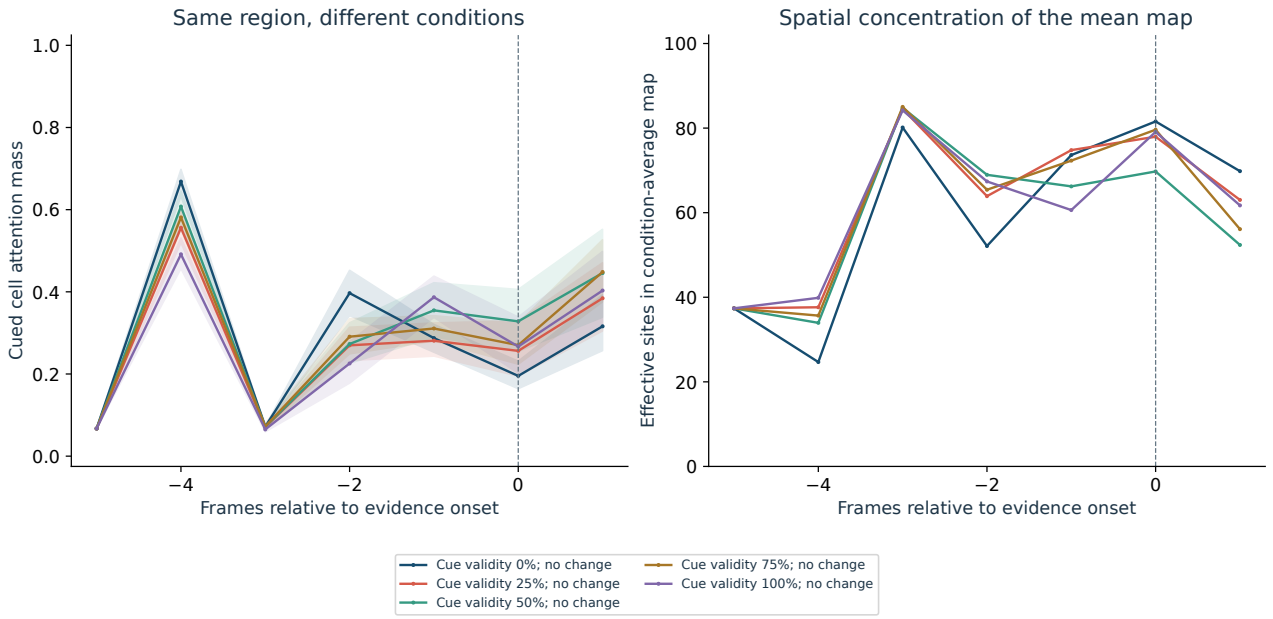


Figure 9: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Cue validity 0%; no change: cued cell mass peaks at 0.668 at original frame 1 (cue), and is 0.316 on the final frame. Cue validity 25%; no change: cued cell mass peaks at 0.556 at original frame 1 (cue), and is 0.384 on the final frame. Cue validity 50%; no change: cued cell mass peaks at 0.608 at original frame 1 (cue), and is 0.446 on the final frame. Cue validity 75%; no change: cued cell mass peaks at 0.581 at original frame 1 (cue), and is 0.448 on the final frame. Cue validity 100%; no change: cued cell mass peaks at 0.491 at original frame 1 (cue), and is 0.403 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.5 Causal sensitivity to spatial and mnemonic interventions

For distractor_+8_evidence, paired accuracy changed by -23.4 percentage points (paired 95% interval [-35.9%, -10.9%]; paired p=0.000535; within-task Holm p=0.02247). There were 21 corrected trials and 51 newly incorrect trials. For target_-2_evidence, paired accuracy changed by +6.2 percentage points (paired 95% interval [0.8%, 11.7%]; paired p=0.05737; within-task Holm p=1). There were 11 corrected trials and 3 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

3.6 Bottom-right cue and top-left counterfactual change

For catch events, bottom-right inhibition at dose -4 changed correct-report frequency by +6.6 percentage points, with paired interval [1.2%, 12.5%]. For cued events, bottom-right inhibition at dose -4 changed

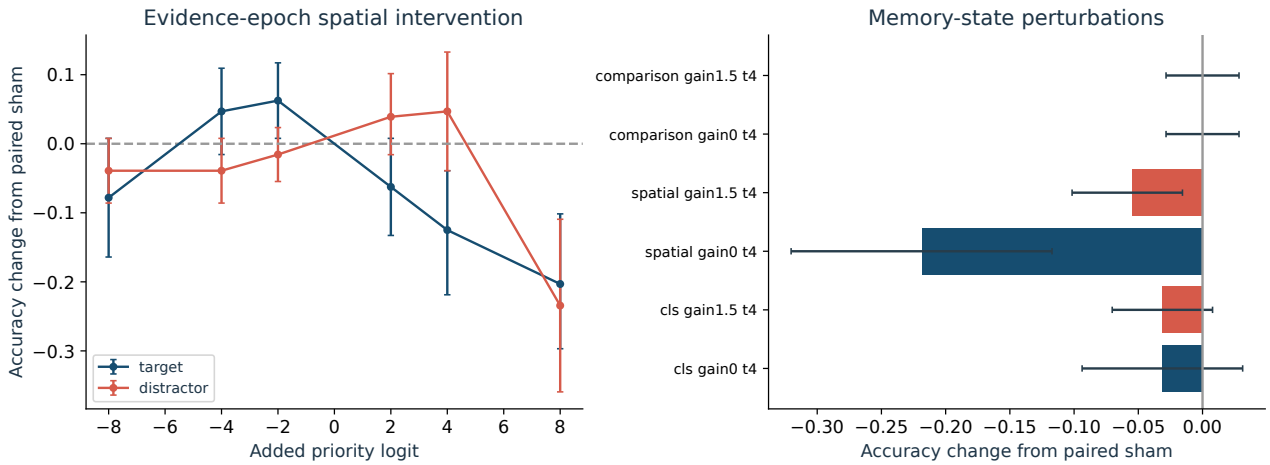


Figure 10: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

correct-report frequency by -59.0 percentage points, with paired interval [-64.8%, -52.7%]. For uncued events, bottom-right inhibition at dose -4 changed correct-report frequency by +3.1 percentage points, with paired interval [0.4%, 5.9%]. The 100 percent cue and uncued-change combination is deliberately outside the native cue-validity rule. A change at top left still has a correct positive label because VDA requires a report for a change anywhere. Catch trials have no physical change and provide the false-alarm control.

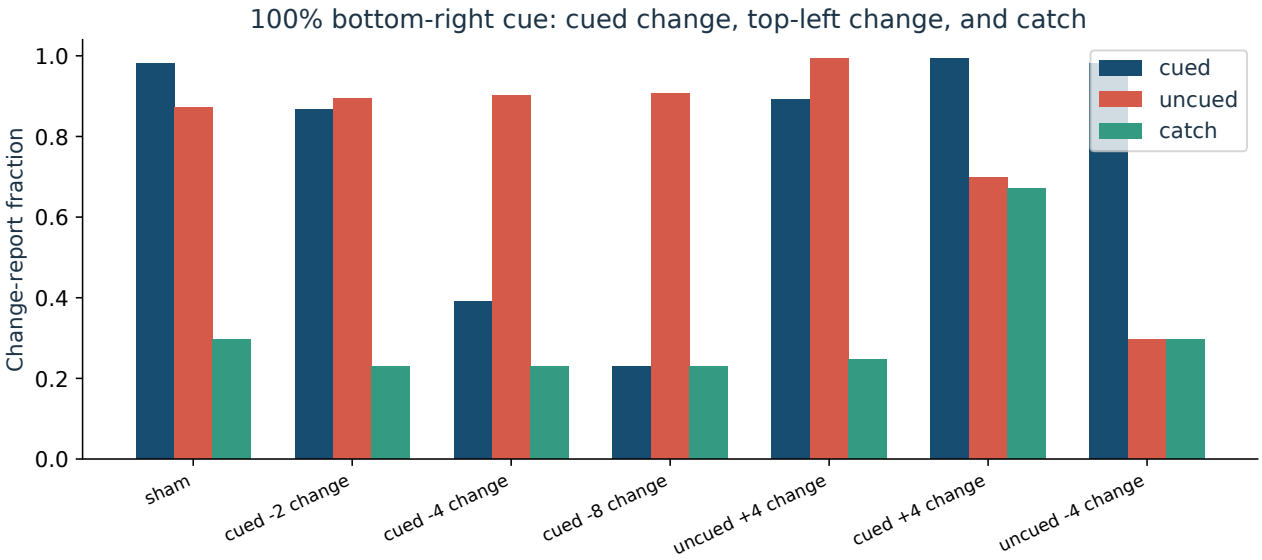


Figure 11: Requested counterfactual experiment. Every movie has a 100 percent bottom-right cue. Change reports are plotted separately for changes at the cue, changes at top left and catches. Priority perturbations occur only on the two change frames. Each event type contains 256 matched nuisance draws per intervention arm.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
TH65	target_-4	+4.7	1
TH65	target_+4	-12.5	0.6912
TH65	distractor_-4	-3.9	1
TH65	distractor_+4	+4.7	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.462	45.8%	0.820
fusion	0.720	66.2%	0.010
spatial	0.940	86.7%	0.005
cls	0.961	92.3%	0.005
comparison	Constant zero on this condition		
attention	0.867	80.5%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

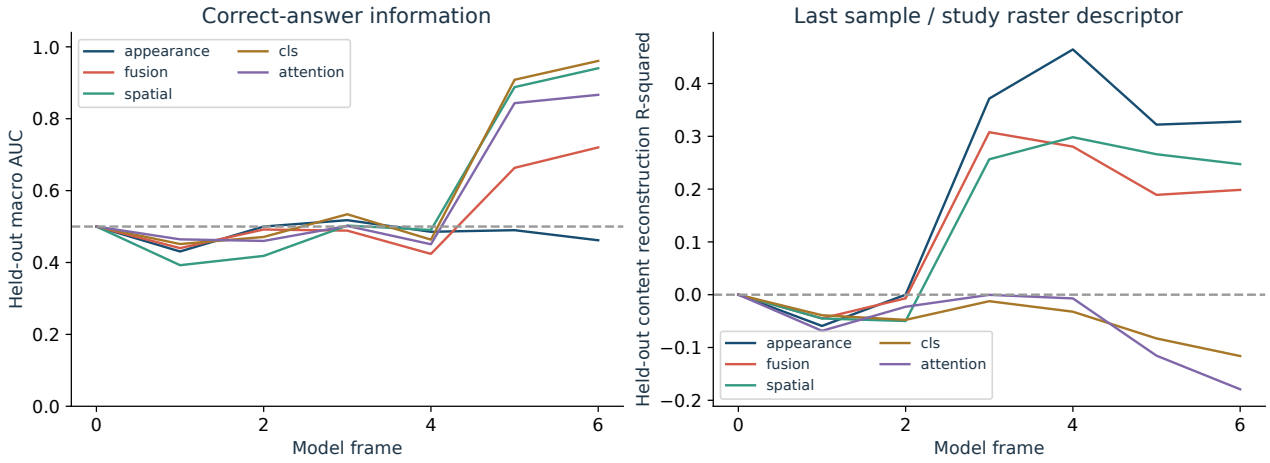


Figure 12: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 0.00 to 12.89 within the retained subspace. The final step displacement was 1.458 standardized RMS units. In spatial memory, the participation ratio ranged from 0.00 to 20.32 within the retained subspace. The final step displacement was 0.450 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 4, coarse sample-content reconstruction had R-squared -0.032 from CLS memory and 0.298 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -3.1 percentage points, spatial, -21.9 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

3.11 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.456	0.468	3200
fusion	0.712	0.528	6400
spatial	0.944	0.579	6400
cls	0.951	0.456	64
comparison	unavailable	unavailable	–
attention	0.777	0.434	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 48 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.12 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

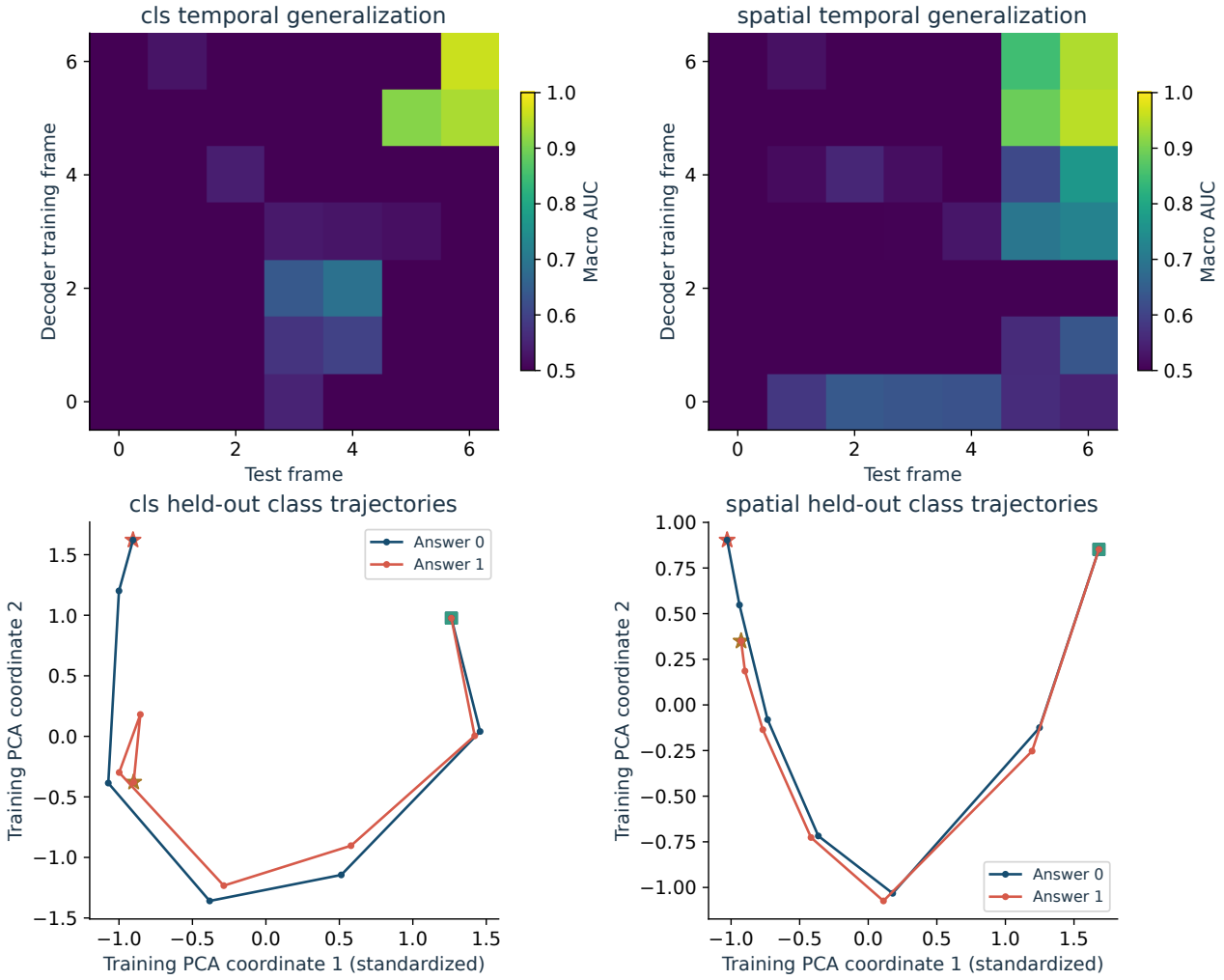


Figure 13: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

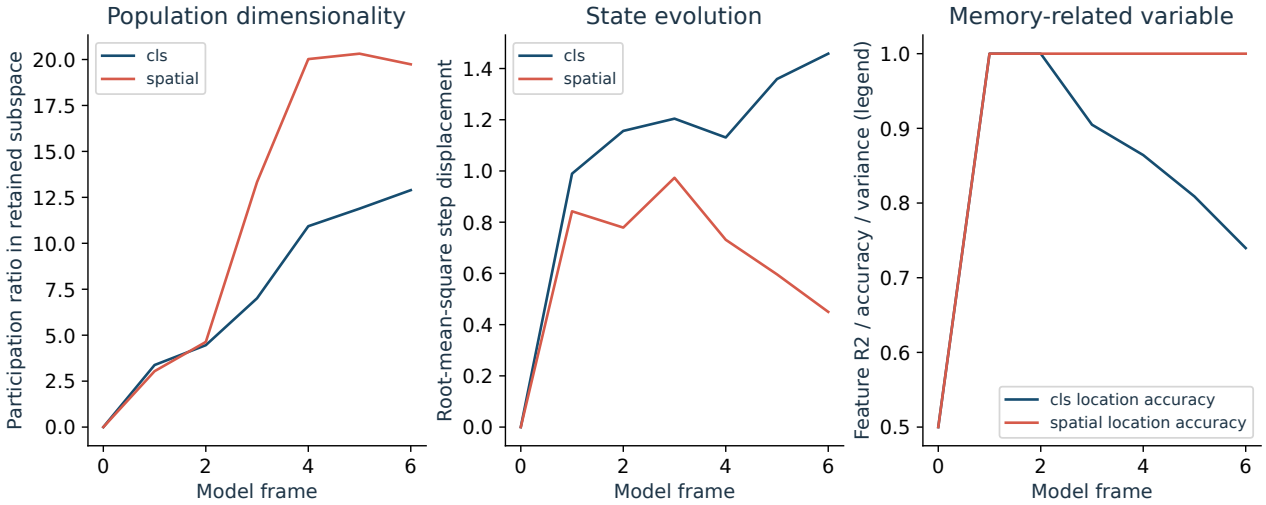


Figure 14: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

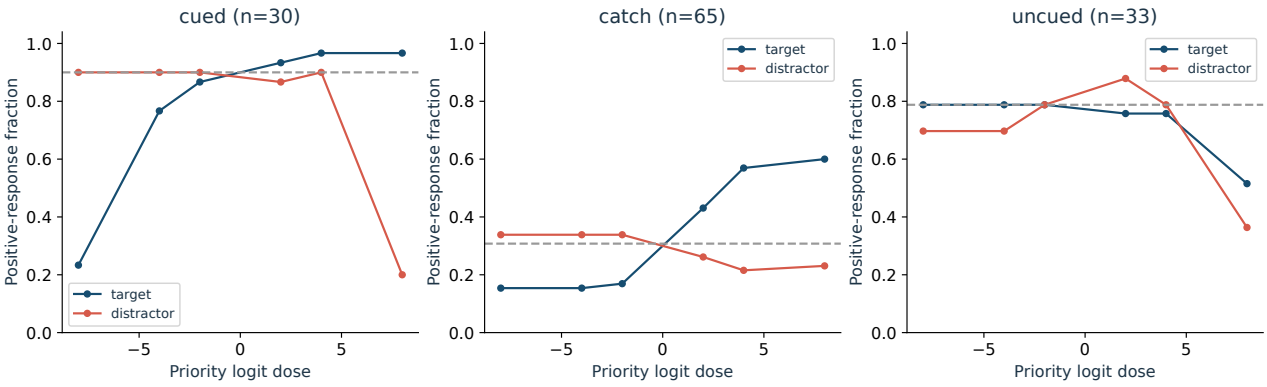


Figure 15: Event-specific causal response functions. Curves show positive reports after evidence-epoch target or alternative-site intervention; the horizontal line is paired sham. Event counts are stated above each panel. A positive report has the task-specific meaning defined in the text.

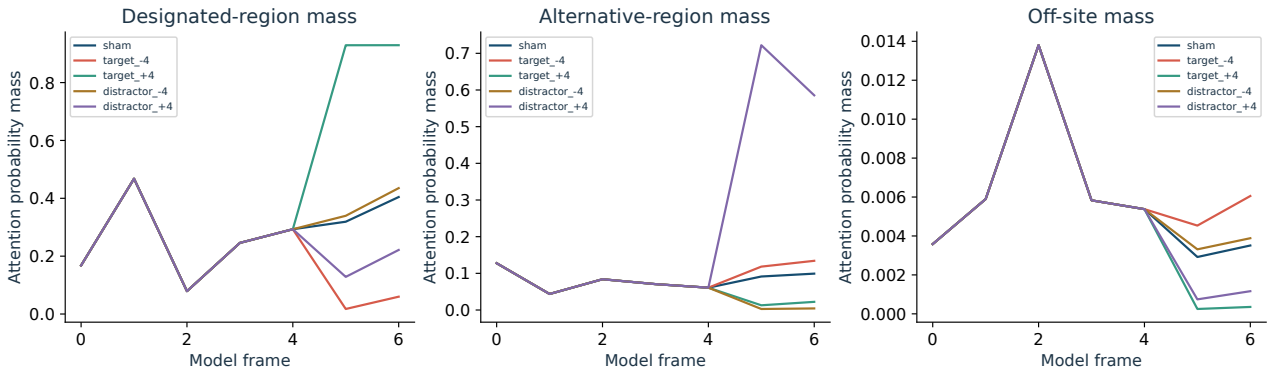


Figure 16: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

3.13 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

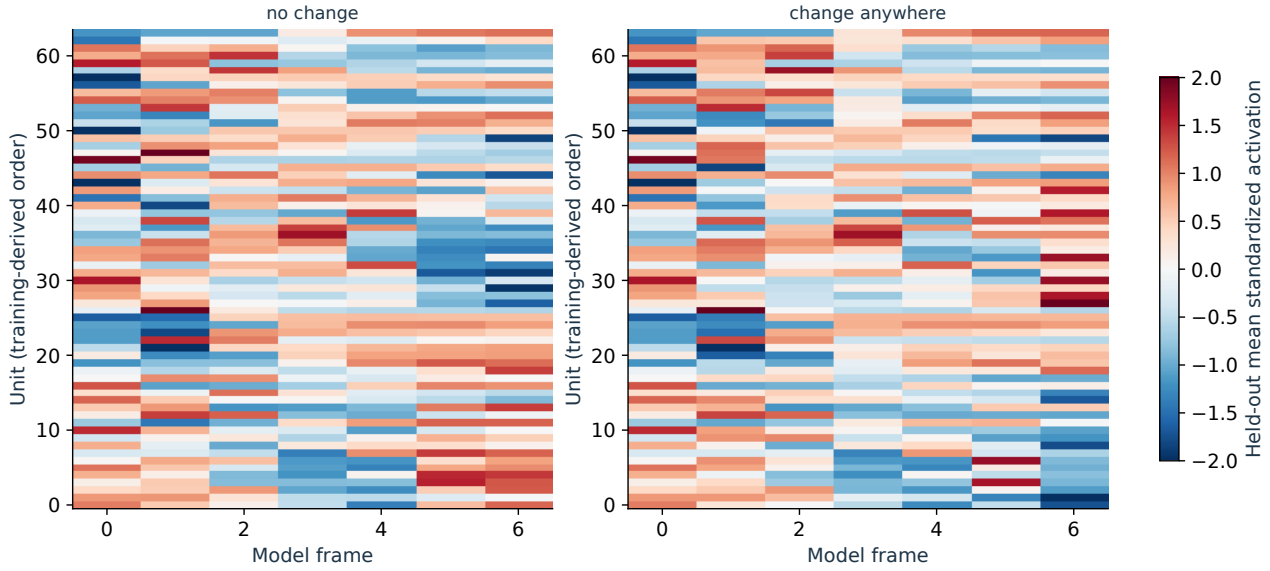


Figure 17: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.14 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional

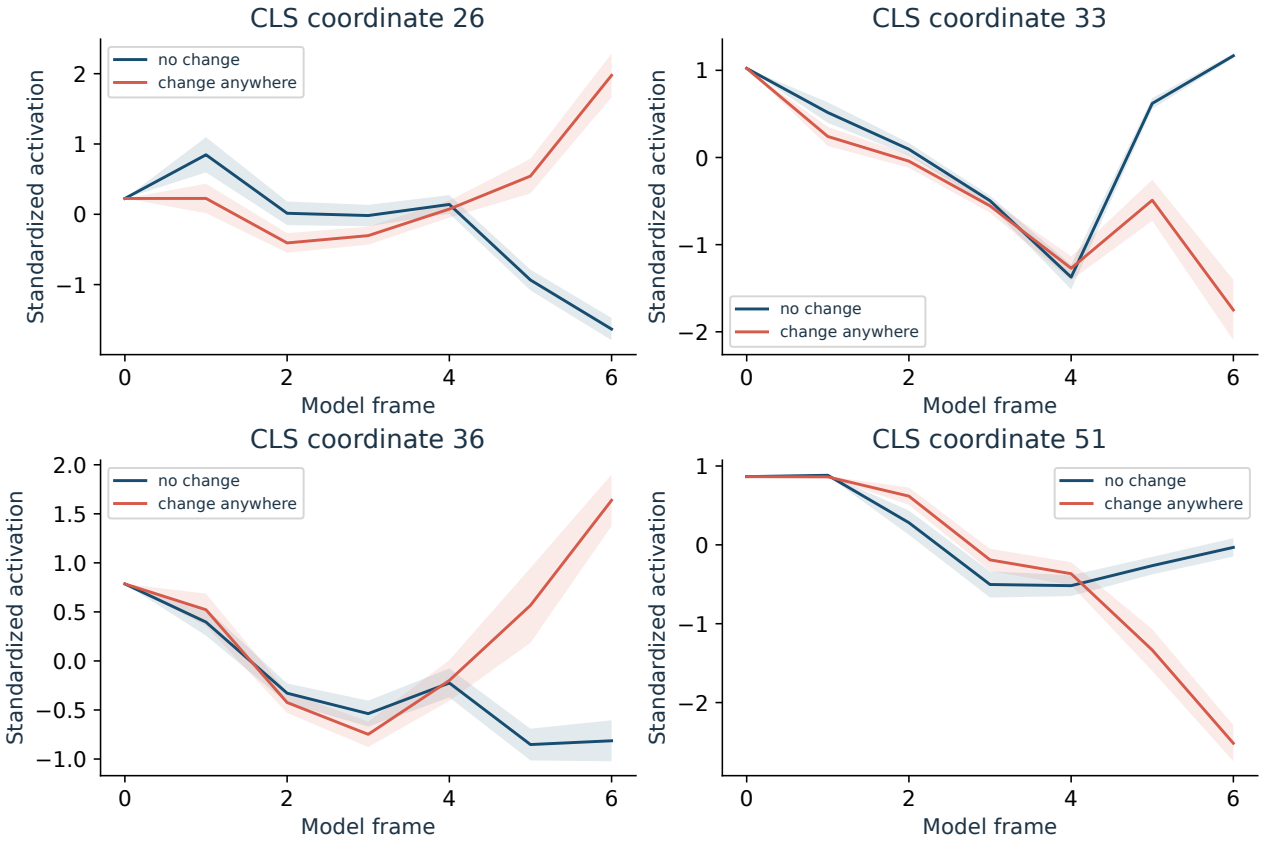


Figure 18: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	45.8%	42.5%	48.3%
fusion	66.2%	68.7%	52.9%
spatial	86.7%	87.3%	100.0%
cls	92.3%	93.8%	74.0%
attention	80.5%	76.9%	89.9%

3.15 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

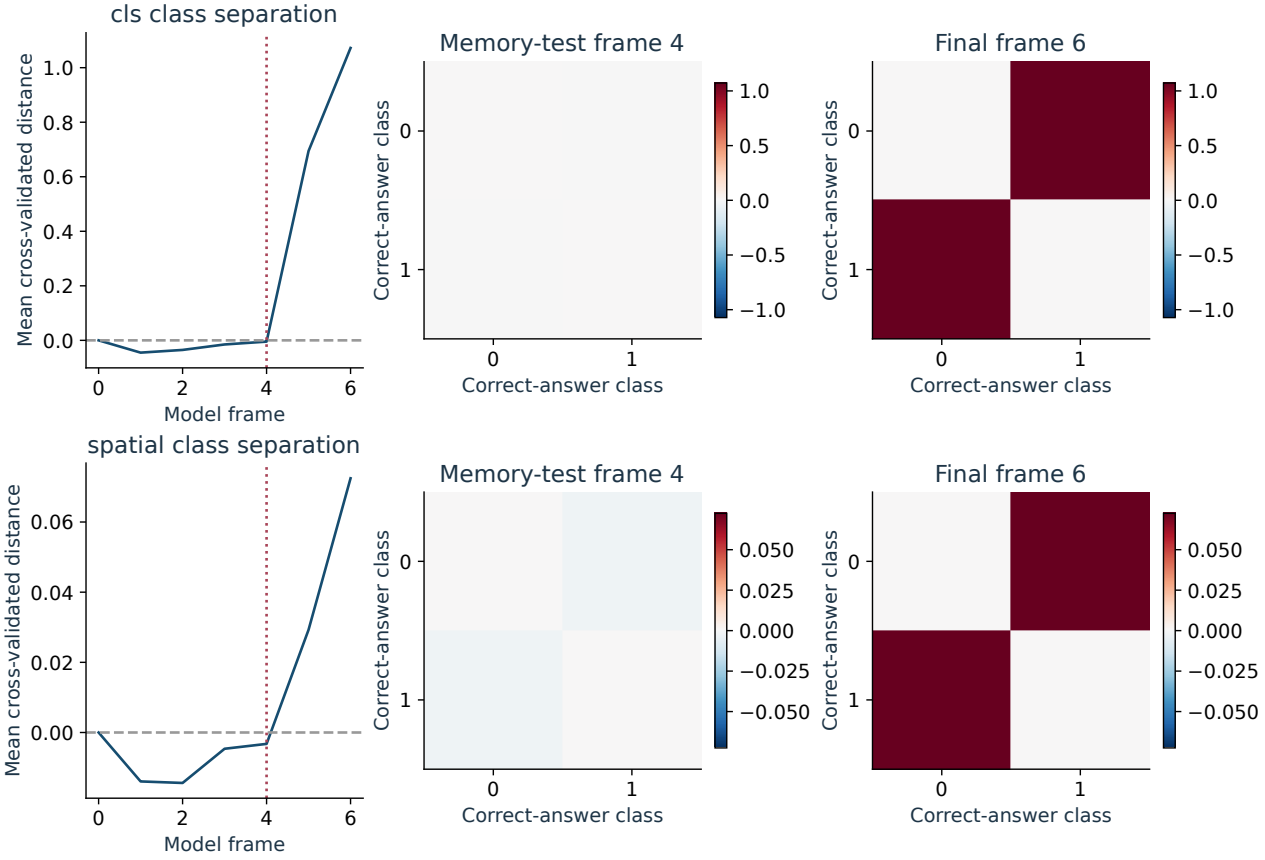


Figure 19: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

3.16 Weaker-signal follow-up to test the improvement prediction

After the 18-degree VDA4 counterfactual showed high uncued baseline detection, an exploratory follow-up tested 2-, 4- and 8-degree changes, plus catches, in all three VDA sizes. Each event-strength cell contains 128 new matched nuisance draws per arm. The bottom-right cue remains certain and perturbations occur during the change frames. Holm adjustment below covers the entire follow-up family within this task. These tests address whether the apparent absence of an uncued benefit at strong signals reflects limited headroom.

Event	Degrees	Sham accuracy	Cue-inhibition change	Holm p
catch	0	68.0%	+7.8 pp	1
cued	2	30.5%	-7.0 pp	1
uncued	2	30.5%	-6.2 pp	1
cued	4	36.7%	-13.3 pp	0.1976
uncued	4	32.0%	-5.5 pp	1
cued	8	58.6%	-35.2 pp	6.251e-08
uncued	8	39.8%	-0.8 pp	1

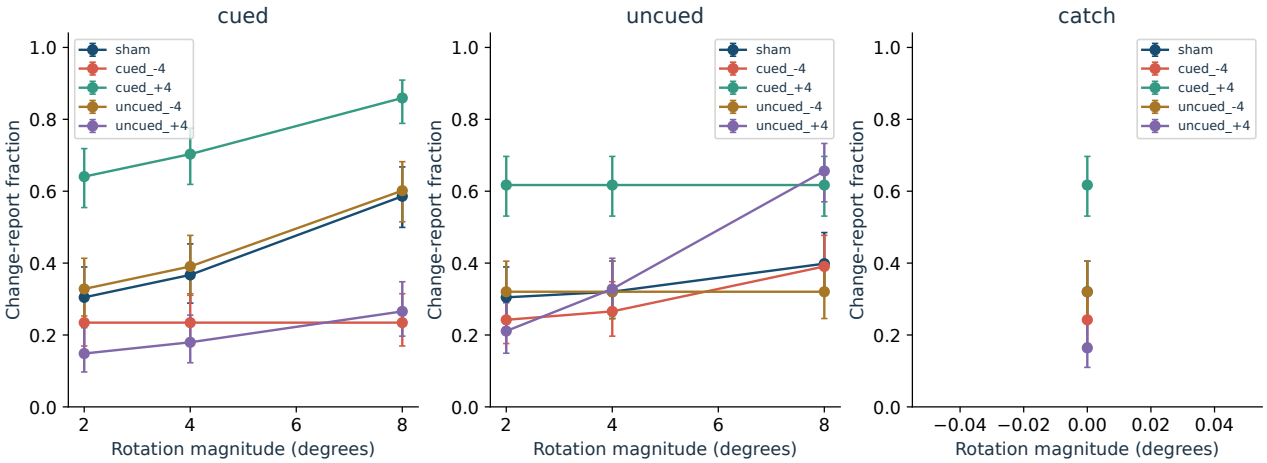


Figure 20: Exploratory weak-change counterfactual. Cued and uncued hit fractions are shown across three magnitudes; the catch panel shows false reports without a physical change. Error bars are Wilson 95 percent intervals. All interventions reuse identical movies within each cell.

3.17 Probability cues, reward value and prechange allocation

The native task varies displayed change-location probability and cue color, which specifies trial value. We stratify the original 384 trials by displayed probability, event location and value. Cued and uncued change reports are hits; reports on catch trials are false alarms. Prechange cue mass is measured at frame 4, before the two test frames, and is expressed relative to a uniform distribution over the 100 sites. Probability intervals use Wilson bounds; attention intervals resample native trials 3,000 times. These native stratifications characterize how the learned observer treats the task instructions, with the paired interventions supplying the causal test.

The model obtained 67.7% of the available native value, calculated as the sum of value on correctly answered trials divided by total offered value. At value 1, accuracy was 69.8% across 116 trials. At value 3, accuracy was 69.6% across 138 trials. At value 5, accuracy was 66.2% across 130 trials.

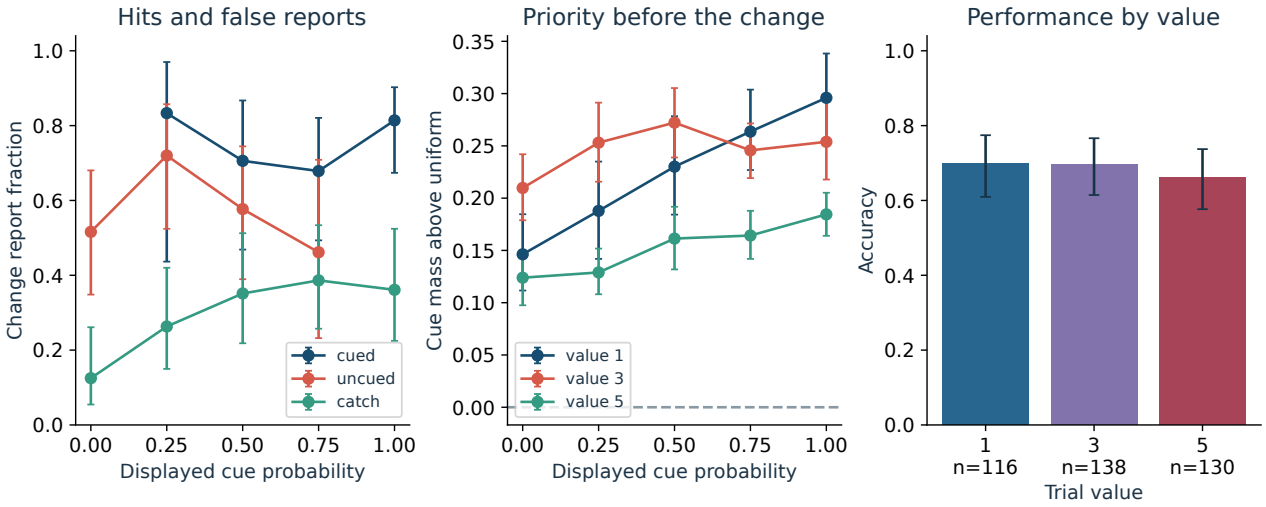


Figure 21: Native probability and value stratification. Left: cued hits, uncued hits and catch false reports. Middle: cue-region attention above uniform at the last prechange sample frame, separated by trial value. Right: accuracy by offered value. Sparse event-probability groups have wide sampling intervals; their sample counts are supplied in the source table.

4 Discussion

The corner-to-corner counterfactual again shows a strong cued detection cost under cue inhibition. The uncued benefit at the strong signal is smaller and uncertain. The weak-change follow-up again produces a cued cost at eight degrees, about 35 percentage points, with no uncued benefit. It therefore does not recover the joint dual effect after correction across the three VDA follow-ups. Nine stimulus cells occupy unequal numbers of the ten-by-ten model sites, so the recorded mass and affected patch counts help interpret a common per-patch logit dose. Decodable change information increases across the visual-to-recurrent hierarchy, and spatial-memory reset impairs the native condition. The combined measurements connect spatial competition to retained sample information while separating local intervention geometry from behavioral sensitivity.

The behavioral result establishes 68.5% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies distractor_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Matched-area interventions and trial-length manipulations would determine how much of a difference from VDA4 is explained by geometry, stimulus competition, or available temporal computation.

4.1 Interpreting these results alongside biological experiments

4.1.1 Reynolds, Chelazzi and Desimone (1999)

The animal experiment motivates comparing responses with and without competing stimuli while holding the target fixed. Our nine-cell array adds competitors but also changes the mapping from stimuli to the 10-by-10 model grid. Region mass must therefore be interpreted with the number of represented sites and their visual occupancy. A common logit dose can affect unequal amounts of evidence across cells. [7]

4.1.2 Basso and Wurtz (1998)

Probability-dependent selection is separable from the number of visible objects. VDA9 can exploit that distinction by keeping all nine cells present and varying only the cue information. The two native corner cue positions also permit matched spatial comparisons. The appropriate neural analogue is a change in preparatory selection for the same display, followed by an event-dependent report. [8]

4.1.3 Bollimunta, Bogadhi and Krauzlis (2018)

The inactivation comparison motivates examining spatially specific detection costs under a fixed covert task. The model has a strong cued cost under inhibition, and weak eight-degree changes lose about 35

percentage points at the cue. The absence of an uncued gain indicates that restoring a larger relative share of attention elsewhere does not automatically restore a useful detector. [9]

4.2 A computational account of the observed behavior

Biased competition is a useful theory when several representations compete for a downstream effect. In the model, normalized spatial weights explicitly implement that competition, while recurrent memory supplies the previous visual state needed to detect change. The interpretation must include geometry: nine stimulus cells divide into a ten-site axis unevenly, and patch centers can sample different fractions of each cell. Increased set size can therefore alter feature quality, storage demand and selection competition together. The present spatial reset cost and improved recurrent answer decoding show that retained processing helps, but they do not assign all extra difficulty to an attentional capacity limit. The condition atlas and manipulation checks supply the needed map-level view before behavioral comparisons are pooled.

4.3 Where this task agrees with the biological evidence

The agreement is cue-sensitive, memory-dependent competition with a reproducible cued suppression cost. The difference is the missing reciprocal improvement and the discrete grid's unequal sampling. Both are scientifically useful: they specify which parts of the shared-selection account work and where stimulus quality or readout can limit its behavioral expression.

4.4 A follow-up experiment motivated by these results

Render nine equally sampled apertures on a grid that gives each one the same number of model sites, and compare with the native resize. Hold target pixels fixed while adding distractors. A competition account predicts a distractor-dependent impairment beyond any change caused by target raster quality.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Morgan J, Albanna B, Herman JP (2025). A recurrent vision transformer shows signatures of primate visual attention. arXiv:2502.10955. https://arxiv.org/abs/2502.10955.

[2] Bollimunta A, Bogadhi AR, Krauzlis RJ (2018). Comparing frontal eye field and superior colliculus contributions to covert spatial attention. Nature Communications 9:3553. https://www.nature.com/articles/s41467-018-06042-2.

[3] Reynolds JH, Heeger DJ (2009). The normalization model of attention. Neuron 61:168-185. https://pubmed.ncbi.nlm.nih.gov/19186161/.

[4] Wolfe JM (2021). Guided Search 6.0: An updated model of visual search. Psychonomic Bulletin and Review 28:1060-1092. https://pubmed.ncbi.nlm.nih.gov/33547630/.

[5] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[6] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[7] Reynolds, Chelazzi and Desimone (1999). Competitive mechanisms subserve attention in macaque areas V2 and V4. Primary source.

[8] Basso and Wurtz (1998). Modulation of neuronal activity in superior colliculus by changes in target probability. Primary source.

[9] Bollimunta, Bogadhi and Krauzlis (2018). Comparing frontal eye field and superior colliculus contributions to covert spatial attention. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
TH8	128	50.8%	0.521	0.539
TH26	128	73.4%	0.829	0.360
TH65	128	81.2%	0.931	0.289

B Every-frame attention atlas

Condition TH65. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 31 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

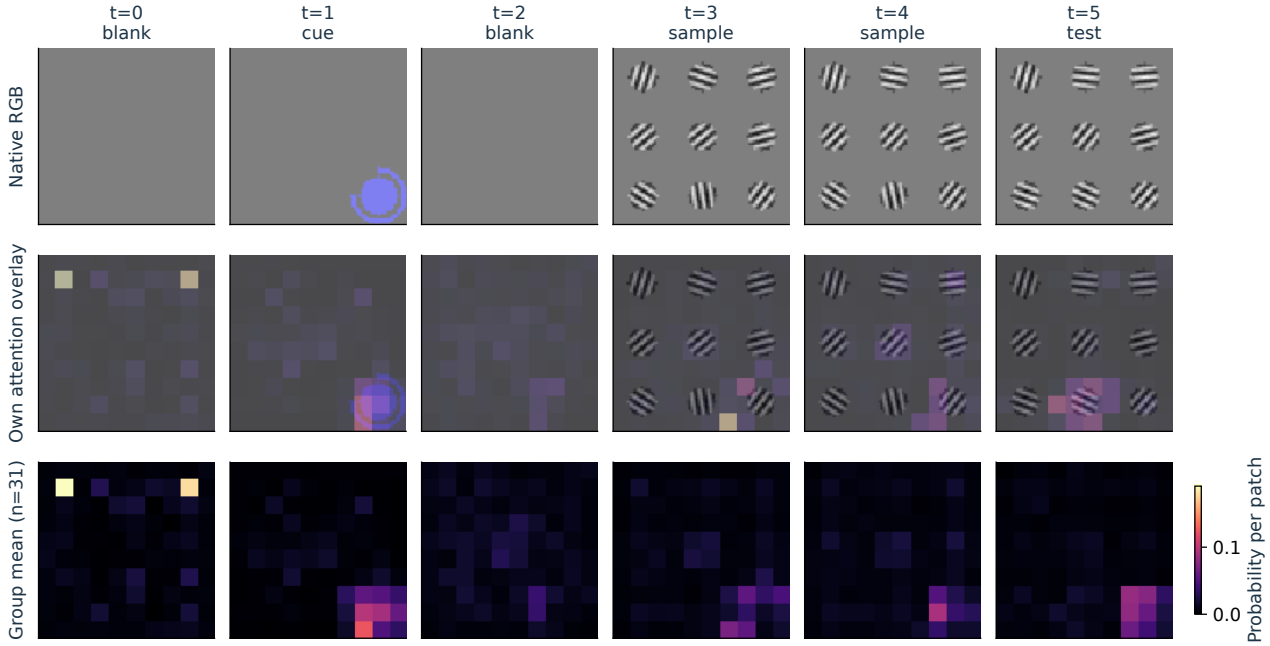


Figure 22: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

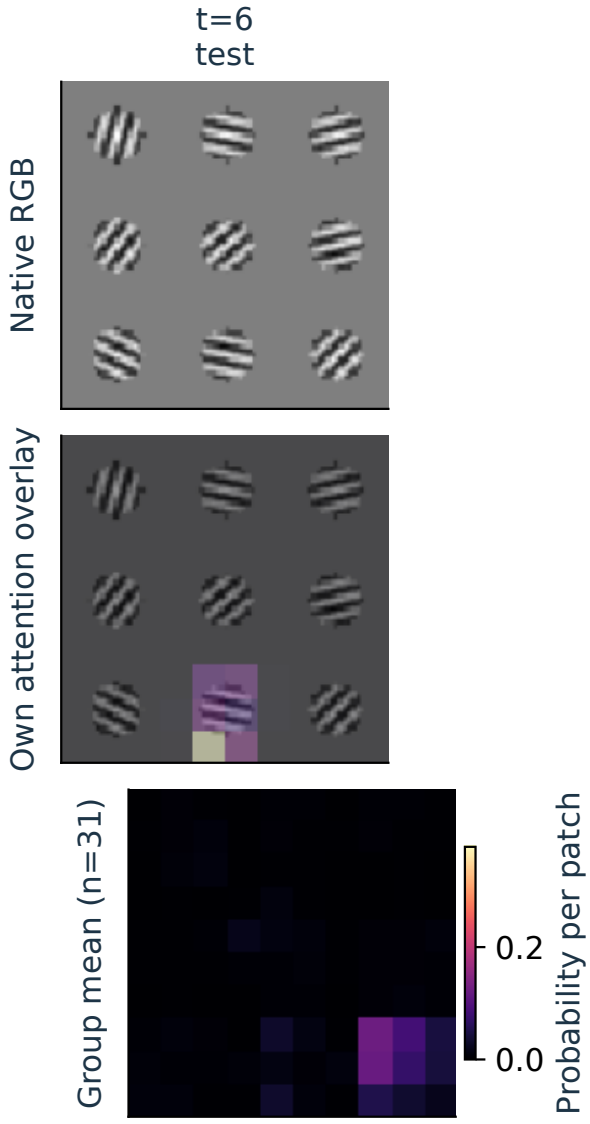


Figure 23: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

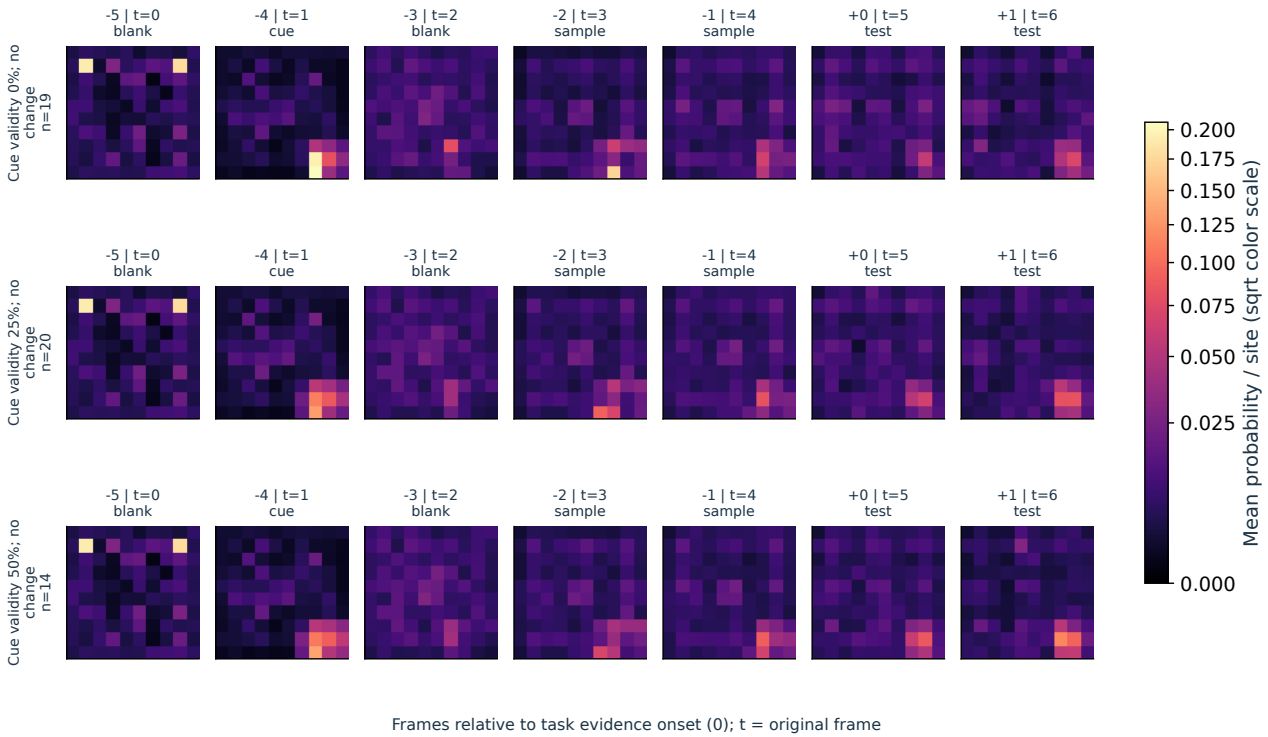


Figure 24: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

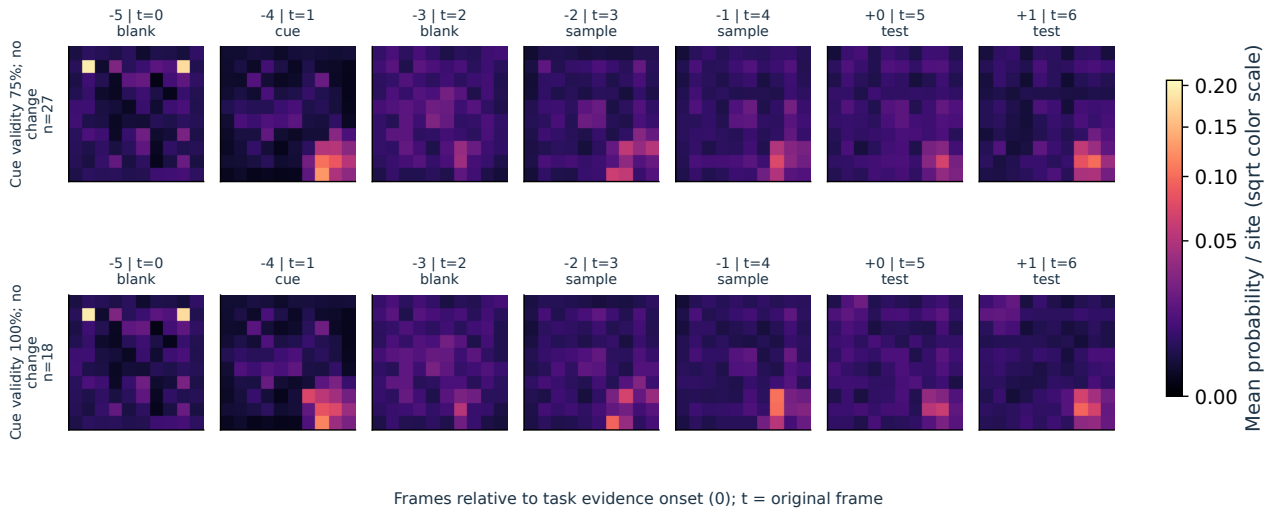


Figure 25: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Shared spatial priority under sixteen-location change competition

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized vda16 in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 3 native conditions and 384 fresh trials, equal-condition accuracy was 64.8% (95% interval [60.2%, 69.5%]). For the representative condition TH65, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 0.894. The largest observed impairment in that condition was spatial_gain0_t4, changing accuracy by -25.0 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

Sixteen Gabor locations place the greatest spatial competition demand on the value-directed sub-battery. Each stimulus occupies a smaller region of the common visual representation. The model must compare local evidence while respecting the probabilistic cue and reporting changes anywhere.

The analyses preserve native cell geometry and distinguish changes at the cued and uncued locations. Set size is fixed within this task, so comparisons to VDA4 and VDA9 form a cross-task load comparison. They also change the spatial sampling of each Gabor, which is measured through the patch allocation maps.

The biological motivation is developed through [1], [2], [3], [4]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

A dense array creates two potentially interacting demands: several locations compete for priority, and each stimulus is represented by fewer spatial samples. The attention intervention directly changes competition while preserving the rendered movie. A benefit at an unexpected location would show that useful evidence was present but underweighted. An attention shift without a behavioral benefit would instead motivate closer examination of the quality of the newly selected representation. This distinction links causal manipulation to population decoding rather than treating a visible movement of the map as sufficient evidence of improved processing.

1.2 Biological experiments motivating this task

Experimental question. Is poorer performance in a dense array caused by selection, representation, memory, or shared variability?

Cohen and Maunsell (2009) Population recordings in V4 during an orientation-change task linked much of the attentional improvement in population sensitivity to reduced shared variability, beyond changes in average firing rates. [7]

Reynolds, Chelazzi and Desimone (1999) For two stimuli within a receptive field, attention shifted the response toward that evoked by the attended item alone. The findings supported biased competition between simultaneously represented stimuli. [8]

Zenon and Krauzlis (2012) SC inactivation impaired attentive behavior while established attentional modulation in visual cortex persisted. Behavioral selection and cortical response enhancement could therefore dissociate. [9]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Did any Gabor change orientation?

Response labels are 0: no change; 1: change anywhere. The native suite contains 3 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use TH65.

1.4 Stimulus sequence and spatial layout

For the representative native condition TH65, the sequence has 7 frames, indexed from zero: frame 0: blank; frame 1: cue; frame 2: blank; frames 3–4: sample; frames 5–6: test. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Native rotation parameter (degrees)	65.0
Native angular noise parameter (degrees)	5.0
Displayed cue validity	0.25, 0.5, 0.75, 1.0
Native trial reward value	1.0, 3.0, 5.0
Original raster width (pixels)	100

A strong cue can concentrate the attention distribution on a small part of a crowded display. Inhibition can release that concentration, but successful detection elsewhere still requires sufficient sensory information at the new location. Thus a redistribution of attention and a behavioral benefit are separate empirical outcomes.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 64.8%, with macro AUC 0.723. The matched final checkpoint changed accuracy by +2.08 percentage points on average across conditions, using 32 same-movie com-

parisons per condition. The lowest native cell was TH8 at 54.7%; the highest was TH65 at 76.6%. These comparisons retain the native sampling design rather than selecting trials after observing success.

In the empirical difficulty challenge, the smallest positive tested strength (2) produced 35.2% correct responses, compared with 99.2% at the largest strength (65). Each point uses 128 trials. The full curve, including intermediate levels and event-specific responses, is shown in Figure 1.

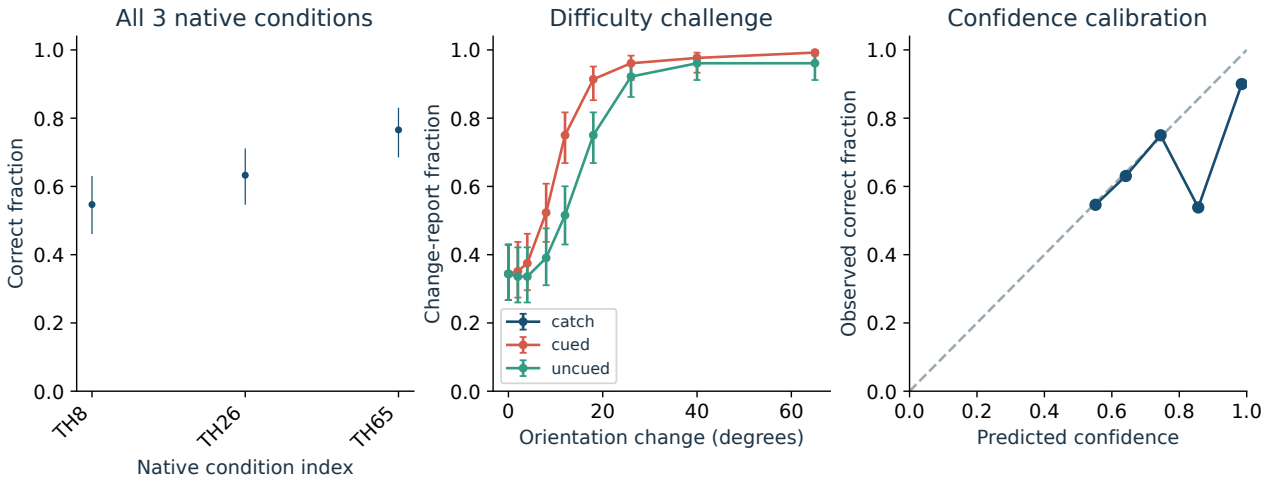


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	89.1%	1.0	78.1%	100.0%
0.7	35.9%	1.0	87.0%	100.0%
0.8	32.8%	2.0	90.5%	100.0%
0.9	28.1%	2.0	97.2%	85.2%

The representative condition TH65 provides 2 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

3.3 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

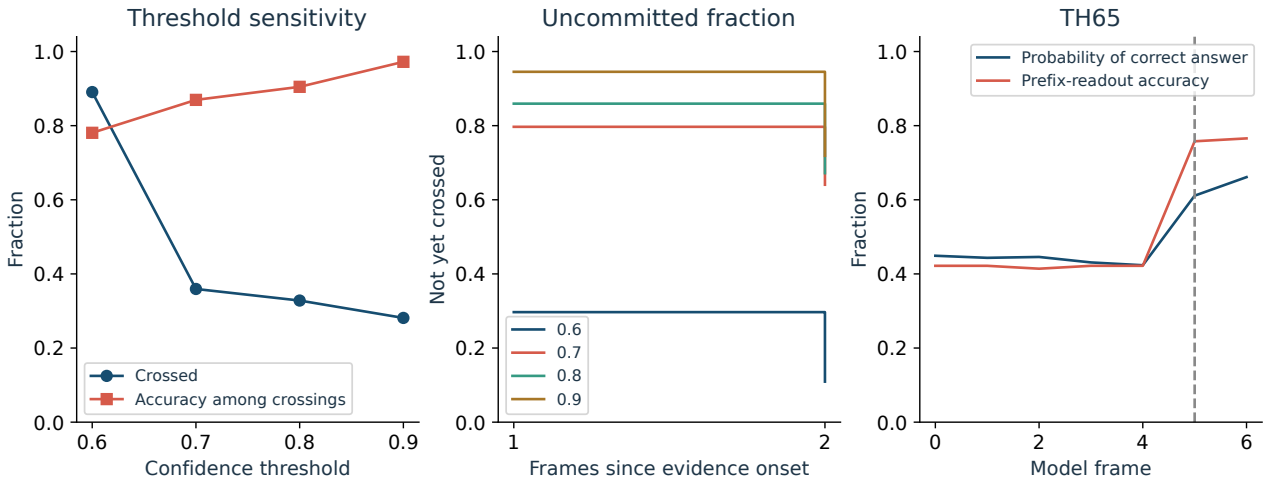


Figure 2: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

At threshold 0.8 across all native presentations, 75 of 384 trials crossed. Of those, 0 changed their answer later, 0 corrected an early error, and 0 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

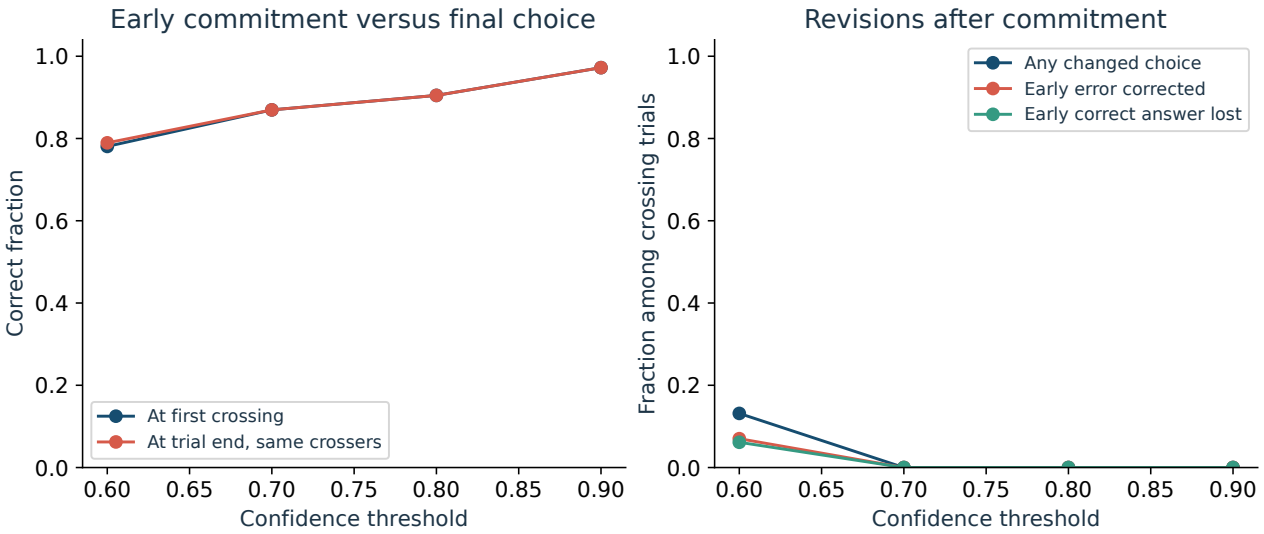


Figure 3: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

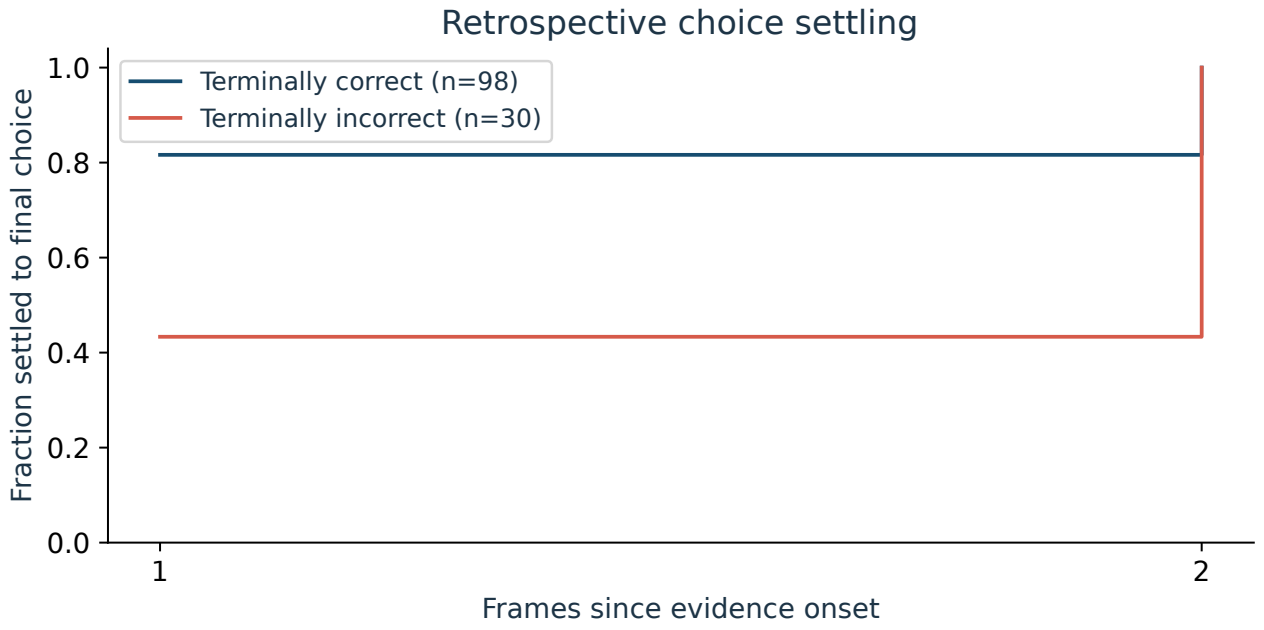


Figure 4: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.4 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The cue is fixed at row 2, column 3 of the four-by-four array. No-change trials are pooled into lower (25/50 percent) and higher (75/100 percent) displayed-validity groups because separate fixed-site validity cells have few observations. Difficulty and reward colors are pooled; the displayed sample counts quantify the available evidence.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Cue validity 25-50%; cue row 2, column 3	9	9
Cue validity 75-100%; cue row 2, column 3	10	10

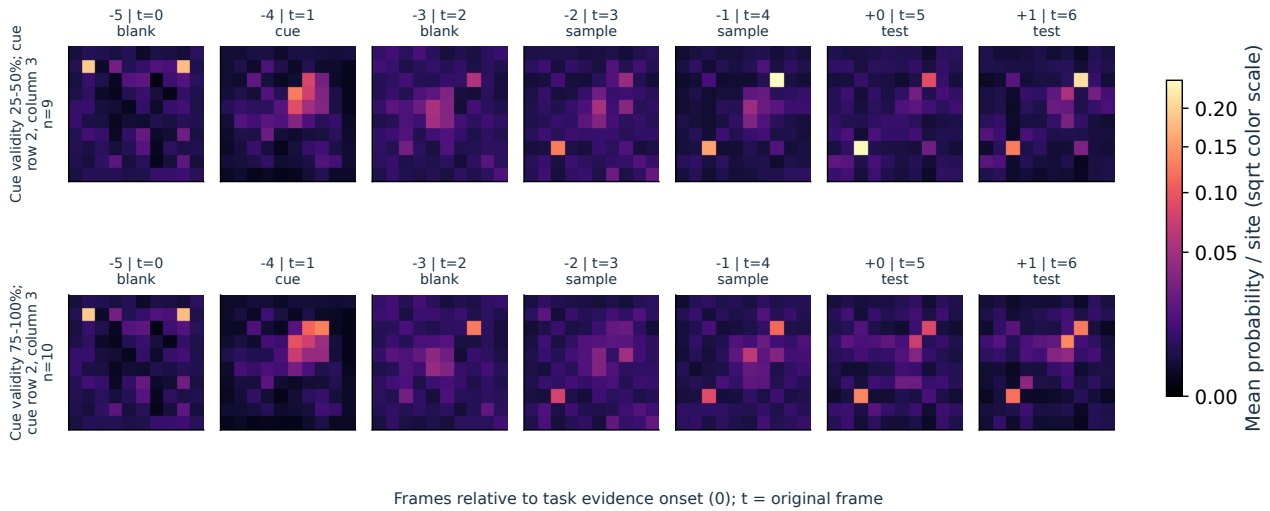


Figure 5: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

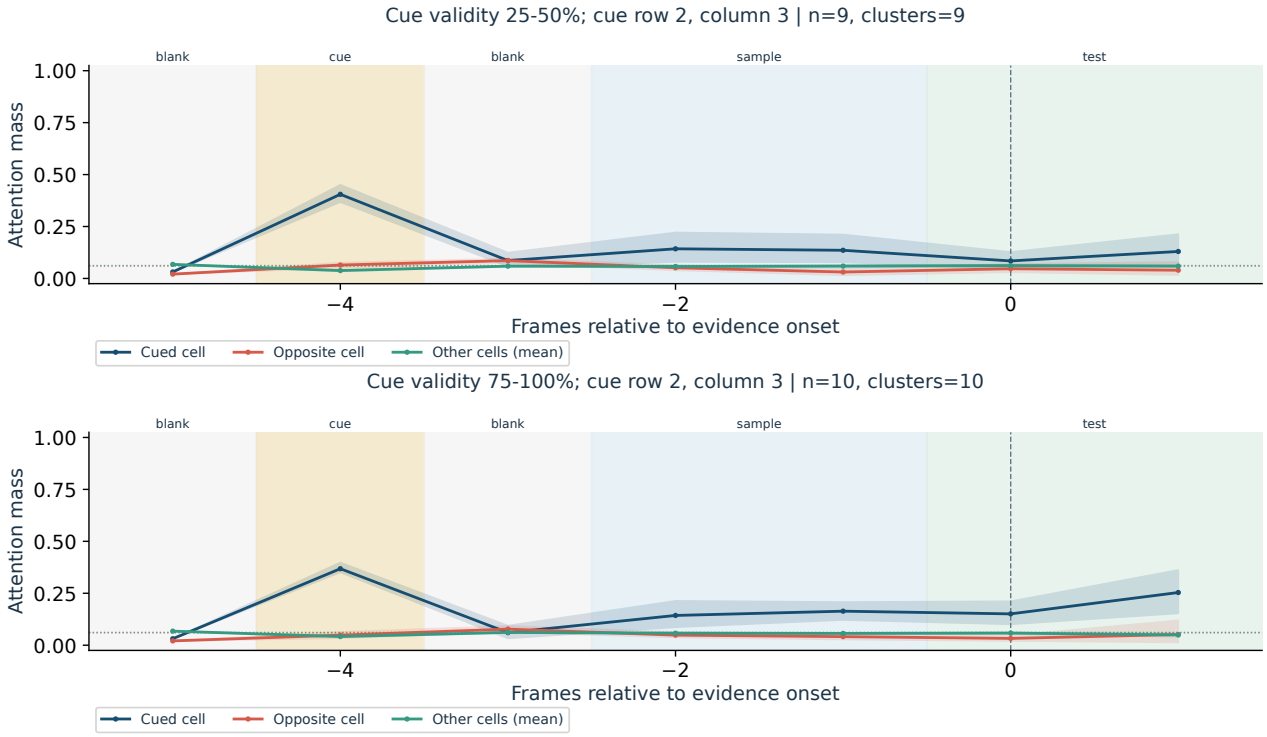


Figure 6: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

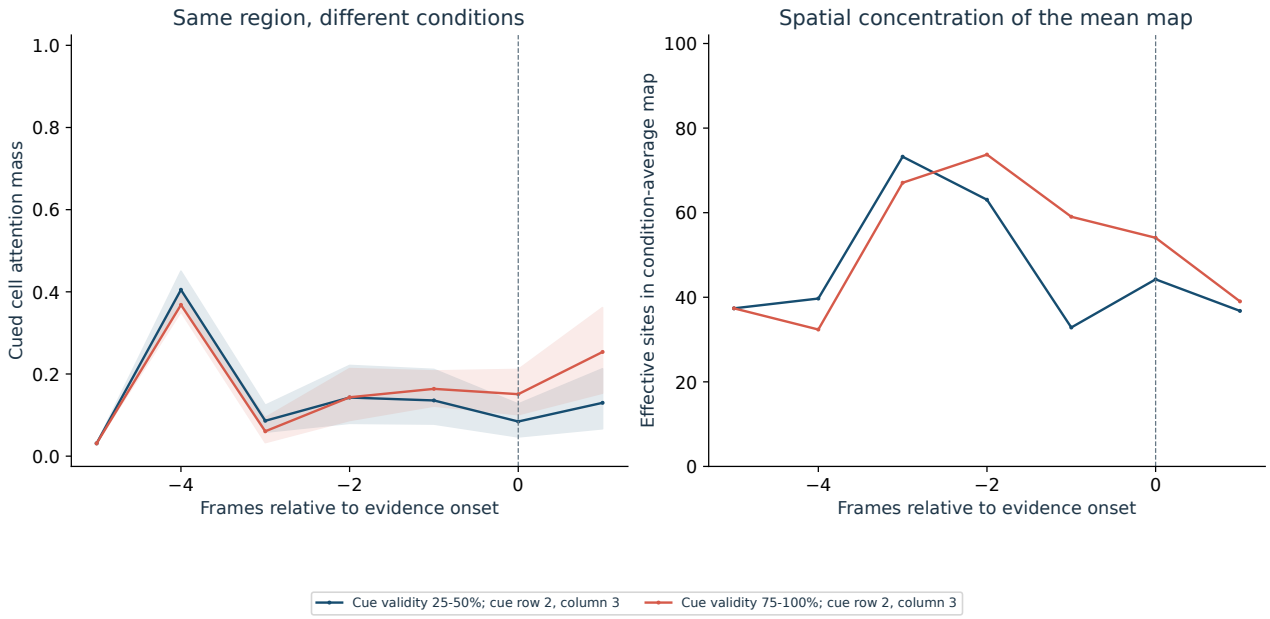


Figure 7: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Cue validity 25-50%; cue row 2, column 3: cued cell mass peaks at 0.405 at original frame 1 (cue), and is 0.130 on the final frame. Cue validity 75-100%; cue row 2, column 3: cued cell mass peaks at 0.368 at original frame 1 (cue), and is 0.254 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.5 Causal sensitivity to spatial and mnemonic interventions

For spatial_gain0_t4, paired accuracy changed by -25.0 percentage points (paired 95% interval [-34.4%, -15.6%]; paired p=9.43e-07; within-task Holm p=3.866e-05). There were 6 corrected trials and 38 newly incorrect trials. For distractor_+2_evidence, paired accuracy changed by +1.6 percentage points (paired 95% interval [-2.3%, 5.5%]; paired p=0.6875; within-task Holm p=1). There were 4 corrected trials and 2 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

3.6 Bottom-right cue and top-left counterfactual change

For catch events, bottom-right inhibition at dose -4 changed correct-report frequency by +13.7 percentage points, with paired interval [9.4%, 18.4%]. For cued events, bottom-right inhibition at dose -4 changed correct-report frequency by -27.3 percentage points, with paired interval [-32.8%, -21.9%]. For uncued events, bottom-right inhibition at dose -4 changed correct-report frequency by -2.7 percentage points, with paired interval [-5.9%, 0.4%]. The 100 percent cue and uncued-change combination is

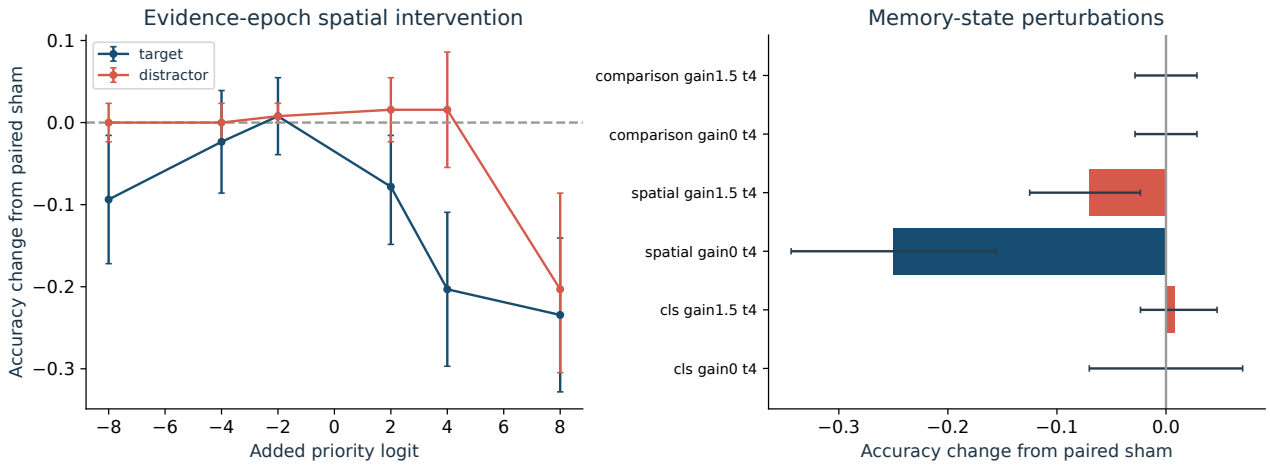


Figure 8: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

deliberately outside the native cue-validity rule. A change at top left still has a correct positive label because VDA requires a report for a change anywhere. Catch trials have no physical change and provide the false-alarm control.

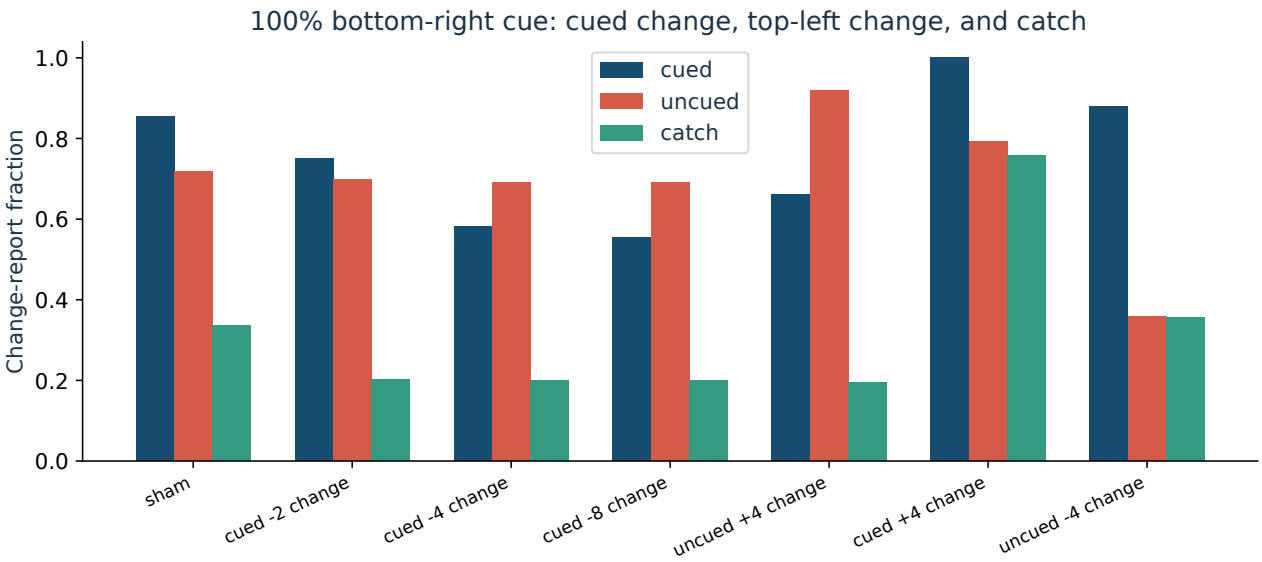


Figure 9: Requested counterfactual experiment. Every movie has a 100 percent bottom-right cue. Change reports are plotted separately for changes at the cue, changes at top left and catches. Priority perturbations occur only on the two change frames. Each event type contains 256 matched nuisance draws per intervention arm.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
TH65	target_-4	-2.3	1
TH65	target_+4	-20.3	0.003171
TH65	distractor_-4	+0.0	1
TH65	distractor_+4	+1.6	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.444	41.3%	0.965
fusion	0.463	47.3%	0.695
spatial	0.804	70.3%	0.005
cls	0.894	79.8%	0.005
comparison	Constant zero on this condition		
attention	0.598	61.7%	0.020

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

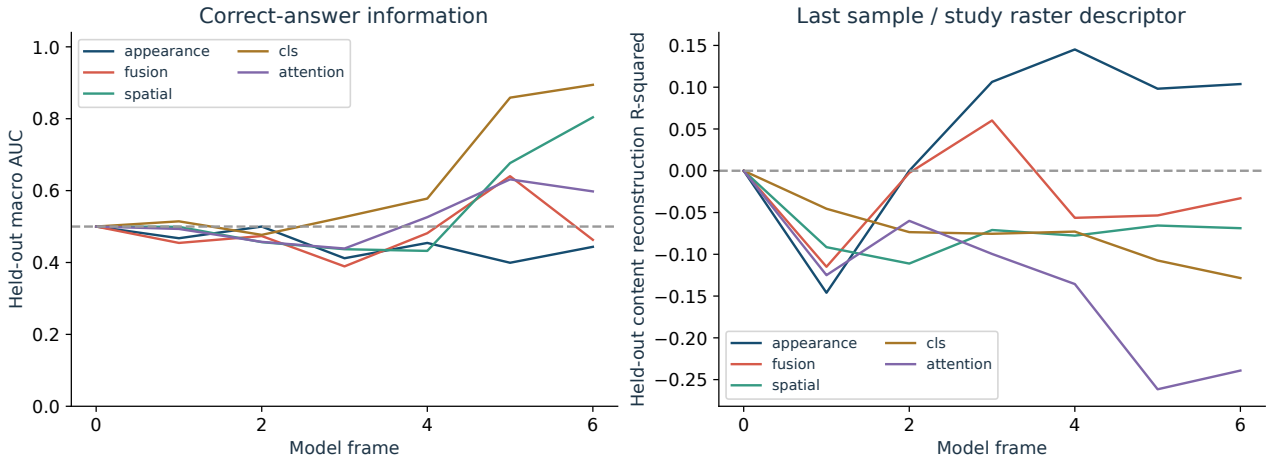


Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 0.00 to 12.34 within the retained subspace. The final step displacement was 1.297 standardized RMS units. In spatial memory, the participation ratio ranged from 0.00 to 21.60 within the retained subspace. The final step displacement was 0.453 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 4, coarse sample-content reconstruction had R-squared -0.073 from CLS memory and -0.078 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, +0.0 percentage points, spatial, -25.0 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

3.11 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.448	0.547	3200
fusion	0.523	0.502	6400
spatial	0.848	0.468	6400
cls	0.906	0.525	64
comparison	unavailable	unavailable	–
attention	0.687	0.478	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 38 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.12 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

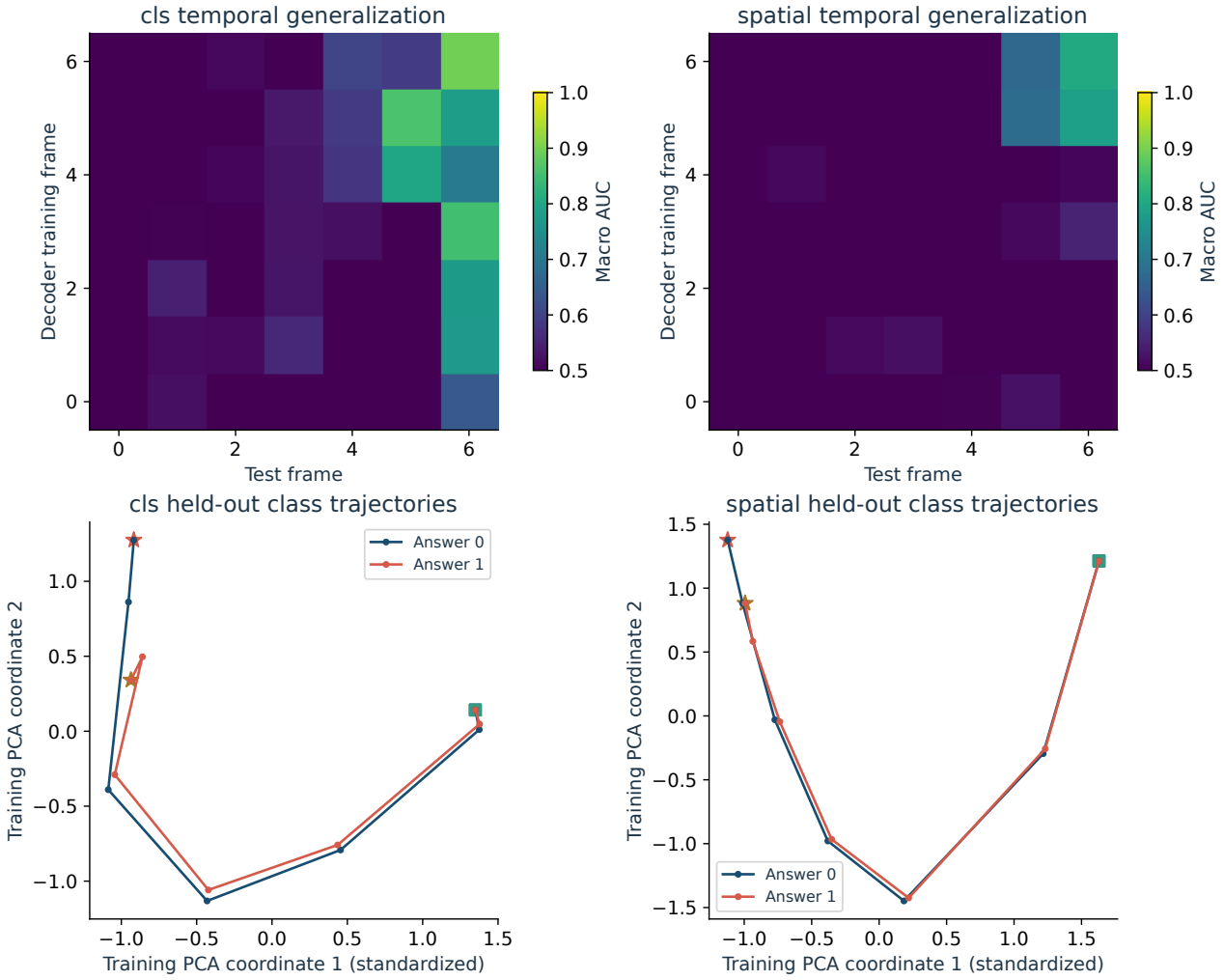


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

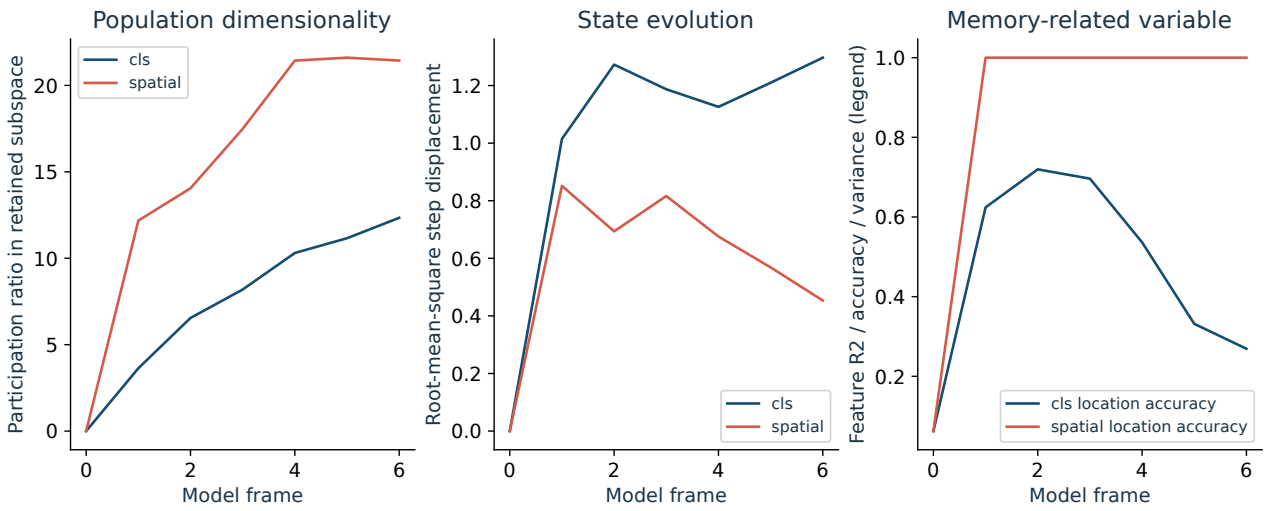


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

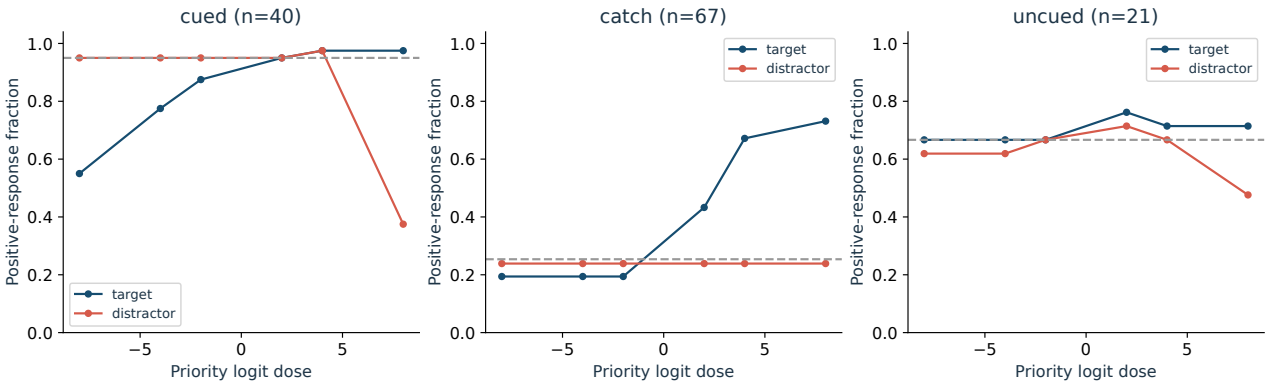


Figure 13: Event-specific causal response functions. Curves show positive reports after evidence-epoch target or alternative-site intervention; the horizontal line is paired sham. Event counts are stated above each panel. A positive report has the task-specific meaning defined in the text.

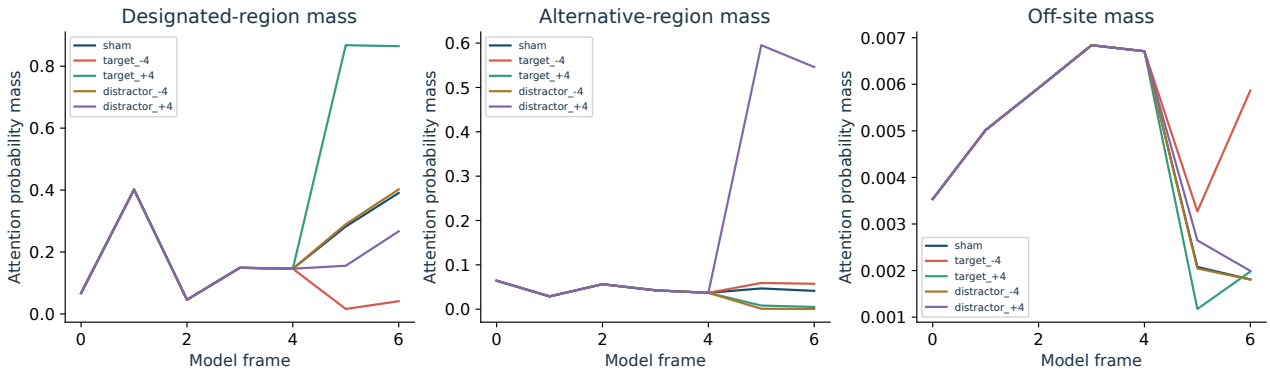


Figure 14: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

3.13 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

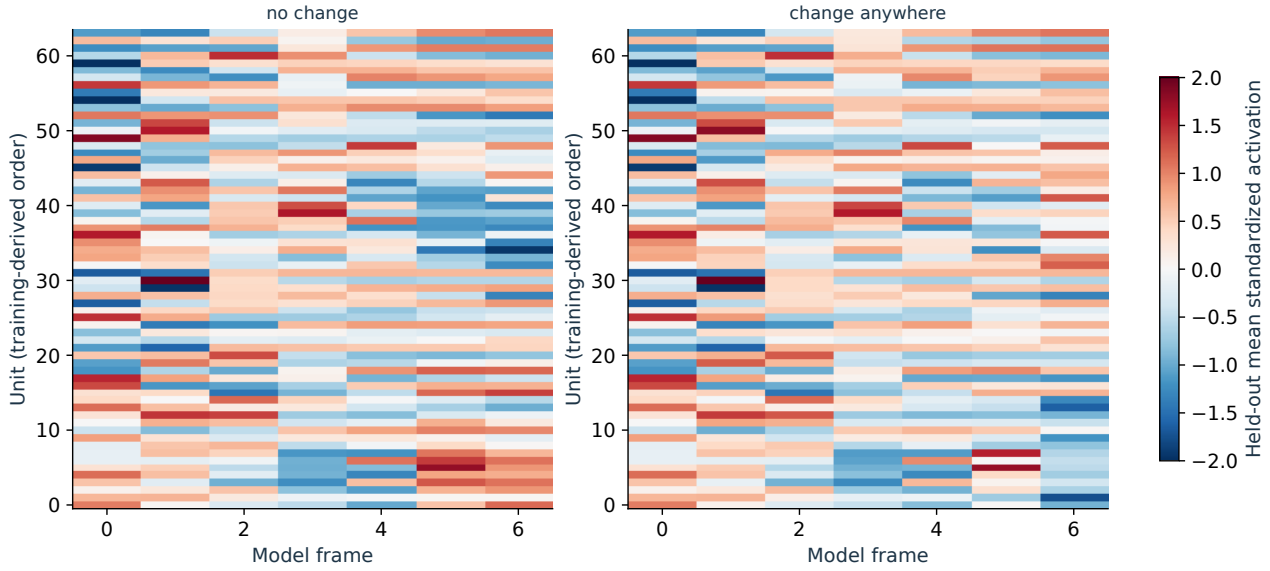


Figure 15: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.14 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional

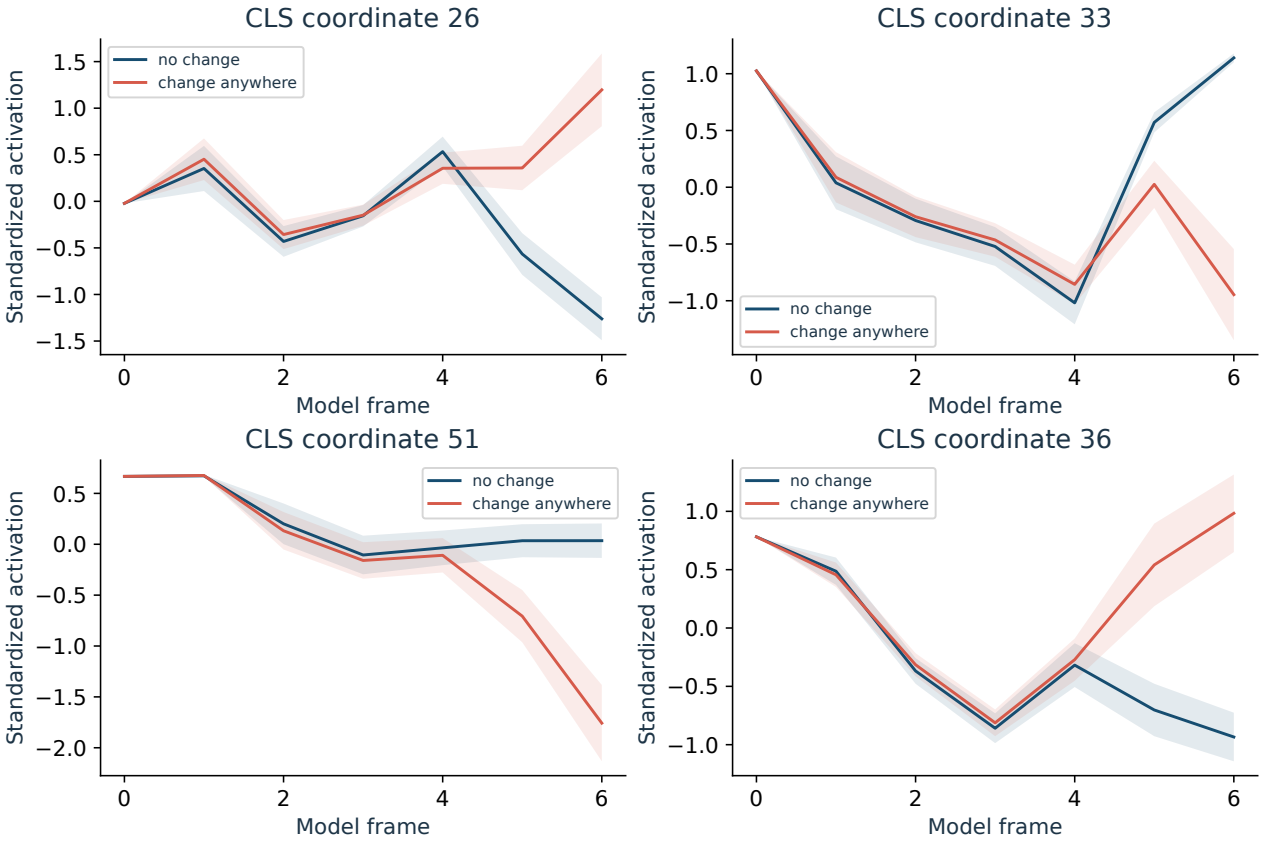


Figure 16: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	41.3%	51.7%	4.6%
fusion	47.3%	51.8%	8.3%
spatial	70.3%	79.2%	100.0%
cls	79.8%	91.2%	26.9%
attention	61.7%	60.3%	60.2%

3.15 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

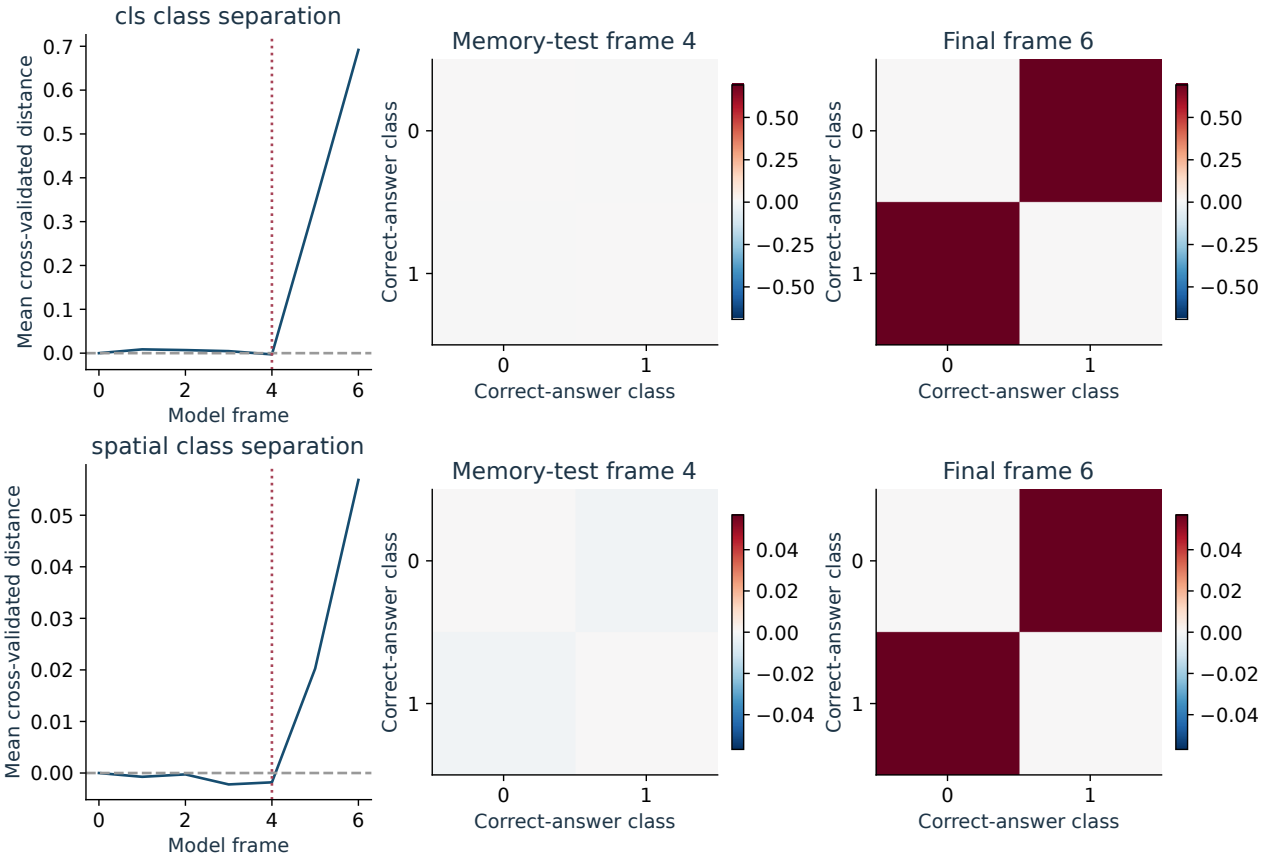


Figure 17: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

3.16 Weaker-signal follow-up to test the improvement prediction

After the 18-degree VDA4 counterfactual showed high uncued baseline detection, an exploratory follow-up tested 2-, 4- and 8-degree changes, plus catches, in all three VDA sizes. Each event-strength cell contains 128 new matched nuisance draws per arm. The bottom-right cue remains certain and perturbations occur during the change frames. Holm adjustment below covers the entire follow-up family within this task. These tests address whether the apparent absence of an uncued benefit at strong signals reflects limited headroom.

Event	Degrees	Sham accuracy	Cue-inhibition change	Holm p
catch	0	73.4%	+6.2 pp	1
cued	2	27.3%	-8.6 pp	0.1162
uncued	2	26.6%	-5.5 pp	1
cued	4	34.4%	-14.1 pp	0.001522
uncued	4	28.9%	-6.2 pp	1
cued	8	54.7%	-28.1 pp	1.222e-09
uncued	8	35.9%	-6.2 pp	1

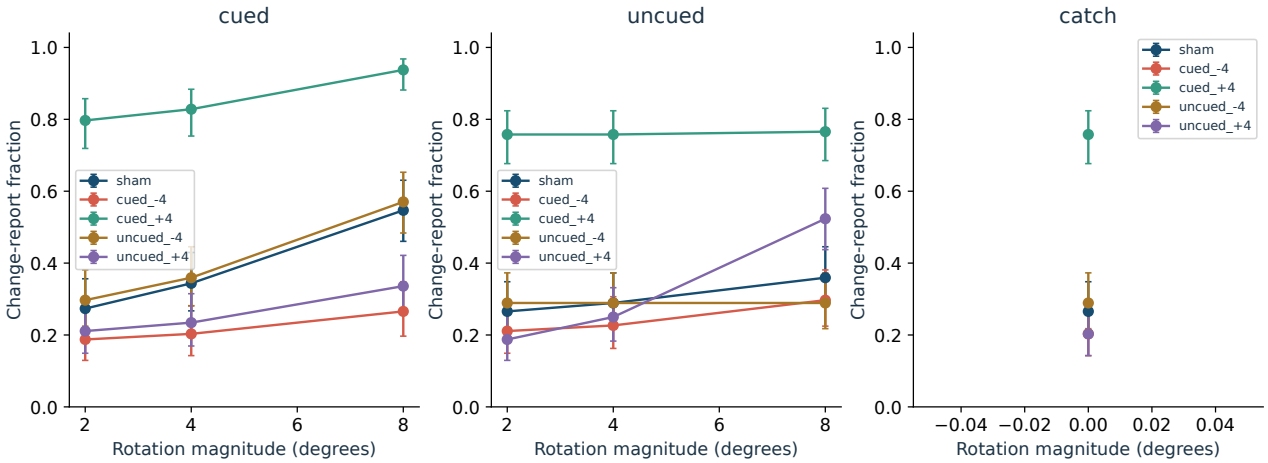


Figure 18: Exploratory weak-change counterfactual. Cued and uncued hit fractions are shown across three magnitudes; the catch panel shows false reports without a physical change. Error bars are Wilson 95 percent intervals. All interventions reuse identical movies within each cell.

3.17 Probability cues, reward value and prechange allocation

The native task varies displayed change-location probability and cue color, which specifies trial value. We stratify the original 384 trials by displayed probability, event location and value. Cued and uncued change reports are hits; reports on catch trials are false alarms. Prechange cue mass is measured at frame 4, before the two test frames, and is expressed relative to a uniform distribution over the 100 sites. Probability intervals use Wilson bounds; attention intervals resample native trials 3,000 times. These native stratifications characterize how the learned observer treats the task instructions, with the paired interventions supplying the causal test.

The model obtained 67.9% of the available native value, calculated as the sum of value on correctly answered trials divided by total offered value. At value 1, accuracy was 54.5% across 123 trials. At value 3, accuracy was 70.5% across 129 trials. At value 5, accuracy was 68.9% across 132 trials.

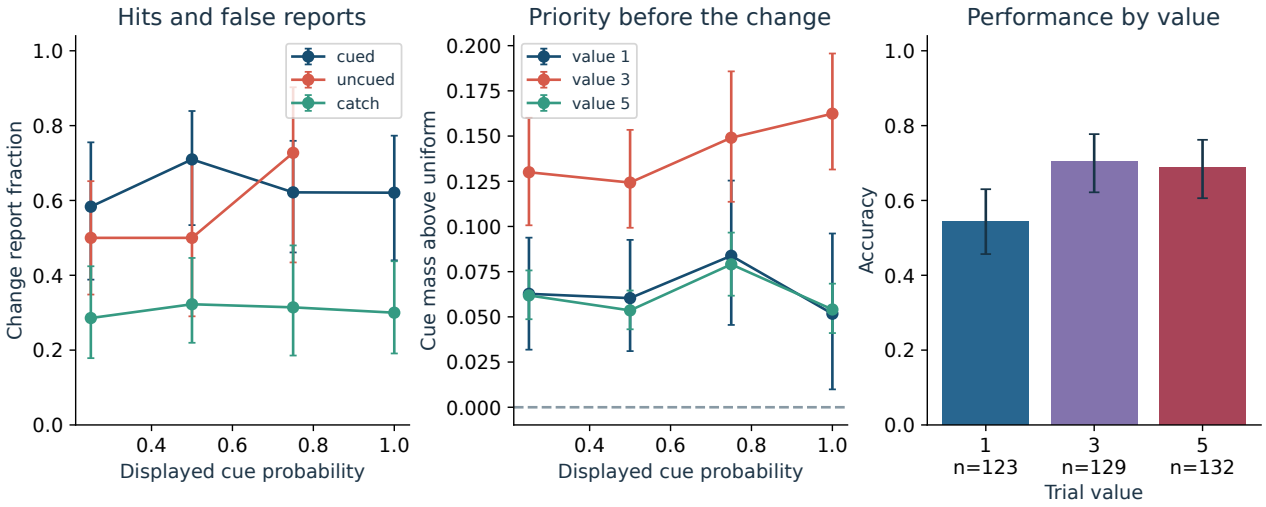


Figure 19: Native probability and value stratification. Left: cued hits, uncued hits and catch false reports. Middle: cue-region attention above uniform at the last prechange sample frame, separated by trial value. Right: accuracy by offered value. Sparse event-probability groups have wide sampling intervals; their sample counts are supplied in the source table.

4 Discussion

The sixteen-location condition combines greater competition with a different sampling of each Gabor on the common grid. Its weaker native discrimination and the absence of a clear uncued gain in the strong-signal cue-inhibition experiment make it especially useful for separating attention redistribution from successful change extraction. Cue inhibition also changes catch performance, so the report criterion must be measured alongside hits. In the weaker-signal follow-up, cued costs at four and eight degrees survive the cross-VDA correction, while the uncued point estimates remain negative and do not show a reliable improvement. The requested dual effect is therefore absent at these tested strengths for this checkpoint. The recurrent populations contain more answer information than the early capacity-matched visual probes, and spatial-memory reset is harmful. These observations identify recurrent processing as useful even when the final detector remains imperfect.

The behavioral result establishes 64.8% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies spatial_gain0_ t4 as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Rendering the same physical Gabor size while changing set size would isolate competition from resolution. Coupling that manipulation to longer sample and comparison periods would test whether more recurrent processing compensates for crowding.

4.1 Interpreting these results alongside biological experiments

4.1.1 Cohen and Maunsell (2009)

The biological result concerns trial-to-trial variability after controlling stimulus condition. Our frozen model is deterministic for an identical movie, so variability across randomly generated movies primarily reflects changing input. Correlations across those movies are signal-dependent correlations. To test the proposed neural mechanism, repeated identical stimuli would need an explicit internal-noise model and a separately defined residual covariance analysis. [7]

4.1.2 Reynolds, Chelazzi and Desimone (1999)

Dense displays create a natural competition challenge. The present model's priority intervention changes access to spatial representations, and its spatial reset harms detection. A target-alone versus target-with-competitors comparison would reveal whether the same target evidence degrades because of other items, independently of the smaller per-item raster in the native sixteen-cell condition. [8]

4.1.3 Zenon and Krauzlis (2012)

A detector can fail even when a visual representation remains enhanced. Here, redistribution under cue inhibition is established, but weak four- and eight-degree experiments show reliable cued costs without an uncued gain. This is a useful model example of the gap between changing a selection signal and obtaining a behavioral improvement. The specific neural mechanism must be judged from representational and decision measurements together. [9]

4.2 A computational account of the observed behavior

Three bottlenecks can produce the same accuracy decline. A sensory bottleneck loses the small orientation change during rasterization or feature extraction. A memory bottleneck loses the relevant prechange orientation. A selection or readout bottleneck retains usable information but fails to compare the right location. The experiments already show recurrent information and a spatial-memory dependency, yet that does not exclude a sensory limitation at the smallest angles. The most direct next analysis would reconstruct prechange and postchange orientations at each cell and compare an external location-specific decoder with the agent's binary response. Geometry and task load should be independently manipulated before attributing the VDA4-to-VDA16 difference to a single capacity parameter.

4.3 Where this task agrees with the biological evidence

The model agrees with a causal role for recurrent spatial information and selective routing in dense change detection. It diverges from the hoped-for dual behavioral effect at the tested strengths. The data currently favor a multi-stage explanation of difficulty over a claim that attention redistribution alone is sufficient.

4.4 A follow-up experiment motivated by these results

Cross item count with fixed item size and fixed retinal positions in a larger canvas. Decode local orientation differences, then repeat paired inhibition with matched catches. Introduce internal variability only as a separately documented model variant if testing residual-noise-correlation theories.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Morgan J, Albanna B, Herman JP (2025). A recurrent vision transformer shows signatures of primate visual attention. arXiv:2502.10955. https://arxiv.org/abs/2502.10955.

[2] Bollimunta A, Bogadhi AR, Krauzlis RJ (2018). Comparing frontal eye field and superior colliculus contributions to covert spatial attention. Nature Communications 9:3553. https://www.nature.com/articles/s41467-018-06042-2.

[3] Reynolds JH, Heeger DJ (2009). The normalization model of attention. Neuron 61:168-185. https://pubmed.ncbi.nlm.nih.gov/19186161/.

[4] Wolfe JM (2021). Guided Search 6.0: An updated model of visual search. Psychonomic Bulletin and Review 28:1060-1092. https://pubmed.ncbi.nlm.nih.gov/33547630/.

[5] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[6] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[7] Cohen and Maunsell (2009). Attention improves performance primarily by reducing interneuronal correlations. Primary source.

[8] Reynolds, Chelazzi and Desimone (1999). Competitive mechanisms subserve attention in macaque areas V2 and V4. Primary source.

[9] Zenon and Krauzlis (2012). Attention deficits without cortical neuronal deficits. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
TH8	128	54.7%	0.543	0.535
TH26	128	63.3%	0.735	0.421
TH65	128	76.6%	0.891	0.346

B Every-frame attention atlas

Condition TH65. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 5 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

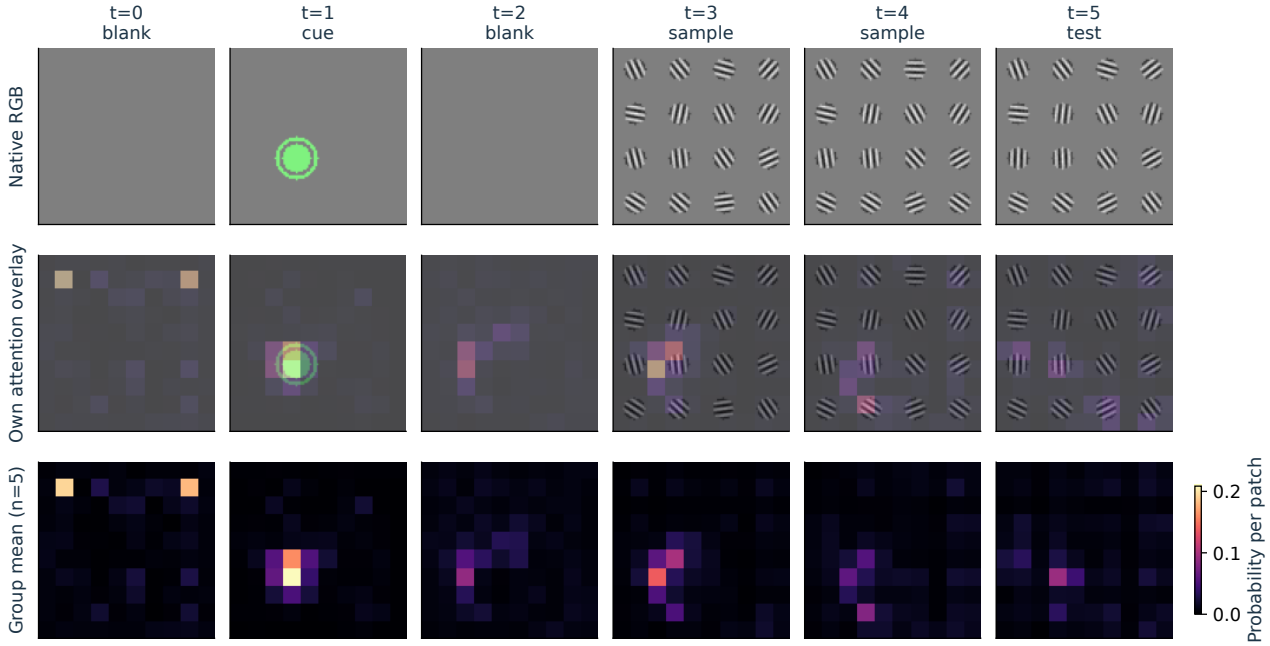


Figure 20: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

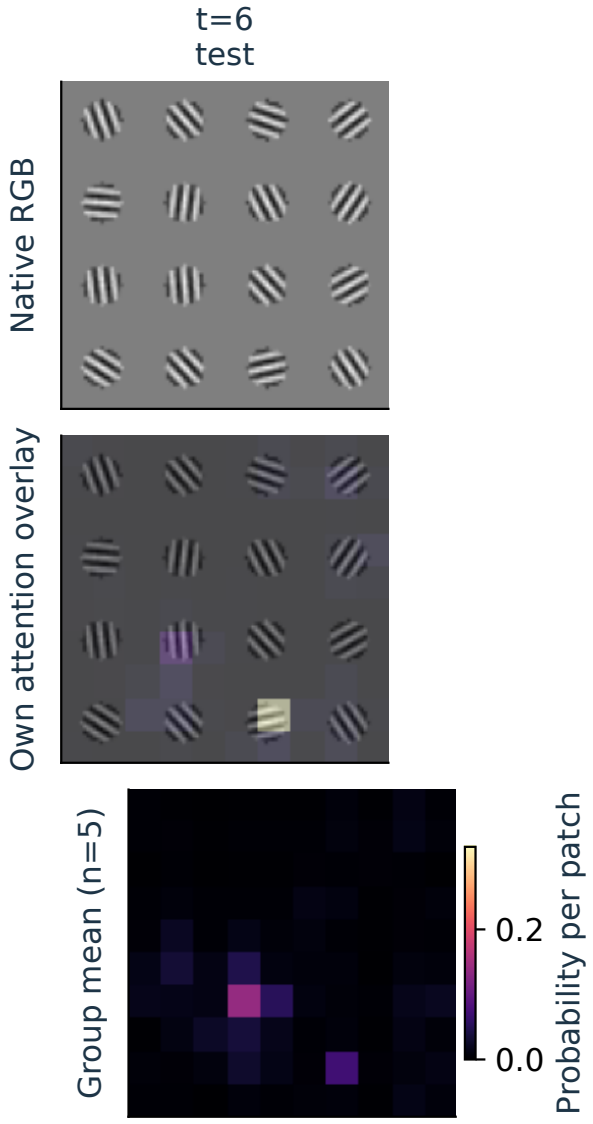


Figure 21: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

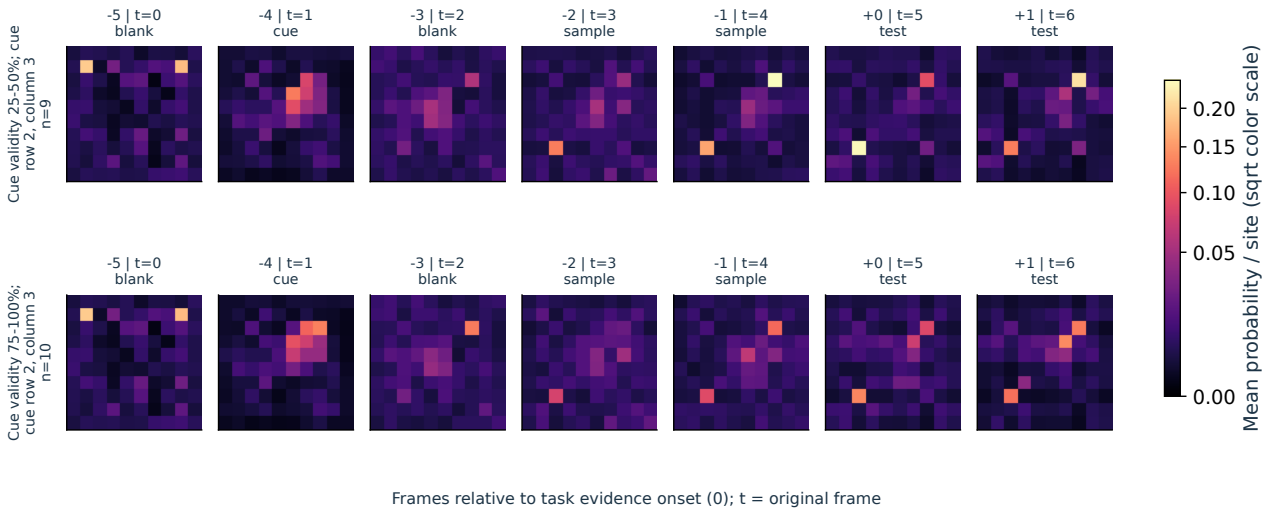


Figure 22: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Object memory, feature conjunctions and the selection of a matching location

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized memory guided search in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 288 native conditions and 36,864 fresh trials, equal-condition accuracy was 53.1% (95% interval [50.9%, 55.4%]). For the representative condition features_distinct_S8_D12_H8, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was distractor_+8_evidence, changing accuracy by -93.0 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The observer first sees an object, retains its appearance, and then reports its location among four candidates. The arrays contain feature combinations, irregular objects or procedurally generated textures. Distinct and similar candidates vary the specificity required of the remembered template.

The full factorial crosses stimulus family, similarity, sample duration, blank delay and array exposure. This is essential for localizing the failure: poor brief-sample performance suggests encoding limits; a selective delay cost suggests retention limits; improvement with longer array exposure suggests retrieval or comparison limits. Family-specific neural analyses retain these distinctions.

The biological motivation is developed through [1], [2], [3]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Chelazzi and colleagues connected remembered target information with visual selection in inferior temporal cortex; feature-integration and Guided Search work motivate considering both templates and competition among candidates. In this model, a remembered object must influence a spatial choice only after an array appears. Encoding an object and using that encoding as a search template are therefore separate achievements. The factorial timing design, content reconstruction, target-location decoding and causal interventions are organized around this separation. A preserved memory with poor target selection would motivate changing how memory participates in the query that selects visual evidence.

1.2 Biological experiments motivating this task

Experimental question. When remembered content is available, why does it sometimes fail to guide the choice of a matching object?

Chelazzi, Miller, Duncan and Desimone (1993) Monkeys remembered a picture and selected its match from a small array. IT showed sample-related delay activity, followed during search by suppression of nontarget responses and dominance of the target representation. [6]

Soto, Heinke, Humphreys and Blanco (2005) A remembered colored shape facilitated line search when it surrounded the target and impaired search when it surrounded a distractor. Effects extended to early responses and the first saccade. [7]

Bichot, Rossi and Desimone (2005) During free visual search, V4 responses favored objects sharing target features across the display. The experiment related feature-based parallel bias to selection during search among complex items. [8]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Where is the object shown in the sample?

Response labels are 0: top-left; 1: top-right; 2: bottom-left; 3: bottom-right. The native suite contains 288 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use features_distinct_S8_D12_H8, features_similar_S8_D12_H8, objects_distinct_S8_D12_H8, objects_similar_S8_D12_H8, fractals_distinct_S8_D12_H8, fractals_similar_S8_D12_H8.

1.4 Stimulus sequence and spatial layout

For the representative native condition features_distinct_S8_D12_H8, the sequence has 29 frames, indexed from zero: frame 0: fixation; frames 1–8: sample; frames 9–20: delay; frames 21–28: search. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Stimulus family	features
Similarity condition	distinct

Stimulus centers in the 100 by 100 input are [[25, 25], [75, 25], [25, 75], [75, 75]]. Coordinate x increases rightward and y downward.

Each candidate is a 28 by 28 raster. The sample appears centrally and the array contains the exact same sample raster translated to one of four locations. Feature stimuli combine color, shape and texture;

irregular objects use radial Fourier contours; fractal stimuli use spectral fractional-noise textures with matched within-support intensity histograms. Similar feature arrays contain repeated individual feature values, requiring conjunction information to select a unique object.

A target-location decoder applied before the array tests a variable the agent cannot yet determine from the visible sample alone. Sample-content reconstruction is therefore the more direct pre-array memory measure. During search, cue-content information must be translated into a spatial selection. Comparing spatial memory, CLS decoding and target attention can expose a separation between storing a template and using it to locate a match.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and

comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression $k^{\mathsf{T}}q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^\delta A/(1 - A + e^\delta A)$. Thus its odds against the remaining image are multiplied by e^δ; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance

probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ±4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated

class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 53.1%, with macro AUC 0.724. The matched final checkpoint changed accuracy by +4.05 percentage points on average across conditions, using 32 same-movie comparisons per condition. The lowest native cell was objects_distinct_S1_D0_H8 at 23.4%; the highest was features_distinct_S1_D0_H4 at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

Marginal accuracy by family was features: 93.6%; objects: 28.8%; fractals: 36.9%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

Marginal accuracy by similarity was distinct: 57.9%; similar: 48.3%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

Marginal accuracy by delay was 0: 53.1%; 4: 53.3%; 12: 52.9%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

Marginal accuracy by sample frames was 1: 52.7%; 2: 53.3%; 4: 53.2%; 8: 53.2%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

Marginal accuracy by exposure was 1: 52.3%; 2: 53.2%; 4: 54.0%; 8: 53.0%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

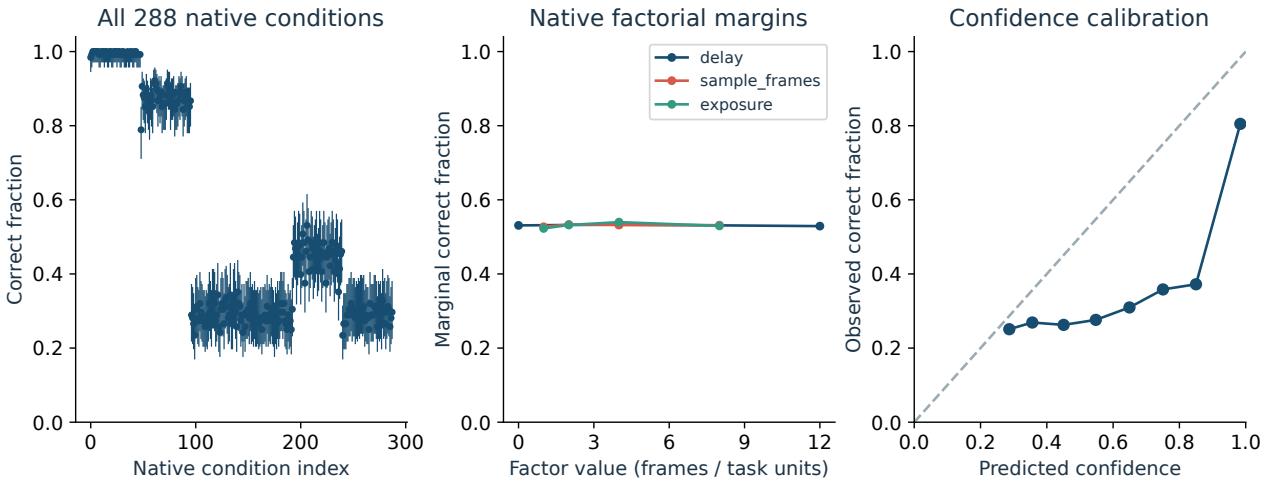


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Matched-scene effects of sample, delay and search time

The native counter-based generator reuses scene identity and target assignment across timing conditions. We compare the longest and shortest values of each timing factor while matching the other factors, then average those paired differences within scene. Static arrays remain pixel-identical; motion arrays retain the same direction and layout seeds and shared frame prefix as duration increases. Bootstrap intervals resample the 128 scene identities. A two-sided sign-flip test uses 9,999 random sign patterns on the scene-level differences, followed by Holm correction over this task's timing comparisons. These exploratory contrasts distinguish changes in encoding opportunity, retention duration and search opportunity.

Group	Factor	Short/long	Change (pp)	Holm p
features / distinct	sample	1/8	-0.2	1
features / distinct	delay	0/12	-0.2	1
features / distinct	exposure	1/8	+0.6	1
features / similar	sample	1/8	+0.8	1
features / similar	delay	0/12	-3.5	0.2556
features / similar	exposure	1/8	+2.2	1
fractals / distinct	sample	1/8	-0.6	1
fractals / distinct	delay	0/12	-0.7	1
fractals / distinct	exposure	1/8	+8.0	0.6307
fractals / similar	sample	1/8	+0.6	1
fractals / similar	delay	0/12	+0.8	1
fractals / similar	exposure	1/8	+0.5	1
objects / distinct	sample	1/8	+1.9	1
objects / distinct	delay	0/12	+3.2	1
objects / distinct	exposure	1/8	-4.3	1
objects / similar	sample	1/8	+0.0	1
objects / similar	delay	0/12	-0.7	1
objects / similar	exposure	1/8	-2.9	1

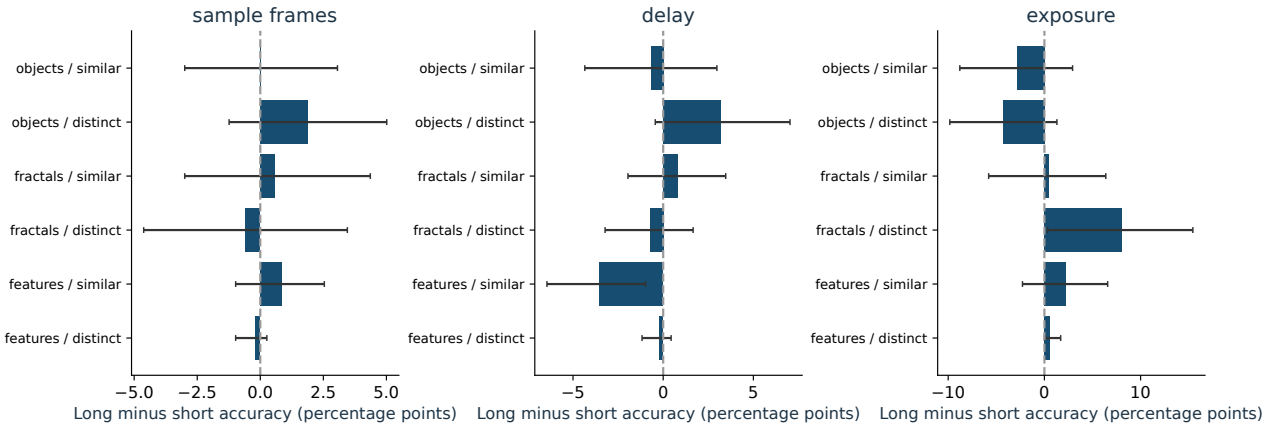


Figure 2: Paired timing effects with scene identity preserved. Each value is the longest-duration accuracy minus the shortest-duration accuracy, averaged over matched remaining factors. Intervals resample complete scenes.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	99.2%	100.0%
0.7	100.0%	1.0	99.2%	96.9%
0.8	100.0%	1.0	99.2%	87.5%
0.9	100.0%	1.0	99.2%	45.3%

The representative condition features_distinct_S8_D12_H8 provides 8 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

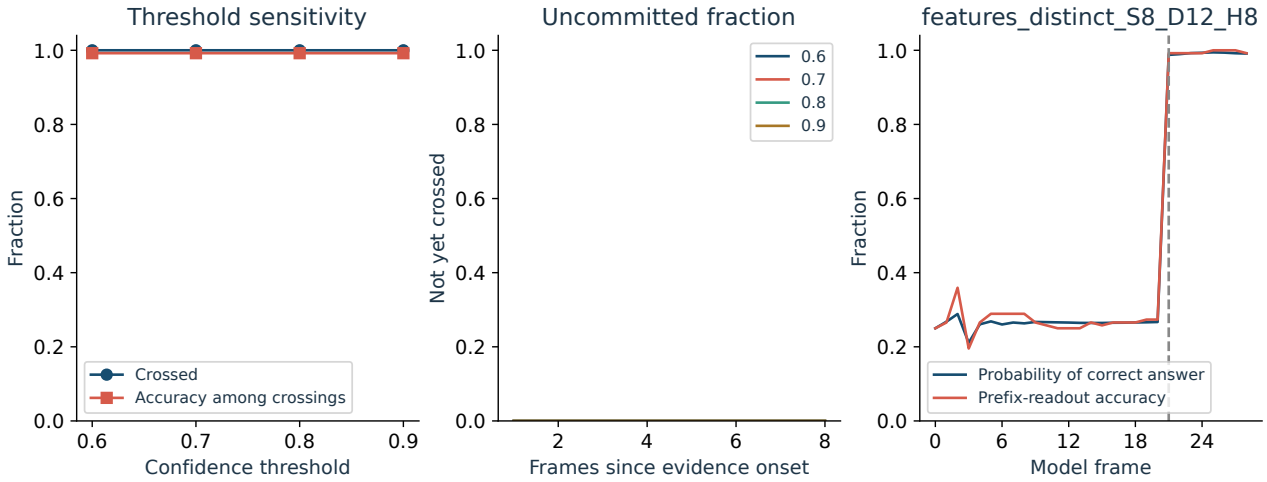


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 22,921 of 36,864 trials crossed. Of those, 1,396 changed their answer later, 549 corrected an early error, and 345 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

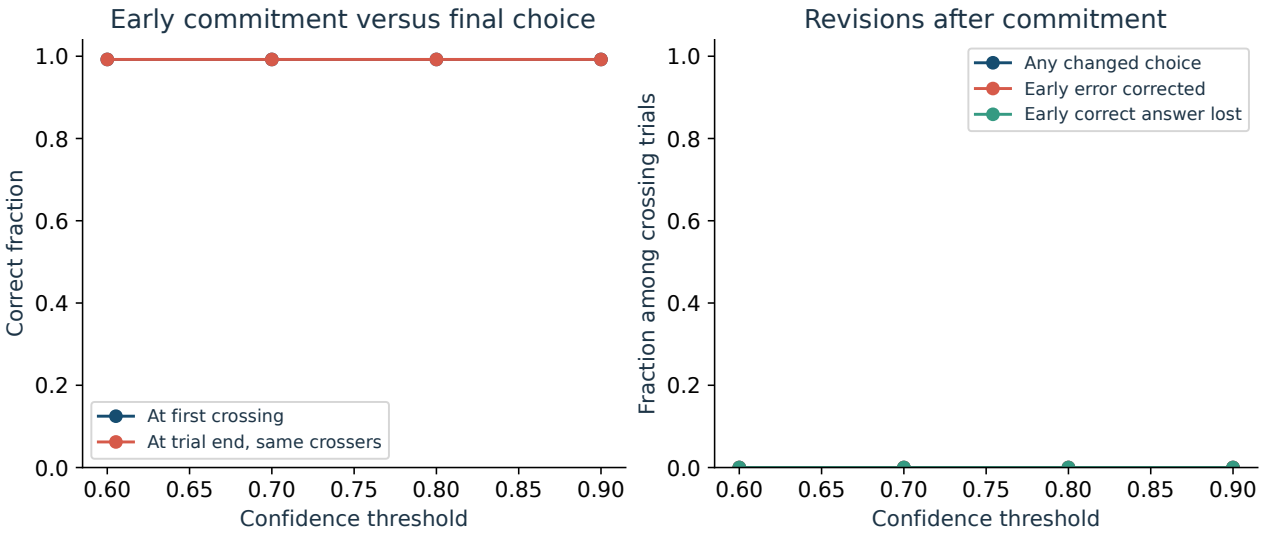


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

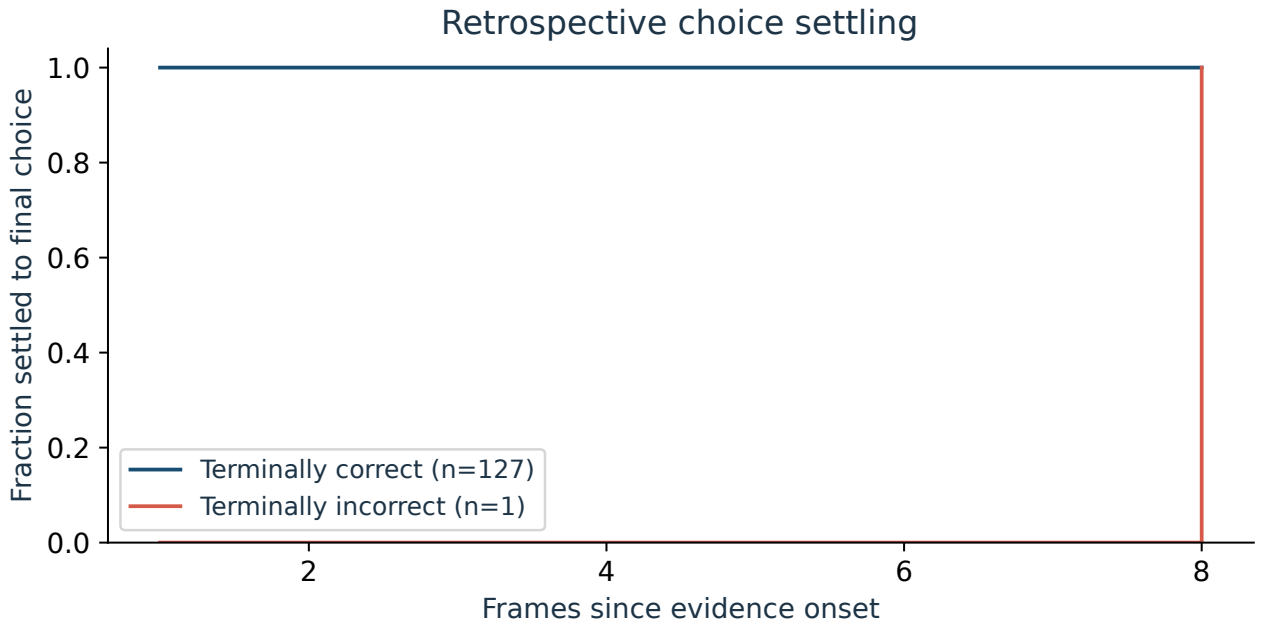


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The correct array location is fixed at top left. Rows cross feature, irregular-object and fractal families with distinct versus similar candidates. Sample duration, blank delay and array exposure are fixed at 8, 12 and 8 frames. This holds timing and spatial geometry constant while changing the demands on remembered-content matching.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Features Distinct	32	32
Features Similar	32	32
Objects Distinct	32	32
Objects Similar	32	32
Fractals Distinct	32	32
Fractals Similar	32	32

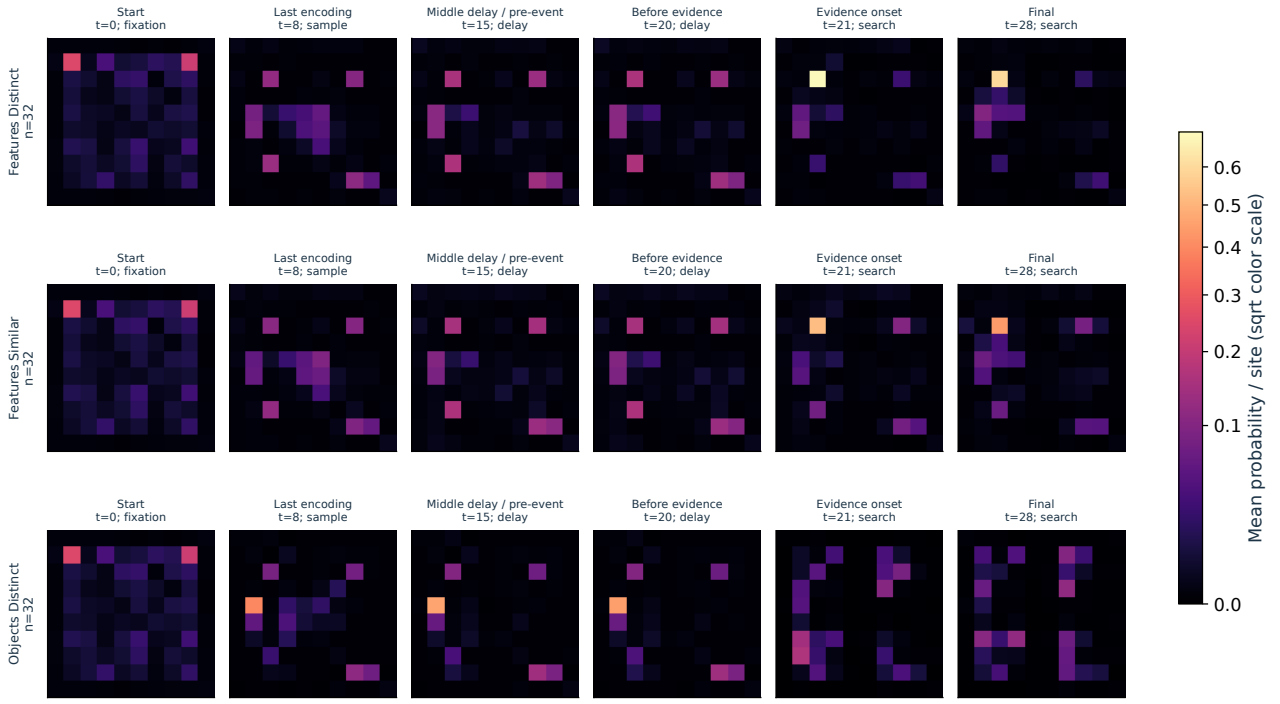


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

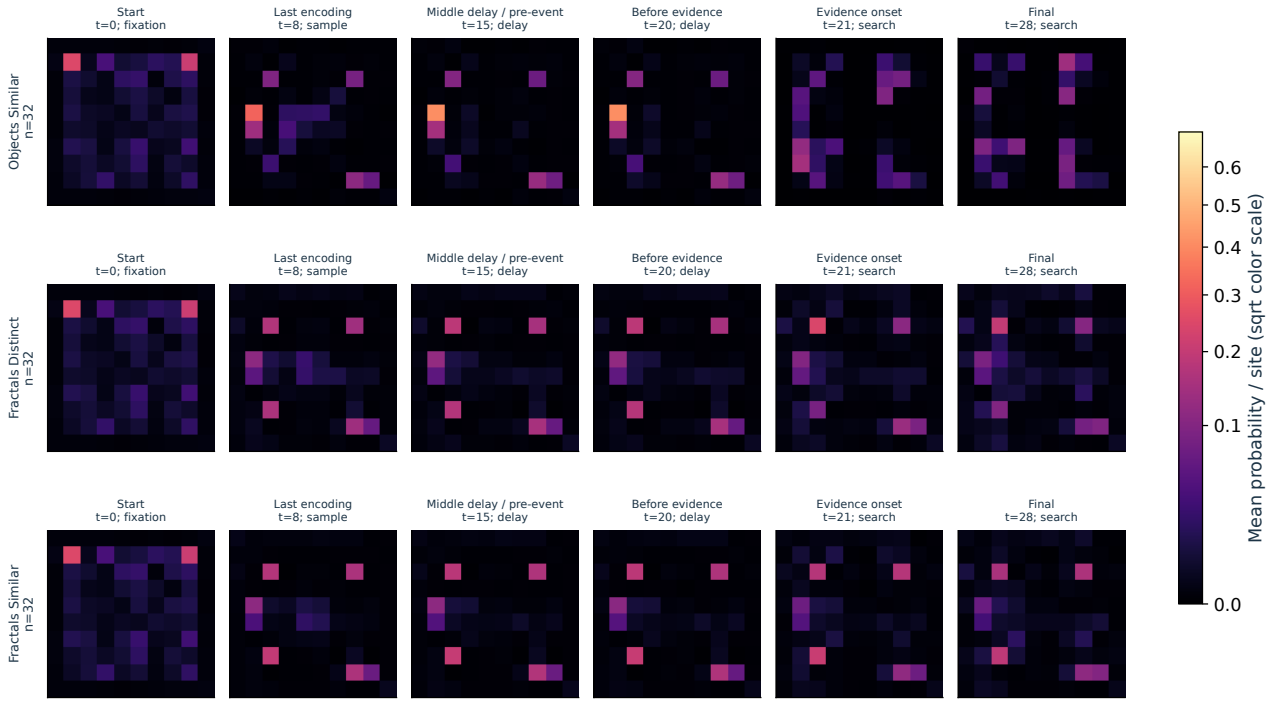


Figure 7: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

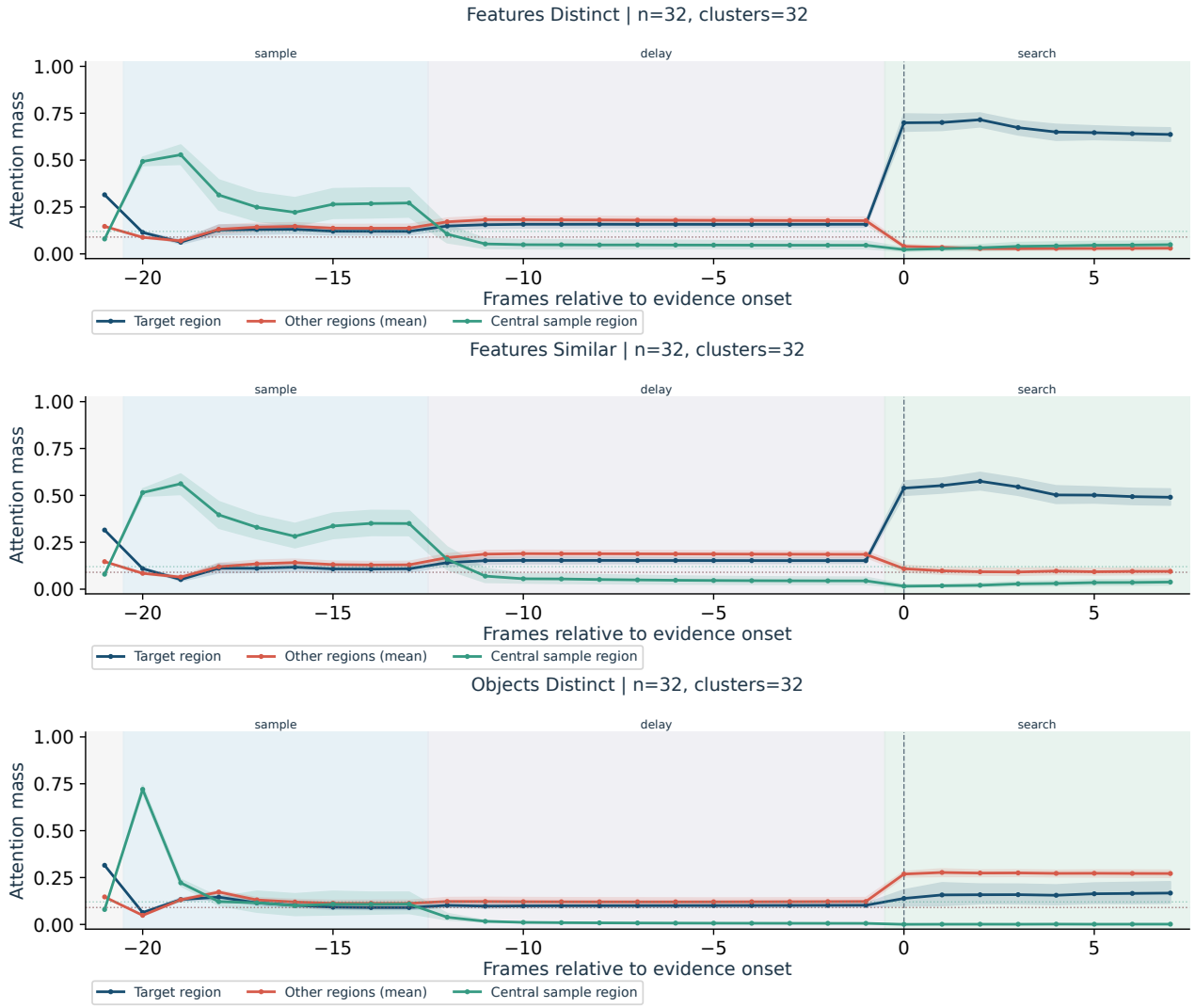


Figure 8: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

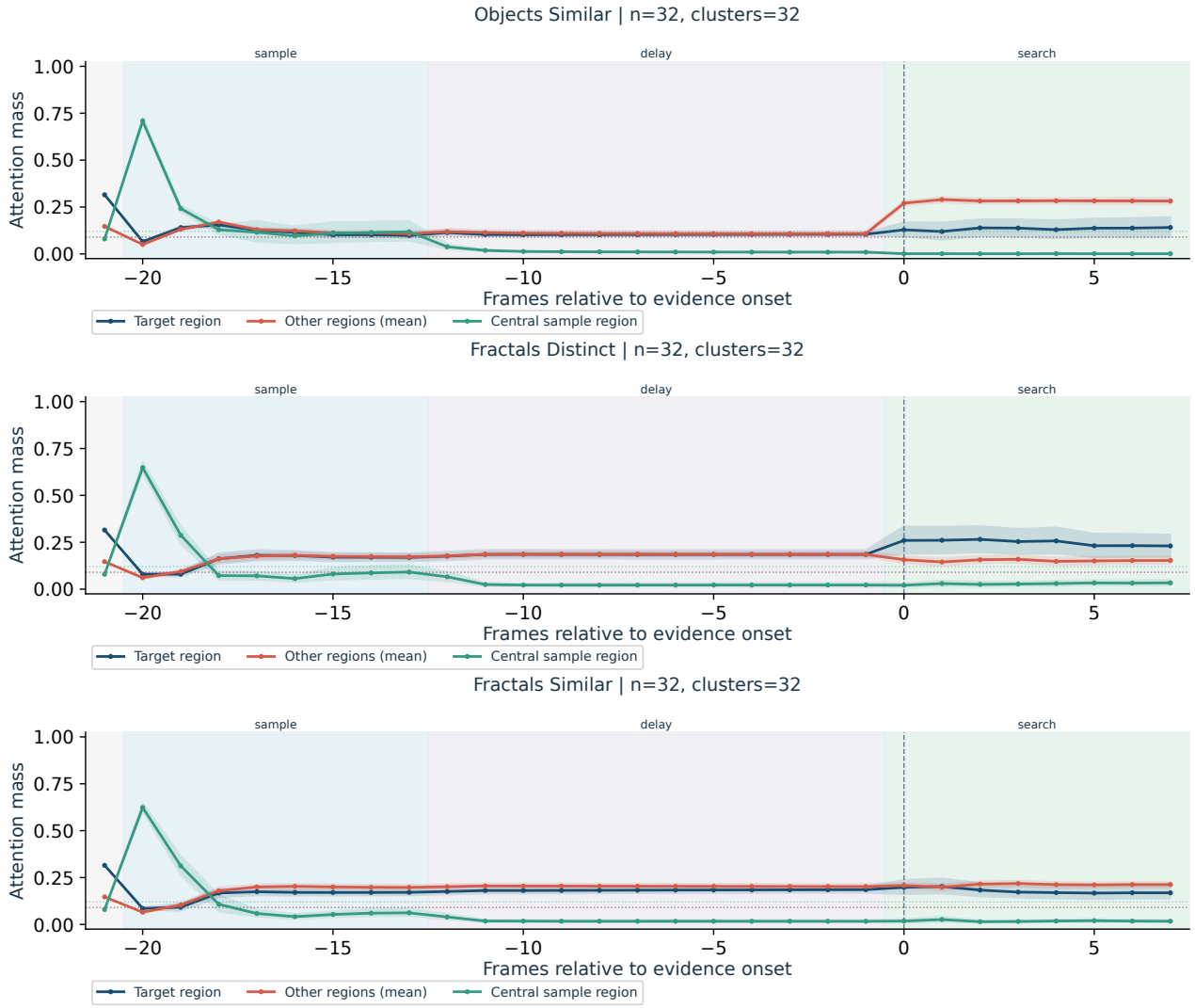


Figure 9: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

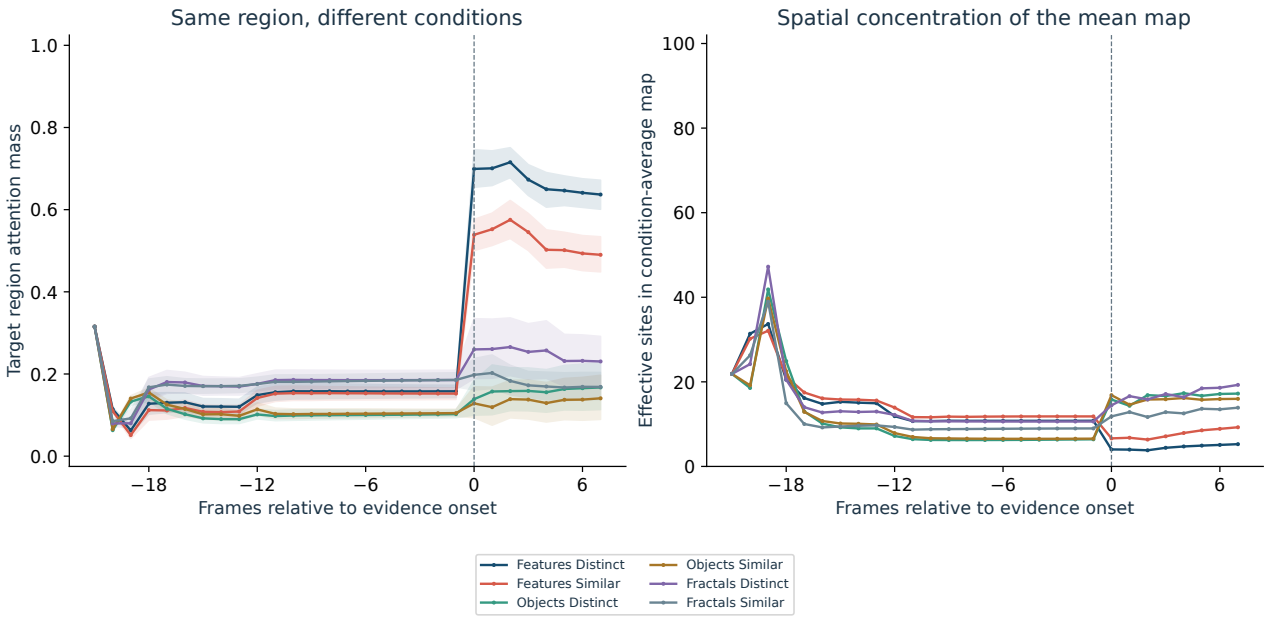


Figure 10: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Features Distinct: target region mass peaks at 0.715 at original frame 23 (search), and is 0.637 on the final frame. Features Similar: target region mass peaks at 0.575 at original frame 23 (search), and is 0.490 on the final frame. Objects Distinct: target region mass peaks at 0.315 at original frame 0 (fixation), and is 0.167 on the final frame. Objects Similar: target region mass peaks at 0.315 at original frame 0 (fixation), and is 0.141 on the final frame. Fractals Distinct: target region mass peaks at 0.315 at original frame 0 (fixation), and is 0.231 on the final frame. Fractals Similar: target region mass peaks at 0.315 at original frame 0 (fixation), and is 0.169 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For distractor_+8_evidence, paired accuracy changed by -93.0 percentage points (paired 95% interval [-96.9%, -88.3%]; paired p=3.009e-36; within-task Holm p=5.597e-34). There were 0 corrected trials and 119 newly incorrect trials. For target_+2_evidence, paired accuracy changed by +0.8 percentage points (paired 95% interval [0.0%, 2.3%]; paired p=1; within-task Holm p=1). There were 1 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

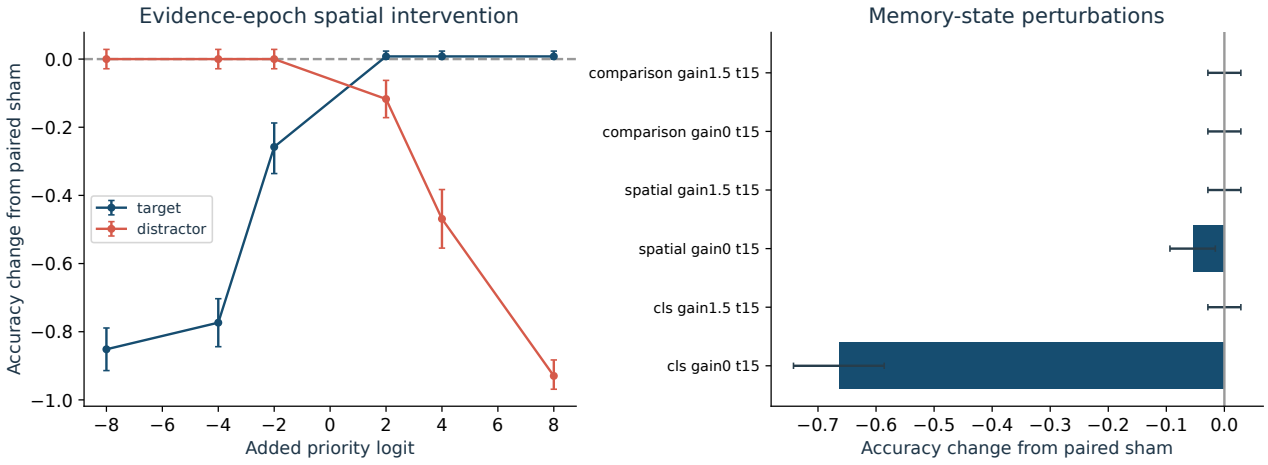


Figure 11: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
features_distinct	target_-4	-77.3	3.282e-28
features_distinct	target_+4	+0.8	1
features_distinct	distractor_-4	+0.0	1
features_distinct	distractor_+4	-46.9	1.648e-16
features_similar	target_-4	-88.3	2.041e-32
features_similar	target_+4	+8.6	0.06445
features_similar	distractor_-4	+0.0	1
features_similar	distractor_+4	-59.4	5.123e-20
fractals_distinct	target_-4	-47.7	8.327e-17
fractals_distinct	target_+4	+50.0	1.052e-17
fractals_distinct	distractor_-4	+13.3	0.001205
fractals_distinct	distractor_+4	-46.9	2.514e-15
fractals_similar	target_-4	-22.7	3.204e-07
fractals_similar	target_+4	+73.4	1.04e-26
fractals_similar	distractor_-4	+12.5	0.05731
fractals_similar	distractor_+4	-25.0	3.463e-07
objects_distinct	target_-4	-10.2	0.6586
objects_distinct	target_+4	+19.5	0.01754
objects_distinct	distractor_-4	+5.5	1
objects_distinct	distractor_+4	+1.6	1
objects_similar	target_-4	-15.6	0.002763
objects_similar	target_+4	+22.7	0.002923
objects_similar	distractor_-4	-3.1	1
objects_similar	distractor_+4	-1.6	1

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.529	27.3%	0.250
fusion	0.528	28.1%	0.190
spatial	0.704	44.5%	0.005
cls	1.000	100.0%	0.005
comparison	0.620	34.4%	0.045
attention	1.000	100.0%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

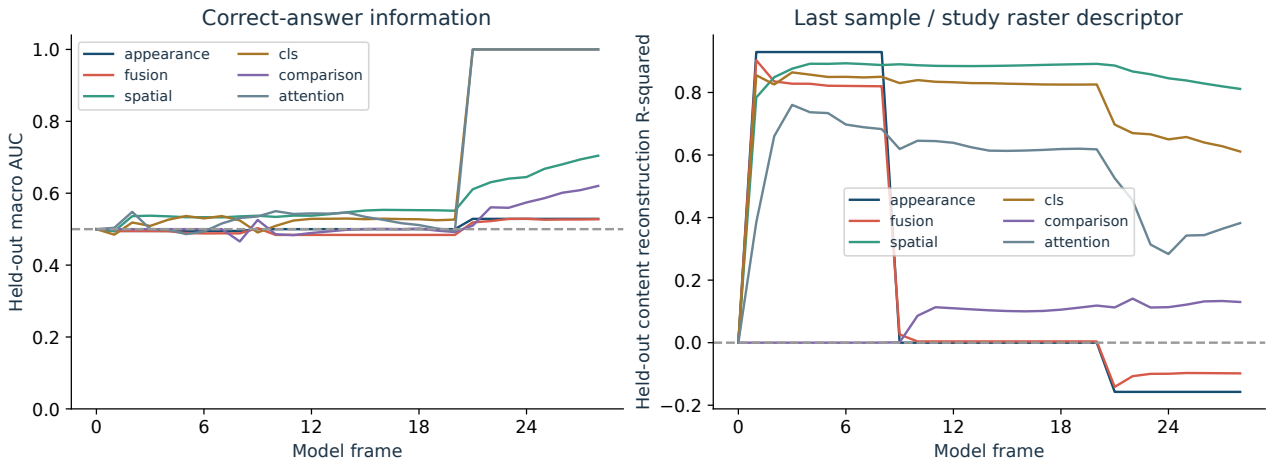


Figure 12: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 0.00 to 13.44 within the retained subspace. The final step displacement was 0.085 standardized RMS units. In spatial memory, the participation ratio ranged from 0.00 to 17.09 within the retained subspace. The final step displacement was 0.079 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 15, coarse sample-content reconstruction had R-squared 0.828 from CLS memory and 0.886 from spatial memory. These values quantify reconstruction of the specified 8-by-8 descriptor on unseen trials.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -66.4 percentage points, spatial, -5.5 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

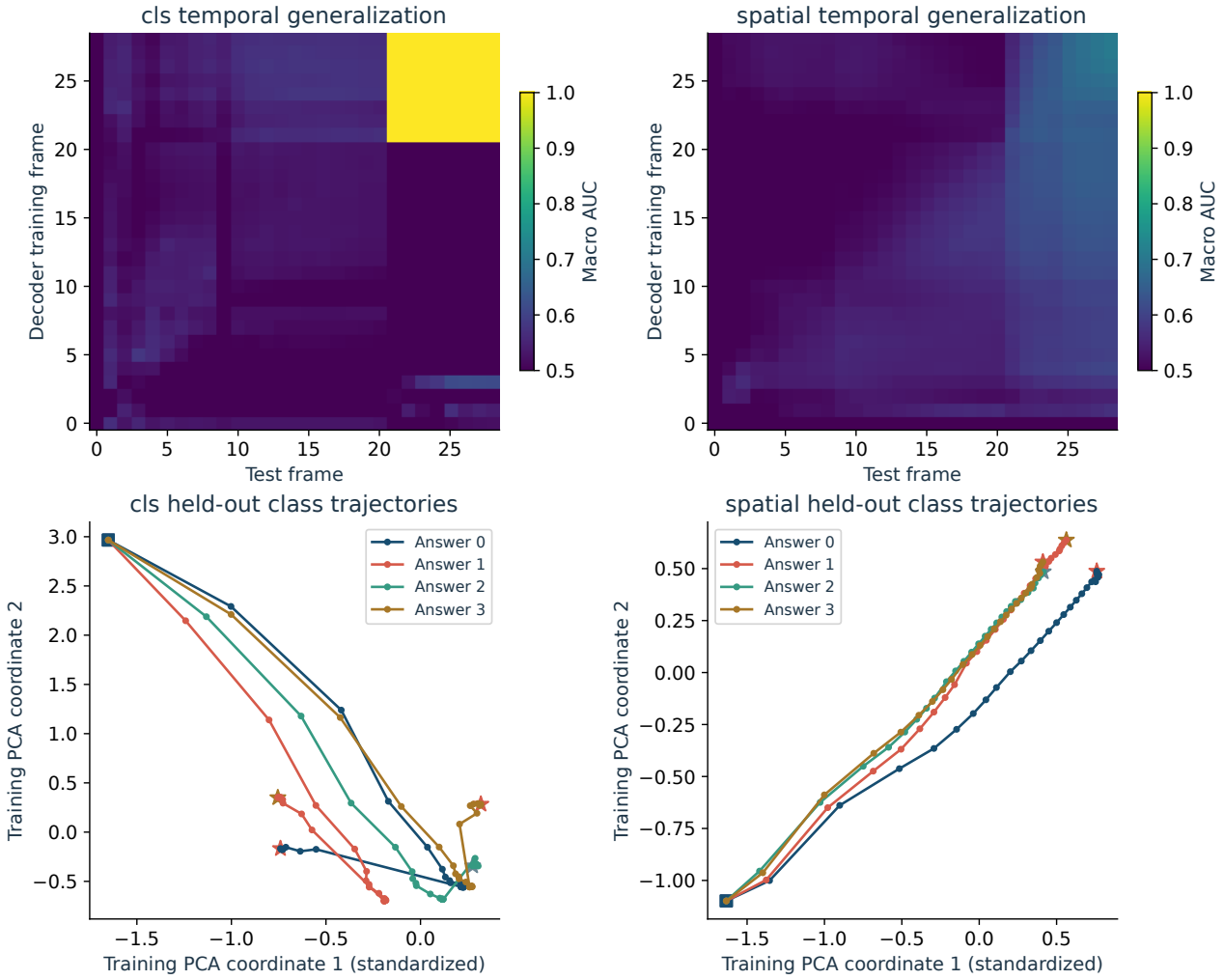


Figure 13: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

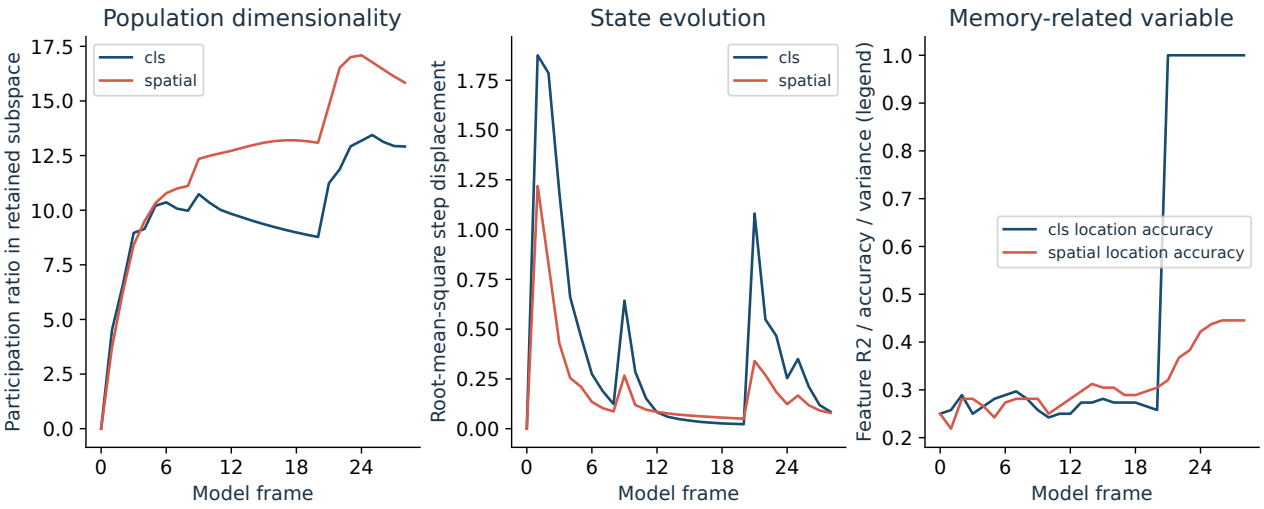


Figure 14: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

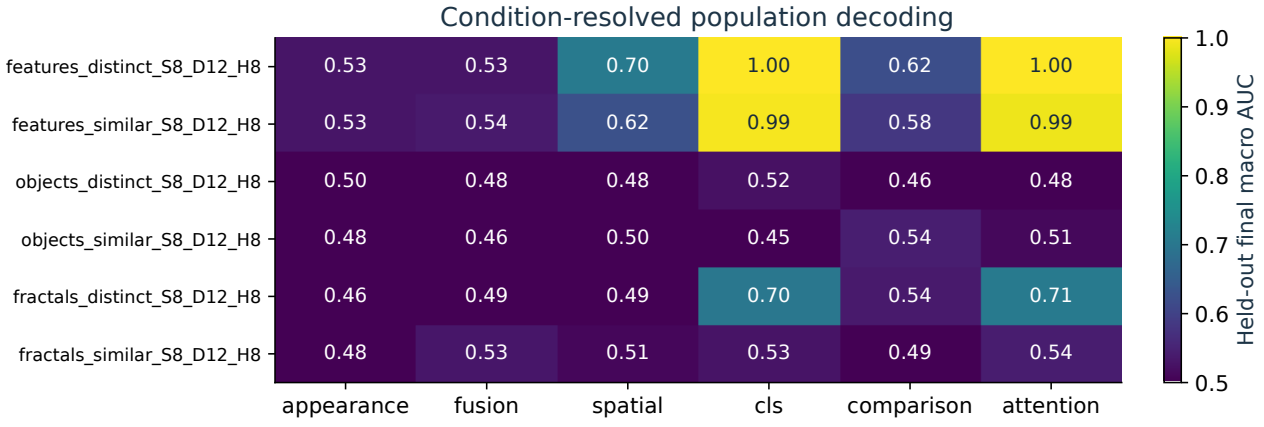


Figure 15: Held-out final answer decoding across selected conditions and recording sites. A constant-zero comparison representation is displayed at chance AUC.

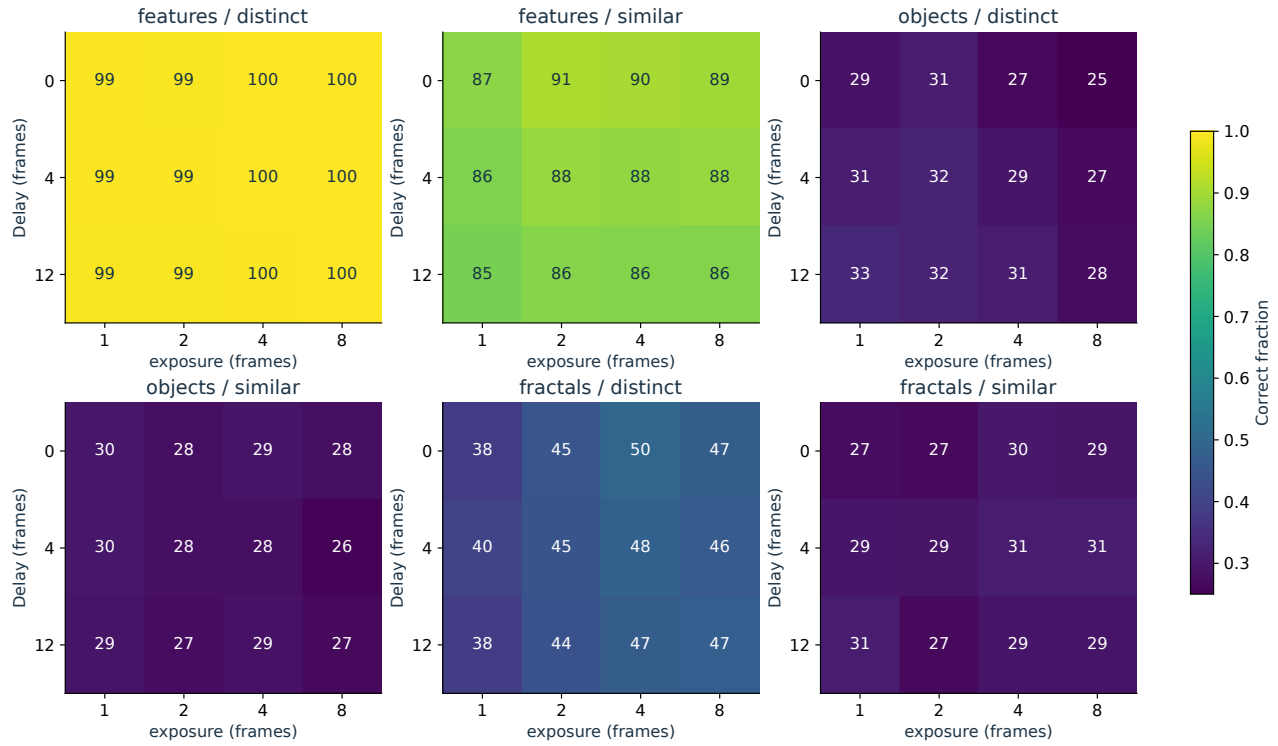


Figure 16: Memory-search factorial. Each cell shows accuracy averaged over the other exposure or sample-duration factor, within the indicated family/similarity or motion-spacing group. Numbers are percent correct; all contributing native conditions contain 128 trials. Delay, encoding time and search time remain distinct factors.

3.11 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.449	unavailable	3200
fusion	0.536	0.511	6400
spatial	0.980	0.493	6400
cls	1.000	0.515	64
comparison	0.595	0.513	735
attention	1.000	0.479	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 54 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

For similar feature-conjunction arrays, there were 767 errors among 6,144 evaluated trials across the timing factorial. Among errors, the selected alternative shared the target color 687 times, shape 0 times and texture 80 times. These are descriptive counts across repeated-scene timing conditions; the scene-clustered analyses supply the uncertainty framework.

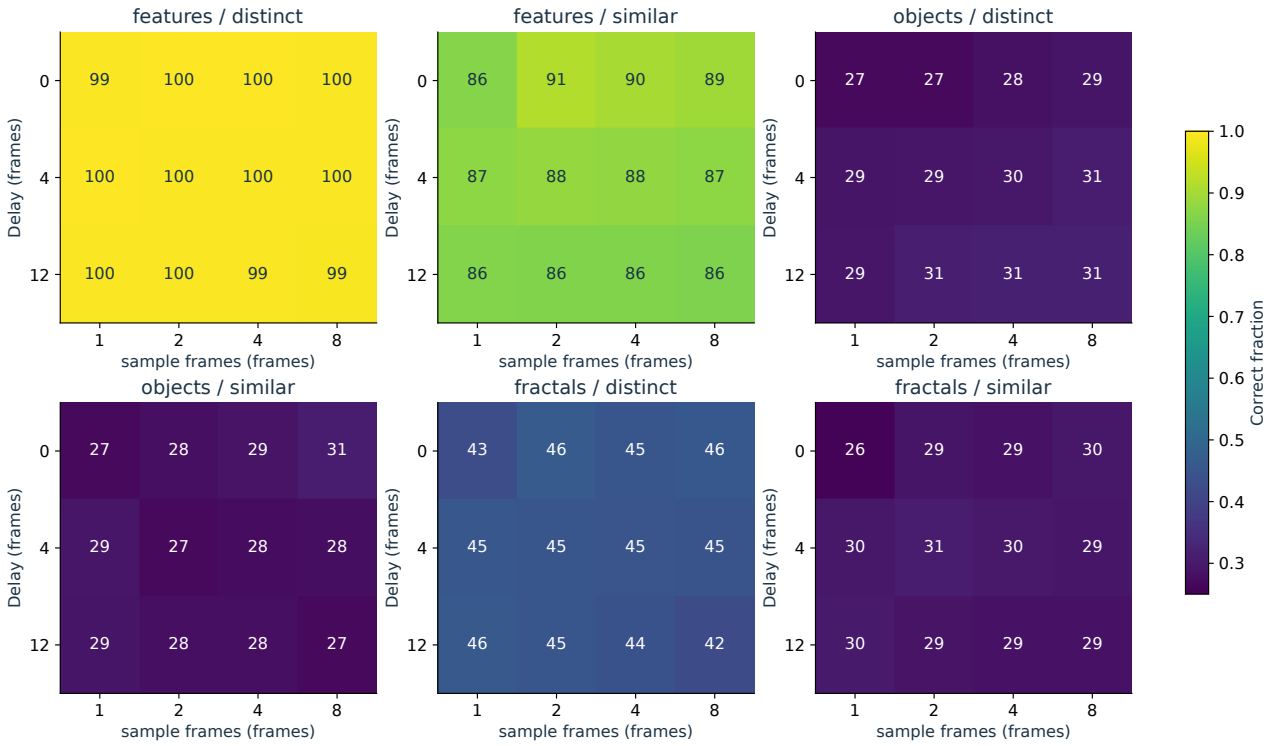


Figure 17: Memory-search factorial. Each cell shows accuracy averaged over the other exposure or sample-duration factor, within the indicated family/similarity or motion-spacing group. Numbers are percent correct; all contributing native conditions contain 128 trials. Delay, encoding time and search time remain distinct factors.

3.12 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check. For features_distinct_S8_D12_H8, a sample-centered encoding dose of -4 changed paired accuracy by -68.8 percentage points on 32 trials (interval [-84.4%, -53.1%]). This manipulation addresses the central sample location, separately from the future array-target location.

For features_distinct_S8_D12_H8, a sample-centered encoding dose of +4 changed paired accuracy by -6.2 percentage points on 32 trials (interval [-15.6%, 0.0%]). This manipulation addresses the central sample location, separately from the future array-target location.

For features_similar_S8_D12_H8, a sample-centered encoding dose of -4 changed paired accuracy by -56.2 percentage points on 32 trials (interval [-71.9%, -37.5%]). This manipulation addresses the central sample location, separately from the future array-target location.

For features_similar_S8_D12_H8, a sample-centered encoding dose of +4 changed paired accuracy by +3.1 percentage points on 32 trials (interval [-6.2%, 12.5%]). This manipulation addresses the central sample location, separately from the future array-target location.

For objects_distinct_S8_D12_H8, a sample-centered encoding dose of -4 changed paired accuracy by +3.1 percentage points on 32 trials (interval [-18.8%, 25.0%]). This manipulation addresses the central sample location, separately from the future array-target location.

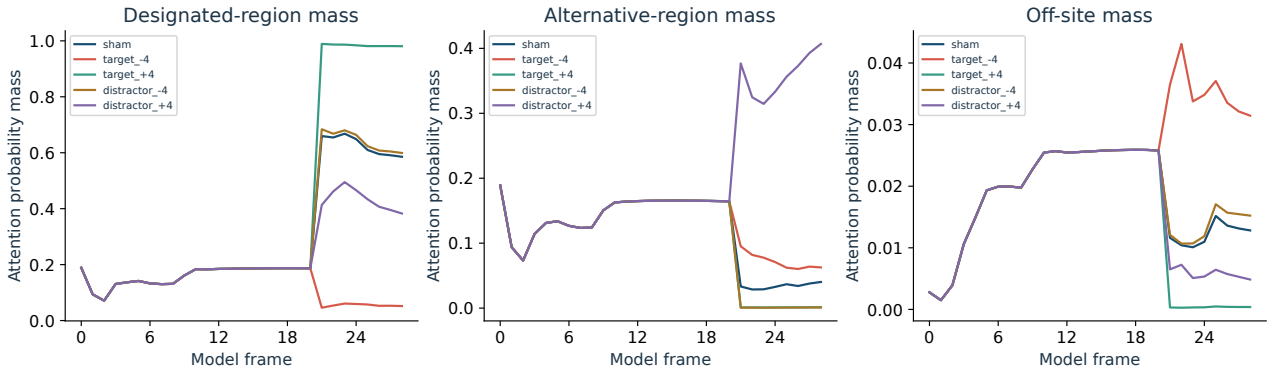


Figure 18: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

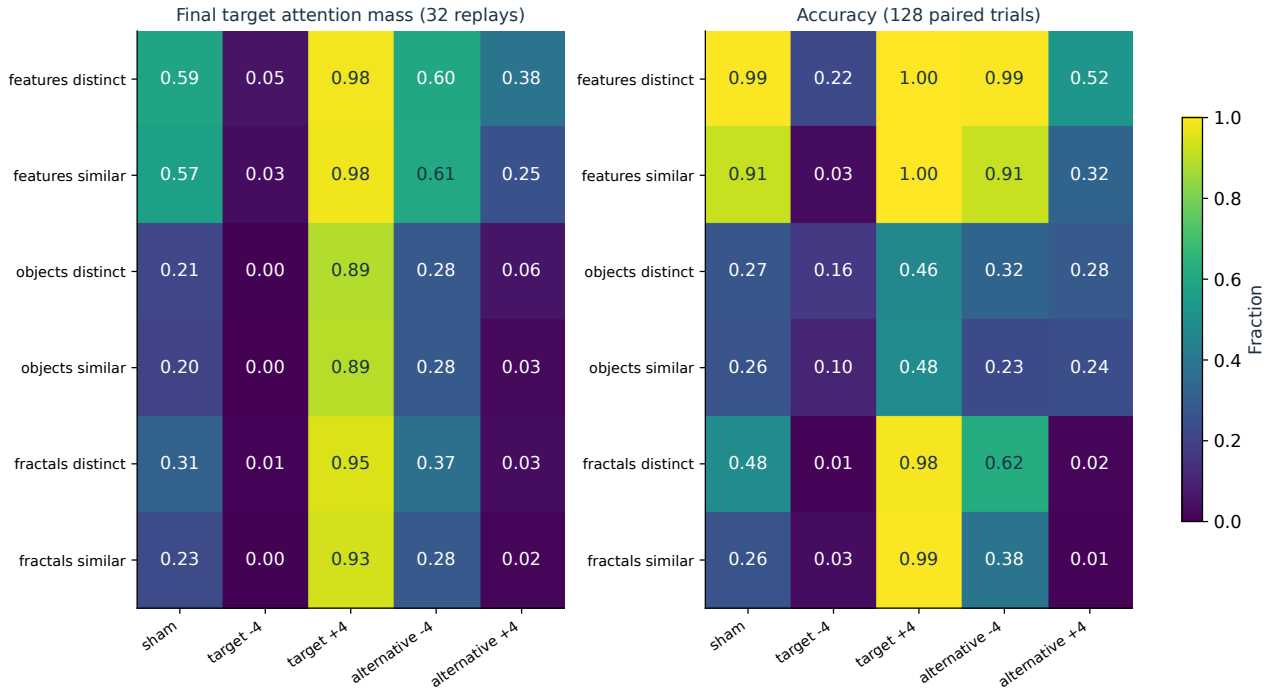


Figure 19: Condition-resolved manipulation checks. The left panel measures final target-region allocation on 32 exact replays; the right measures accuracy on all 128 paired intervention trials. All displayed spatial doses have magnitude four and act during evidence frames. Rows preserve each selected condition, making redistribution and behavioral rescue separately visible.

For objects_distinct_S8_D12_H8, a sample-centered encoding dose of +4 changed paired accuracy by +3.1 percentage points on 32 trials (interval [-9.4%, 18.8%]). This manipulation addresses the central sample location, separately from the future array-target location.

For objects_similar_S8_D12_H8, a sample-centered encoding dose of -4 changed paired accuracy by +3.1 percentage points on 32 trials (interval [-18.8%, 25.0%]). This manipulation addresses the central sample location, separately from the future array-target location.

For objects_similar_S8_D12_H8, a sample-centered encoding dose of +4 changed paired accuracy by +6.2 percentage points on 32 trials (interval [-6.2%, 18.8%]). This manipulation addresses the central sample location, separately from the future array-target location.

For fractals_distinct_S8_D12_H8, a sample-centered encoding dose of -4 changed paired accuracy by -25.0 percentage points on 32 trials (interval [-40.6%, -12.5%]). This manipulation addresses the central sample location, separately from the future array-target location.

For fractals_distinct_S8_D12_H8, a sample-centered encoding dose of +4 changed paired accuracy by -18.8 percentage points on 32 trials (interval [-37.5%, 0.0%]). This manipulation addresses the central sample location, separately from the future array-target location.

For fractals_similar_S8_D12_H8, a sample-centered encoding dose of -4 changed paired accuracy by -3.1 percentage points on 32 trials (interval [-25.0%, 18.8%]). This manipulation addresses the central sample location, separately from the future array-target location.

For fractals_similar_S8_D12_H8, a sample-centered encoding dose of +4 changed paired accuracy by +3.1 percentage points on 32 trials (interval [-18.8%, 21.9%]). This manipulation addresses the central sample location, separately from the future array-target location.

3.13 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

3.14 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	27.3%	25.8%	27.3%
fusion	28.1%	26.6%	28.1%
spatial	44.5%	43.8%	44.5%
cls	100.0%	100.0%	100.0%
comparison	34.4%	35.2%	34.4%
attention	100.0%	100.0%	100.0%

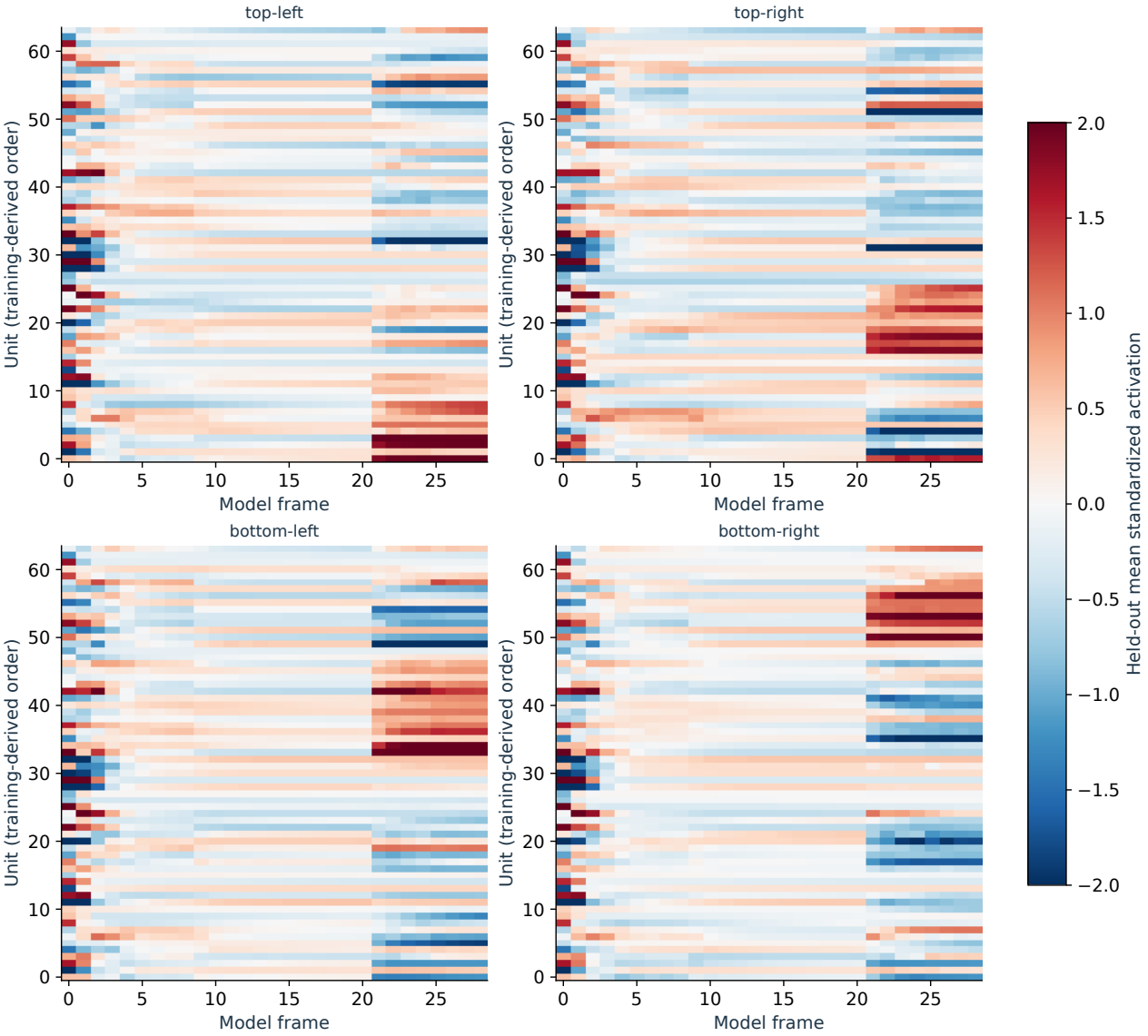


Figure 20: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

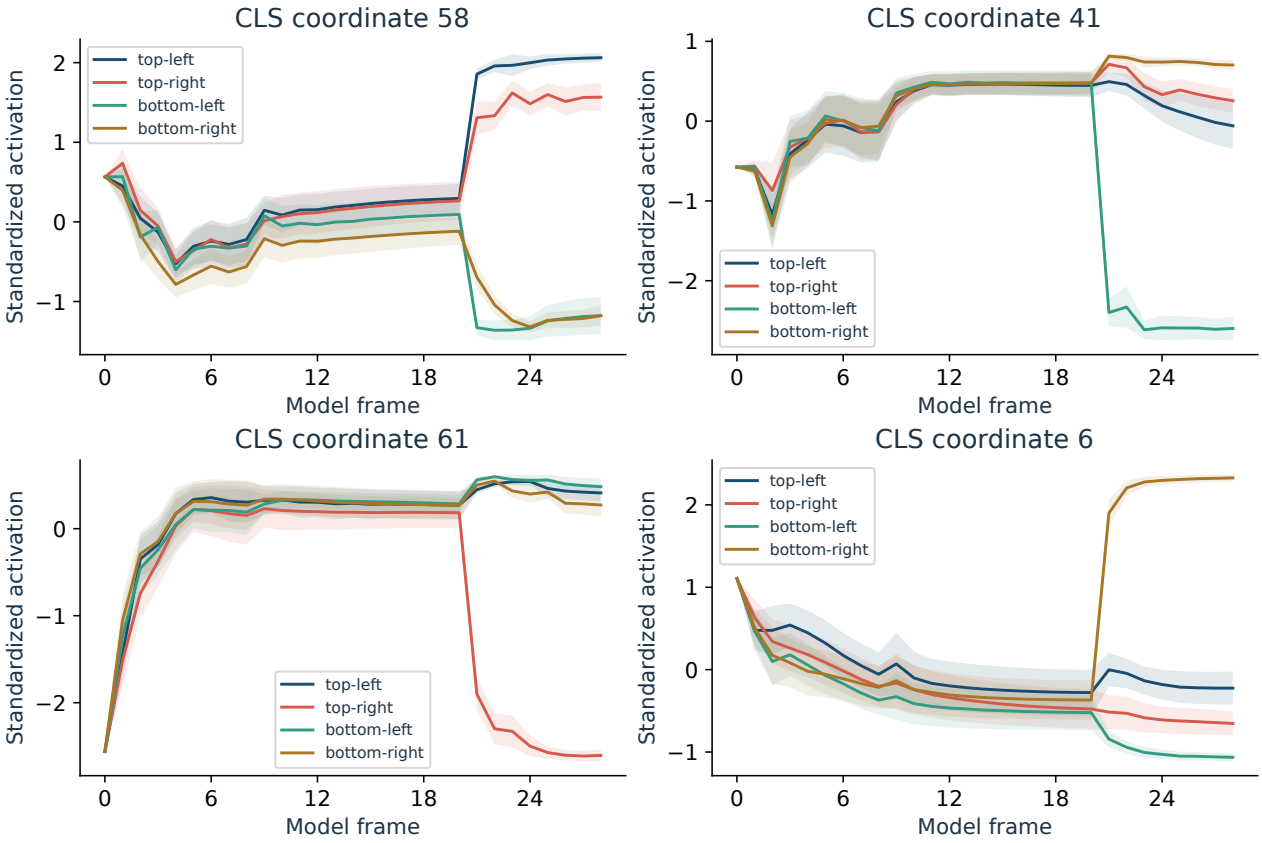


Figure 21: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

3.15 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\text{fit}} d_j^{\text{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

3.16 Checkpoint robustness within each stimulus family

The final 220,637 checkpoint is compared with the selected 204,289 checkpoint on the same first 32 native scenes in each family. Each scene is evaluated across all 48 timing settings in that family. Differences are averaged over timing within scene, and intervals resample the 32 paired scenes 3,000 times. These rows therefore refer to a matched subset of the native trials, separately from the independent neural-recording population below.

Family	Selected	Final	Change interval (pp)
features_distinct	99.8%	97.9%	[-6.4, +0.6]

Family	Selected	Final	Change interval (pp)
features_similar	80.3%	82.2%	[-11.5, +15.2]
fractals_distinct	46.5%	58.0%	[-2.9, +25.5]
fractals_similar	34.2%	34.8%	[-16.4, +17.7]
objects_distinct	26.7%	39.4%	[-3.0, +28.0]
objects_similar	24.3%	23.9%	[-11.0, +9.7]

3.17 Reconstructing the complete remembered object

The full-crop analysis predicts all 28 by 28 RGB sample pixels, giving 2,352 continuous outputs. Targets are regenerated from the exact neural-trial seeds and their stored raster hashes are verified. Full-coordinate ridge mappings use 192 fitting trials and 128 independent test trials per condition; the penalty is chosen from 0.001, 0.01, 0.1, 1 and 10 using an inner 75/25 fitting split. Separate mappings at encoding, middle delay, first search and final frames compare appearance, global memory and spatial memory. R-squared uses the fitting-mean sample as the baseline. The first held-out example in each condition is shown; images are clipped only for display, while scores use the unbounded predictions.

For features_distinct_S8_D12_H8, middle-delay full-crop R-squared was 0.759 from CLS memory and 0.922 from spatial memory. For features_similar_S8_D12_H8, middle-delay full-crop R-squared was 0.724 from CLS memory and 0.918 from spatial memory. For objects_distinct_S8_D12_H8, middle-delay full-crop R-squared was 0.378 from CLS memory and 0.723 from spatial memory. For objects_similar_S8_D12_H8, middle-delay full-crop R-squared was 0.414 from CLS memory and 0.724 from spatial memory. For fractals_distinct_S8_D12_H8, middle-delay full-crop R-squared was 0.064 from CLS memory and 0.271 from spatial memory. For fractals_similar_S8_D12_H8, middle-delay full-crop R-squared was 0.052 from CLS memory and 0.241 from spatial memory.

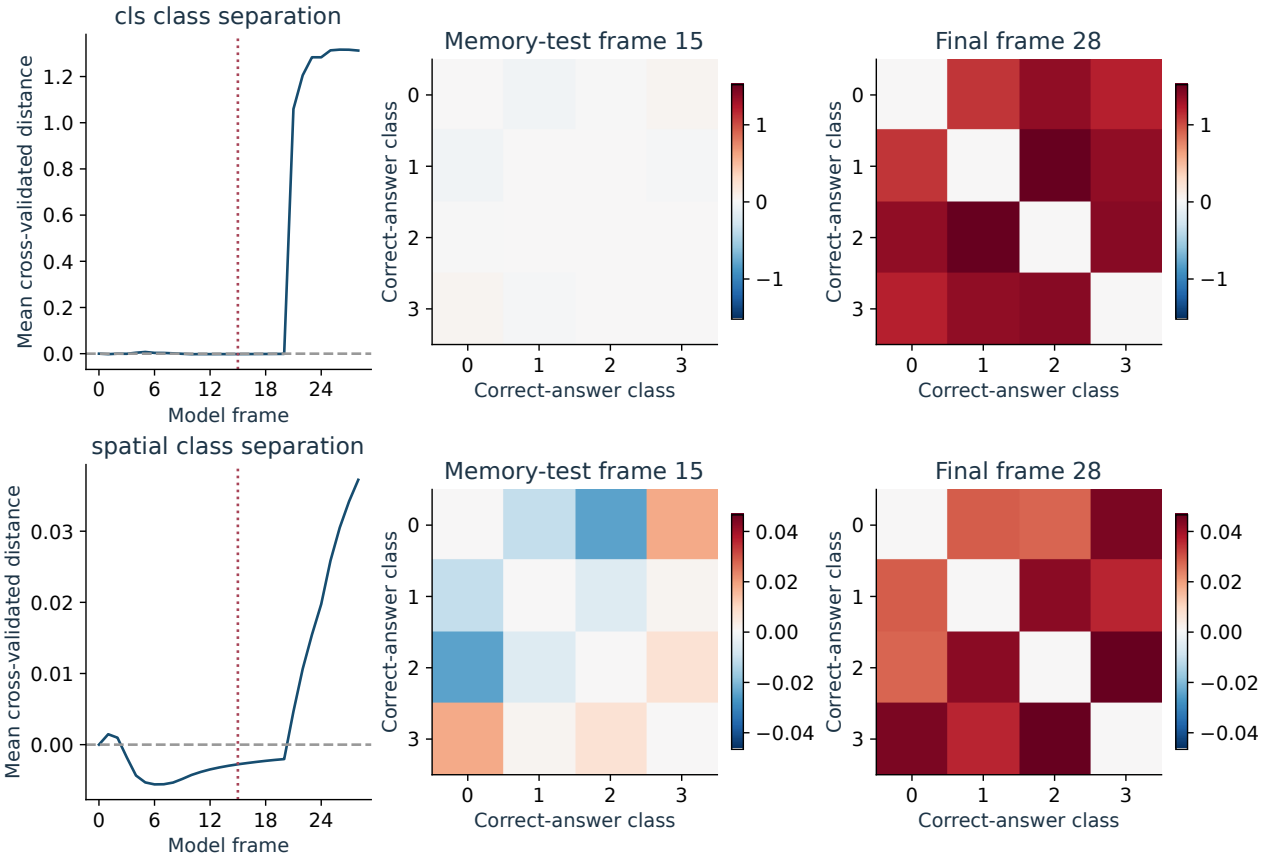


Figure 22: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

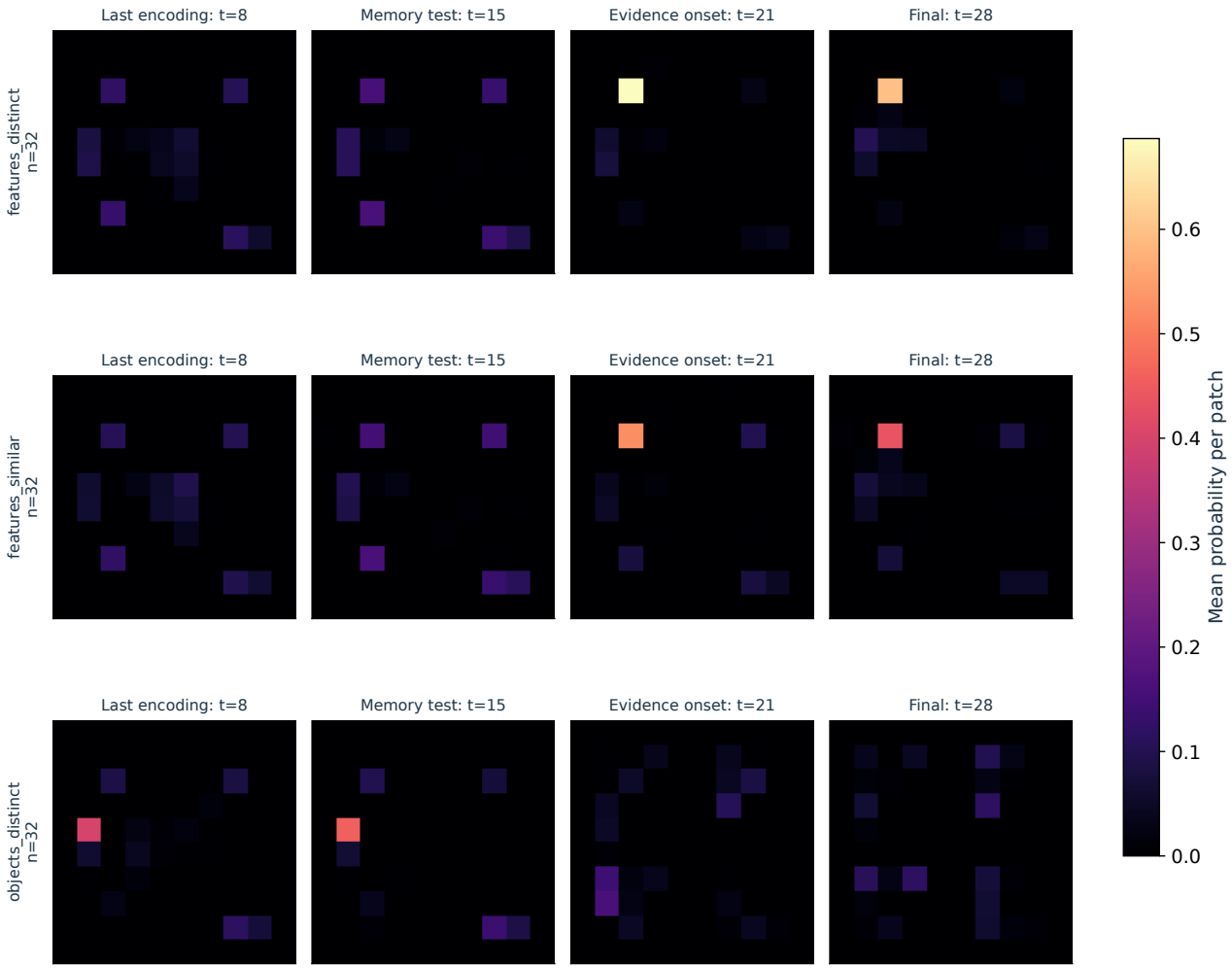


Figure 23: Condition-resolved attention at four temporal landmarks. Memory-search and pop-out rows average trials with top-left targets. Image-recognition rows average seen-probe trials. Each map is shown in the original image layout with a common probability scale across conditions; the row label gives the number of averaged trials. The memory-test frame is the middle delay, or the frame before evidence when a blank delay is absent.

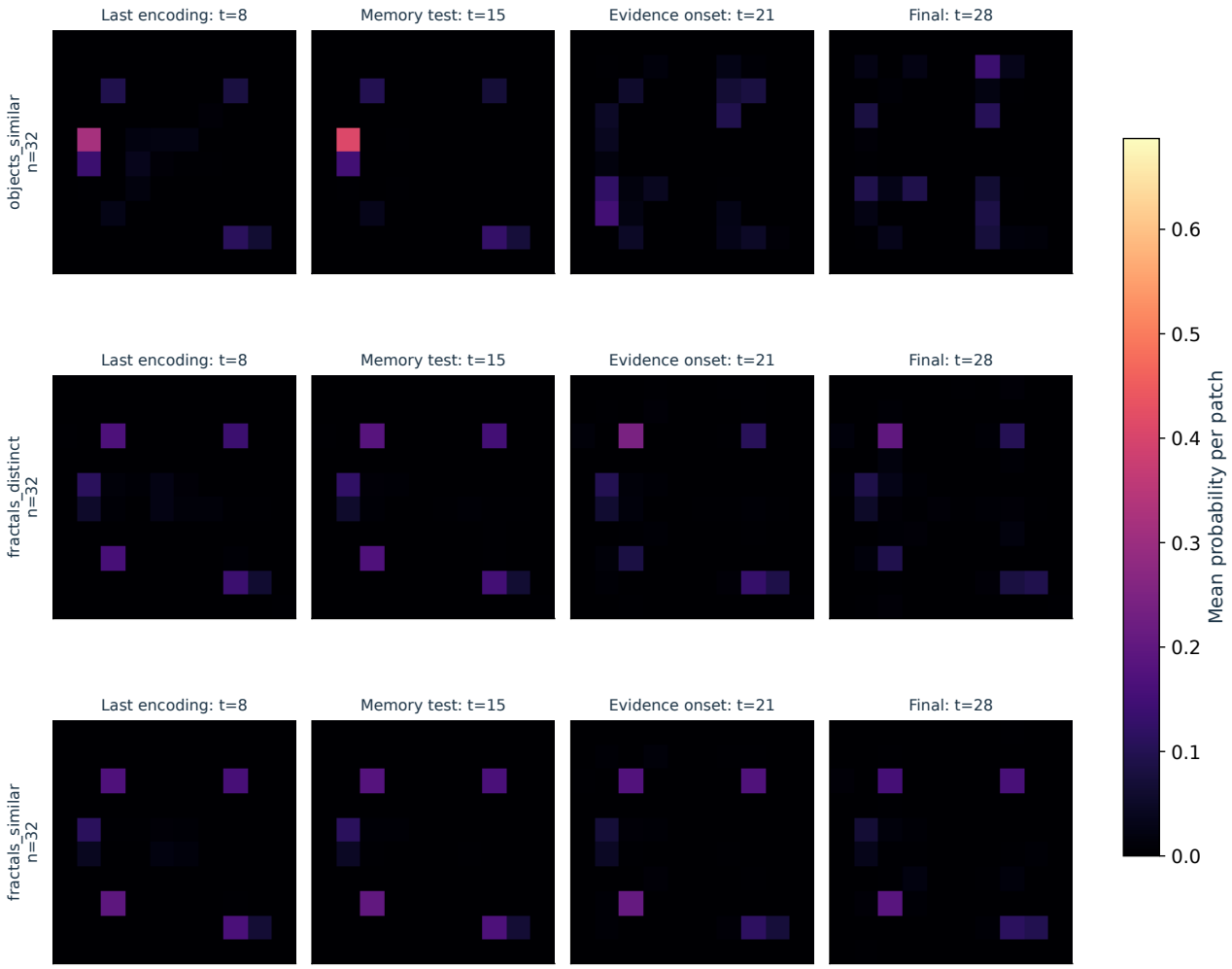


Figure 24: Condition-resolved attention at four temporal landmarks. Memory-search and pop-out rows average trials with top-left targets. Image-recognition rows average seen-probe trials. Each map is shown in the original image layout with a common probability scale across conditions; the row label gives the number of averaged trials. The memory-test frame is the middle delay, or the frame before evidence when a blank delay is absent.

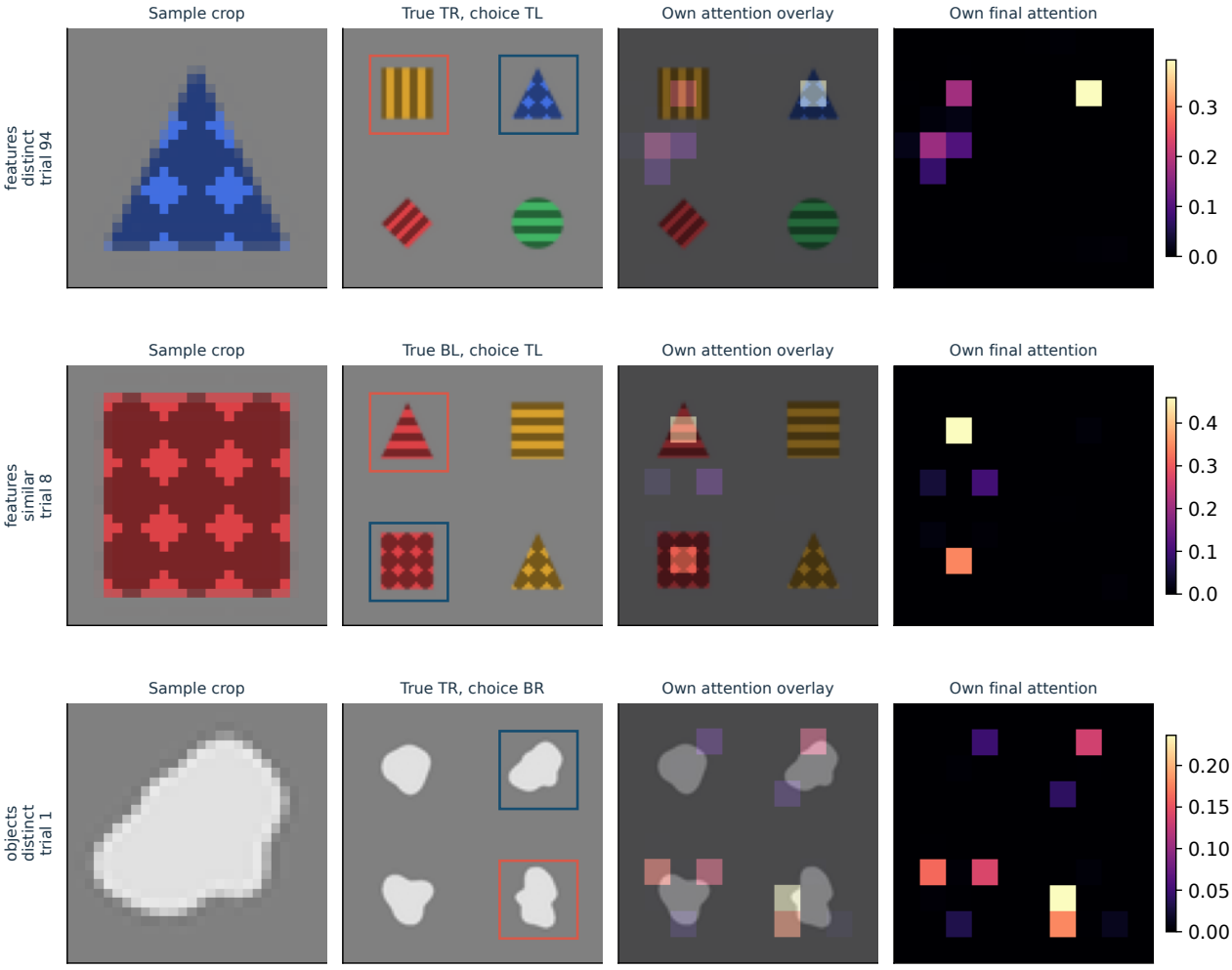


Figure 25: Concrete search examples from all six family/similarity conditions. Each row uses the first error in the selected condition, or the first trial if all 128 were correct. The enlarged sample crop is followed by the exact array, the same trial's attention overlay, and its final attention map. Blue marks the target and coral the chosen location. Attention scales are shown separately for each example. These illustrate failure structure; the factorial results quantify its frequency.

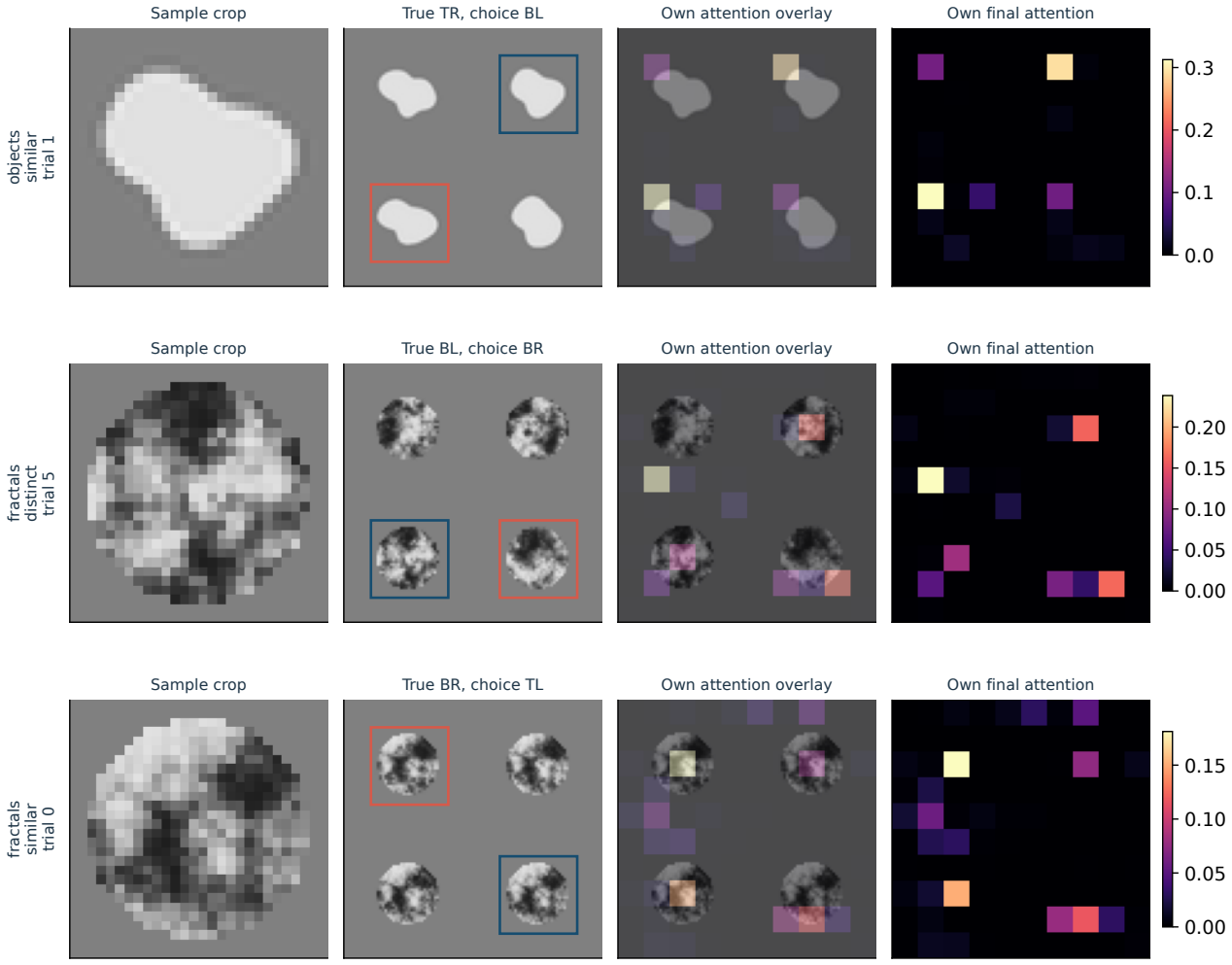


Figure 26: Concrete search examples from all six family/similarity conditions. Each row uses the first error in the selected condition, or the first trial if all 128 were correct. The enlarged sample crop is followed by the exact array, the same trial's attention overlay, and its final attention map. Blue marks the target and coral the chosen location. Attention scales are shown separately for each example. These illustrate failure structure; the factorial results quantify its frequency.

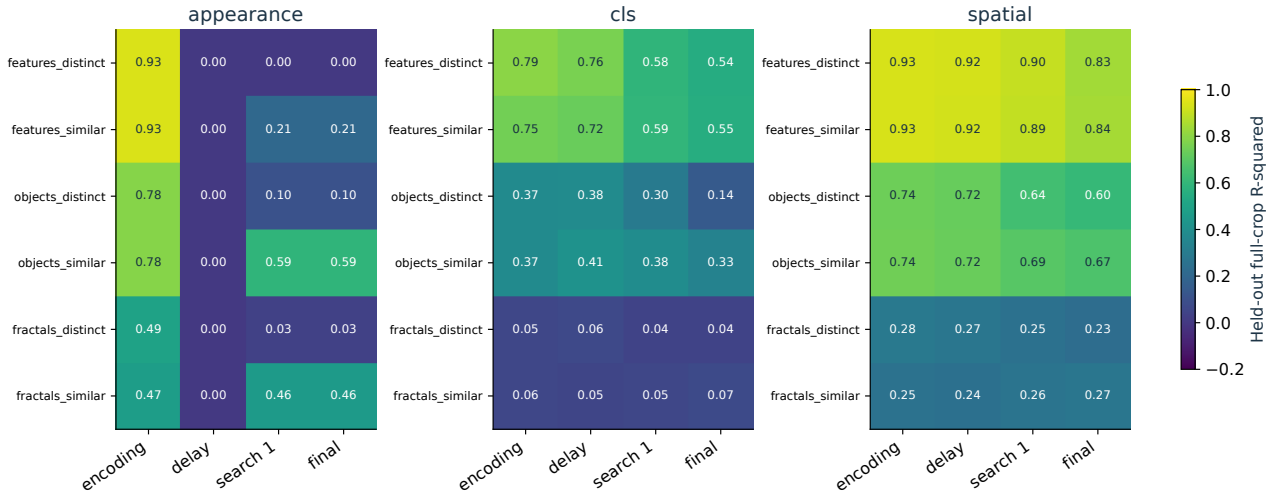


Figure 27: Full-resolution object content reconstructed from the original coordinate widths. Rows preserve family and similarity, and columns preserve temporal stage. Blank-frame appearance provides a comparison with stored content in recurrent states.

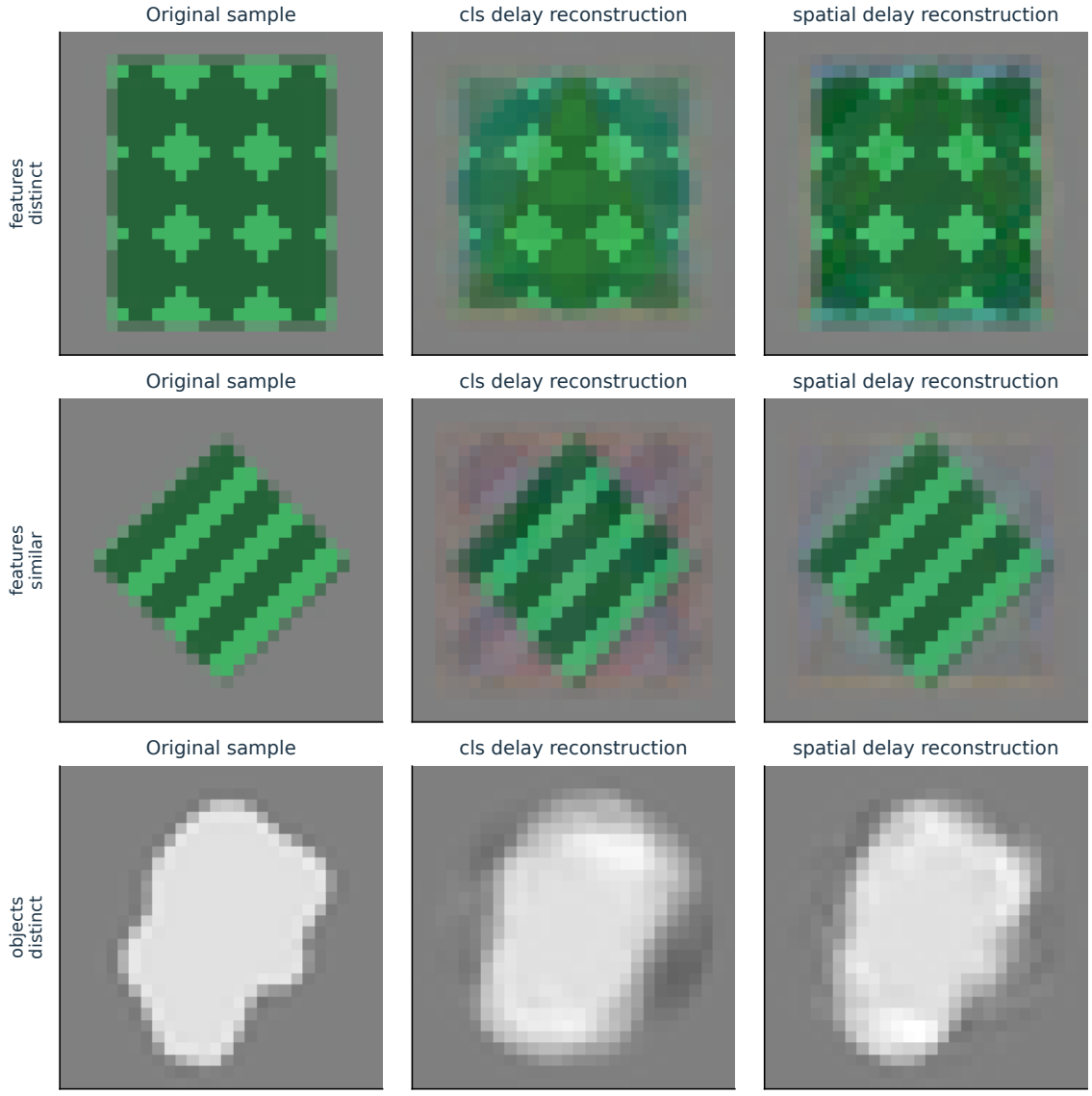


Figure 28: The first held-out sample and its linear reconstruction from middle-delay global and spatial memory. These examples were fixed by trial index, independently of reconstruction quality.

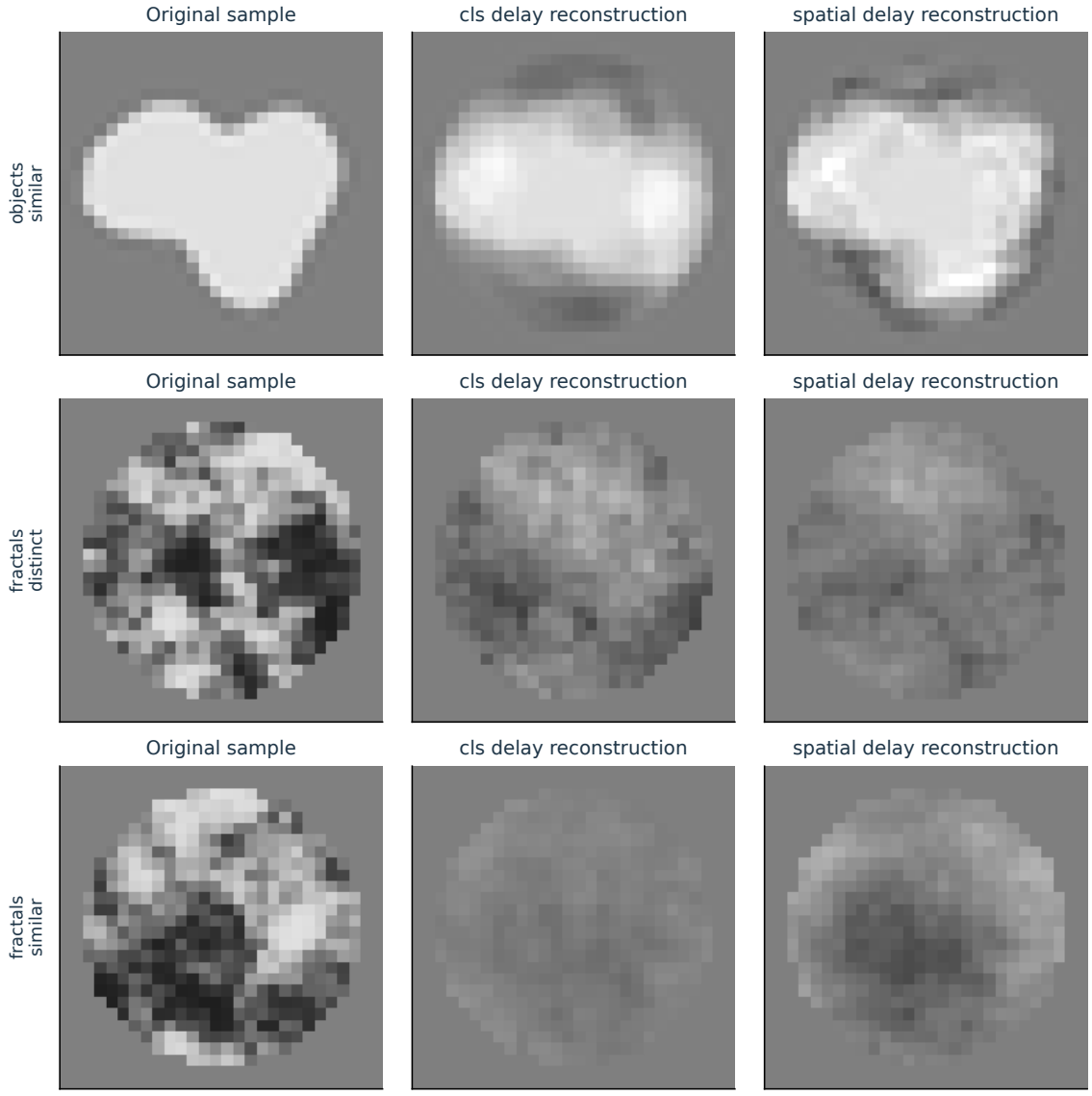


Figure 29: The first held-out sample and its linear reconstruction from middle-delay global and spatial memory. These examples were fixed by trial index, independently of reconstruction quality.

3.18 Can the decoded memory identify the matching object?

An external readout compares each reconstructed sample with the four actual candidate rasters and chooses the candidate with the smallest mean squared pixel difference. The reconstruction mapping sees only the recorded state; candidate rasters enter the subsequent matching calculation. The same 128 held-out scenes supply the original agent choices and the external choices. Paired comparisons use exact discordance tests and Holm correction over all 72 region-by-time-by-condition comparisons. The table emphasizes the middle blank delay, before array presentation, so this test isolates information retained from the sample.

Condition	Agent	CLS match	Spatial match	Spatial Holm p
features_distinct	100.0%	100.0%	100.0%	1
features_similar	88.3%	100.0%	100.0%	0.002563
objects_distinct	32.0%	90.6%	100.0%	8.53e-25
objects_similar	20.3%	52.3%	95.3%	4.186e-26
fractals_distinct	46.1%	63.3%	87.5%	2.582e-10
fractals_similar	29.7%	29.7%	45.3%	0.5057

These results separate retained content from its use by the learned decision computation. For object arrays and distinct fractals, an external comparison of decoded spatial memory with visible candidates substantially exceeds the original agent's performance. Thus those failures coexist with usable sample information in spatial memory. Similar fractals remain difficult under both readouts, indicating a further representational or matching limitation for this condition. The external readout has access to an explicit pixel-distance comparison; its performance establishes an available information route, while implementing that route inside the agent is a separate architectural experiment.

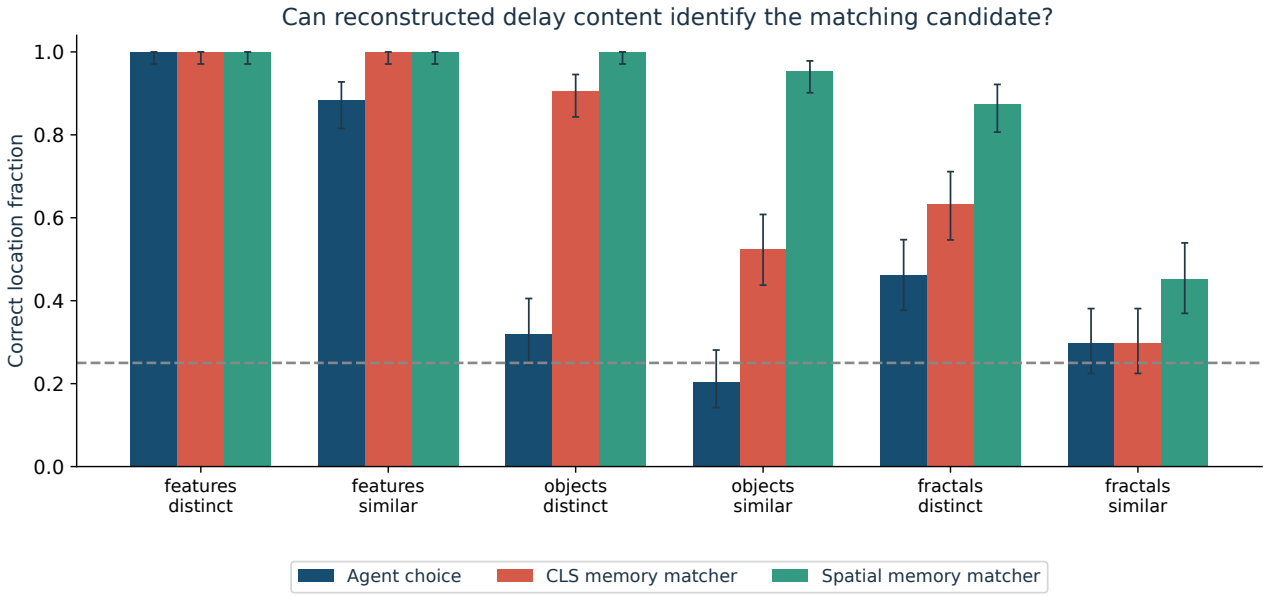


Figure 30: Original agent choices versus external candidate matching from reconstructed middle-delay content. Dashed line: 25 percent chance; error bars: Wilson 95 percent intervals. All three readouts use the same 128 held-out scenes in each condition; the external readouts reconstruct the sample before seeing the search array.

3.19 Matching a decoded memory to decoded visual candidates

A further exploratory readout checks whether the current array encoder also preserves the needed information. For each candidate, its centered three-by-three grid neighborhood provides 288 appearance coordinates (nine sites times 32 channels) at the first array frame. Within each family/similarity condition, one linear ridge decoder shared across the four candidate locations predicts the 2,352 candidate RGB values from those coordinates. A separate probe pair is fitted for each of the six conditions. Four candidates in each of 192 fitting scenes give 768 fitting examples. Penalty selection splits whole scenes 75/25, keeping their four candidates together. The 128 test scenes remain independent, and the fitting objective uses raster reconstruction rather than behavioral labels.

The separately fitted middle-delay sample decoder and the current-array decoder then supply two decoded pixel representations. Their squared-distance comparison chooses the most similar candidate. This joint readout consumes internal representations at test time. An exact-sample plus decoded-array control separately measures the candidate decoder's usefulness. The six spatial-memory joint readouts are compared with original agent choices on the same scenes, with Holm correction across those six paired tests. This follow-up was chosen after observing successful memory reconstruction.

Family	Candidate R^2	Exact sample	Joint spatial	Holm p
features_distinct	0.933	100.0%	100.0%	1
features_similar	0.918	100.0%	100.0%	0.0001831
objects_distinct	0.690	100.0%	100.0%	7.755e-26
objects_similar	0.687	93.0%	93.0%	2.553e-24
fractals_distinct	0.493	99.2%	86.7%	1.076e-09
fractals_similar	0.461	73.4%	52.3%	0.0002272

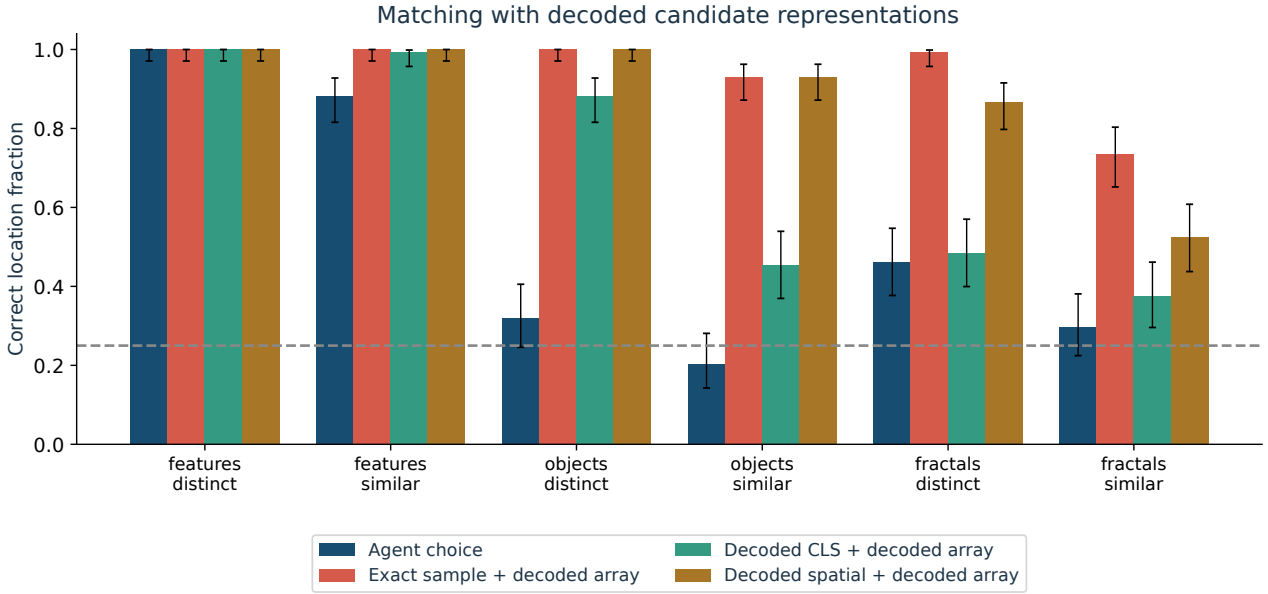


Figure 31: External matching from decoded memory and decoded current-array features. Exact sample plus decoded array isolates the candidate readout. CLS and spatial joint readouts use decoded templates from the blank delay and decoded candidates from the first array frame. Agent performance is measured on the same 128 independent scenes; error bars are Wilson 95 percent intervals and chance is 25 percent.

3.20 Identical-array, changed-memory experiment

For each selected condition, 64 new scenes were each replayed four times. Only the sampled candidate changed; all search-array pixels were verified identical across the four movies. Every location served once as the correct answer within each scene. An additional replay replaced the sample with uniform midpoint gray. Intervals resample whole scenes, keeping their four queries together; all-perfect or all-zero accuracy boundaries use a Wilson interval on the 64 scene units. The template probability gain subtracts mean off-diagonal report probability from mean diagonal probability; the attention gain applies the same diagonal-minus-off-diagonal comparison to final spatial masses. These contrasts measure how a change in the remembered sample changes the response to a fixed array.

For features_distinct_S8_D12_H8, accuracy was 99.6% (scene-bootstrap interval [98.8%, 100.0%]); 98.4% of scenes had all four queries answered correctly. Choice varied across queries in 100.0% of scenes. Template probability gain was 0.991 (95 percent interval [0.983, 0.996]) and final attention gain was 0.603 (interval [0.583, 0.624]). The sample-blank response is fixed within each array and therefore scores exactly 25 percent across the four balanced candidate labels.

For features_similar_S8_D12_H8, accuracy was 86.7% (scene-bootstrap interval [82.8%, 90.6%]); 53.1% of scenes had all four queries answered correctly. Choice varied across queries in 100.0% of scenes. Template probability gain was 0.827 (95 percent interval [0.784, 0.868]) and final attention gain was 0.444 (interval [0.420, 0.467]). The sample-blank response is fixed within each array and therefore scores exactly 25 percent across the four balanced candidate labels.

For fractals_distinct_S8_D12_H8, accuracy was 45.3% (scene-bootstrap interval [39.5%, 51.2%]); 3.1% of scenes had all four queries answered correctly. Choice varied across queries in 95.3% of scenes. Template probability gain was 0.250 (95 percent interval [0.186, 0.313]) and final attention gain was 0.136 (interval [0.116, 0.156]). The sample-blank response is fixed within each array and therefore scores exactly 25 percent across the four balanced candidate labels.

For fractals_similar_S8_D12_H8, accuracy was 29.3% (scene-bootstrap interval [25.0%, 33.6%]); 0.0% of scenes had all four queries answered correctly. Choice varied across queries in 73.4% of scenes. Template probability gain was 0.049 (95 percent interval [0.013, 0.089]) and final attention gain was 0.020 (interval [0.011, 0.029]). The sample-blank response is fixed within each array and therefore scores exactly 25 percent across the four balanced candidate labels.

For objects_distinct_S8_D12_H8, accuracy was 32.4% (scene-bootstrap interval [28.9%, 36.3%]); 0.0% of scenes had all four queries answered correctly. Choice varied across queries in 62.5% of scenes. Template probability gain was 0.061 (95 percent interval [0.038, 0.087]) and final attention gain was 0.003 (interval [-0.005, 0.011]). The sample-blank response is fixed within each array and therefore scores exactly 25 percent across the four balanced candidate labels.

For objects_similar_S8_D12_H8, accuracy was 26.2% (scene-bootstrap interval [23.4%, 28.9%]); 0.0% of scenes had all four queries answered correctly. Choice varied across queries in 43.8% of scenes. Template probability gain was 0.013 (95 percent interval [-0.001, 0.028]) and final attention gain was 0.000 (interval [-0.004, 0.005]). The sample-blank response is fixed within each array and therefore scores exactly 25 percent across the four balanced candidate labels.

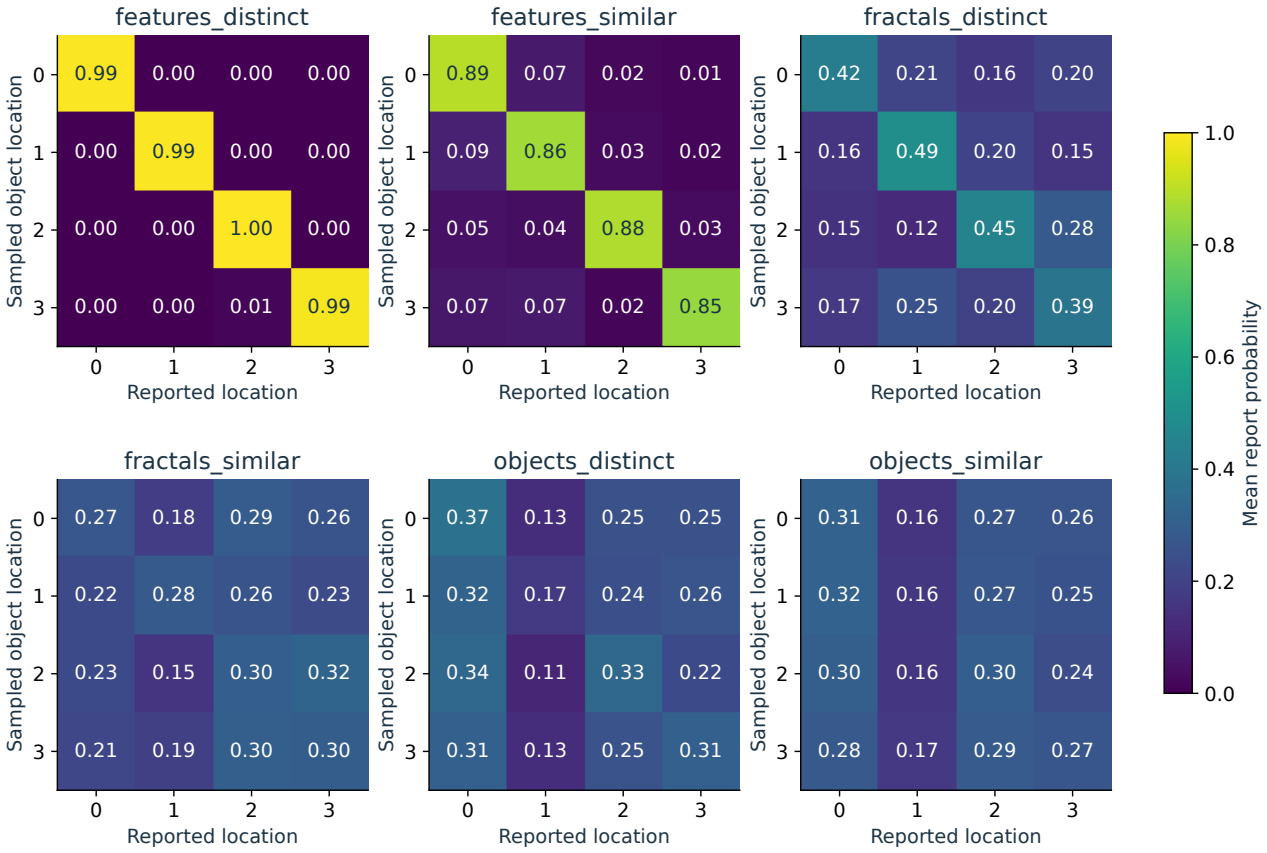


Figure 32: Causal template dependence with array pixels held fixed. Rows identify the candidate used as the remembered sample; columns identify report location. A diagonal pattern indicates that memory changes the selected answer appropriately. Each panel averages 64 matched scenes.

4 Discussion

The family-resolved results expose a selective failure that a task-wide average conceals. Distinct feature objects are matched almost perfectly, feature conjunctions remain substantially above chance, and irregular contour objects are much harder. The matched timing contrasts found no reliable rescue after correction across the tested timing comparisons. Full-resolution reconstruction provides a more specific explanation: spatial memory retains enough object information for an external pixel-distance matcher to achieve 95–100 percent accuracy in the two object conditions, whereas the agent makes much poorer choices on the same scenes. A follow-up also reconstructs the current candidates from local appearance features. Combining those decoded candidates with decoded spatial memory retains 100 percent accuracy for distinct objects and about 93 percent for similar objects, localizing a substantial deficit to the use of available representations. Joint decoding also improves distinct and similar fractals, although similar-fractal accuracy remains about 52 percent. Probe pairs are fitted separately within each stimulus condition, so these comparisons establish condition-specific information availability. Thus errors in using stored content and limitations of the stored or decoded representation coexist across stimulus families. The direct map check adds a second distinction: target stimulation raises final object-target allocation from about 21 percent to about 89 percent on the replay subset, while the full paired object experiments show only a partial behavioral rescue. Correct spatial allocation alone therefore does not restore successful object comparison. The changed-sample, identical-array experiment independently measures whether the agent uses its remembered sample to control choice and spatial allocation. Together, these measurements motivate a direct route from distributed spatial memory into the search computation, while identifying similar fractals as a separate demanding test of representational precision.

The behavioral result establishes 53.1% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies distractor_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

The matched-array query experiment establishes how the sample affects selection with search pixels fixed. Candidate-matching decoders and targeted memory-to-query interventions can now test which retained features mediate that effect. Additional object transformations would assess tolerance beyond the exact-raster matching rule.

4.1 Interpreting these results alongside biological experiments

4.1.1 Chelazzi, Miller, Duncan and Desimone (1993)

The basic sequence closely matches the model task: sample, delay, then an array containing the match. The animal result links remembered object content to later target selection. Our strongest comparison is the separation between retained information and its use. Irregular-object content can be reconstructed well enough for an external matcher to succeed, while the trained agent remains much poorer on the same scenes. [6]

4.1.2 Soto, Heinke, Humphreys and Blanco (2005)

Human memory can bias attention even when the remembered item is incidental to the current target. Our identical-array experiment changes only the preceding sample, giving a direct test of memory-dependent choice and allocation. A model could change its answer through downstream comparison while leaving its attention map similar; measuring both outcomes distinguishes guidance of selection from use of memory after selection. [7]

4.1.3 Bichot, Rossi and Desimone (2005)

Feature-based selection can favor partial target matches before an object is finally identified. Our feature-family success and irregular-object weakness make that distinction useful. Plot allocation to candidates sharing individual features, then separately assess the unique conjunction match. The existing four-location output can express a correct choice even when the internal priority pattern distributes weight across several partial matches. [8]

4.2 A computational account of the observed behavior

Guided-search theory separates a representation of what is sought, a priority map that selects candidates, and the comparison that establishes a match. The present architecture retains detailed spatial memory but derives its global attention query from previous global memory. This creates a possible access bottleneck: stored object detail may remain readable externally without entering the query or nonlinear comparator in a useful form. The external decoded-memory plus decoded-candidate matcher performs near perfectly for distinct objects and very well for similar objects. Yet forcing attention toward the target yields only a partial rescue. Together, these observations locate a difficulty beyond simple loss of the sample or failure to raise target allocation. Similar fractals retain a lower external-matcher ceiling, adding a representational precision limit for that family. [9]

4.3 Where this task agrees with the biological evidence

The model agrees with successful memory-guided selection for feature-defined objects and a useful memory trace across families. It differs sharply from a uniform object-template account: rich stored information is not consistently converted into a correct match. This is a mechanistic failure worth emphasizing rather than averaging away in the task-wide score.

4.4 A follow-up experiment motivated by these results

Compare memory-to-query access, candidate encoding and the final comparator using matched scenes and independently fitted probes. A direct spatial-memory query route predicts larger improvements for irregular objects than for already-solved feature stimuli. Evaluate that prediction on the separate fresh-architecture experiment, with the present frozen model retained as the baseline.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing

classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Chelazzi L, Miller EK, Duncan J, Desimone R (1993). A neural basis for visual search in inferior temporal cortex. Nature 363:345-347. https://pubmed.ncbi.nlm.nih.gov/8497317/.

[2] Treisman AM, Gelade G (1980). A feature-integration theory of attention. Cognitive Psychology 12:97-136. https://pubmed.ncbi.nlm.nih.gov/7351125/.

[3] Wolfe JM (2021). Guided Search 6.0: An updated model of visual search. Psychonomic Bulletin and Review 28:1060-1092. https://pubmed.ncbi.nlm.nih.gov/33547630/.

[4] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[5] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[6] Chelazzi, Miller, Duncan and Desimone (1993). A neural basis for visual search in inferior temporal cortex. Primary source.

[7] Soto, Heinke, Humphreys and Blanco (2005). Early, involuntary top-down guidance of attention from working memory. Primary source.

[8] Bichot, Rossi and Desimone (2005). Parallel and serial neural mechanisms for visual search in macaque area V4. Primary source.

[9] Wolfe (2021). Guided Search 6.0: An updated model of visual search. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
features_distinct_S1_D0_H1	128	98.4%	0.999	0.049
features_distinct_S1_D0_H2	128	99.2%	1.000	0.015
features_distinct_S1_D0_H4	128	100.0%	1.000	0.001
features_distinct_S1_D0_H8	128	100.0%	1.000	0.001
features_distinct_S1_D4_H1	128	100.0%	1.000	0.004
features_distinct_S1_D4_H2	128	100.0%	1.000	0.001
features_distinct_S1_D4_H4	128	100.0%	1.000	0.001
features_distinct_S1_D4_H8	128	100.0%	1.000	0.001
features_distinct_S1_D12_H1	128	99.2%	1.000	0.011
features_distinct_S1_D12_H2	128	99.2%	1.000	0.007
features_distinct_S1_D12_H4	128	100.0%	1.000	0.001
features_distinct_S1_D12_H8	128	100.0%	1.000	0.002
features_distinct_S2_D0_H1	128	100.0%	1.000	0.002
features_distinct_S2_D0_H2	128	99.2%	1.000	0.008
features_distinct_S2_D0_H4	128	100.0%	1.000	0.000
features_distinct_S2_D0_H8	128	100.0%	1.000	0.002
features_distinct_S2_D4_H1	128	99.2%	1.000	0.007
features_distinct_S2_D4_H2	128	99.2%	1.000	0.004
features_distinct_S2_D4_H4	128	100.0%	1.000	0.000
features_distinct_S2_D4_H8	128	100.0%	1.000	0.001
features_distinct_S2_D12_H1	128	99.2%	1.000	0.012
features_distinct_S2_D12_H2	128	99.2%	1.000	0.012
features_distinct_S2_D12_H4	128	100.0%	1.000	0.004
features_distinct_S2_D12_H8	128	100.0%	1.000	0.003
features_distinct_S4_D0_H1	128	100.0%	1.000	0.003
features_distinct_S4_D0_H2	128	99.2%	1.000	0.006
features_distinct_S4_D0_H4	128	100.0%	1.000	0.000
features_distinct_S4_D0_H8	128	100.0%	1.000	0.000
features_distinct_S4_D4_H1	128	99.2%	1.000	0.009
features_distinct_S4_D4_H2	128	99.2%	1.000	0.007
features_distinct_S4_D4_H4	128	100.0%	1.000	0.001
features_distinct_S4_D4_H8	128	100.0%	1.000	0.001
features_distinct_S4_D12_H1	128	99.2%	1.000	0.013
features_distinct_S4_D12_H2	128	99.2%	1.000	0.013
features_distinct_S4_D12_H4	128	99.2%	1.000	0.004
features_distinct_S4_D12_H8	128	100.0%	1.000	0.004
features_distinct_S8_D0_H1	128	99.2%	1.000	0.010
features_distinct_S8_D0_H2	128	99.2%	1.000	0.011
features_distinct_S8_D0_H4	128	100.0%	1.000	0.001
features_distinct_S8_D0_H8	128	100.0%	1.000	0.001
features_distinct_S8_D4_H1	128	99.2%	1.000	0.011
features_distinct_S8_D4_H2	128	99.2%	1.000	0.011
features_distinct_S8_D4_H4	128	100.0%	1.000	0.003
features_distinct_S8_D4_H8	128	100.0%	1.000	0.003
features_distinct_S8_D12_H1	128	99.2%	1.000	0.013
features_distinct_S8_D12_H2	128	99.2%	1.000	0.014
features_distinct_S8_D12_H4	128	99.2%	1.000	0.006
features_distinct_S8_D12_H8	128	99.2%	1.000	0.005
features_similar_S1_D0_H1	128	78.9%	0.963	0.308
features_similar_S1_D0_H2	128	90.6%	0.986	0.154
features_similar_S1_D0_H4	128	88.3%	0.990	0.169

Condition	n	Accuracy	AUC	Brier
features_similar_S1_D0_H8	128	87.5%	0.984	0.205
features_similar_S1_D4_H1	128	85.2%	0.978	0.231
features_similar_S1_D4_H2	128	89.8%	0.988	0.167
features_similar_S1_D4_H4	128	87.5%	0.986	0.202
features_similar_S1_D4_H8	128	85.9%	0.982	0.212
features_similar_S1_D12_H1	128	83.6%	0.980	0.250
features_similar_S1_D12_H2	128	88.3%	0.984	0.184
features_similar_S1_D12_H4	128	86.7%	0.985	0.216
features_similar_S1_D12_H8	128	85.2%	0.981	0.227
features_similar_S2_D0_H1	128	91.4%	0.986	0.150
features_similar_S2_D0_H2	128	92.2%	0.989	0.138
features_similar_S2_D0_H4	128	91.4%	0.993	0.146
features_similar_S2_D0_H8	128	89.8%	0.987	0.173
features_similar_S2_D4_H1	128	86.7%	0.986	0.212
features_similar_S2_D4_H2	128	89.1%	0.990	0.162
features_similar_S2_D4_H4	128	87.5%	0.991	0.166
features_similar_S2_D4_H8	128	89.1%	0.988	0.183
features_similar_S2_D12_H1	128	85.2%	0.984	0.226
features_similar_S2_D12_H2	128	85.9%	0.986	0.193
features_similar_S2_D12_H4	128	85.9%	0.989	0.189
features_similar_S2_D12_H8	128	85.9%	0.987	0.210
features_similar_S4_D0_H1	128	88.3%	0.988	0.158
features_similar_S4_D0_H2	128	91.4%	0.993	0.125
features_similar_S4_D0_H4	128	90.6%	0.993	0.143
features_similar_S4_D0_H8	128	90.6%	0.988	0.170
features_similar_S4_D4_H1	128	85.9%	0.986	0.221
features_similar_S4_D4_H2	128	88.3%	0.989	0.173
features_similar_S4_D4_H4	128	88.3%	0.991	0.170
features_similar_S4_D4_H8	128	89.1%	0.988	0.184
features_similar_S4_D12_H1	128	84.4%	0.985	0.232
features_similar_S4_D12_H2	128	85.9%	0.985	0.198
features_similar_S4_D12_H4	128	85.9%	0.989	0.193
features_similar_S4_D12_H8	128	86.7%	0.987	0.210
features_similar_S8_D0_H1	128	89.1%	0.986	0.179
features_similar_S8_D0_H2	128	88.3%	0.990	0.157
features_similar_S8_D0_H4	128	90.6%	0.991	0.147
features_similar_S8_D0_H8	128	89.1%	0.988	0.175
features_similar_S8_D4_H1	128	84.4%	0.987	0.222
features_similar_S8_D4_H2	128	86.7%	0.988	0.184
features_similar_S8_D4_H4	128	88.3%	0.990	0.173
features_similar_S8_D4_H8	128	89.1%	0.988	0.191
features_similar_S8_D12_H1	128	85.2%	0.985	0.236
features_similar_S8_D12_H2	128	85.2%	0.985	0.201
features_similar_S8_D12_H4	128	85.2%	0.988	0.197
features_similar_S8_D12_H8	128	86.7%	0.987	0.212
objects_distinct_S1_D0_H1	128	28.9%	0.543	0.751
objects_distinct_S1_D0_H2	128	28.1%	0.511	0.816
objects_distinct_S1_D0_H4	128	26.6%	0.534	0.856
objects_distinct_S1_D0_H8	128	23.4%	0.519	0.871
objects_distinct_S1_D4_H1	128	30.5%	0.556	0.808
objects_distinct_S1_D4_H2	128	31.2%	0.563	0.862
objects_distinct_S1_D4_H4	128	28.1%	0.560	0.897
objects_distinct_S1_D4_H8	128	27.3%	0.549	0.916
objects_distinct_S1_D12_H1	128	32.0%	0.572	0.803

Condition	n	Accuracy	AUC	Brier
objects_distinct_S1_D12_H2	128	29.7%	0.567	0.862
objects_distinct_S1_D12_H4	128	28.9%	0.566	0.891
objects_distinct_S1_D12_H8	128	27.3%	0.557	0.904
objects_distinct_S2_D0_H1	128	27.3%	0.532	0.767
objects_distinct_S2_D0_H2	128	25.8%	0.500	0.865
objects_distinct_S2_D0_H4	128	27.3%	0.534	0.889
objects_distinct_S2_D0_H8	128	26.6%	0.519	0.899
objects_distinct_S2_D4_H1	128	28.9%	0.552	0.823
objects_distinct_S2_D4_H2	128	32.0%	0.568	0.859
objects_distinct_S2_D4_H4	128	29.7%	0.565	0.897
objects_distinct_S2_D4_H8	128	27.3%	0.545	0.921
objects_distinct_S2_D12_H1	128	32.8%	0.562	0.809
objects_distinct_S2_D12_H2	128	32.8%	0.571	0.856
objects_distinct_S2_D12_H4	128	32.0%	0.573	0.882
objects_distinct_S2_D12_H8	128	27.3%	0.554	0.900
objects_distinct_S4_D0_H1	128	28.9%	0.538	0.795
objects_distinct_S4_D0_H2	128	34.4%	0.568	0.844
objects_distinct_S4_D0_H4	128	25.8%	0.551	0.912
objects_distinct_S4_D0_H8	128	24.2%	0.532	0.933
objects_distinct_S4_D4_H1	128	30.5%	0.550	0.842
objects_distinct_S4_D4_H2	128	32.0%	0.569	0.876
objects_distinct_S4_D4_H4	128	30.5%	0.564	0.910
objects_distinct_S4_D4_H8	128	26.6%	0.549	0.926
objects_distinct_S4_D12_H1	128	32.8%	0.560	0.818
objects_distinct_S4_D12_H2	128	33.6%	0.571	0.853
objects_distinct_S4_D12_H4	128	31.2%	0.576	0.880
objects_distinct_S4_D12_H8	128	28.1%	0.560	0.898
objects_distinct_S8_D0_H1	128	28.9%	0.557	0.812
objects_distinct_S8_D0_H2	128	34.4%	0.575	0.850
objects_distinct_S8_D0_H4	128	26.6%	0.561	0.910
objects_distinct_S8_D0_H8	128	26.6%	0.548	0.921
objects_distinct_S8_D4_H1	128	32.8%	0.560	0.831
objects_distinct_S8_D4_H2	128	34.4%	0.576	0.858
objects_distinct_S8_D4_H4	128	29.7%	0.576	0.896
objects_distinct_S8_D4_H8	128	25.8%	0.560	0.911
objects_distinct_S8_D12_H1	128	35.2%	0.562	0.812
objects_distinct_S8_D12_H2	128	32.0%	0.579	0.839
objects_distinct_S8_D12_H4	128	31.2%	0.583	0.869
objects_distinct_S8_D12_H8	128	27.3%	0.571	0.888
objects_similar_S1_D0_H1	128	28.9%	0.495	0.760
objects_similar_S1_D0_H2	128	24.2%	0.516	0.802
objects_similar_S1_D0_H4	128	28.9%	0.525	0.832
objects_similar_S1_D0_H8	128	25.8%	0.504	0.862
objects_similar_S1_D4_H1	128	29.7%	0.527	0.881
objects_similar_S1_D4_H2	128	29.7%	0.552	0.888
objects_similar_S1_D4_H4	128	30.5%	0.545	0.929
objects_similar_S1_D4_H8	128	28.1%	0.539	0.934
objects_similar_S1_D12_H1	128	32.0%	0.546	0.864
objects_similar_S1_D12_H2	128	28.1%	0.557	0.885
objects_similar_S1_D12_H4	128	28.9%	0.549	0.933
objects_similar_S1_D12_H8	128	27.3%	0.537	0.940
objects_similar_S2_D0_H1	128	29.7%	0.511	0.773
objects_similar_S2_D0_H2	128	27.3%	0.517	0.817
objects_similar_S2_D0_H4	128	27.3%	0.537	0.846

Condition	n	Accuracy	AUC	Brier
objects_similar_S2_D0_H8	128	28.1%	0.520	0.867
objects_similar_S2_D4_H1	128	26.6%	0.541	0.866
objects_similar_S2_D4_H2	128	27.3%	0.554	0.907
objects_similar_S2_D4_H4	128	28.1%	0.547	0.940
objects_similar_S2_D4_H8	128	25.0%	0.540	0.942
objects_similar_S2_D12_H1	128	28.1%	0.553	0.855
objects_similar_S2_D12_H2	128	28.1%	0.557	0.896
objects_similar_S2_D12_H4	128	28.9%	0.554	0.932
objects_similar_S2_D12_H8	128	27.3%	0.543	0.930
objects_similar_S4_D0_H1	128	31.2%	0.543	0.809
objects_similar_S4_D0_H2	128	27.3%	0.535	0.890
objects_similar_S4_D0_H4	128	28.9%	0.535	0.907
objects_similar_S4_D0_H8	128	28.1%	0.529	0.915
objects_similar_S4_D4_H1	128	30.5%	0.544	0.874
objects_similar_S4_D4_H2	128	28.1%	0.549	0.917
objects_similar_S4_D4_H4	128	27.3%	0.545	0.951
objects_similar_S4_D4_H8	128	24.2%	0.537	0.948
objects_similar_S4_D12_H1	128	28.1%	0.557	0.854
objects_similar_S4_D12_H2	128	26.6%	0.554	0.897
objects_similar_S4_D12_H4	128	29.7%	0.551	0.934
objects_similar_S4_D12_H8	128	27.3%	0.537	0.932
objects_similar_S8_D0_H1	128	29.7%	0.551	0.841
objects_similar_S8_D0_H2	128	32.0%	0.547	0.891
objects_similar_S8_D0_H4	128	32.0%	0.557	0.912
objects_similar_S8_D0_H8	128	29.7%	0.547	0.918
objects_similar_S8_D4_H1	128	32.0%	0.547	0.866
objects_similar_S8_D4_H2	128	27.3%	0.552	0.908
objects_similar_S8_D4_H4	128	26.6%	0.546	0.942
objects_similar_S8_D4_H8	128	25.0%	0.539	0.944
objects_similar_S8_D12_H1	128	28.9%	0.555	0.843
objects_similar_S8_D12_H2	128	26.6%	0.553	0.887
objects_similar_S8_D12_H4	128	27.3%	0.551	0.923
objects_similar_S8_D12_H8	128	25.0%	0.539	0.926
fractals_distinct_S1_D0_H1	128	30.5%	0.615	0.825
fractals_distinct_S1_D0_H2	128	44.5%	0.723	0.842
fractals_distinct_S1_D0_H4	128	48.4%	0.753	0.822
fractals_distinct_S1_D0_H8	128	46.9%	0.759	0.797
fractals_distinct_S1_D4_H1	128	39.8%	0.680	0.988
fractals_distinct_S1_D4_H2	128	45.3%	0.712	0.911
fractals_distinct_S1_D4_H4	128	48.4%	0.733	0.879
fractals_distinct_S1_D4_H8	128	47.7%	0.727	0.853
fractals_distinct_S1_D12_H1	128	39.8%	0.693	0.953
fractals_distinct_S1_D12_H2	128	44.5%	0.716	0.874
fractals_distinct_S1_D12_H4	128	50.8%	0.732	0.853
fractals_distinct_S1_D12_H8	128	48.4%	0.727	0.853
fractals_distinct_S2_D0_H1	128	37.5%	0.688	0.909
fractals_distinct_S2_D0_H2	128	46.1%	0.756	0.819
fractals_distinct_S2_D0_H4	128	53.1%	0.768	0.777
fractals_distinct_S2_D0_H8	128	49.2%	0.773	0.775
fractals_distinct_S2_D4_H1	128	40.6%	0.701	0.941
fractals_distinct_S2_D4_H2	128	44.5%	0.736	0.865
fractals_distinct_S2_D4_H4	128	48.4%	0.758	0.832
fractals_distinct_S2_D4_H8	128	46.9%	0.753	0.824
fractals_distinct_S2_D12_H1	128	41.4%	0.712	0.928

Condition	n	Accuracy	AUC	Brier
fractals_distinct_S2_D12_H2	128	44.5%	0.739	0.848
fractals_distinct_S2_D12_H4	128	46.1%	0.752	0.837
fractals_distinct_S2_D12_H8	128	48.4%	0.742	0.830
fractals_distinct_S4_D0_H1	128	42.2%	0.697	0.939
fractals_distinct_S4_D0_H2	128	43.8%	0.732	0.891
fractals_distinct_S4_D0_H4	128	48.4%	0.756	0.839
fractals_distinct_S4_D0_H8	128	46.1%	0.752	0.834
fractals_distinct_S4_D4_H1	128	40.6%	0.708	0.936
fractals_distinct_S4_D4_H2	128	44.5%	0.738	0.858
fractals_distinct_S4_D4_H4	128	48.4%	0.756	0.832
fractals_distinct_S4_D4_H8	128	45.3%	0.746	0.830
fractals_distinct_S4_D12_H1	128	37.5%	0.713	0.929
fractals_distinct_S4_D12_H2	128	44.5%	0.740	0.846
fractals_distinct_S4_D12_H4	128	46.1%	0.751	0.838
fractals_distinct_S4_D12_H8	128	46.9%	0.739	0.845
fractals_distinct_S8_D0_H1	128	42.2%	0.700	0.935
fractals_distinct_S8_D0_H2	128	45.3%	0.729	0.875
fractals_distinct_S8_D0_H4	128	48.4%	0.750	0.839
fractals_distinct_S8_D0_H8	128	46.1%	0.747	0.813
fractals_distinct_S8_D4_H1	128	40.6%	0.709	0.937
fractals_distinct_S8_D4_H2	128	43.8%	0.742	0.849
fractals_distinct_S8_D4_H4	128	47.7%	0.757	0.828
fractals_distinct_S8_D4_H8	128	46.1%	0.746	0.819
fractals_distinct_S8_D12_H1	128	35.2%	0.712	0.939
fractals_distinct_S8_D12_H2	128	41.4%	0.743	0.849
fractals_distinct_S8_D12_H4	128	45.3%	0.751	0.837
fractals_distinct_S8_D12_H8	128	46.1%	0.740	0.831
fractals_similar_S1_D0_H1	128	23.4%	0.481	0.932
fractals_similar_S1_D0_H2	128	26.6%	0.521	1.140
fractals_similar_S1_D0_H4	128	26.6%	0.532	1.139
fractals_similar_S1_D0_H8	128	26.6%	0.541	1.097
fractals_similar_S1_D4_H1	128	28.9%	0.509	1.159
fractals_similar_S1_D4_H2	128	28.9%	0.518	1.133
fractals_similar_S1_D4_H4	128	30.5%	0.520	1.120
fractals_similar_S1_D4_H8	128	30.5%	0.542	1.080
fractals_similar_S1_D12_H1	128	32.0%	0.519	1.138
fractals_similar_S1_D12_H2	128	28.9%	0.526	1.103
fractals_similar_S1_D12_H4	128	30.5%	0.530	1.106
fractals_similar_S1_D12_H8	128	30.5%	0.542	1.078
fractals_similar_S2_D0_H1	128	28.1%	0.524	1.091
fractals_similar_S2_D0_H2	128	29.7%	0.523	1.134
fractals_similar_S2_D0_H4	128	28.9%	0.536	1.101
fractals_similar_S2_D0_H8	128	30.5%	0.545	1.072
fractals_similar_S2_D4_H1	128	29.7%	0.519	1.117
fractals_similar_S2_D4_H2	128	31.2%	0.520	1.134
fractals_similar_S2_D4_H4	128	31.2%	0.534	1.087
fractals_similar_S2_D4_H8	128	30.5%	0.549	1.051
fractals_similar_S2_D12_H1	128	31.2%	0.531	1.116
fractals_similar_S2_D12_H2	128	27.3%	0.527	1.124
fractals_similar_S2_D12_H4	128	28.9%	0.541	1.078
fractals_similar_S2_D12_H8	128	28.1%	0.552	1.050
fractals_similar_S4_D0_H1	128	28.1%	0.509	1.129
fractals_similar_S4_D0_H2	128	25.0%	0.514	1.167
fractals_similar_S4_D0_H4	128	32.0%	0.538	1.090

Condition	n	Accuracy	AUC	Brier
fractals_similar_S4_D0_H8	128	28.9%	0.548	1.065
fractals_similar_S4_D4_H1	128	28.9%	0.523	1.113
fractals_similar_S4_D4_H2	128	28.9%	0.520	1.123
fractals_similar_S4_D4_H4	128	31.2%	0.536	1.086
fractals_similar_S4_D4_H8	128	31.2%	0.549	1.053
fractals_similar_S4_D12_H1	128	29.7%	0.531	1.112
fractals_similar_S4_D12_H2	128	25.8%	0.527	1.121
fractals_similar_S4_D12_H4	128	29.7%	0.541	1.076
fractals_similar_S4_D12_H8	128	28.9%	0.551	1.043
fractals_similar_S8_D0_H1	128	29.7%	0.520	1.140
fractals_similar_S8_D0_H2	128	26.6%	0.523	1.137
fractals_similar_S8_D0_H4	128	32.8%	0.544	1.081
fractals_similar_S8_D0_H8	128	29.7%	0.551	1.060
fractals_similar_S8_D4_H1	128	30.5%	0.526	1.121
fractals_similar_S8_D4_H2	128	26.6%	0.523	1.125
fractals_similar_S8_D4_H4	128	29.7%	0.539	1.082
fractals_similar_S8_D4_H8	128	31.2%	0.550	1.048
fractals_similar_S8_D12_H1	128	30.5%	0.531	1.115
fractals_similar_S8_D12_H2	128	25.8%	0.531	1.117
fractals_similar_S8_D12_H4	128	28.1%	0.542	1.070
fractals_similar_S8_D12_H8	128	29.7%	0.552	1.035

B Every-frame attention atlas

Condition features_distinct_S8_D12_H8. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 32 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

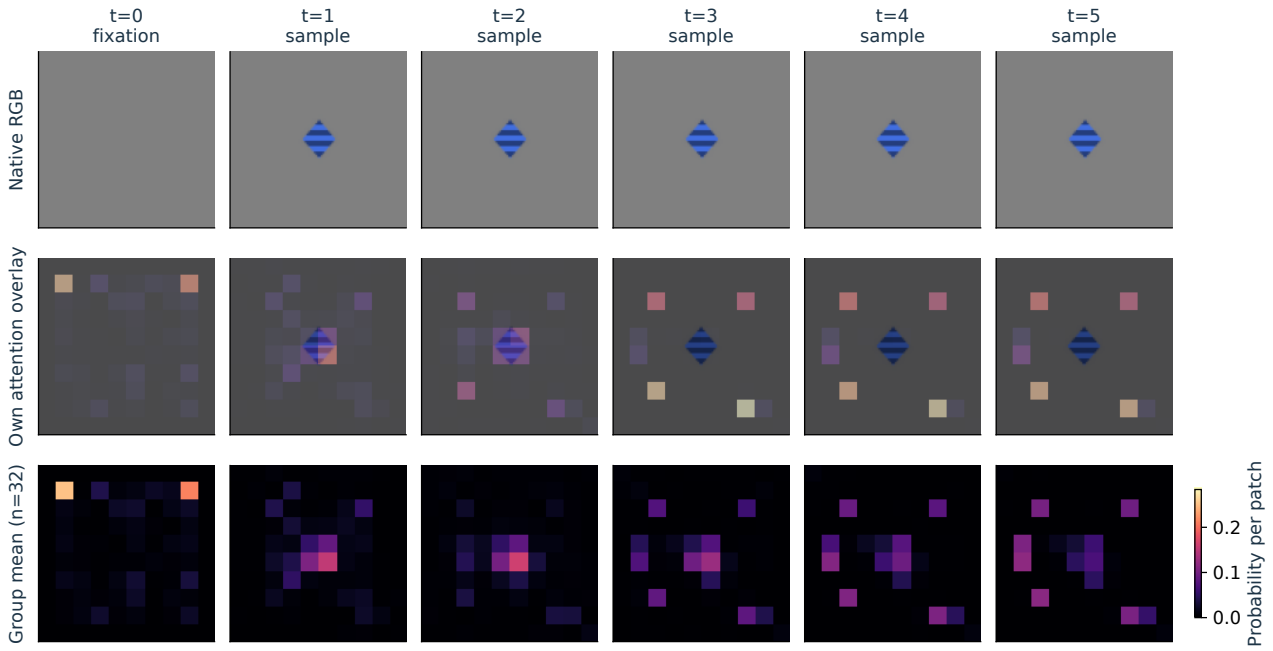


Figure 33: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

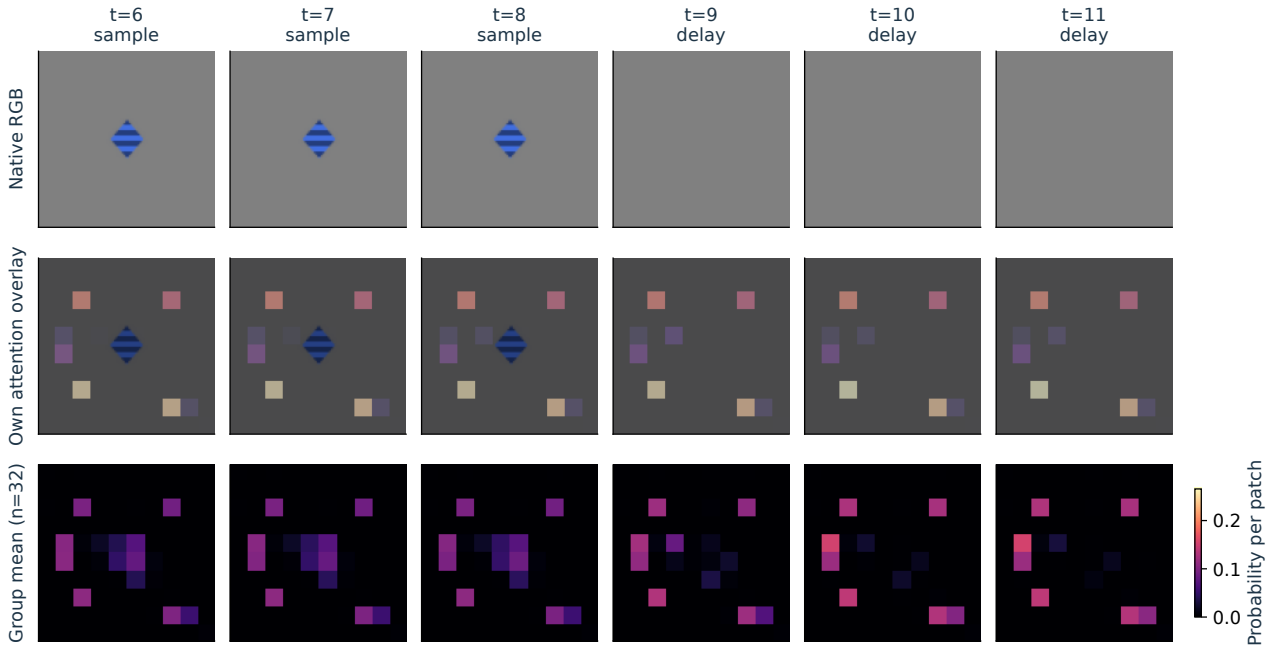


Figure 34: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

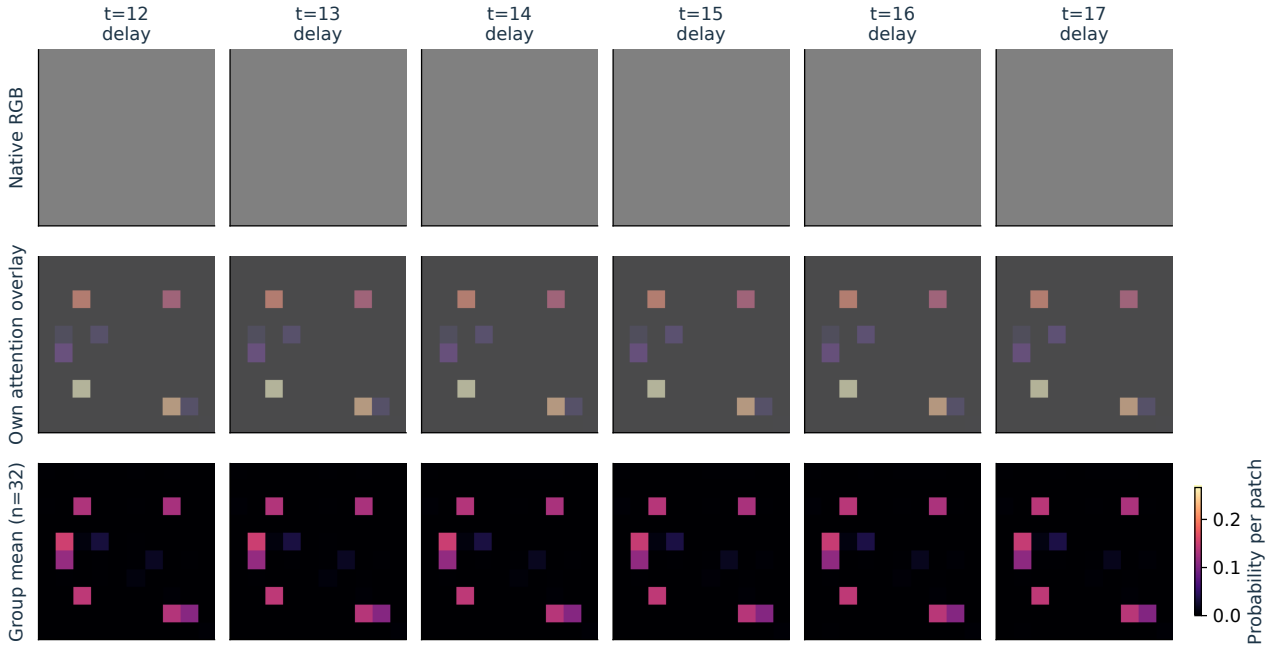


Figure 35: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

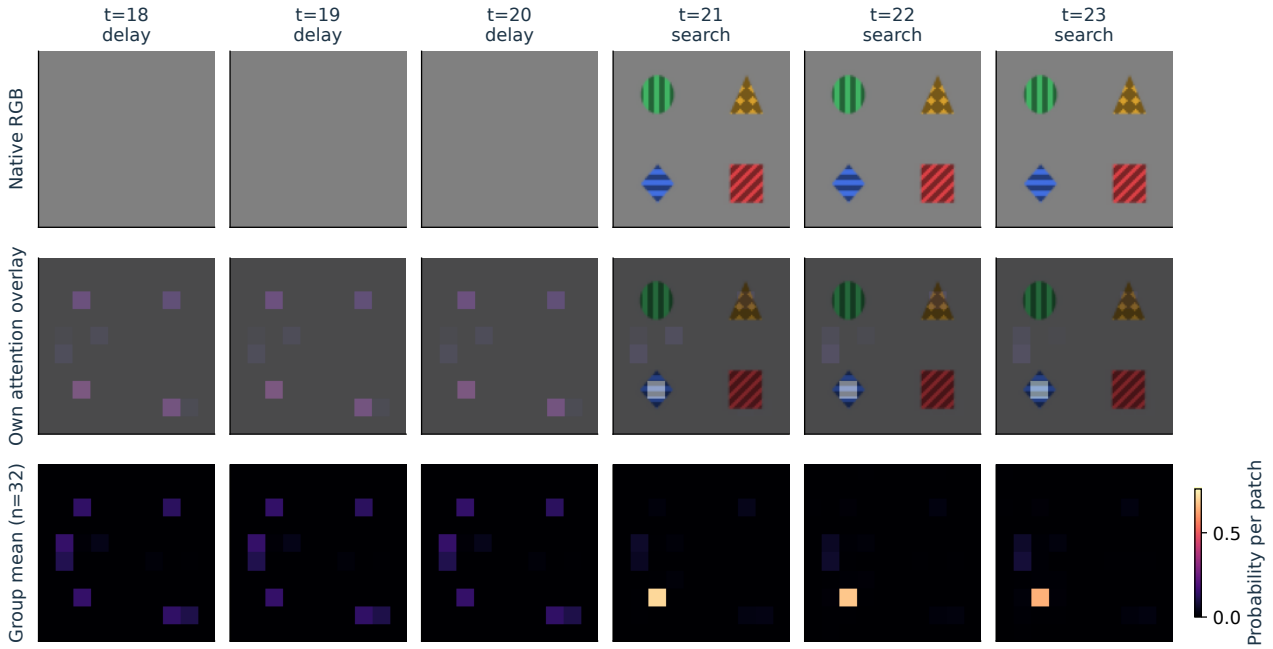


Figure 36: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

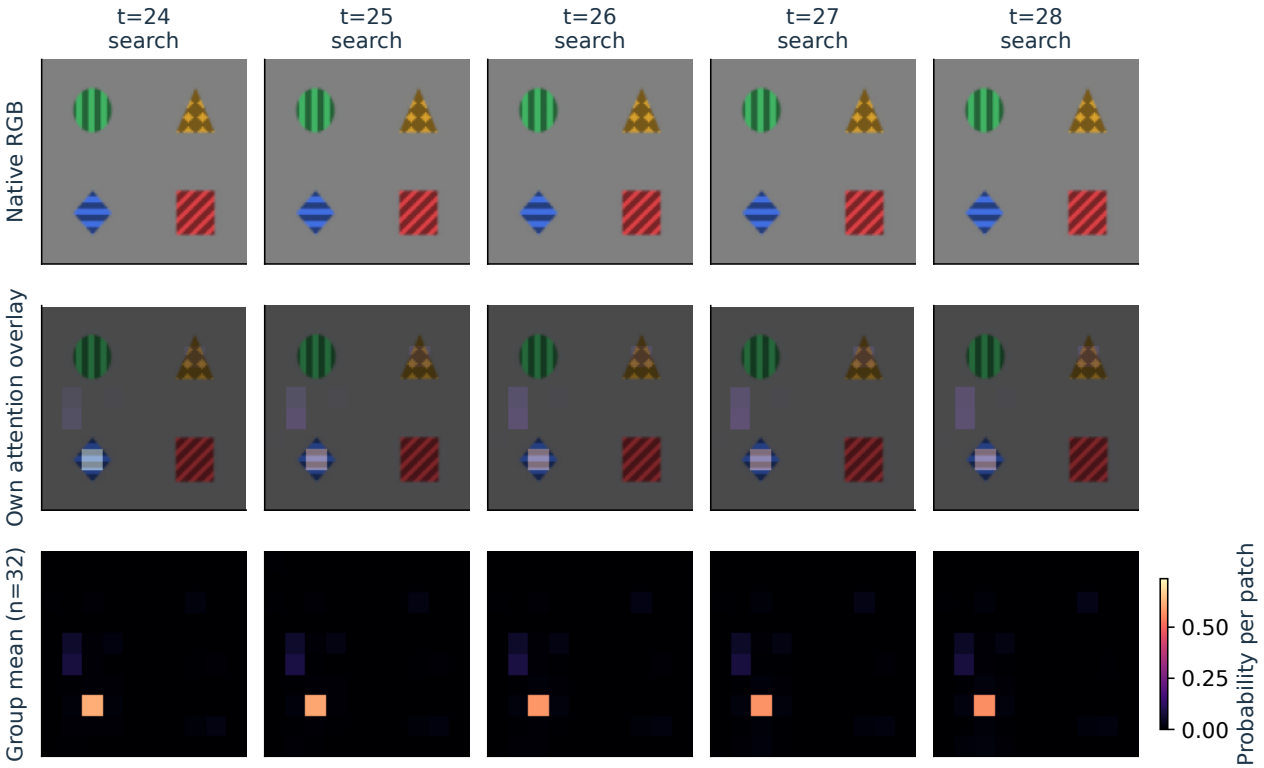


Figure 37: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

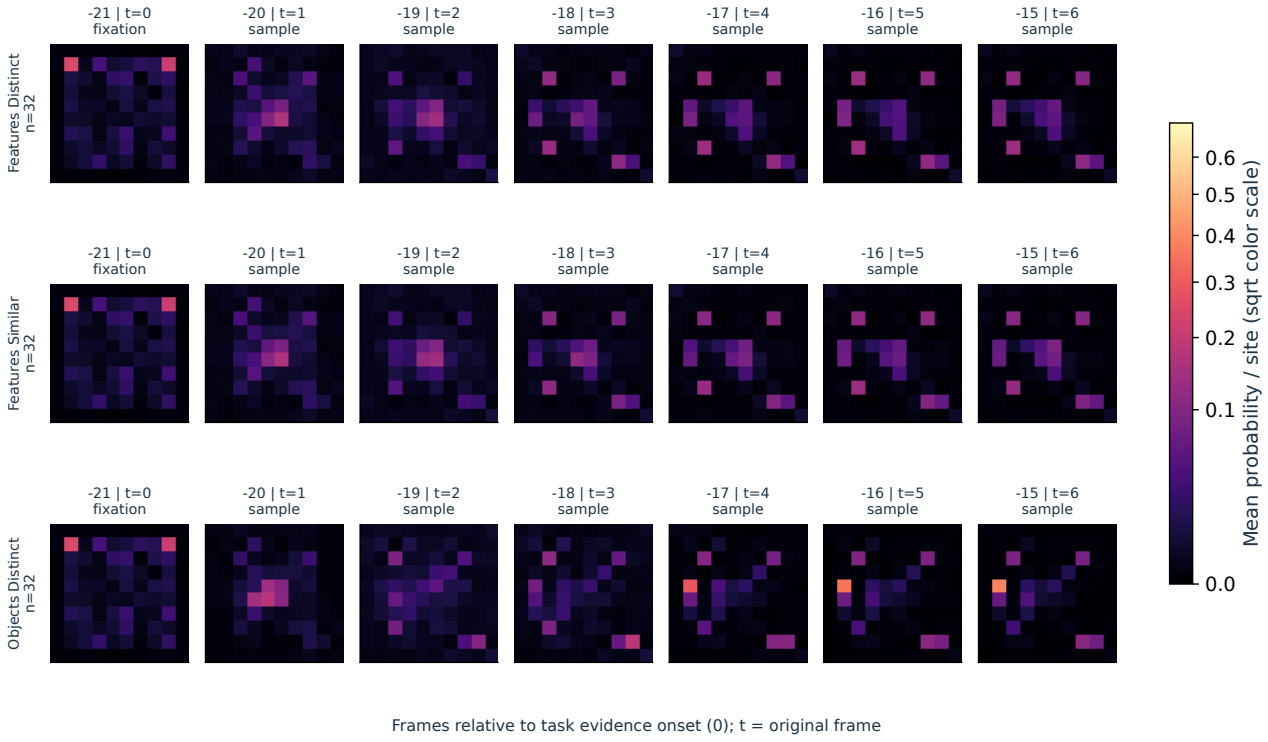


Figure 38: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

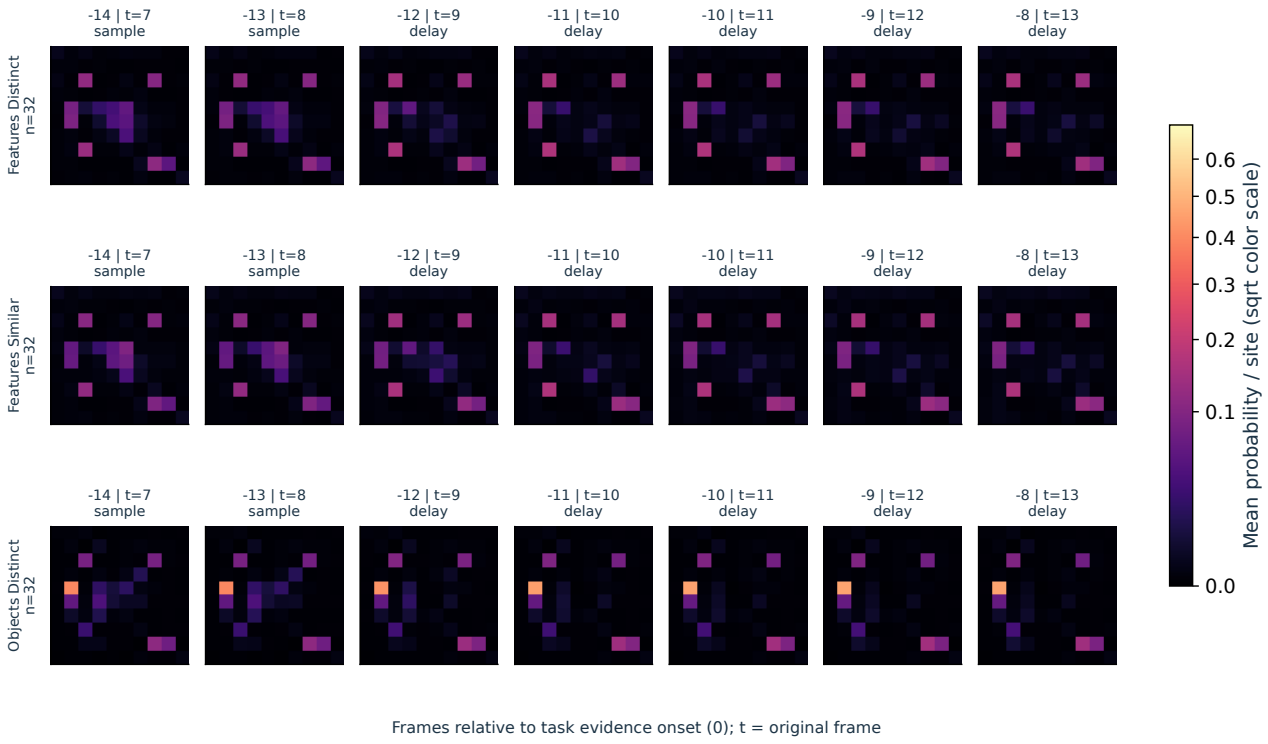


Figure 39: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

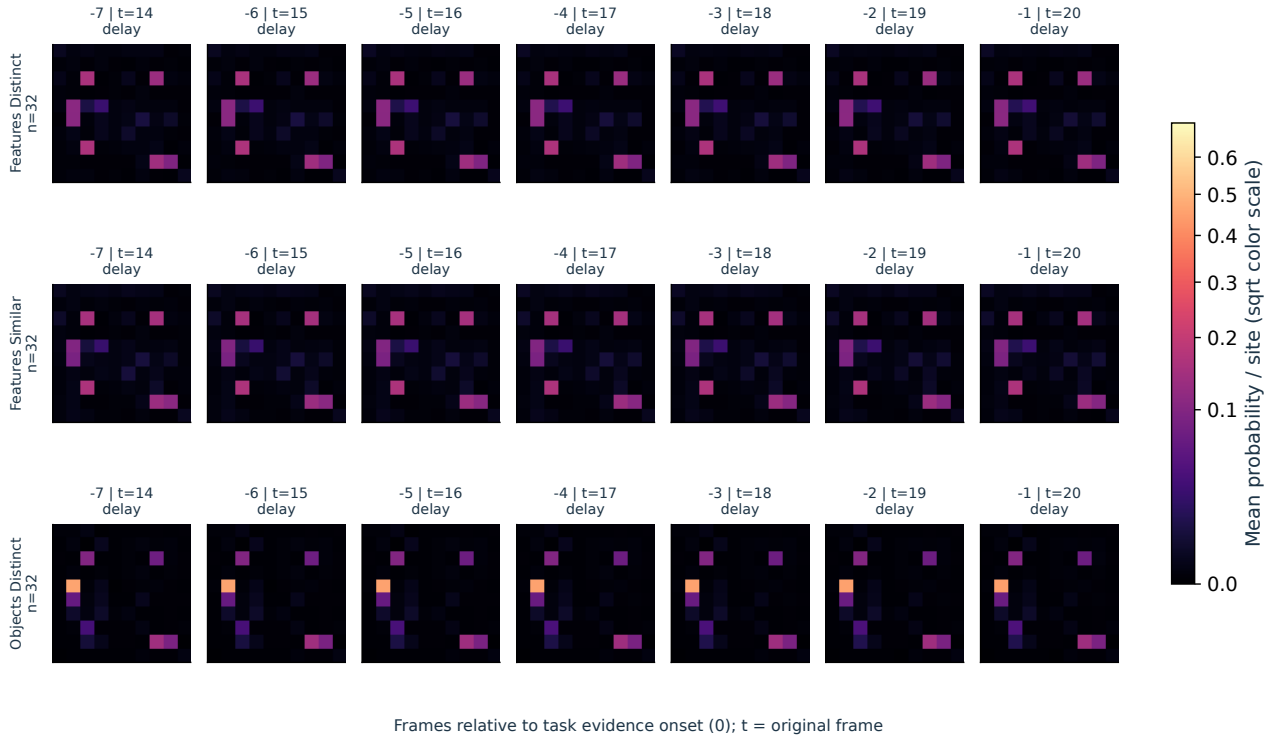


Figure 40: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

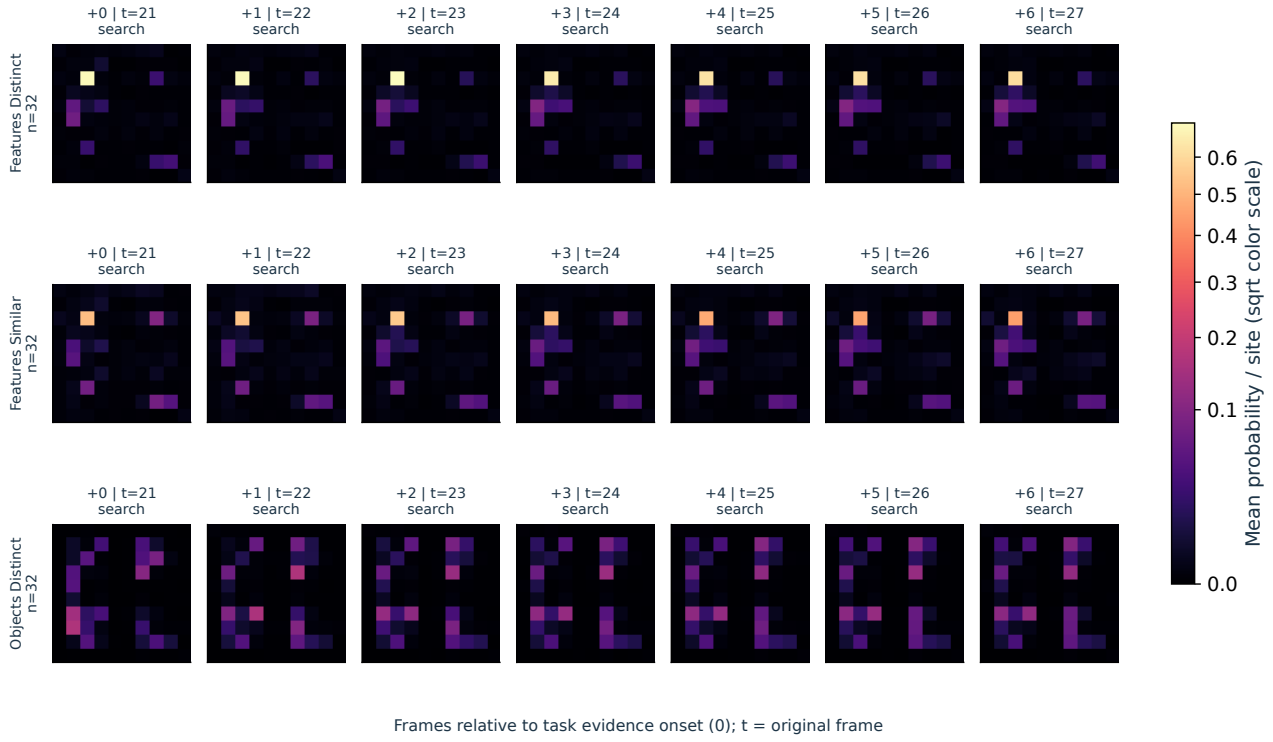


Figure 41: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

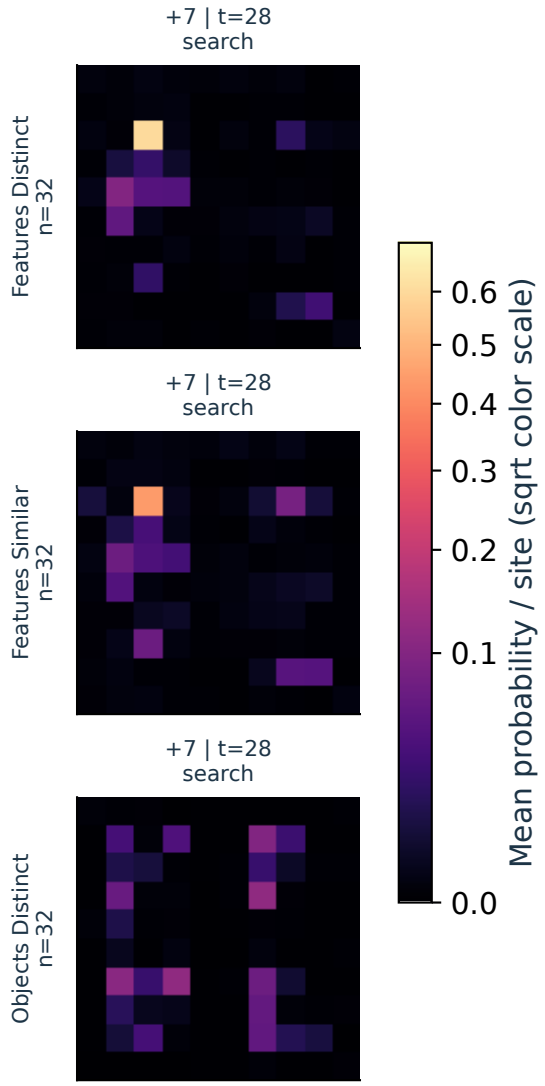


Figure 42: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

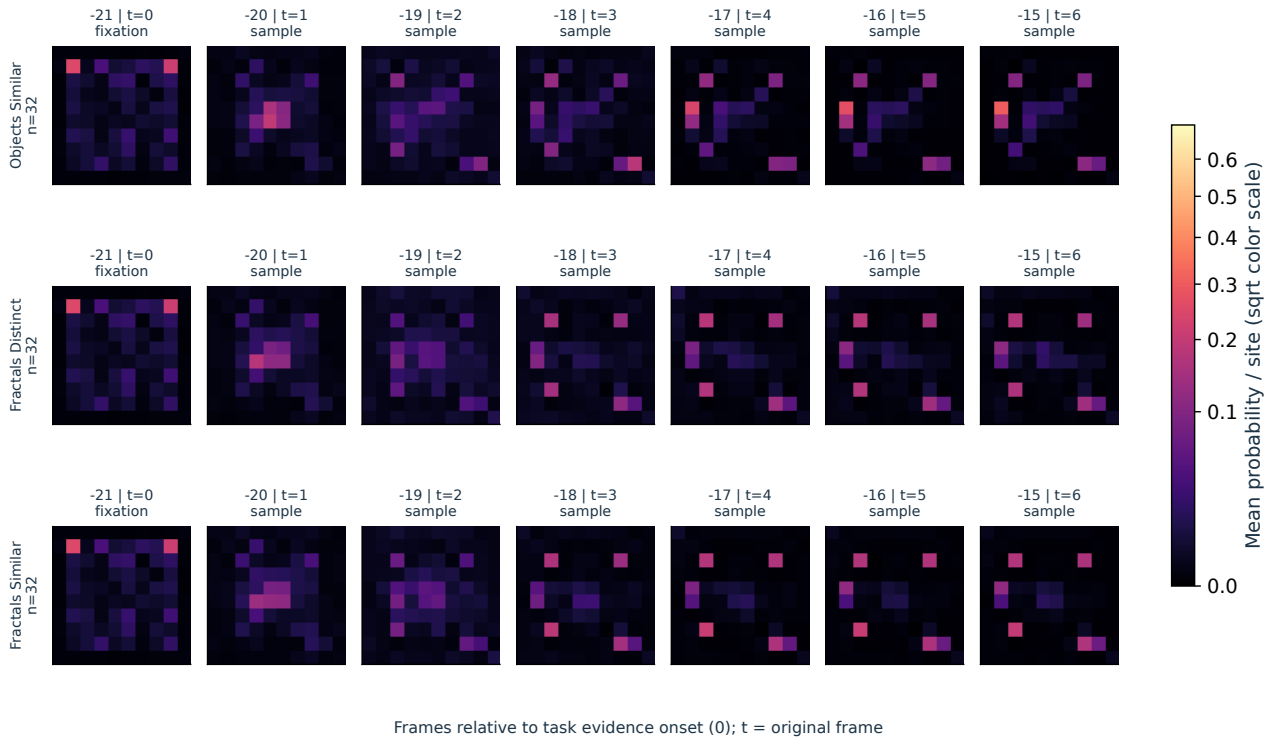


Figure 43: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

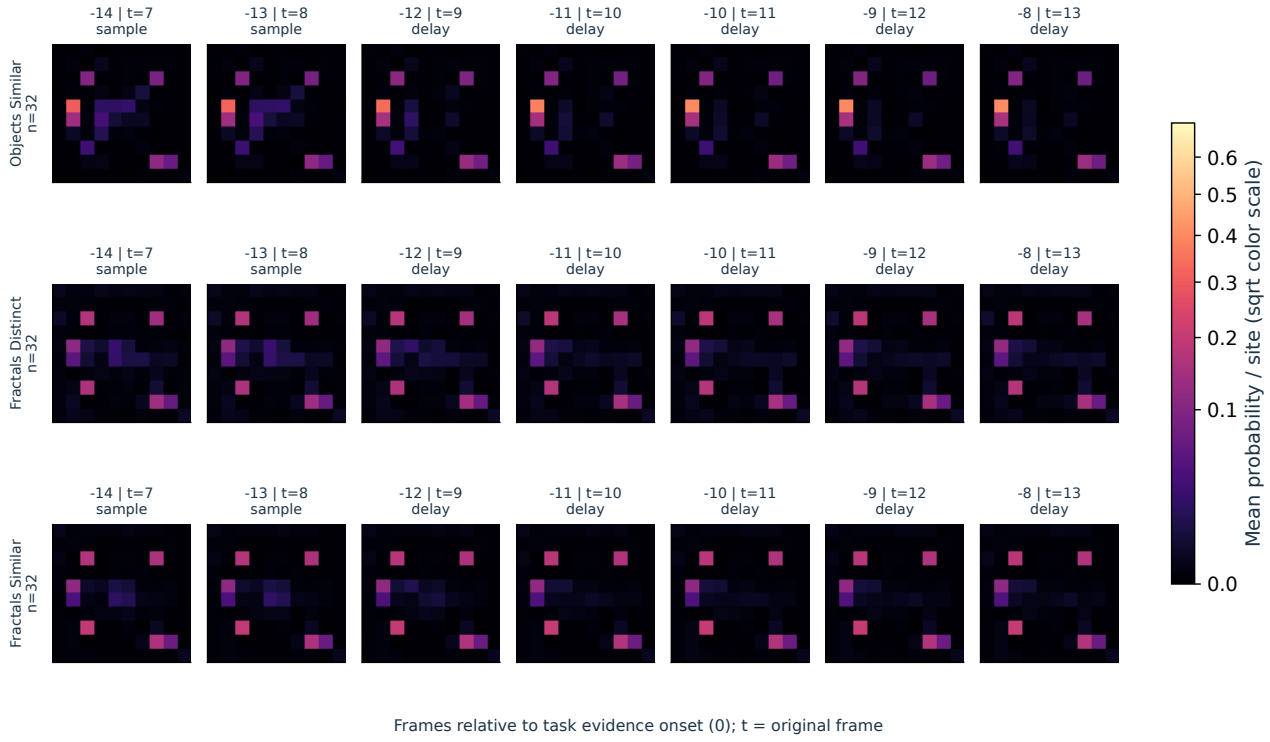


Figure 44: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

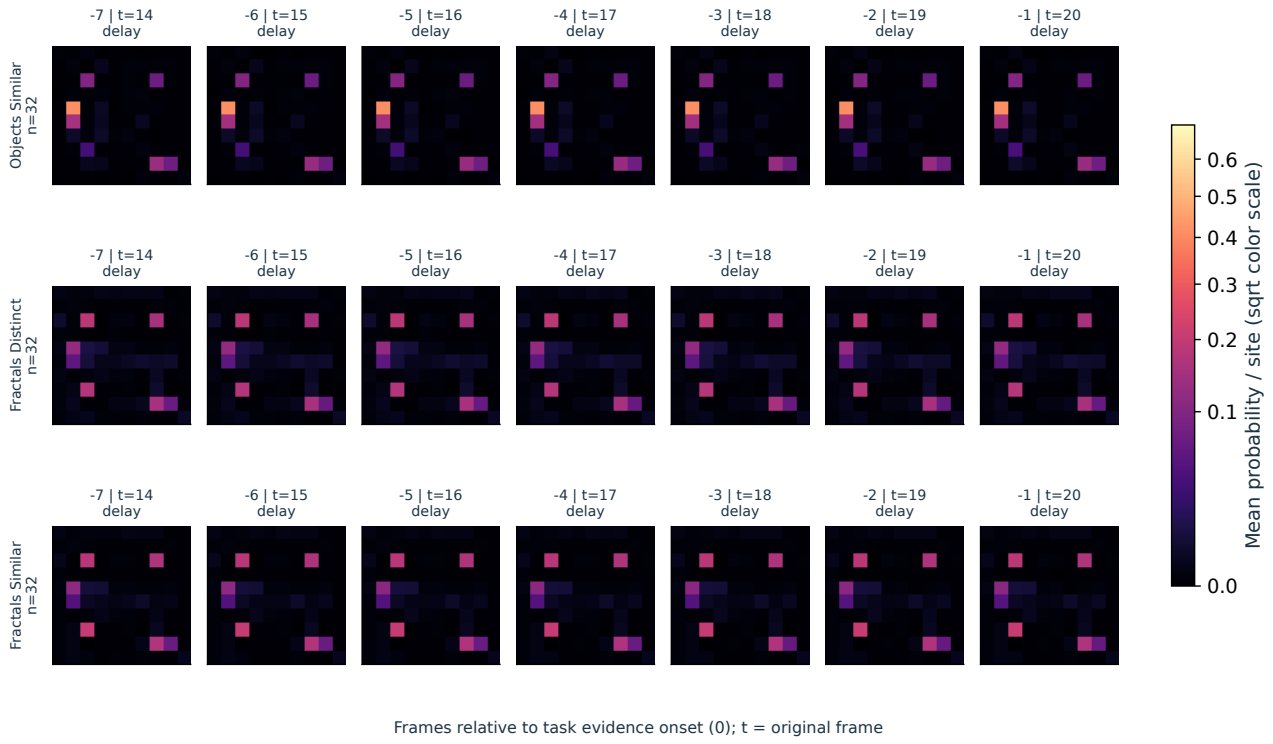


Figure 45: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

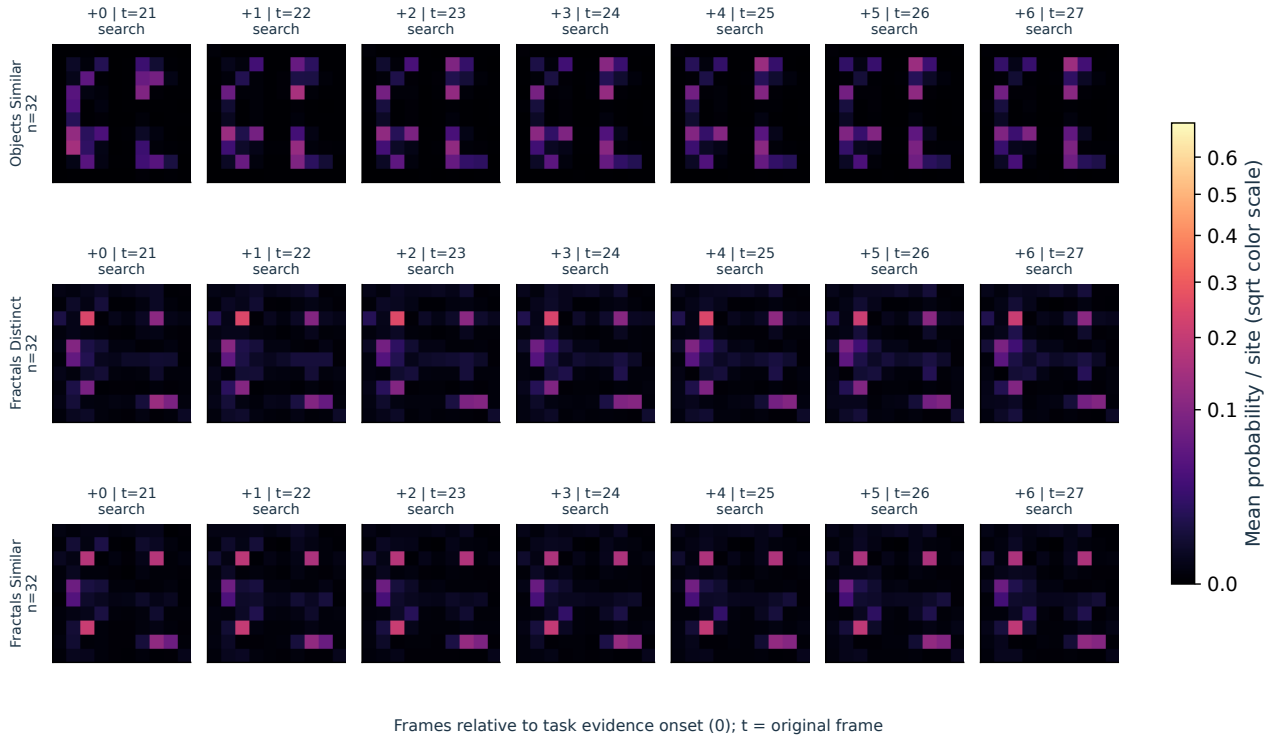


Figure 46: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

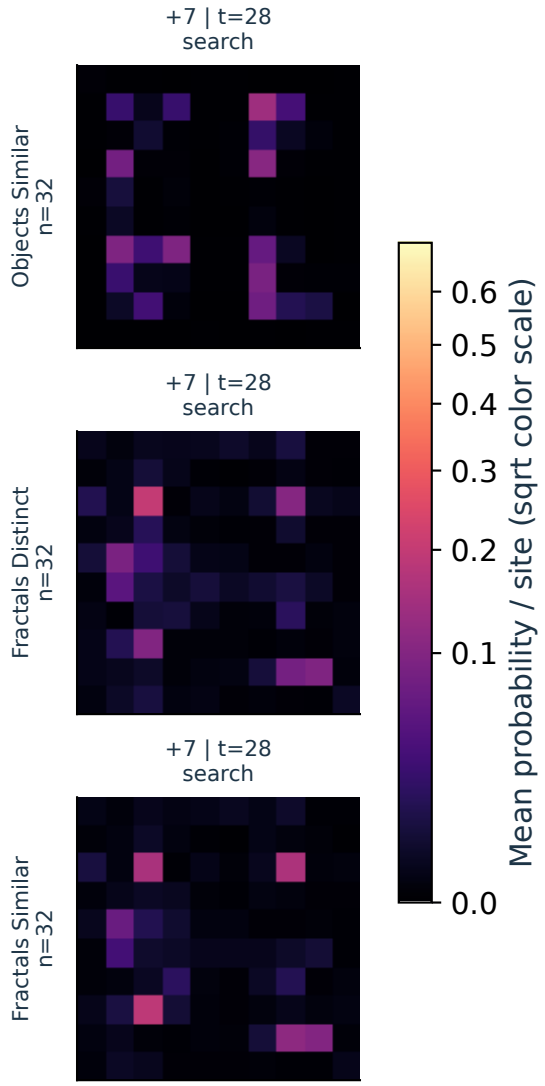


Figure 47: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Feature contrast, exposure time and location reports in pop-out search

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized popout search in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 24 native conditions and 3,072 fresh trials, equal-condition accuracy was 86.4% (95% interval [85.2%, 87.5%]). For the representative condition color_X12_H8, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 1.000. The largest observed impairment in that condition was distractor_+8_evidence, changing accuracy by -93.8 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

One item differs from the other three in color or orientation. The observer reports its location. There is no preceding sample to remember, so the task isolates selection from simultaneous feature contrast in an array.

We cross feature difference with array exposure. A benefit from exposure duration can arise from more stable selection or more complete readout even when the target is physically present from the first array frame. We therefore examine both time-resolved attention and the evolving location code.

The biological motivation is developed through [1], [2]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Feature-integration and Guided Search theories distinguish feature-based guidance from the later identification or response process. This task provides a controlled case with one differing feature among three matching distractors. Exposure duration can affect either the emergence of a useful priority distribution or the downstream conversion of that distribution into a location report. Recording both attention and recurrent answer information allows these possibilities to be compared. Because the target is present from the first array frame, a delayed location code can be interpreted against an unchanging physical stimulus.

1.2 Biological experiments motivating this task

Experimental question. How does a physically distinctive item become a selected location over successive processing steps?

Schall and Hanes (1993) Initial FEF visual responses did not distinguish target from distractor. Activity subsequently evolved to identify the target before the eye movement, separating stimulus onset responses from selection. [5]

Bichot, Rossi and Desimone (2005) During free visual search, V4 responses favored objects sharing target features across the display. The experiment related feature-based parallel bias to selection during search among complex items. [6]

Ling, Liu and Carrasco (2009) External-noise motion experiments separated spatial from feature-based attention. Threshold-versus-noise functions supported different combinations of population gain and tuning, giving behavioral tests of attentional mechanisms. [7]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Where is the item that differs from the other three?

Response labels are 0: top-left; 1: top-right; 2: bottom-left; 3: bottom-right. The native suite contains 24 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use color_X12_H8, orientation_X5_H8.

1.4 Stimulus sequence and spatial layout

For the representative native condition color_X12_H8, the sequence has 9 frames, indexed from zero: frame 0: fixation; frames 1–8: search. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Singleton feature difference	12

Stimulus centers in the 100 by 100 input are [[25, 25], [75, 25], [25, 75], [75, 75]]. Coordinate x increases rightward and y downward.

The task uses four locations in every trial. Its behavioral curves establish sensitivity and exposure dependence at that set size. The attention interventions reveal whether directing priority toward or away from the target changes localization, while decoding tests whether the target is represented before it dominates the selected state.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 86.4%, with macro AUC 0.938. The matched final checkpoint changed accuracy by -4.17 percentage points on average across conditions, using 32 same-movie comparisons per condition. The lowest native cell was orientation_X5_H1 at 25.0%; the highest was color_

X12_H2 at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

Marginal accuracy by exposure was 1: 56.8%; 2: 95.3%; 4: 96.6%; 8: 96.7%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

Marginal accuracy by difference was 12: 87.1%; 36: 98.0%; 72: 100.0%; 5: 69.7%; 15: 80.7%; 45: 82.6%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

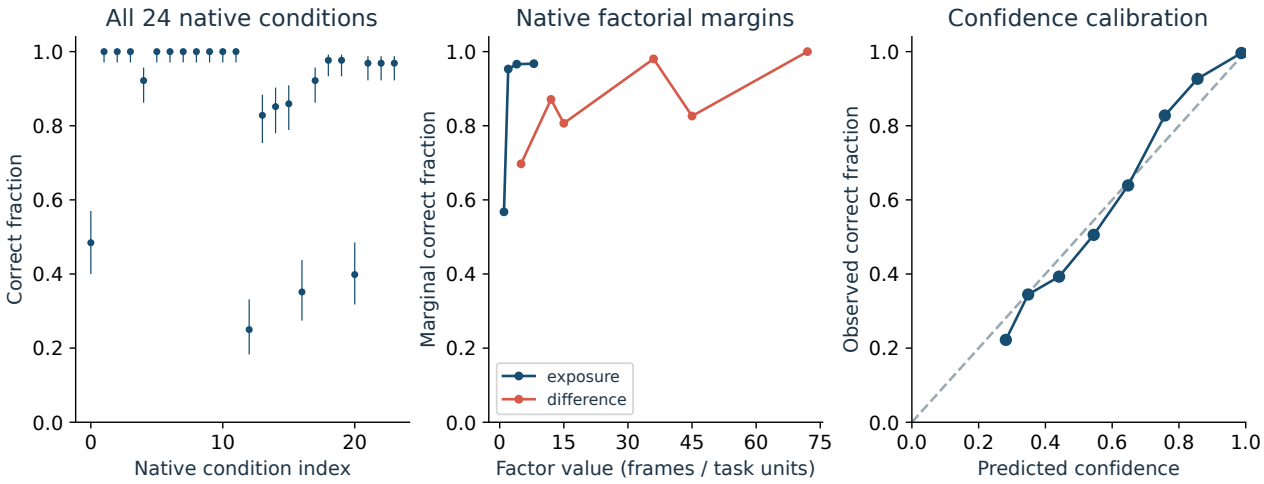


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Matched-scene effects of sample, delay and search time

The native counter-based generator reuses scene identity and target assignment across timing conditions. We compare the longest and shortest values of each timing factor while matching the other factors, then average those paired differences within scene. Static arrays remain pixel-identical; motion arrays retain the same direction and layout seeds and shared frame prefix as duration increases. Bootstrap intervals resample the 128 scene identities. A two-sided sign-flip test uses 9,999 random sign patterns on the scene-level differences, followed by Holm correction over this task's timing comparisons. These exploratory contrasts distinguish changes in encoding opportunity, retention duration and search opportunity.

Group	Factor	Short/long	Change (pp)	Holm p
color / 12	exposure	1/8	+51.6	0.0006
color / 36	exposure	1/8	+7.8	0.0044
color / 72	exposure	1/8	+0.0	1
orientation / 5	exposure	1/8	+60.9	0.0006
orientation / 15	exposure	1/8	+62.5	0.0006
orientation / 45	exposure	1/8	+57.0	0.0006

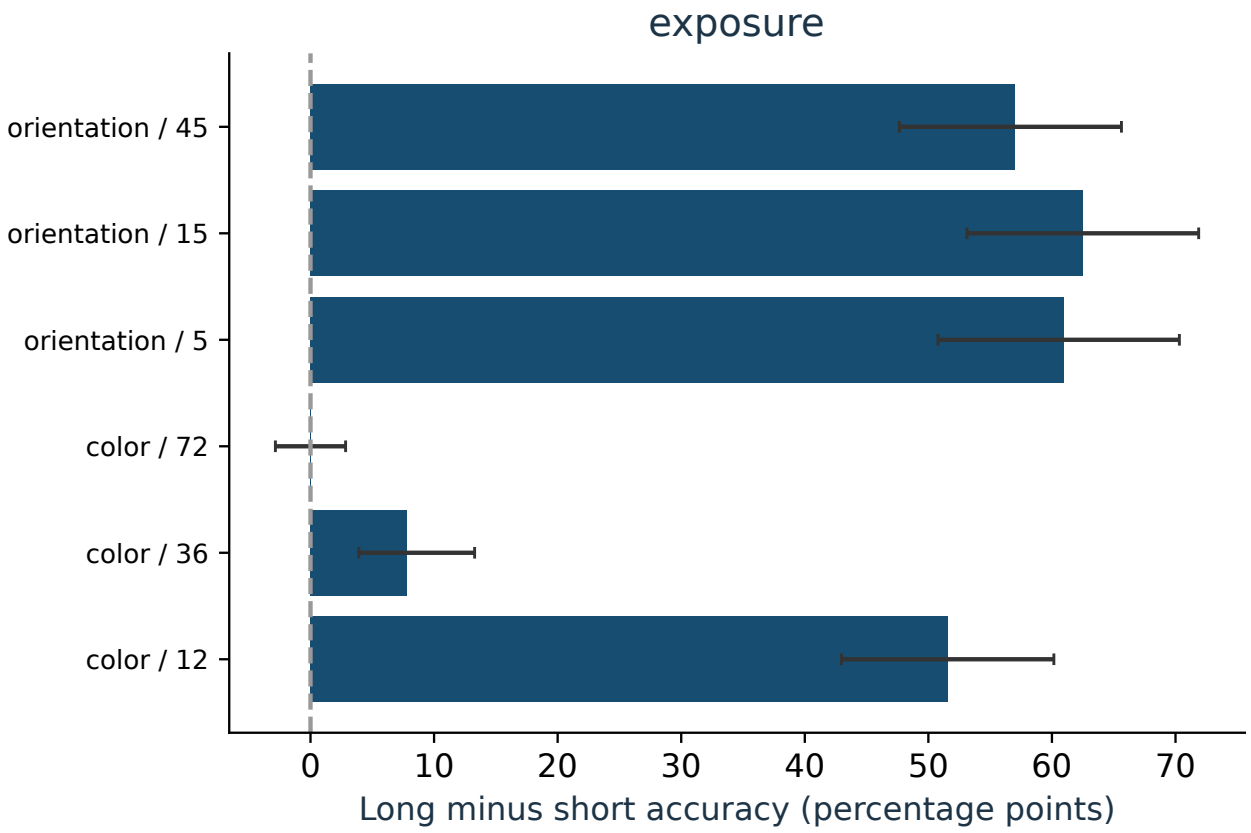


Figure 2: Paired timing effects with scene identity preserved. Each value is the longest-duration accuracy minus the shortest-duration accuracy, averaged over matched remaining factors. Intervals resample complete scenes.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	2.0	96.1%	100.0%
0.7	100.0%	2.0	99.2%	0.0%
0.8	100.0%	2.0	100.0%	0.0%
0.9	100.0%	2.0	100.0%	0.0%

The representative condition color_X12_H8 provides 8 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

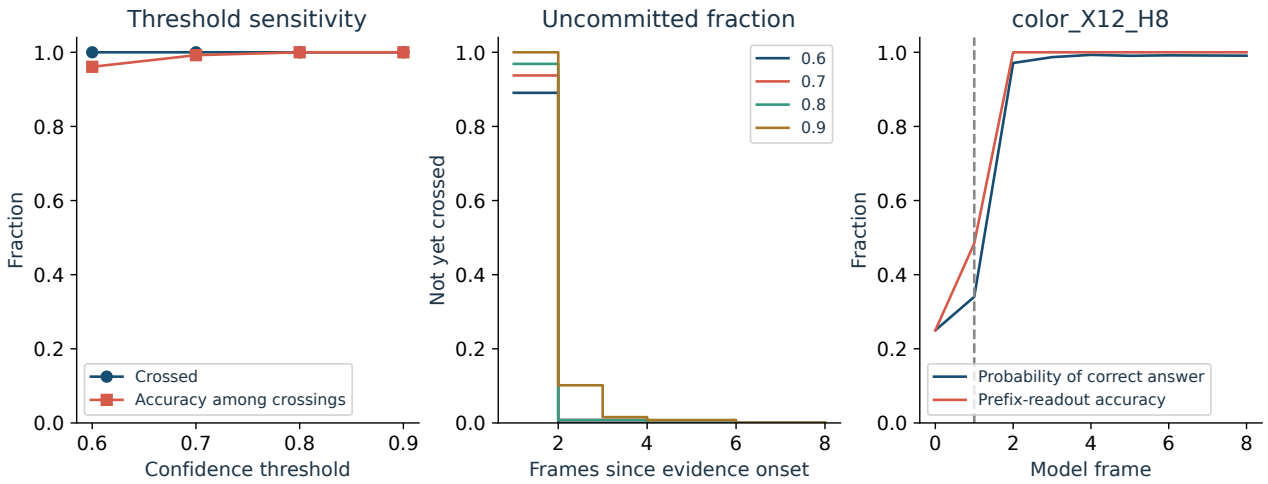


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 2,384 of 3,072 trials crossed. Of those, 15 changed their answer later, 12 corrected an early error, and 2 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

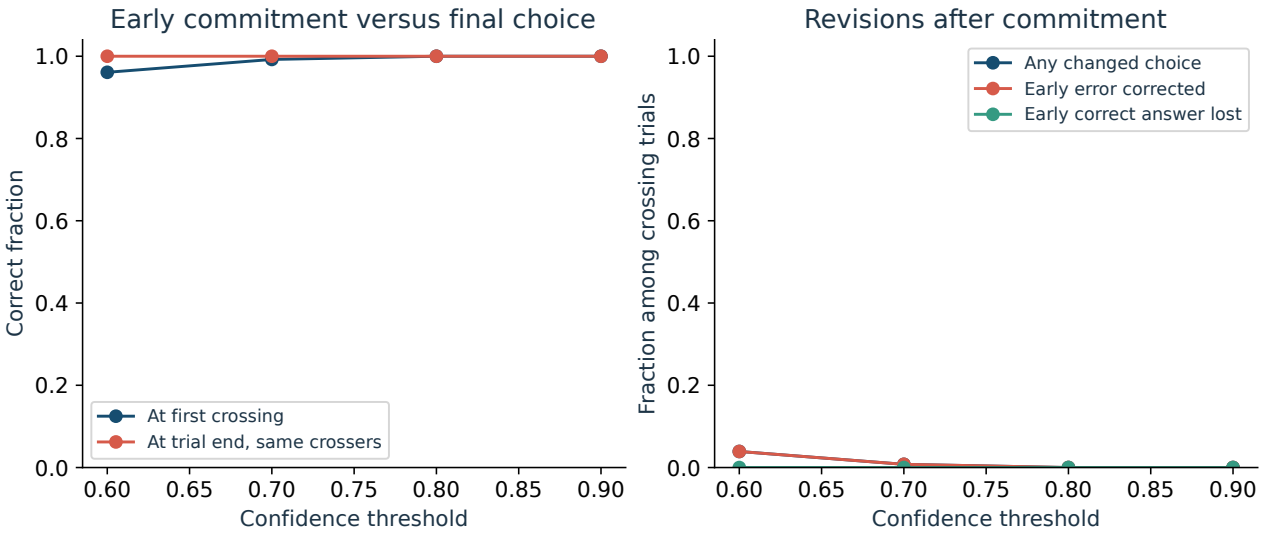


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

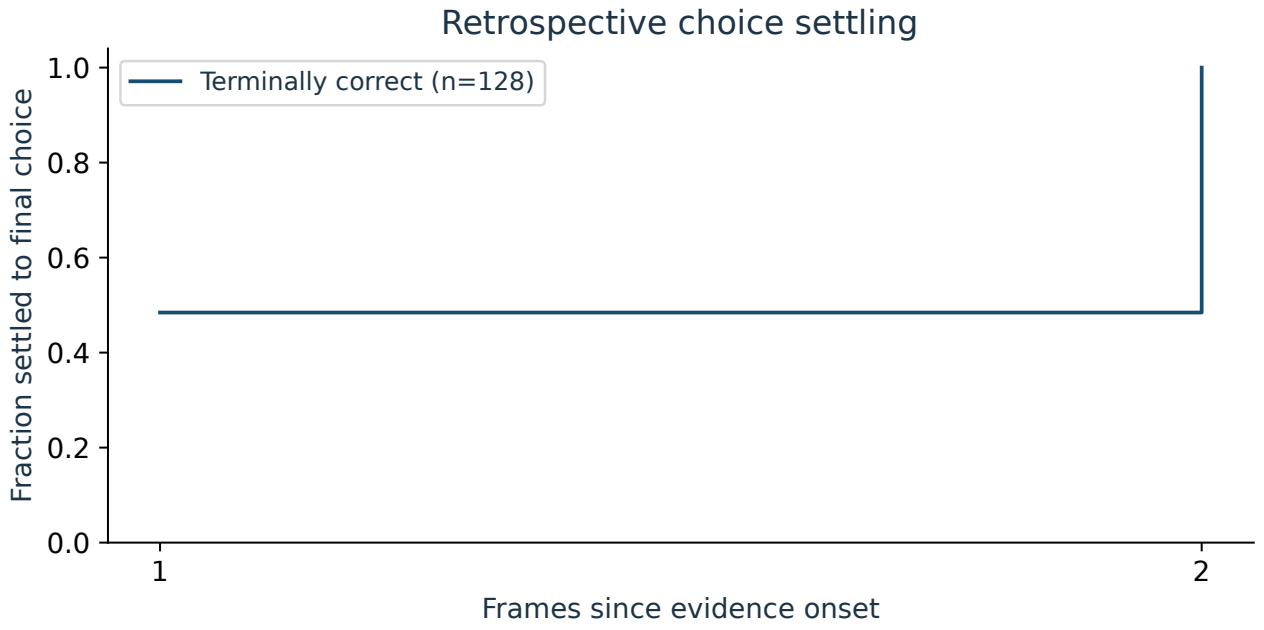


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The singleton is fixed at top left and array exposure is eight frames. Rows compare weak and strong color differences (12 and 72 renderer units) and weak and strong orientation differences (5 and 45 degrees).

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Color difference 12	32	32
Color difference 72	32	32
Orientation difference 5	32	32
Orientation difference 45	32	32

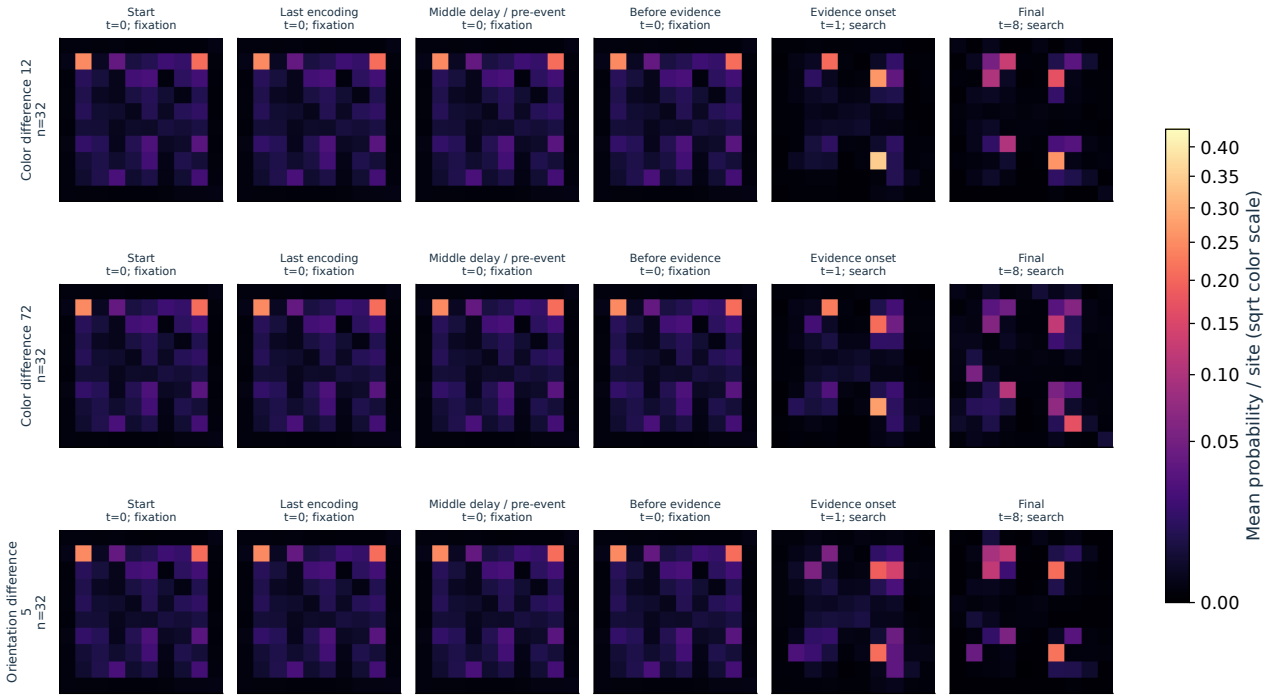


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

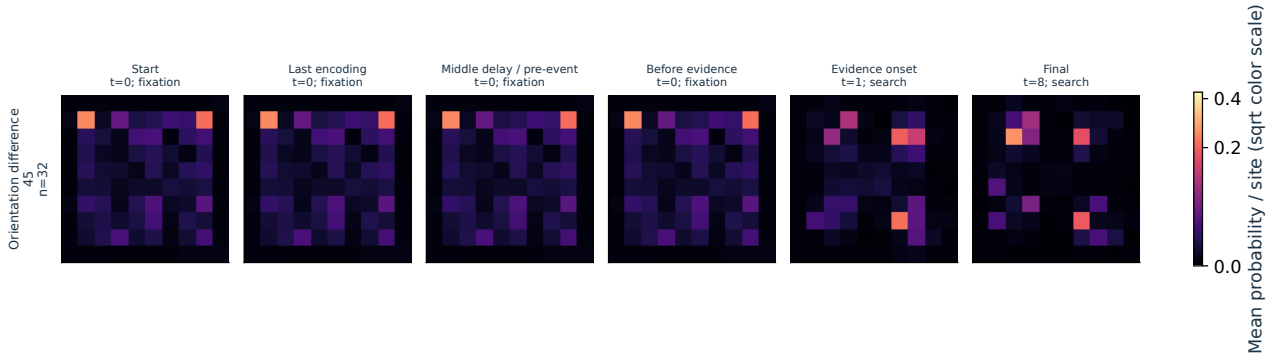


Figure 7: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

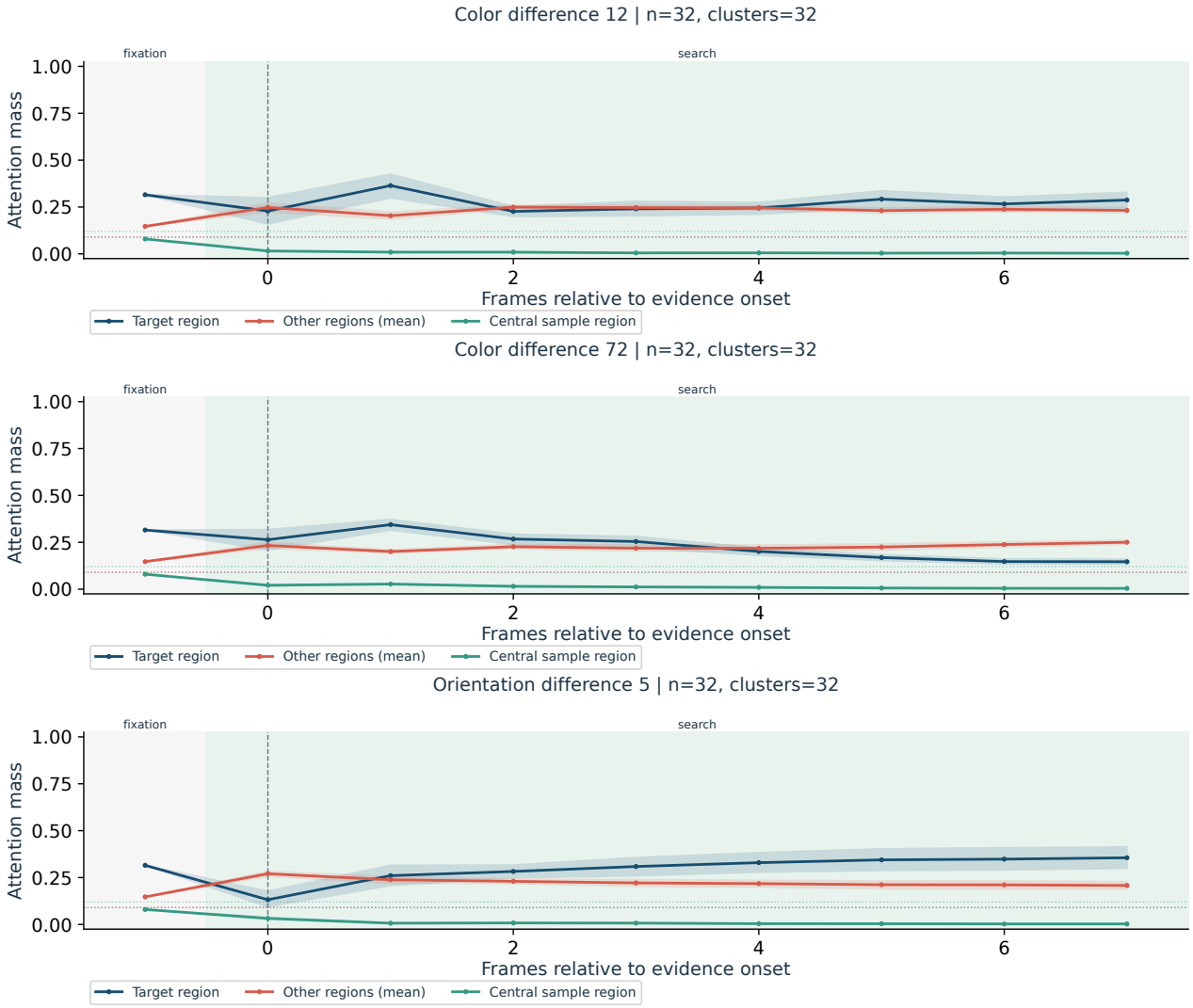


Figure 8: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

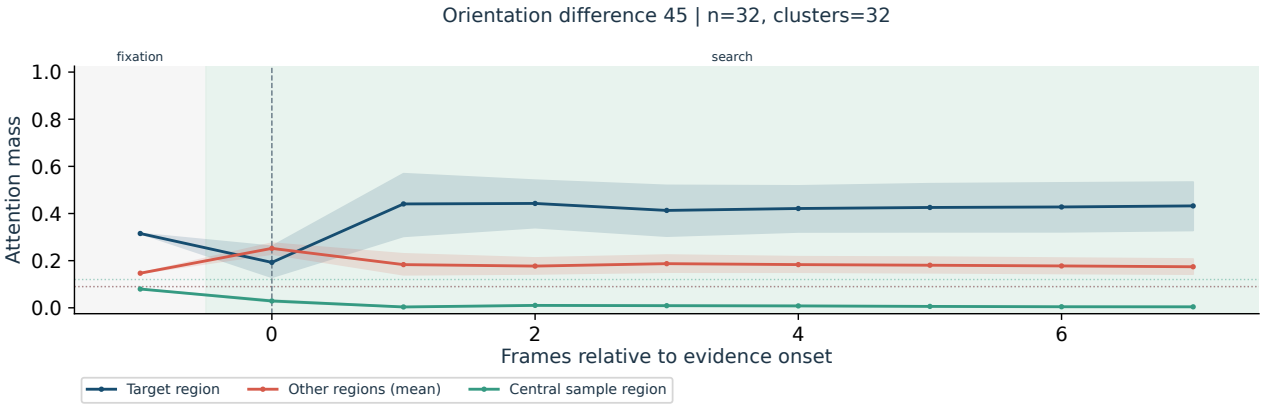


Figure 9: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

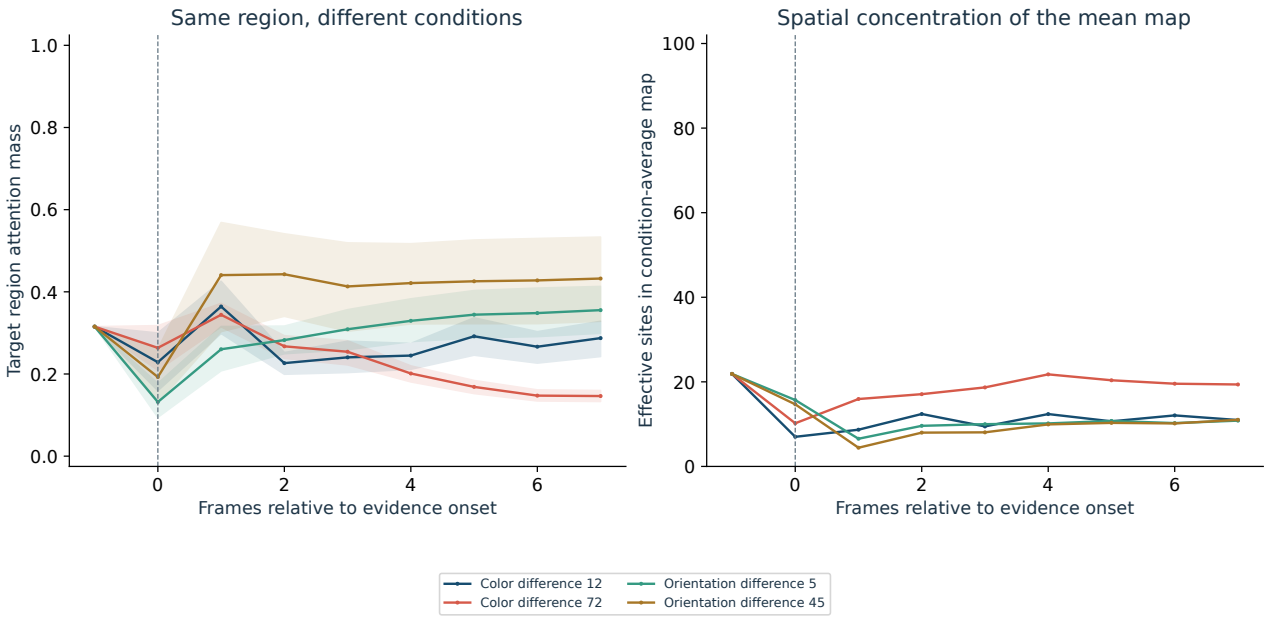


Figure 10: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Color difference 12: target region mass peaks at 0.364 at original frame 2 (search), and is 0.287 on the final frame. Color difference 72: target region mass peaks at 0.344 at original frame 2 (search), and is 0.146 on the final frame. Orientation difference 5: target region mass peaks at 0.355 at original frame 8 (search), and is 0.355 on the final frame. Orientation difference 45: target region mass peaks at 0.443 at original frame 3 (search), and is 0.432 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For distractor_+8_evidence, paired accuracy changed by -93.8 percentage points (paired 95% interval [-97.7%, -89.1%]; paired p=1.505e-36; within-task Holm p=9.329e-35). There were 0 corrected trials and 120 newly incorrect trials. For offsite_-4_evidence, paired accuracy changed by +0.0 percentage points (paired 95% interval [-2.8%, 2.8%]; paired p=1; within-task Holm p=1). There were 0 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

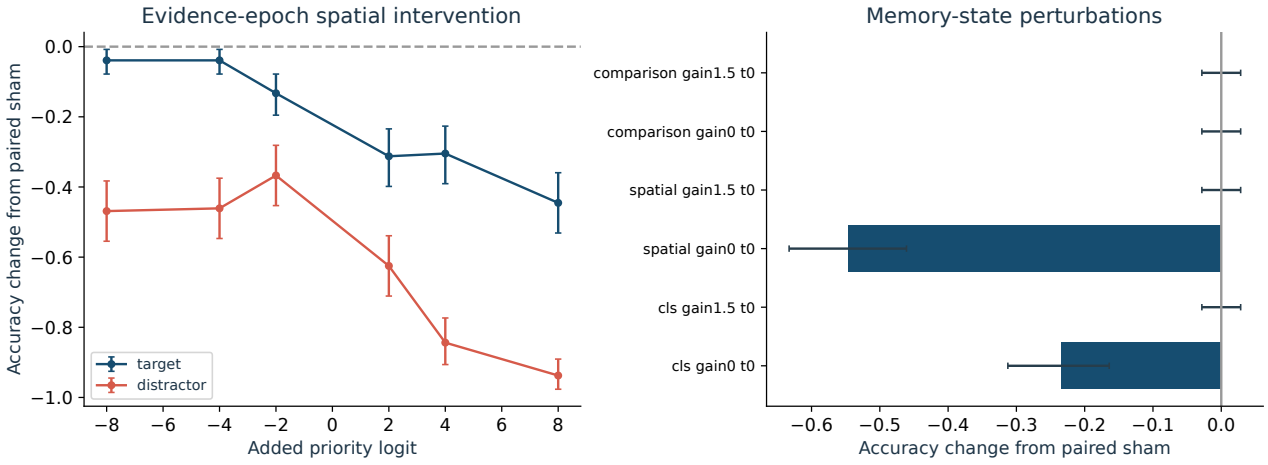


Figure 11: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
color_X12_H8	target_-4	-3.9	1
color_X12_H8	target_+4	-30.5	3.274e-10
color_X12_H8	distractor_-4	-46.1	3.261e-16
color_X12_H8	distractor_+4	-84.4	6.471e-31
orientation_X5_H8	target_-4	-0.8	1
orientation_X5_H8	target_+4	-2.3	1
orientation_X5_H8	distractor_-4	-60.2	2.621e-20
orientation_X5_H8	distractor_+4	-90.6	2.6e-33

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.520	44.5%	0.005
fusion	0.525	42.2%	0.010
spatial	0.998	96.9%	0.005
cls	1.000	100.0%	0.005
comparison	0.645	34.4%	0.310
attention	0.999	100.0%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 0.00 to 15.44 within the retained subspace. The final step displacement was 0.471 standardized RMS units. In spatial memory, the participation ratio ranged from 0.00 to 15.39 within the retained subspace. The final step displacement was 0.359 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -23.4 percentage points, spatial, -54.7 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

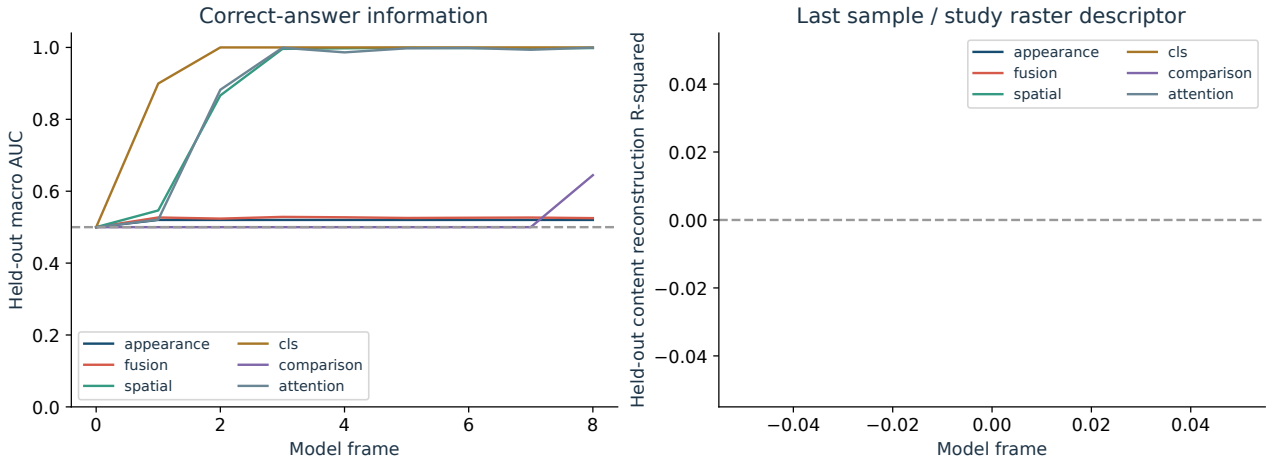


Figure 12: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

3.10 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

3.11 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.518	unavailable	3200
fusion	0.525	unavailable	6400
spatial	1.000	unavailable	6400
cls	1.000	unavailable	64
comparison	0.645	unavailable	735
attention	0.999	unavailable	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 32 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

3.12 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

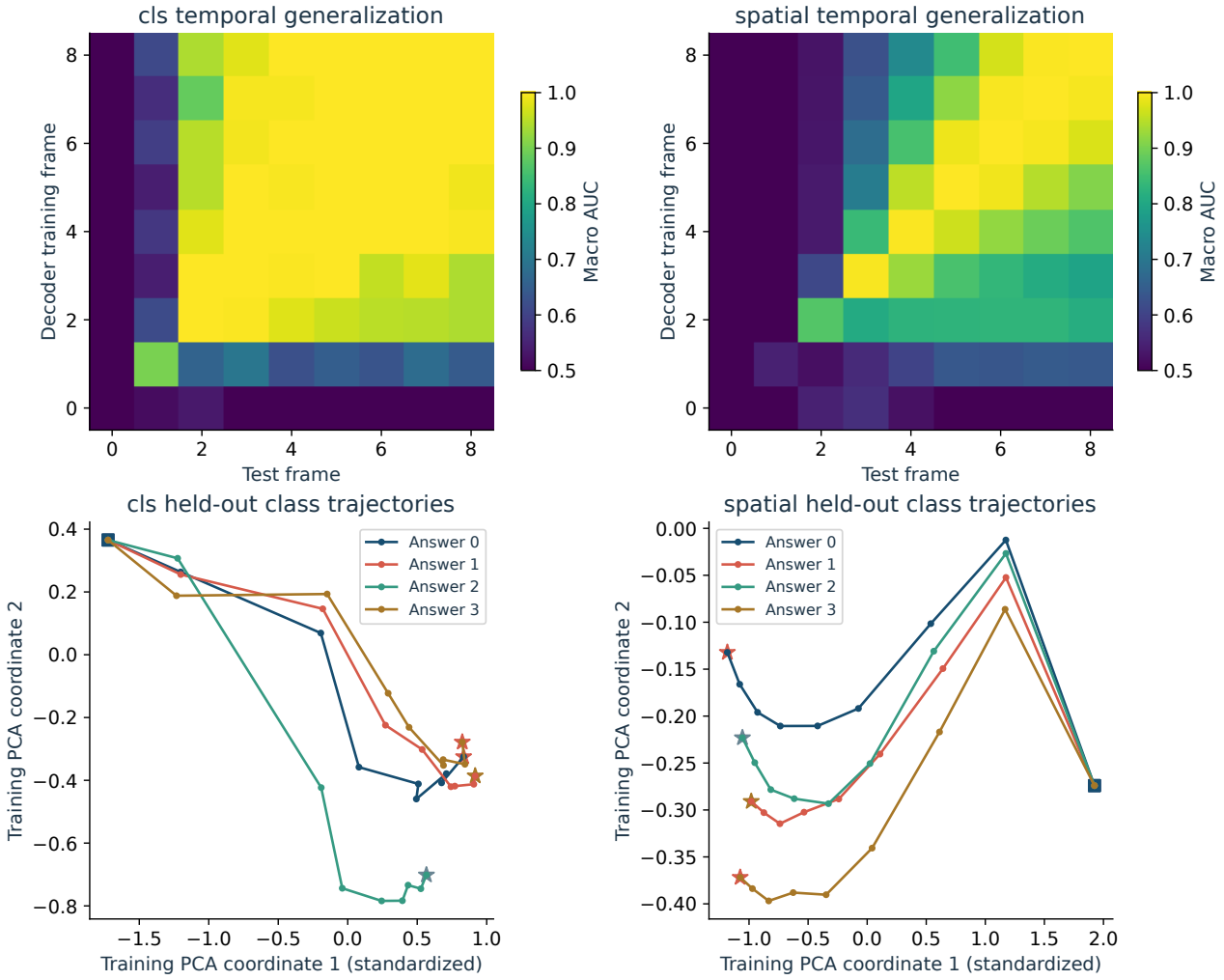


Figure 13: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

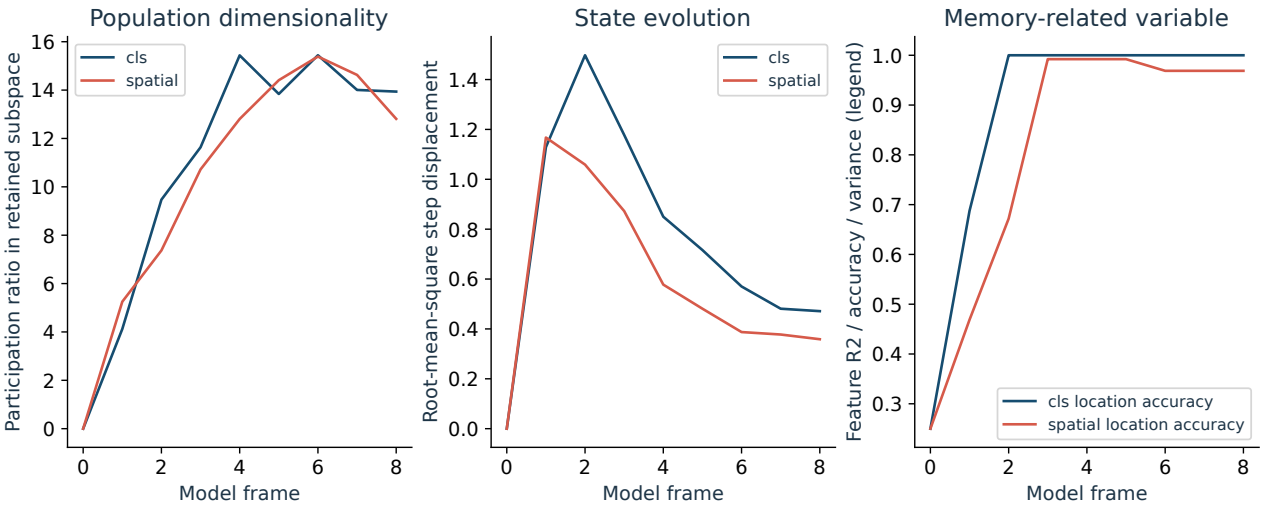


Figure 14: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

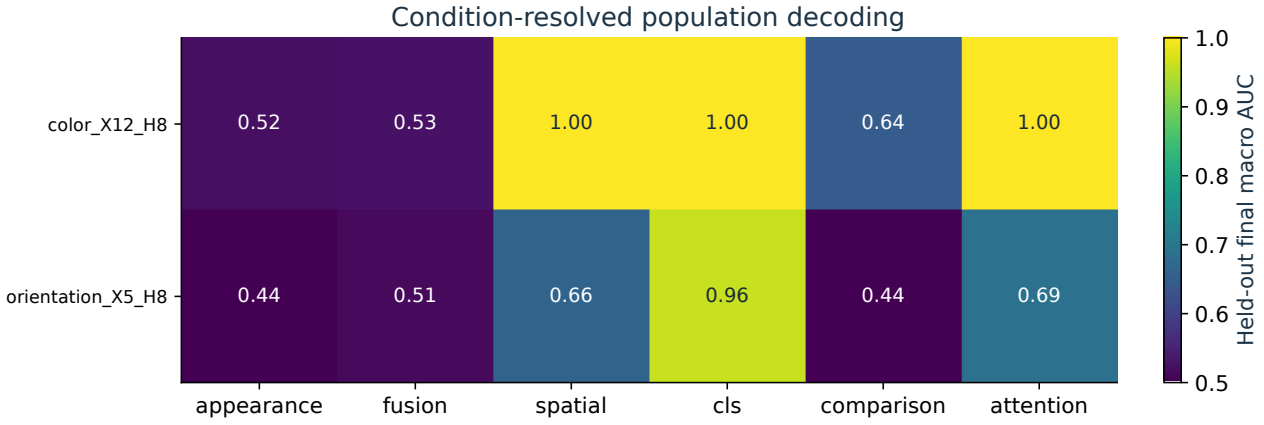


Figure 15: Held-out final answer decoding across selected conditions and recording sites. A constant-zero comparison representation is displayed at chance AUC.

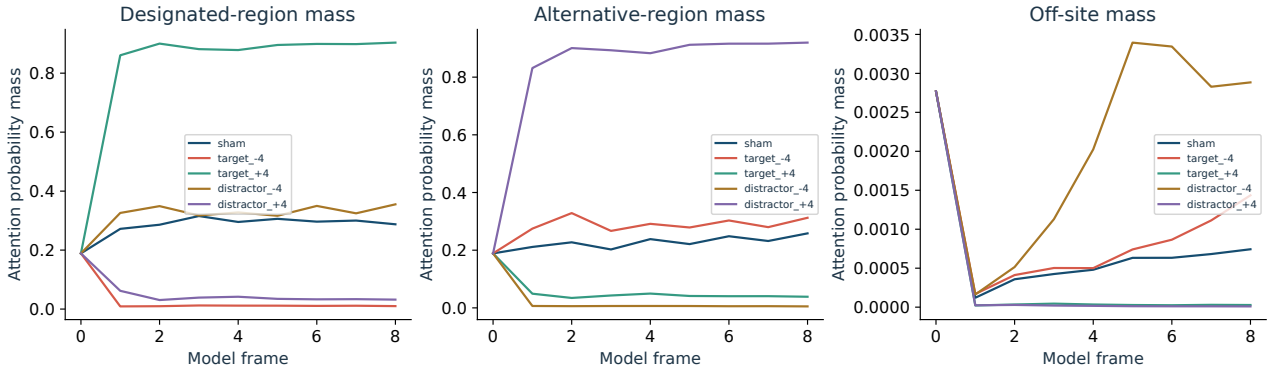


Figure 16: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

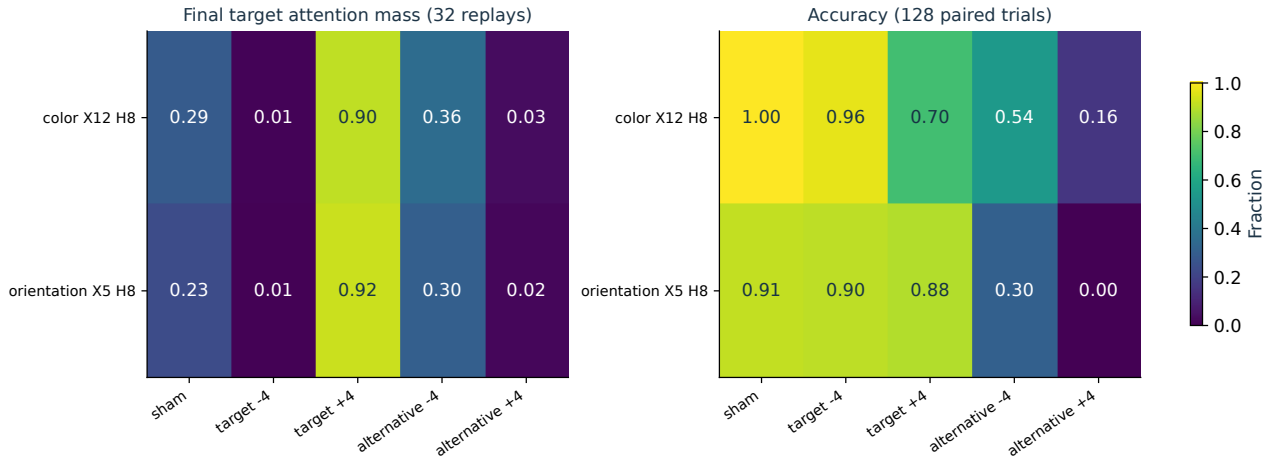


Figure 17: Condition-resolved manipulation checks. The left panel measures final target-region allocation on 32 exact replays; the right measures accuracy on all 128 paired intervention trials. All displayed spatial doses have magnitude four and act during evidence frames. Rows preserve each selected condition, making redistribution and behavioral rescue separately visible.

3.13 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

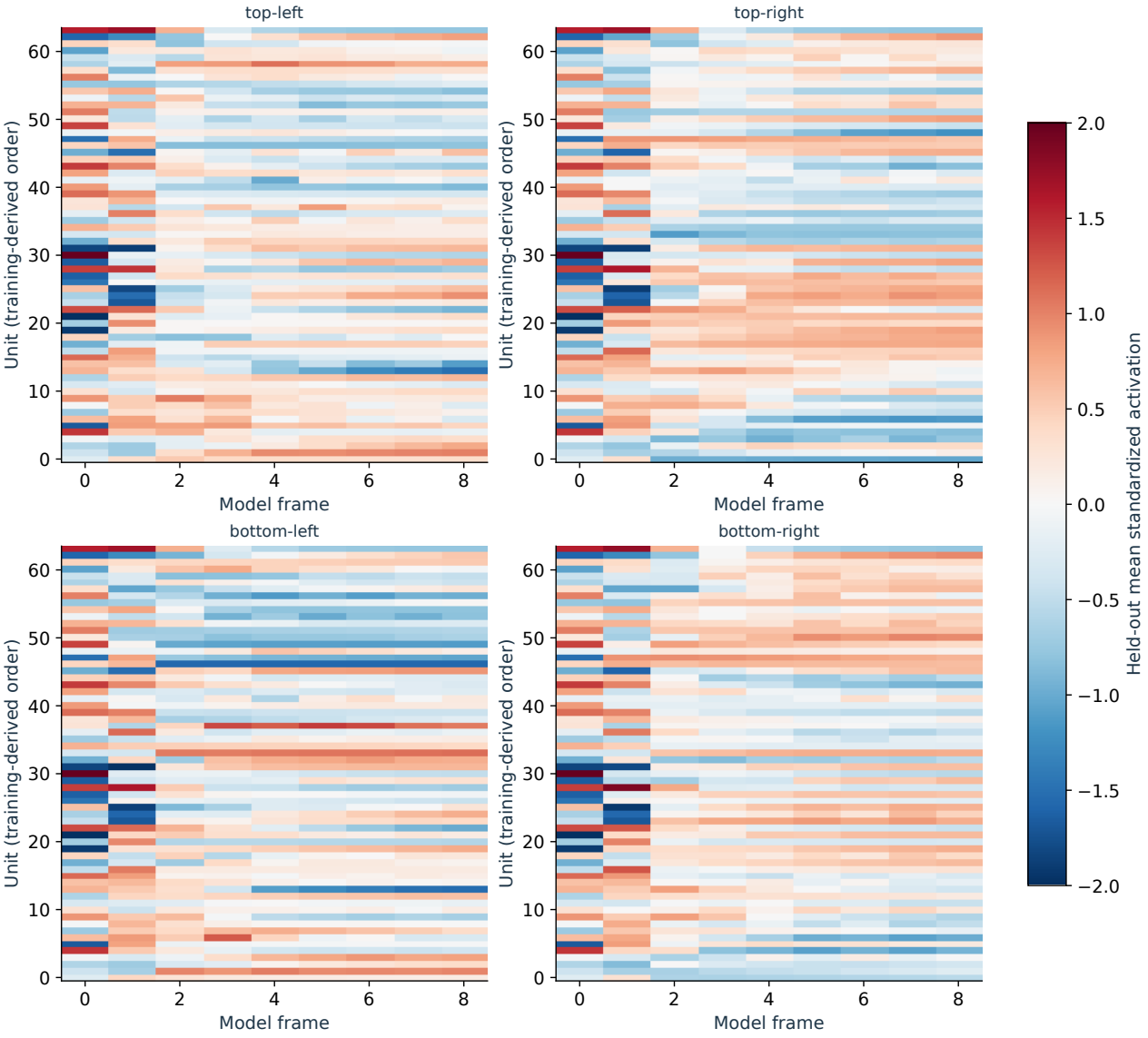


Figure 18: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

3.14 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional

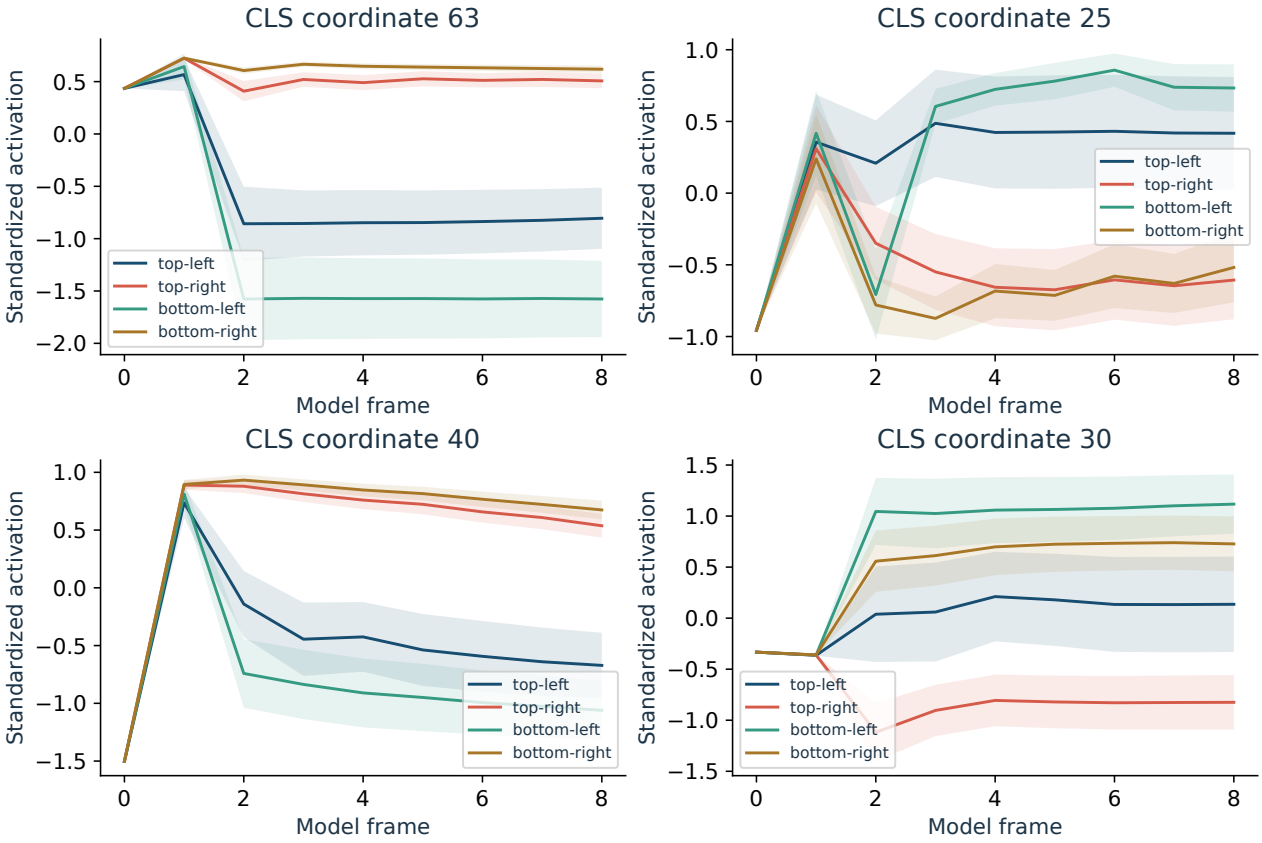


Figure 19: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	44.5%	44.5%	44.5%
fusion	42.2%	42.2%	42.2%
spatial	96.9%	96.9%	96.9%
cls	100.0%	100.0%	100.0%
comparison	34.4%	34.4%	34.4%
attention	100.0%	100.0%	100.0%

3.15 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained.

Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

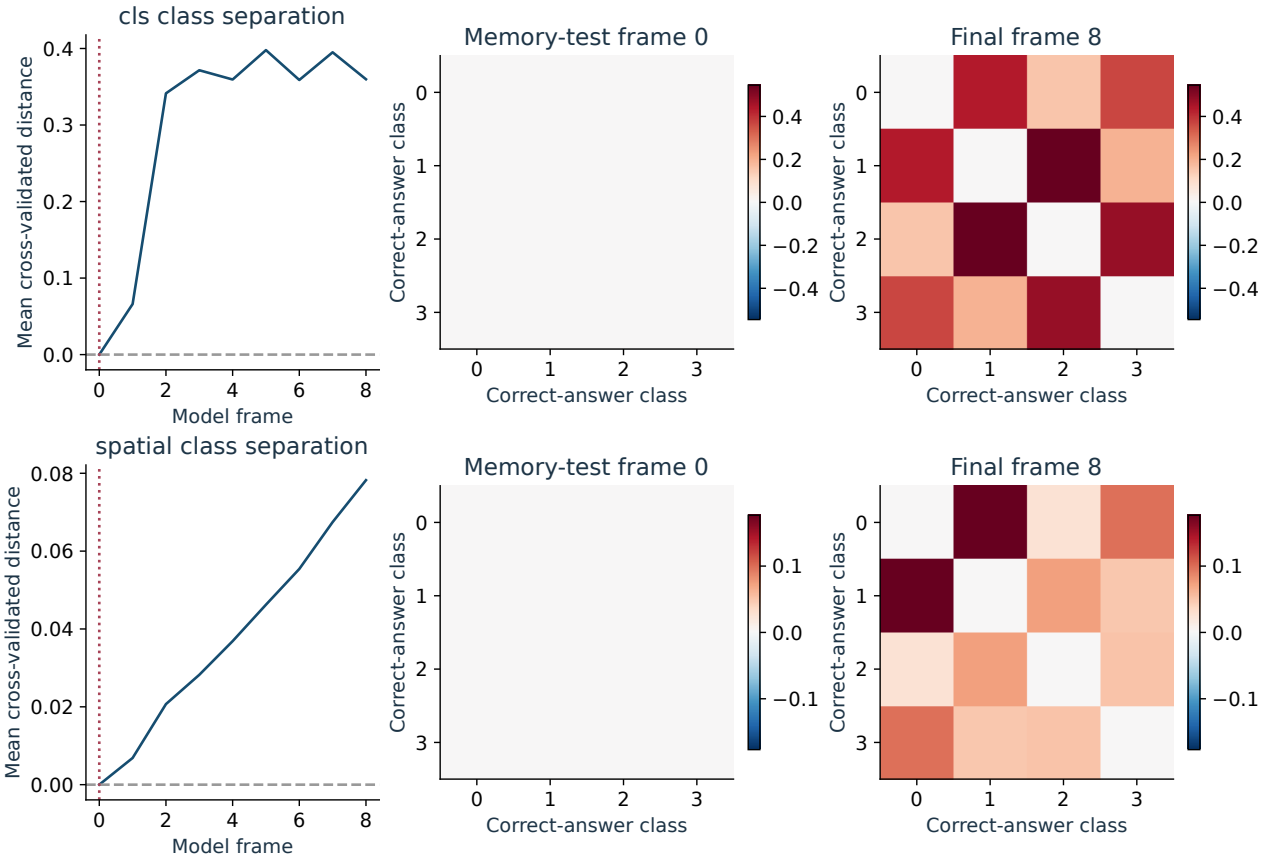


Figure 20: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

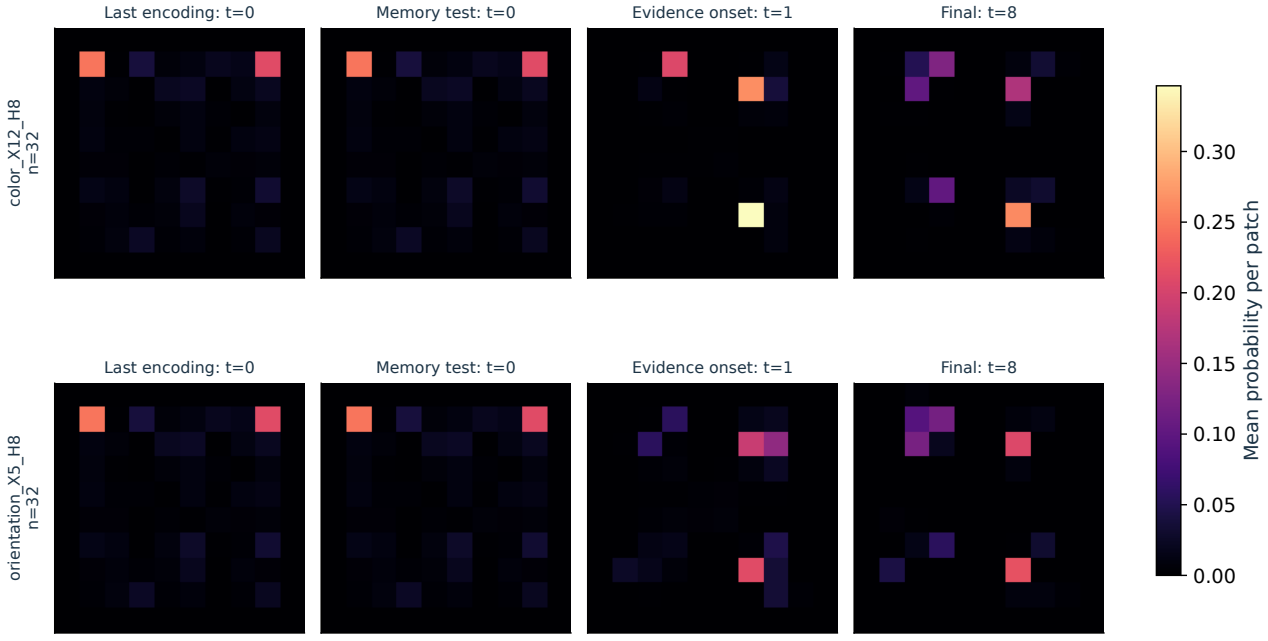


Figure 21: Condition-resolved attention at four temporal landmarks. Memory-search and pop-out rows average trials with top-left targets. Image-recognition rows average seen-probe trials. Each map is shown in the original image layout with a common probability scale across conditions; the row label gives the number of averaged trials. The memory-test frame is the middle delay, or the frame before evidence when a blank delay is absent.

4 Discussion

Additional processing of the same static array produces a large behavioral benefit. At the smallest color difference, accuracy rises from about 48 percent after one array frame to 100 percent after two. At a five-degree orientation difference it rises from 25 percent to about 83 percent after two frames, reaching about 86 percent after eight. The matched one-versus-eight-frame comparisons survive correction in five of the six feature-difference groups; the largest color difference is already at ceiling. Thus a physically available singleton is not necessarily converted into a correct location report on the first processing step. Repeated frames permit both recurrent state updates and re-evaluation of the visual pathways. The priority and state interventions then ask which parts of that processing support selection. Since item count is fixed at four, these curves characterize feature discrimination and temporal processing within that array size.

The behavioral result establishes 86.4% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies distractor_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

A true search-efficiency experiment should vary set size, distractor heterogeneity and target prevalence. That would allow slopes of decision time against item count to be compared across feature conditions.

4.1 Interpreting these results alongside biological experiments

4.1.1 Schall and Hanes (1993)

The animal experiment separates an initial visual response from later target selection. Our repeated static array likewise produces a large one-to-two-frame improvement for weak singletons. This is a functional parallel in the emergence of selection. The network's frames are computation steps, so the comparison should concern ordering of processes and feature difficulty before assigning a biological time scale. [5]

4.1.2 Bichot, Rossi and Desimone (2005)

Parallel feature bias and later selection can coexist. In the model, all sites are represented each frame and one normalized priority distribution weights their contribution. A later correct location report therefore need not imply serial inspection of every item. The time course of target-minus-distractor allocation and local feature evidence can reveal whether an early distributed signal becomes a concentrated selection. [6]

4.1.3 Ling, Liu and Carrasco (2009)

External-noise methods provide a way to separate weak feature evidence from poor selection. A small singleton difference can produce a subtle signal even when the display contains only four objects. Varying orientation noise around the distractor mean, separately from the target difference, would reveal whether extra model frames increase effective gain or improve exclusion of irrelevant feature variation. [7]

4.2 A computational account of the observed behavior

A priority-map account combines local distinctiveness with task-relevant feature information. Repeated frames allow the same static evidence to update recurrent state and feed back into the next selection step. In the present data, the weakest color singleton rises from roughly 48 percent after one frame to perfect performance after two; a five-degree orientation singleton rises from chance to about 83 percent. These are large processing-time effects despite an unchanged visible array. Guided Search supplies a framework in which selection and identification unfold over time, while the model provides an explicit recurrent implementation. To test classical search efficiency, however, item count must vary: fixed four-item trials measure processing duration and feature separation, not a reaction-time slope per added distractor. [8]

4.3 Where this task agrees with the biological evidence

The agreement is a time-dependent transition from visible feature information to useful target selection, especially for weak differences. The model's discrete forced report and fixed set size differ from free-viewing or variable-size search experiments. These differences specify the next tests rather than weakening the observed temporal benefit.

4.4 A follow-up experiment motivated by these results

Vary set size while holding item size, spacing and feature difference fixed. Record confidence crossing, final choice and target allocation. Compare target-present and target-absent trials, and test whether additional distractors affect identification time after controlling the singleton's local contrast.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Treisman AM, Gelade G (1980). A feature-integration theory of attention. Cognitive Psychology 12:97-136. https://pubmed.ncbi.nlm.nih.gov/7351125/.

[2] Wolfe JM (2021). Guided Search 6.0: An updated model of visual search. Psychonomic Bulletin and Review 28:1060-1092. https://pubmed.ncbi.nlm.nih.gov/33547630/.

[3] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[4] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[5] Schall and Hanes (1993). Neural basis of saccade target selection in frontal eye field during visual search. Primary source.

[6] Bichot, Rossi and Desimone (2005). Parallel and serial neural mechanisms for visual search in macaque area V4. Primary source.

[7] Ling, Liu and Carrasco (2009). How spatial and feature-based attention affect the gain and tuning of population responses. Primary source.

[8] Wolfe (2021). Guided Search 6.0: An updated model of visual search. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
color_X12_H1	128	48.4%	0.785	0.641
color_X12_H2	128	100.0%	1.000	0.007
color_X12_H4	128	100.0%	1.000	0.000
color_X12_H8	128	100.0%	1.000	0.000
color_X36_H1	128	92.2%	0.997	0.172
color_X36_H2	128	100.0%	1.000	0.000
color_X36_H4	128	100.0%	1.000	0.000
color_X36_H8	128	100.0%	1.000	0.000
color_X72_H1	128	100.0%	1.000	0.020
color_X72_H2	128	100.0%	1.000	0.000
color_X72_H4	128	100.0%	1.000	0.000
color_X72_H8	128	100.0%	1.000	0.000
orientation_X5_H1	128	25.0%	0.518	0.780
orientation_X5_H2	128	82.8%	0.975	0.219
orientation_X5_H4	128	85.2%	0.976	0.192
orientation_X5_H8	128	85.9%	0.972	0.190
orientation_X15_H1	128	35.2%	0.619	0.739
orientation_X15_H2	128	92.2%	0.991	0.088
orientation_X15_H4	128	97.7%	1.000	0.030
orientation_X15_H8	128	97.7%	0.999	0.049
orientation_X45_H1	128	39.8%	0.692	0.686
orientation_X45_H2	128	96.9%	0.995	0.053
orientation_X45_H4	128	96.9%	0.999	0.043
orientation_X45_H8	128	96.9%	0.998	0.053

B Every-frame attention atlas

Condition color_X12_H8. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 32 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

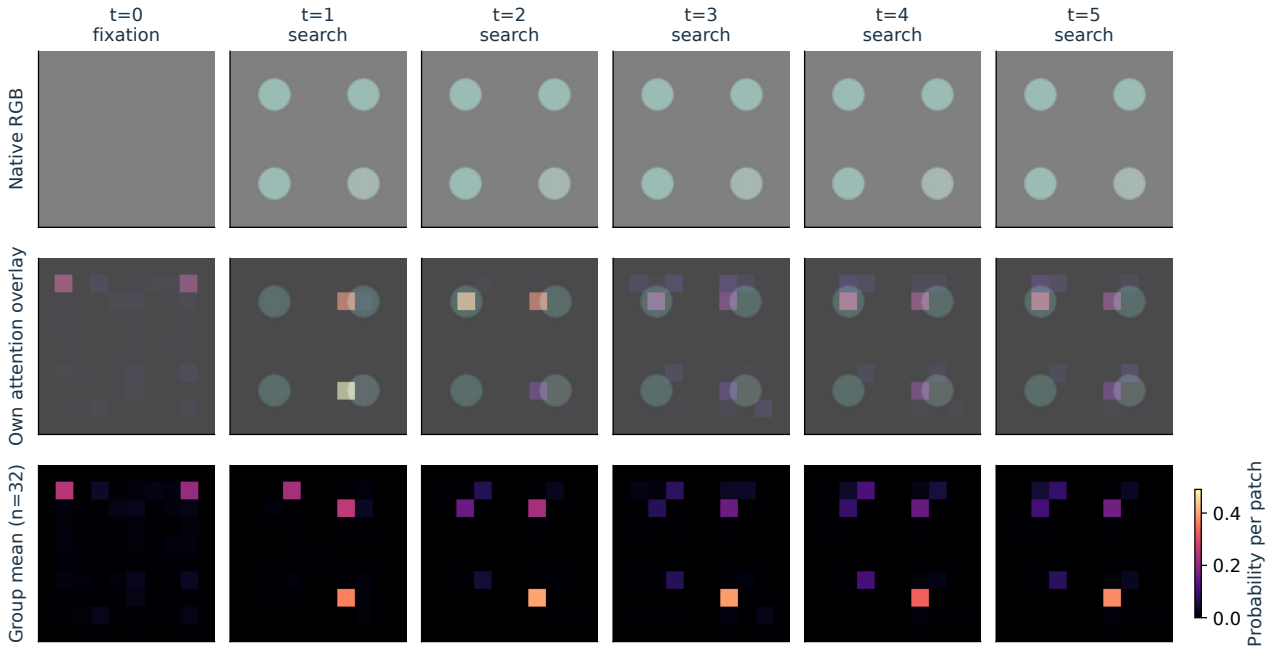


Figure 22: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

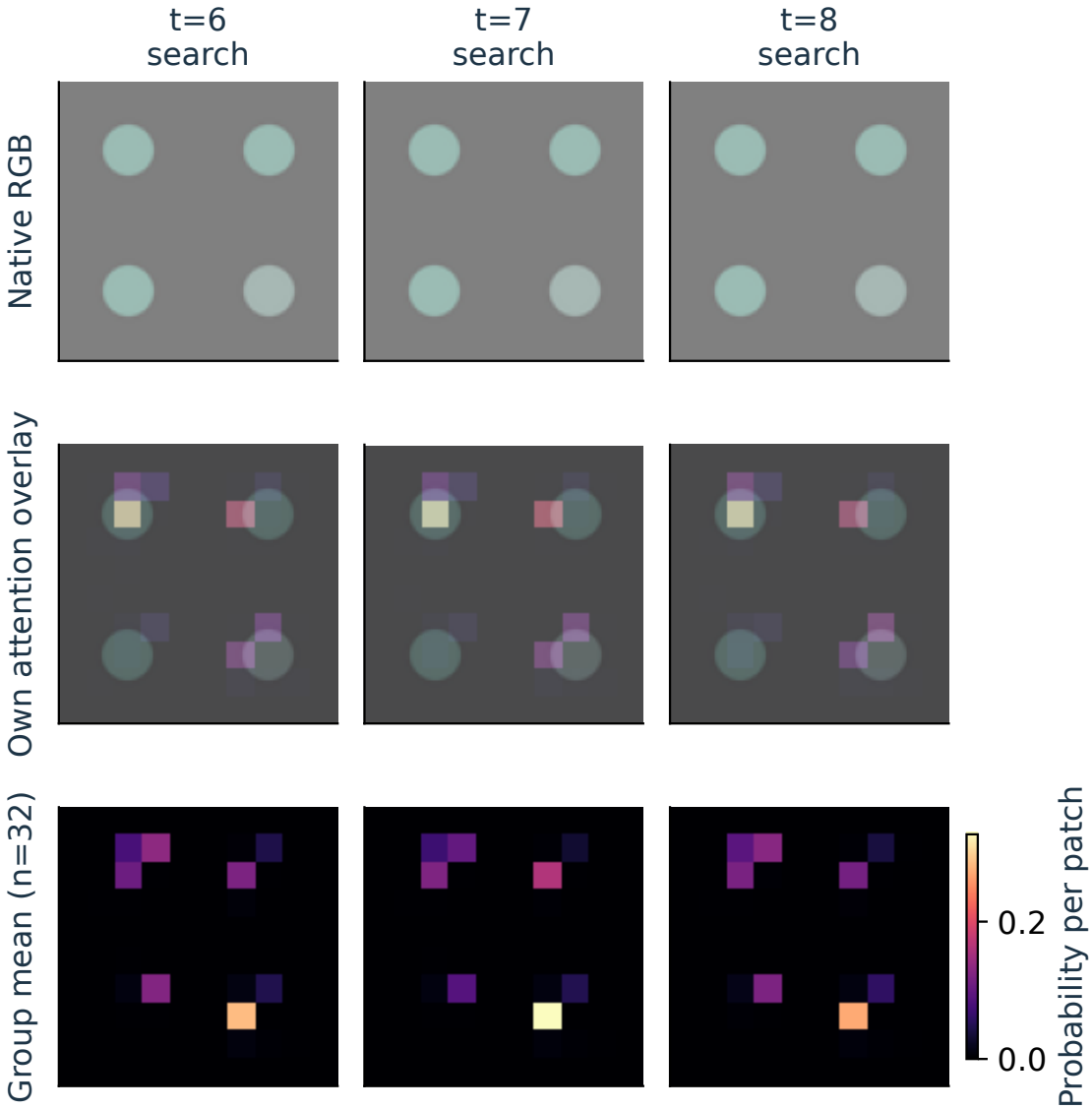


Figure 23: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

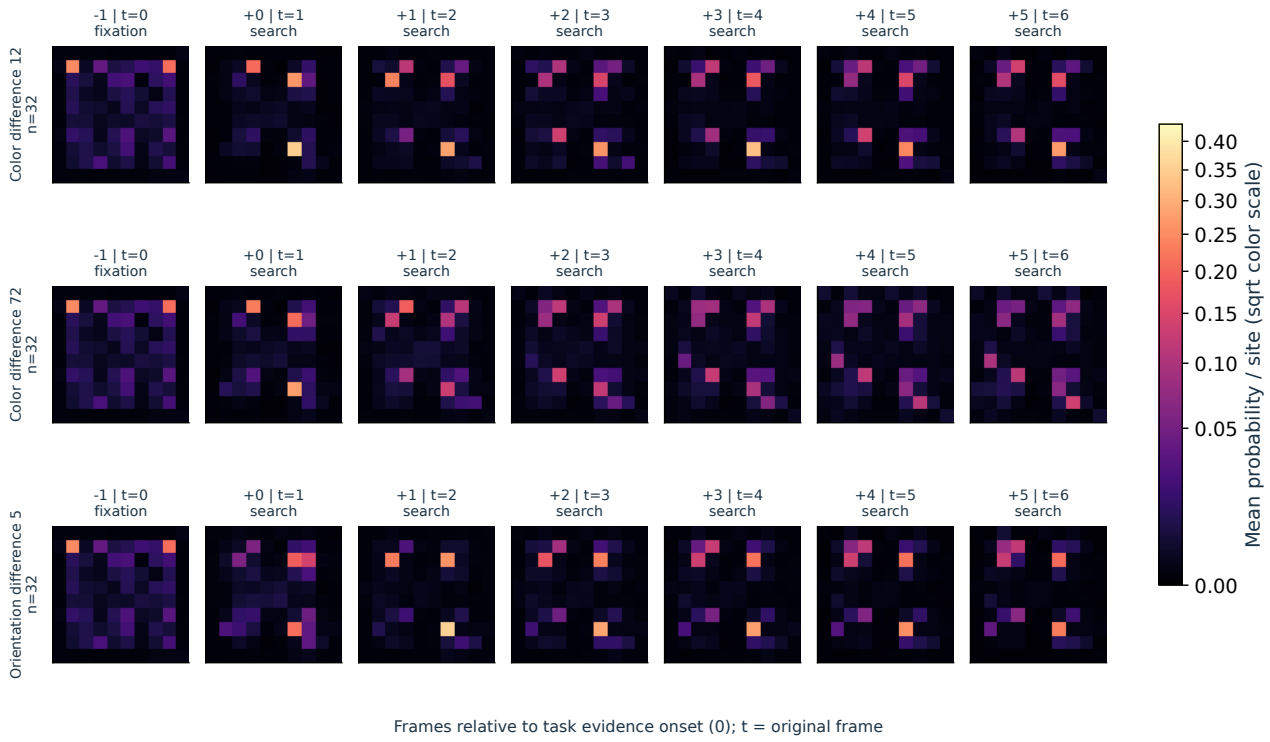


Figure 24: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

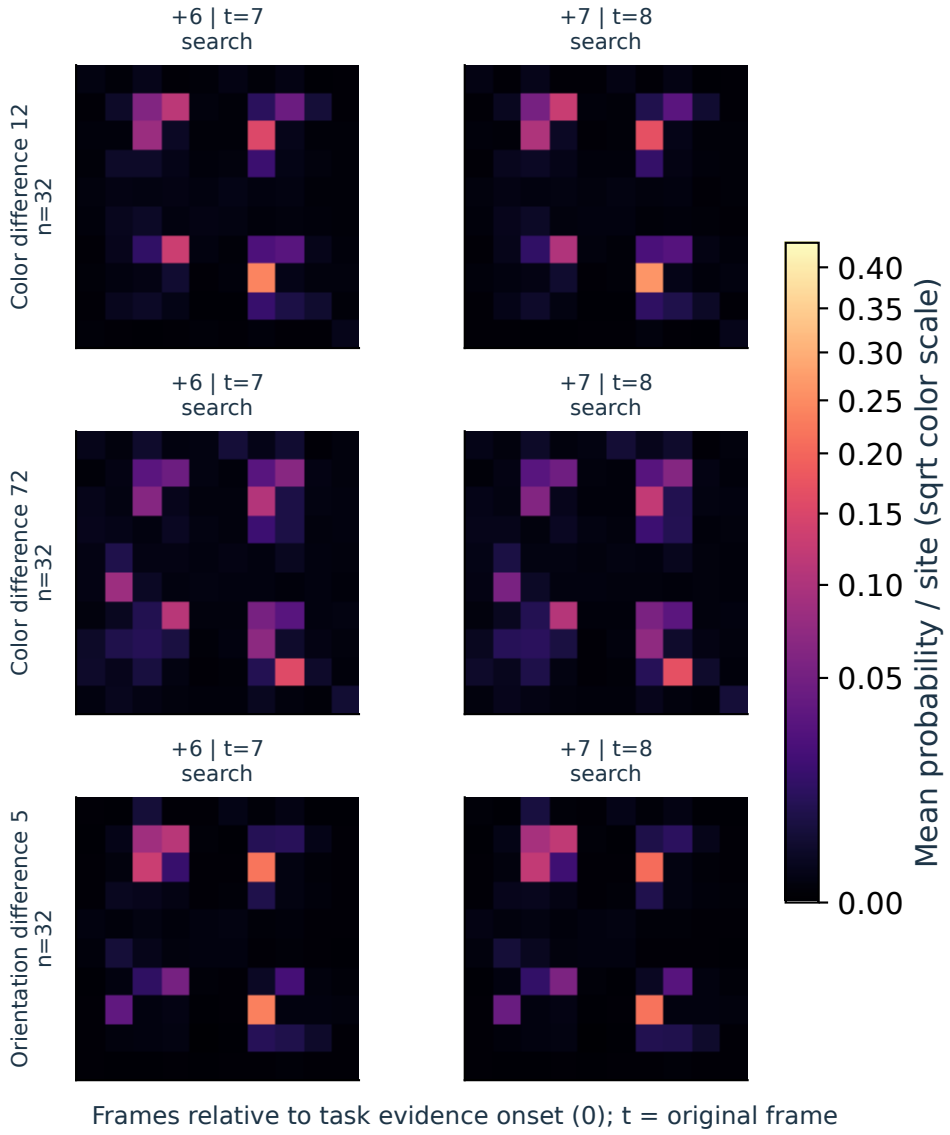


Figure 25: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

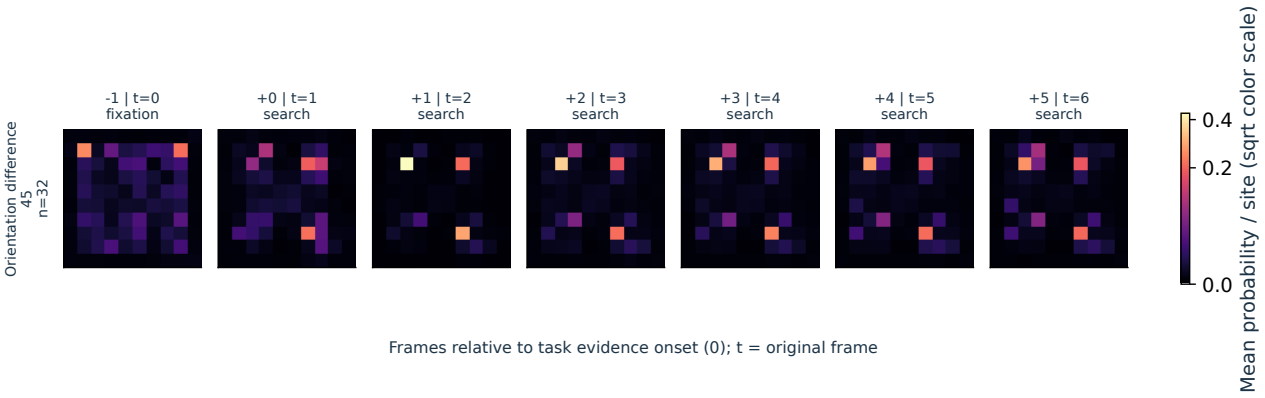


Figure 26: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

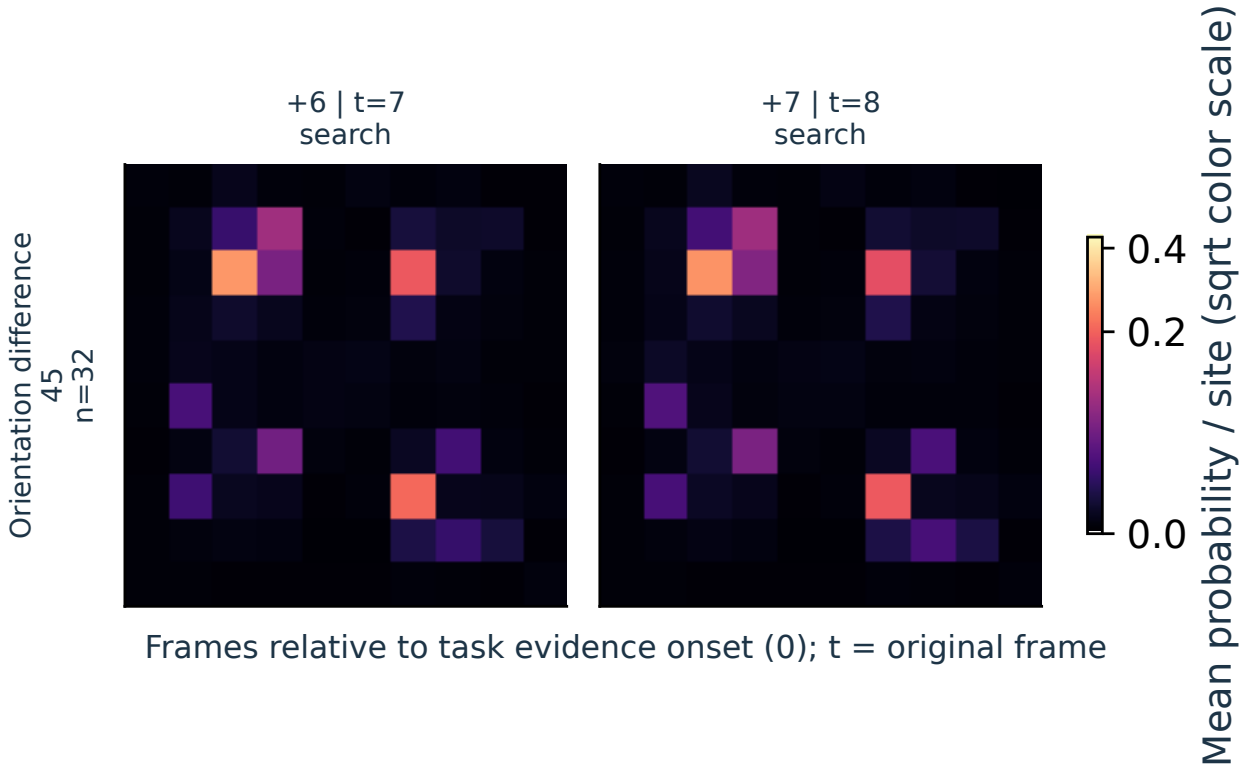


Figure 27: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.

Remembered motion as a template for selection among moving alternatives

Visual Attention and Working Memory Project
Computational experiment series: previous trained model

18 October 2026

Abstract

We characterized motion memory search in a frozen recurrent visual observer using behavioral challenges, spatial-priority interventions, memory perturbations and population decoding. Across 81 native conditions and 10,368 fresh trials, equal-condition accuracy was 97.7% (95% interval [96.6%, 98.8%]). For the representative condition motion_A20_S12_D12_H12, the strongest final answer readout among the tested regions was cls, with held-out macro AUC 0.996. The largest observed impairment in that condition was distractor_+8_evidence, changing accuracy by -96.1 percentage points. The analyses relate task competence to information availability, spatial selection and state dependence, with paired interventions and independent probe trials.

1 Introduction and scientific questions

The sample is a moving-dot patch. After a blank delay, four independent moving-dot patches appear, and the observer must locate the one with the remembered direction. The dot layouts differ between sample and array, so an exact raster match cannot provide the solution.

Direction spacing controls the similarity of alternatives. Sample duration, retention delay and search duration are independently varied. The remembered continuous direction is decoded as cosine and sine of its angle, alongside final location and coarse sample-image content. Those targets distinguish motion memory from memory for a particular dot layout.

The biological motivation is developed through [1], [2], [3]. The present experiments use the model as an accessible computational subject: its choices, internal population coordinates and interventions can be measured on exactly matched movies.

1.1 Neuroscientific rationale

Motion physiology supplies a basis for representing direction, while memory-guided search motivates using a remembered feature to select among alternatives. Their combination is demanding because the sample and candidates have independent dot arrangements. The useful memory must describe motion in a form that survives this change in image realization. A circular direction decoder directly tests that information, while a location decoder tests the later use of the template. The delay and exposure manipulations then distinguish retention of direction from the process of finding the matching moving patch.

1.2 Biological experiments motivating this task

Experimental question. How can a remembered motion direction guide selection of a new moving pattern at a new location?

Zaksas and Pasternak (2006) Monkeys compared two motion directions across a delay. MT and PFC carried direction information with transient delay signals; test responses depended on the remembered sample and related to the eventual decision. [6]

Chelazzi, Miller, Duncan and Desimone (1993) Monkeys remembered a picture and selected its match from a small array. IT showed sample-related delay activity, followed during search by suppression of nontarget responses and dominance of the target representation. [7]

Ling, Liu and Carrasco (2009) External-noise motion experiments separated spatial from feature-based attention. Threshold-versus-noise functions supported different combinations of population gain and tuning, giving behavioral tests of attentional mechanisms. [8]

These experiments motivate the behavioral, spatial-selection and population measurements reported below. The discussion returns to each study to compare its findings with the measured behavior and internal states of this observer.

1.3 Task definition and prediction structure

Which patch moves in the remembered sample direction?

Response labels are 0: top-left; 1: top-right; 2: bottom-left; 3: bottom-right. The native suite contains 81 conditions for this task. A complete condition table follows the main results. Detailed neural and intervention measurements use motion_A20_S12_D12_H12, motion_A90_S12_D12_H12.

1.4 Stimulus sequence and spatial layout

For the representative native condition motion_A20_S12_D12_H12, the sequence has 37 frames, indexed from zero: frame 0: fixation; frames 1–12: sample; frames 13–24: delay; frames 25–36: search. The complete condition table specifies the other tested timing and difficulty settings.

The following values are observed in the 128 native trials of this representative condition. Continuous nuisance ranges summarize those draws.

Stimulus property	Observed values
Candidate direction spacing (degrees)	20

Stimulus centers in the 100 by 100 input are [[25, 25], [75, 25], [25, 75], [75, 75]]. Coordinate x increases rightward and y downward.

The motion pathway must create a representation that generalizes across dot realizations, and the recurrent system must maintain it until new candidates appear. A difference between direction decodability and localization accuracy would identify a retrieval or routing bottleneck. Attention manipulations at the correct array location and an alternative quantify causal competition during matching.

2 Methods

2.1 Computational subject and experimental units

The experimental subject is the previously trained 19-task model, with the main analysis fixed at update 204,289. This checkpoint was selected by the existing equal-task monitoring criterion before the present analysis. Its SHA-256 identifier is recorded in the reproducibility table. The final saved checkpoint at update 220,637 provides a matched robustness comparison. Model parameters are frozen throughout these experiments. A trial is a freshly generated native movie, its prescribed response, and the model's output. Independent procedural draws provide the sampling unit; resampling intervals quantify uncertainty for this trained subject under the specified stimulus distribution.

The 19-task training stage inherited the 16-task backbone and output heads, added three search-task heads, and continued optimization with Adam at learning rate 3×10^{-4}, moment coefficients $(0.9, 0.999)$, numerical epsilon 10^{-8} and no weight decay. Effective batches contained 32 trials, accumulated in microbatches of four. The objective combined terminal cross-entropy with a 0.05-weighted pixel-prediction loss; VDA terminal cross-entropy additionally used the native reward value divided by three. Active-parameter gradient norm was clipped at 100. The selected checkpoint was saved during two-times static-memory-search sampling; the final checkpoint includes later four-times sampling. Checkpoint indices refer to this continuation stage, which itself inherited earlier learning.

Each native condition receives 128 new test trials. The final-checkpoint comparison replays the first 32 of those movies under the same generator namespace. Probe fitting uses 192 separate validation-split trials per selected condition and evaluation uses 128 test-split trials. Natural-image probe fitting and evaluation retain separate source-image splits. The native four-trial generator calls are preserved even when their tensors are combined into eight-trial GPU forwards. Inference uses float32; dense appearance, fusion and spatial recordings are stored in float16, while global memory, comparison memory and probabilities retain float32. Stimulus metadata defines analyses and interventions; only native pixels and the requested task head enter the observer's ordinary forward computation.

2.2 Architecture and recording sites

At each model frame, a 100 by 100 RGB (red, green and blue) image is represented at 100 sites arranged on a 10 by 10 grid. Each site carries 32 learned appearance channels, local RGB and contrast statistics, and temporal correspondence-derived features. A learned linear fusion maps these to 64 channels. The spatial recurrent state has 64 channels at each grid site; the global recurrent state, denoted c_t and called CLS memory in the code, has 64 channels. CLS denotes the global classification-token pathway. A separate 735-dimensional comparison-memory vector accumulates selected local temporal-comparison evidence. The task head reads a learned 64-dimensional transformation of the concatenated global and comparison memories.

Let $v_{t,s}$ be the fused visual vector at site s, $h_{t-1,s}$ the preceding spatial memory, and c_{t-1} the preceding global memory. The key and query are

$$k_{t,s} = \mathrm{unit}\{W_k[\mathrm{LN}_v(v_{t,s}) + \mathrm{LN}_h(h_{t-1,s})]\}, \tag{1}$$

$$q_t = \mathrm{unit}\{W_q\,\mathrm{LN}_q(c_{t-1} + e_{\mathrm{CLS}})\}, \tag{2}$$

$$\ell_{t,s} = \kappa\, k_{t,s}^{\mathsf{T}} q_t, \qquad \alpha_{t,s} = \frac{\exp \ell_{t,s}}{\sum_{u=1}^{100} \exp \ell_{t,u}}. \tag{3}$$

Here W_k, W_q are learned matrices, e_{CLS} is a learned token vector, LN denotes layer normalization across channels, and unit divides a vector by its Euclidean norm with a numerical floor of 10^{-6}. The expression

$k^\mathsf{T} q$ is the key-query dot product. The scale is $\kappa = 1 + 15\sigma(\rho)$, where ρ is a learned scalar and σ is the sigmoid function. Its value in the selected checkpoint is approximately 7.7525. The distribution α supplies the actual spatial-selection weights. There is one global query, so averaging the attention matrix over queries returns this same vector. The same spatial weights route learned visual values and temporal-comparison channels and influence spatial-memory writes.

Recordings are taken from appearance features, fused visual tokens, the spatial recurrent field, the global recurrent state, comparison memory and the attention distribution. We verify exact final-output and attention parity between the recording implementation and the authoritative model on all 19 tasks. Intermediate readouts are verified against forwards truncated at the corresponding frame. These controls establish that recording preserves the observer's original computation.

2.3 Behavioral and signal-detection measures

The model's response is the largest terminal class probability. Accuracy is the fraction of prescribed responses selected. Balanced accuracy averages recall across response classes. Macro AUC averages one-versus-rest receiver-operating-characteristic areas using continuous probabilities. Each native condition receives equal weight in the task summary. Wilson intervals accompany individual-condition accuracy; task intervals resample stimulus clusters within the fixed condition set. Matched procedural scenes remain together across timing conditions, and repeated photographs are grouped by image identity. A Wilson boundary interval is used when every observed response is correct or incorrect.

For binary judgements with both classes present, let H and F be hit and false-positive rates. With h, f observed positive reports and n_1, n_0 opportunities, the finite-sample rates are $\widetilde{H} = (h + 1/2)/(n_1 + 1)$ and $\widetilde{F} = (f + 1/2)/(n_0 + 1)$. We report

$$d' = \Phi^{-1}(\widetilde{H}) - \Phi^{-1}(\widetilde{F}), \qquad c = -\tfrac{1}{2}[\Phi^{-1}(\widetilde{H}) + \Phi^{-1}(\widetilde{F})]. \tag{4}$$

Φ^{-1} is the inverse standard-normal cumulative distribution. Sensitivity d' measures separation under this signal-detection convention; criterion c summarizes positive-response tendency. This scalar criterion is distinct from the time-indexed global-memory vector c_t. Labels and the meaning of a positive report are specified separately for each task. The Brier score in the condition tables is the trial-averaged sum $\sum_k [p(k) - \mathbf{1}(y = k)]^2$, where y is the prescribed class and $\mathbf{1}(y = k)$ is one for that class and zero otherwise; smaller values indicate better probability predictions. Difficulty challenges use the existing audited renderers. Zero-strength observations retain their scheduled labels as an ambiguity control. The complete native factorial, empirical difficulty points and response probabilities are retained in machine-readable files.

2.4 Threshold-defined chronometry

The unchanged terminal readout is applied to each recorded prefix state. If $p_t(k)$ is its probability for class k, evidence becomes available at frame t_0, and θ is a fixed confidence threshold, then

$$\tau_\theta = \min\{t - t_0 + 1 : t \geq t_0,\ \max_k p_t(k) \geq \theta\}. \tag{5}$$

The chosen class is the argmax at that first crossing. We use thresholds 0.6, 0.7, 0.8 and 0.9, fixed before inspection of outcomes. Trials without a crossing are right-censored at movie end. We report the crossing fraction, median frames among crossings, accuracy of crossing decisions and the fraction already above threshold before task evidence. Time is measured in model frames; two-observation tasks have only one post-evidence frame. This defines a reproducible computational commitment time for the trained readout.

2.5 Spatial priority and causal manipulations

Attention mass in a region S is $A_t(S) = \sum_{s \in S} \alpha_{t,s}$. Spatial concentration is summarized by effective patch count, $N_{\text{eff}} = \exp[-\sum_s \alpha_s \log \alpha_s]$, which equals 100 for a uniform distribution and approaches one for selection of a single patch. Spatial masks use model-site centers at pixel coordinates $4.5 + 10j$, where $j = 0, \ldots, 9$ on either axis. For point arrays, the designated mask surrounds the task-defined target center; the alternative is the next center in the renderer's location ordering, with wraparound. The radius is 16 pixels for search-array centers and 12 pixels for the other point arrays. Thus the alternative mask is a fixed geometric comparison location and need not coincide with the particular foil that changed. For VDA, masks contain the model-site centers mapped into each native cell by the exact nearest-neighbor resize; the alternative is the cell obtained by reversing the row-major cell index. Native VDA9 cues in this implementation occupy the two opposite corners, so these two masks are distinct. Full-field tasks use the top-left and bottom-right image quadrants as designated and alternative regions. Their designated sites are spatial test regions rather than inferred target objects. Off-site controls use a single site outside the two regional masks; the VDA control additionally minimizes Gabor occupancy.

Computational stimulation and inhibition add a signed dose δ to the priority logits at the selected sites and frames:

$$\alpha_{t,s}^{(\delta)} = \text{softmax}_s\{\ell_{t,s} + \delta\, \mathbf{1}[s \in S]\mathbf{1}[t \in E]\}. \tag{6}$$

E identifies encoding, delay or evidence frames. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. Positive doses raise relative priority and negative doses lower it. The dose is measured in logit units. If a selected region has attention mass A before adding a uniform regional dose, its immediate new mass is $A' = e^{\delta}A/(1 - A + e^{\delta}A)$. Thus its odds against the remaining image are multiplied by e^{δ}; a dose of four multiplies these odds by approximately 54.6. Later frames can additionally change through recurrent feedback. Evidence-epoch doses are $\pm 2, \pm 4, \pm 8$ at designated and alternative sites; encoding and delay use ± 4. A missing epoch is explicitly recorded as zero active intervention trials. Off-site and global shifts provide controls. Every arm reuses identical input pixels. Adding the same constant to all logits must leave the softmax unchanged, and this invariance is checked during collection.

Memory manipulations replace one recurrent state by g times that state after processing the middle delay frame, or after the frame immediately before evidence when no blank delay exists. The next frame is then processed from that modified state. Gains $g = 0$ and $g = 1.5$ respectively reset and amplify the global state, spatial state or comparison memory, one component at a time. The movie then continues normally. Paired accuracy differences compare each arm with the same-trial sham. Intervals use 3,000 paired resamples; zero-discordance outcomes use a conservative exact bound on the discordance probability, and all-gain/all-loss boundaries use Wilson intervals; exact discordant-pair binomial tests assess changes in correctness for independent procedural trials. For natural photographs, repeated probe or base-image identities remain grouped in bootstrap resamples, and 9,999 paired source-image sign flips test the accuracy contrast. The saved test-method field distinguishes those Monte Carlo cluster tests from the exact trial-level tests. Holm adjustment is applied within task, and the four designated ± 4 evidence comparisons also enter one family across tasks and selected conditions.

2.6 Independent-trial decoding and population geometry

Each region is transformed using probe-fitting trials only. Representations wider than 256 coordinates undergo a seeded, label-independent random projection to 256 dimensions. A fixed training-derived principal-component basis then retains at most 32 dimensions and is shared across time. Coordinates are standardized from the fitting data. This capacity-matched representation is used for external linear probes; the original agent remains frozen.

For a design matrix X, target matrix Y and fixed ridge penalty $\lambda_{\text{ridge}} = 10$, the fitted mapping minimizes

$$\|Y - XW - \mathbf{1}b^{\mathsf{T}}\|_F^2 + \lambda_{\text{ridge}}\|W\|_F^2. \tag{7}$$

Each row of X represents one trial and each column one retained activation coordinate. Each row of Y contains that trial's desired decoding target. The matrix W contains fitted weights and b contains output intercepts; $\mathbf{1}$ is a column of ones. The squared Frobenius norm $\|\cdot\|_F^2$ sums squared entries of a matrix. The first term therefore measures prediction error, while the second penalizes large fitted weights. Classification targets are one-hot answer labels, the model's eventual choice, and cue or target location where available. Continuous targets include an 8 by 8 RGB descriptor of the last sample/study image and circular orientation or motion variables when provided by the task. Continuous reconstruction is scored against a predictor using the fitting-set mean. The resulting R^2 can be negative when the fitted mapping predicts new trials worse than that baseline.

For the spatial and global recurrent states, a decoder fitted at each time is evaluated at every time on independent test trials. The vertical axis is training time and the horizontal axis is test time. Diagonal performance measures available information at the same phase; off-diagonal performance tests transfer of the fitted linear readout between phases. A final-time label-permutation test uses 199 fitting-label shuffles, with unchanged held-out labels. Population trajectories use the same training-derived axes, and their class means are computed from held-out trials. Participation ratio is $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ for covariance eigenvalues λ_j within the retained subspace. Frame-to-frame root-mean-square (RMS) displacement measures how rapidly the state evolves in those coordinates.

2.7 Full-state controls and unit-level selectivity

As a control for dimensionality reduction, external ridge classifiers are also fitted directly to every recorded coordinate at the final frame and at the memory-test frame (the middle blank-delay frame, or the frame immediately before evidence when no blank delay exists). The ridge penalty is selected from 0.001, 0.01, 0.1, 1 and 10 using a class-stratified 75/25 split inside the probe-fitting set. Standardization is fitted on the inner fitting subset during selection and on the full fitting set for the final held-out evaluation. Each standardized full-coordinate vector is additionally divided by the square root of its coordinate count; these penalty values refer to that normalized feature scale. The full-crop reconstruction and candidate-reconstruction regressions use the same scaling convention. These full-state probes supplement the fixed-capacity temporal analysis. Global-memory unit selectivity is assessed by response-class differences on held-out trials, with Holm correction across the 64 units. Cross-validated class-centroid distances multiply independently estimated fitting and test centroid differences, standardized by fitting-set channel dispersion; negative estimates are retained.

For native factorial summaries, bootstrap clusters retain the same procedural scene across matched timing conditions. Photo repeats are grouped by probe or base image identity within condition. This preserves the dependence generated by reusing stimuli. The saved condition-level results allow every marginal to be reconstructed and interactions to be examined directly.

3 Results

3.1 Native behavior, difficulty and checkpoint robustness

Native equal-condition accuracy was 97.7%, with macro AUC 0.999. The matched final checkpoint changed accuracy by +0.58 percentage points on average across conditions, using 32 same-movie comparisons per condition. The lowest native cell was motion_A20_S6_D4_H3 at 94.5%; the highest was

motion_A90_S3_D0_H6 at 100.0%. These comparisons retain the native sampling design rather than selecting trials after observing success.

Marginal accuracy by delay was 0: 98.0%; 4: 97.7%; 12: 97.5%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

Marginal accuracy by sample frames was 3: 97.8%; 6: 97.7%; 12: 97.7%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

Marginal accuracy by exposure was 3: 97.4%; 6: 97.9%; 12: 97.9%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

Marginal accuracy by spacing was 20: 96.5%; 45: 97.0%; 90: 99.7%. These marginal summaries average over the other native factors; the condition table preserves their interactions.

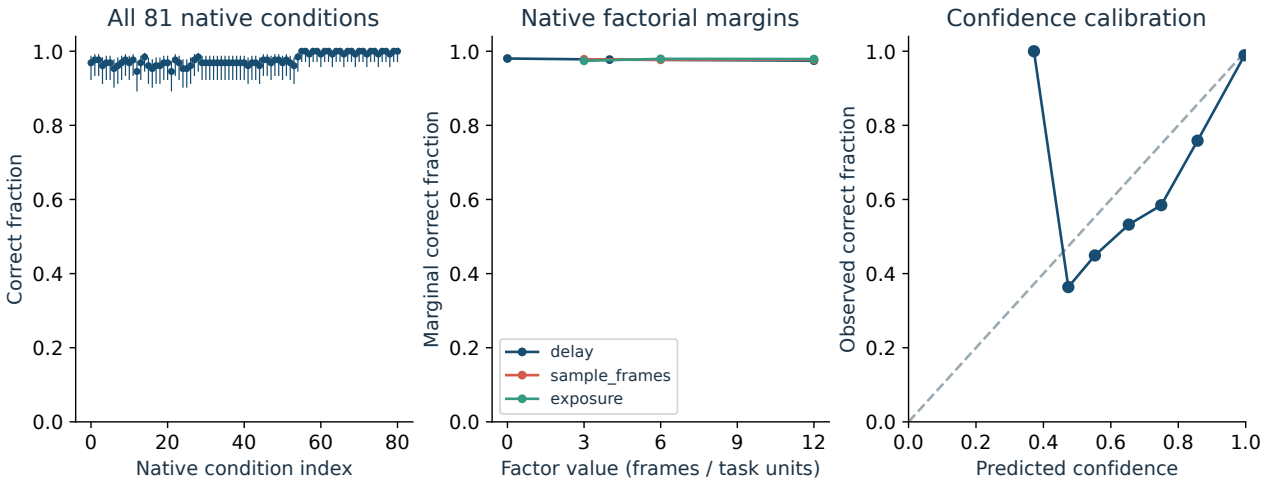


Figure 1: Behavioral phenotype. Left: accuracy and Wilson intervals for every native condition. Middle: empirical signal-strength functions or native factorial margins. Right: observed accuracy within confidence bins, against the identity line. Zero-strength challenges assess responses to ambiguous physical evidence. Native cells contain 128 trials each.

3.2 Matched-scene effects of sample, delay and search time

The native counter-based generator reuses scene identity and target assignment across timing conditions. We compare the longest and shortest values of each timing factor while matching the other factors, then average those paired differences within scene. Static arrays remain pixel-identical; motion arrays retain the same direction and layout seeds and shared frame prefix as duration increases. Bootstrap intervals resample the 128 scene identities. A two-sided sign-flip test uses 9,999 random sign patterns on the scene-level differences, followed by Holm correction over this task's timing comparisons. These exploratory contrasts distinguish changes in encoding opportunity, retention duration and search opportunity.

Group	Factor	Short/long	Change (pp)	Holm p
20	sample	3/12	-0.5	1
20	delay	0/12	-1.3	1
20	exposure	3/12	+1.2	1
45	sample	3/12	+0.1	1
45	delay	0/12	-0.5	1
45	exposure	3/12	-0.6	1
90	sample	3/12	+0.1	1
90	delay	0/12	+0.1	1
90	exposure	3/12	+0.9	1

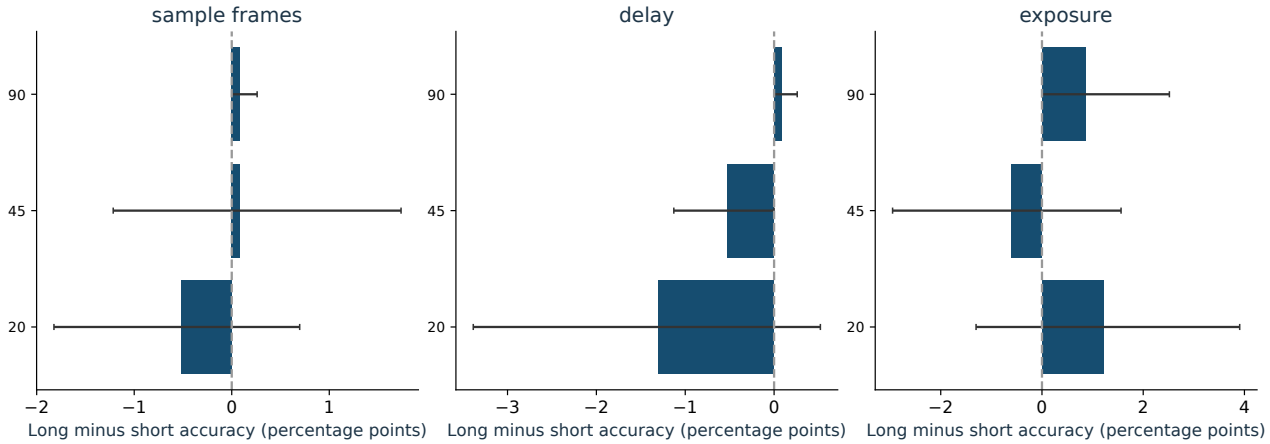


Figure 2: Paired timing effects with scene identity preserved. Each value is the longest-duration accuracy minus the shortest-duration accuracy, averaged over matched remaining factors. Intervals resample complete scenes.

3.3 Time course and commitment thresholds

Threshold	Crossed	Median frames	Crossing accuracy	Premature
0.6	100.0%	1.0	34.4%	100.0%
0.7	100.0%	1.0	43.8%	100.0%
0.8	100.0%	2.0	57.0%	100.0%
0.9	100.0%	2.0	67.2%	100.0%

The representative condition motion_A20_S12_D12_H12 provides 12 post-evidence frames. Median latency is conditional on crossing; the uncommitted-fraction plot retains all trials. The threshold sweep

makes the speed–coverage tradeoff explicit. A rise in correct-answer probability over time can be compared with the development of location and answer decodability.

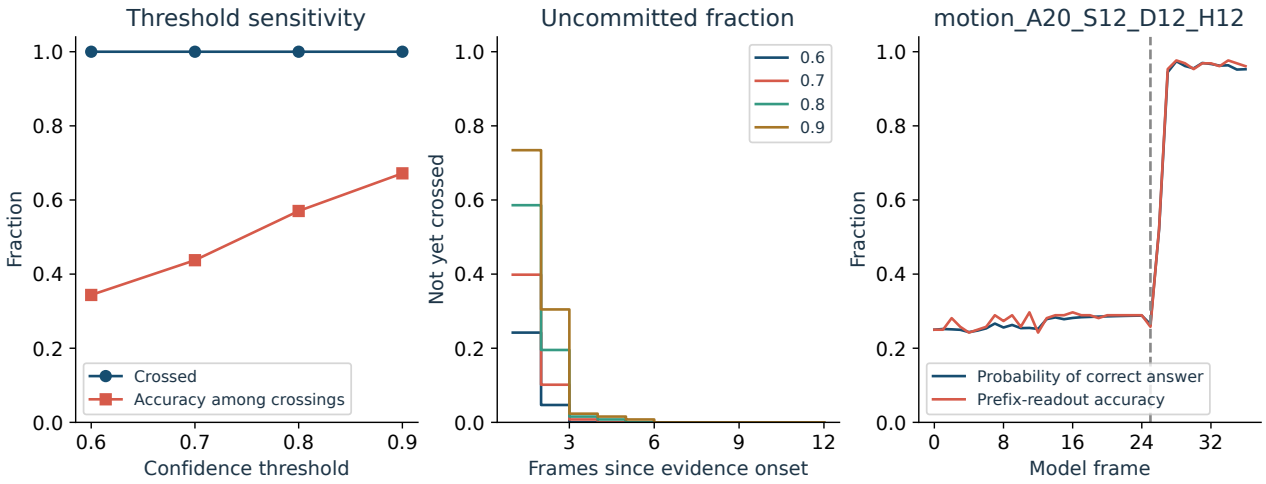


Figure 3: Threshold-defined timing. Left: the fraction reaching each confidence threshold and accuracy among those crossings. Middle: the fraction still uncommitted after each available evidence frame. Right: probability assigned to the correct response and accuracy of the unchanged head applied to prefix states. The dashed vertical line marks evidence onset.

3.4 Does an early confident answer persist?

A confidence threshold defines a hypothetical commitment rule. We test that rule directly by comparing its first post-evidence choice with the unchanged terminal choice on the same movie. A rescued error is an incorrect early commitment followed by a correct final answer; a spoiled answer changes from correct to incorrect. For four-class tasks, a revision can also exchange one incorrect alternative for another. These measurements determine whether terminal accuracy transfers to early stopping.

At threshold 0.8 across all native presentations, 10,346 of 10,368 trials crossed. Of those, 4,257 changed their answer later, 4,168 corrected an early error, and 44 lost an initially correct answer. These pooled counts are descriptive; the representative condition is shown separately below.

We additionally measure retrospective choice-settling latency: the first post-evidence frame from which the argmax remains equal to its terminal value for the rest of the movie. Frame counts start at one for the first available evidence frame. This measure uses future observations to locate the final revision. It describes the dynamics of the existing computation; it is distinct from the causal first-threshold stopping rule. Terminally correct and incorrect trajectories are kept separate.

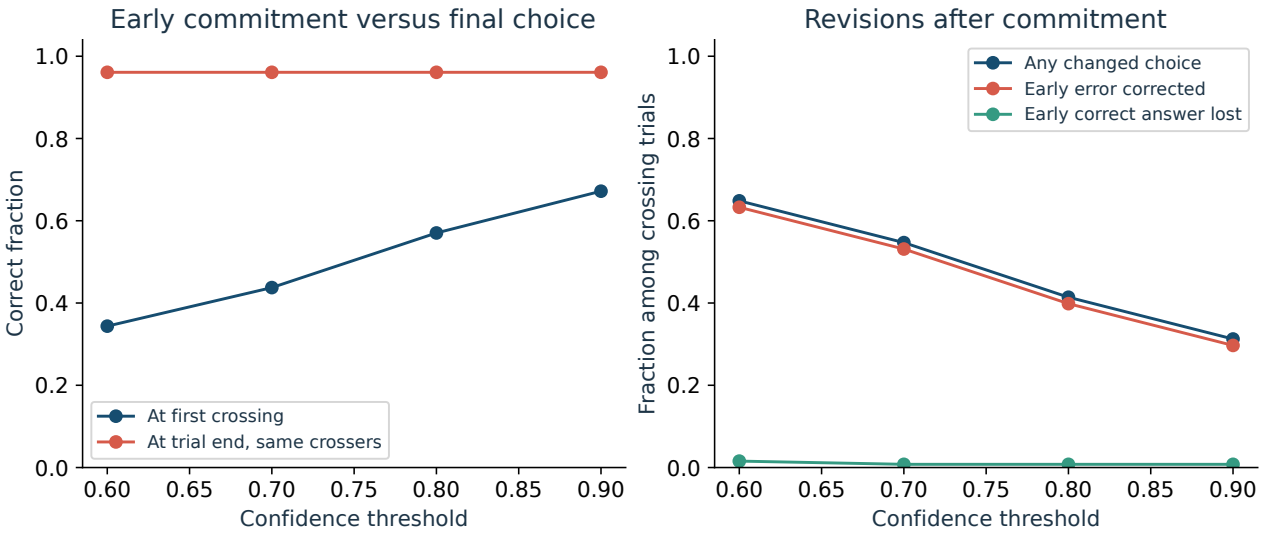


Figure 4: Validity of first-threshold commitment in the representative native condition. Early and terminal accuracy are compared on exactly the same crossing trials at each threshold; the second panel separates kinds of later revision. Trials that never cross remain censored in the preceding analysis.

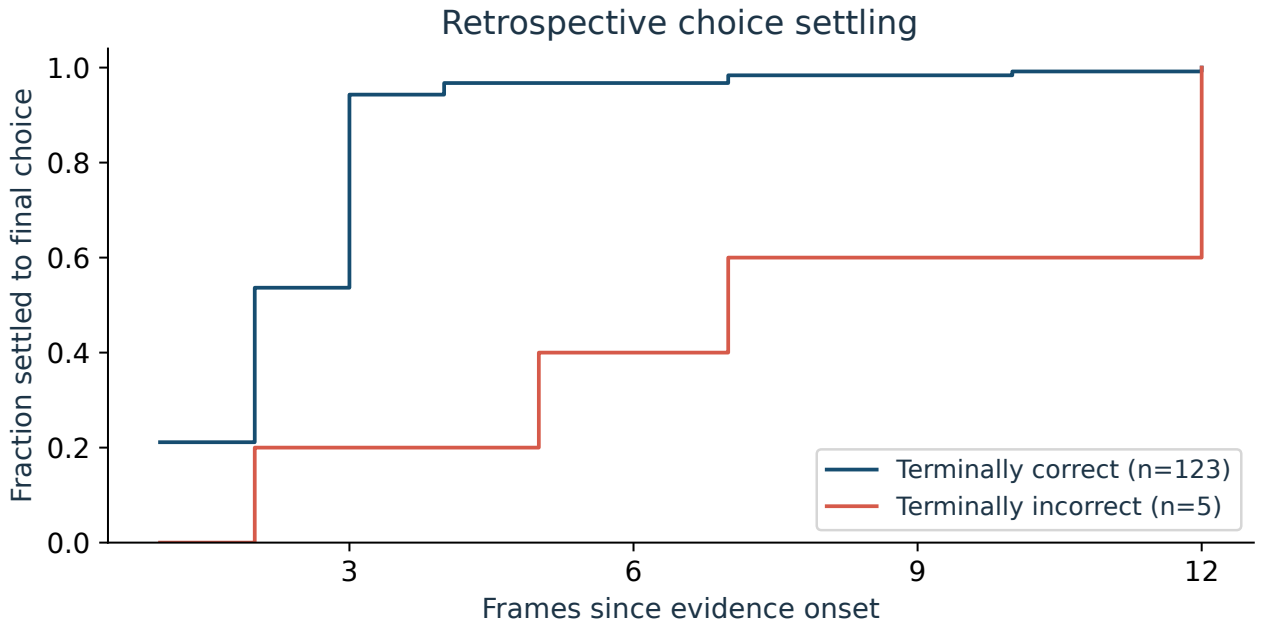


Figure 5: Cumulative distribution of retrospective choice-settling times in the representative condition. A choice is settled once all subsequent prefix classifications agree with the terminal classification. Correct and incorrect terminal decisions can both settle, and are plotted separately.

3.5 Attention across conditions and time

The analysis is organized around a direct visual question: where does the shared priority signal go as the trial unfolds, and how does that trajectory depend on the task condition? Rows retain fixed experimental conditions and physical locations. Columns follow time. The arrangement follows the heatmap-plus-time-course logic of Morgan, Albanna and Herman (2025), Figure 4 (https://arxiv.org/html/2502.10955v1).

The matching candidate is fixed at top left. Rows compare 20-, 45- and 90-degree candidate separation at twelve sample frames, twelve delay frames and twelve array frames.

What each plotted quantity means. For trial i, frame t and image patch p, α_{itp} is the attention probability assigned to that patch; its sum over the 100 patches is one. For a condition containing n trials, the map shows $\bar{\alpha}_{tp} = n^{-1} \sum_i \alpha_{itp}$. The 100 entries are placed back into their original 10-by-10 image grid without spatial rotation. A line for region R shows the trial-average mass $M_R(t) = n^{-1} \sum_i \sum_{p \in R} \alpha_{itp}$. A line labeled "other regions (mean) averages their separate masses rather than summing all alternatives. A dotted horizontal line marks the mass the same region would receive under uniform attention, its patch count divided by 100.

Target apertures use the same patch-center geometry throughout a task: circular masks with a 16-pixel radius for search arrays and a 12-pixel radius for the other localized tasks. VDA uses its native stimulus-cell boundaries mapped onto the 100-site grid. The central sample region is a 16-pixel-radius circle excluding candidate apertures. The source tables retain the exact region weights.

Every curve includes all selected trials, independently of whether the model answered correctly. Shaded bands are pointwise 95 percent intervals from 2,000 whole-trial bootstrap resamples. Repeated photograph identities are resampled together. These bands describe each condition separately. Phase-colored backgrounds mark encoding, retention and comparison; frame zero on the horizontal axis is the task-specific evidence onset, or the first motion transition in the duration task. Original frame numbers remain visible on the maps.

Map colors share one scale across every condition and time point within this task. A square-root color transform makes small weights visible; the colorbar retains probability units. The overview uses named temporal landmarks, with explicit original frame numbers when trial lengths differ. The appendix supplies every recorded frame, and the accompanying HTML viewer pairs an exact trial movie with its own attention overlay and the condition-average map.

Condition	Trials	Clusters
Direction separation 20 degrees	32	32
Direction separation 45 degrees	32	32
Direction separation 90 degrees	32	32

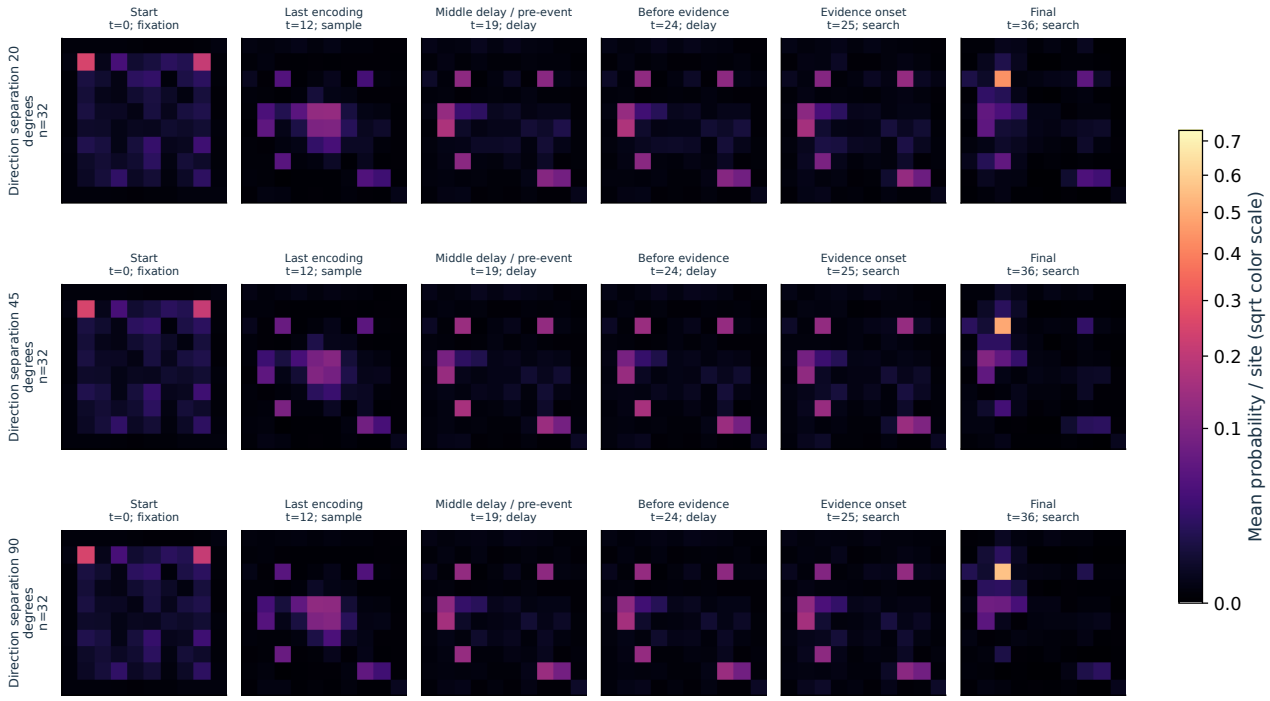


Figure 6: Condition-by-time spatial allocation. Each row averages its stated trial group at a fixed physical location. The common square-root color scale is labeled in probability per patch. Original frame t and phase identify each observation; on full-series panels the signed number additionally gives time relative to evidence onset.

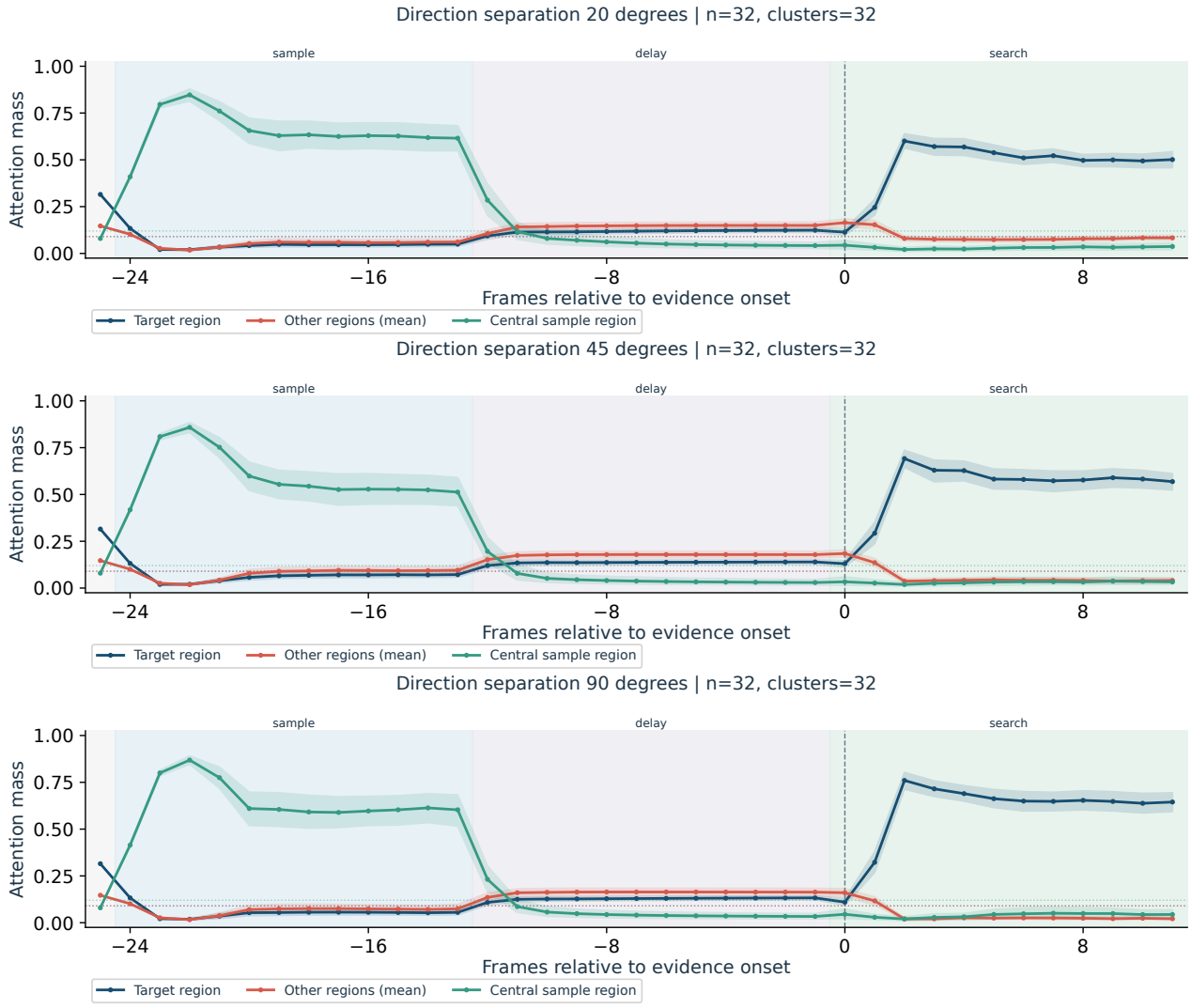


Figure 7: Regional attention through the entire trial, separately for each condition. Solid traces are mean probability masses; shading gives pointwise 95 percent cluster-bootstrap intervals. Dotted lines show uniform allocation for each region. Background bands and labels identify trial phases. The vertical dashed line is the alignment event.

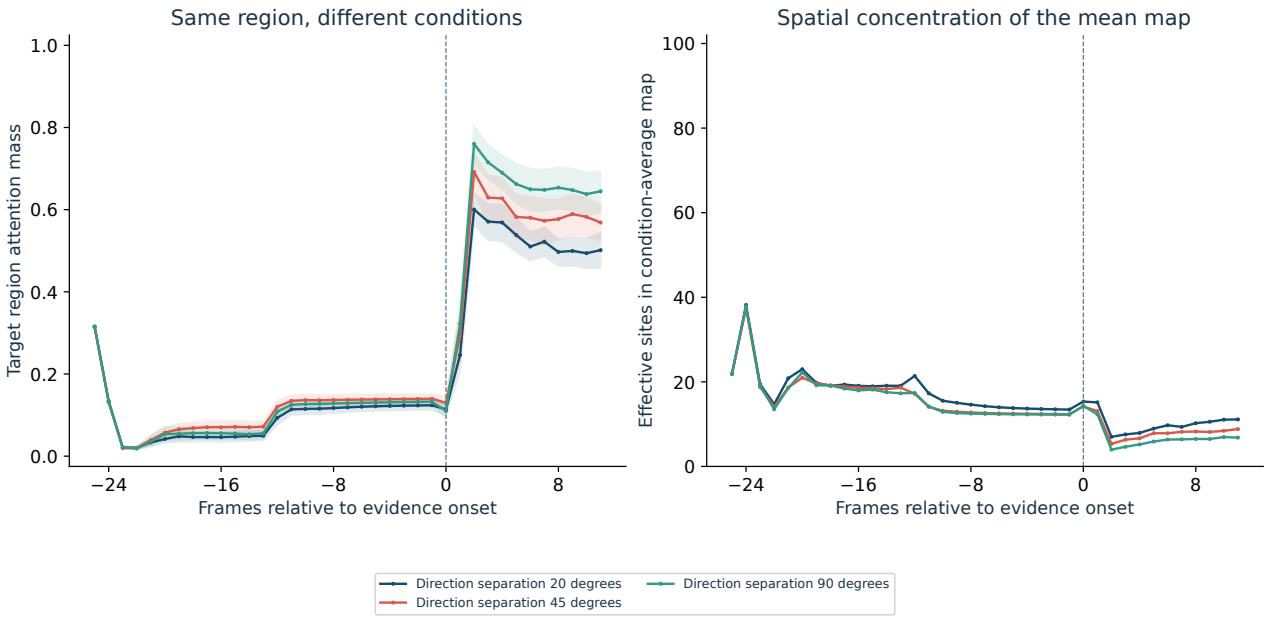


Figure 8: Direct comparison of conditions on the same axes. Left: attention at the first named region, with pointwise uncertainty. Right: concentration of the condition-average map, measured as the exponential of its entropy. A value near 100 means a broad mean map; smaller values mean concentration on fewer sites. This summary is computed after averaging maps and is distinct from averaging each trial's concentration.

Reading the observed trajectories. Direction separation 20 degrees: target region mass peaks at 0.600 at original frame 27 (search), and is 0.501 on the final frame. Direction separation 45 degrees: target region mass peaks at 0.692 at original frame 27 (search), and is 0.569 on the final frame. Direction separation 90 degrees: target region mass peaks at 0.760 at original frame 27 (search), and is 0.645 on the final frame. These summaries locate the observed allocation in time; the neighboring intervention experiments test its causal contribution to behavior.

3.6 Causal sensitivity to spatial and mnemonic interventions

For distractor_+8_evidence, paired accuracy changed by -96.1 percentage points (paired 95% interval [-99.2%, -92.2%]; paired p=1.881e-37; within-task Holm p=1.166e-35). There were 0 corrected trials and 123 newly incorrect trials. For target_+2_evidence, paired accuracy changed by +3.9 percentage points (paired 95% interval [0.8%, 7.8%]; paired p=0.0625; within-task Holm p=1). There were 5 corrected trials and 0 newly incorrect trials. These extrema describe the observed exploratory sweep. The four predesignated spatial comparisons at dose magnitude four are evaluated jointly across the full series in the integrated synthesis.

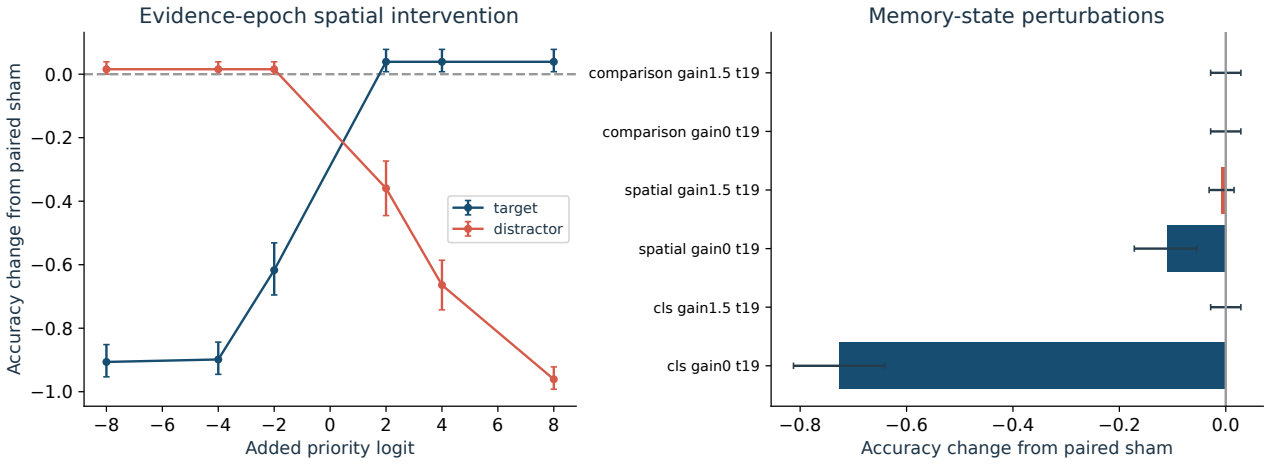


Figure 9: Matched causal effects in the representative condition. Left: changes from sham across signed spatial-priority doses during available evidence. Right: resetting or amplifying one memory component at the specified frame. Intervals resample matched trials, preserving sham–intervention pairing. Target and distractor labels denote the task geometry; full-field tasks use fixed quadrants.

3.7 Predesignated spatial comparisons across the full series

The following dose-four evidence comparisons belong to one family of 108 tests across all nineteen tasks and selected conditions. The Holm-adjusted values therefore account for the complete primary spatial family. Positive changes indicate benefit and negative changes indicate impairment relative to the same-movie sham.

Condition	Intervention	Change (pp)	Global Holm p
motion_A20_S12_D12_H12	target_-4	-89.8	5.152e-33
motion_A20_S12_D12_H12	target_+4	+3.9	1
motion_A20_S12_D12_H12	distractor_-4	+1.6	1
motion_A20_S12_D12_H12	distractor_+4	-66.4	5.222e-24
motion_A90_S12_D12_H12	target_-4	-70.3	1.648e-25
motion_A90_S12_D12_H12	target_+4	+0.8	1
motion_A90_S12_D12_H12	distractor_-4	+0.8	1
motion_A90_S12_D12_H12	distractor_+4	-46.1	3.261e-16

3.8 Readable information across representation levels

Region	Final AUC	Balanced accuracy	Permutation p
appearance	0.499	25.8%	0.395
fusion	0.509	24.2%	0.635
spatial	0.679	41.4%	0.005
cls	0.996	95.3%	0.005
comparison	0.626	32.0%	0.105
attention	0.986	94.5%	0.005

All table entries are evaluated on held-out trials. Final-time permutation p-values quantify how unusual the observed decoding is relative to shuffled fitting labels; the full regional panel is exploratory. The capacity-matched probes quantify readable information in the specified projected subspaces. Choice, location and continuous-feature targets are retained in the accompanying JSON results.

3.9 Mnemonic population geometry and code stability

In cls memory, the participation ratio ranged from 0.00 to 14.97 within the retained subspace. The final step displacement was 0.279 standardized RMS units. In spatial memory, the participation ratio ranged from 0.00 to 18.68 within the retained subspace. The final step displacement was 0.178 standardized RMS units. The time-generalization matrices distinguish a stable readout from information that is readable only by a phase-specific decoder. Broad off-diagonal AUC supports persistence of a common linear discriminant; a strong diagonal with weaker transfer supports changing code geometry. Trajectories use one set of axes fitted to independent fitting trials, so apparent movement is not caused by refitting the axes at each frame.

At memory-test frame 19, reconstruction of the sample motion direction, represented by its cosine and sine had held-out R-squared 0.759 in CLS memory and 0.868 in spatial memory. This continuous target describes stimulus content, while the final-answer decoder tests the response required after the subsequent evidence.



Figure 10: Information across six recording sites. The answer-decoding curves report held-out macro AUC. Sample-content curves report reconstruction of a coarse last-sample or last-study image descriptor relative to a fitting-set-mean predictor. Each region uses an independently fitted training-only projection, shared across time; the agent itself is unchanged.

At the same intervention boundary, resetting each component changed paired accuracy as follows: cls, -72.7 percentage points, spatial, -10.9 percentage points, comparison, +0.0 percentage points. Comparing these causal effects with the decoding curves distinguishes information that is recoverable from a state from information whose removal at that instant affects the subsequent choice.

3.10 Condition-resolved mechanisms

The additional panels retain differences between selected stimulus families and timing conditions. Family averages summarize a balanced native factorial; cells remain available individually in the appendix and source data. The decoding comparison uses the same external fitting protocol for every listed condition.

3.11 Full-state probe control and population selectivity

Region	Full final AUC	Memory-test AUC	Coordinates
appearance	0.477	unavailable	3200
fusion	0.545	0.442	6400
spatial	0.929	0.473	6400
cls	0.998	0.475	64
comparison	0.645	0.459	735
attention	0.984	0.460	100

The full-state control uses the original coordinate width, while the main temporal figures retain at most 32 components. Both evaluations use independent held-out trials. At the final frame, 51 of 64 global-memory units had response-class selectivity at within-population Holm p below 0.05. This is selectivity for the task answer in a recurrent activation coordinate. The complete unit effects and cross-validated centroid-distance trajectories are stored in `deep_analysis.json`.

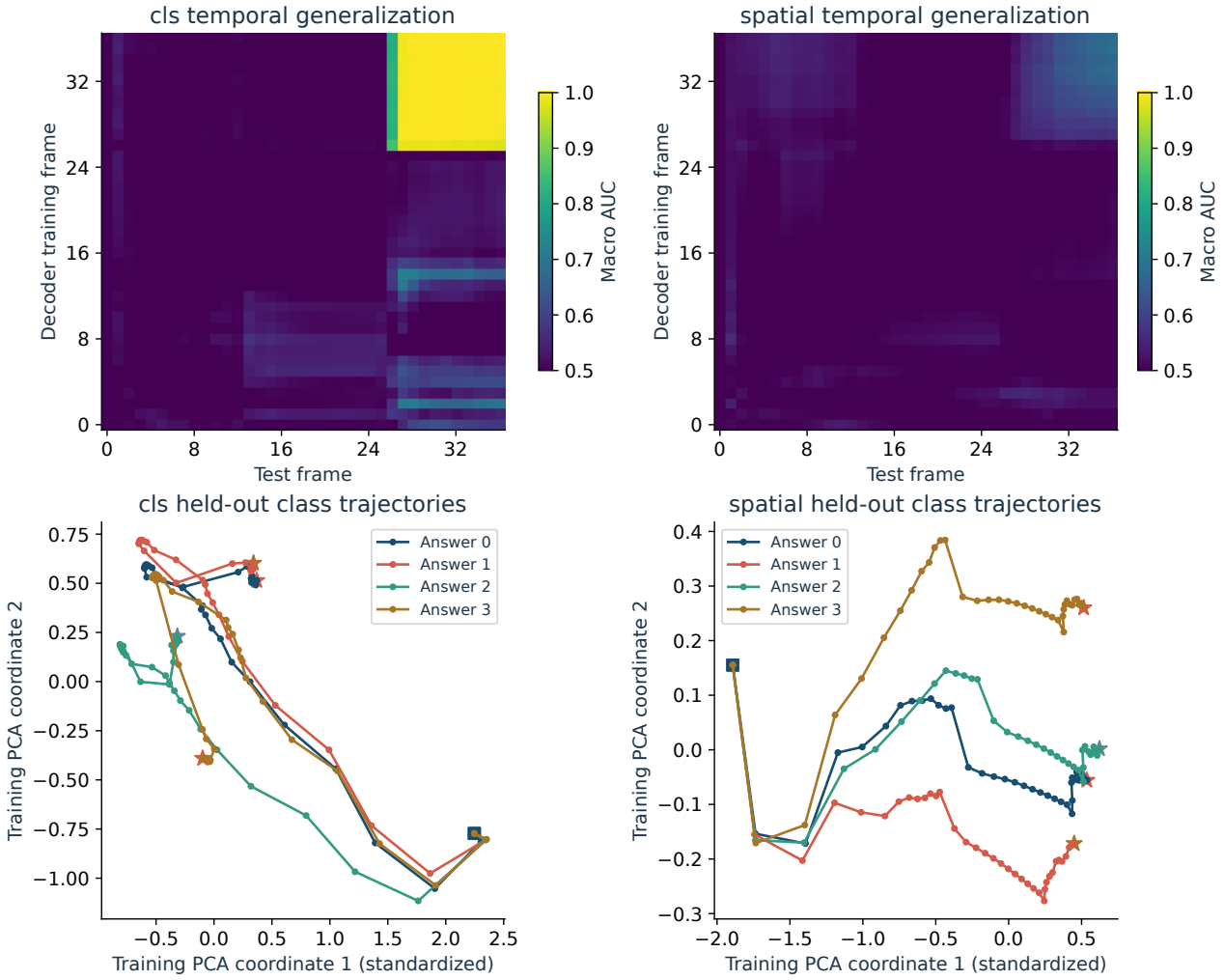


Figure 11: Temporal generalization and held-out population trajectories. Top: decoders fitted separately at each frame and tested across all frames; chance AUC is 0.5. Bottom: class-mean trajectories in a shared training-derived coordinate system, with squares marking starts and stars marking ends. Spatial and CLS populations are analyzed separately.

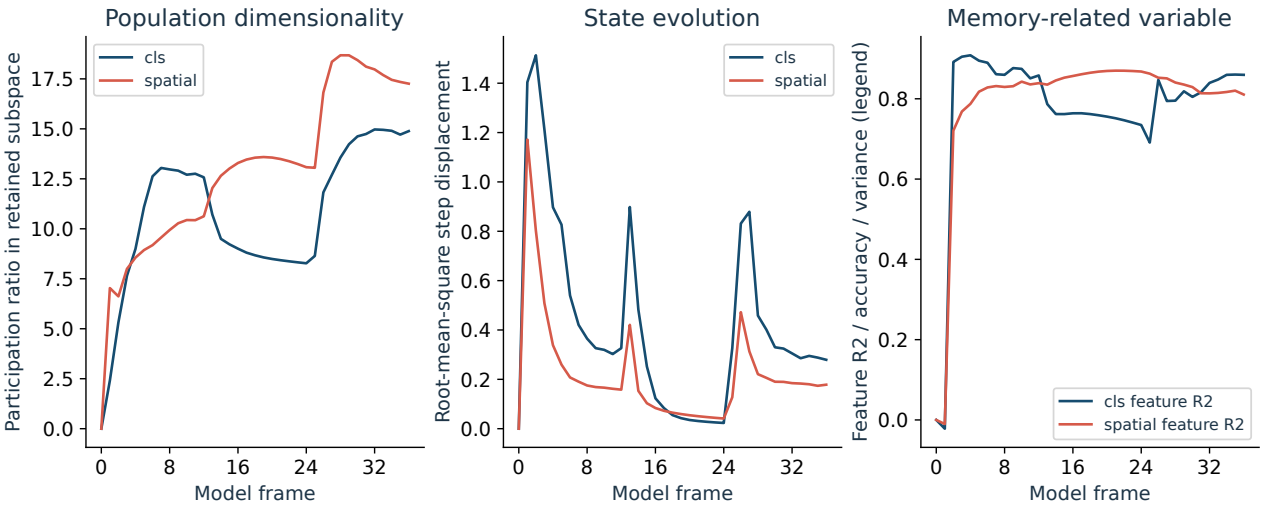


Figure 12: Population dynamics and mnemonic variables. Left: covariance participation ratio within the retained subspace. Middle: frame-to-frame RMS state displacement. Right: continuous remembered-feature reconstruction when supplied by the task, otherwise location decoding or response-class centroid variance as stated by the legend. Each quantity is tied to the original sequence clock.

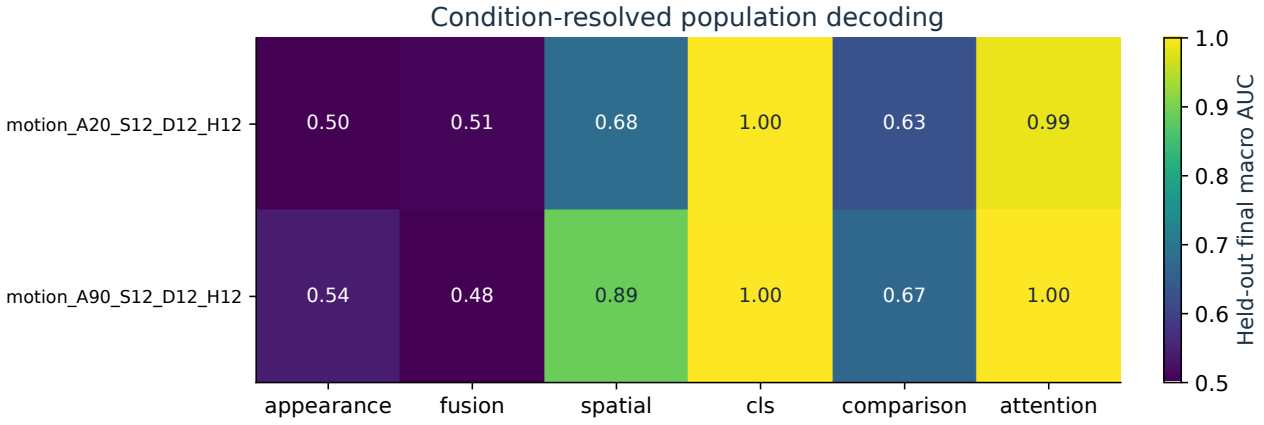


Figure 13: Held-out final answer decoding across selected conditions and recording sites. A constant-zero comparison representation is displayed at chance AUC.

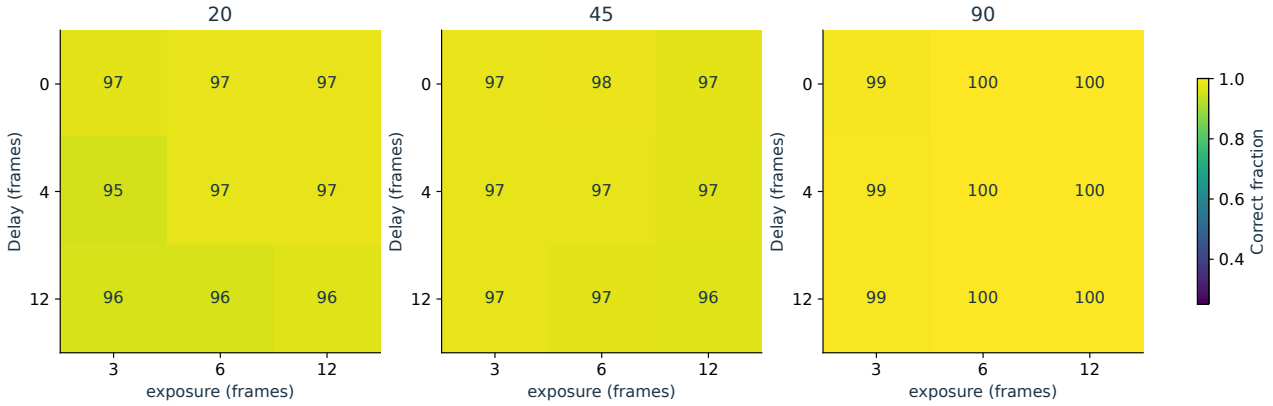


Figure 14: Memory-search factorial. Each cell shows accuracy averaged over the other exposure or sample-duration factor, within the indicated family/similarity or motion-spacing group. Numbers are percent correct; all contributing native conditions contain 128 trials. Delay, encoding time and search time remain distinct factors.

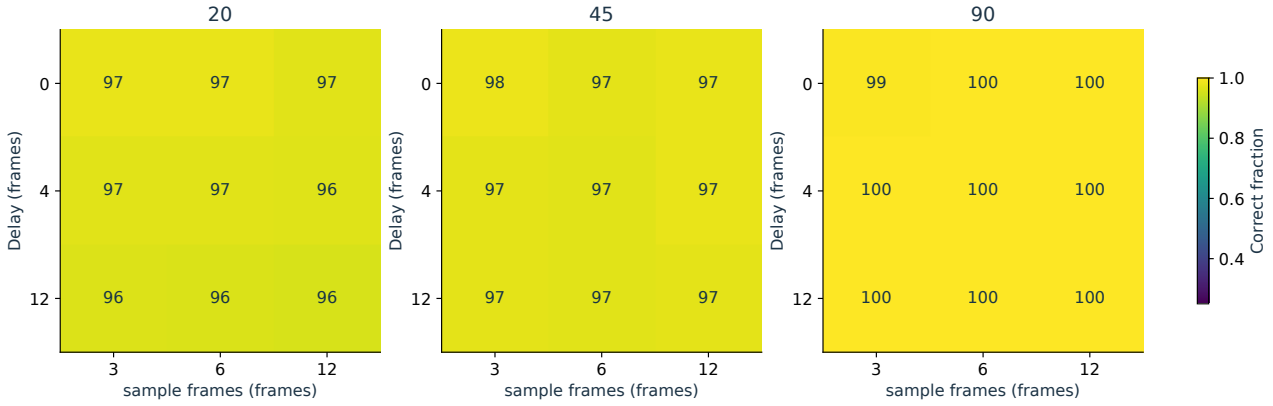


Figure 15: Memory-search factorial. Each cell shows accuracy averaged over the other exposure or sample-duration factor, within the indicated family/similarity or motion-spacing group. Numbers are percent correct; all contributing native conditions contain 128 trials. Delay, encoding time and search time remain distinct factors.

3.12 Direct verification of attention redistribution

On 32 exactly replayed trials per selected condition, all six priority arms reproduced their original behavioral probabilities within numerical tolerance. The following figure measures the induced changes in spatial allocation at every frame. This subset supplies a direct map-level manipulation check.

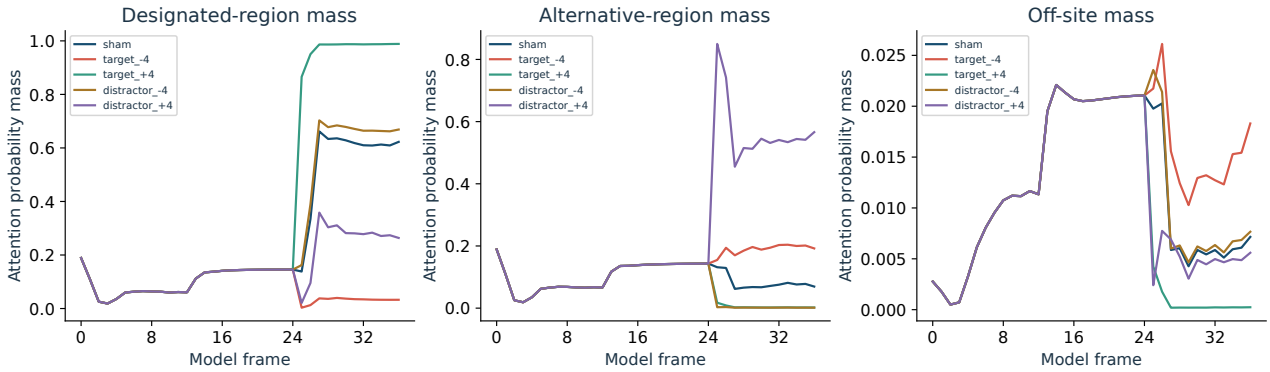


Figure 16: Matched map-level verification. Mean designated, alternative and off-site attention mass under sham and four dose-four evidence interventions. These are the first 32 exact movies from the primary intervention experiment, not a new behavioral sample.

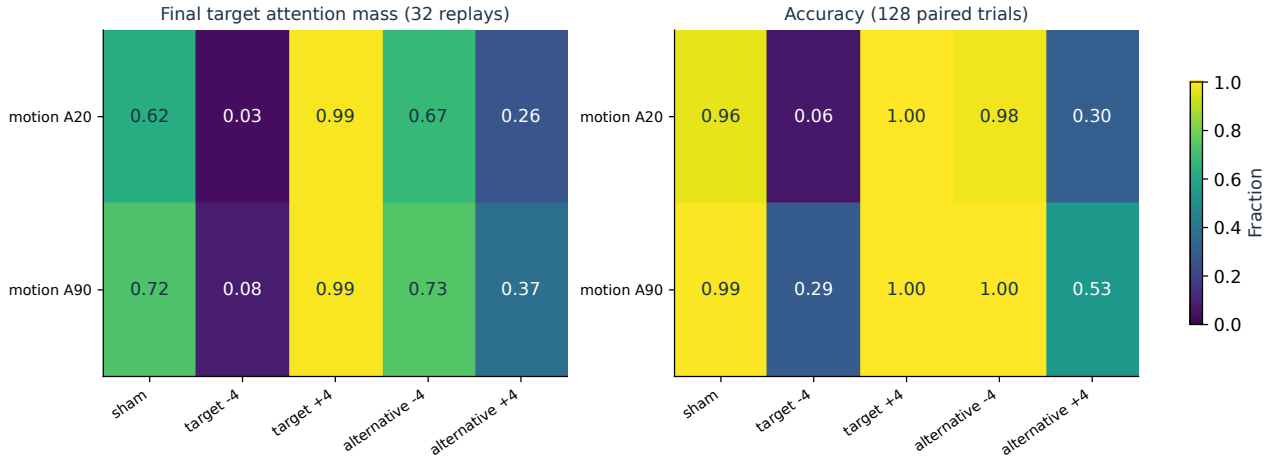


Figure 17: Condition-resolved manipulation checks. The left panel measures final target-region allocation on 32 exact replays; the right measures accuracy on all 128 paired intervention trials. All displayed spatial doses have magnitude four and act during evidence frames. Rows preserve each selected condition, making redistribution and behavioral rescue separately visible.

For motion_A20_S12_D12_H12, a sample-centered encoding dose of -4 changed paired accuracy by -56.2 percentage points on 32 trials (interval [-71.9%, -40.5%]). This manipulation addresses the central sample location, separately from the future array-target location.

For motion_A20_S12_D12_H12, a sample-centered encoding dose of +4 changed paired accuracy by -3.1 percentage points on 32 trials (interval [-9.4%, 0.0%]). This manipulation addresses the central sample location, separately from the future array-target location.

For motion_A90_S12_D12_H12, a sample-centered encoding dose of -4 changed paired accuracy by -34.4 percentage points on 32 trials (interval [-50.0%, -18.8%]). This manipulation addresses the central

sample location, separately from the future array-target location.

For motion_A90_S12_D12_H12, a sample-centered encoding dose of +4 changed paired accuracy by +0.0 percentage points on 32 trials (interval [-10.9%, 10.9%]). This manipulation addresses the central sample location, separately from the future array-target location.

3.13 Held-out population response profiles

The 64 CLS coordinates are ordered by the class with the largest final mean response on fitting trials, with within-class order determined by between-class response variance. The displayed means use independent test trials. Four example coordinates are selected by the largest between-class variance on fitting trials; their held-out trajectories are shown with 1.96 standard-error bands. Activation coordinates use the fitting-set mean and standard deviation across trials and time.

3.14 Correct answers, actual choices and spatial instructions

The correct-answer decoder asks what the stimulus requires; a separate decoder predicts the model's actual terminal choice. They use the same independent fitting/test split and capacity-matched regional features, but distinct target labels. Errors allow these two kinds of information to separate. A location decoder uses the cue index when available, otherwise the instructed or target location. Balanced accuracy averages recall over the labels represented in the held-out population. When every response is correct, answer and choice labels coincide.

Region	Correct answer	Actual choice	Cue/target location
appearance	25.8%	24.4%	25.8%
fusion	24.2%	28.7%	24.2%
spatial	41.4%	40.5%	41.4%
cls	95.3%	97.6%	95.3%
comparison	32.0%	36.0%	32.0%
attention	94.5%	96.9%	94.5%

3.15 Class geometry in the full recurrent states

For classes a and b, let d_j^{fit} and d_j^{test} be their mean-activation differences in coordinate j, estimated from independent fitting and test populations and divided by the fitting-set standard deviation at that frame (with a 0.05 floor). The cross-validated distance is $D_{ab} = M^{-1} \sum_{j=1}^{M} d_j^{\mathrm{fit}} d_j^{\mathrm{test}}$, where M is 64 for global memory and 6,400 for spatial memory. Independent differences remove the positive within-sample noise contribution present when one squared mean difference is used. Negative estimates are retained. Class codes follow the task response labels. This analysis uses every recorded recurrent coordinate and complements the low-dimensional trajectory visualization.

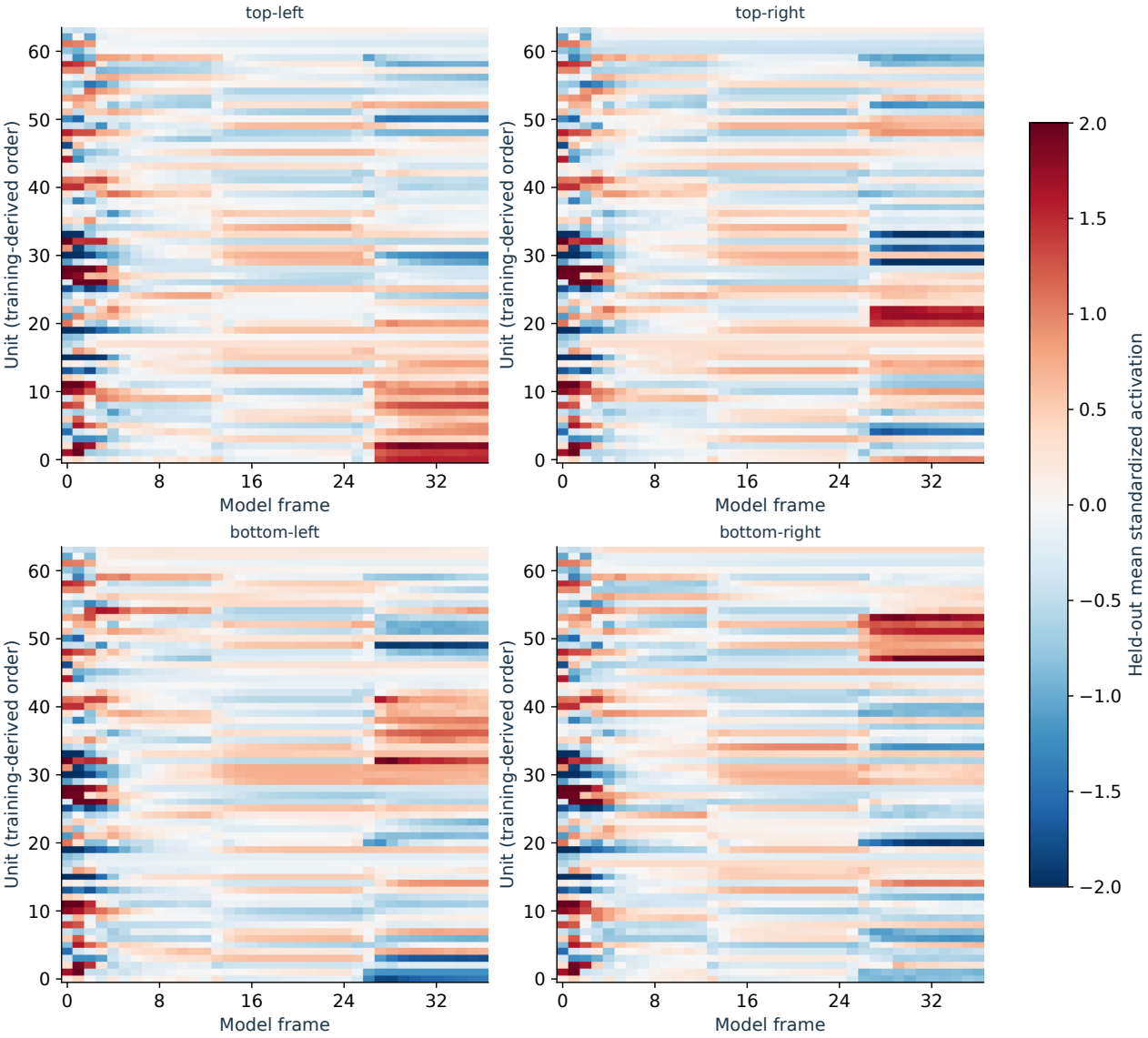


Figure 18: All 64 global-memory coordinates, shown separately for each correct-answer class. Unit ordering and standardization are learned from the fitting population; displayed activity comes from held-out trials. Colors represent activation, not firing rate.

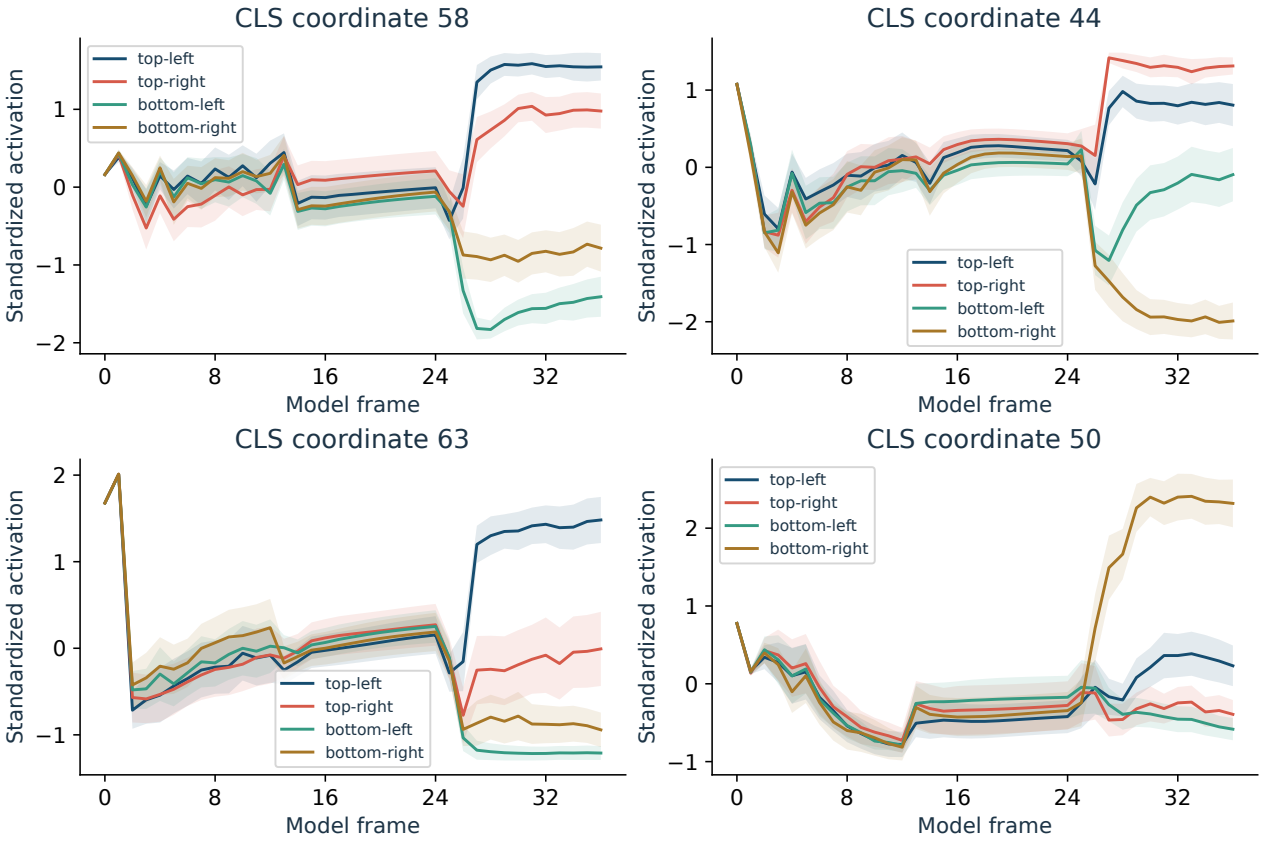


Figure 19: Four population coordinates selected on fitting trials and evaluated on independent trials. Curves separate response classes; shaded bands describe uncertainty in the mean activation.

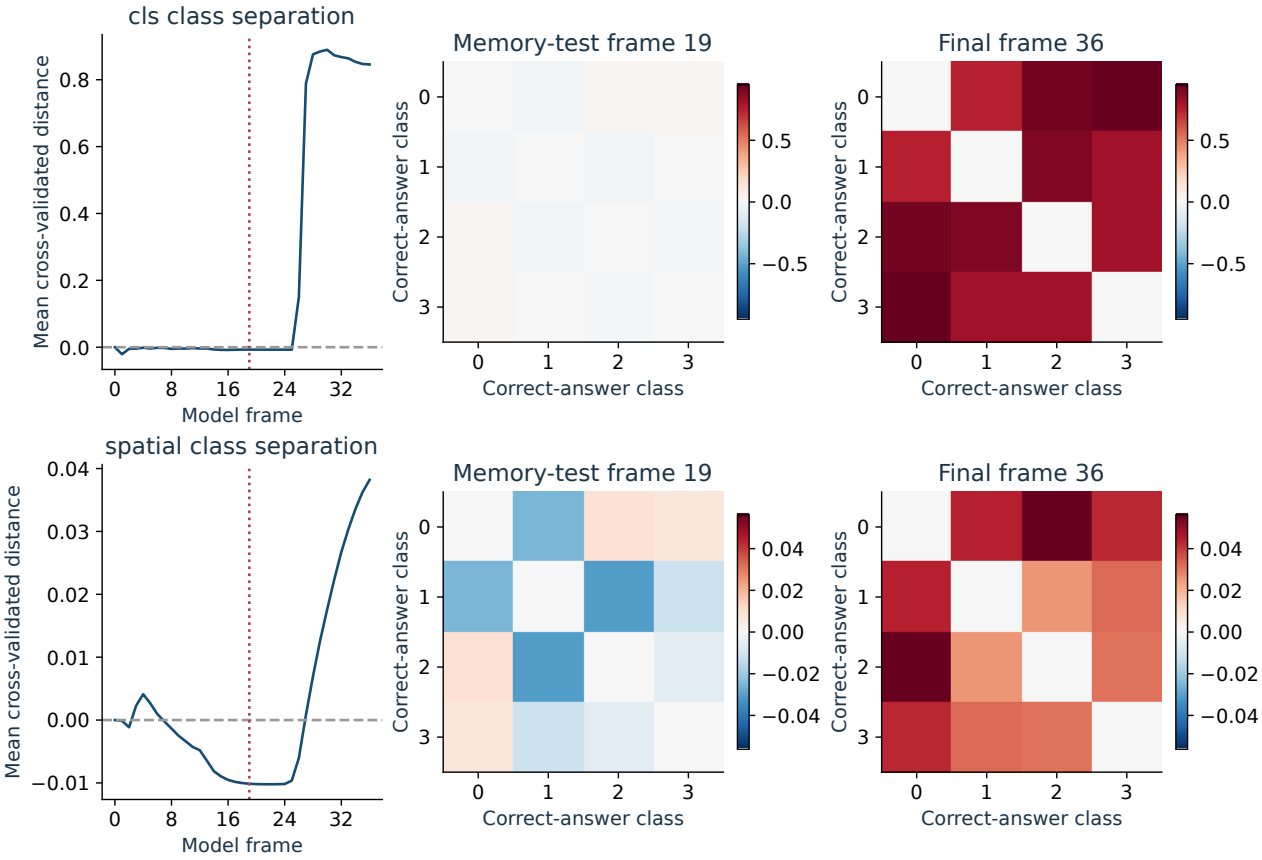


Figure 20: Cross-validated class geometry of the complete recurrent states. Left: mean off-diagonal distance over time, with the memory-test frame marked. Middle and right: class-pair distance matrices at the memory-test and final frames. Color scales are shared between the two matrices within each population. Zero is the null separation baseline; negative estimates remain visible.

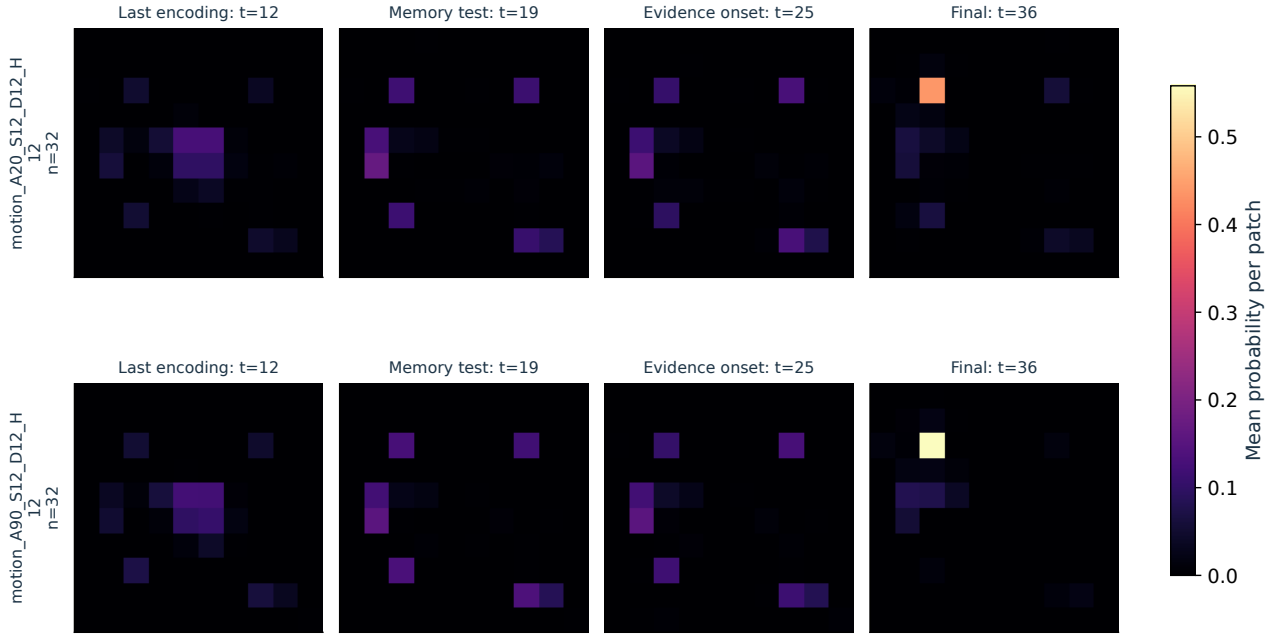


Figure 21: Condition-resolved attention at four temporal landmarks. Memory-search and pop-out rows average trials with top-left targets. Image-recognition rows average seen-probe trials. Each map is shown in the original image layout with a common probability scale across conditions; the row label gives the number of averaged trials. The memory-test frame is the middle delay, or the frame before evidence when a blank delay is absent.

3.16 Identical-array, changed-memory experiment

For each selected condition, 64 new scenes were each replayed four times. Only the sampled candidate changed; all search-array pixels were verified identical across the four movies. Every location served once as the correct answer within each scene. An additional replay replaced the sample with uniform midpoint gray. Intervals resample whole scenes, keeping their four queries together; all-perfect or all-zero accuracy boundaries use a Wilson interval on the 64 scene units. The template probability gain subtracts mean off-diagonal report probability from mean diagonal probability; the attention gain applies the same diagonal-minus-off-diagonal comparison to final spatial masses. These contrasts measure how a change in the remembered sample changes the response to a fixed array.

For motion_A20_S12_D12_H12, accuracy was 95.3% (scene-bootstrap interval [93.0%, 97.7%]); 81.2% of scenes had all four queries answered correctly. Choice varied across queries in 100.0% of scenes. Template probability gain was 0.925 (95 percent interval [0.896, 0.951]) and final attention gain was 0.517 (interval [0.497, 0.537]). The sample-blank response is fixed within each array and therefore scores exactly 25 percent across the four balanced candidate labels.

For motion_A90_S12_D12_H12, accuracy was 99.2% (scene-bootstrap interval [98.0%, 100.0%]); 96.9% of scenes had all four queries answered correctly. Choice varied across queries in 100.0% of scenes. Template probability gain was 0.987 (95 percent interval [0.971, 0.998]) and final attention gain was 0.703 (interval [0.685, 0.719]). The sample-blank response is fixed within each array and therefore scores exactly 25 percent across the four balanced candidate labels.

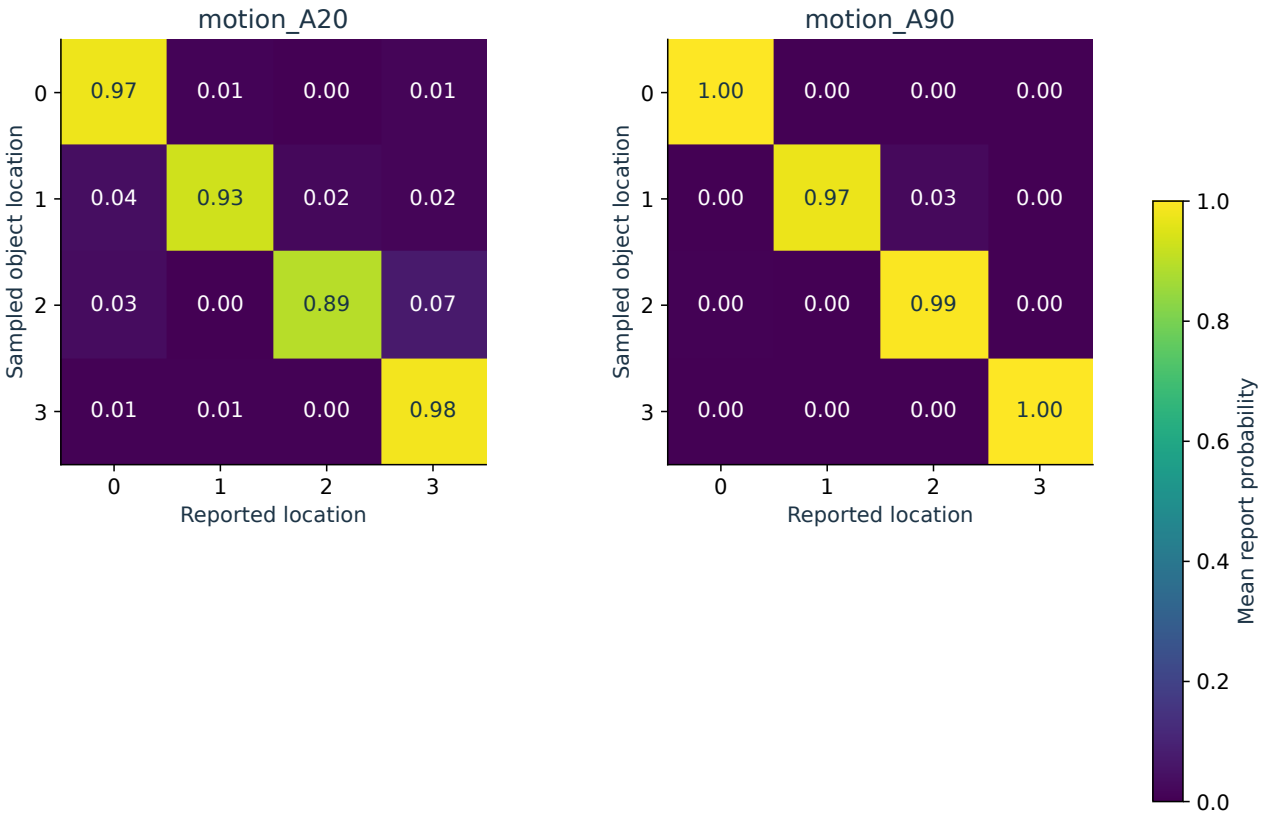


Figure 22: Causal template dependence with array pixels held fixed. Rows identify the candidate used as the remembered sample; columns identify report location. A diagonal pattern indicates that memory changes the selected answer appropriately. Each panel averages 64 matched scenes.

4 Discussion

Across 81 native conditions, mean accuracy is about 97.7 percent. The paired timing contrasts show no corrected benefit from extending the tested sample or array durations, consistent with high performance at the shorter settings. The sample and search candidates use independent dot layouts, so the task requires a motion template that survives a change in visual realization. At the middle delay, a capacity-matched linear decoder reconstructs the remembered direction with R-squared about 0.76 from global memory and 0.87 from spatial memory. Resetting global memory there reduces accuracy by about 73 percentage points in both selected direction-spacing conditions, whereas the spatial reset is less damaging. Target-priority inhibition during search separately costs about 70 to 90 percentage points. This combination identifies readable retained content, strong dependence on global memory, and a causal selection requirement in a successful search computation. Continuous direction reconstruction tests retention of that template, while the final location decoder tests its use after the candidates appear. Matched timing effects separate sample evidence from search evidence and intervening retention. The identical-array query switch provides a direct causal test of template dependence: only the preceding sample direction changes. Comparing its behavioral probability contrast with its attention contrast reveals whether memory influences spatial selection, the downstream decision, or both.

The behavioral result establishes 97.7% native accuracy for this trained subject across the tested condition distribution. The information result identifies cls as the strongest final regional readout in the representative condition under the fixed probe procedure. The causal result identifies distractor_+8_ evidence as the largest observed impairment in that condition. These three findings concern distinct levels of explanation: successful behavior, recoverable information and dependence on a manipulated component. Their conjunction is more informative than any one alone.

The most useful interpretation of a null manipulation is determined by its interval and by whether it altered an active component at a relevant time. Zero-active-frame interventions are identified in the data, and comparison memory can be exactly zero in short sequences. The shared softmax also makes priority changes competitive: increasing one region necessarily changes the relative mass elsewhere. Thus the causal object is the complete redistribution induced by the specified intervention.

Novel speeds, coherence levels and direction spacings would test the invariances of the motion template. Extending the identical-array query experiment to those transformations would test whether memory-guided selection follows direction across changed motion statistics.

4.1 Interpreting these results alongside biological experiments

4.1.1 Zaksas and Pasternak (2006)

The shared computation is retaining direction across a delay and comparing later motion with that content. Our independent sample and candidate dot layouts force transfer beyond the same dot arrangement. Mid-delay direction decoding and the large global-reset cost provide complementary evidence for a retained motion template. The biological task compared successive directions, whereas ours must choose among simultaneous locations. [6]

4.1.2 Chelazzi, Miller, Duncan and Desimone (1993)

The visual-search structure adds selection among candidates to memory comparison. Here the template is motion rather than a complex static picture. The strong performance across direction-spacing and

timing conditions shows that this architecture can use one kind of remembered content to guide a location decision, even while irregular static-object search remains difficult. [7]

4.1.3 Ling, Liu and Carrasco (2009)

Feature-based motion attention offers a bridge from remembered direction to distributed selection. Candidate directions close to the remembered sample should produce more competition than distant ones. Measuring target and neighbor allocation as a function of angular separation would connect the template's precision to spatial selection, while an external-noise manipulation would distinguish tuning precision from simple gain. [8]

4.2 A computational account of the observed behavior

A circular template can represent direction through its sine and cosine, allowing similarity to depend on angular distance rather than an arbitrary numeric label. The model need not use exactly this two-coordinate code; it can retain a distributed representation with the same geometry. At the middle delay, external direction reconstruction explains substantial variance in both global and spatial states. Resetting global memory produces a much larger behavioral cost than the spatial reset at that moment, identifying a particularly important carrier for subsequent matching. During search, target inhibition creates a separate major cost. These observations link content retention and spatial selection in a successful computation, rather than relying solely on a final correct answer. [9]

4.3 Where this task agrees with the biological evidence

The model agrees with an abstract motion memory that transfers across dot realizations and supports later selection. Its success contrasts with static object search and argues against a single universal failure of memory or attention. The likely difference concerns how each content type is encoded and made accessible to the query and comparison stages.

4.4 A follow-up experiment motivated by these results

Match motion and static search for candidate similarity and baseline difficulty. Compare sample-switch effects on allocation, memory-to-query decodability and paired state interventions. Test whether adding uncertainty to motion templates recreates the static-search pattern of retained information with poor use.

4.5 Scope of inference

This series studies one trained parameter realization. Its uncertainty intervals describe stimulus sampling for that realization. Threshold-defined latencies are in model frames and reflect the existing classifier applied to prefix states. Spatial doses are changes to computational priority logits, and memory perturbations act on recurrent tensors. Independent training seeds, calibrated behavioral clocks and biological intervention models are the next levels of replication. The empirical results, code and task definitions are sufficient to reproduce the computational experiments reported here.

References

[1] Salzman CD, Britten KH, Newsome WT (1990). Cortical microstimulation influences perceptual judgements of motion direction. Nature 346:174-177. https://pubmed.ncbi.nlm.nih.gov/2366872/.

[2] Chelazzi L, Miller EK, Duncan J, Desimone R (1993). A neural basis for visual search in inferior temporal cortex. Nature 363:345-347. https://pubmed.ncbi.nlm.nih.gov/8497317/.

[3] Funahashi S, Bruce CJ, Goldman-Rakic PS (1989). Mnemonic coding of visual space in the monkey dorsolateral prefrontal cortex. Journal of Neurophysiology 61:331-349. https://pubmed.ncbi.nlm.nih.gov/2918358/.

[4] King JR, Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18:203-210. https://pmc.ncbi.nlm.nih.gov/articles/PMC5635958/.

[5] Stokes MG et al. (2013). Dynamic coding for cognitive control in prefrontal cortex. Neuron 78:364-375. https://pubmed.ncbi.nlm.nih.gov/23562541/.

[6] Zaksas and Pasternak (2006). Directional signals in the prefrontal cortex and in area MT during a working memory for visual motion task. Primary source.

[7] Chelazzi, Miller, Duncan and Desimone (1993). A neural basis for visual search in inferior temporal cortex. Primary source.

[8] Ling, Liu and Carrasco (2009). How spatial and feature-based attention affect the gain and tuning of population responses. Primary source.

[9] Wolfe (2021). Guided Search 6.0: An updated model of visual search. Primary source.

A Complete native condition results

Condition	n	Accuracy	AUC	Brier
motion_A20_S3_D0_H3	128	96.9%	0.998	0.058
motion_A20_S3_D0_H6	128	97.7%	0.999	0.035
motion_A20_S3_D0_H12	128	97.7%	0.999	0.042
motion_A20_S3_D4_H3	128	96.1%	0.998	0.056
motion_A20_S3_D4_H6	128	96.9%	0.998	0.049
motion_A20_S3_D4_H12	128	96.9%	0.999	0.039
motion_A20_S3_D12_H3	128	95.3%	0.997	0.073
motion_A20_S3_D12_H6	128	96.1%	0.996	0.063
motion_A20_S3_D12_H12	128	96.9%	0.999	0.043
motion_A20_S6_D0_H3	128	97.7%	0.998	0.051
motion_A20_S6_D0_H6	128	96.9%	0.999	0.046
motion_A20_S6_D0_H12	128	97.7%	0.999	0.043
motion_A20_S6_D4_H3	128	94.5%	0.999	0.060
motion_A20_S6_D4_H6	128	96.9%	0.999	0.052
motion_A20_S6_D4_H12	128	98.4%	0.999	0.037
motion_A20_S6_D12_H3	128	96.1%	0.998	0.064
motion_A20_S6_D12_H6	128	95.3%	0.998	0.067
motion_A20_S6_D12_H12	128	96.1%	0.999	0.046
motion_A20_S12_D0_H3	128	96.1%	0.998	0.060
motion_A20_S12_D0_H6	128	96.9%	0.999	0.045
motion_A20_S12_D0_H12	128	96.9%	0.999	0.044
motion_A20_S12_D4_H3	128	94.5%	0.999	0.065
motion_A20_S12_D4_H6	128	97.7%	0.999	0.048
motion_A20_S12_D4_H12	128	96.9%	0.999	0.046
motion_A20_S12_D12_H3	128	95.3%	0.998	0.069
motion_A20_S12_D12_H6	128	95.3%	0.999	0.063
motion_A20_S12_D12_H12	128	96.1%	0.999	0.058
motion_A45_S3_D0_H3	128	97.7%	0.999	0.042
motion_A45_S3_D0_H6	128	98.4%	1.000	0.034
motion_A45_S3_D0_H12	128	96.9%	0.998	0.046
motion_A45_S3_D4_H3	128	96.9%	0.999	0.049
motion_A45_S3_D4_H6	128	96.9%	1.000	0.059
motion_A45_S3_D4_H12	128	96.9%	0.998	0.055
motion_A45_S3_D12_H3	128	96.9%	0.998	0.048
motion_A45_S3_D12_H6	128	96.9%	0.999	0.063
motion_A45_S3_D12_H12	128	96.9%	0.998	0.055
motion_A45_S6_D0_H3	128	96.9%	1.000	0.047
motion_A45_S6_D0_H6	128	96.9%	1.000	0.058
motion_A45_S6_D0_H12	128	96.9%	0.997	0.055
motion_A45_S6_D4_H3	128	96.9%	0.999	0.048
motion_A45_S6_D4_H6	128	96.9%	0.999	0.058
motion_A45_S6_D4_H12	128	96.1%	0.997	0.054
motion_A45_S6_D12_H3	128	96.9%	0.998	0.044
motion_A45_S6_D12_H6	128	96.9%	0.998	0.060
motion_A45_S6_D12_H12	128	96.1%	0.997	0.057
motion_A45_S12_D0_H3	128	97.7%	1.000	0.031
motion_A45_S12_D0_H6	128	97.7%	0.999	0.046
motion_A45_S12_D0_H12	128	96.9%	0.997	0.050
motion_A45_S12_D4_H3	128	97.7%	0.999	0.036
motion_A45_S12_D4_H6	128	97.7%	0.999	0.050
motion_A45_S12_D4_H12	128	96.9%	0.997	0.053

Condition	n	Accuracy	AUC	Brier
motion_A45_S12_D12_H3	128	97.7%	0.999	0.038
motion_A45_S12_D12_H6	128	96.9%	0.998	0.059
motion_A45_S12_D12_H12	128	96.1%	0.996	0.059
motion_A90_S3_D0_H3	128	98.4%	0.999	0.026
motion_A90_S3_D0_H6	128	100.0%	1.000	0.000
motion_A90_S3_D0_H12	128	100.0%	1.000	0.000
motion_A90_S3_D4_H3	128	99.2%	1.000	0.015
motion_A90_S3_D4_H6	128	100.0%	1.000	0.000
motion_A90_S3_D4_H12	128	100.0%	1.000	0.000
motion_A90_S3_D12_H3	128	99.2%	1.000	0.015
motion_A90_S3_D12_H6	128	100.0%	1.000	0.000
motion_A90_S3_D12_H12	128	100.0%	1.000	0.000
motion_A90_S6_D0_H3	128	99.2%	1.000	0.016
motion_A90_S6_D0_H6	128	100.0%	1.000	0.000
motion_A90_S6_D0_H12	128	100.0%	1.000	0.000
motion_A90_S6_D4_H3	128	99.2%	1.000	0.015
motion_A90_S6_D4_H6	128	100.0%	1.000	0.000
motion_A90_S6_D4_H12	128	100.0%	1.000	0.000
motion_A90_S6_D12_H3	128	99.2%	1.000	0.015
motion_A90_S6_D12_H6	128	100.0%	1.000	0.000
motion_A90_S6_D12_H12	128	100.0%	1.000	0.000
motion_A90_S12_D0_H3	128	99.2%	1.000	0.014
motion_A90_S12_D0_H6	128	100.0%	1.000	0.000
motion_A90_S12_D0_H12	128	100.0%	1.000	0.000
motion_A90_S12_D4_H3	128	99.2%	1.000	0.015
motion_A90_S12_D4_H6	128	100.0%	1.000	0.000
motion_A90_S12_D4_H12	128	100.0%	1.000	0.000
motion_A90_S12_D12_H3	128	99.2%	1.000	0.014
motion_A90_S12_D12_H6	128	100.0%	1.000	0.000
motion_A90_S12_D12_H12	128	100.0%	1.000	0.000

B Every-frame attention atlas

Condition motion_A20_S12_D12_H12. The first two rows show one exact native trial and its own attention overlay. The third row averages attention over 32 trials sharing the selected location/cue and response label. Every model frame is included. Averaging unrelated object identities is restricted to the attention map; the displayed RGB movie remains one real trial.

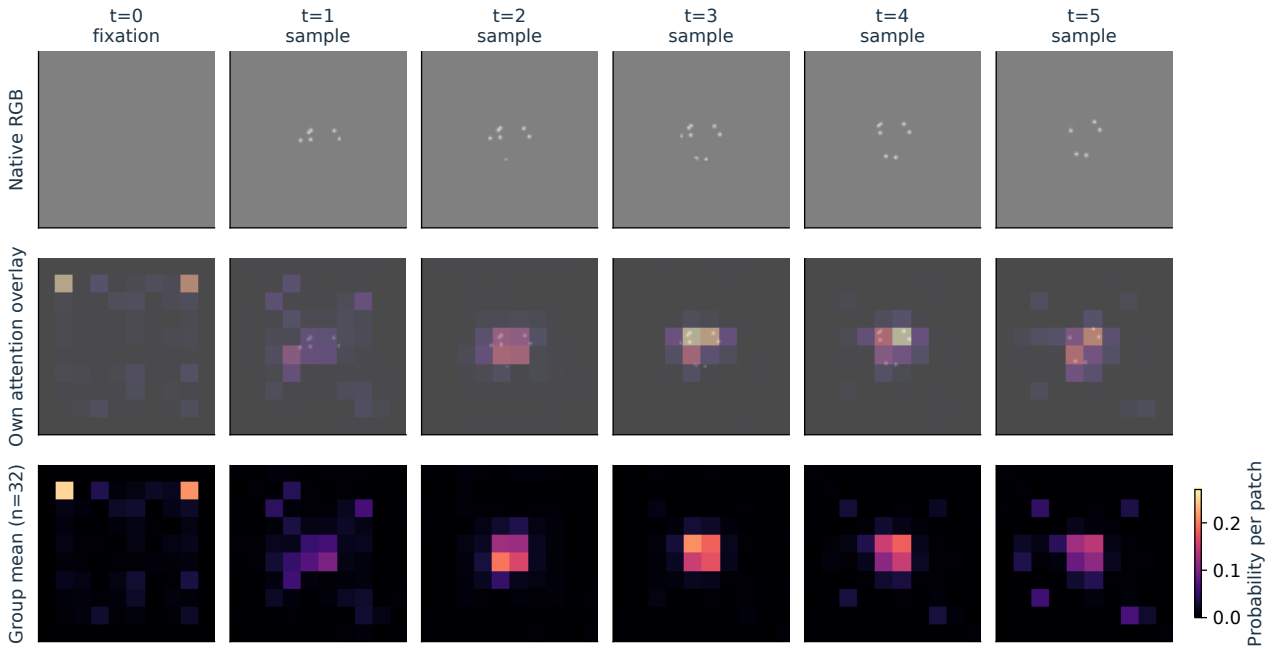


Figure 23: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

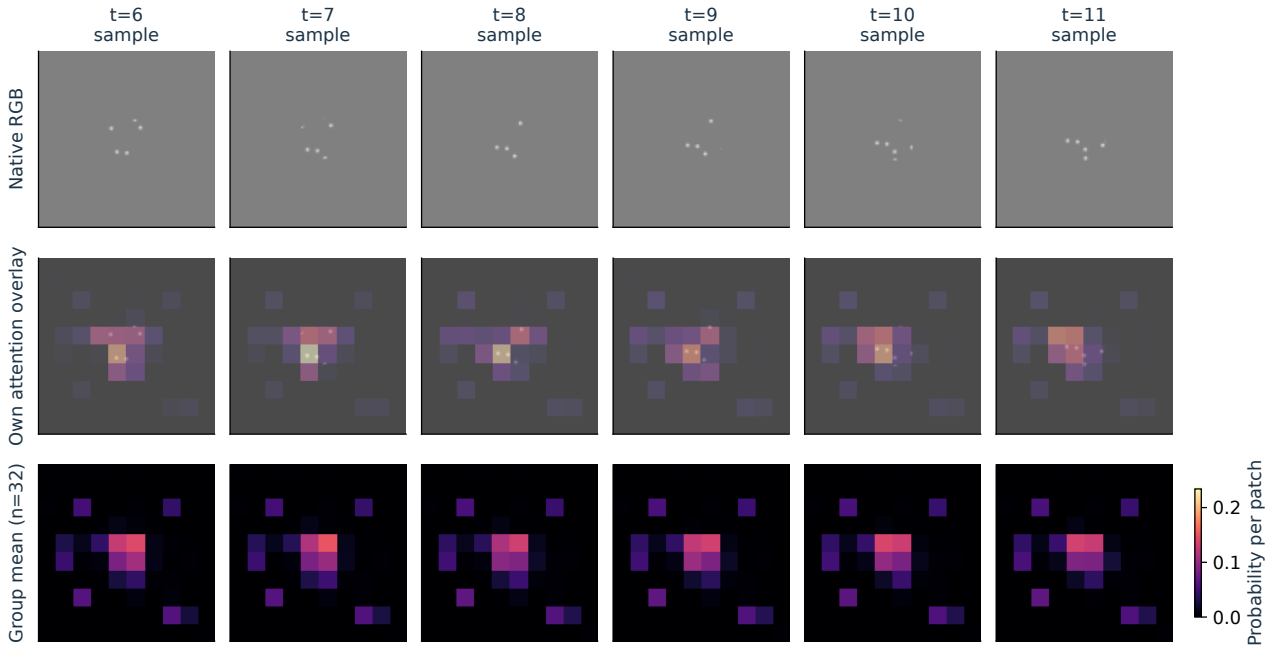


Figure 24: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

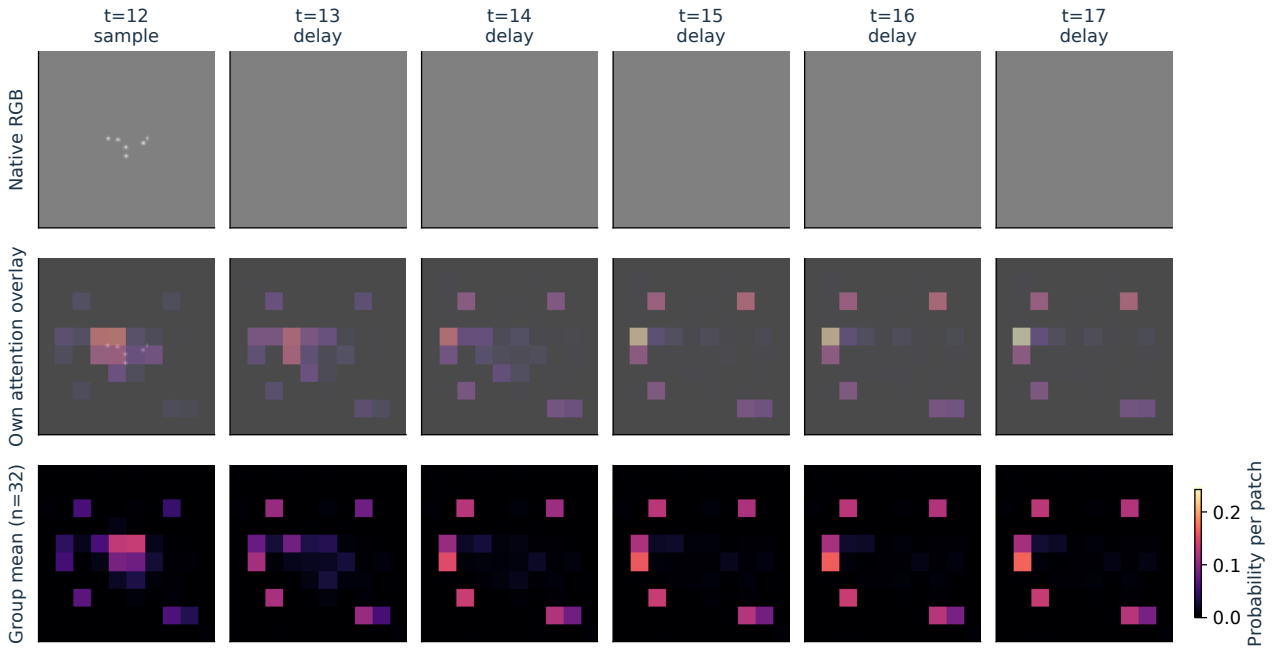


Figure 25: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

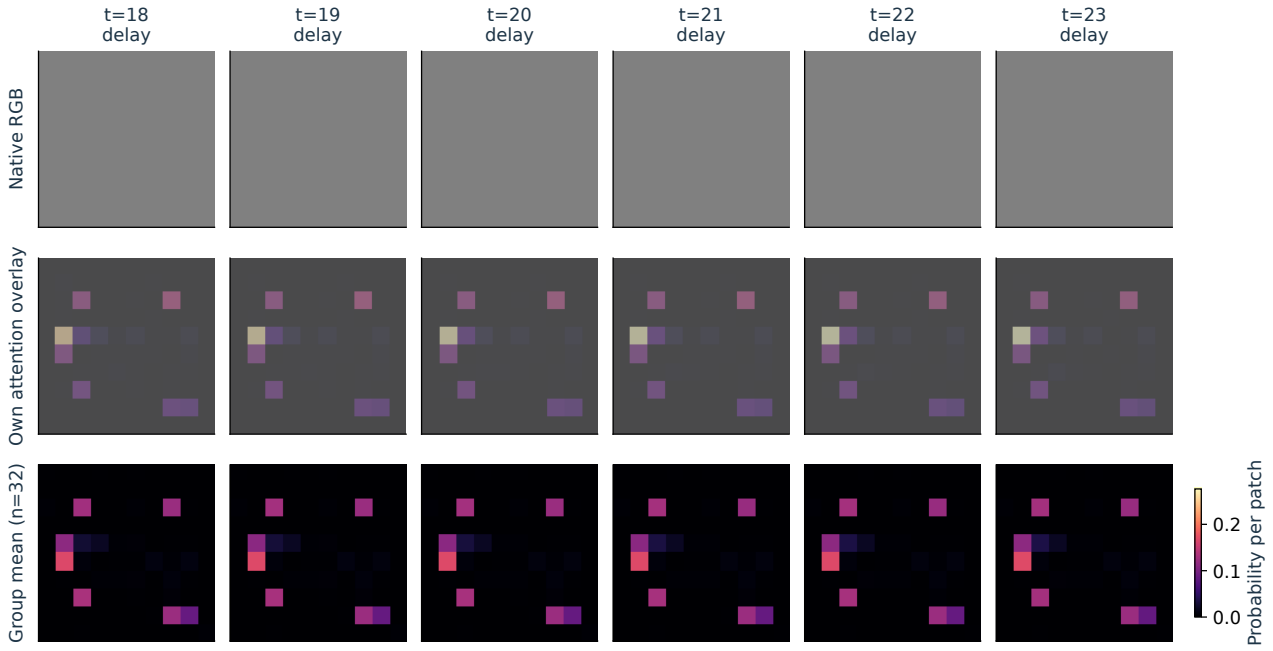


Figure 26: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

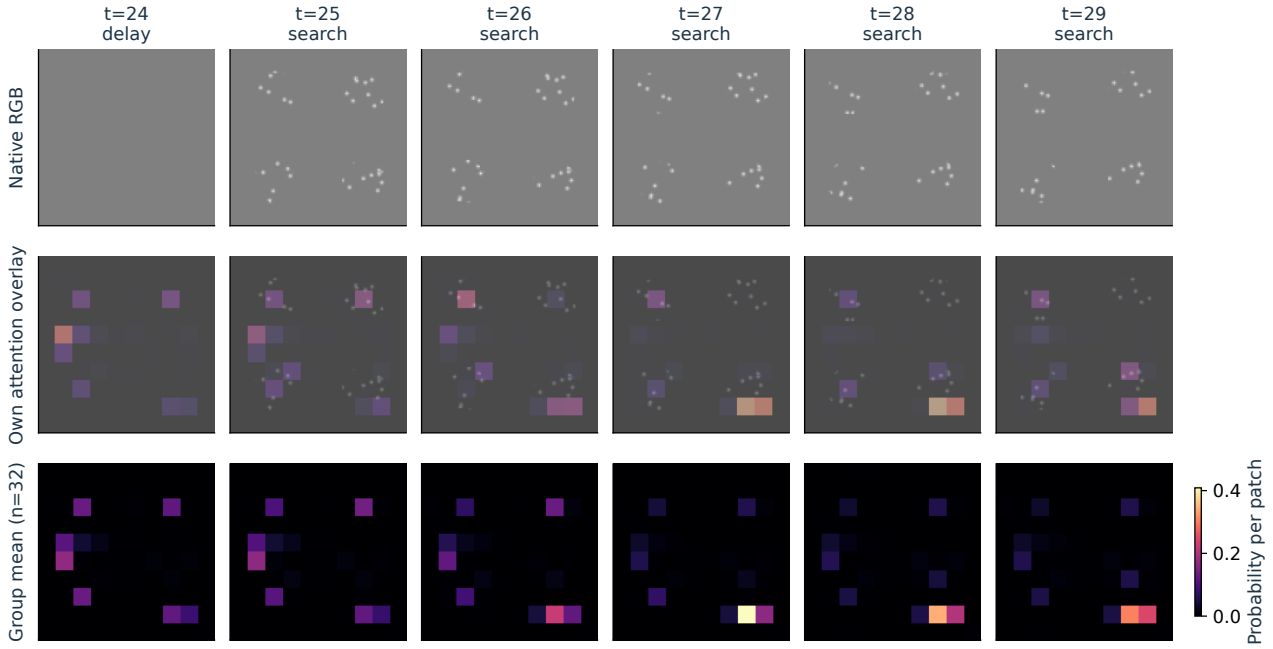


Figure 27: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

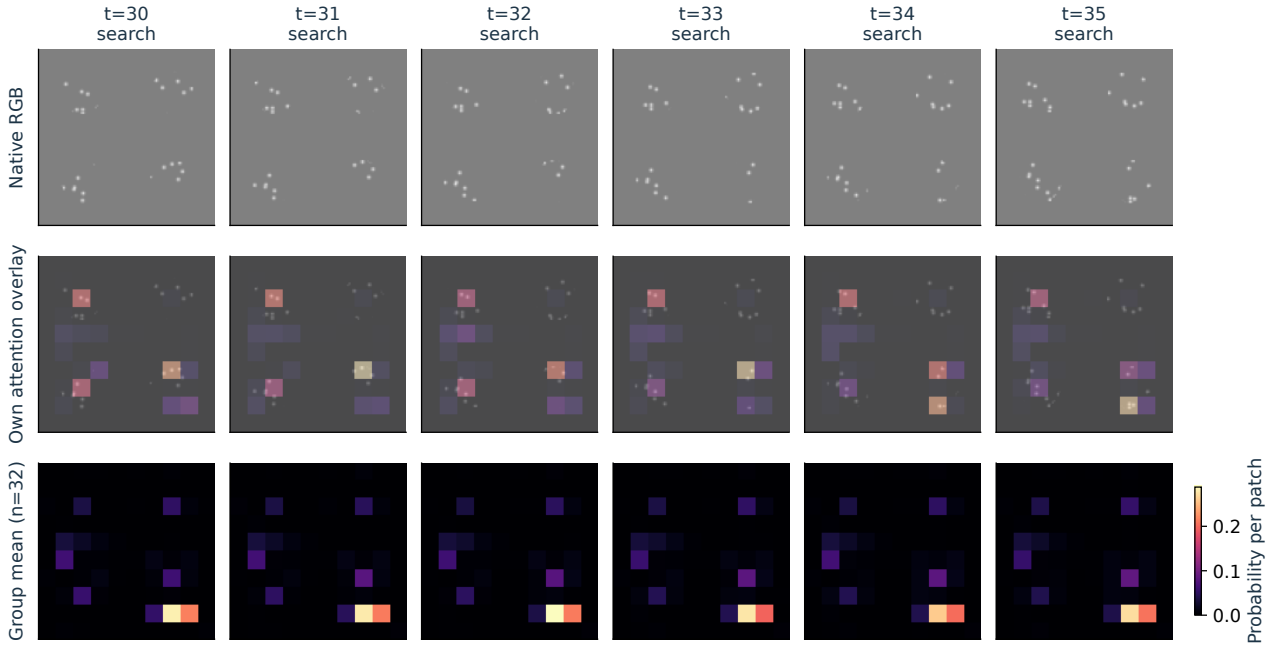


Figure 28: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

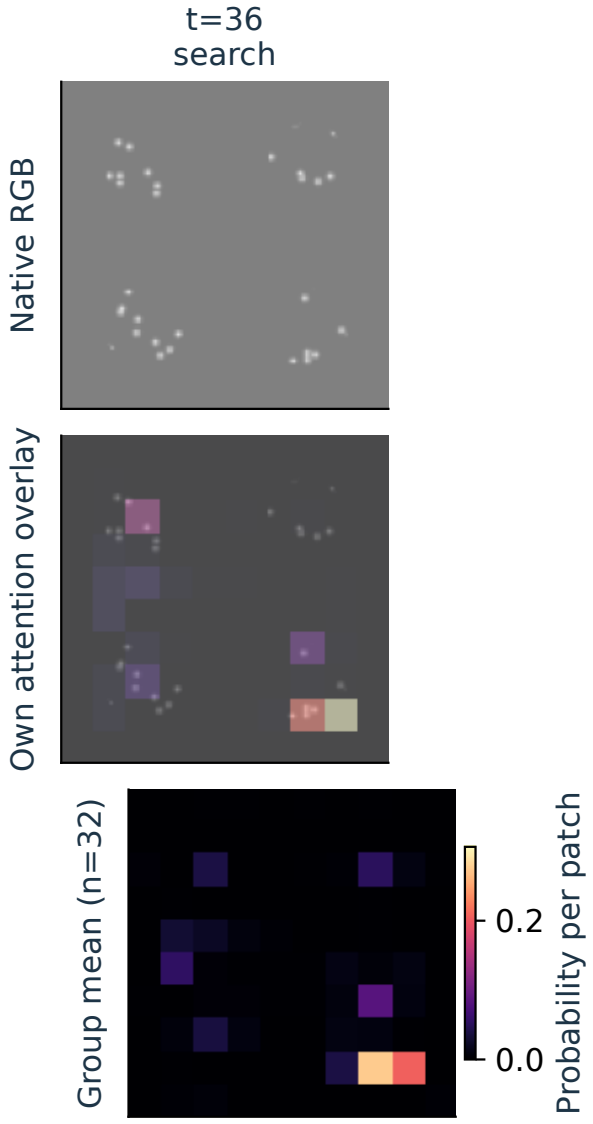


Figure 29: Native frames, own attention overlay and condition-group spatial projection. Shared color scaling within each panel represents attention probability per patch.

C Condition-by-time attention: all frames

Each panel retains the same condition definitions and color scale as the main attention section. Frames are aligned to the task-specific evidence onset. Gray cells mark times outside a particular trial, rather than missing observations. All observed frames are displayed.

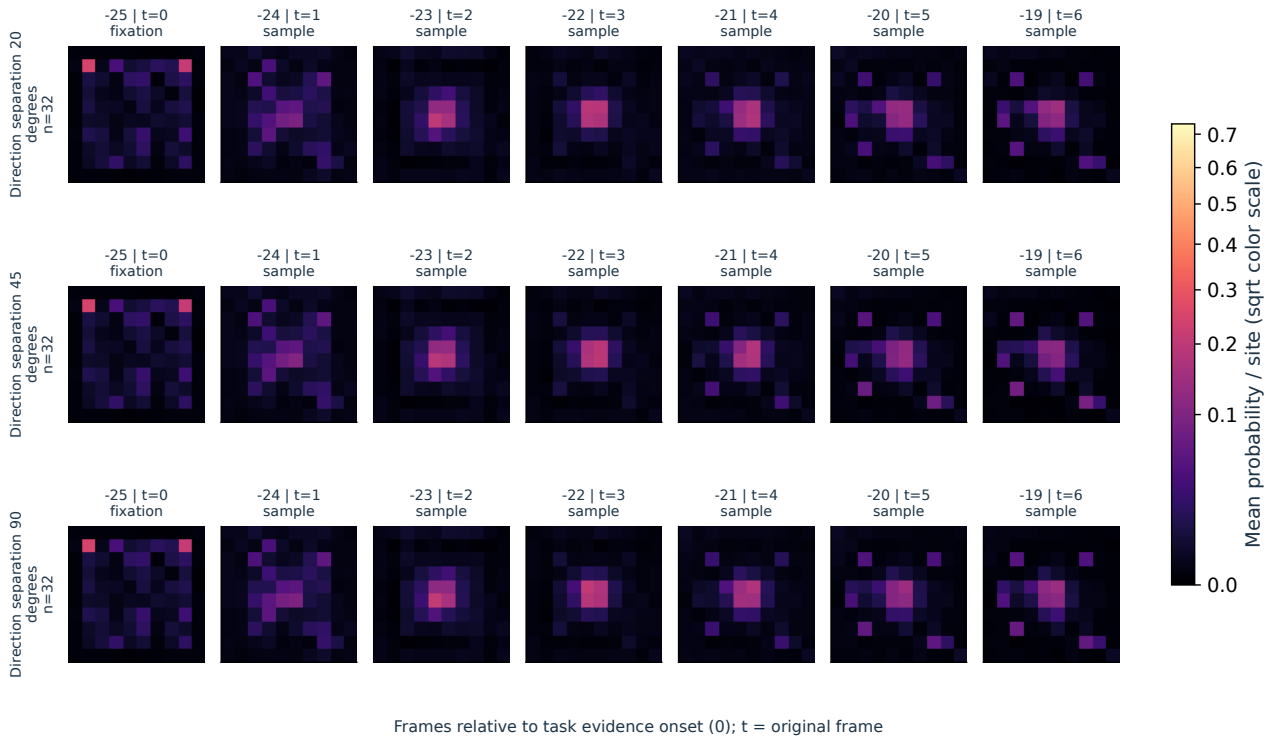


Figure 30: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

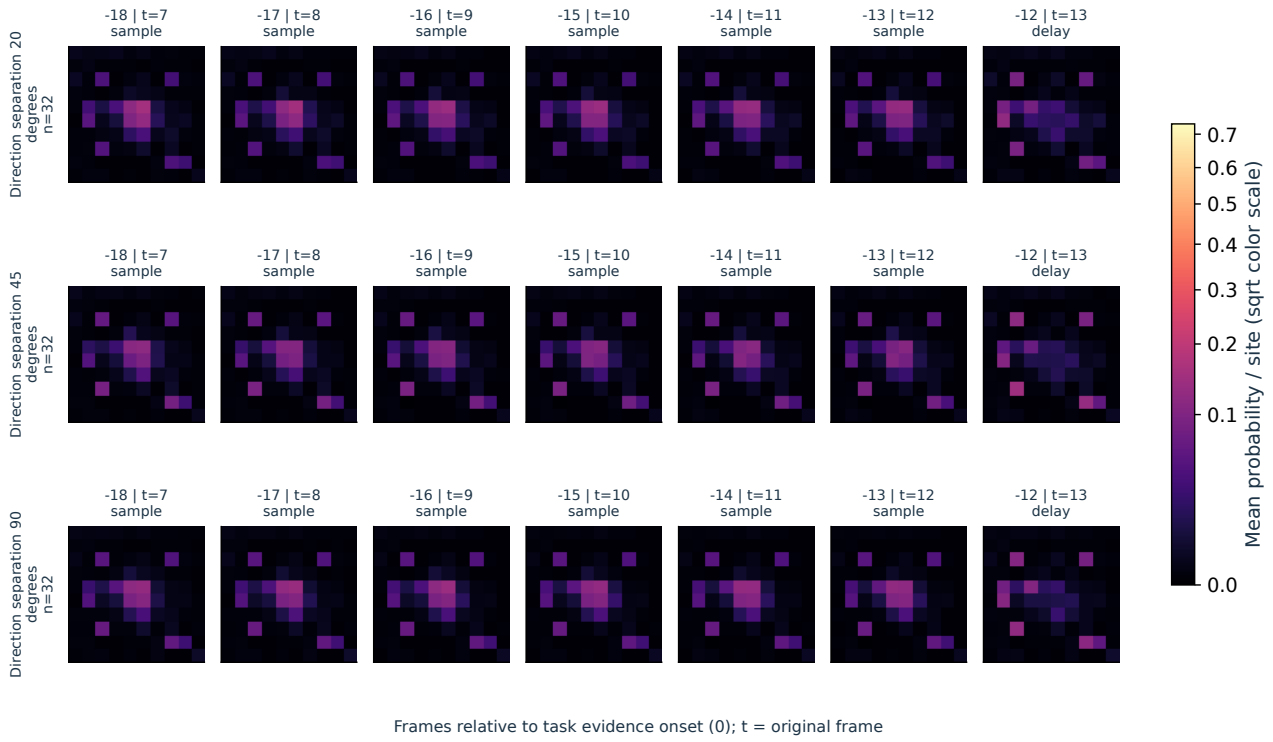


Figure 31: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

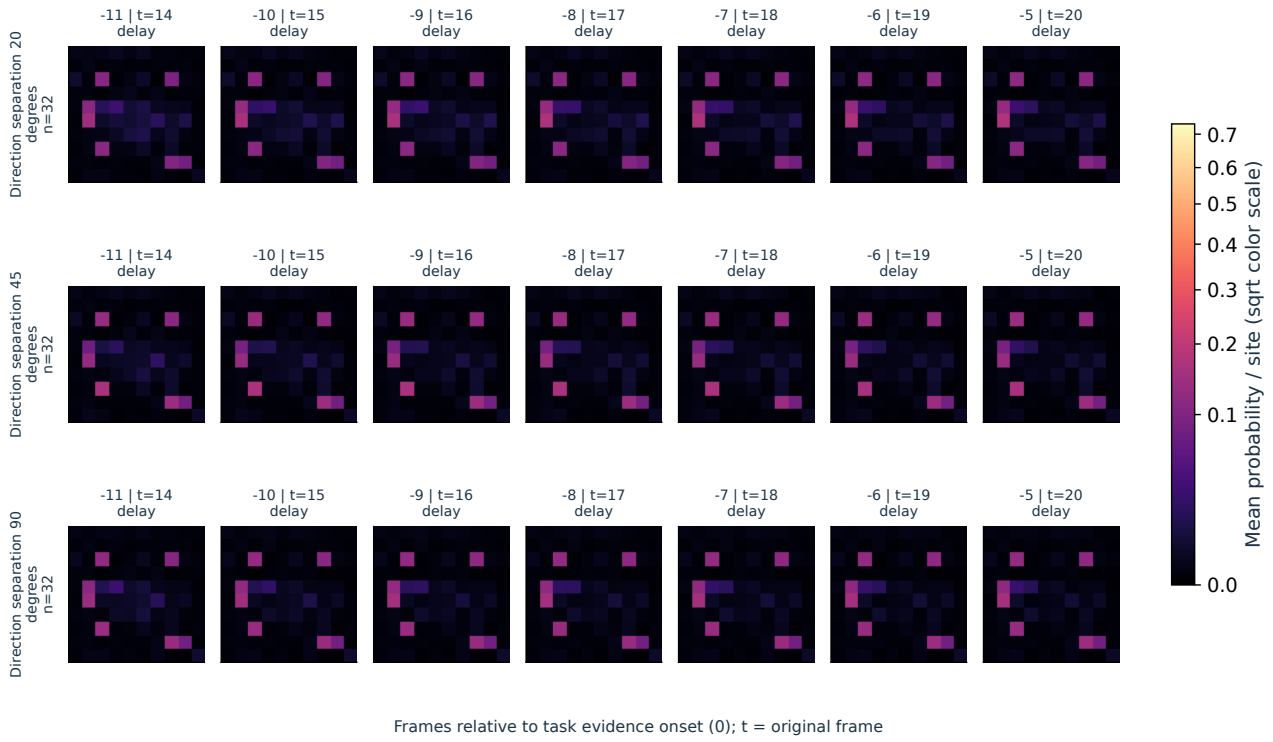


Figure 32: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

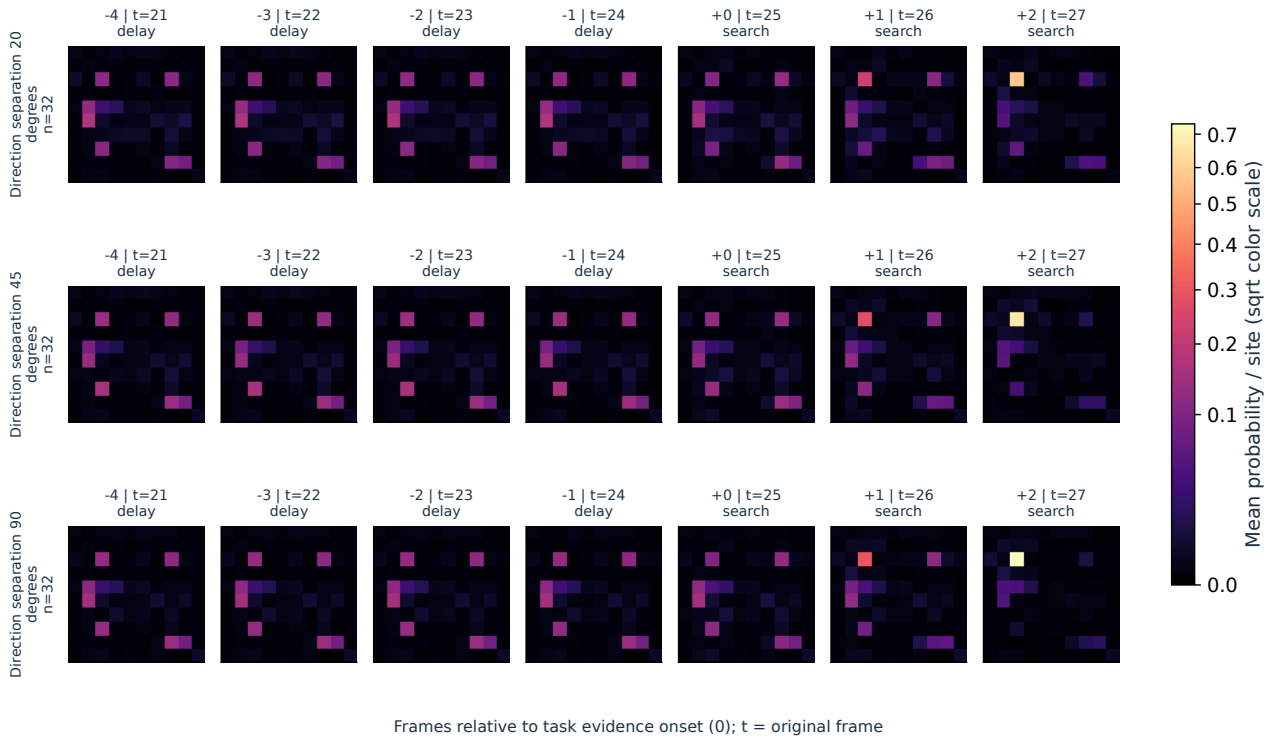


Figure 33: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

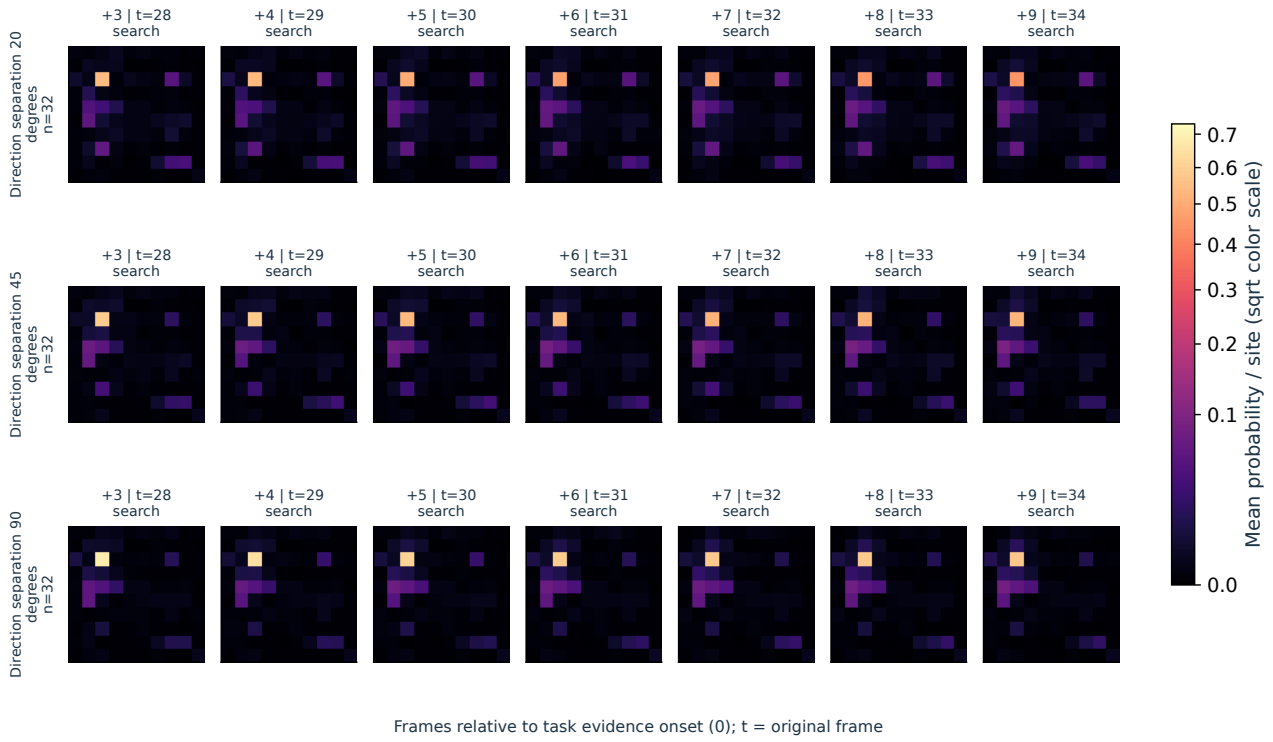


Figure 34: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

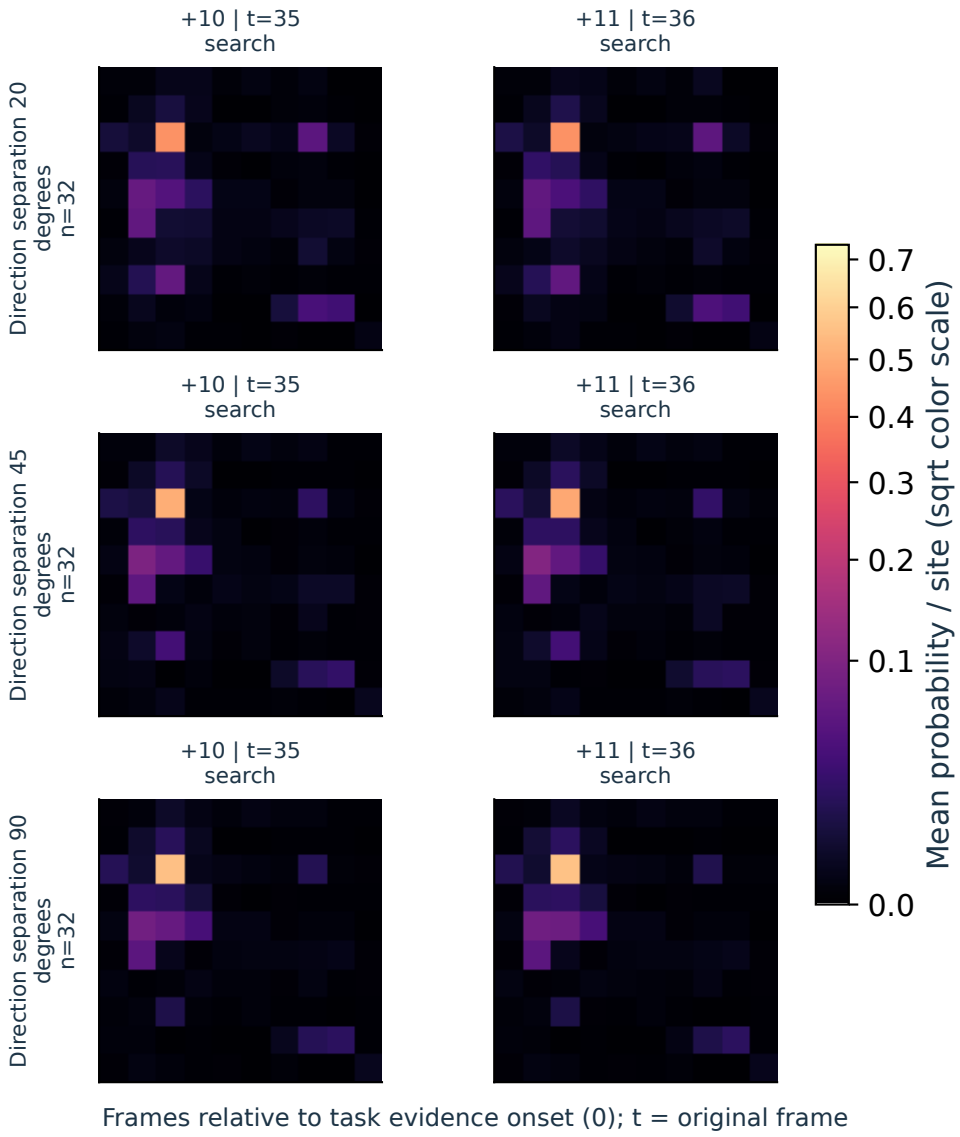


Figure 35: Complete condition-average spatial trajectory. Rows are conditions, columns are aligned frames, and each title also records the original frame and task phase. Sample counts and a shared probability scale accompany every panel.

D Reproducibility record

Main checkpoint: update 204,289. Final robustness checkpoint: update 220,637.

best: `596bc351783b3ddbfa8d37a9cf8d6514`

`6d1333dcc0097d40f0f75665b6451bcf`

latest: `c13338ff6958c5dbb617914da538bca1`

`39944b89122b848b72885ed3241995d4`

Data namespace 205903; probe fitting 205904; probe testing 205905; interventions 205906; difficulty challenges 205907; VDA counterfactuals 205908. The analysis package is `SecondPass/TaskNeuroscience`. The protocol, source hashes, parity checks, per-trial NPZ files, metadata, statistical JSON files and editable LaTeX sources are retained beside this paper. Figures and text are generated directly from completed analysis results.